

Mathematical Logic: From Formal Language to Gödel and Cohen
Volumes I–X

Student Edition

Contents

How to read the series	vii
Notation and conventions	ix
I Volume I: Propositional and First-Order Logic	1
Preface	2
1 Syntax and propositional semantics	3
2 Propositional proof systems	8
3 First-order syntax, substitution, and equality	12
4 Structures and satisfaction	16
5 Formal deduction	20
6 Henkin completeness	25
7 Compactness and Löwenheim–Skolem	30
End-of-volume perspective	35
II Volume II: Computability and Arithmetized Syntax	36
Preface	37
8 Primitive recursion	38
9 Coding finite objects	42
10 Recursive and computably enumerable relations	46
11 Weak arithmetic	49

12 Representability	53
13 Gödel numbering of syntax	57
14 Proof and provability predicates	61
End-of-volume perspective	64
III Volume III: Incompleteness, Provability, and Truth	65
Preface	66
15 Self-reference	67
16 Gödel's first theorem	72
17 Rosser and essential undecidability	77
18 Consistency and soundness notions	82
19 Formalized provability	86
20 Löb and second incompleteness	90
21 Truth, reflection, and iterated consistency	95
End-of-volume perspective	100
IV Volume IV: ZF Set Theory and the Transfinite	101
Preface	102
22 ZF axioms and elementary constructions	103
23 Natural numbers and recursion	108
24 Foundation and rank	112
25 Ordinals	117
26 Ordinal arithmetic	121
27 Cardinals without assuming choice	126
28 Well-founded relations and collapse	129

End-of-volume perspective	133
V Volume V: Choice, Models, Satisfaction, and Absoluteness	134
Preface	135
29 Axiom of Choice and equivalents	136
30 Cardinal arithmetic with choice	139
31 Satisfaction for set-sized structures	143
32 Absoluteness	147
33 Reflection	151
34 Elementary hulls	154
35 Inner models and the Skolem paradox	158
End-of-volume perspective	162
VI Volume VI: The Constructible Universe	163
Preface	164
36 Definability and $\text{Def}(M)$	165
37 The L -hierarchy	169
38 Uniform definability and absoluteness of L	173
39 ZF axioms in L , part I	179
40 ZF axioms in L , part II	184
41 Canonical well-order and Choice	188
42 Condensation	192
43 GCH in L	199
End-of-volume perspective	204

VII	Volume VII: Forcing I: Generic Extensions and the Forcing Theorem	205
	Preface	206
	44 Posets, dense sets, and generics	207
	45 Names and valuation	212
	46 Atomic forcing	217
	47 General forcing relation	222
	48 Truth and generic extension theorem	230
	49 ZFC in generic extensions	237
	End-of-volume perspective	244
VIII	Volume VIII: Forcing II: Boolean Values, Preservation, and Relative Consistency	245
	Preface	246
	50 Separative quotients and Boolean completion	247
	51 Boolean-valued semantics	254
	52 Boolean-valued ZFC	263
	53 Consistency transfer	277
	54 Chain conditions and closure	283
	55 Nice names and preservation bookkeeping	287
	End-of-volume perspective	291
IX	Volume IX: Cohen Forcing and Independence of CH	292
	Preface	293
	56 Adding Cohen reals	294
	57 The ccc	299
	58 Lower bound for the continuum	303

59 Upper bound by nice names	307
60 CH and its failure	312
61 Homogeneity and relative consistency	316
End-of-volume perspective	320
 X Volume X: Symmetric Extensions and Independence of Choice	 321
Preface	322
62 Fraenkel–Mostowski precursor	323
63 Automorphisms of forcing and names	327
64 Symmetric and hereditarily symmetric names	332
65 ZF in a symmetric extension, elementary axioms	349
66 ZF in a symmetric extension, hard axioms	355
67 The basic Cohen symmetric model	362
68 Failure of Choice	367
End-of-volume perspective	379
Glossary of recurring terms and notation	380

How to read the series

The ten volumes form two large arcs. Volumes I–III develop logic, effective syntax, and incompleteness. Volumes IV–X develop set theory, constructibility, forcing, and the independence of CH and AC. The books are cumulative, but not every chapter has to be read with the same intensity on a first pass.

For a first route to Gödel’s theorems, read Volume I through completeness, then the coding and provability chapters of Volume II, followed by all of Volume III. For a first route to Cohen’s theorems, read the transfinite core of Volume IV, the satisfaction/absoluteness/hull chapters of Volume V, then Volumes VI–X in order.

Major theorem roadmap

The following landmarks indicate where the central proof dependencies close. Page numbers refer to the unified binder.

First-order completeness (p. 28)

Henkin witnesses and the quotient term model convert consistency into satisfiability.

Downward Löwenheim–Skolem (p. 33)

Skolem closure plus Tarski–Vaught gives elementary submodels of controlled size.

Diagonal lemma (p. 68)

Represented substitution produces formal self-reference.

Gödel I (p. 75)

The canonical Gödel sentence is undecided under the original consistency hypotheses.

Löb’s theorem (p. 90) and Gödel II (p. 92)

The derivability conditions turn arithmetized provability into the second incompleteness theorem.

Reflection (p. 152) and Skolem hulls (p. 155)

These are the model-theoretic tools later used in constructibility.

$L \models \text{ZF}$ (p. 186)

Separation, Power Set, Collection, and Replacement are verified inside the constructible universe.

Condensation (p. 195) and $L \models \text{GCH}$ (p. 202)

Small elementary hulls collapse to earlier L -levels, yielding the cardinal localization needed for GCH.

Truth Lemma (p. 230) and Fundamental Theorem of Forcing (p. 232)

Truth in a generic extension is matched with the ground-model forcing relation.

Formal consistency transfer (p. 280)

Boolean-valued/finite-fragment semantics supplies the genuine $\text{Con}(T) \Rightarrow \text{Con}(T + \varphi)$ bridge.

Cohen continuum calculation (p. 313)

ccc preservation, coordinate reals, and nice-name counting yield $2^{\aleph_0} = \aleph_2$ in the target extension.

Symmetric Power Set (p. 356) and failure of Choice (p. 372)

The symmetric submodel first satisfies ZF; only then does the finite-support argument produce $\neg\text{AC}$.

Conventions for hypotheses

Whenever a model M is assumed transitive, externally countable, or a model of Choice, that hypothesis is meant literally and is not inferred merely from consistency. Likewise, consistency, ω -consistency, Σ_1 -soundness, and full soundness are distinct notions. Later chapters rely on these distinctions.

Notation and conventions

Throughout the series, structures are written with roman letters such as M, N , while theories are usually T, S . Semantic consequence is $T \models \varphi$ and formal derivability is $T \vdash \varphi$. The symbol $\text{Con}(T)$ always denotes the canonical arithmetical consistency sentence attached to the proof predicate fixed in Volume II; it is not an informal metalinguistic abbreviation.

For set theory, Ord is the class of ordinals, V_α is the cumulative hierarchy, and L_α is the constructible hierarchy. Cardinal successors are written κ^+ , always in the model currently under discussion unless a superscript such as κ^{+L} is displayed.

For forcing, the convention is that $q \leq p$ means that q is stronger than p . Every forcing notion used in Volumes VII–X has a largest (weakest) condition $1_{\mathbb{P}}$; when a given poset has no largest condition, one is adjoined before names are defined. A generic extension is $M[G]$; canonical ground-model names are $\check{x}$. In Boolean-valued semantics, $\|\varphi\|$ is the Boolean truth value of φ . In symmetric extensions, N denotes the symmetric submodel of the ambient generic extension.

Historical names are used only after the underlying assertion has been stated precisely. Source notes indicate provenance and useful comparison points; they are never substitutes for proofs in the text.

A five-question checklist for reading a dense formula

When a displayed line feels opaque, pause before manipulating it and answer these five questions.

1. **What is the type of each object?** Is it a formula, a natural-number code, a set, an ordinal, a forcing name, a condition, a Boolean value, or an actual object of a model?
2. **Which universe or theory is speaking?** Distinguish assertions inside T , truth in a structure M , truth in an extension $M[G]$, and statements in the surrounding metatheory.
3. **What operation occurs on this line?** Substitution, recursion, coding, valuation, forcing, collapse, taking a supremum, or cardinal counting should be named before it is used.
4. **What licenses the operation?** Look for the induction hypothesis, freshness condition, rank decrease, density, genericity, transitivity, chain condition, or consistency assumption.
5. **What is the local goal?** A long proof becomes easier when each displayed transformation is read as solving one small typed subproblem rather than as a mysterious global leap.

Part I

Volume I: Propositional and First-Order Logic

Preface

We begin with the formal distinction between syntax and semantics. Every recursive definition is made explicit, because later Gödel coding depends on the exact shape of syntax.

Position in the series

This opening volume supplies the proof-theoretic and model-theoretic language used everywhere else. Its completeness and Löwenheim–Skolem results are prerequisites for the metamathematical and set-theoretic model constructions that follow.

Sources and further reading

The presentation follows the standard Henkin route to completeness, with the term-model proof written so that equality and witness introduction are visible. For historical context see [1]; for a modern systematic treatment see [2].

Notation for this volume

$\models$ denotes semantic consequence and $\vdash$ formal derivability. Structures are denoted by M, N ; assignments by s .

How to read symbolic arguments in this volume

When a formula appears, read it in layers rather than as one visual block. First identify the *ambient language*: are we speaking inside a formal theory, about a structure, or in the surrounding metatheory? Next mark the objects being manipulated (formulas, codes, sets, names, conditions, ordinals) and the operation being applied to them. Finally check the side conditions—free variables, rank bounds, compatibility, genericity, transitivity, or consistency hypotheses. Whenever a displayed derivation changes level, the text says so explicitly. A displayed line is therefore meant to be read like a line of well-commented code: the symbols perform the operation, and the surrounding prose records its type, preconditions, and purpose.

Chapter 1

Syntax and propositional semantics

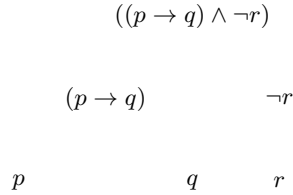


Figure 1.1: A parse tree makes structural recursion visible. A recursive definition evaluates the leaves first and then moves upward through the constructors.

Definitions and setup

Formal definitions used in this chapter

Propositional formulas and notation. Fix a countable set PropVar of propositional variables. In the surrounding metatheory define finite construction stages by

$$\text{Form}_{\text{prop}}^{(0)} := \text{PropVar},$$

$$\text{Form}_{\text{prop}}^{(n+1)} := \text{Form}_{\text{prop}}^{(n)} \cup \{\neg\varphi : \varphi \in \text{Form}_{\text{prop}}^{(n)}\} \cup \{(\varphi \circ \psi) : \varphi, \psi \in \text{Form}_{\text{prop}}^{(n)}, \circ \in \{\wedge, \vee, \rightarrow\}\}.$$

Then

$$\text{Form}_{\text{prop}} := \bigcup_{n < \omega} \text{Form}_{\text{prop}}^{(n)}. \quad (\text{PF})$$

A *propositional formula* is an element of this explicitly constructed set. The symbol $\leftrightarrow$ is *not* an additional primitive connective. Throughout the series it is the syntactic abbreviation

$$\varphi \leftrightarrow \psi \quad := \quad (\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi). \quad (\text{BI})$$

Whenever a proof code or a recursive syntax operation encounters a displayed biconditional, (BI) is expanded before coding; there is no separate biconditional constructor tag.

Parse tree and structural notions. A *parse tree* for a formula is a finite rooted tree witnessing one construction of the formula from variables by the displayed constructors. Its root is the formula and the children of a non-atomic node are the immediate constituent formula(s) used at that construction step. A formula is *atomic* exactly when it is a propositional variable; otherwise it is non-atomic. For a chosen parse tree, the *outermost connective* is the connective labeling its root. Lemma 1.1 proves that the parse tree, and hence the outermost connective and immediate constituents, are unique.

Literals and syntactic normal forms. A literal is a propositional variable or its negation. A *disjunctive normal form* (DNF) is a disjunction of conjunctions of literals, and a *conjunctive normal form* (CNF) is a conjunction of disjunctions of literals.

Base truth assignment. A base truth assignment is a function

$$v_0 : \text{PropVar} \longrightarrow \{0, 1\}.$$

The structural-recursion theorem below proves that every such v_0 has a unique truth-functional extension to all formulas. The term *valuation* and semantic consequence notation are introduced only after that existence–uniqueness theorem.

Terminating algorithm and decision procedure. A terminating algorithm is a finite, mechanically specified sequence of elementary steps that halts on every allowed input. A decision procedure for a yes/no property P is a terminating algorithm that returns “yes” exactly on inputs satisfying P and “no” on all other allowed inputs.

Before using the familiar word “least,” verify it from the stage construction. Every propositional variable lies in $\text{Form}_{\text{prop}}^{(0)}$, hence in $\text{Form}_{\text{prop}}$. If $\varphi, \psi \in \text{Form}_{\text{prop}}$, choose N large enough that both lie in $\text{Form}_{\text{prop}}^{(N)}$; then their negation/binary combinations lie in $\text{Form}_{\text{prop}}^{(N+1)}$. Thus $\text{Form}_{\text{prop}}$ is closed under every constructor. Conversely, if a set S contains PropVar and is closed under every constructor, induction on n gives

$$\text{Form}_{\text{prop}}^{(n)} \subseteq S \quad (n < \omega),$$

so (PF) gives $\text{Form}_{\text{prop}} \subseteq S$. Therefore (PF) really is the least constructor-closed set containing the variables; no hidden least-fixed-point theorem is being used.

A propositional language begins with a set of propositional variables and a fixed finite list of connectives. Formula (PF) constructs formulas in finite stages and the preceding verification owns the closure/minimality claim. Unique readability comes next; structural recursion then turns a base truth assignment into a valuation. Only after that theorem will semantic consequence be introduced. Syntactic derivability has not yet been introduced.

How this chapter fits together

The first task is to make the recursive nature of formulas completely explicit. Unique readability is what turns an informal string of symbols into a uniquely determined tree. Structural induction and recursion are proved only after that uniqueness result. Once the recursion theorem is available, a base truth assignment extends uniquely to a valuation and the semantic notation $\Gamma \models \varphi$ can be introduced without circularity. Syntactic derivability comes only in the next chapter.

Lemma 1.1 (Unique readability for propositional formulas). *Let $\text{Form}_{\text{prop}}$ be the formulas generated from propositional variables by $\neg, \wedge, \vee, \rightarrow$. Every $\varphi \in \text{Form}_{\text{prop}}$ has a unique parse tree. Equivalently, if φ is non-atomic, then its outermost connective and its immediate constituent formula(s) are uniquely determined.*

Proof. We prove uniqueness of the immediate constituents by induction on the length of the written formula, without first invoking any recursively defined complexity function. If φ is a variable, there is no decomposition. If the first symbol is $\neg$, the formation convention forces $\varphi = \neg\psi$, and the remaining well-formed substring ψ is determined uniquely. If the formula is built with a binary connective, matching-parenthesis depth determines a unique outer split

$$\varphi = (\psi \circ \chi).$$

Any second construction must have the same outer connective and the same split point; the induction hypothesis then gives the same parses of ψ and χ . Thus every formula has exactly one parse tree. Recursive notions such as formula height are introduced only through the structural-recursion theorem below, now that the unique immediate constituents have been proved to exist. *Point specific to this result.* For propositional

formulas, the decisive point is that the outermost connective and its matching subexpressions are recoverable from the written string. The induction therefore proves uniqueness of the parse itself, not merely existence of some parse. $\square$

Theorem 1.2 (Structural induction and structural recursion on formulas). *Let P be a property of propositional formulas. If P holds for every propositional variable and is preserved by each formation operation, then $P(\varphi)$ holds for every formula φ . Moreover, given values on variables and prescribed operations corresponding to the connectives, there is a unique function on formulas satisfying the associated recursive clauses.*

Proof. For structural induction, prove by ordinary induction on the construction stage that every member of $\text{Form}_{\text{prop}}^{(n)}$ satisfies P . The stage 0 case is the hypothesis on variables. At stage $n + 1$, old formulas retain the property and every newly constructed formula has it because the relevant immediate constituents lie in stage n . Taking the union over $n < \omega$ proves P for all formulas.

For recursion, prescribe a value a_p for each variable p , an operation N for negation, and operations $B_\circ$ for the binary connectives. For a formula φ , let

$$s(\varphi) := \min\{n < \omega : \varphi \in \text{Form}_{\text{prop}}^{(n)}\};$$

this minimum exists because the displayed set of stages is nonempty. Define F by induction on $s(\varphi)$. At stage 0, put $F(p) = a_p$. If a non-atomic formula first appears at stage $n + 1$, unique readability gives its single outer constructor and immediate constituents, all of which occur by stage n ; put

$$F(\neg\psi) = N(F(\psi)), \quad F(\psi \circ \chi) = B_\circ(F(\psi), F(\chi)).$$

Thus every formula receives one value. A second function satisfying the same clauses agrees with F by induction on the least construction stage, proving uniqueness. *Point specific to this result.* The recursion part needs uniqueness of the immediate constituents: once the outer constructor is known, the value of the recursively defined function is forced by the already determined values on those constituents. This is why unique readability logically precedes structural recursion. $\square$

Formal definitions used in this chapter

Valuation. Given a base truth assignment $v_0 : \text{PropVar} \rightarrow \{0, 1\}$, Theorem 1.2 supplies a unique function

$$v : \text{Form}_{\text{prop}} \longrightarrow \{0, 1\}$$

which extends v_0 and satisfies

$$v(\neg\varphi) = 1 - v(\varphi), \quad v(\varphi \wedge \psi) = \min\{v(\varphi), v(\psi)\},$$

$$v(\varphi \vee \psi) = \max\{v(\varphi), v(\psi)\}, \quad v(\varphi \rightarrow \psi) = 0 \iff (v(\varphi), v(\psi)) = (1, 0).$$

This unique extension is the *valuation determined by v_0* . Write $v \models \varphi$ when $v(\varphi) = 1$, and $v \models \Gamma$ when $v \models \gamma$ for every $\gamma \in \Gamma$.

Semantic consequence and validity.

$$\Gamma \models \varphi \iff \forall v (v \models \Gamma \Rightarrow v \models \varphi).$$

The notation $\models \varphi$ abbreviates $\emptyset \models \varphi$; such a formula is a *tautology* or *valid formula*.

Satisfiable. A set Γ of formulas is satisfiable when some valuation v satisfies every member, i.e. $v \models \Gamma$.

Truth table and empty normal-form conventions. If $p_1, \dots, p_n$ are the propositional variables occurring in a finite list of formulas, its truth table is the finite table with one row for each base assignment $v_0 : \{p_1, \dots, p_n\} \rightarrow \{0, 1\}$, recording the values supplied by its unique valuation extension. The empty conjunction has value 1 under every valuation and the empty disjunction has value 0; these conventions cover the constant-true and constant-false normal-form edge cases.

Theorem 1.3 (Disjunctive and conjunctive normal-form theorems). *For every propositional formula φ there exist a disjunctive normal form D_φ and a conjunctive normal form C_φ , built from literals using only $\wedge$ and $\vee$, such that $\models \varphi \leftrightarrow D_\varphi$ and $\models \varphi \leftrightarrow C_\varphi$.*

Proof. Let $p_1, \dots, p_n$ be the variables occurring in φ . For each valuation $v \in 2^n$, define the minterm

$$m_v := \bigwedge_{i=1}^n \ell_i^v, \quad \ell_i^v = \begin{cases} p_i, & v(p_i) = 1, \\ \neg p_i, & v(p_i) = 0. \end{cases}$$

Then m_v is true under exactly the valuation v . Hence

$$D_\varphi := \bigvee_{v \models \varphi} m_v$$

has the same truth table as φ , so $\models \varphi \leftrightarrow D_\varphi$. Dually, for each falsifying valuation let

$$c_v := \bigvee_{i=1}^n \neg \ell_i^v;$$

this clause is false exactly at v . Therefore

$$C_\varphi := \bigwedge_{v \not\models \varphi} c_v$$

is equivalent to φ . Empty disjunction/conjunction conventions cover contradictions and tautologies. Both constructions are finite because φ contains only finitely many variables. *Further explanation.* A useful way to separate the two normal forms is to notice what the distribution step is doing. DNF records the valuations on which the formula is true, one conjunction of literals per satisfying row; CNF records the valuations on which it is false, one disjunction excluding each falsifying row. Thus the semantic construction and the syntactic distribution construction lead to the same endpoint. This also explains why the theorem concerns finite formulas: only finitely many propositional variables occur in a fixed formula. $\square$

Proposition 1.4 (Finite truth-table decision procedure for propositional consequence). *For every finite set Γ of propositional formulas and every formula φ , there is a terminating algorithm that decides whether $\Gamma \models \varphi$: enumerate the finitely many valuations of the variables occurring in $\Gamma \cup \{\varphi\}$ and test whether any valuation satisfies $\Gamma \cup \{\neg\varphi\}$.*

Proof. Let V be the finite set of propositional variables occurring in $\Gamma \cup \{\varphi\}$. If $|V| = n$, there are exactly 2^n assignments $v : V \rightarrow \{0, 1\}$. For each assignment, recursively evaluate every formula using the truth functions of the connectives. The algorithm outputs “yes” iff

$$\forall v : V \rightarrow 2 \quad \left[(\forall \gamma \in \Gamma \ v(\gamma) = 1) \implies v(\varphi) = 1 \right].$$

Equivalently, it searches for a valuation satisfying $\Gamma \cup \{\neg\varphi\}$. Coincidence for propositional evaluation shows that extending v arbitrarily to variables outside V cannot alter the formulas being tested. Thus the finite search is exhaustive, terminates after 2^n rows, and decides $\Gamma \models \varphi$. *Further explanation.* Correctness depends on restricting attention to the variables that actually occur. Extending a valuation on $p_1, \dots, p_n$ arbitrarily to all other propositional variables cannot change the truth value of any formula in $\Gamma \cup \{\varphi\}$. Hence the finite table is exhaustive for the consequence question at hand. The procedure therefore proves decidability of finite propositional consequence without making any appeal to the later completeness theorem. $\square$

Worked example

Worked Example 1.5 (A truth table becomes an algorithm). Consider $\varphi = (p \rightarrow q) \wedge (p \vee r)$. Eliminate the implication to obtain $(\neg p \vee q) \wedge (p \vee r)$, already a CNF. With variables p, q, r , only eight valuations need inspection. If a finite set Γ uses the same variables, semantic consequence is decided by checking precisely those rows on which every member of Γ is true. This small calculation is the finite prototype of later decision and coding arguments.

A useful warning

Do not infer proof-theoretic completeness from the existence of a truth-table procedure. Deciding whether a finite propositional consequence is valid is a semantic fact; proving that every valid consequence has a derivation is a separate theorem.

Looking ahead

The next chapter, *Propositional proof systems*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 2

Propositional proof systems

Definitions and setup

Formal definitions used in this chapter

Chosen propositional calculus. To make $\vdash_{\text{Prop}}$ completely explicit, the logical axioms are *all propositional tautologies* (as defined semantically in Chapter 1), and the sole inference rule is modus ponens

$$\frac{\alpha \quad \alpha \rightarrow \beta}{\beta}.$$

Because Chapter 1 gives a terminating truth-table test for tautologyhood, the axiom set is decidable. This semantically presented Hilbert calculus is chosen only to keep the proof system exact and economical; all later syntactic arguments use only finiteness of derivations, tautological axiom instances, and modus ponens.

Derivation and derivability. A derivation of φ from Γ is a finite sequence ending in φ whose lines are members of Γ , propositional tautologies, or follow from earlier lines by modus ponens. Write $\Gamma \vdash_{\text{Prop}} \varphi$ when such a derivation exists.

Contradiction and consistency. Fix a propositional variable p_0 and put

$$\perp := p_0 \wedge \neg p_0.$$

A set Γ is *consistent* when $\Gamma \not\vdash_{\text{Prop}} \perp$; it is inconsistent otherwise. This is the definition used below; any equivalent characterizations are proved from the calculus rather than built into the terminology.

Complete and maximal consistent. A consistent set Δ is *complete* when for every formula φ , exactly one of φ and $\neg\varphi$ belongs to Δ . It is *maximal consistent* when no proper superset is consistent. Deductive closure of the maximal extension constructed below is a conclusion of Lemma 2.3, not part of these definitions.

Fix one Hilbert-style propositional calculus. A derivation from Γ is a finite sequence whose lines are logical axioms, members of Γ , or follow from earlier lines by modus ponens. Write $\Gamma \vdash \varphi$ if such a derivation ends in φ . A set of formulas is syntactically consistent if no contradiction is derivable. In the definition above, completeness is a membership condition: exactly one of φ and $\neg\varphi$ belongs to the set. For the maximal extension constructed below, Lemma 2.3 proves deductive closure, so membership and derivability coincide there. A maximal consistent set is a consistent set with no proper consistent extension. These notions concern formal proofs and must not be confused with satisfiability, which is semantic.

How this chapter fits together

We now introduce a formal calculus and ask whether it matches the semantics just developed. Soundness runs from proofs to truth. Completeness runs in the opposite direction and is subtler: a consistent theory is extended to a maximal one, and that maximal theory itself determines a valuation. Compactness then falls out because formal proofs are finite objects. This chapter is the first place where the syntactic/semantic distinction does real work.

Theorem 2.1 (Soundness of the chosen propositional calculus). *For every set Γ of propositional formulas and every formula φ , if $\Gamma \vdash_{\text{Prop}} \varphi$, then $\Gamma \models \varphi$.*

Proof. We induct on the length of a derivation

$$\psi_0, \dots, \psi_m = \varphi$$

from Γ . If $\psi_i \in \Gamma$, every valuation satisfying Γ satisfies ψ_i . If ψ_i is an instance of a propositional axiom scheme, a finite truth-table check of that scheme shows $\models \psi_i$. The only inference rule is modus ponens. If earlier lines are α and $\alpha \rightarrow \beta$, the induction hypothesis gives, for every $v \models \Gamma$,

$$v(\alpha) = 1, \quad v(\alpha \rightarrow \beta) = 1.$$

The truth table for implication then forces $v(\beta) = 1$. Thus each line of the derivation is true under every valuation satisfying Γ , and in particular

$$\Gamma \vdash_{\text{Prop}} \varphi \implies \Gamma \models \varphi.$$

No quantifier rule or first-order side condition occurs in this argument. *Point specific to this result.* Here the induction is on derivation length. The content is semantic preservation: axioms are tautologies and each rule sends truth-preserving premises to a truth-preserving conclusion. No maximal-consistency construction is used in this direction. $\square$

Lemma 2.2 (Propositional deduction lemma). *For every set Γ of propositional formulas and formulas α, β ,*

$$\Gamma \cup \{\alpha\} \vdash_{\text{Prop}} \beta \implies \Gamma \vdash_{\text{Prop}} \alpha \rightarrow \beta.$$

Conversely, if $\Gamma \vdash_{\text{Prop}} \alpha \rightarrow \beta$, then $\Gamma \cup \{\alpha\} \vdash_{\text{Prop}} \beta$.

Proof. For the forward implication, induct on a derivation of β from $\Gamma \cup \{\alpha\}$. If a line θ is a tautology or belongs to Γ , then $\theta \rightarrow (\alpha \rightarrow \theta)$ is a tautology, so $\Gamma \vdash_{\text{Prop}} \alpha \rightarrow \theta$. If the line is α , use the tautology $\alpha \rightarrow \alpha$. If θ follows by modus ponens from earlier lines ρ and $\rho \rightarrow \theta$, the induction hypothesis gives

$$\Gamma \vdash_{\text{Prop}} \alpha \rightarrow \rho, \quad \Gamma \vdash_{\text{Prop}} \alpha \rightarrow (\rho \rightarrow \theta).$$

The propositional tautology

$$(\alpha \rightarrow (\rho \rightarrow \theta)) \rightarrow ((\alpha \rightarrow \rho) \rightarrow (\alpha \rightarrow \theta))$$

and two applications of modus ponens yield $\Gamma \vdash_{\text{Prop}} \alpha \rightarrow \theta$. Apply this to the last line β . The converse is one application of modus ponens after using α as an added assumption. $\square$

Lemma 2.3 (Maximal consistent extension for propositional theories). *If Γ is a consistent set of propositional formulas, then there exists a consistent set $\Delta \supseteq \Gamma$ such that for every propositional formula φ , exactly one of φ and $\neg\varphi$ belongs to Δ . Such a Δ is maximal consistent. Moreover it is deductively closed, and for every formula θ ,*

$$\Delta \vdash_{\text{Prop}} \theta \iff \theta \in \Delta.$$

Proof. Enumerate all propositional formulas as $\varphi_0, \varphi_1, \dots$. Put $\Gamma_0 = \Gamma$ and recursively define

$$\Gamma_{n+1} = \begin{cases} \Gamma_n \cup \{\varphi_n\}, & \text{if this set is consistent,} \\ \Gamma_n \cup \{\neg\varphi_n\}, & \text{otherwise.} \end{cases}$$

At least one choice is consistent: if both extensions were inconsistent, the deduction theorem would give

$$\Gamma_n \vdash \neg\varphi_n \quad \text{and} \quad \Gamma_n \vdash \varphi_n,$$

contradicting consistency of Γ_n . Let $\Delta = \bigcup_n \Gamma_n$. A contradiction derived from Δ uses only finitely many assumptions and hence already occurs in some Γ_n , impossible. Thus Δ is consistent. Every formula is decided at its enumeration stage, so exactly one of $\varphi, \neg\varphi$ belongs to Δ . Any proper extension would add the opposite of a decided formula and become inconsistent; hence Δ is maximal consistent.

It remains to prove the asserted deductive closure. If $\theta \in \Delta$, then $\Delta \vdash_{\text{Prop}} \theta$ by the assumption rule. Conversely, suppose $\Delta \vdash_{\text{Prop}} \theta$ but $\theta \notin \Delta$. Completeness of Δ gives $\neg\theta \in \Delta$, hence $\Delta \vdash_{\text{Prop}} \neg\theta$. The propositional tautology

$$\theta \rightarrow (\neg\theta \rightarrow \perp)$$

and two applications of modus ponens yield $\Delta \vdash_{\text{Prop}} \perp$, contradicting consistency. Therefore $\theta \in \Delta$. Thus membership and derivability agree in the constructed maximal consistent set. *Point specific to this result.* The

extension step uses the elementary dichotomy that, for consistent T , at least one of $T \cup \{\varphi\}$ and $T \cup \{\neg\varphi\}$ is consistent. Finiteness of proofs is then what makes the union of the stage-by-stage construction consistent. $\square$

Theorem 2.4 (Completeness of propositional logic). *For every set Γ of propositional formulas and every formula φ , $\Gamma \models \varphi$ implies $\Gamma \vdash_{\text{Prop}} \varphi$. Hence syntactic and semantic consequence coincide for the chosen propositional calculus.*

How to read the symbols: $\Gamma \models \varphi$ versus $\Gamma \vdash \varphi$

The symbol $\models$ belongs to semantics: it quantifies over valuations. The symbol $\vdash$ belongs to syntax: it asserts the existence of a finite derivation. Completeness proves that the semantic condition implies the syntactic one. The proof does not transform a truth table into a derivation directly; instead it assumes $\Gamma \not\models \varphi$, constructs a valuation satisfying $\Gamma \cup \{\neg\varphi\}$, and contradicts $\Gamma \models \varphi$.

Proof. Assume $\Gamma \models \varphi$ but $\Gamma \not\models \varphi$. By the propositional deduction theorem,

$$\Gamma \cup \{\neg\varphi\}$$

is consistent. Extend it to a maximal consistent set Δ . Define a valuation on variables by

$$v(p) = 1 \iff p \in \Delta.$$

We prove the truth lemma

$$v(\psi) = 1 \iff \psi \in \Delta \tag{*}$$

by structural induction. For negation, maximal consistency gives $\neg\psi \in \Delta$ iff $\psi \notin \Delta$. For each binary connective, the corresponding propositional tautologies and closure of Δ under modus ponens give exactly the truth-table clause. Thus (*) holds. Since $\Gamma \subseteq \Delta$ and $\neg\varphi \in \Delta$, the valuation satisfies Γ and falsifies φ , contradicting $\Gamma \models \varphi$. Therefore $\Gamma \vdash \varphi$. *Further explanation.* The key point is the truth lemma for the

valuation extracted from T^* . Maximal consistency gives exactly the Boolean closure properties needed by the induction: for instance, $\varphi \wedge \psi \in T^*$ iff both conjuncts belong to T^* , and $\neg\varphi \in T^*$ iff $\varphi \notin T^*$. Once that equivalence is established for every formula, the contradiction argument converts the semantic hypothesis $\Gamma \models \varphi$ into a formal derivation. Thus completeness is not an enumeration of proofs; it is a model construction from failure of derivability. $\square$

Theorem 2.5 (Compactness of propositional logic). *A set Γ of propositional formulas is satisfiable if and only if every finite subset $\Gamma_0 \subseteq \Gamma$ is satisfiable.*

Proof. Assume every finite subset of Γ is satisfiable. If Γ were inconsistent, there would be a finite formal derivation of a contradiction from Γ . Only finitely many assumptions $\gamma_1, \dots, \gamma_n \in \Gamma$ occur in that derivation, so

$$\{\gamma_1, \dots, \gamma_n\} \vdash \perp.$$

By soundness, this finite subset would be unsatisfiable, contrary to hypothesis. Hence Γ is consistent. Propositional completeness supplies a valuation v with

$$v \models \Gamma.$$

Conversely, if Γ has a satisfying valuation, the same valuation satisfies every finite subset. Thus

$$\Gamma \text{ satisfiable} \iff \forall \Gamma_0 [(\Gamma_0 \subseteq \Gamma \ \& \ \Gamma_0 \text{ finite}) \rightarrow \Gamma_0 \text{ satisfiable}].$$

The proof isolates the key compactness mechanism: formal derivations are finite. *Point specific to this result.*

The compactness conclusion is a corollary of proof finiteness plus completeness: if a whole theory produced a contradiction, one finite set of premises occurring in that proof would already do so. The argument is therefore different in direction from the maximal-consistency construction that establishes completeness. $\square$

Worked example

Worked Example 2.6 (From a maximal theory to a valuation). Let T^* be maximal consistent and define $v(p) = 1$ exactly when $p \in T^*$. If $\neg\psi \in T^*$, consistency prevents $\psi \in T^*$; maximality gives the converse. Similar closure facts handle $\wedge$ and $\vee$. An induction therefore proves $v(\psi) = 1$ iff $\psi \in T^*$. The abstract maximal set has become a concrete semantic object.

A useful warning

Maximal consistency is not the same as containing every formula. It means deciding every formula by exactly one of φ and $\neg\varphi$, while remaining closed under the deductive calculus.

Looking ahead

The next chapter, *First-order syntax, substitution, and equality*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 3

First-order syntax, substitution, and equality

Definitions and setup

Formal definitions used in this chapter

First-order language. A first-order language $\mathcal{L}$ consists of pairwise disjoint sets of constant symbols, function symbols, and relation symbols, with a specified positive finite arity for every function and relation symbol, together with variables, equality, the connectives $\neg, \wedge, \vee, \rightarrow$, quantifiers $\forall, \exists$, and punctuation. The earlier biconditional notation $\varphi \leftrightarrow \psi$ remains the abbreviation (BI), not a new formula constructor.

Terms. In the surrounding metatheory let

$$\text{Term}_{\mathcal{L}}^{(0)} := \text{Var} \cup \text{Const}_{\mathcal{L}},$$

and recursively let $\text{Term}_{\mathcal{L}}^{(n+1)}$ be $\text{Term}_{\mathcal{L}}^{(n)}$ together with every string $f(t_1, \dots, t_k)$ for which f is a k -ary function symbol and $t_1, \dots, t_k \in \text{Term}_{\mathcal{L}}^{(n)}$. Define

$$\text{Term}_{\mathcal{L}} := \bigcup_{n < \omega} \text{Term}_{\mathcal{L}}^{(n)}. \quad (\text{FT})$$

Its members are the $\mathcal{L}$ -terms. A term is *closed* when it contains no variables.

Atomic formulas and formulas. An *atomic $\mathcal{L}$ -formula* is $R(t_1, \dots, t_k)$, where R is a k -ary relation symbol and the t_i are $\mathcal{L}$ -terms, or an equality $s = t$ of $\mathcal{L}$ -terms. Let $\text{Form}_{\mathcal{L}}^{(0)}$ be the set of atomic formulas. Having defined $\text{Form}_{\mathcal{L}}^{(n)}$, let $\text{Form}_{\mathcal{L}}^{(n+1)}$ contain it together with

$$\neg\varphi, \quad (\varphi \wedge \psi), \quad (\varphi \vee \psi), \quad (\varphi \rightarrow \psi), \quad \forall x \varphi, \quad \exists x \varphi$$

for $\varphi, \psi \in \text{Form}_{\mathcal{L}}^{(n)}$ and variables x . Define

$$\text{Form}_{\mathcal{L}} := \bigcup_{n < \omega} \text{Form}_{\mathcal{L}}^{(n)}. \quad (\text{FF})$$

Its members are the $\mathcal{L}$ -formulas. Thus the earlier descriptions “least set of terms/formulas closed under the constructors” are consequences of the explicit finite-stage constructions (FT)–(FF), not unstated existence assumptions.

Free and bound occurrences. In $Qx \psi$, with $Q \in \{\forall, \exists\}$, the displayed quantifier binds every occurrence of x in ψ not already lying inside the scope of a nearer quantifier on x . An occurrence is *bound* if some

quantifier binds it and *free* otherwise. $FV(t)$ and $FV(\varphi)$ are the sets of variables occurring in the term t and occurring free in the formula φ , respectively.

Fresh variable, alpha-renaming, and alpha-equivalence. A variable y is *fresh* for a displayed finite collection of terms/formulas when it does not occur in any member of that collection. If $Qx\psi$ is a quantified subformula and y is fresh for ψ , an *alpha-renaming step* replaces that subformula by $Qy\psi[y/x]_{\text{bd}}$, where $\psi[y/x]_{\text{bd}}$ replaces precisely the occurrences of x bound by that displayed quantifier. Two formulas are *alpha-equivalent* when one can be obtained from the other by finitely many alpha-renaming steps and their inverses.

Congruence on terms. If A is a set of $\mathcal{L}$ -terms, an equivalence relation E on A is a *term congruence* when, for every n -ary function symbol f , whenever $s_i E t_i$ for $1 \leq i \leq n$ and all displayed terms lie in A , one has $f(s_1, \dots, s_n) E f(t_1, \dots, t_n)$. This is the exact property needed for function symbols to descend to equivalence classes. Relation symbols require a separate invariance verification, supplied later when the term model is constructed.

The formation clauses above are finite-stage inductive definitions: no strings other than those obtained by finitely many applications of the displayed rules count as terms or formulas. The next two results first establish unique readability and structural recursion. Only after those results is the permanent notation for capture-avoiding substitution introduced. Alpha-renaming is already a syntactic relation on bound occurrences and will be used in the quantifier branch of that recursion.

How this chapter fits together

First-order syntax introduces two complications absent propositionally: variables can be bound, and equality identifies terms semantically even when their strings differ. We therefore prove unique readability and structural recursion before defining capture-avoiding substitution, then state its freshness conditions explicitly. Equality is handled later, after derivability has been defined: provable equality becomes a congruence, so the eventual term model uses equivalence classes of terms rather than raw strings.

Lemma 3.1 (Unique readability for first-order terms and formulas). *Fix a first-order language $\mathcal{L}$. Every $\mathcal{L}$ -term and every $\mathcal{L}$ -formula has a unique parse tree; in particular, the outermost constructor and immediate constituent term(s) or formula(s) are uniquely determined.*

Proof. Define terms and formulas simultaneously by their formation trees. For terms, variables and constants are atomic and $f(t_1, \dots, t_n)$ has immediate subterms $t_1, \dots, t_n$. For formulas, atomic formulas have the form $R(t_1, \dots, t_n)$ or $t = s$, Boolean connectives attach formula constituents, and $\forall x\psi$ or $\exists x\psi$ has immediate constituent ψ . We prove by induction on string length that the outermost syntactic category and constructor are unique. Parenthesis depth determines the split points of argument lists and binary connectives; a leading quantifier determines its bound variable and scope. The induction hypothesis gives unique parses of all immediate constituents. Hence every term and formula has a unique syntax tree. This makes free/bound occurrence and subformula unambiguous and supplies the decomposition needed for the structural-recursion theorem below; substitution itself is introduced only after that theorem. *Point specific to this result.* First-order syntax adds terms, quantifiers, and variable binding, but the uniqueness argument still begins by recovering the outer constructor. Parentheses and arities determine the immediate term/formula constituents; bound-variable scope is part of that recovered parse. $\square$

Theorem 3.2 (Structural recursion on terms and formulas). *Any assignment of values to variables together with recursive clauses for the function, relation, connective, and quantifier constructors extends uniquely to recursively defined functions on all $\mathcal{L}$ -terms and formulas, provided the clauses depend only on immediate constituents.*

Proof. Let target sets and operations be specified for each syntactic constructor. Define a function on terms by

$$F(x) = a_x, \quad F(f(t_1, \dots, t_n)) = F_f(F(t_1), \dots, F(t_n)).$$

Unique readability makes this recursive clause unambiguous, and induction on term height proves existence and uniqueness. For formulas, prescribe values on atomic formulas and operations $N, B_\circ, Q_x$ corresponding to negation, binary connectives, and quantification; then set

$$G(\neg\psi) = N(G(\psi)), \quad G(\psi \circ \chi) = B_\circ(G(\psi), G(\chi)), \quad G(\forall x\psi) = Q_x^\forall(G(\psi)), \quad G(\exists x\psi) = Q_x^\exists(G(\psi)).$$

Again unique readability identifies the single applicable clause. Structural induction is the associated proof principle: a property holding for atomic syntax and preserved by every constructor holds for all terms/formulas. *Point specific to this result.* Because formulas contain terms and terms may occur inside atomic formulas, the clean formulation is a mutual recursion on the two syntactic classes. The recursion is legitimate only after unique readability guarantees a unique decomposition at each step. $\square$

Formal definitions used in this chapter

Capture-avoiding substitution on terms. With structural recursion now established, define $t[s/x]$ by

$$x[s/x] = s, \quad y[s/x] = y \ (y \neq x), \quad c[s/x] = c,$$

and

$$f(t_1, \dots, t_n)[s/x] := f(t_1[s/x], \dots, t_n[s/x]).$$

Capture-avoiding substitution on formulas. Define $\varphi[s/x]$ by structural recursion. Atomic formulas are handled by substituting in their constituent terms, and the operation commutes with $\neg, \wedge, \vee, \rightarrow$. For a quantified formula $Qy\psi$, with $Q \in \{\forall, \exists\}$:

1. if $y = x$, put $(Qx\psi)[s/x] := Qx\psi$;
2. if $y \neq x$ and $y \notin \text{FV}(s)$, put $(Qy\psi)[s/x] := Qy(\psi[s/x])$;
3. if $y \neq x$ and $y \in \text{FV}(s)$ while the substitution enters the scope, first choose a variable z fresh for ψ, s, x , alpha-rename the displayed binder Qy to Qz , and then apply the preceding clause.

The resulting operation is *capture avoiding*: no variable free in the inserted term becomes bound. We now use the permanent notation $\varphi[s/x]$ and $t[s/x]$ for these recursively defined operations.

Lemma 3.3 (Capture-avoiding substitution and substitution composition). *Let $x \neq y$, let s, t be terms, and let φ be a formula. Before substitution, alpha-rename bound variables of φ so that no bound variable occurring on a substitution path belongs to $\text{FV}(s) \cup \text{FV}(t)$. Then capture-avoiding substitution is well defined and*

$$\varphi[s/x][t/y] = \varphi[s[t/y]/x, t/y],$$

where the right-hand side is simultaneous substitution: free occurrences of x are replaced by $s[t/y]$, free occurrences of y are replaced by t , and no substitution is performed inside the replacement terms after they are inserted.

Commented symbolic trace: capture-avoiding substitution

Read $\varphi[s/x][t/y]$ from left to right. First replace free occurrences of x by s ; only after that replacement do we substitute t for the free occurrences of y . A bound variable may have to be renamed before either step so that a free variable of the inserted term does not become accidentally bound. That renaming is not cosmetic: it preserves the set of free variables and therefore preserves the intended semantics.

Proof. Define substitution on terms by structural recursion. For formulas, recurse through the Boolean connectives. At a quantified formula $Qz\psi$, where $Q \in \{\forall, \exists\}$, first alpha-rename z when necessary so that

$$z \notin \text{FV}(s) \cup \text{FV}(t) \cup \{x, y\}.$$

After this renaming, recursion under the quantifier cannot capture a variable that was free in either replacement term. Thus the capture-avoiding operation is defined at every node of the syntax tree.

For the composition formula, argue by structural induction on φ . The variable case exhibits the three possibilities. If the variable is x , the left side is

$$x[s/x][t/y] = s[t/y],$$

which is exactly the simultaneous replacement assigned to x . If it is y , then

$$y[s/x][t/y] = t,$$

and any other variable is unchanged. Function and relation symbols reduce componentwise to the term induction hypothesis. Boolean connectives inherit the equality from their immediate subformulas.

For a quantified formula $Qz\psi$, use the alpha-renamed representative with z fresh as displayed above. Both sides leave the binder Qz fixed and perform the two substitutions only in ψ . The induction hypothesis gives

$$\psi[s/x][t/y] = \psi[s[t/y]/x, t/y],$$

so prefixing the same quantifier yields the desired equality for $Qz\psi$. Alpha-equivalent preliminary renamings lead to alpha-equivalent outputs, so the construction is independent of the particular fresh name chosen. This proves both well-definedness and the stated composition law without an unspecified “freshness condition.” $\square$

How to read the symbols: why the equality quotient theorem is postponed

The quotient of closed terms by provable equality will be needed for the Henkin term model. At this point, however, the first-order derivability relation $T \vdash \varphi$ has not yet been defined. To respect definition-before-use, the equality-congruence theorem and its proof are therefore placed in Chapter 5 immediately after the first-order calculus is fixed. The present chapter supplies only the syntactic substitution machinery on which that later theorem depends.

Worked example

Worked Example 3.4 (Why capture avoidance is necessary). In $\forall y R(x, y)$, substituting the term y naively for x gives $\forall y R(y, y)$, where the formerly free y has been captured. Rename the bound variable first: $\forall z R(x, z)$, then substitute to obtain $\forall z R(y, z)$. The two results have very different meanings; freshness is therefore a mathematical condition, not typographical etiquette.

A useful warning

Never identify two closed terms merely because the intended interpretation makes them equal. The term model is built from syntax, so equality must first be shown to be a provable congruence and then quotiented.

Looking ahead

The next chapter, *Structures and satisfaction*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 4

Structures and satisfaction

Definitions and setup

Formal definitions used in this chapter

$\mathcal{L}$ -structure and its domain. An $\mathcal{L}$ -structure M has a nonempty domain $|M|$. Each constant symbol c is interpreted by an element $c^M \in |M|$; each n -ary function symbol f by a function $f^M : |M|^n \rightarrow |M|$; and each n -ary relation symbol R by a set $R^M \subseteq |M|^n$.

Assignment and assignment update. An assignment in M is a function $a : \text{Var} \rightarrow |M|$. For $m \in |M|$, the updated assignment $a[x \mapsto m]$ agrees with a away from x and sends x to m .

Term evaluation. The value $t^M[a] \in |M|$ of a term t is defined recursively by

$$x^M[a] = a(x), \quad c^M[a] = c^M,$$

$$(f(t_1, \dots, t_n))^M[a] = f^M(t_1^M[a], \dots, t_n^M[a]).$$

Tarski satisfaction recursion. For formulas φ, ψ , define $M, a \models \varphi$ recursively by

$$M, a \models (t = s) \iff t^M[a] = s^M[a],$$

$$M, a \models R(t_1, \dots, t_n) \iff (t_1^M[a], \dots, t_n^M[a]) \in R^M,$$

$$M, a \models \neg\varphi \iff M, a \not\models \varphi, \quad M, a \models (\varphi \wedge \psi) \iff (M, a \models \varphi \text{ \& } M, a \models \psi),$$

$$M, a \models (\varphi \vee \psi) \iff (M, a \models \varphi \text{ or } M, a \models \psi),$$

$$M, a \models (\varphi \rightarrow \psi) \iff (M, a \not\models \varphi \text{ or } M, a \models \psi),$$

$$M, a \models \exists x \varphi \iff \exists m \in |M| \ M, a[x \mapsto m] \models \varphi,$$

$$M, a \models \forall x \varphi \iff \forall m \in |M| \ M, a[x \mapsto m] \models \varphi.$$

If σ is a sentence, define

$$M \models \sigma \iff \forall a : \text{Var} \rightarrow |M| \ (M, a \models \sigma).$$

The Coincidence Lemma below proves that, because a sentence has no free variables, this is equivalent to truth under any one assignment. Assignment-independence is therefore a theorem, not part of the recursive definition.

Isomorphism. For $\mathcal{L}$ -structures M, N , an isomorphism $h : M \cong N$ is a bijection $h : |M| \rightarrow |N|$ such that for every constant c , every n -ary function symbol f , every n -ary relation symbol R , and every tuple $\bar{m} \in |M|^n$ of the required arity,

$$h(c^M) = c^N, \quad h(f^M(\bar{m})) = f^N(h\bar{m}),$$

$$\bar{m} \in R^M \iff h\bar{m} \in R^N.$$

Here $h\bar{m}$ means componentwise application of h to the tuple.

Expansion and reduct. If $\mathcal{L}_0 \subseteq \mathcal{L}$, an $\mathcal{L}$ -structure M is an expansion of an $\mathcal{L}_0$ -structure M_0 when they have the same domain and every symbol of $\mathcal{L}_0$ has the same interpretation in both. The reduct $M \upharpoonright \mathcal{L}_0$ is the $\mathcal{L}_0$ -structure obtained from M by retaining exactly those interpretations whose symbols belong to $\mathcal{L}_0$.

An $\mathcal{L}$ -structure M consists of a nonempty domain together with interpretations of every symbol in the signature. A variable assignment s maps variables to elements of M . Terms receive values $t^M[s]$ by recursion. Satisfaction $M, s \models \varphi$ is then defined recursively; for a sentence the assignment is irrelevant and we write $M \models \varphi$. An isomorphism is exactly the bijection satisfying the constant/function/relation preservation clauses displayed in the definition block above. The more general notion of an embedding is deliberately postponed until Chapter 7, where it is needed for diagrams. A reduct forgets interpretations of symbols, while an expansion adds them.

How this chapter fits together

Satisfaction is a recursion on formulas, but the recursion interacts with assignments. The substitution theorem connects syntactic replacement with semantic reassignment; the coincidence lemma says truth depends only on values of free variables. These two facts make the quantifier clauses manageable and are reused whenever formulas are transported between structures, reducts, expansions, and isomorphic copies.

Theorem 4.1 (Term-evaluation substitution theorem). *For every $\mathcal{L}$ -structure M , assignment a , term t , term s , and variable x ,*

$$(t[s/x])^M[a] = t^M[a(x := s^M[a])].$$

The analogous substitution lemma holds for formulas:

$$M, a \models \varphi[s/x] \iff M, a(x := s^M[a]) \models \varphi,$$

whenever the substitution is capture avoiding.

Proof. We prove by induction on the term t that for every structure M and assignment a ,

$$(t[s/x])^M[a] = t^M[a(x \mapsto s^M[a])]. \quad (1)$$

For a variable this is immediate from the two assignment cases; for $f(t_1, \dots, t_n)$, apply the induction hypothesis to each argument and then the interpretation f^M . The formula substitution theorem follows by structural induction on φ :

$$M, a \models \varphi[s/x] \iff M, a(x \mapsto s^M[a]) \models \varphi. \quad (2)$$

Atomic formulas use (1), Boolean connectives use the induction hypothesis, and quantifiers use capture avoidance to ensure the modified assignments agree on the correct free variables. Equations (1)–(2) are the semantic correctness theorem for syntactic substitution. *Point specific to this result.* The term case is the

engine of the theorem: evaluating $t[s/x]$ under an assignment equals evaluating t after changing the value of x to the value of s . The formula substitution lemma later inherits this identity in its atomic case. $\square$

Lemma 4.2 (Coincidence lemma). *If assignments a, b agree on every variable free in a term t , then $t^M[a] = t^M[b]$. If they agree on every variable free in a formula φ , then*

$$M, a \models \varphi \iff M, b \models \varphi.$$

Proof. Let a, b be assignments in a structure M . First prove by induction on terms that if a and b agree on all variables occurring in t , then

$$t^M[a] = t^M[b].$$

For formulas, prove by structural induction that agreement on the free variables of φ implies

$$M, a \models \varphi \iff M, b \models \varphi.$$

Atomic formulas use the term result. Boolean cases are direct. If $\varphi = \exists x\psi$, any witness assignment $a(x \mapsto m)$ and $b(x \mapsto m)$ still agree on the free variables of ψ ; the induction hypothesis transfers truth of ψ . Thus bound-variable assignments and variables absent from the formula cannot affect satisfaction. This is the coincidence lemma. *Point specific to this result.* For coincidence, the induction tracks free variables rather than substitution. At the quantifier step, assignments may differ at the newly bound variable; the induction hypothesis applies because every other free coordinate still agrees. $\square$

Theorem 4.3 (Isomorphism invariance of satisfaction). *If $h : M \cong N$ is an isomorphism of $\mathcal{L}$ -structures, then for every formula $\varphi(\bar{x})$ and every tuple $\bar{a} \in M$,*

$$M \models \varphi(\bar{a}) \iff N \models \varphi(h\bar{a}).$$

Proof. Let $h : M \rightarrow N$ be an isomorphism. Extend it to assignments by $(h \circ a)(x) = h(a(x))$. Induct on terms to prove

$$h(t^M[a]) = t^N[h \circ a]. \quad (1)$$

For a function application, (1) follows from preservation of the function interpretation by h . Now induct on formulas. For an atomic relation,

$$M \models R(t_1, \dots, t_n)[a] \iff N \models R(t_1, \dots, t_n)[h \circ a]$$

by (1) and preservation/reflection of R . Equality is similar; Boolean cases follow immediately. For $\exists x\psi$, a witness $m \in M$ is carried to $h(m) \in N$, and surjectivity of h gives the converse. Therefore

$$M, a \models \varphi \iff N, h \circ a \models \varphi.$$

Point specific to this result. The key extra statement is the term identity $h(t^M(\bar{a})) = t^N(h\bar{a})$. Once proved by term induction, it transports atomic truth; the formula induction then shows that quantification is preserved because an isomorphism is bijective. $\square$

Proposition 4.4 (Expansion/reduct compatibility of satisfaction). *Let $\mathcal{L}_0 \subseteq \mathcal{L}$, let M be an $\mathcal{L}$ -structure, and let $M \upharpoonright \mathcal{L}_0$ be its reduct. For every $\mathcal{L}_0$ -formula φ and assignment a ,*

$$M, a \models \varphi \iff M \upharpoonright \mathcal{L}_0, a \models \varphi.$$

Equivalently, expanding a structure does not change the truth of formulas in the old language.

Proof. Suppose $L \subseteq L'$, M' is an L' -structure, and $M = M' \upharpoonright L$ is its reduct. We prove by induction on L -terms that

$$t^{M'}[a] = t^M[a],$$

because every symbol occurring in t has the same interpretation in the reduct. Induction on L -formulas then gives

$$M', a \models \varphi \iff M, a \models \varphi.$$

Atomic cases use the term identity; Boolean and quantifier clauses quantify over the same underlying set. Conversely, an expansion may choose arbitrary interpretations only for symbols in $L' \setminus L$; it cannot change the truth of an L -formula. Thus satisfaction of formulas in the smaller signature is invariant under expansion/reduct. *Point specific to this result.* Only symbols occurring in the smaller language can affect the value of its terms or formulas. Thus an expansion and its reduct give identical interpretations to every symbol visible to the formula, and the satisfaction recursion yields the same truth value. $\square$

Worked example

Worked Example 4.5 (Substitution as reassignment). Let $t(x) = f(x)$ and let s assign a to x . If a term u has value b under s , then evaluating $t[u/x]$ under s gives $f^M(b)$. The substitution lemma says the same value is obtained by evaluating t under the assignment $s[x \mapsto b]$. The formula version is the same principle with truth values instead of term values.

A useful warning

The quantifier step always changes one assignment coordinate. Omitting that bookkeeping is the most common source of false substitution and isomorphism arguments.

Looking ahead

The next chapter, *Formal deduction*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 5

Formal deduction

Definitions and setup

Formal definitions used in this chapter

First-order theory, sentence, and countable language. A first-order theory is a set of sentences in a fixed language $\mathcal{L}$; a sentence is a formula with no free variables. A language is *countable* when its set of nonlogical symbols is finite or countably infinite, so its finite strings and sentences can be enumerated by ω .

The chosen first-order calculus. In the axiom displays below, every occurrence of $\leftrightarrow$ is first expanded by (BI). A derivation may use:

1. every substitution instance of a propositional tautology from Chapter 2;
2. the universal-instantiation axiom

$$\forall x \varphi(x) \rightarrow \varphi(t)$$

whenever t is free for x in φ (that is, the capture-avoiding substitution $\varphi[t/x]$ is permitted);

3. the existential-duality axiom

$$\exists x \varphi \leftrightarrow \neg \forall x \neg \varphi;$$

thus the primitive symbol $\exists$ is proof-theoretically tied to the universal quantifier rather than being left as an unrelated syntactic constructor;

4. the quantifier-distribution axiom

$$\forall x(\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \forall x\psi)$$

when $x \notin \text{FV}(\varphi)$;

5. equality axioms consisting of $t = t$ and the substitutivity scheme

$$s = t \rightarrow (\varphi(s) \leftrightarrow \varphi(t)),$$

where $\varphi(s)$ and $\varphi(t)$ denote capture-avoiding substitution into one displayed free argument position;

6. modus ponens and generalization $\varphi/\forall x\varphi$.

For a derivation from assumptions Γ , generalization is subject to the eigenvariable restriction below. This fixed calculus is the meaning of $\Gamma \vdash \varphi$ throughout the rest of Volume I.

Eigenvariable restriction. In a derivation from assumptions Γ , the generalization rule may infer $\forall x\varphi$ from φ only when x is not free in any undischarged assumption on which that occurrence of φ depends. This is the side condition used by the deduction theorem.

Syntactic and semantic consequence. $\Gamma \vdash \varphi$ means that the chosen calculus has a finite derivation of φ from assumptions in Γ . The semantic notation $\Gamma \models \varphi$ means that every $\mathcal{L}$ -structure satisfying every sentence of Γ satisfies φ .

Consistency, completeness, maximal consistency, and deductive closure. A theory T is *consistent* when there is no sentence θ with both $T \vdash \theta$ and $T \vdash \neg\theta$. It is *syntactically complete* when for every sentence θ , either $T \vdash \theta$ or $T \vdash \neg\theta$. A consistent set T is *maximal consistent* when no proper extension of it is consistent. A theory T is *deductively closed* when $T \vdash \theta$ implies $\theta \in T$ for every sentence θ . The Lindenbaum theorem below proves that its constructed maximal consistent extension is deductively closed; until that theorem is proved, membership $\theta \in T$ and derivability $T \vdash \theta$ are not identified by definition.

Closed-term algebra and quotient structure. The *closed-term algebra* of $\mathcal{L}$ is the set of all closed $\mathcal{L}$ -terms equipped with the operations induced by the function symbols: an n -ary f sends $(t_1, \dots, t_n)$ to $f(t_1, \dots, t_n)$. If an equivalence relation E is a term congruence, these function operations descend to the quotient classes. To obtain an $\mathcal{L}$ -structure one must additionally verify that every relation symbol is invariant under replacing arguments by E -equivalent terms.

The first-order calculus extends the propositional calculus by quantifier axioms/rules and equality axioms. We fix the eigenvariable side conditions explicitly: generalization may not bind a variable that is free in an undischarged assumption when the deduction theorem is being used. A theory is simply a set of sentences in the fixed language. Consistency, completeness, and derivability are syntactic notions. The deduction theorem is a transformation of finite derivations; it does not assert semantic implication until soundness has been proved.

How this chapter fits together

The formal calculus now has to support the later completeness construction. Soundness is proved line by line, while the deduction theorem converts a derivation with an extra assumption into an implication. Consistency extension and Lindenbaum's lemma then prepare a theory that decides every sentence. Side conditions on generalization are essential because a free variable in an undischarged assumption cannot be varied arbitrarily.

Theorem 5.1 (Equality congruence and quotient-term construction). *Let T be a theory with equality and let $\sim$ be the relation on closed terms defined by $s \sim t \iff T \vdash s = t$. Then $\sim$ is an equivalence relation and a term congruence for every function symbol. Moreover, every relation symbol is invariant under replacing any argument by a $\sim$ -equivalent closed term. Consequently the quotient of the closed-term algebra by $\sim$ carries a well-defined $\mathcal{L}$ -structure.*

Proof. On closed terms define

$$s \sim t \iff T \vdash s = t.$$

Equality axioms give reflexivity, symmetry, and transitivity, so $\sim$ is an equivalence relation. Substitutivity gives congruence: if $s_i \sim t_i$ for $1 \leq i \leq n$, then

$$T \vdash f(s_1, \dots, s_n) = f(t_1, \dots, t_n),$$

and for a relation symbol R ,

$$T \vdash R(\bar{s}) \leftrightarrow R(\bar{t}).$$

Hence functions and relations are well defined on equivalence classes $[t]$. The quotient of closed terms by $\sim$ is therefore a legitimate first-order structure. This quotient is essential in the Henkin model: syntactically different terms that T proves equal represent one element rather than two. *Further explanation.* There are two well-definedness checks. First, if $s_i \sim t_i$, substitutivity proves that a function symbol has the same equivalence class whether it is applied to the s_i 's or the t_i 's. Second, the same argument for relation symbols shows that truth of an atomic relation depends only on the equivalence classes of its arguments. These checks are what make the quotient a genuine structure rather than merely a set of equivalence classes. $\square$

Theorem 5.2 (Soundness of first-order deduction). *For every first-order theory Γ and sentence φ , if $\Gamma \vdash \varphi$ in the chosen first-order calculus, then every $\mathcal{L}$ -structure satisfying Γ also satisfies φ ; that is, $\Gamma \vdash \varphi \Rightarrow \Gamma \models \varphi$.*

Proof. Induct on the length of a first-order derivation from Γ . Propositional logical axiom instances are valid under every assignment by the truth-functional semantic clauses, and equality axioms are valid because equality in a structure is actual identity. Universal instantiation and quantifier distribution are verified directly from the Tarski clauses and their stated free-variable conditions.

The existential-duality scheme in the fixed calculus also requires its own check. For every structure M , assignment a , and formula ψ ,

$$\begin{aligned} M, a \models \exists x \psi &\iff \text{for some } m \in M, M, a[x \mapsto m] \models \psi \\ &\iff \neg(\text{for every } m \in M, M, a[x \mapsto m] \models \neg\psi) \\ &\iff M, a \models \neg\forall x \neg\psi. \end{aligned} \quad (\text{ED-Sound})$$

Hence the (BI)-expanded biconditional $\exists x \psi \leftrightarrow \neg\forall x \neg\psi$ is true under every structure and assignment. This is exactly the additional logical-axiom scheme later recognized by the code predicate `LogAx` in Volume II.

Assumptions in Γ hold in every model of Γ . Modus ponens preserves truth. For generalization, suppose the rule passes from ψ to $\forall x \psi$ and the calculus requires that x not occur free in any undischarged assumption. If $M, a \models \Gamma$, then for every $m \in M$, the coincidence lemma gives

$$M, a(x \mapsto m) \models \Gamma.$$

By the induction hypothesis,

$$M, a(x \mapsto m) \models \psi.$$

Since m was arbitrary, $M, a \models \forall x \psi$. Hence

$$\Gamma \vdash \varphi \implies \Gamma \models \varphi.$$

Point specific to this result. The quantifier rules are the only new semantic issue beyond propositional soundness. Their eigenvariable and free-variable side conditions ensure that varying a quantified coordinate cannot change an undischarged assumption, which is exactly what the semantic proof requires. $\square$

Lemma 5.3 (Derived existential introduction). *Let t be free for x in $\varphi(x)$. In the chosen first-order calculus,*

$$\Gamma \vdash \varphi(t) \implies \Gamma \vdash \exists x \varphi(x). \quad (\text{EI})$$

Proof. Apply universal instantiation to the formula $\neg\varphi$:

$$\forall x \neg\varphi(x) \rightarrow \neg\varphi(t).$$

The propositional tautology

$$(A \rightarrow \neg B) \rightarrow (B \rightarrow \neg A)$$

with $A = \forall x \neg\varphi(x)$ and $B = \varphi(t)$, followed by modus ponens, gives

$$\Gamma \vdash \varphi(t) \rightarrow \neg\forall x \neg\varphi(x).$$

Using the assumed derivation of $\varphi(t)$, modus ponens yields

$$\Gamma \vdash \neg\forall x \neg\varphi(x).$$

The existential-duality axiom supplies

$$\neg\forall x \neg\varphi(x) \rightarrow \exists x \varphi(x),$$

so one final application of modus ponens proves (EI). $\square$

Theorem 5.4 (Deduction theorem in the chosen calculus). *For sentences φ, ψ and theory Γ , subject to the eigenvariable restrictions of the chosen calculus,*

$$\Gamma \cup \{\varphi\} \vdash \psi \iff \Gamma \vdash \varphi \rightarrow \psi.$$

Proof. We prove the nontrivial direction by induction on a derivation of ψ from $\Gamma \cup \{\varphi\}$. If a line α is an axiom or belongs to Γ , propositional tautologies give

$$\Gamma \vdash \varphi \rightarrow \alpha.$$

If $\alpha = \varphi$, derive $\varphi \rightarrow \varphi$. If β follows from α and $\alpha \rightarrow \beta$ by modus ponens, combine the induction hypotheses using

$$(\varphi \rightarrow (\alpha \rightarrow \beta)) \rightarrow ((\varphi \rightarrow \alpha) \rightarrow (\varphi \rightarrow \beta)).$$

For generalization, the eigenvariable restriction guarantees that the quantified variable is not free in φ or the undischarged assumptions, so the quantifier axiom lifts the induction hypothesis. Thus

$$\Gamma \cup \{\varphi\} \vdash \psi \implies \Gamma \vdash \varphi \rightarrow \psi.$$

The converse is one application of modus ponens. *Further explanation.* The generalization case is the only place where a careless formulation fails. If x occurs free in the assumption φ , then from a derivation of ψ under that assumption one cannot in general infer $\forall x \psi$ while keeping the assumption fixed. The side condition prevents exactly this change of meaning. With it in place, the induction converts every line of the original derivation into a theorem of the form $\varphi \rightarrow \chi$, so the final line is $T \vdash \varphi \rightarrow \psi$. $\square$

Lemma 5.5 (Consistency extension lemma). *If Γ is consistent and φ is any sentence, then at least one of $\Gamma \cup \{\varphi\}$ and $\Gamma \cup \{\neg\varphi\}$ is consistent. In particular, if $\Gamma \not\vdash \varphi$, then $\Gamma \cup \{\neg\varphi\}$ is consistent.*

Proof. Let T be consistent and let σ be a sentence not decided by T . If both

$$T \cup \{\sigma\} \quad \text{and} \quad T \cup \{\neg\sigma\}$$

were inconsistent, the deduction theorem would give

$$T \vdash \neg\sigma \quad \text{and} \quad T \vdash \neg\neg\sigma,$$

and classical logic yields $T \vdash \sigma$, contradicting consistency. Hence at least one of the two one-sentence extensions is consistent. Repeating this decision step along an enumeration of formulas and taking unions at limit stages yields consistent extensions. The finiteness of proofs is what preserves consistency at unions: any contradiction in the union appears at a finite stage. This lemma is the local consistency fact used by the Lindenbaum construction. *Further explanation.* For the extension step, suppose both $T \cup \{\varphi\}$ and $T \cup \{\neg\varphi\}$ were inconsistent. By the deduction theorem the first would give $T \vdash \neg\varphi$ and the second $T \vdash \neg\neg\varphi$, hence T itself would be inconsistent. Therefore at least one of the two one-sentence extensions preserves consistency. This elementary dichotomy is the engine behind the Lindenbaum construction and later Henkin completions. $\square$

Theorem 5.6 (Lindenbaum maximal-consistency and deductive-closure theorem). *Every consistent first-order theory Γ in a countable language has an extension $\Delta \supseteq \Gamma$ such that:*

1. Δ is consistent;
2. for every sentence φ , exactly one of φ and $\neg\varphi$ belongs to Δ ;
3. Δ is maximal consistent;
4. Δ is deductively closed, and for every sentence θ ,

$$\Delta \vdash \theta \iff \theta \in \Delta. \tag{LC}$$

Proof. Enumerate all sentences as $\sigma_0, \sigma_1, \dots$. Starting from consistent $T_0 = T$, use the consistency-extension lemma to choose

$$T_{n+1} = T_n \cup \{\sigma_n\} \quad \text{or} \quad T_n \cup \{\neg\sigma_n\}$$

so that T_{n+1} remains consistent. Set

$$T^* := \bigcup_{n < \omega} T_n.$$

A contradiction from T^* uses finitely many assumptions and hence occurs in some T_n , impossible. Every sentence is decided when it is enumerated, so T^* is complete. If $S \supsetneq T^*$ were consistent, choose $\sigma \in S \setminus T^*$; completeness gives $\neg\sigma \in T^* \subseteq S$, contradiction. Thus T^* is maximal consistent.

It remains to prove deductive closure rather than assume it. If $\theta \in T^*$, then $T^* \vdash \theta$ by the assumption rule. Conversely suppose $T^* \vdash \theta$ but $\theta \notin T^*$. Sentence-decision gives $\neg\theta \in T^*$, hence $T^* \vdash \neg\theta$ by the assumption rule. This contradicts consistency. Therefore

$$T^* \vdash \theta \iff \theta \in T^*,$$

which proves (LC) and deductive closure. *Point specific to this result.* At each enumeration stage we decide one formula while preserving consistency. Completeness of the resulting theory means syntactic decision of every sentence, not semantic completeness of the proof system; that latter theorem comes only after the Henkin model is built. $\square$

Worked example

Worked Example 5.7 (A side condition that matters). From the assumption $P(x)$ one may not in general infer $\forall x P(x)$: the assumption may be true only under the present assignment to x . This is why generalization requires the quantified variable not to occur free in the undischarged assumptions. The deduction theorem preserves precisely this restriction.

A useful warning

When a proof uses generalization, check the eigenvariable condition before applying the semantic argument. Dropping it makes the formal system unsound.

Looking ahead

The next chapter, *Henkin completeness*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 6

Henkin completeness

Definitions and setup

Formal definitions used in this chapter

Language expansion and fresh constant. A language expansion enlarges $\mathcal{L}$ by new nonlogical symbols while retaining all old symbols. A constant c is *fresh* for a theory/formula when it does not occur in the original language expressions under discussion.

Henkin witness axiom, Henkin constant, and Henkin theory. For an existential formula $\exists x \varphi(x)$ and a fresh constant c_φ , its Henkin witness axiom is

$$\exists x \varphi(x) \rightarrow \varphi(c_\varphi).$$

The fresh constant symbol c_φ assigned to this witness axiom is called a *Henkin constant*. Because it is a constant symbol, it is itself a closed term. A theory T has the *Henkin witness property* when for every existential sentence,

$$T \vdash \exists x \varphi(x) \implies \exists \text{ closed term } t \ T \vdash \varphi(t).$$

A *Henkin theory* is a theory with this property.

Provable-equality quotient and term model. For a consistent theory T with equality, define on closed terms

$$s \sim_T t \iff T \vdash s = t, \quad [t]_T := \{s : s \sim_T t\}.$$

The equality-congruence theorem from Chapter 5 makes the quotient set of closed terms into an $\mathcal{L}$ -structure M_T by

$$\begin{aligned} f^{M_T}([t_1]_T, \dots, [t_n]_T) &:= [f(t_1, \dots, t_n)]_T, \\ M_T \models R([t_1]_T, \dots, [t_n]_T) &\iff T \vdash R(t_1, \dots, t_n), \end{aligned}$$

and $c^{M_T} = [c]_T$ for constants.

Notation for the completed Henkin theory. T_H denotes an intermediate consistent theory after witness axioms have been added. The symbol T^H denotes the *final* complete, consistent Henkin extension. No T^* notation is used for this final object in Chapter 6.

A Henkin theory has enough names for existential witnesses: for every formula $\exists x \varphi(x)$ there is a constant c such that the theory contains $\exists x \varphi(x) \rightarrow \varphi(c)$. A complete Henkin theory is also syntactically complete. Its term model has equivalence classes of closed terms as elements, with two terms identified exactly when the theory proves them equal. The central truth lemma says that semantic truth in this term model agrees with membership/provability in the complete Henkin theory.

How this chapter fits together

Completeness is proved by manufacturing a model from a consistent theory. Henkin constants provide witnesses for existential formulas; maximal consistency decides every sentence; provable equality is quotiented; and the truth lemma identifies semantic truth in the term model with membership in the complete Henkin theory. The proof should be read as a construction with four stages: witnesses, completion, quotient, and truth lemma.

Lemma 6.1 (Henkin witness extension preserves consistency). *Let T be a consistent theory in a language $\mathcal{L}$, let $\varphi(x)$ be an $\mathcal{L}$ -formula, and let $c \notin \mathcal{L}$ be a new constant symbol. Then*

$$T \cup \{\exists x \varphi(x) \rightarrow \varphi(c)\}$$

is consistent. Iterating the construction with a fresh constant for each enumerated existential formula preserves consistency at every finite stage and at the union stage.

Proof. Let T be consistent, let $\exists x \varphi(x)$ be a sentence, and let c be a new constant. Add the Henkin axiom

$$H : \quad \exists x \varphi(x) \rightarrow \varphi(c).$$

Suppose $T \cup \{H\}$ were inconsistent. By the deduction theorem,

$$T \vdash \neg H.$$

Classical propositional reasoning gives

$$T \vdash \exists x \varphi(x) \quad \text{and} \quad T \vdash \neg \varphi(c).$$

Because c does not occur in T or φ , replace it by a fresh variable and generalize to obtain

$$T \vdash \forall x \neg \varphi(x),$$

contradicting the existential theorem. Therefore adjoining one fresh witness axiom preserves consistency. Iterating this construction over an enumeration preserves consistency because every contradiction is finite.

Further explanation. Freshness of c is indispensable. It lets us replace the new constant by a variable after deriving $\neg \varphi(c)$, because no previous axiom or assumption can depend on that constant. The contradiction therefore returns to the old language and conflicts with $\exists x \varphi(x)$. Repeating the step over an enumeration of existential formulas is safe because any finite contradiction in the union would already appear at a finite stage. $\square$

Proposition 6.2 (Construction of a complete Henkin theory). *Every consistent theory T in a countable language has a consistent extension T^H in a countable expansion of the language such that (i) T^H is complete, and (ii) for every sentence $\exists x \varphi(x)$, if $T^H \vdash \exists x \varphi(x)$, then $T^H \vdash \varphi(c)$ for some closed Henkin constant c .*

Proof. We first carry out a Henkinization that is closed under the new constants it introduces. Put

$$\mathcal{L}_n := \mathcal{L} \cup \{c_0, \dots, c_{n-1}\} \quad (n < \omega),$$

where the c_n 's are pairwise distinct new constant symbols, and let $\mathcal{L}_\omega := \bigcup_{n < \omega} \mathcal{L}_n$. For each $i < \omega$, fix an enumeration

$$E_{i,0}, E_{i,1}, \dots$$

of all existential $\mathcal{L}_i$ -sentences, writing $E_{i,j} = \exists x \varphi_{i,j}(x)$. Fix also a diagonal enumeration $n \mapsto (i_n, j_n)$ of $\omega \times \omega$ such that $i_n \leq n$ for every n and every pair (i, j) occurs at some stage.

Starting with $T_0 = T$, define recursively

$$T_{n+1} := T_n \cup \{E_{i_n, j_n} \rightarrow \varphi_{i_n, j_n}(c_n)\}. \quad (\text{Hn})$$

Because $i_n \leq n$, the sentence E_{i_n, j_n} already belongs to the current language $\mathcal{L}_n$, whereas $c_n \notin \mathcal{L}_n$. Lemma 6.1 therefore applies at every stage and proves that each T_n is consistent. Set

$$T_H := \bigcup_{n < \omega} T_n.$$

If T_H were inconsistent, a finite contradiction proof would use only finitely many axioms from the union and hence would already occur in some T_N , impossible. Thus T_H is consistent.

We next verify the witness property for the *whole* union language. Let $E = \exists x \varphi(x)$ be any existential $\mathcal{L}_\omega$ -sentence. Since a formula is a finite string, all constants occurring in E lie in some $\mathcal{L}_i$. Hence $E = E_{i,j}$ for some j , and the pair (i, j) is processed at some stage n . The corresponding axiom

$$E \rightarrow \varphi(c_n)$$

belongs to T_H . Consequently

$$T_H \vdash E \implies T_H \vdash \varphi(c_n)$$

by modus ponens. Thus T_H is a consistent Henkin theory in the countable language $\mathcal{L}_\omega$.

Now apply Theorem 5.6 in the fixed countable language $\mathcal{L}_\omega$. It gives a complete, maximal consistent, deductively closed extension

$$T^H \supseteq T_H.$$

Every witness axiom of T_H remains an axiom of T^H ; therefore, if $T^H \vdash E$, deductive closure gives $E \in T^H$, and the corresponding witness axiom together with modus ponens yields a closed Henkin witness. Hence T^H is simultaneously complete, consistent, deductively closed, and Henkin.

Further explanation. The witness stage and the Lindenbaum-completion stage really may be performed separately, but only after the witness stage has been dovetailed over formulas in all of the successively enlarged languages. The diagonal schedule above is the missing closure mechanism: an existential sentence mentioning an earlier Henkin constant eventually appears in some $\mathcal{L}_i$ -enumeration and receives a still fresher witness. Once this has been done, later completion cannot destroy the witness axioms already present. $\square$

Lemma 6.3 (Truth lemma for the Henkin term model). *Let T^H be a complete, consistent, deductively closed Henkin theory, and let M_{T^H} be its quotient term model. For every formula $\varphi(\bar{x})$ and closed terms $\bar{t}$,*

$$M_{T^H} \models \varphi([\bar{t}]) \iff T^H \vdash \varphi(\bar{t}).$$

How to read the symbols: the Henkin truth lemma

The left side $M_{T^H} \models \varphi([\bar{t}])$ is a semantic statement about the term model. The right side $T^H \vdash \varphi(\bar{t})$ is a syntactic statement about the complete Henkin theory. The brackets $[t]$ matter because the model element is an equivalence class of closed terms modulo provable equality, whereas t itself is a piece of syntax. The induction proves that these two levels agree formula by formula.

Proof. Let the universe of the term model M_{T^H} be equivalence classes $[t]$ of closed terms modulo

$$s \sim t \iff T^H \vdash s = t.$$

Interpret functions by term formation and relations by

$$M_{T^H} \models R([t_1], \dots, [t_n]) \iff T^H \vdash R(t_1, \dots, t_n).$$

Equality congruence makes these definitions independent of representatives. We prove by induction on φ the stated truth lemma in the derivability form

$$M_{T^H} \models \varphi([\bar{t}]) \iff T^H \vdash \varphi(\bar{t}). \quad (\text{HTL})$$

Whenever it is convenient to write membership instead, deductive closure from Theorem 5.6 gives $T^H \vdash \theta \iff \theta \in T^H$ for sentences θ . Atomic cases are definitions; Boolean cases use completeness, consistency, and this membership/derivability equivalence. For $\exists x\psi$, if the left side holds, some closed term witnesses it and the induction hypothesis gives the corresponding derivable instance; the derived existential-introduction Lemma 5.3 yields the required existential theorem. Conversely, if $T^H \vdash \exists x\psi$, the Henkin property supplies a witness constant c with $T^H \vdash \psi(c)$, and the induction hypothesis gives a model witness. *Further explanation.*

The existential step is the reason for having constructed a Henkin theory. If the model satisfies $\exists x\psi(x, \bar{t})$, some closed term class $[s]$ witnesses it, and the induction hypothesis yields $T^H \vdash \psi(s, \bar{t})$. Conversely, if $T^H \vdash \exists x\psi(x, \bar{t})$, the witness property gives a designated closed term c with $T^H \vdash \psi(c, \bar{t})$. The induction hypothesis then turns that derivation back into a model witness. All other connectives follow from completeness and consistency, with Theorem 5.6 supplying the proved equivalence between derivability and membership whenever the membership presentation is used. $\square$

Theorem 6.4 (Gödel completeness theorem for first-order logic with equality). *For every first-order theory Γ with equality and every sentence φ ,*

$$\Gamma \models \varphi \iff \Gamma \vdash \varphi.$$

Equivalently, a first-order theory is syntactically consistent if and only if it has a model.

Commented symbolic trace: the completeness contradiction

The proof has three typed steps:

1. $T \not\vdash \varphi$ is syntactic information.
2. By the deduction theorem, this makes $T \cup \{\neg\varphi\}$ syntactically consistent.
3. The Henkin construction converts that consistency into a semantic model of $T \cup \{\neg\varphi\}$, contradicting the semantic assumption $T \models \varphi$.

Keeping these types separate prevents the circular mistake of using completeness in order to prove completeness.

Proof. Assume $\Gamma \models \varphi$ and suppose $\Gamma \not\vdash \varphi$. By the deduction theorem,

$$\Gamma \cup \{\neg\varphi\}$$

is consistent. Extend this theory to a complete Henkin theory H , and form its equality-quotiented term model M_H . The Henkin truth lemma gives

$$M_H \models H.$$

In particular,

$$M_H \models \Gamma \quad \text{and} \quad M_H \models \neg\varphi,$$

contradicting $\Gamma \models \varphi$. Therefore $\Gamma \vdash \varphi$. Combining this with soundness yields

$$\Gamma \vdash \varphi \iff \Gamma \models \varphi.$$

The equality axioms are already built into the quotient term model, so no separate equality-completeness argument is hidden. *Further explanation.* This proof is most transparent in contrapositive form. Failure of derivability says that adding $\neg\varphi$ does not create a contradiction; Henkinization converts that syntactic consistency into a concrete model of $T \cup \{\neg\varphi\}$. Such a model is precisely a counterexample to semantic consequence. Equality causes no additional gap because the term model was already quotiented by provable equality, so the equality axioms are represented by genuine identity in the constructed structure. $\square$

Remark 6.5 (Source note). Gödel proved first-order completeness in 1930; the witness-constant proof used here follows the later Henkin organization. Compare [3, 1, 2].

Worked example

Worked Example 6.6 (A tiny Henkin witness). Suppose the language contains a unary predicate P and the theory contains $\exists x P(x)$. Add a fresh constant c together with $\exists x P(x) \rightarrow P(c)$. In any complete Henkin extension containing the existential sentence, $P(c)$ is present. The class of c in the term model is therefore an actual witness. The general construction simply performs this operation for every existential formula.

A useful warning

A term model with equality cannot use raw closed terms. Without quotienting by provable equality, the equality axioms may be true syntactically but false in the constructed structure.

Looking ahead

The next chapter, *Compactness and Löwenheim–Skolem*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 7

Compactness and Löwenheim–Skolem

Definitions and setup

Formal definitions used in this chapter

Expanded language $\mathcal{L}(M)$. If M is an $\mathcal{L}$ -structure,

$$\mathcal{L}(M) := \mathcal{L} \cup \{c_a : a \in M\},$$

where each c_a is a new constant symbol. The canonical $\mathcal{L}(M)$ -expansion of M interprets c_a as a .

Substructure. An $\mathcal{L}$ -structure N is a substructure of M , written $N \subseteq_{\mathcal{L}} M$, when its domain is a subset of M , it contains every interpreted constant, it is closed under every interpreted function, and every relation in N is the restriction of the corresponding relation in M .

Embedding and elementary embedding. An embedding $e : M \hookrightarrow N$ is an injective map preserving constants and functions and preserving and reflecting every relation (including equality). It is *elementary* when for every formula $\varphi(\bar{x})$ and tuple $\bar{a} \in M$,

$$M \models \varphi(\bar{a}) \iff N \models \varphi(e\bar{a}).$$

Elementary substructure. $N \preceq M$ means that $N \subseteq_{\mathcal{L}} M$ and the inclusion map $N \hookrightarrow M$ is elementary. Equivalently, every formula with parameters from N has the same truth value in N and M .

Atomic and elementary diagrams. The atomic diagram

$$\text{Diag}_{\text{at}}(M)$$

is the set of all atomic and negated atomic $\mathcal{L}(M)$ -sentences true in the canonical expansion of M . The elementary diagram

$$\text{Diag}_{\text{el}}(M)$$

is the set of *all* $\mathcal{L}(M)$ -sentences true in that expansion.

Skolem function and Skolem hull. For a formula $\exists x \varphi(x, \bar{y})$ interpreted in M , a Skolem function is a chosen function f_{φ} on parameter tuples such that

$$M \models \exists x \varphi(x, \bar{a}) \implies M \models \varphi(f_{\varphi}(\bar{a}), \bar{a}).$$

Fix the family $\mathcal{F}_{\text{Sk}}$ consisting of the interpreted basic functions of M together with all chosen Skolem functions. For $A \subseteq M$, put

$$H_0 = A \cup \{c^M : c \text{ is a constant symbol of } \mathcal{L}\},$$

and recursively

$$H_{n+1} = H_n \cup \{f(\bar{a}) : f \in \mathcal{F}_{\text{Sk}}, \bar{a} \in H_n^{\text{arity}(f)}\}.$$

The *Skolem hull* of A in M is

$$\text{Hull}_{\text{Sk}}^M(A) := \bigcup_{n < \omega} H_n.$$

This explicit construction is closed under every function in $\mathcal{F}_{\text{Sk}}$. Moreover, if $B \subseteq M$ contains A , contains the interpreted constants, and is closed under $\mathcal{F}_{\text{Sk}}$, then induction on n gives $H_n \subseteq B$; hence $\text{Hull}_{\text{Sk}}^M(A) \subseteq B$. Thus the displayed union really is the least such closed subset rather than an existence claim hidden inside the word “hull.”

Tarski–Vaught witness condition. If $N \subseteq_{\mathcal{L}} M$, say that N satisfies the *Tarski–Vaught witness condition* when, for every formula $\varphi(x, \bar{y})$ and every tuple $\bar{a} \in N$,

$$M \models \exists x \varphi(x, \bar{a}) \implies \exists b \in N \, M \models \varphi(b, \bar{a}).$$

The theorem below proves that this witness condition is equivalent to $N \preccurlyeq M$; the equivalence is not taken as part of the definition.

Ambient cardinality convention used in the Löwenheim–Skolem bounds. The model-size arguments in this chapter are carried out in the surrounding ZFC metatheory, not inside the object theory under study. Here $|X| \leq |Y|$ means that there is an injection $X \rightarrow Y$, and $|X| = |Y|$ means that there is a bijection. An *ambient ordinal* is a transitive set well ordered by membership, and an *ambient cardinal* is an ordinal not bijective with any smaller ordinal. By ambient Choice, every set is bijective with a unique such cardinal; that cardinal is denoted $|X|$. The same ambient Choice principle is also used below to choose, simultaneously for each existential formula, a Skolem witness function on all parameter tuples for which a witness exists. The set of ambient natural numbers is ω , and $\aleph_0 := |\omega|$. A set is *countable* when $|X| \leq \aleph_0$. For ambient cardinals, $\max\{|A|, |\mathcal{L}|, \aleph_0\}$ is the least of the three cardinal bounds that is greater than or equal to each of them. Volume IV later develops these set-theoretic notions internally; here they are explicit metatheoretic prerequisites for the Löwenheim–Skolem size estimates.

Compactness states that a first-order theory has a model exactly when every finite subtheory has a model. The diagram of a structure is obtained by adding a constant name for each element and recording the atomic facts true of those names; the elementary diagram records all first-order facts with parameters. An elementary substructure $A \preccurlyeq M$ is a substructure in which every formula with parameters from A has the same truth value as in M . Löwenheim–Skolem theorems control the possible cardinalities of elementary models; they do not say that an externally small model regards all of its sets as small.

How this chapter fits together

Completeness immediately yields compactness because a contradiction has a finite proof. Compactness, diagrams, and Skolem hulls then control model size. Downward Löwenheim–Skolem constructs small elementary substructures; upward Löwenheim–Skolem uses compactness to force many distinct elements. The resulting non-categoricity phenomena are a feature of first-order expressiveness rather than a defect in the proofs.

Theorem 7.1 (First-order compactness theorem). *A first-order theory Γ is satisfiable if and only if every finite subset $\Gamma_0 \subseteq \Gamma$ is satisfiable.*

Proof. Suppose every finite subset of a first-order theory T is satisfiable. If T were inconsistent in the sense fixed in Chapter 4, there would be a sentence θ for which both $T \vdash \theta$ and $T \vdash \neg\theta$. The two finite derivations use only finitely many assumptions, so some finite $T_0 \subseteq T$ satisfies

$$T_0 \vdash \theta, \quad T_0 \vdash \neg\theta.$$

By soundness no structure can satisfy T_0 , contrary to finite satisfiability. Thus T is consistent. Notice that no propositional symbol $\perp$ is imported into the first-order language here. By the first-order Completeness

Theorem, T has a model. The converse is immediate: a model of T is also a model of every finite subset. Hence

$$T \text{ satisfiable} \iff \forall T_0 [(T_0 \subseteq T \ \& \ T_0 \text{ finite}) \rightarrow T_0 \text{ satisfiable}].$$

Unlike propositional compactness, the conclusion may concern an infinite structure and cannot be obtained by a finite truth table; completeness supplies the model. *Point specific to this result.* The first-order theorem is obtained only after Henkin completeness is available. A contradictory derivation from an infinite theory uses finitely many assumptions, so finite satisfiability implies syntactic consistency; completeness then supplies a model of the entire theory. $\square$

Remark 7.2 (Source note). Compactness is presented here as a corollary of completeness plus finiteness of formal derivations; see [2].

Lemma 7.3 (Diagram lemma). *Let M be an $\mathcal{L}$ -structure. If an $\mathcal{L}(M)$ -structure N satisfies the atomic diagram of M , then the map $a \mapsto c_a^N$ is an embedding $M \hookrightarrow N$. If N satisfies the elementary diagram of M , this embedding is elementary.*

Proof. Expand the language by a constant c_a for each $a \in M$. The atomic diagram $\text{Diag}_{\text{at}}(M)$ contains every atomic or negated atomic sentence with these constants that is true in M . If an expanded structure N satisfies the diagram, define

$$e : M \rightarrow N, \quad e(a) = c_a^N.$$

The equality sentences in the diagram make e injective. Function and relation sentences show that e preserves all symbols, so it is an embedding. If instead N satisfies the elementary diagram, which contains every first-order sentence $\varphi(c_{a_1}, \dots, c_{a_n})$ true in M , then

$$M \models \varphi(\bar{a}) \iff N \models \varphi(e\bar{a}),$$

so e is elementary. Conversely any (elementary) embedding makes the corresponding diagram true. *Further explanation.* For the elementary diagram, the converse direction is equally important. If $j : M \rightarrow N$ is elementary and N interprets each new constant c_a as $j(a)$, then every sentence in the elementary diagram is true in N . Thus realizing the elementary diagram and admitting an elementary embedding are equivalent descriptions of the same situation. The atomic diagram gives only an embedding because it records only the atomic structure. $\square$

Theorem 7.4 (Tarski–Vaught criterion). *Let $N \subseteq_{\mathcal{L}} M$ be a substructure. Then*

$$N \preceq M$$

if and only if N satisfies the Tarski–Vaught witness condition: for every formula $\varphi(x, \bar{y})$ and every tuple $\bar{a} \in N$,

$$M \models \exists x \varphi(x, \bar{a}) \implies \exists b \in N \ M \models \varphi(b, \bar{a}).$$

Proof. If $N \preceq M$ and $M \models \exists x \varphi(x, \bar{a})$ with $\bar{a} \in N$, then elementarity gives

$$N \models \exists x \varphi(x, \bar{a}).$$

By the Tarski semantics for $\exists$, some $b \in N$ satisfies $N \models \varphi(b, \bar{a})$; elementarity again yields $M \models \varphi(b, \bar{a})$.

Conversely assume the witness condition. We prove by induction on formula complexity that, for every tuple $\bar{a} \in N$,

$$N \models \psi(\bar{a}) \iff M \models \psi(\bar{a}). \tag{TV}$$

Atomic formulas agree because $N \subseteq_{\mathcal{L}} M$ is a substructure. The Boolean connectives follow immediately from the induction hypothesis. For the existential step, write $\psi(\bar{y}) = \exists x \chi(x, \bar{y})$.

If $N \models \exists x \chi(x, \bar{a})$, choose $b \in N$ with $N \models \chi(b, \bar{a})$. The induction hypothesis gives $M \models \chi(b, \bar{a})$, hence $M \models \exists x \chi(x, \bar{a})$.

Conversely, if $M \models \exists x \chi(x, \bar{a})$, the witness condition gives $b \in N$ with $M \models \chi(b, \bar{a})$. By the induction hypothesis, $N \models \chi(b, \bar{a})$, so $N \models \exists x \chi(x, \bar{a})$. Thus (TV) holds for every formula, which is exactly $N \preceq M$. $\square$

Theorem 7.5 (Downward Löwenheim–Skolem theorem). *Let M be an infinite $\mathcal{L}$ -structure and $A \subseteq M$. Then there is an elementary substructure $N \preceq M$ with $A \subseteq N$ and*

$$|N| \leq \max\{|A|, |\mathcal{L}|, \aleph_0\}.$$

How to read the symbols: what the Skolem hull is doing

The hull construction is a closure algorithm. Start with the parameters you insist on keeping. Repeatedly apply every basic function symbol and every chosen Skolem function. The union after ω stages is small because each stage adds only values of finitely many arguments, and it is elementary because every existential statement true in the large structure has a chosen witness produced by a Skolem function inside the hull.

Proof. Choose one Skolem function for every existential formula and let

$$H = \text{Hull}_{\text{Sk}}^M(A)$$

be the explicitly defined finite-stage hull from the setup. If $|\mathcal{L}| = \lambda$, then

$$|H| \leq \max(|A|, \lambda, \aleph_0).$$

By construction, whenever

$$M \models \exists x \varphi(x, \bar{a}) \quad (\bar{a} \in H),$$

the chosen Skolem function produces a witness in H . Tarski–Vaught therefore gives

$$H \preceq M.$$

Taking the reduct to the original language yields the required elementary substructure of the stated cardinal bound. *Further explanation.* The cardinal estimate comes from the closure process itself. At each finite stage there are at most $\max(|A|, |\mathcal{L}|, \aleph_0)$ tuples to which the countably/ $|\mathcal{L}|$ -many Skolem functions can be applied, so that bound is preserved. Taking the union of ω stages does not increase it. Tarski–Vaught then supplies elementarity, so the construction controls size without sacrificing any first-order truths with parameters from the smaller structure. $\square$

Theorem 7.6 (Upward Löwenheim–Skolem theorem). *If a theory T in the language $\mathcal{L}$ has an infinite model, then for every infinite cardinal*

$$\kappa \geq \max\{|\mathcal{L}|, \aleph_0\}$$

it has a model of cardinality exactly κ .

Proof. Add new constants c_α for $\alpha < \kappa$ and the sentences

$$c_\alpha \neq c_\beta \quad (\alpha \neq \beta).$$

Every finite subset of the expanded theory is satisfiable in an infinite model of T , because only finitely many new constants must be interpreted as distinct elements. By Compactness there is therefore an expanded model M^* with at least κ elements.

Apply the already proved Downward Löwenheim–Skolem theorem to M^* , taking as the prescribed parameter set

$$A = \{c_\alpha^{M^*} : \alpha < \kappa\}.$$

The expanded language has cardinality at most κ , so there is $N^* \preceq M^*$ containing A with

$$|N^*| \leq \kappa.$$

Because the elements $c_\alpha^{M^*}$ are pairwise distinct and all belong to N^* ,

$$|N^*| \geq \kappa.$$

Hence $|N^*| = \kappa$. Forgetting the added constants gives a model of the original theory T of cardinality exactly κ .

Dependency check. Tarski–Vaught is no longer hidden inside this proof: it was proved before Downward Löwenheim–Skolem, and Downward Löwenheim–Skolem is the only elementarity theorem used here. $\square$

Worked example

Worked Example 7.7 (A nonstandard element by compactness). Work in any arithmetic language containing a constant 0 and successor symbol S , and write $\bar{n} := S^n 0$ for the numeral attached to the external natural number n . Add a new constant c together with the sentences

$$c \neq \bar{n} \quad (n \in \omega).$$

Every finite subset is satisfiable in the standard natural-number structure: interpret c by a natural number whose numeral is not among the finitely many excluded numerals. Compactness therefore gives a model containing an element c^M different from every standard numeral $\bar{n}^M$. Hence the model has an element not named by any standard numeral; this is the precise nonstandard-element conclusion, and it uses no previously undefined order symbol $<$.

A useful warning

Downward Löwenheim–Skolem says that if M is an infinite structure in language $\mathcal{L}$ and $A \subseteq M$, then there is $N \preccurlyeq M$ containing A with $|N| \leq \max(|A|, |\mathcal{L}|, \aleph_0)$; it does not say that sets externally countable in the ambient universe are internally seen as countable.

End-of-volume perspective

The logical language is now mature enough to build and compare models. The two indispensable bridges are completeness (syntax to semantics) and compactness (finite satisfiability to global satisfiability).

Part II

Volume II: Computability and Arithmetized Syntax

Preface

The aim is to make finite syntax manipulable inside arithmetic. Concrete encodings are treated as mathematical objects rather than as an appeal to “effective coding.”

Position in the series

Volume II turns finite syntax into arithmetic data. Its coding and representability results are the technical input for the fixed-point and provability arguments of Volume III.

Sources and further reading

The arithmetization is arranged to feed directly into the incompleteness proofs of Volume III. The historical source is Gödel’s 1931 paper [4]; modern background may be found in [2].

Notation for this volume

$\ulcorner \varphi \urcorner$ denotes the Gödel number of φ . $\text{Prf}_T(p, q)$ is the proof predicate and $\text{Prov}_T(q) \equiv \exists p \text{Prf}_T(p, q)$.

How to read symbolic arguments in this volume

When a formula appears, read it in layers rather than as one visual block. First identify the *ambient language*: are we speaking inside a formal theory, about a structure, or in the surrounding metatheory? Next mark the objects being manipulated (formulas, codes, sets, names, conditions, ordinals) and the operation being applied to them. Finally check the side conditions—free variables, rank bounds, compatibility, genericity, transitivity, or consistency hypotheses. Whenever a displayed derivation changes level, the text says so explicitly. A displayed line is therefore meant to be read like a line of well-commented code: the symbols perform the operation, and the surrounding prose records its type, preconditions, and purpose.

Chapter 8

Primitive recursion

Definitions and setup

Formal definitions used in this chapter

Initial functions. For each arity $n \geq 1$, the initial numerical functions are the zero function $Z^n(\bar{x}) = 0$, the successor function $S(x) = x + 1$, and the coordinate projections

$$\pi_i^n(x_1, \dots, x_n) = x_i \quad (1 \leq i \leq n).$$

Here $\bar{x}$ abbreviates a finite tuple of natural-number variables.

Composition. If $g_1, \dots, g_k : \mathbb{N}^m \rightarrow \mathbb{N}$ and $h : \mathbb{N}^k \rightarrow \mathbb{N}$, their composition is

$$f(\bar{x}) = h(g_1(\bar{x}), \dots, g_k(\bar{x})).$$

Primitive-recursion scheme. Given $g : \mathbb{N}^m \rightarrow \mathbb{N}$ and $h : \mathbb{N}^{m+2} \rightarrow \mathbb{N}$, the ordinary recursion theorem on $\mathbb{N}$ gives a unique $f : \mathbb{N}^{m+1} \rightarrow \mathbb{N}$ satisfying

$$f(\bar{x}, 0) = g(\bar{x}), \quad f(\bar{x}, n+1) = h(\bar{x}, n, f(\bar{x}, n)).$$

The *primitive-recursion constructor* is the operation that takes the pair (g, h) to this function f .

Primitive-recursive function. A finite *primitive-recursive construction tree* is a finite rooted tree whose leaves are initial functions and whose internal nodes are labeled either by a well-typed composition constructor or by a well-typed primitive-recursion constructor applied to the functions represented by its children. A numerical function is *primitive recursive* when it is represented by some such finite construction tree. The class of all primitive-recursive numerical functions is denoted $\mathcal{PR}$. Equivalently, $\mathcal{PR}$ is the closure of the initial functions under finitely many applications of composition and primitive recursion; the finite-tree definition makes the word “generated” explicit and does not hide an existence claim inside “least class.”

Characteristic function and primitive-recursive relation. For a relation $R \subseteq \mathbb{N}^k$, its characteristic function is $\chi_R(\bar{x}) = 1$ when $R(\bar{x})$ holds and 0 otherwise. The relation is *primitive recursive* when $\chi_R \in \mathcal{PR}$.

Bounded quantification and bounded search. If $R(\bar{x}, y)$ is a numerical relation and $z \in \mathbb{N}$, then

$$(\exists y < z)R(\bar{x}, y), \quad (\forall y < z)R(\bar{x}, y)$$

are its bounded existential and universal quantifications. The bounded minimum $\mu y < z R(\bar{x}, y)$ is the least $y < z$ satisfying R , with the fixed default value z if no such y exists.

Pairing and coordinate decoding. A pairing system consists of an injective map $\langle \cdot, \cdot \rangle : \mathbb{N}^2 \rightarrow \mathbb{N}$ and functions $\pi_0, \pi_1 : \mathbb{N} \rightarrow \mathbb{N}$ satisfying

$$\pi_0(\langle x, y \rangle) = x, \quad \pi_1(\langle x, y \rangle) = y.$$

When all three functions are primitive recursive we call it a primitive-recursive pairing system.

Primitive-recursive functions are exactly the functions represented by finite construction trees built from zero, successor, and projections using composition and primitive recursion. The point is not merely that such functions are computable, but that their computations can be established by bounded finite data. A relation is primitive recursive when its characteristic function is primitive recursive. Bounded sums, products, and searches will be derived rather than added as new primitives. Pairing functions let several numerical parameters be packaged into one code.

How this chapter fits together

Primitive recursion supplies a deliberately restrictive language of effective constructions. The point is not merely that familiar numerical functions are computable, but that their computation histories can later be represented inside arithmetic. Closure under composition, recursion, bounded sums, bounded products, and bounded search gives a toolbox from which the coding functions of the next chapters are built.

Theorem 8.1 (Closure of primitive-recursive functions under composition). *If $g_1, \dots, g_k : \mathbb{N}^m \rightarrow \mathbb{N}$ and $h : \mathbb{N}^k \rightarrow \mathbb{N}$ are primitive recursive, then the composition $f(\bar{x}) = h(g_1(\bar{x}), \dots, g_k(\bar{x}))$ is primitive recursive.*

Proof. By definition, membership in $\mathcal{PR}$ means possession of a finite primitive-recursive construction tree. Assume

$$g_i \in \mathcal{PR} \quad (1 \leq i \leq k), \quad h \in \mathcal{PR}.$$

The function in the statement is

$$f(\bar{x}) = h(g_1(\bar{x}), \dots, g_k(\bar{x})).$$

This is exactly one application of the composition clause in the definition of $\mathcal{PR}$. More formally, if C_h is a primitive-recursive program for h and C_i is one for g_i , the composition constructor produces the program

$$\bar{x} \mapsto (g_1(\bar{x}), \dots, g_k(\bar{x})) \mapsto h(g_1(\bar{x}), \dots, g_k(\bar{x})).$$

No minimization or unbounded search is introduced. Therefore the resulting function remains in $\mathcal{PR}$. Notice that this proof uses the closure clause itself; it does not assert that every computable composition is primitive recursive. The latter would require first proving that the component functions already lie in $\mathcal{PR}$.

Point specific to this result. Composition is one of the defining closure operations: substitute already primitive-recursive outputs into a primitive-recursive outer function. The point of recording it explicitly is that later coding constructions are nested compositions of projections, arithmetic, and bounded case distinctions. $\square$

Theorem 8.2 (Closure under primitive recursion). *If $g : \mathbb{N}^m \rightarrow \mathbb{N}$ and $h : \mathbb{N}^{m+2} \rightarrow \mathbb{N}$ are primitive recursive, then the uniquely defined function f satisfying*

$$f(\bar{x}, 0) = g(\bar{x}), \quad f(\bar{x}, n+1) = h(\bar{x}, n, f(\bar{x}, n))$$

is primitive recursive.

Proof. Let $g : \mathbb{N}^m \rightarrow \mathbb{N}$ and $h : \mathbb{N}^{m+2} \rightarrow \mathbb{N}$ be primitive recursive. By the primitive-recursion constructor there is a unique function $f : \mathbb{N}^{m+1} \rightarrow \mathbb{N}$ satisfying

$$f(\bar{x}, 0) = g(\bar{x}), \quad f(\bar{x}, n+1) = h(\bar{x}, n, f(\bar{x}, n)).$$

The construction is itself one of the defining closure operations of $\mathcal{PR}$, so $f \in \mathcal{PR}$. Uniqueness can also be checked independently by induction on n : if f, f' satisfy the two equations, then they agree at 0; and from $f(\bar{x}, n) = f'(\bar{x}, n)$ the successor equation gives

$$f(\bar{x}, n+1) = h(\bar{x}, n, f(\bar{x}, n)) = h(\bar{x}, n, f'(\bar{x}, n)) = f'(\bar{x}, n+1).$$

This uniqueness is useful later, because arithmetic will represent a primitive-recursive computation by a finite sequence and we must know that the represented output is determined by the input. $\square$

Lemma 8.3 (Primitive-recursive bounded search). *If $R(\bar{x}, y)$ is a primitive-recursive relation, then the bounded minimization function*

$$\mu y < z R(\bar{x}, y)$$

(with a fixed default value when no witness exists) is primitive recursive. Consequently bounded existential and universal quantification preserve primitive recursiveness.

Proof. Let $R(\bar{x}, y)$ be a primitive-recursive predicate and let $b(\bar{x})$ be a primitive-recursive bound. Define the bounded-search function

$$\mu_{y < b(\bar{x})} R(\bar{x}, y) = \begin{cases} \min\{y < b(\bar{x}) : R(\bar{x}, y)\}, & \text{if such } y \text{ exists,} \\ b(\bar{x}), & \text{otherwise.} \end{cases}$$

Construct it by primitive recursion on $n \leq b(\bar{x})$. Let $s(\bar{x}, 0) = b(\bar{x})$, and at stage $n + 1$ replace the stored value by n exactly when no smaller witness has been stored and $R(\bar{x}, n)$ holds. The tests “ $s = b$ ”, “ $n < b$ ”, and R are primitive recursive, and case distinction on a primitive-recursive predicate is primitive recursive. Hence s is primitive recursive. Evaluating it at the primitive-recursive bound $b(\bar{x})$ gives the required bounded minimum. The essential point is that the search range is bounded in advance; unrestricted minimization would leave the primitive-recursive class.

Point specific to this result. Bounded search is total because the search interval is finite. A primitive recursion can carry the least successful index found so far, so no unbounded minimization principle is being introduced. $\square$

Proposition 8.4 (Primitive-recursive pairing and projection functions). *There are primitive-recursive functions $\langle x, y \rangle$, π_0 , π_1 such that*

$$\pi_0(\langle x, y \rangle) = x, \quad \pi_1(\langle x, y \rangle) = y$$

for all $x, y \in \mathbb{N}$.

Proof. Use Cantor’s pairing polynomial

$$\langle x, y \rangle = \frac{(x + y)(x + y + 1)}{2} + y.$$

Addition and multiplication are primitive recursive. The numerator is always even, and integer division by 2 can be implemented by bounded search for the unique $q \leq (x + y)(x + y + 1)$ with $2q$ equal to that numerator. Thus pairing is primitive recursive. To recover (x, y) from z , first find the unique $w \leq z$ such that

$$\frac{w(w + 1)}{2} \leq z < \frac{(w + 1)(w + 2)}{2}.$$

This is bounded search. Put

$$y = z - \frac{w(w + 1)}{2}, \quad x = w - y.$$

Truncated subtraction and the preceding bounded searches are primitive recursive, so both projections $\pi_0(z) = x$ and $\pi_1(z) = y$ are primitive recursive. Direct substitution verifies

$$\pi_0(\langle x, y \rangle) = x, \quad \pi_1(\langle x, y \rangle) = y.$$

These identities, rather than the particular polynomial, are what later coding arguments use.

Point specific to this result. For pairing, it is not enough that a bijection $\mathbb{N}^2 \rightarrow \mathbb{N}$ exists; both encoding and decoding must be primitive recursive. The inverse coordinates are found by searches bounded by the code itself, preserving primitive recursiveness. $\square$

Worked example

Worked Example 8.5 (A bounded search). If $R(x, y)$ is primitive recursive, define $m(x, z)$ to be the least $y < z$ with $R(x, y)$, or z if there is none. A recursion scans $0, 1, \dots, z - 1$ while storing the first successful index. Because the search is bounded in advance, the construction remains primitive recursive.

A useful warning

Unbounded minimization is not primitive recursion. Later c.e. predicates may use an existentially quantified computation witness, but that is conceptually different from a bounded primitive-recursive search.

Looking ahead

The next chapter, *Coding finite objects*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 9

Coding finite objects

Definitions and setup

Formal definitions used in this chapter

Finite-sequence coding package. A *finite-sequence coding package* consists of a numerical predicate $\text{Seq}(s)$, a length function $\text{lh}(s)$, and a component function $(s)_i$ such that, whenever $\text{Seq}(s)$ holds, $\text{lh}(s)$ is the represented length and $(s)_i$ is the represented i -th entry for $i < \text{lh}(s)$, and every finite tuple has a code. The package is *primitive recursive* when these operations, together with the constructor operations subsequently used, are primitive recursive. The theorem immediately below constructs one explicit package. Only after that theorem is proved do the symbols Seq , lh , $(s)_i$ refer to the fixed package used throughout the rest of the book.

Finite strings and labelled rooted trees. A finite string over a fixed coded alphabet is a finite sequence of symbol codes. A finite rooted tree is a finite set of node codes with a distinguished root and a parent map assigning to each nonroot node its unique immediate predecessor; a labelled tree additionally assigns a symbol code to each node.

Primitive-recursive coding of finite data. A coding is called primitive recursive when its well-formedness predicates, component-extraction functions, and the constructor operations used in this volume are primitive recursive. Thus “effective coding” below means primitive-recursive coding in this explicit sense.

Coded proof calculus and finite derivation. A *coded proof calculus* consists of: (i) a numerical coding of its formulas; (ii) a primitive-recursive predicate recognizing formula codes; and (iii) primitive-recursive local predicates recognizing axiom instances and checking each inference rule from the codes of its premises and conclusion. Such a calculus is called *primitive-recursively checkable*. A finite derivation in a coded calculus is a finite sequence of formula codes together with finite justification records for its lines. This chapter proves the conditional coding lemma for any such calculus; Chapter 6 constructs the required Gödel coding for the concrete first-order calculus fixed in Volume I.

A Gödel-style code for a finite object must support effective recognition and component extraction. We use a primitive-recursive coding of finite sequences and then encode strings, trees, and derivations as tagged sequences. The length function, the i -th component function, concatenation, and bounded modification are required to be primitive recursive. The precise numerical values of the codes are irrelevant; the uniform primitive-recursive operations on them are the load-bearing content.

How this chapter fits together

Gödel coding needs more than a pairing function: formulas and proofs are variable-length finite objects. We therefore encode finite sequences with primitive-recursive access to their length and coordinates. Once sequence manipulation is available, finite strings, trees, substitutions, and derivations can all be treated arithmetically.

Theorem 9.1 (Primitive-recursive coding and decoding of finite sequences). *There is a primitive-recursive coding of finite sequences by natural numbers together with primitive-recursive functions $\text{lh}(s)$ and $(s)_i$ such that every finite sequence $\langle a_0, \dots, a_{n-1} \rangle$ has a code s with $\text{lh}(s) = n$ and $(s)_i = a_i$ for $i < n$.*

How to read the symbols: sequence codes are data structures

A natural number is being used as a serialized finite list. The coding theorem must supply three operations analogous to a programming interface: create a code, recover its length, and read its i -th entry. Later arithmetization never relies on the visual form of the code; it relies only on the fact that these operations are primitive recursive and hence representable in weak arithmetic.

Proof. Use the primitive-recursive pairing function $\langle x, y \rangle$ and its primitive-recursive coordinate projections π_0, π_1 from the preceding chapter. Encode a finite sequence by a length together with a nested-pair payload. Define

$$\text{nil} := 0, \quad \text{cons}(a, r) := \langle a, r \rangle.$$

For a tuple $\bar{a} = \langle a_0, \dots, a_{n-1} \rangle$, define its payload backwards by

$$r_n := 0, \quad r_i := \langle a_i, r_{i+1} \rangle \quad (i < n),$$

and let

$$\text{code}(\bar{a}) := \langle n, r_0 \rangle. \tag{S1}$$

Thus existence of a code is completely explicit.

For an arbitrary number s , put

$$\text{lh}(s) := \pi_0(s), \quad \text{tail}_0(s) := \pi_1(s), \quad \text{tail}_{i+1}(s) := \pi_1(\text{tail}_i(s)).$$

The bounded iteration $(s, i) \mapsto \text{tail}_i(s)$ is primitive recursive by primitive recursion on i . Define

$$(s)_i := \begin{cases} \pi_0(\text{tail}_i(s)), & i < \text{lh}(s), \\ 0, & i \geq \text{lh}(s), \end{cases} \tag{S2}$$

and

$$\text{Seq}(s) \iff \text{tail}_{\text{lh}(s)}(s) = 0. \tag{S3}$$

Bounded comparison, pairing, projections, and bounded iteration are primitive recursive, so (S2)–(S3) are primitive recursive. For the explicit code in (S1), exactly n tail operations reach 0, and induction on $i < n$ gives

$$\text{lh}(\text{code}(\bar{a})) = n, \quad (\text{code}(\bar{a}))_i = a_i.$$

Hence every finite sequence has a code and the required decoding operations are primitive recursive.

The same nested representation also supplies a primitive-recursive constructor. If r is a payload and a is a new first entry, $\text{cons}(a, r) = \langle a, r \rangle$ is primitive recursive. Consequently finite payloads can be rebuilt by bounded recursion from their decoded entries. This constructor property is recorded explicitly because the later concatenation and replacement lemmas use it; no Chinese-remainder or unrecorded search bound is needed.

Point specific to this result. The load-bearing fact is not merely that finite sequences can be injected into $\mathbb{N}$. The code carries a primitive-recursive length, primitive-recursive random access to every bounded coordinate, and a primitive-recursive cons operation from which the later finite-data constructors are built. $\square$

Proposition 9.2 (Primitive-recursive coding of finite trees and strings). *Finite labelled strings and finite rooted trees over a fixed finite or computably coded alphabet admit Gödel codings for which the predicates “ s codes such a string/tree” and the basic constructor, predecessor, and label operations are primitive recursive.*

Proof. A finite string over a finite alphabet is a finite sequence of symbol codes, so the sequence coding theorem already gives primitive-recursive coding and decoding. A finite rooted labelled tree can be coded by a finite sequence

$$\langle \langle \ell_i, p_i \rangle : i < n \rangle,$$

where ℓ_i is the label of node i and $p_i < i$ is the parent index, with a distinguished root convention. The predicate saying that such a code is a tree is bounded: check for every $i < n$ that the parent index has the required range, that the root is unique, and that the local arities agree with the labels. Bounded universal quantification over a primitive-recursive predicate is primitive recursive. Consequently “ t codes a finite labelled tree”, “node i has label a ”, and “ j is a child of i ” are primitive-recursive relations. Syntax trees will later be a special case in which the labels are logical symbols and the arity test enforces the grammar.

Point specific to this result. Trees and strings are reduced to finite sequences by storing node tags, parent data, and labels. Local well-formedness is then a bounded property of the code, so later syntactic parse trees can be checked arithmetically. $\square$

Lemma 9.3 (Bounded sequence manipulation is primitive recursive). *For the chosen sequence coding, the operations of taking length, extracting a component, concatenating two coded sequences, replacing a bounded component, and checking a bounded local property of all components are primitive recursive.*

Proof. Write a canonical sequence code as $s = \langle n, r \rangle$, where r is its nested-pair payload. Length and component extraction are primitive recursive by the coding theorem. We now give the constructors rather than appealing to an unspecified search bound.

Define the payload-append operation by primitive recursion on the number of entries to copy:

$$A(0, r, t) := t, \quad A(k+1, r, t) := \langle \pi_0(r), A(k, \pi_1(r), t) \rangle. \quad (\text{S4})$$

If $s = \langle n, r \rangle$ and $u = \langle m, t \rangle$ are sequence codes, define

$$\text{Concat}(s, u) := \langle n + m, A(n, r, t) \rangle. \quad (\text{S5})$$

Induction on n verifies that (S5) decodes to the concatenation of the two represented sequences. Because (S4) is primitive recursion using the primitive-recursive pairing projections, Concat is primitive recursive.

For replacement, define a payload transformation $R(k, j, a, r)$ by recursion on k : copy the current head unless the current position is j , in which case insert a , and recurse on the tail. Equivalently, one may reconstruct a length- n payload from the primitive-recursive component function

$$b_i = \begin{cases} a, & i = j, \\ (s)_i, & i \neq j. \end{cases}$$

using the same bounded cons recursion. Initial segments are obtained by rebuilding only the first $k \leq n$ components. Membership among components and any local property $P((s)_i)$ are checked by bounded existential or universal quantification over $i < n$. Closure of primitive-recursive relations under bounded quantification therefore gives all the stated tests.

Hence length, extraction, concatenation, bounded replacement, initial-segment formation, and bounded componentwise checks are primitive recursive for the fixed coding. Every operation has now been given by an explicit bounded/primitive recursion; no hidden minimization or unstated numerical bound is used.

Point specific to this result. The nested-pair payload makes reconstruction constructive. Concatenation literally copies one finite payload onto another by recursion on the recorded length, so the later proof- and syntax-manipulation routines remain inside $\mathcal{PR}$. $\square$

Proposition 9.4 (Coding finite derivations in a primitive-recursively checkable calculus). *Fix a coded proof calculus that is primitive-recursively checkable in the sense just defined. Then there is a primitive-recursive relation $\text{Deriv}(p, q)$ saying that p codes a finite derivation whose last formula has code q .*

Proof. Fix numerical codes for formulas, axiom instances, and inference-rule tags. Code a finite derivation

$$\varphi_0, \dots, \varphi_{n-1}$$

as a finite sequence d of line records. A line record contains the formula code and enough bounded auxiliary data to certify its justification: an axiom tag, or the indices of earlier premises used by an inference rule. Define

$$\text{LocOK}(d, i)$$

to mean that line $i < \text{lh}(d)$ is either a permitted axiom or follows from earlier lines according to one of the finitely many rules. Formula recognition and checking an axiom schema are primitive recursive once syntax is coded; the referenced indices are bounded by i . Hence LocOK is primitive recursive. Then

$$\text{Deriv}(d) \iff \text{Seq}(d) \wedge \forall i < \text{lh}(d) \text{ LocOK}(d, i)$$

is a bounded universal condition and therefore primitive recursive. Thus finite formal derivations admit effective numerical coding whose local correctness can be checked primitive recursively.

Point specific to this result. A derivation code stores a finite formula sequence together with enough local justification data to check each line. This makes proof verification a finite arithmetic computation rather than a meta-level appeal to the existence of a derivation. $\square$

Worked example

Worked Example 9.5 (Encoding a short sequence). A sequence code is useful only if arithmetic can recover its entries. For a sequence $(3, 1, 4)$, the chosen β -style code packages parameters b, c so that $\beta(b, c, 0) = 3$, $\beta(b, c, 1) = 1$, and $\beta(b, c, 2) = 4$. The numerical value of the code is unimportant; what matters is that the relation “the i -th entry is a ” is primitive recursive.

A useful warning

Do not confuse existence of an external coding scheme with representability of its decoding operations inside arithmetic. Both levels are needed later.

Looking ahead

The next chapter, *Recursive and computably enumerable relations*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 10

Recursive and computably enumerable relations

Definitions and setup

Formal definitions used in this chapter

Register-machine computation. For definiteness, a register machine has finitely many registers containing natural numbers and a finite instruction list built from: set a register to 0; increment a register; conditionally decrement a nonzero register and jump; unconditional jump; and halt. An input tuple is placed in designated input registers and, when the machine halts, the designated output register is read.

Partial computable and computable functions. A partial function $f : \mathbb{N}^k \rightarrow \mathbb{N}$ is *partial computable* if some register machine halts with output $f(\bar{n})$ exactly on inputs $\bar{n}$ in the domain of f . It is *computable* (or recursive) if its domain is all of $\mathbb{N}^k$.

Recursive relation. A relation $R \subseteq \mathbb{N}^k$ is recursive when its characteristic function is a total computable function.

Computably enumerable relation. A relation $R \subseteq \mathbb{N}^k$ is computably enumerable (c.e.) when there is a register machine that halts on input $\bar{n}$ exactly when $R(\bar{n})$ holds. The existential primitive-recursive certificate characterization is *not* part of this definition; it is proved later as the Normal-form Characterization Theorem before that form is used load-bearingly.

Effective enumeration and c.e. theory. An effective enumeration of a set $A \subseteq \mathbb{N}$ is a partial computable function $e : \mathbb{N} \rightarrow \mathbb{N}$ whose range is exactly A ; repetitions are allowed, and the empty partial function enumerates the empty set. A formal theory T is c.e. when the set of Gödel numbers of its nonlogical axioms has such an enumeration.

Substitution into a numerical relation. If $R \subseteq \mathbb{N}^r$ and $f_1, \dots, f_r : \mathbb{N}^k \rightarrow \mathbb{N}$, the substitution of the functions f_i into R is the relation $S(\bar{x}) \iff R(f_1(\bar{x}), \dots, f_r(\bar{x}))$. Thus “recursive/computable substitution” below has this literal meaning.

A relation is recursive when there is an algorithm deciding both positive and negative instances. It is computably enumerable (c.e.) when a positive search halts exactly on the positive instances, in the register-machine sense fixed above. A partial recursive function may fail to halt; primitive-recursive functions are total. These distinctions matter because a formal theory may have a c.e. axiom set even when axiom membership is not decidable. The Normal-form Characterization Theorem below will prove, rather than assume, the later certificate description by finite accepting computations.

How this chapter fits together

The distinction between decidable and merely enumerable relations is central to incompleteness. A recursive relation has a terminating decision procedure; a c.e. relation has a terminating positive search when the answer is yes. Finite computation histories turn the latter into existential statements over primitive-recursive predicates, which is exactly the form needed for c.e. theories.

Theorem 10.1 (Closure properties of recursive relations). *The class of recursive relations is closed under complement, finite conjunction/disjunction, substitution of recursive functions, and bounded quantification.*

Proof. A relation $R \subseteq \mathbb{N}^k$ is recursive when its characteristic function χ_R is total recursive. If R, S are recursive, then Boolean combinations are recursive because

$$\chi_{\neg R} = 1 - \chi_R, \quad \chi_{R \cap S} = \chi_R \chi_S, \quad \chi_{R \cup S} = \text{sg}(\chi_R + \chi_S),$$

where $\text{sg}(0) = 0$ and $\text{sg}(n + 1) = 1$. For bounded existential quantification define

$$Q(\bar{x}) \iff \exists y < b(\bar{x}) R(\bar{x}, y).$$

A terminating algorithm checks $y = 0, 1, \dots, b(\bar{x}) - 1$, so Q is recursive whenever b and R are recursive. Bounded universal quantification follows by complement. Substitution of total recursive functions into a recursive relation is recursive by composition. These closure facts are algorithmic: every construction displayed above terminates on every input, which is the defining difference from merely c.e. relations.

Point specific to this result. For recursive relations, both positive and negative membership terminate. To decide the complement, run the same terminating decision procedure and reverse its yes/no output; and conjunction/disjunction are decided by combining the terminating decision procedures. $\square$

Theorem 10.2 (Closure properties of computably enumerable relations). *The class of computably enumerable relations is closed under finite conjunction/disjunction, existential quantification, and computable substitution. A relation is recursive exactly when both it and its complement are computably enumerable.*

Proof. A relation $R(\bar{x})$ is computably enumerable when some register-machine search halts exactly on the tuples in R . The closure argument here uses only that machine formulation; the existential primitive-recursive certificate normal form is proved in the next theorem.

If R and S are c.e., dovetail their searches. For the union, halt when either search succeeds; for the intersection, run both searches in parallel and halt once both have succeeded. Thus $R \cup S$ and $R \cap S$ are c.e. If f is total computable, then

$$Q(\bar{x}) \iff R(f(\bar{x}))$$

is c.e. by first computing $f(\bar{x})$ and then running the enumerator for R . Existential projection preserves c.e.-ness: for $\exists y R(\bar{x}, y)$, dovetail the searches for $R(\bar{x}, 0), R(\bar{x}, 1), \dots$. Complement is deliberately absent from this list: a c.e. relation need not have c.e. complement.

Point specific to this result. For c.e. relations one cannot in general wait for negative evidence. Union is handled by dovetailing enumerations, intersection by waiting until an item has appeared in both, and existential projection by dovetailing over witnesses. $\square$

Theorem 10.3 (Normal-form characterization of c.e. relations in the form needed later). *A relation $R(\bar{x})$ is computably enumerable if and only if there is a primitive-recursive relation $C(\bar{x}, y)$ such that*

$$R(\bar{x}) \iff \exists y C(\bar{x}, y).$$

The witness y may be taken to code a finite accepting computation.

Proof. Let $R(\bar{x})$ be c.e. Choose a machine M halting exactly when $R(\bar{x})$ holds. A finite computation history of M is a finite sequence of configurations. Because each machine transition is local and effective, there is a primitive-recursive predicate

$$C_M(\bar{x}, s)$$

saying that s codes a valid halting computation of M on input $\bar{x}$. Hence

$$R(\bar{x}) \iff \exists s C_M(\bar{x}, s).$$

Conversely, if C is primitive recursive, an algorithm enumerates $s = 0, 1, 2, \dots$ and halts at the first s satisfying $C(\bar{x}, s)$. Therefore every relation of the displayed existential primitive-recursive form is c.e. This normal form is the exact bridge needed later: the existential variable s will become a numerical witness that a c.e. axiom has appeared or that a finite computation/proof certificate is correct. □

Proposition 10.4 (Effective enumeration of finite proof searches). *For every c.e. theory T and bound b , the finite search through all proof certificates of code at most b is effective; equivalently, the relation “some T -proof of φ has code $\leq b$ ” is recursive uniformly in b and $\ulcorner \varphi \urcorner$.*

Proof. Enumerate all finite sequences of natural numbers effectively, for example in increasing code order. For each candidate d , run the primitive-recursive predicate $\text{Deriv}(d)$. If it holds, output the code of the last formula of d , together with d if a proof witness is desired. Because every genuine finite derivation has some natural-number code, it eventually appears in this enumeration. Thus the relation

$$\text{Thm}(q) \iff \exists d [\text{Deriv}(d) \wedge \text{Last}(d) = q]$$

is c.e. More generally, if the nonlogical axiom set is itself c.e., attach to each axiom line an enumeration-stage witness; local proof checking then remains primitive recursive. The enumeration is therefore effective even though membership in the theorem set need not be decidable.

Point specific to this result. Because candidate proof codes are natural numbers and local proof correctness is decidable once the relevant certificate data are supplied, one can enumerate all successful finite proofs by scanning codes in increasing order. □

Worked example

Worked Example 10.5 (Dovetailing two searches). If both R and its complement are c.e., run the two enumerations in alternating steps. Exactly one must eventually exhibit the input, so the combined procedure halts with the correct yes/no answer. Thus a relation is recursive precisely when it and its complement are c.e. in the setting used here.

A useful warning

An enumerator need not tell us that a sentence will never appear. That asymmetry is why c.e. theorem sets can be effectively listed without being decidable.

Looking ahead

The next chapter, *Weak arithmetic*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 11

Weak arithmetic

Definitions and setup

Formal definitions used in this chapter

Language of arithmetic and numerals. The language $\mathcal{L}_A = \{0, S, +, \cdot\}$ has a constant 0, unary successor S , and binary addition and multiplication. The numeral for $n \in \mathbb{N}$ is $\bar{n} = S^n 0$.

Arithmetic order and bounded (Δ_0) formulas. The symbol $<$ is an abbreviation in the language $\mathcal{L}_A = \{0, S, +, \cdot\}$:

$$x < y \quad \Longleftrightarrow \quad \exists z (x + Sz = y).$$

For a term t not containing x , bounded quantifiers are abbreviations

$$\exists x < t \varphi \quad \Longleftrightarrow \quad \exists x (x < t \wedge \varphi),$$

$$\forall x < t \varphi \quad \Longleftrightarrow \quad \forall x (x < t \rightarrow \varphi).$$

A formula is *bounded*, or Δ_0 , when every quantifier occurs in one of these bounded forms.

Σ_1 formula. In this volume a formula is Σ_1 when it has the syntactic form $\exists y_1 \cdots \exists y_r \delta$, where $\delta \in \Delta_0$ (allowing $r = 0$). When a later formula is only provably equivalent to such a formula, the relevant base theory will be named explicitly.

Robinson arithmetic Q . Q is the $\mathcal{L}_A$ -theory with axioms

$$Sx \neq 0, \quad Sx = Sy \rightarrow x = y, \quad x \neq 0 \rightarrow \exists y (x = Sy),$$

$$x + 0 = x, \quad x + Sy = S(x + y), \quad x \cdot 0 = 0, \quad x \cdot Sy = x \cdot y + x.$$

It has no induction schema.

Induction schema, $I\Sigma_1$, and PA. For a formula $\varphi(x, \bar{z})$, its induction axiom is

$$(\varphi(0, \bar{z}) \wedge \forall x (\varphi(x, \bar{z}) \rightarrow \varphi(Sx, \bar{z}))) \rightarrow \forall x \varphi(x, \bar{z}).$$

For this book, $I\Sigma_1$ denotes Q plus all induction axioms whose induction formula is Σ_1 , while Peano Arithmetic PA is Q plus induction for every $\mathcal{L}_A$ -formula.

Standard natural numbers and tuples. Throughout the metatheoretic statements in this volume, a *standard natural number* is an actual element of the external set $\mathbb{N}$, as opposed to a possibly nonstandard element of a model of arithmetic. A *standard tuple* is a finite tuple $\mathbf{n} = (n_1, \dots, n_k) \in \mathbb{N}^k$ of such external natural numbers; its numeral tuple is $\bar{\mathbf{n}} = (\bar{n}_1, \dots, \bar{n}_k)$.

Numeralwise verification. A theory verifies a numerical relation numeralwise when it proves the correct closed instance, or its negation, after replacing each standard input by its numeral.

Robinson arithmetic Q is a finitely axiomatized first-order theory of $0, S, +, \times$ containing basic defining equations but no induction scheme. A bounded (Δ_0) formula has all quantifiers explicitly bounded by terms. A Σ_1 formula begins with existential number quantifiers followed by a bounded matrix. The key fact used later is instance-wise: true finite primitive-recursive calculations can be verified on numerals in a weak theory, and true Σ_1 statements have concrete witnesses.

How this chapter fits together

Robinson arithmetic Q is intentionally weak: it has enough arithmetic to verify each concrete finite computation but lacks induction as a general axiom scheme. This makes it a useful baseline for incompleteness. The key fact is instance-wise Σ_1 -completeness: a true existential statement with a bounded computable matrix has concrete witnesses whose calculation Q can verify.

Proposition 11.1 (Basic arithmetic derivations in Robinson arithmetic Q). *Robinson arithmetic Q proves every true closed instance of the defining equations for successor, addition, and multiplication on numerals; in particular, for concrete $m, n \in \mathbb{N}$, Q proves the correct numeral equations for $\bar{m} + \bar{n}$, $\bar{m} \cdot \bar{n}$, and inequality of distinct numerals.*

Proof. Robinson arithmetic Q has the equations

$$\begin{aligned} x + 0 &= x, & x + S(y) &= S(x + y), \\ x \cdot 0 &= 0, & x \cdot S(y) &= x \cdot y + x, \end{aligned}$$

together with the three successor axioms displayed in the definition of Q . Fix standard numerals $\bar{n} = S^n 0$. For concrete m, n , repeated use of the recursion equations yields finite derivations

$$Q \vdash \bar{m} + \bar{n} = \overline{m + n}, \quad Q \vdash \bar{m} \cdot \bar{n} = \overline{mn}.$$

The proof is an external induction on the standard integer n , not an induction carried out inside Q . For example,

$$\bar{m} + \overline{n + 1} = \bar{m} + S(\bar{n}) = S(\bar{m} + \bar{n}) = S(\overline{m + n}) = \overline{m + n + 1}.$$

Likewise the successor axioms prove distinct standard numerals unequal. These finite numeral calculations are the only arithmetic strength needed for many later correctness-on-standard-codes arguments.

Point specific to this result. The claim concerns concrete closed calculations, not induction in Q . Each numeral computation is reduced finitely by the defining equations for successor, addition, and multiplication. $\square$

Lemma 11.2 (Numeralwise correctness of Q for bounded primitive-recursive calculations). *Let $R(\bar{x})$ be a bounded relation represented by a Δ_0 formula $\rho_R(\bar{x})$, and let $\bar{n} = (n_1, \dots, n_k) \in \omega^k$. If $R(\bar{n})$ is true, then*

$$Q \vdash \rho_R(\bar{n}_1, \dots, \bar{n}_k),$$

and if $R(\bar{n})$ is false, then

$$Q \vdash \neg \rho_R(\bar{n}_1, \dots, \bar{n}_k).$$

Thus Q correctly verifies every closed bounded computation occurring in the coding developed here.

Proof. Let $R(\bar{x})$ be a bounded relation obtained from addition, multiplication, order, and bounded quantifiers. Fix a standard tuple $\bar{n}$. Every bounded quantifier in $R(\bar{n})$ ranges over a finite standard set. Evaluate the atomic arithmetic expressions using the numeral calculations already proved in Q . Then eliminate bounded quantifiers by a finite case analysis. Thus, if $R(\bar{n})$ is true in $\mathbb{N}$, one obtains a finite derivation

$$Q \vdash R(\bar{n}),$$

and if it is false,

$$Q \vdash \neg R(\bar{n}).$$

No induction schema is invoked inside Q : the metatheoretic evaluation supplies a finite proof for this particular tuple. This “numeralwise correctness” is exactly what is later used when the primitive-recursive proof predicate is evaluated at an actual standard natural-number code of a proof.

Point specific to this result. Numeralwise correctness is external-to-internal: for each actual primitive-recursive computation on fixed numerals, one constructs a finite Q -derivation of its correct value. The theorem does not assert a single universally quantified induction principle inside Q . $\square$

Theorem 11.3 (Sigma-1 completeness of Q in the required form). *If σ is a true Σ_1 sentence of arithmetic, then $Q \vdash \sigma$. More generally, every true Σ_1 formula is provable in Q after substituting numerals for its free variables.*

Commented symbolic trace: why true Σ_1 sentences are provable in Q

A true Σ_1 sentence has the form $\exists \bar{x} \delta(\bar{x})$ with bounded δ . Externally choose the actual natural-number witnesses $\bar{n}$. The bounded matrix $\delta(\bar{n})$ reduces to a finite arithmetic computation, and Q proves that concrete computation. Only then do we use existential introduction inside the formal proof. Thus the proof moves from an external witness to an internal derivation; it does not assume that Q can search for the witness by induction.

Proof. Let σ be a true Σ_1 sentence. Put it in the form

$$\sigma \equiv \exists x_1 \cdots \exists x_k \delta(x_1, \dots, x_k),$$

where δ is bounded. Since $\mathbb{N} \models \sigma$, choose standard witnesses $n_1, \dots, n_k$ with

$$\mathbb{N} \models \delta(n_1, \dots, n_k).$$

By numeralwise correctness,

$$Q \vdash \delta(\bar{n}_1, \dots, \bar{n}_k).$$

Apply existential introduction successively to obtain

$$Q \vdash \exists x_1 \cdots \exists x_k \delta(x_1, \dots, x_k),$$

which is $Q \vdash \sigma$. The proof is one-way: truth of a Σ_1 sentence yields provability in Q . We have not claimed that every sentence provable in an arbitrary extension of Q is true; that would be a soundness assumption on the extension, not Σ_1 -completeness of Q .

Point specific to this result. A true Σ_1 sentence comes with actual numerical witnesses. After substituting their numerals, the bounded matrix is a finite computation that Q can verify; existential introduction then proves the original sentence. $\square$

Proposition 11.4 (Comparison of Q , $I\Sigma_1$, and PA for later metamathematics). *For the metamathematical arguments of Volumes II–III, Q suffices for numeralwise verification and Σ_1 -completeness, while induction principles used to reason uniformly about coded proofs are explicitly assigned to stronger fragments such as $I\Sigma_1$ or PA. No later theorem may use an induction scheme not present in its stated base theory.*

Proof. The theories are separated by the induction they provide. Robinson arithmetic Q has no induction schema; it suffices for verifying individual standard computations. A bounded-arithmetic fragment adds induction only for a specified syntactic class, while $I\Sigma_1$ contains induction for all Σ_1 formulas. Peano arithmetic contains induction for every first-order arithmetic formula. Consequently

$$Q \subseteq I\Sigma_1 \subseteq \text{PA}$$

for the standard presentations. Later arguments are tagged accordingly: construction and numeralwise verification of Prf_T need only weak arithmetic; formalized closure of proofs and the Hilbert–Bernays–Löb derivability conditions are most cleanly carried out in $I\Sigma_1$ or PA. The point of this proposition is therefore not an abstract inclusion alone, but a dependency rule: whenever an internal induction is used, the proof must explicitly move to a theory containing the corresponding induction scheme.

Point specific to this result. The comparison isolates which later arguments need only numeralwise computation in Q and which require induction available in stronger arithmetic. This prevents the incompleteness proofs from silently borrowing PA-strength at coding steps where Q suffices. $\square$

Worked example

Worked Example 11.5 (Verifying a concrete computation in Q). To prove a true sentence such as $\bar{2} + \bar{3} = \bar{5}$, repeatedly use the defining equations for addition until the closed term reduces to the expected numeral. A true Σ_1 sentence first supplies actual numerical witnesses; the bounded matrix is then reduced by the same kind of finite calculation.

A useful warning

Σ_1 -completeness of Q is not full soundness, completeness, or induction. It is a one-way theorem about true Σ_1 sentences.

Looking ahead

The next chapter, *Representability*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 12

Representability

Definitions and setup

Formal definitions used in this chapter

Numeral tuple. For $\mathbf{n} = (n_1, \dots, n_k) \in \mathbb{N}^k$, write $\bar{\mathbf{n}} = (\bar{n}_1, \dots, \bar{n}_k)$.

Representation of a numerical function. A formula $F_f(\bar{x}, y)$ represents $f : \mathbb{N}^k \rightarrow \mathbb{N}$ in a theory T in the strong numeralwise sense used here when, for every standard $\mathbf{n}$ and $m = f(\mathbf{n})$,

$$T \vdash \forall y (F_f(\bar{\mathbf{n}}, y) \leftrightarrow y = \bar{m}).$$

Strong representation of a relation. A formula $\rho_R(\bar{x})$ strongly represents $R \subseteq \mathbb{N}^k$ in T when for every standard $\mathbf{n}$,

$$R(\mathbf{n}) \Rightarrow T \vdash \rho_R(\bar{\mathbf{n}}), \quad \neg R(\mathbf{n}) \Rightarrow T \vdash \neg \rho_R(\bar{\mathbf{n}}).$$

Weak representation of a c.e. relation. In this volume, ρ_R weakly represents a c.e. relation R when

$$R(\mathbf{n}) \iff \mathbb{N} \models \rho_R(\bar{\mathbf{n}})$$

for every standard $\mathbf{n}$, and every positive instance satisfies $R(\mathbf{n}) \Rightarrow Q \vdash \rho_R(\bar{\mathbf{n}})$. No uniform proof of the negative instances is required.

A numerical function f is represented in a theory T by a formula $F(\bar{x}, y)$ when, for each standard input, the theory proves the correct output instance and proves uniqueness in the strength required by later arguments. A relation is represented by a formula whose numeral instances are provable when true and whose negations are provable when false in the strong form. Representability is a theorem about formal arithmetic, stronger than the external fact that a function is computable.

How this chapter fits together

Representability is the bridge from external computation to internal arithmetic. A primitive-recursive computation is encoded by a finite sequence satisfying local transition equations; arithmetic can then assert the existence of such a sequence. This is why recursion on natural numbers becomes an existential arithmetic formula about a coded history.

Theorem 12.1 (Representability of primitive-recursive functions). *For every primitive-recursive function $f : \mathbb{N}^k \rightarrow \mathbb{N}$ there is an arithmetic formula $F_f(\bar{x}, y)$ such that, for every standard tuple $\mathbf{n} = (n_1, \dots, n_k)$ and $m = f(\mathbf{n})$,*

$$Q \vdash \forall y (F_f(\bar{\mathbf{n}}, y) \leftrightarrow y = \bar{m}).$$

Thus the graph formula proves both the correct output and uniqueness of that output for each standard input tuple.

Proof. We induct on the generation of the primitive-recursive function f .

For the zero, successor, and projection functions use respectively

$$y = 0, \quad y = Sx_1, \quad y = x_i.$$

For each standard input tuple, Q proves the corresponding closed equation and proves that no different numeral can satisfy it.

Suppose

$$f(\bar{x}) = h(g_1(\bar{x}), \dots, g_r(\bar{x}))$$

and graph formulas $G_i(\bar{x}, z_i)$ and $H(\bar{z}, y)$ already strongly represent the component functions. Put

$$F_f(\bar{x}, y) := \exists z_1 \cdots z_r \left(\bigwedge_{i=1}^r G_i(\bar{x}, z_i) \wedge H(\bar{z}, y) \right).$$

Fix a standard tuple $\mathbf{n}$. The induction hypotheses identify each intermediate variable z_i with the unique numeral $\overline{g_i(\mathbf{n})}$, after which the hypothesis for H identifies y with $\overline{f(\mathbf{n})}$. Hence Q proves the required biconditional for this input.

For primitive recursion, with

$$f(\bar{x}, 0) = g(\bar{x}), \quad f(\bar{x}, n+1) = h(\bar{x}, n, f(\bar{x}, n)),$$

let $F_f(\bar{x}, n, y)$ assert the existence of a coded sequence s of length $n+1$ whose first entry satisfies the graph formula for g , whose successive entries satisfy the graph formula for h , and whose last entry is y . Sequence coding and bounded component extraction are primitive recursive, so this is an arithmetic formula. For a fixed standard n , the recursion equations can be unrolled through exactly n steps inside Q . The induction hypotheses at those finitely many steps prove both existence of the actual computation sequence and uniqueness of every entry. Therefore, if $m = f(\mathbf{n}, n)$,

$$Q \vdash \forall y (F_f(\bar{\mathbf{n}}, \bar{n}, y) \leftrightarrow y = \bar{m}).$$

This is strong numeralwise representability. No uniform induction principle inside Q is being claimed; for each standard input, only a finite derivation obtained by unrolling the corresponding finite computation is required. $\square$

Theorem 12.2 (Representability of primitive-recursive relations). *For every primitive-recursive relation $R \subseteq \mathbb{N}^k$ there is an arithmetic formula $\rho_R(\bar{x})$ such that for every $\bar{n}$,*

$$R(\bar{n}) \Rightarrow Q \vdash \rho_R(\bar{\mathbf{n}}), \quad \neg R(\bar{n}) \Rightarrow Q \vdash \neg \rho_R(\bar{\mathbf{n}}).$$

Proof. Let $R(\bar{x})$ be primitive recursive. Its characteristic function χ_R is primitive recursive, so by the preceding theorem there is a formula $C_R(\bar{x}, y)$ representing the graph of χ_R . Define

$$\rho_R(\bar{x}) := C_R(\bar{x}, \bar{1}).$$

If $R(\bar{n})$ holds in $\mathbb{N}$, then $\chi_R(\bar{n}) = 1$, and representability gives

$$Q \vdash \rho_R(\bar{\mathbf{n}}).$$

If $R(\bar{n})$ fails, then $\chi_R(\bar{n}) = 0$; uniqueness of the represented output together with $\bar{0} \neq \bar{1}$ gives

$$Q \vdash \neg \rho_R(\bar{\mathbf{n}}).$$

Thus primitive-recursive relations are strongly numeralwise representable. This two-sided standard-instance correctness is what lets arithmetic later verify that a particular natural number is, or is not, a correct proof code.

Point specific to this result. For a primitive-recursive relation, represent its characteristic function and assert that its value is 1. Numeralwise correctness then gives proofs of true instances and, in the strong form used here, proofs of negated false instances. $\square$

Theorem 12.3 (Representability of recursive relations). *Every recursive relation is strongly representable in Q in the same numeralwise sense as a primitive-recursive relation.*

Proof. If R is recursive, its characteristic function is total recursive. Choose a terminating computation model for that function. The relation

$$C(\bar{x}, y, s)$$

that says “ s codes a halting computation on input $\bar{x}$ with output y ” is primitive recursive. Therefore

$$F_R(\bar{x}, y) := \exists s C(\bar{x}, y, s)$$

represents the graph of the characteristic function. Determinism of the computation gives uniqueness of the output, and termination guarantees that for every standard input some computation witness exists. Define

$$\rho_R(\bar{x}) := F_R(\bar{x}, 1).$$

Then true and false standard instances are proved exactly as for primitive-recursive relations. The difference is that the existential computation witness s need not be bounded by a primitive-recursive function of the input; the representing formula is therefore generally more complex than the primitive-recursive case.

Point specific to this result. A recursive relation has terminating positive/negative computation. Representing its decision procedure therefore transfers both outcomes into arithmetic, which is stronger than merely representing successful enumerations. $\square$

Proposition 12.4 (Weak representability of c.e. relations). *If $R(\bar{x})$ is computably enumerable, then there is a Σ_1 formula $\rho_R(\bar{x})$ such that*

$$R(\bar{n}) \Rightarrow Q \vdash \rho_R(\bar{\mathbf{n}}),$$

and $\mathbb{N} \models \rho_R(\bar{\mathbf{n}})$ exactly when $R(\bar{n})$.

Proof. Let $R(\bar{x})$ be c.e. By the c.e. normal form there is a primitive-recursive relation $C(\bar{x}, s)$ with

$$R(\bar{x}) \iff \exists s C(\bar{x}, s).$$

Let $\gamma_C(\bar{x}, s)$ strongly represent C . Put

$$\rho_R(\bar{x}) := \exists s \gamma_C(\bar{x}, s).$$

If $R(\bar{n})$ is true, choose an actual witness m . Then

$$Q \vdash \gamma_C(\bar{\mathbf{n}}, \bar{m}),$$

so $Q \vdash \rho_R(\bar{\mathbf{n}})$. This is the positive, or weak, representability direction. If $R(\bar{n})$ is false, there is no single finite witness to its failure; proving $\neg \rho_R(\bar{\mathbf{n}})$ may require much stronger information. Hence c.e. relations are represented in the one-sided sense required for theoremhood and provability predicates.

Point specific to this result. For a c.e. relation $R(\bar{x}) \iff \exists y C(\bar{x}, y)$, represent the primitive-recursive computation predicate C and existentially quantify the witness. Positive instances are provable; no corresponding theorem for negative instances is claimed. $\square$

Worked example

Worked Example 12.5 (Representing addition). The graph of addition is represented by the formula expressing the recursive equations $x + 0 = x$ and $x + S(y) = S(x + y)$. For a general primitive-recursive function, the same idea is packaged into a coded computation sequence: the first entry is the initial value, each later entry satisfies the recursive transition, and the last entry is the output.

A useful warning

Representability is a property of a formula relative to a theory; it is stronger than merely saying that the intended function is computable in the metatheory.

Looking ahead

The next chapter, *Gödel numbering of syntax*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 13

Gödel numbering of syntax

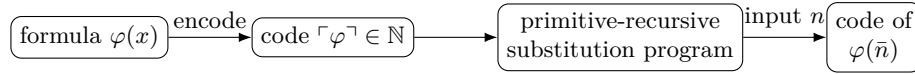


Figure 13.1: Arithmetization separates an expression from its numerical code. The arithmetic theory manipulates the codes.

Definitions and setup

Formal definitions used in this chapter

Gödel numbering. A Gödel numbering is an injective map from the finite terms and formulas of the fixed language to $\mathbb{N}$ such that the predicates for syntactic category and the constructor/destructor operations used below are primitive recursive. We write $\ulcorner \varphi \urcorner$ for the code of φ .

Code predicates and constituent operations. $\text{Term}(n)$ and $\text{Form}(n)$ mean that n codes a term or formula. Immediate-constituent, outermost-constructor, and complexity functions return a fixed default value, say 0, on malformed inputs, so that they are total numerical functions.

Free-variable and substitution functions on codes. $\text{Free}(v, n)$ means that variable code v occurs free in the formula coded by n . $\text{Sub}(n, v, m)$ is the code of the capture-avoiding result of substituting the term coded by m for the free occurrences of variable v in the formula coded by n , again with a fixed default on malformed inputs.

Logical-axiom and rule recognizers for the fixed Volume-I calculus. $\text{LogAx}(n)$ means that n codes exactly one of the following:

1. a first-order substitution instance of a propositional tautology;
2. a universal-instantiation axiom $\forall x \varphi \rightarrow \varphi[t/x]$, with the coded free-for/capture-avoidance condition satisfied;
3. an existential-duality axiom, with the Volume-I biconditional abbreviation expanded before coding, namely

$$(\exists x \varphi \rightarrow \neg \forall x \neg \varphi) \wedge (\neg \forall x \neg \varphi \rightarrow \exists x \varphi);$$

4. a quantifier-distribution axiom $\forall x(\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \forall x \psi)$ with $x \notin \text{FV}(\varphi)$;
5. an equality-reflexivity or equality-substitutivity axiom from the fixed calculus, again after expanding every displayed $\leftrightarrow$ by (BI).

Thus LogAx recognizes only the constructor tags already present in the Volume-I syntax; it never assumes a separate biconditional tag. The predicate $\text{MP}(a, b, c)$ says that a, b, c code formulas $\alpha, \alpha \rightarrow \beta, \beta$, respectively. The predicate $\text{Gen}(a, v, b)$ says that a, b code $\alpha, \forall v \alpha$. For proofs from a theory, all nonlogical

assumptions are sentences, so the Volume-I eigenvariable side condition for generalization is automatic; for a general derivation with open assumptions, a finite dependency record must also be stored and checked against Free. These exact recognizers, rather than an unnamed “standard calculus,” are the ones used in the proof certificates below.

Diagonal function. Let v_0 be the distinguished one-variable slot and let $\text{Num}(n) = \ulcorner \bar{n} \urcorner$ be the code of the numeral for n . Define

$$d(n) = \text{Sub}(n, v_0, \text{Num}(n)).$$

Recursively axiomatized theory. A theory T is recursively axiomatized when the set of Gödel codes of its nonlogical axioms is a recursive relation; its decision predicate is denoted $\text{Ax}_T(n)$.

A Gödel numbering is an injective coding of syntactic objects by natural numbers with primitive-recursive formation and manipulation predicates. We write $\ulcorner \varphi \urcorner$ for the code of φ and $\bar{n}$ for the numeral naming n inside arithmetic. The substitution function $\text{Sub}(m, n)$ is numerical: given the code of a formula and a number, it returns the code obtained by substituting the corresponding numeral. This numerical function is what makes diagonalization formalizable.

How this chapter fits together

With sequence coding in place, syntax itself becomes arithmetic data. Formation, free-variable occurrence, substitution, and diagonalization must be shown primitive recursive, not merely described as mechanical. The diagonal function will later feed a formula its own code, so this chapter carries the technical burden behind self-reference.

Theorem 13.1 (Primitive-recursive coding of terms and formulas). *For the fixed Gödel numbering, the predicates “ n codes a term”, “ n codes a formula”, and the functions extracting immediate constituents, complexity, and outermost constructor are primitive recursive.*

Proof. Assign distinct numerical tags to variables, function symbols, relation symbols, connectives, and quantifiers. Define codes recursively, for example

$$\ulcorner f(t_1, \dots, t_k) \urcorner = \langle \text{tag}(f), \ulcorner t_1 \urcorner, \dots, \ulcorner t_k \urcorner \rangle,$$

For quantified formulas use separate constructor tags:

$$\ulcorner Qv \varphi \urcorner = \langle \text{tag}(Q), v, \ulcorner \varphi \urcorner \rangle, \quad Q \in \{\forall, \exists\}.$$

Boolean and atomic formulas are coded by their own distinct tags in the same way. Because pairing and finite-sequence coding are primitive recursive, forming the code of a compound expression from the codes of its immediate constituents is primitive recursive. Recognition proceeds by course-of-values recursion on the code: inspect the outer tag, decode the finite list of constituents, and recursively verify that each constituent has the required syntactic category and arity. Hence

$$\text{Term}(n), \quad \text{Form}(n)$$

are primitive-recursive predicates. The exact numerical coding is conventional; later arguments depend only on these primitive-recursive formation and decoding operations, not on a particular choice of tags.

Point specific to this result. The constructors are tagged so arithmetic can distinguish variables, function applications, atomic formulas, connectives, and quantifiers. Course-of-values recursion on the code then recognizes well-formed expressions. $\square$

Theorem 13.2 (Primitive-recursive free-variable, substitution, and diagonalization functions). *For the fixed Gödel numbering, free-variable occurrence, capture-avoiding substitution on codes, and the diagonal function*

$$d(n) = \text{Sub}(n, v_0, \text{Num}(n))$$

are primitive recursive as total functions under the default-value convention above.

How to read the symbols: codes versus the expressions they encode

If $n = \ulcorner \varphi \urcorner$, then n is a natural number and φ is a formula. The substitution function operates on the number n , decodes just enough syntax to locate the relevant variable occurrences, and returns another number coding the substituted formula. The diagonal function is a special case of this numerical program. This distinction between an expression and its code is the type discipline needed for Gödel's self-reference construction.

Proof. Work by recursion on the syntax code. The predicate $\text{Free}(v, n)$ is computed from the outer constructor: at an atomic formula it is the corresponding free-variable test on terms; Boolean connectives take unions of the free-variable sets; and

$$\text{Free}(v, \ulcorner Qw \varphi \urcorner) \iff v \neq w \wedge \text{Free}(v, \ulcorner \varphi \urcorner), \quad Q \in \{\forall, \exists\}.$$

For substitution, recursively descend through the code. At a quantified formula $Qw \varphi$, $Q \in \{\forall, \exists\}$, if w would capture a free variable of the substituted term, choose the least fresh variable code and first rename the bound variable. Freshness is a bounded syntactic check. Thus

$$\text{Sub}(n, v, m)$$

– the code obtained by substituting the numeral/term coded by m for free occurrences of variable v in formula code n – is primitive recursive. The diagonal function is the special case

$$d(n) = \text{Sub}(n, v_0, \text{Num}(n)),$$

so it too is primitive recursive.

Point specific to this result. Free-variable occurrence and substitution descend recursively through the parse code, with a bound-variable renaming clause for capture avoidance. Diagonalization is then the special substitution that inserts the numeral of the input code into its designated free-variable position. $\square$

Proposition 13.3 (Primitive-recursive checking of the fixed logical calculus). *The predicates*

$$\text{LogAx}(n), \quad \text{MP}(a, b, c), \quad \text{Gen}(a, v, b)$$

defined in the setup are primitive recursive. If open-assumption derivations are coded with finite dependency records, the Volume-I eigenvariable condition is also primitive-recursively checkable.

Proof. The outer constructor and immediate constituents of a formula code are primitive recursive by Theorem 13.1. Propositional-tautology instances are recognized by extracting the finite propositional skeleton and running its finite truth table. Universal instantiation and equality substitution use the primitive-recursive capture-avoiding substitution operation of Theorem 13.2; the free-for condition and the side condition $x \notin \text{FV}(\varphi)$ are checked by the primitive-recursive free-variable predicate. Existential duality has one fixed parse pattern, so its recognizer is immediate. Equality reflexivity and the displayed substitutivity scheme likewise have primitive-recursively recognizable parse patterns.

For modus ponens, decode the three formulas and check that the second has outer constructor $\rightarrow$ with the first and third as its immediate constituents. For generalization, decode the conclusion and check that it is $\forall v$ applied to the premise. In a proof from a theory the assumptions are sentences, hence no variable is free in them. In the more general open-assumption coding, the stored finite dependency set is scanned and $\text{Free}(v, \cdot)$ is checked on each dependent assumption. All scans are bounded by the finite proof record. Hence every displayed recognizer is primitive recursive. $\square$

Proposition 13.4 (Primitive-recursive coding of axioms for recursive theories). *If T is recursively axiomatized, the predicate $\text{Ax}_T(n)$ saying that n is the Gödel code of a nonlogical axiom of T may be chosen recursive; hence its use in checking finite proof codes is effective.*

Proof. For a recursively axiomatized theory T , the numerical predicate $Ax_T(n)$ deciding whether n is the code of a nonlogical axiom is recursive by definition. A recursive decision procedure has a primitive-recursive finite computation-history relation $C_T(n, b, s)$, where $b \in \{0, 1\}$ is the claimed yes/no output and s codes a complete halting computation. Hence

$$Ax_T(n) \iff \exists s C_T(n, 1, s).$$

For proof checking we therefore store, with each line asserted to be a nonlogical axiom, a finite certificate s witnessing the positive computation. Logical axiom instances are checked by their already primitive-recursive syntactic predicates. The local predicate saying that a line is a correctly certified axiom line is then primitive recursive in the enlarged proof record. The same certificate idea is used in the next chapter for an arbitrary c.e. axiom set, where the witness certifies that the enumerator has produced the axiom.

Point specific to this result. Recursive axiomhood is decidable, but decidability alone does not make the bare decision predicate primitive recursive. The finite computation certificate is what makes each local proof-line check primitive recursive; the later c.e. case uses an enumeration witness for the same reason. $\square$

Worked example

Worked Example 13.5 (Computing a diagonal code). Suppose n codes a one-variable formula $\psi(x)$. The diagonal function first constructs the numeral $\bar{n}$, then applies the already coded capture-avoiding substitution operation to replace the distinguished free variable by $\bar{n}$. Its output is the code of $\psi(\bar{n})$. Every component of this pipeline is primitive recursive.

A useful warning

A Gödel number has no intrinsic mathematical meaning. Meaning enters through the primitive-recursive decoding predicates that recognize well-formed terms, formulas, and substitutions.

Looking ahead

The next chapter, *Proof and provability predicates*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 14

Proof and provability predicates

Definitions and setup

Formal definitions used in this chapter

Proof certificate and proof-code relation. For a c.e. theory T , a proof certificate is a coded finite derivation in which each line carries enough finite justification data to verify either a logical axiom via LogAx, a witnessed occurrence of a c.e. theory axiom, or one of the fixed inference rules via the primitive-recursive MP/Gen checks. The numerical relation $\text{Prf}_T(p, q)$ means that p is such a correct certificate whose last formula has Gödel code q .

Representing proof formula. After the numerical primitive-recursive relation Prf_T has been represented in arithmetic, the same symbol $\text{Prf}_T(x, y)$ denotes its representing formula; the argument type (numbers in the metatheory versus variables/numerals in arithmetic) determines which meaning is intended.

Provability predicate and theoremhood. Define

$$\text{Prov}_T(q) := \exists p \text{Prf}_T(p, q).$$

For a sentence φ , $T \vdash \varphi$ means that an actual finite T -derivation of φ exists. Thus the theorem set of T is $\{\ulcorner \varphi \urcorner : T \vdash \varphi\}$.

Fix an effectively axiomatized theory T . $\text{Prf}_T(p, q)$ is the arithmetic relation saying that p codes a correct finite T -proof whose final formula has code q . The provability predicate is $\text{Prov}_T(q) \equiv \exists p \text{Prf}_T(p, q)$. Externally, $T \vdash \varphi$ means an actual finite proof exists; internally, $\text{Prov}_T(\ulcorner \varphi \urcorner)$ is an arithmetic sentence. The distinction between these two levels drives all of Volume III.

How this chapter fits together

A c.e. axiom set requires slightly more care than a recursive one: a proof line claiming to be an axiom carries a finite witness that the axiom enumerator produced it. With those witnesses included in the proof certificate, checking a purported proof becomes primitive recursive. Provability is then the existential statement that some code passes this check.

Theorem 14.1 (Effective proof predicate for c.e. theories). *If T is computably enumerable, there is a primitive-recursive certificate relation $\text{Prf}_T(p, q)$ such that*

$$\mathbb{N} \models \text{Prf}_T(p, q)$$

if and only if p codes a correct finite T -derivation whose last formula has Gödel number q .

How to read the symbols: what a proof code certifies

The predicate $\text{Prf}_T(p, q)$ should be read as a verifier, not as an oracle. Its input p is a proposed finite proof certificate and q is the code of the claimed final formula. The verifier checks each line locally: axiom instance, witnessed theory axiom, or valid inference from earlier lines. Because each check is finite and primitive recursive, arithmetic can represent the verifier itself.

Proof. Let T be c.e. and choose a primitive-recursive enumeration relation $E_T(a, s)$ for its axioms. A certified proof code p is a finite sequence of line records. For each line i , the record contains a formula code and a justification. Define $\text{LineOK}_T(p, i)$ by a finite disjunction:

$$\begin{array}{ll} \text{(logical axiom)} & \text{LogAx}((p)_i), \\ \text{(theory axiom)} & \exists s \leq p \ E_T((p)_i, s) \text{ with the stored witness } s, \\ \text{(rule)} & \text{the stored rule data pass the MP/Gen checks.} \end{array}$$

For an open-assumption derivation, the generalization record additionally carries the finite dependency data needed for the eigenvariable check described above. All quantifiers involved in inspecting the finite proof record are bounded by the code or its length. Hence LineOK_T is primitive recursive, and so is

$$\text{Prf}_T(p, q) \iff \text{Seq}(p) \wedge \text{lh}(p) > 0 \wedge (p)_{\text{lh}(p)-1} = q \wedge \forall i < \text{lh}(p) \ \text{LineOK}_T(p, i).$$

Thus even for a c.e. theory, proof correctness has a primitive-recursive certificate relation.

Point specific to this result. The certificate attached to each theory-axiom line is the critical device: it turns c.e. axiom membership into a finitely checkable claim. Once every line has finitely checkable justification data, the whole proof predicate is primitive recursive. $\square$

Theorem 14.2 (Arithmetical representability of $\text{Prf}_T(x, y)$). *The primitive-recursive relation $\text{Prf}_T(p, q)$ has an arithmetic representing formula, again denoted $\text{Prf}_T(x, y)$, that may be chosen Σ_1 and for which every true and every false standard numeral instance is correctly proved in Robinson arithmetic Q .*

Proof. The numerical relation $\text{Prf}_T(p, q)$ constructed above is primitive recursive. By the representability theorem for primitive-recursive relations, there is an arithmetic formula, which we denote by the same symbol,

$$\text{Prf}_T(x, y),$$

with the following standard-instance correctness: for every $m, n \in \mathbb{N}$,

$$\begin{aligned} \text{Prf}_T(m, n) &\Rightarrow Q \vdash \text{Prf}_T(\bar{m}, \bar{n}), \\ \neg \text{Prf}_T(m, n) &\Rightarrow Q \vdash \neg \text{Prf}_T(\bar{m}, \bar{n}). \end{aligned}$$

The first argument codes a finite proof certificate; the second codes its concluding formula. This is stronger than merely saying “proofhood is definable”: later incompleteness arguments use the first displayed implication for an actual proof code and, in Rosser’s argument, the second implication for finitely many smaller nonproof codes.

Point specific to this result. Having shown the external certificate relation primitive recursive, apply the representability theorem to obtain an arithmetic formula $\text{Prf}_T(p, q)$. This is the precise bridge from finite metamathematical proofs to formulas manipulated inside arithmetic. $\square$

Theorem 14.3 (Definability and basic complexity of $\text{Prov}_T(y)$). *Define*

$$\text{Prov}_T(y) \equiv \exists x \text{Prf}_T(x, y).$$

Then Prov_T is a Σ_1 formula representing theoremhood of T : for every sentence φ ,

$$T \vdash \varphi \iff \mathbb{N} \models \text{Prov}_T(\ulcorner \varphi \urcorner).$$

Proof. Define the canonical provability predicate by existentially quantifying the represented proof relation:

$$\text{Prov}_T(y) := \exists x \text{Prf}_T(x, y).$$

By the preceding theorem we choose the representing formula $\text{Prf}_T(x, y)$ to be Σ_1 . Adding the leading existential quantifier preserves the Σ_1 form, so Prov_T is Σ_1 . For every sentence φ , if $T \vdash \varphi$, choose an actual standard natural-number code of a proof n . Then representability gives

$$Q \vdash \text{Prf}_T(\bar{n}, \ulcorner \varphi \urcorner),$$

and hence, by existential introduction,

$$Q \vdash \text{Prov}_T(\ulcorner \varphi \urcorner).$$

If $Q \subseteq T$, the same provability assertion is therefore derivable in T . This is the external-to-internal bridge used in the first derivability condition; no converse $\text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi$ is asserted here.

Point specific to this result. Define $\text{Prov}_T(q) \equiv \exists p \text{Prf}_T(p, q)$. Since the matrix is primitive recursive and hence representable by a bounded/low-complexity predicate in the chosen coding, provability is Σ_1 in the required presentation. $\square$

Worked example

Worked Example 14.4 (A proof certificate). For a three-line derivation, the code contains the three formula codes together with tags recording whether each line is a logical axiom, a witnessed theory axiom, or follows by a rule from earlier line numbers. Proof checking inspects only finite local data. Consequently $\text{Prf}_T(p, q)$ is arithmetically tame even when membership in the axiom set is only c.e.

A useful warning

The formula $\text{Prov}_T(q) \equiv \exists p \text{Prf}_T(p, q)$ expresses existence of a proof code; it does not itself assert that the proved sentence is true.

End-of-volume perspective

Syntax and formal proofs have now become arithmetical objects. The next volume will exploit that internal coding to construct sentences that talk, indirectly but precisely, about their own provability.

Part III

Volume III: Incompleteness, Provability, and Truth

Preface

Self-reference enters only after proof checking and substitution have been arithmetized. The volume keeps consistency, soundness, and truth sharply separated.

Position in the series

Volume III is the first major endpoint of the series: Gödel incompleteness, Löb’s theorem, Gödel’s second theorem, and Tarski undefinability. The remainder of the series shifts from arithmetic metamathematics to set-theoretic independence.

Sources and further reading

The main proof families are Gödel’s incompleteness argument [4], Rosser’s consistency-only refinement [5], and Löb’s theorem [6].

Notation for this volume

The symbols G_T and R_T will be assigned, respectively, to the canonical diagonal outputs constructed in Chapters 2 and 3 of this volume. Until those constructions are proved, no existence is built into that notation. $\text{Con}(T)$ denotes the canonical arithmetical statement $\neg\text{Prov}_T(\ulcorner 0 = 1 \urcorner)$.

How to read symbolic arguments in this volume

When a formula appears, read it in layers rather than as one visual block. First identify the *ambient language*: are we speaking inside a formal theory, about a structure, or in the surrounding metatheory? Next mark the objects being manipulated (formulas, codes, sets, names, conditions, ordinals) and the operation being applied to them. Finally check the side conditions—free variables, rank bounds, compatibility, genericity, transitivity, or consistency hypotheses. Whenever a displayed derivation changes level, the text says so explicitly. A displayed line is therefore meant to be read like a line of well-commented code: the symbols perform the operation, and the surrounding prose records its type, preconditions, and purpose.

Chapter 15

Self-reference

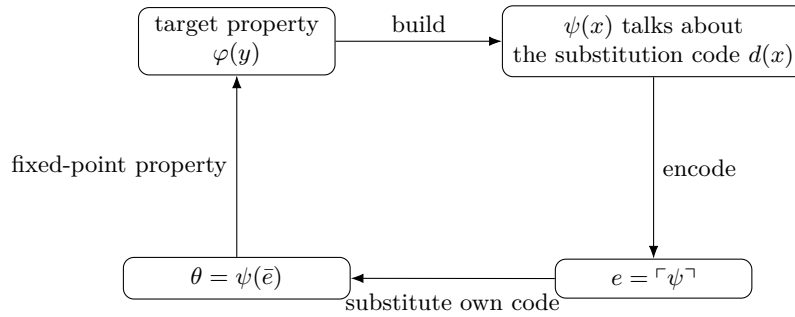


Figure 15.1: The diagonal construction is a feedback loop on codes, not a sentence literally containing itself. The bottom arrow performs the self-substitution; the left arrow records the property thereby obtained.

Definitions and setup

Formal definitions used in this chapter

Recalled code-level substitution. From Volume II, $\text{Sub}(n, v, m)$ is the total primitive-recursive function coding capture-avoiding substitution of the term coded by m for the free occurrences of variable code v in the formula coded by n , and $\text{Num}(n) = \ulcorner \bar{n} \urcorner$ is the code of the numeral for n . The diagonal function is

$$d(n) = \text{Sub}(n, v_0, \text{Num}(n)).$$

Graph formula for a numerical function. A formula $D(x, y)$ strongly represents the graph of d over Q when, for every standard n ,

$$Q \vdash \forall y (D(\bar{n}, y) \leftrightarrow y = \overline{d(n)}).$$

Fixed point. If T is a theory in the language of arithmetic and $\varphi(y)$ is a formula with at most the displayed free variable, a sentence θ is a fixed point of φ over T when

$$T \vdash \theta \leftrightarrow \varphi(\ulcorner \theta \urcorner).$$

The expression $\ulcorner \theta \urcorner$ is the numeral/code naming the finite syntactic expression θ , not the sentence itself.

Uniform fixed point with parameters. For a formula $\varphi(y, \bar{z})$, a uniform fixed point is a formula $\theta(\bar{z})$ satisfying

$$T \vdash \forall \bar{z} [\theta(\bar{z}) \leftrightarrow \varphi(\ulcorner \theta \urcorner, \bar{z})],$$

where only the code variable is diagonalized and the parameter variables $\bar{z}$ remain free.

Effective construction from a code. A syntactic construction is called effective here when a primitive-recursive transformation of Gödel codes computes the output code from the input code.

Self-reference is implemented syntactically, not by an English paradox. A fixed point for a one-variable formula $\varphi(x)$ is a sentence θ such that the base theory proves $\theta \leftrightarrow \varphi(\ulcorner \theta \urcorner)$. The construction uses the represented substitution function. We reserve the word diagonalization for this coding mechanism and do not assume that arbitrary semantic predicates can be named inside arithmetic.

How this chapter fits together

The diagonal lemma is best understood as a composition theorem for syntax. A formula does not literally contain its own infinitely long description; instead, arithmetic represents the finite substitution function sending a formula code to the code obtained after inserting a numeral. Feeding the code of a carefully chosen formula into that function creates the fixed point.

Lemma 15.1 (Substitution/diagonal function lemma). *The diagonal substitution function*

$$d(n) = \text{Sub}(n, v_0, \text{Num}(n))$$

is primitive recursive. Hence there is a graph formula $D(x, y)$ such that for every standard n ,

$$Q \vdash \forall y (D(\bar{n}, y) \leftrightarrow y = \overline{d(n)}).$$

The universal biconditional, not merely correctness on pairs of standard numerals, is the form used in the diagonal lemma.

Proof.

Formal derivation spine

Formal spine: represent the numerical diagonal function strongly.

$$d(n) = \text{Sub}(n, v_0, \text{Num}(n)), \quad Q \vdash \forall y [D(\bar{n}, y) \leftrightarrow y = \overline{d(n)}].$$

Volume II proves that the code-level substitution function

$$\text{Sub}(e, i, m)$$

is primitive recursive, as is the numeral-code function $n \mapsto \text{Num}(n)$. Closure under composition therefore gives

$$d(n) := \text{Sub}(n, 0, \text{Num}(n))$$

as a primitive-recursive numerical function.

Apply strong representability of primitive-recursive functions. There is an arithmetic formula $D(x, y)$ such that, for each standard n ,

$$Q \vdash \forall y (D(\bar{n}, y) \leftrightarrow y = \overline{d(n)}). \quad (1)$$

Equation (1) says more than $Q \vdash D(\bar{n}, \overline{d(n)})$: it excludes nonstandard alternative witnesses for y when the first argument is the fixed numeral $\bar{n}$. That uniqueness is exactly what permits an existential quantifier over $D(\bar{n}, y)$ to be replaced, inside Q , by substitution of the single numeral $\overline{d(n)}$. $\square$

Theorem 15.2 (Fixed-point (diagonal) lemma). *Let T be any theory in the language of arithmetic with $Q \subseteq T$. For every arithmetic formula $\varphi(y)$ with at most the displayed free variable, there exists an arithmetic sentence θ such that*

$$T \vdash \theta \leftrightarrow \varphi(\ulcorner \theta \urcorner).$$

The construction is effective from a Gödel code of φ .

Uses: the strong representability of the diagonal substitution function and the preceding diagonal-graph lemma.

Commented symbolic trace: the diagonal lemma as a typed program

Given a one-variable formula $\varphi(y)$, form a new formula $\psi(x)$ that says: “apply the substitution program to x , obtaining code y , and then $\varphi(y)$ holds.” Now feed ψ ’s own code $e = \ulcorner \psi \urcorner$ into ψ . The numerical identity $d(e) = \ulcorner \psi(\bar{e}) \urcorner$ is what turns ordinary substitution into self-reference. Nothing in the construction asks a formula to literally contain itself as a substring.

Proof.

Formal derivation spine

Formal spine: feed a formula its own code.

$$\psi(x) := \exists y (D(x, y) \wedge \varphi(y)), \quad e := \ulcorner \psi \urcorner, \quad \theta := \psi(\bar{e}), \quad d(e) = \ulcorner \theta \urcorner.$$

Let $D(x, y)$ be the strongly representing graph formula from the preceding lemma. Given $\varphi(y)$, define

$$\psi(x) := \exists y (D(x, y) \wedge \varphi(y)).$$

Let $e := \ulcorner \psi \urcorner$ and put

$$\theta := \psi(\bar{e}).$$

By the definition of the numerical diagonal function,

$$d(e) = \ulcorner \psi(\bar{e}) \urcorner = \ulcorner \theta \urcorner. \tag{2}$$

Strong representability gives

$$Q \vdash \forall y (D(\bar{e}, y) \leftrightarrow y = \overline{d(e)}). \tag{3}$$

Using (3) in the definition of θ , first-order logic proves in Q

$$\theta \leftrightarrow \varphi(\overline{d(e)}). \tag{4}$$

The two numerals in (2) are literally the same numeral in the chosen coding, so (4) is

$$Q \vdash \theta \leftrightarrow \varphi(\ulcorner \theta \urcorner).$$

Since $Q \subseteq T$, the same biconditional is provable in T . Every step occurs at a specified level: $d(e)$ is an external numerical computation, (3) is its internal arithmetic representation, and the final displayed formula is a theorem of T . $\square$

Remark 15.3 (Source note). The fixed-point mechanism is the standard arithmetized diagonal argument underlying Gödel’s 1931 paper; compare [4].

Proposition 15.4 (Uniform diagonal lemma with parameters). *Let T be a theory in the language of arithmetic with $Q \subseteq T$. For every formula $\varphi(y, \bar{z})$ there is a formula $\theta(\bar{z})$ such that*

$$T \vdash \forall \bar{z} [\theta(\bar{z}) \leftrightarrow \varphi(\ulcorner \theta \urcorner, \bar{z})].$$

Only the code variable is diagonalized; the parameters $\bar{z}$ remain free.

Proof.

Formal derivation spine

Formal spine: diagonalize only the code variable.

$$\psi(x, \bar{z}) := \exists u (D(x, u) \wedge \varphi(u, \bar{z})), \quad T \vdash \forall \bar{z} [\theta(\bar{z}) \leftrightarrow \varphi(\ulcorner \theta \urcorner, \bar{z})].$$

Keep the parameter tuple $\bar{z}$ free and diagonalize only the code variable. Define

$$\psi(x, \bar{z}) := \exists u (D(x, u) \wedge \varphi(u, \bar{z})), \quad e := \ulcorner \psi \urcorner,$$

and set

$$\theta(\bar{z}) := \psi(\bar{e}, \bar{z}).$$

The numerical substitution identity is independent of any values assigned to $\bar{z}$:

$$d(e) = \ulcorner \theta \urcorner. \tag{1}$$

Strong representability gives

$$Q \vdash \forall u (D(\bar{e}, u) \leftrightarrow u = \overline{d(e)}). \tag{2}$$

Substituting (2) into the defining formula for θ , while leaving $\bar{z}$ free, yields

$$Q \vdash \forall \bar{z} (\theta(\bar{z}) \leftrightarrow \varphi(\overline{d(e)}, \bar{z})).$$

Now use (1) to identify the displayed numeral with $\ulcorner \theta \urcorner$. Because $Q \subseteq T$,

$$T \vdash \forall \bar{z} (\theta(\bar{z}) \leftrightarrow \varphi(\ulcorner \theta \urcorner, \bar{z})).$$

The code is the code of the open formula $\theta(\bar{z})$; it does not vary with the parameter values. $\square$

Worked Theorem 15.5 (Explicit construction of a self-referential arithmetic sentence). *Given a concrete arithmetic predicate $\varphi(y)$, the diagonal construction effectively produces a specific sentence θ and a formal T -derivation of $\theta \leftrightarrow \varphi(\ulcorner \theta \urcorner)$.*

Proof.

Formal derivation spine

Formal spine for Explicit construction of a self-referential arithmetic sentence.

$$\begin{aligned} a &:= \ulcorner \varphi(y) \urcorner, \\ \psi(x) &:= \exists u (D(x, u) \wedge \varphi(u)). \\ m &:= d(e) = \text{Sub}(e, 0, \text{Num}(e)). \end{aligned}$$

This is the algorithmic version of the preceding theorem. Starting from the concrete code

$$a := \ulcorner \varphi(y) \urcorner,$$

compute the code e of

$$\psi(x) := \exists u (D(x, u) \wedge \varphi(u)).$$

Because formula formation and substitution on codes are primitive recursive, e is effectively computable from a . Next compute

$$m := d(e) = \text{Sub}(e, 0, \text{Num}(e)).$$

Decode m ; by construction it is the code of the closed sentence

$$\theta := \psi(\bar{e}).$$

The represented computation gives

$$T \vdash D(\bar{e}, \bar{m}),$$

so unfolding the definition of ψ yields

$$T \vdash \theta \leftrightarrow \varphi(\bar{m}).$$

Finally $m = \ulcorner \theta \urcorner$, hence

$$T \vdash \theta \leftrightarrow \varphi(\ulcorner \theta \urcorner).$$

Every step is an effective operation on finite codes. *Specialization to this result.* To obtain an explicit self-referential sentence, choose a concrete one-variable target predicate and follow the fixed-point construction literally. The point is that no quotation device is added to the object language: every reference to syntax is mediated by numerals and the represented substitution relation.

Detailed formal verification. The coding argument is carried out on natural numbers. Every syntactic constructor has a primitive-recursive code operation, so substitution is obtained by course-of-values recursion on the parse code. The crucial closure statement is

$$\text{primitive recursive syntax operations} \implies \text{representable numerical graph in } Q.$$

Whenever a formula code is transformed, the proof identifies the corresponding numerical function first and only then replaces its graph by an arithmetic formula. This prevents the common type error of treating a formula as if it were its Gödel number. $\square$

Worked example

Worked Example 15.6 (A schematic fixed point). Given a one-variable formula $\varphi(y)$, let $\psi(x)$ say that φ holds of the result of diagonalizing x . Substitute the numeral for $\ulcorner\psi\urcorner$ into ψ itself. The resulting sentence θ is provably equivalent to $\varphi(\ulcorner\theta\urcorner)$. The apparent circularity is replaced by an ordinary finite substitution calculation.

A useful warning

The fixed-point lemma is syntactic. No truth predicate and no semantic liar sentence is used in its construction.

Looking ahead

The next chapter, *Gödel's first theorem*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 16

Gödel's first theorem

Definitions and setup

Formal definitions used in this chapter

Effectively axiomatized theory. In this volume this means a c.e. theory: the Gödel codes of the nonlogical axioms are computably enumerable. An *extension* $T \supseteq Q$ contains every axiom, and hence every theorem, of Q .

Consistency and incompleteness. T is consistent when $T \not\vdash 0 = 1$. It is complete when, for every sentence σ of its language, either $T \vdash \sigma$ or $T \vdash \neg\sigma$; otherwise it is incomplete.

Canonical provability predicate and Gödel fixed-point property. $\text{Prov}_T(y) \equiv \exists p \text{Prf}_T(p, y)$ is the proof predicate fixed in Volume II. A sentence θ is a *Gödel fixed point for T* when

$$T \vdash \theta \leftrightarrow \neg \text{Prov}_T(\ulcorner \theta \urcorner).$$

This is only a property of a candidate sentence. Proposition 16.1 proves existence; after that proof we fix the output of the canonical diagonal construction and denote it by G_T . **ω -consistency.** T is ω -consistent when there is no formula $\varphi(x)$ such that

$$T \vdash \exists x \varphi(x) \quad \text{and} \quad T \vdash \neg \varphi(\bar{n}) \text{ for every standard } n \in \mathbb{N}.$$

For an effectively axiomatized theory T , the proposition below will construct a fixed point of $\neg \text{Prov}_T(x)$; only after that construction will it be named G_T . Consistency here means exactly $T \not\vdash 0 = 1$, as defined above. ω -consistency is stronger: T cannot prove an existential statement while refuting every numeral instance of its matrix. The original Gödel proof uses ordinary consistency for one direction and ω -consistency for the other; the next chapter replaces the latter with Rosser's argument.

How this chapter fits together

The first proposition applies diagonalization to non-provability and thereby constructs the sentence that will thereafter be denoted G_T . The two halves of the classical argument have different hypotheses. After that construction, consistency alone rules out a proof of G_T ; Gödel's original argument uses the stronger ω -consistency assumption to rule out a proof of $\neg G_T$. Rosser's refinement in the next chapter removes that asymmetry.

Proposition 16.1 (Existence of a Gödel fixed point). *Let T be c.e. and extend Q . There is a sentence θ satisfying*

$$T \vdash \theta \leftrightarrow \neg \text{Prov}_T(\ulcorner \theta \urcorner).$$

How to read the symbols: what the fixed point says and where it says it

The proposition asserts existence of an ordinary arithmetic sentence θ . The displayed equivalence

$$T \vdash \theta \leftrightarrow \neg \text{Prov}_T(\ulcorner \theta \urcorner)$$

is a theorem of T about the numeral coding θ . Outside the theory, we interpret Prov_T as the canonical proof predicate. Keep those levels separate: the fixed-point theorem gives the internal equivalence; consistency is an external hypothesis used later to infer unprovability.

Proof.

Formal derivation spine

Formal spine: diagonalize the non-provability predicate.

$$\varphi(y) := \neg \text{Prov}_T(y), \quad T \vdash \theta \leftrightarrow \varphi(\ulcorner \theta \urcorner).$$

Apply the diagonal lemma to $\varphi(y) \equiv \neg \text{Prov}_T(y)$. We obtain a sentence θ with

$$T \vdash \theta \leftrightarrow \neg \text{Prov}_T(\ulcorner \theta \urcorner).$$

This is exactly the Gödel fixed-point property defined above. The construction depends only on the effective proof predicate and diagonal substitution, both established earlier.

Further explanation. The target predicate is $\neg \text{Prov}_T(y)$, not an informal English sentence. The diagonal lemma supplies a sentence whose own code occupies the argument place y . Saying later that the chosen fixed point is true requires an external hypothesis about the actual proof relation on standard natural numbers; that semantic conclusion is not part of the fixed-point construction itself. $\square$

Formal definitions used in this chapter

Canonical Gödel sentence. Fix the output θ of the canonical diagonal construction used in Proposition 16.1 and denote it from now on by G_T . Thus, by the proposition already proved,

$$T \vdash G_T \leftrightarrow \neg \text{Prov}_T(\ulcorner G_T \urcorner).$$

Theorem 16.2 (Consistency implies T does not prove G_T). *Let T be a consistent c.e. extension of Q , and let G_T be its canonical Gödel sentence. Then*

$$T \not\vdash G_T.$$

Proof.

Formal derivation spine

Formal spine: consistency excludes a proof of the Gödel sentence.

$$T \vdash G_T \ \& \ n \text{ codes that proof} \implies T \vdash \text{Prf}_T(\bar{n}, \ulcorner G_T \urcorner) \implies T \vdash \text{Prov}_T(\ulcorner G_T \urcorner).$$

But the fixed point gives

$$T \vdash G_T \rightarrow \neg \text{Prov}_T(\ulcorner G_T \urcorner),$$

so $T \vdash G_T$ would make T inconsistent.

Assume T is consistent and $T \vdash G_T$. Let n be the actual code of such a proof. Representability of the proof relation yields $T \vdash \text{Prf}_T(\bar{n}, \ulcorner G_T \urcorner)$, hence $T \vdash \text{Prov}_T(\ulcorner G_T \urcorner)$. On the other hand the fixed-point equivalence and $T \vdash G_T$ yield $T \vdash \neg \text{Prov}_T(\ulcorner G_T \urcorner)$. Thus T proves a contradiction, contrary to consistency. Therefore $T \not\vdash G_T$.

Point specific to this result. This half uses only ordinary consistency: an actual proof of G_T can be arithmetized into a proof of $\text{Prov}_T(\ulcorner G_T \urcorner)$, while the fixed-point equivalence gives its negation. No soundness assumption is used.

Further explanation. Notice the two levels in the contradiction. The natural number n is an external, actual proof code. Representability lets the theory verify the finite fact that n is a proof, producing the internal sentence $\text{Prov}_T(\ulcorner G_T \urcorner)$. The fixed point converts the assumed internal theorem G_T into the negation of that same provability assertion. Consistency is therefore enough because the argument starts from an actual derivation, not from an assumption that every theorem of T is true. $\square$

Theorem 16.3 (Omega-consistency implies T does not prove $\neg G_T$). *Let T be a c.e. extension of Q that is ω -consistent. For its canonical Gödel sentence,*

$$T \not\vdash \neg G_T.$$

Commented symbolic trace: where ω -consistency enters

Assuming $T \vdash \neg G_T$ gives one internal existential statement: $T \vdash \exists p \text{Prf}_T(p, \ulcorner G_T \urcorner)$. First, ω -consistency implies ordinary consistency: otherwise explosion would produce an existential statement together with the negation of every numeral instance. Ordinary consistency then tells us that no standard natural number actually codes a proof of G_T ; numeralwise correctness therefore gives $T \vdash \neg \text{Prf}_T(\bar{n}, \ulcorner G_T \urcorner)$ for every standard n . ω -consistency forbids exactly this pattern: proving an existential while refuting every standard numeral instance.

Proof.

Formal derivation spine

Formal spine for Omega-consistency implies T does not prove $\neg G_T$.

$$T \vdash G_T \leftrightarrow \neg \text{Prov}_T(\ulcorner G_T \urcorner).$$

$$T \vdash \text{Prov}_T(\ulcorner G_T \urcorner).$$

$$T \vdash \exists p \text{Prf}_T(p, \ulcorner G_T \urcorner).$$

Assume that T is ω -consistent. We first record the ordinary consistency needed below. If T were inconsistent, then by classical explosion, for the formula $\theta(x) \equiv (x = x)$ we would have

$$T \vdash \exists x \theta(x)$$

and also

$$T \vdash \neg \theta(\bar{n}) \quad \text{for every standard } n \in \mathbb{N}.$$

This is forbidden by the definition of ω -consistency. Hence T is consistent.

Now suppose, toward a contradiction, that $T \vdash \neg G_T$. From the fixed-point equivalence

$$T \vdash G_T \leftrightarrow \neg \text{Prov}_T(\ulcorner G_T \urcorner).$$

The right-to-left half of the equivalence is

$$T \vdash \neg \text{Prov}_T(\ulcorner G_T \urcorner) \rightarrow G_T.$$

Together with the assumed theorem $T \vdash \neg G_T$, classical propositional reasoning gives

$$T \vdash \neg \neg \text{Prov}_T(\ulcorner G_T \urcorner),$$

and double-negation elimination in the fixed classical calculus yields

$$T \vdash \text{Prov}_T(\ulcorner G_T \urcorner).$$

Now unfold the definition of the provability predicate: $\text{Prov}_T(q)$ is the existential statement saying that some number codes a T -proof of the formula with code q . Thus the previous line is exactly

$$T \vdash \exists p \text{Prf}_T(p, \ulcorner G_T \urcorner).$$

On the other hand, the ordinary consistency just derived and Theorem 16.2 imply $T \not\vdash G_T$. Hence no standard natural number is in fact the code of a T -proof of G_T . For each standard n , the primitive-recursive proof predicate correctly recognizes this finite fact, so numeralwise representability gives

$$T \vdash \neg \text{Prf}_T(\bar{n}, \ulcorner G_T \urcorner).$$

Thus T proves an existential statement $\exists p \text{Prf}_T(p, \ulcorner G_T \urcorner)$ while proving the negation of every standard numeral instance. This is exactly the configuration forbidden by ω -consistency. Therefore $T \not\vdash \neg G_T$.

Why the stronger hypothesis enters. The first half of Gödel's argument turns an actual proof into an internal provability assertion and therefore needs only consistency. The present half must rule out a theory that proves "some proof code exists" while rejecting each standard candidate separately; ω -consistency is used precisely at that final step. $\square$

Theorem 16.4 (Gödel's original first incompleteness theorem). *Every effectively axiomatized ω -consistent theory $T \supseteq Q$ is incomplete: there is a sentence G_T for which*

$$T \not\vdash G_T \quad \text{and} \quad T \not\vdash \neg G_T.$$

Uses: the fixed-point lemma, representability of the proof predicate, and the two preceding unprovability arguments.

Proof.

Formal derivation spine

Formal spine: combine the two asymmetric halves of Gödel I.

$$\omega\text{-Con}(T) \implies \text{Con}(T),$$

$$\text{Con}(T) \implies T \not\vdash G_T, \quad \omega\text{-Con}(T) \implies T \not\vdash \neg G_T.$$

Therefore

$$T \not\vdash G_T \quad \& \quad T \not\vdash \neg G_T.$$

Combine the preceding two propositions. Under consistency, G_T is not provable; under ω -consistency, its negation is not provable. Therefore an effectively axiomatized theory satisfying the stated arithmetical strength and ω -consistency is incomplete. The proof records exactly where the stronger hypothesis enters: only the refutation half of Gödel's original sentence, not the unprovability half.

Further explanation. The hypotheses should be read asymmetrically. Ordinary consistency already excludes $T \vdash G_T$. The stronger ω -consistency assumption is used only to exclude $T \vdash \neg G_T$, because that case produces an existential proof assertion together with the negation of each standard numeral instance. Rosser's refinement will replace this second argument, which is why modern formulations of the first incompleteness theorem usually require only ordinary consistency for a suitably strengthened self-referential sentence. $\square$

Remark 16.5 (Source note). This is the hypothesis structure of Gödel's original first incompleteness argument: consistency for the unprovability of G_T , and the stronger ω -consistency assumption for the unprovability of $\neg G_T$; see [4].

Worked example

Worked Example 16.6 (What a proof of the Gödel sentence would cause). If n were an actual T -proof of G_T , representability would let T verify $\text{Prf}_T(\bar{n}, \ulcorner G_T \urcorner)$, hence $\text{Prov}_T(\ulcorner G_T \urcorner)$. But the fixed-point equivalence turns G_T itself into $\neg \text{Prov}_T(\ulcorner G_T \urcorner)$. Thus a proof of G_T would manufacture an outright contradiction inside T .

A useful warning

Do not replace ω -consistency by ordinary consistency in Gödel's original refutation argument. That strengthening is exactly what Rosser's proof is designed to avoid.

Looking ahead

The next chapter, *Rosser and essential undecidability*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 17

Rosser and essential undecidability

Definitions and setup

Formal definitions used in this chapter

Proof comparison. For a sentence code q , let $\text{Neg}(q)$ denote the primitive-recursive code of its negation. The Rosser comparison relation is

$$\text{RPrf}_T(p, q) :\iff \text{Prf}_T(p, q) \wedge \forall r \leq p \neg \text{Prf}_T(r, \text{Neg}(q)).$$

Thus a positive instance records a proof of the sentence together with the absence of any no-larger proof of its negation.

Rosser fixed-point property. A sentence θ is a *Rosser fixed point* for T when

$$T \vdash \theta \leftrightarrow \forall p (\text{Prf}_T(p, \ulcorner \theta \urcorner) \rightarrow \exists q \leq p \text{Prf}_T(q, \text{Neg}(\ulcorner \theta \urcorner))).$$

This definition specifies a property but does not assert that such a sentence exists. Proposition 17.2 proves existence; after that proof its canonical diagonal output is denoted by R_T . **Decidable and undecidable theorem sets.** The theorem set $\text{Thm}(T) = \{\ulcorner \sigma \urcorner : T \vdash \sigma\}$ is decidable when its characteristic function is computable; otherwise it is undecidable. A theory is called decidable or undecidable according to its theorem set.

Recursive theory and essential undecidability. A recursive theory has a recursive axiom set. A theory T is essentially undecidable when every consistent recursive extension of T is undecidable.

Rosser's idea compares the numerical sizes of a proof of a sentence and a proof of its negation. Because “there is a proof with code at most n ” is bounded, the weak arithmetic can verify the finite comparison needed in the least-proof argument. The proposition below will construct the required self-referential fixed point, after which ordinary consistency replaces ω -consistency. Essential undecidability means that every consistent c.e. extension of a given weak theory is incomplete.

How this chapter fits together

Rosser compares proof codes for a sentence and its negation. The sentence is engineered so that any proof on one side predicts a shorter proof on the other. Minimal proof codes then turn ordinary consistency into the required contradiction. Essential undecidability follows because every consistent c.e. extension of Q inherits this obstruction.

Lemma 17.1 (Rosser proof-comparison predicate). *There is a primitive-recursive relation $RPrf_T(p, q)$ comparing proof codes so that the statement “ p is a proof of the sentence coded by q , and no code $r \leq p$ is a proof of its negation” is arithmetically representable.*

Proof.

Formal derivation spine

Formal spine for Rosser proof-comparison predicate.

$$RPrf_T(p, q) :\iff Prf_T(p, q) \wedge \forall r \leq p \neg Prf_T(r, \text{Neg}(q)).$$

$$RPrf_T \in PR.$$

$$Prf_T(\bar{q}, \ulcorner \sigma \urcorner) \quad \text{or} \quad \neg Prf_T(\bar{q}, \ulcorner \sigma \urcorner) \quad (q < p).$$

Let $\text{Neg}(q)$ be the primitive-recursive function sending the code of a sentence to the code of its negation. Define

$$RPrf_T(p, q) :\iff Prf_T(p, q) \wedge \forall r \leq p \neg Prf_T(r, \text{Neg}(q)).$$

The first conjunct is primitive recursive by Volume II. The function Neg is primitive recursive because it is one syntactic constructor, and bounded universal quantification preserves primitive recursiveness. Therefore

$$RPrf_T \in PR.$$

Representability provides an arithmetic formula, again written $RPrf_T(p, q)$, that correctly decides every numeral instance in Q . The relation does not yet assert a self-referential sentence; it merely formalizes the finite comparison “ p proves the sentence coded by q , and no proof of its negation has code at most p .” The next result diagonalizes this comparison predicate. *Specialization to this result.* The Rosser comparison relation is bounded in the competing proof code: given a candidate proof of one side, arithmetic checks whether a smaller proof of the negation exists. Because the search is bounded by the candidate code, weak arithmetic can verify each concrete comparison needed in the minimal-proof argument. $\square$

Proposition 17.2 (Existence of a Rosser fixed point). *For every c.e. theory $T \supseteq Q$ there is a sentence θ such that*

$$T \vdash \theta \leftrightarrow \forall p (Prf_T(p, \ulcorner \theta \urcorner) \rightarrow \exists q \leq p Prf_T(q, \text{Neg}(\ulcorner \theta \urcorner))).$$

Proof.

Formal derivation spine

Formal spine: diagonalize the Rosser comparison formula.

$$\rho(y) := \forall p (Prf_T(p, y) \rightarrow \exists q \leq p Prf_T(q, \text{Neg}(y))).$$

$$T \vdash \theta \leftrightarrow \rho(\ulcorner \theta \urcorner).$$

Use the proof-comparison relation to form the one-variable formula

$$\rho(y) := \forall p (Prf_T(p, y) \rightarrow \exists q \leq p Prf_T(q, \text{Neg}(y))).$$

By the fixed-point lemma there is a sentence θ such that

$$T \vdash \theta \leftrightarrow \rho(\ulcorner \theta \urcorner).$$

Expanding ρ , this is

$$T \vdash \theta \leftrightarrow \forall p (Prf_T(p, \ulcorner \theta \urcorner) \rightarrow \exists q \leq p Prf_T(q, \text{Neg}(\ulcorner \theta \urcorner))).$$

This is exactly the Rosser fixed-point property defined above. The construction uses only effective proof coding and diagonalization; consistency is not needed for existence. Consistency enters only in the following theorem,

where least proof codes are compared. *Specialization to this result.* Diagonalize the Rosser comparison predicate rather than ordinary non-provability. The resulting sentence says that every proof of itself is accompanied by a no-larger proof of its negation, which is precisely the asymmetry needed for the least-proof contradiction. $\square$

Formal definitions used in this chapter

Canonical Rosser sentence. Fix the output θ of the canonical diagonal construction used in Proposition 17.2 and denote it from now on by R_T . By Proposition 17.2,

$$T \vdash R_T \leftrightarrow \forall p (\text{Prf}_T(p, \ulcorner R_T \urcorner) \rightarrow \exists q \leq p \text{Prf}_T(q, \ulcorner \neg R_T \urcorner)).$$

Theorem 17.3 (Rosser incompleteness theorem from ordinary consistency). *If $T \supseteq Q$ is c.e. and consistent, then its Rosser sentence is undecidable in T :*

$$T \not\vdash R_T \quad \text{and} \quad T \not\vdash \neg R_T.$$

Proof.

Formal derivation spine

Formal spine: the two least-proof contradictions. For the Rosser sentence R ,

$$T \vdash R \leftrightarrow \forall p (\text{Prf}_T(p, \ulcorner R \urcorner) \rightarrow \exists q \leq p \text{Prf}_T(q, \ulcorner \neg R \urcorner)). \quad (\text{R})$$

A least proof of R produces a no-larger proof of $\neg R$; a least proof of $\neg R$, together with the finite verification of all smaller codes, makes T prove the right-hand side of (R).

Assume first that $T \vdash R$, and let m be the least code of a T -proof of R . Since m is an actual proof code, numeralwise representability gives

$$T \vdash \text{Prf}_T(\bar{m}, \ulcorner R \urcorner). \quad (1)$$

From (R) and (1), T proves

$$\exists q \leq \bar{m} \text{Prf}_T(q, \ulcorner \neg R \urcorner). \quad (2)$$

The bounded formula in (2) is true only if some standard $q \leq m$ is an actual proof code of $\neg R$: bounded quantifiers below a standard numeral range over finitely many standard candidates, and the proof predicate is correct on each such candidate. Thus T would have proofs of both R and $\neg R$, contrary to consistency. Hence $T \not\vdash R$.

Now assume that $T \vdash \neg R$, and let n be the least code of such a proof. For every standard $m < n$, consistency implies that m is not a proof of R . Correctness of the primitive-recursive proof predicate yields, for each of the finitely many $m < n$,

$$Q \vdash \neg \text{Prf}_T(\bar{m}, \ulcorner R \urcorner).$$

Combining these finitely many derivations gives

$$Q \vdash \forall m < \bar{n} \neg \text{Prf}_T(m, \ulcorner R \urcorner). \quad (3)$$

Because n is an actual proof of $\neg R$,

$$Q \vdash \text{Prf}_T(\bar{n}, \ulcorner \neg R \urcorner). \quad (4)$$

We now verify the right-hand side of (R) inside Q . Fix p . If $p < \bar{n}$, (3) makes the antecedent $\text{Prf}_T(p, \ulcorner R \urcorner)$ false. If $\bar{n} \leq p$, the witness $q = \bar{n}$, together with (4), satisfies the consequent. Since Q proves the bounded order dichotomy

$$p < \bar{n} \vee \bar{n} \leq p,$$

it follows that Q , hence T , proves the Rosser condition in (R). Therefore $T \vdash R$, contradicting $T \vdash \neg R$ and consistency. Thus $T \not\vdash \neg R$. Ordinary consistency suffices in both directions; no soundness or ω -consistency assumption is used. $\square$

Remark 17.4 (Source note). Rosser's refinement replaces the ω -consistency half by a bounded comparison of proof codes, reducing the hypothesis to ordinary consistency; see [5].

Theorem 17.5 (Essential undecidability of Q and its effective consequences). *Every consistent c.e. extension $T \supseteq Q$ is incomplete and has undecidable theorem set. Consequently Q is essentially undecidable: no consistent recursive theory extending Q is complete.*

Proof.

Formal derivation spine

Formal spine: essential undecidability. For every consistent c.e. $S \supseteq Q$, Rosser gives

$$S \not\vdash R_S \quad \& \quad S \not\vdash \neg R_S,$$

so S is incomplete. If the theorem set of such an S were recursive, a recursive greedy completion would produce a consistent complete c.e. extension $S^* \supseteq Q$, contradicting Rosser again.

Let $S \supseteq Q$ be consistent and computably enumerable. Rosser's theorem, applied to S , produces a sentence R_S with

$$S \not\vdash R_S, \quad S \not\vdash \neg R_S.$$

Thus every consistent c.e. extension of Q is incomplete.

It remains to prove the stronger decision-theoretic assertion. Suppose, toward a contradiction, that the theorem set

$$\text{Thm}(S) := \{\ulcorner \sigma \urcorner : S \vdash \sigma\}$$

is recursive. Enumerate all sentences as $\varphi_0, \varphi_1, \dots$, and construct finite sets F_n recursively. Put $F_0 = \emptyset$. Having constructed F_n , decide whether

$$S \cup F_n \cup \{\varphi_n\}$$

is consistent. Since F_n is finite, the Deduction Theorem gives the exact test

$$S \cup F_n \cup \{\varphi_n\} \text{ inconsistent} \iff S \vdash \neg(\bigwedge F_n \wedge \varphi_n).$$

The right-hand side is decidable because $\text{Thm}(S)$ is recursive. If the extension is consistent, set $F_{n+1} = F_n \cup \{\varphi_n\}$; otherwise set $F_{n+1} = F_n \cup \{\neg\varphi_n\}$. The second choice is consistent because the first choice was detected inconsistent and $S \cup F_n$ is consistent.

Let

$$S^* := S \cup \bigcup_{n < \omega} F_n.$$

Every finite subset of S^* lies in some consistent stage, so S^* is consistent. By construction, for every sentence φ_n , exactly one of $\varphi_n, \neg\varphi_n$ is added, so S^* is complete. The construction is recursive from the assumed decision procedure for $\text{Thm}(S)$; hence S^* is a computably enumerable extension of Q . Rosser's theorem applied to S^* says that S^* is incomplete, a contradiction. Therefore $\text{Thm}(S)$ is undecidable.

Consequently no consistent recursive theory extending Q can be complete, and indeed no consistent c.e. extension of Q has a decidable theorem set. This is the essential-undecidability property used later. $\square$

Worked example

Worked Example 17.6 (The least-proof trick). Assume a Rosser sentence R has a least proof code m . Its defining property yields a proof of $\neg R$ with code no larger than m , contradicting consistency. Conversely, if $\neg R$ has least proof code n , the finitely many smaller candidate proofs can be checked one by one inside weak arithmetic, forcing the Rosser condition and hence R .

A useful warning

The argument uses the arithmetic ordering of proof codes, not an assumption that proofs arrive in chronological order from some enumerator.

Looking ahead

The next chapter, *Consistency and soundness notions*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 18

Consistency and soundness notions

Definitions and setup

Formal definitions used in this chapter

Standard model and external truth. $\mathbb{N}$ is the structure with domain the natural numbers and the intended interpretations of $0, S, +, \cdot$. “ $\mathbb{N} \models \sigma$ ” is an external semantic assertion in the surrounding mathematics, not a formula asserted inside T .

Consistency. T is consistent when $T \not\vdash 0 = 1$.

Soundness. T is sound for $\mathbb{N}$ when $T \vdash \sigma \Rightarrow \mathbb{N} \models \sigma$ for every arithmetic sentence σ .

Σ_1 -soundness and 1-consistency. In this book these are synonymous: T is Σ_1 -sound (or 1-consistent) when no false Σ_1 sentence is provable, equivalently

$$T \vdash \sigma \text{ and } \sigma \in \Sigma_1 \implies \mathbb{N} \models \sigma.$$

ω -consistency. Recall that T is ω -consistent when no formula $\varphi(x)$ has $T \vdash \exists x \varphi(x)$ together with $T \vdash \neg \varphi(\bar{n})$ for every standard n .

Consistency, ω -consistency, Σ_1 -soundness (called 1-consistency in this book), and full soundness are distinct levels of hypothesis. Thus 1-consistency and Σ_1 -soundness are two names for the same condition here, not two separate rungs. Soundness refers to truth in the standard model; consistency is purely syntactic. The implications among the distinct levels are one-way in general. We use the weakest condition sufficient for each theorem and state explicitly when an external truth claim about the Gödel sentence requires more than mere consistency.

How this chapter fits together

Several nearby notions are often conflated. Consistency forbids proving a contradiction. In this book 1-consistency is exactly the same condition as Σ_1 -soundness: no false Σ_1 sentence is provable. ω -consistency rules out proving an existential while refuting every numeral instance. Full soundness says every theorem is true in the intended model. The implication diagram matters because different incompleteness arguments use different points on it.

Proposition 18.1 (Implication chain among soundness, Sigma-1 soundness, 1-consistency, omega-consistency, and consistency). *For extensions of Q , the following implications hold:*

$$\text{sound} \Rightarrow \omega\text{-consistent} \Rightarrow 1\text{-consistent} \iff \Sigma_1\text{-sound} \Rightarrow \text{consistent}.$$

Here 1-consistency means that no false Σ_1 sentence is provable.

Proof.

Formal derivation spine

Formal spine for Implication chain among soundness, Sigma-1 soundness, 1-consistency, omega-consistency, and consistency.

$$\begin{aligned} \text{Con}(T) &\iff T \not\vdash 0 = 1, \\ T \vdash \sigma \wedge \sigma \in \Sigma_1 &\implies \mathbb{N} \models \sigma. \end{aligned}$$

By definition, 1-consistency and Σ_1 -soundness are synonymous in this book. We therefore prove only the genuine implications between distinct conditions.

If T is sound and were ω -inconsistent, there would be a formula $\varphi(x)$ with

$$T \vdash \exists x \varphi(x) \quad \text{and} \quad T \vdash \neg \varphi(\bar{n}) \text{ for every standard } n.$$

Soundness would make all these sentences true in $\mathbb{N}$. The existential sentence would then have some standard witness n , contradicting $\mathbb{N} \models \neg \varphi(\bar{n})$. Hence soundness implies ω -consistency.

Now assume $T \supseteq Q$ is ω -consistent and suppose, toward a contradiction, that T is not Σ_1 -sound. Then T proves a false Σ_1 sentence

$$\sigma \equiv \exists x \delta(x),$$

with $\delta \in \Delta_0$. Since σ is false in $\mathbb{N}$, every standard instance $\delta(\bar{n})$ is false. Numeralwise Δ_0 -correctness of Robinson arithmetic, proved in Volume II, gives

$$Q \vdash \neg \delta(\bar{n})$$

for every standard n , and hence the same is true in T . Together with $T \vdash \exists x \delta(x)$, this is exactly the forbidden ω -inconsistency pattern. Thus T is Σ_1 -sound, equivalently 1-consistent.

Finally, Σ_1 -soundness implies consistency because an inconsistent theory proves the false bounded (hence Σ_1) sentence $0 = 1$. The reverse implications are not used without additional hypotheses; later examples such as consistent unsound extensions show why the distinctions matter. $\square$

How to read the symbols: where strictness is proved

The implication chain above is already available, but the separation “consistent does not imply Σ_1 -sound” uses Gödel’s second incompleteness theorem. To keep proof dependencies forward-clean, that counterexample is proved in Chapter 6, immediately after Gödel II, rather than being used here before its proof.

Theorem 18.2 (External truth of canonical Gödel/Rosser sentences under the precise hypotheses). *Let $T \supseteq Q$ be a consistent computably enumerable theory, and use the canonical proof predicate fixed in Volume II. Then the canonical Gödel sentence G_T and the canonical Rosser sentence R_T are both true in the standard model:*

$$\mathbb{N} \models G_T \wedge R_T.$$

These are external semantic conclusions and are not assertions that either sentence is provable in T .

Proof.

Formal derivation spine

Formal spine: external absence of proof codes plus a Q -provable fixed point.

$$\begin{aligned} T \not\vdash G_T &\implies \mathbb{N} \models \neg \text{Prov}_T(\ulcorner G_T \urcorner), \\ Q \vdash G_T &\leftrightarrow \neg \text{Prov}_T(\ulcorner G_T \urcorner) \implies \mathbb{N} \models G_T. \end{aligned}$$

By consistency and the first half of Gödel's theorem,

$$T \not\vdash G_T.$$

Therefore no standard natural number codes a T -proof of G_T , so correctness of the canonical proof predicate on standard numbers gives

$$\mathbb{N} \models \neg \text{Prov}_T(\ulcorner G_T \urcorner). \quad (1)$$

The fixed-point biconditional was proved already in Q , not merely in the possibly unsound extension T :

$$Q \vdash G_T \leftrightarrow \neg \text{Prov}_T(\ulcorner G_T \urcorner). \quad (2)$$

Every axiom of Q is true in $\mathbb{N}$, so soundness applied to (2), together with (1), yields

$$\mathbb{N} \models G_T.$$

This explicitly avoids the invalid inference that an arbitrary theorem of a consistent theory must be true in $\mathbb{N}$. For the Rosser sentence, consistency and Rosser's theorem give

$$T \not\vdash R_T \quad \text{and} \quad T \not\vdash \neg R_T.$$

Hence for every standard p , $\mathbb{N} \models \neg \text{Prf}_T(p, \ulcorner R_T \urcorner)$. The right-hand side of the Rosser fixed-point formula

$$\forall p \left(\text{Prf}_T(p, \ulcorner R_T \urcorner) \rightarrow \exists q \leq p \text{Prf}_T(q, \ulcorner \neg R_T \urcorner) \right)$$

is therefore true in $\mathbb{N}$. Its biconditional with R_T is again provable in Q , hence true in $\mathbb{N}$. Consequently $\mathbb{N} \models R_T$. $\square$

Proposition 18.3 (Why syntactic consistency and semantic truth are distinct notions). *For a theory T , syntactic consistency is the assertion that no contradiction is derivable in T , whereas truth of a sentence φ is relative to a structure such as $\mathbb{N}$. In particular, a consistent theory may contain false sentences, and a true sentence may be unprovable in a consistent incomplete theory.*

Proof.

Formal derivation spine

Formal spine for Why syntactic consistency and semantic truth are distinct notions.

$$T \text{ is syntactically consistent} \iff T \not\vdash 0 = 1.$$

$$\mathbb{N} \models \varphi.$$

$$T \not\vdash \neg \varphi$$

Syntactic consistency is a property of the derivability relation and makes sense for every first-order theory, whether or not its axioms have been given an effective coding:

$$T \text{ is syntactically consistent} \iff T \not\vdash 0 = 1.$$

When T is later equipped with the fixed computable proof coding, this metatheoretic condition is represented by the arithmetical sentence $\text{Con}(T)$. Semantic truth is evaluated in a structure:

$$\mathbb{N} \models \varphi.$$

The two predicates have different arguments and live at different levels. In particular, from

$$T \not\vdash \neg \varphi$$

one cannot infer $\mathbb{N} \models \varphi$ without an additional semantic assumption. Conversely, $\mathbb{N} \models \varphi$ need not imply $T \vdash \varphi$ once T is incomplete. Gödel's sentence gives the canonical example under the hypotheses established above:

$$T \text{ consistent} \implies \mathbb{N} \models G_T \quad \text{and} \quad T \not\vdash G_T.$$

Thus consistency, truth, and provability are formally distinct notions whose relationships require separate theorems. *Point specific to this result.* Consistency says only that no contradiction is derivable in the formal system. Truth refers to the standard structure (or another specified model). Incompleteness exploits precisely the fact that these notions need not coincide for an effectively axiomatized theory. $\square$

Worked example

Worked Example 18.4 (A consistent but unsound extension from the Gödel sentence). Let $T \supseteq Q$ be a consistent c.e. theory and let G_T be its canonical Gödel sentence. The first incompleteness theorem gives $T \not\vdash G_T$. Hence $T + \neg G_T$ is consistent: if the extension proved a contradiction, the Deduction Theorem would give $T \vdash G_T$. The external-truth theorem above gives $\mathbb{N} \models G_T$, so $\neg G_T$ is false in the standard model. Thus $T + \neg G_T$ is a concrete consistent but unsound extension, obtained using only results already proved at this point.

A useful warning

Whenever a proof says “therefore the sentence is true,” identify which soundness hypothesis licenses the move. Unprovability alone is not truth.

Looking ahead

The next chapter, *Formalized provability*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 19

Formalized provability

Definitions and setup

Formal definitions used in this chapter

Fix a c.e. theory $T \supseteq I\Sigma_1$ and its canonical Σ_1 provability predicate $\text{Prov}_T(y) \equiv \exists p \text{Prf}_T(p, y)$. The labels D1–D3 below *define three properties* that an arithmetized provability predicate may satisfy. They are not assumed here; Theorems 19.1–19.3 prove that the canonical predicate satisfies them.

D1 (proof certification property). For every sentence φ ,

$$T \vdash \varphi \implies T \vdash \text{Prov}_T(\ulcorner \varphi \urcorner).$$

D2 (formalized modus-ponens property). For all sentences φ, ψ ,

$$T \vdash \text{Prov}_T(\ulcorner \varphi \rightarrow \psi \urcorner) \rightarrow (\text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \text{Prov}_T(\ulcorner \psi \urcorner)).$$

D3 (positive-introspection property). For every sentence φ ,

$$T \vdash \text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \text{Prov}_T(\ulcorner \text{Prov}_T(\ulcorner \varphi \urcorner) \urcorner).$$

The phrase *Hilbert–Bernays–Löb derivability conditions* means precisely D1–D3 above.

The Hilbert–Bernays–Löb derivability conditions are internal closure properties of the chosen provability predicate. D1 internalizes actual theoremhood, D2 internalizes modus ponens, and D3 says that provability is itself provably provable. They are proved from primitive-recursive transformations of proof codes. These are intensional properties: an arithmetical predicate with the same standard extension need not satisfy them.

How this chapter fits together

The Hilbert–Bernays–Löb conditions are not abstract assumptions here; they are consequences of explicit primitive-recursive transformations of proof codes. One transformation records a known proof, another combines two proofs by modus ponens, and a third turns a proof into a proof that such a proof exists. These internal closure properties are the engine of Löb’s theorem and Gödel II.

Theorem 19.1 (First Hilbert–Bernays–Löb derivability condition). *Let $T \supseteq I\Sigma_1$ be c.e. and let $\text{Prf}_T(p, q)$ be the canonical primitive-recursive proof predicate fixed in Volume II. For every sentence φ ,*

$$T \vdash \varphi \implies T \vdash \text{Prov}_T(\ulcorner \varphi \urcorner), \quad \text{Prov}_T(q) := \exists p \text{Prf}_T(p, q).$$

This is derivability condition D1.

How to read the symbols: D1, D2, and D3 are three different programs on proof codes

D1 turns an externally known proof of A into an internal proof that A is provable. D2 represents the primitive-recursive program that concatenates a proof of $A \rightarrow B$ with a proof of A and appends modus ponens. D3 represents the program that takes a proof of A and produces a proof of the existential sentence asserting that such a proof exists. Remembering the underlying proof-code programs makes the three formulas much less mysterious.

Proof.

Formal derivation spine

Formal spine: D1 inserts one concrete proof code. If n is the code of an actual T -proof of φ , then

$$T \vdash \text{Prf}_T(\bar{n}, \ulcorner \varphi \urcorner) \implies T \vdash \exists p \text{Prf}_T(p, \ulcorner \varphi \urcorner).$$

Assume externally that $T \vdash \varphi$. A formal proof is a finite object, so choose its actual natural-number code n . The relation $\text{Prf}_T(p, q)$ was chosen to represent the primitive-recursive proof-checking relation. Consequently the true closed computation “ n is a T -proof of φ ” is verified inside the arithmetical base:

$$T \vdash \text{Prf}_T(\bar{n}, \ulcorner \varphi \urcorner). \quad (1)$$

Existential introduction applied to (1) gives

$$T \vdash \exists p \text{Prf}_T(p, \ulcorner \varphi \urcorner),$$

which is exactly

$$T \vdash \text{Prov}_T(\ulcorner \varphi \urcorner).$$

The implication in D1 is metamathematical: it begins with an externally known derivation and uses its concrete code. D1 does not assert the internal formula $\varphi \rightarrow \text{Prov}_T(\ulcorner \varphi \urcorner)$ for arbitrary φ . $\square$

Theorem 19.2 (Second derivability condition). *For all sentences φ, ψ , the canonical provability predicate satisfies*

$$T \vdash \text{Prov}_T(\ulcorner \varphi \rightarrow \psi \urcorner) \rightarrow (\text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \text{Prov}_T(\ulcorner \psi \urcorner)).$$

This is derivability condition (D2).

Proof.

Formal derivation spine

Formal spine: D2 represents the modus-ponens program. There is a primitive-recursive function c such that

$$\text{Prf}_T(p, \ulcorner \varphi \rightarrow \psi \urcorner) \wedge \text{Prf}_T(q, \ulcorner \varphi \urcorner) \rightarrow \text{Prf}_T(c(p, q), \ulcorner \psi \urcorner). \quad (1)$$

Existentially quantify the two input proof codes.

Given coded proofs p of $\varphi \rightarrow \psi$ and q of φ , form a new coded proof by concatenating the two proof sequences and appending one modus-ponens line with conclusion ψ . Sequence concatenation, index shifting, and the addition of the last inference line are primitive recursive. Hence the code-transformer $c(p, q)$ is primitive recursive.

The proof checker verifies the transformed sequence uniformly, so the arithmetical base proves (1). Now reason inside T . Suppose

$$\text{Prov}_T(\ulcorner \varphi \rightarrow \psi \urcorner) \quad \text{and} \quad \text{Prov}_T(\ulcorner \varphi \urcorner).$$

Unfolding Prov_T , choose proof-code witnesses p, q . Formula (1) gives

$$\text{Prf}_T(c(p, q), \ulcorner \psi \urcorner),$$

and existential introduction yields $\text{Prov}_T(\ulcorner \psi \urcorner)$. Discharging the two assumptions by propositional logic gives

$$T \vdash \text{Prov}_T(\ulcorner \varphi \rightarrow \psi \urcorner) \rightarrow (\text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \text{Prov}_T(\ulcorner \psi \urcorner)).$$

This is D2. Its content is precisely the internal representability of one syntactic operation: combining two proofs by modus ponens. $\square$

Theorem 19.3 (Third derivability condition). *For every sentence φ , the canonical provability predicate satisfies*

$$T \vdash \text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \text{Prov}_T(\ulcorner \text{Prov}_T(\ulcorner \varphi \urcorner) \urcorner).$$

This is derivability condition (D3).

Proof.

Formal derivation spine

Formal spine: D3 verifies a proof certificate inside another proof. There is a primitive-recursive function v such that

$$\text{Prf}_T(p, \ulcorner \varphi \urcorner) \rightarrow \text{Prf}_T(v(p), \ulcorner \text{Prov}_T(\ulcorner \varphi \urcorner) \urcorner). \quad (1)$$

Existentially quantify p .

Fix a code p of a T -proof of φ . Because proof checking is primitive recursive, there is an effective finite derivation in the base arithmetic of the closed statement

$$\text{Prf}_T(\bar{p}, \ulcorner \varphi \urcorner).$$

Append existential introduction to obtain

$$\exists r \text{Prf}_T(r, \ulcorner \varphi \urcorner),$$

namely $\text{Prov}_T(\ulcorner \varphi \urcorner)$. Coding this uniformly defines a primitive-recursive transformation $v(p)$. Representability of the transformation gives (1) inside T .

Now suppose $\text{Prov}_T(\ulcorner \varphi \urcorner)$, so some p satisfies $\text{Prf}_T(p, \ulcorner \varphi \urcorner)$. Formula (1) supplies a proof code $v(p)$ for the provability sentence. Therefore

$$T \vdash \text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \text{Prov}_T(\ulcorner \text{Prov}_T(\ulcorner \varphi \urcorner) \urcorner).$$

Unlike D1, D3 is wholly internal: the antecedent itself asserts the existence of a proof code, and the primitive-recursive transformer converts any such witness into a proof of that existential assertion. $\square$

Lemma 19.4 (Formalized fixed-point argument for provability predicates). *Let $T \supseteq I\Sigma_1$ have the canonical provability predicate satisfying D1–D3, and suppose*

$$T \vdash \theta \leftrightarrow \chi(\ulcorner \theta \urcorner).$$

Then the equivalence can be internalized at the provability level:

$$T \vdash \text{Prov}_T(\ulcorner \theta \urcorner) \leftrightarrow \text{Prov}_T(\ulcorner \chi(\ulcorner \theta \urcorner) \urcorner).$$

When χ itself contains Prov_T , D2 and D3 may therefore be applied wholly inside T to the fixed-point equivalence.

Proof.

Formal derivation spine

Formal spine for Formalized fixed-point argument for provability predicates.

$$T \vdash \theta \leftrightarrow \chi(\ulcorner \theta \urcorner). \quad (\text{F1})$$

$$\begin{array}{c}
T \vdash \text{Prov}_T(\ulcorner \theta \rightarrow \chi(\ulcorner \theta \urcorner) \urcorner) \\
T \vdash \text{Prov}_T(\ulcorner \chi(\ulcorner \theta \urcorner) \rightarrow \theta \urcorner).
\end{array}$$

Let

$$T \vdash \theta \leftrightarrow \chi(\ulcorner \theta \urcorner). \quad (\text{F1})$$

Split (F1) into the two implications. By D1,

$$T \vdash \text{Prov}_T(\ulcorner \theta \rightarrow \chi(\ulcorner \theta \urcorner) \urcorner)$$

and

$$T \vdash \text{Prov}_T(\ulcorner \chi(\ulcorner \theta \urcorner) \rightarrow \theta \urcorner).$$

Applying D2 to these theorems gives

$$T \vdash \text{Prov}_T(\ulcorner \theta \urcorner) \rightarrow \text{Prov}_T(\ulcorner \chi(\ulcorner \theta \urcorner) \urcorner) \quad (\text{F2})$$

and the reverse implication. Hence

$$T \vdash \text{Prov}_T(\ulcorner \theta \urcorner) \leftrightarrow \text{Prov}_T(\ulcorner \chi(\ulcorner \theta \urcorner) \urcorner).$$

If χ itself contains Prov_T , D3 may now be applied to the provability terms appearing in (F2). This is the precise sense in which the fixed-point calculation has been internalized. *Further explanation.* The formalized fixed-point step must itself be carried out inside the arithmetic theory. One first proves that the syntactic transformation sending a code to the code of its diagonal instance is primitive recursive and hence represented. The theory can then reason about the code transformation and derive the fixed-point biconditional, rather than relying on an external meta-level substitution. This distinction is essential in the proof of Löb's theorem, where provability assertions about the fixed point are manipulated internally. $\square$

Worked example

Worked Example 19.5 (Combining proof codes). Given a code p for a proof of $A \rightarrow B$ and a code q for a proof of A , there is a primitive-recursive function $m(p, q)$ coding the concatenated derivation ending with B by modus ponens. Representing m inside arithmetic yields the second derivability condition for Prov_T .

A useful warning

The derivability conditions apply to the chosen intensional proof predicate. An extensionally equivalent formula for theoremhood need not satisfy them.

Looking ahead

The next chapter, *Löb and second incompleteness*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 20

Löb and second incompleteness

Definitions and setup

Formal definitions used in this chapter

Canonical consistency sentence. For a theory T with the fixed canonical provability predicate,

$$\text{Con}(T) := \neg \text{Prov}_T(\ulcorner 0 = 1 \urcorner).$$

It says, arithmetically, that no natural number codes a T -proof of the chosen contradiction $0 = 1$.

Löb premise. A sentence φ satisfies the Löb premise over T when

$$T \vdash \text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi.$$

Theory class used for Gödel II. Unless a result states otherwise, the second-incompleteness arguments in this chapter assume that T is a consistent c.e. extension of $I\Sigma_1$ equipped with the canonical proof predicate satisfying D1–D3.

Löb's theorem concerns sentences A for which the theory proves $\text{Prov}(\ulcorner A \urcorner) \rightarrow A$. The second incompleteness theorem is obtained by applying the same derivability machinery to the canonical consistency sentence $\text{Con}(T) = \neg \text{Prov}_T(\ulcorner 0 = 1 \urcorner)$. We fix one canonical proof predicate for the whole argument; changing the coding by an intensionally unnatural definition can change which internal consistency sentence is obtained.

How this chapter fits together

Löb's theorem turns an internal reflection implication $\text{Prov}_T(\ulcorner A \urcorner) \rightarrow A$ into a proof of A . Gödel II is the special case obtained by encoding consistency as the absence of a proof of contradiction. The proof is a formalized self-reference calculation, so every use of Prov_T must remain inside the theory until the final external consistency argument.

Theorem 20.1 (Löb's theorem). *Let T be a c.e. extension of $I\Sigma_1$ whose canonical provability predicate satisfies D1–D3. For every sentence φ , if*

$$T \vdash \text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi,$$

then $T \vdash \varphi$.

Uses: the diagonal lemma and derivability conditions D1–D3.

Commented symbolic trace: Löb's theorem without skipping the bookkeeping

Track three sentences: the target A , the fixed point B , and the provability statements about them. The fixed point is chosen so that $B \leftrightarrow (\text{Prov}(B) \rightarrow A)$. D2 moves implications through Prov ; D3 supplies $\text{Prov}(B) \rightarrow \text{Prov}(\text{Prov}(B))$; the assumed reflection implication $\text{Prov}(A) \rightarrow A$ removes the last provability operator. Only after these internal steps do we use D1 to turn the resulting proof of B into $\text{Prov}(B)$. Reading the proof as a pipeline prevents the nested Prov 's from becoming visual noise.

Proof.

Formal derivation spine

Formal spine for Löb's theorem.

$$T \vdash B \leftrightarrow (\text{Prov}(\ulcorner B \urcorner) \rightarrow A).$$

$$T \vdash B \leftrightarrow (\text{Prov}_T(\ulcorner B \urcorner) \rightarrow \varphi). \quad (\text{L1})$$

$$T \vdash \text{Prov}_T(\ulcorner B \urcorner) \rightarrow \text{Prov}_T(\ulcorner \text{Prov}_T(\ulcorner B \urcorner) \urcorner \rightarrow \varphi \urcorner). \quad (\text{L2})$$

Proof map. Keep the two levels visible. The diagonal lemma produces B inside arithmetic. The derivability conditions then allow T to reason about proofs of the implications involving B . The final use of the first derivability condition occurs only after $T \vdash B$ has been obtained externally from the preceding derivation.

Let $T \vdash \text{Prov}(\ulcorner A \urcorner) \rightarrow A$. By the diagonal lemma choose B with

$$T \vdash B \leftrightarrow (\text{Prov}(\ulcorner B \urcorner) \rightarrow A).$$

Using the derivability conditions, formalize the implication $B \rightarrow (\text{Prov}(B) \rightarrow A)$ to obtain $\text{Prov}(B) \rightarrow \text{Prov}(\text{Prov}(B) \rightarrow A)$, and hence $\text{Prov}(B) \rightarrow (\text{Prov}(\text{Prov}(B)) \rightarrow \text{Prov}(A))$. The third derivability condition gives $\text{Prov}(B) \rightarrow \text{Prov}(\text{Prov}(B))$, so $T \vdash \text{Prov}(B) \rightarrow \text{Prov}(A)$. Combining the assumed $\text{Prov}(A) \rightarrow A$ yields $T \vdash \text{Prov}(B) \rightarrow A$, hence $T \vdash B$ by the fixed point. The first derivability condition gives $T \vdash \text{Prov}(B)$, and therefore $T \vdash A$.

Detailed formal verification. For completeness, write the Löb calculation as a derivation inside T . Let B satisfy

$$T \vdash B \leftrightarrow (\text{Prov}_T(\ulcorner B \urcorner) \rightarrow \varphi). \quad (\text{L1})$$

By D1 applied to the forward implication in (L1), and then D2,

$$T \vdash \text{Prov}_T(\ulcorner B \urcorner) \rightarrow \text{Prov}_T(\ulcorner \text{Prov}_T(\ulcorner B \urcorner) \urcorner \rightarrow \varphi \urcorner). \quad (\text{L2})$$

D2 again yields

$$T \vdash \text{Prov}_T(\ulcorner B \urcorner) \rightarrow (\text{Prov}_T(\ulcorner \text{Prov}_T(\ulcorner B \urcorner) \urcorner) \rightarrow \text{Prov}_T(\ulcorner \varphi \urcorner)). \quad (\text{L3})$$

D3 supplies the middle premise, hence

$$T \vdash \text{Prov}_T(\ulcorner B \urcorner) \rightarrow \text{Prov}_T(\ulcorner \varphi \urcorner). \quad (\text{L4})$$

Using the hypothesis $T \vdash \text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi$, we obtain

$$T \vdash \text{Prov}_T(\ulcorner B \urcorner) \rightarrow \varphi.$$

By (L1), $T \vdash B$; D1 gives $T \vdash \text{Prov}_T(\ulcorner B \urcorner)$, and modus ponens finally gives $T \vdash \varphi$. Every displayed implication is internal to T . $\square$

Remark 20.2 (Source note). This proof follows the standard derivability-condition proof of Löb's theorem; compare the original note [6].

Theorem 20.3 (Gödel’s second incompleteness theorem for c.e. extensions of $I\Sigma_1$). *Let T be a consistent c.e. extension of $I\Sigma_1$ with the canonical provability predicate. Writing*

$$\text{Con}(T) := \neg \text{Prov}_T(\ulcorner 0 = 1 \urcorner),$$

one has

$$T \not\vdash \text{Con}(T).$$

Uses: the formalized provability calculus, Löb’s theorem, and the consistency hypothesis on the chosen theory.

How to read the symbols: Gödel II is about one fixed consistency sentence

Here $\text{Con}(T)$ abbreviates $\neg \text{Prov}_T(\ulcorner 0 = 1 \urcorner)$ for the canonical provability predicate already constructed; $0 = 1$ is the fixed arithmetic contradiction chosen in the definition above. The theorem is not the vague statement that “a theory cannot know it is consistent.” It says that, under the stated effectiveness and strength hypotheses, a consistent T does not prove this particular arithmetical sentence expressing absence of a coded T -proof of contradiction.

Proof.

Formal derivation spine

Formal spine: derivability conditions turn consistency into the Gödel sentence. Let $P := \text{Prov}_T(\ulcorner G_T \urcorner)$ and $C := \text{Prov}_T(\ulcorner 0 = 1 \urcorner)$. Then

$$T \vdash P \rightarrow \text{Prov}_T(\ulcorner \neg P \urcorner), \quad T \vdash P \rightarrow \text{Prov}_T(\ulcorner P \urcorner),$$

so D2 gives $T \vdash P \rightarrow C$. Hence

$$T \vdash \neg C \rightarrow \neg P \rightarrow G_T.$$

Let $G = G_T$ and abbreviate

$$P := \text{Prov}_T(\ulcorner G \urcorner), \quad C := \text{Prov}_T(\ulcorner 0 = 1 \urcorner).$$

The fixed-point equivalence yields

$$T \vdash G \leftrightarrow \neg P. \tag{1}$$

In particular, $T \vdash G \rightarrow \neg P$. By D1, T proves the provability of this implication; D2 then internalizes modus ponens and gives

$$T \vdash P \rightarrow \text{Prov}_T(\ulcorner \neg P \urcorner). \tag{2}$$

D3 gives

$$T \vdash P \rightarrow \text{Prov}_T(\ulcorner P \urcorner). \tag{3}$$

The propositional tautology $P \rightarrow (\neg P \rightarrow 0 = 1)$ has a T -proof. Applying D1 and D2 twice shows

$$T \vdash \text{Prov}_T(\ulcorner P \urcorner) \rightarrow (\text{Prov}_T(\ulcorner \neg P \urcorner) \rightarrow C). \tag{4}$$

Combining (2), (3), and (4), we obtain

$$T \vdash P \rightarrow C. \tag{5}$$

Classical contraposition inside T gives

$$T \vdash \neg C \rightarrow \neg P. \tag{6}$$

By the reverse direction of (1), $T \vdash \neg P \rightarrow G$, and therefore

$$T \vdash \text{Con}(T) \rightarrow G, \quad \text{Con}(T) := \neg \text{Prov}_T(\ulcorner 0 = 1 \urcorner). \tag{7}$$

If the consistent theory T proved $\text{Con}(T)$, then (7) would give $T \vdash G$, contradicting the already proved theorem that consistency implies $T \not\vdash G$. Hence $T \not\vdash \text{Con}(T)$. Every internal step is licensed by D1–D3 and propositional logic; no semantic soundness assumption is hidden in the argument. $\square$

Remark 20.4 (Source note). The second incompleteness theorem is tied to the canonical intensional provability predicate developed in this volume; compare [4, 6].

A first separation consequence. The following proposition was deferred from the earlier comparison of soundness notions because its proof depends on Gödel II.

Proposition 20.5 (Strictness/counterexample analysis for the consistency notions). *Let $T \supseteq I\Sigma_1$ be c.e. and consistent. Then*

$$T^* := T + \neg\text{Con}(T)$$

is consistent but not Σ_1 -sound: consistency follows because otherwise $T \vdash \text{Con}(T)$, contrary to Gödel II, while $\neg\text{Con}(T)$ is a false Σ_1 -sentence in $\mathbb{N}$. Consequently ordinary consistency does not imply Σ_1 -soundness (equivalently, 1-consistency in the terminology fixed here). The other nonreverse implications discussed below are separated by similarly explicit examples.

Proof.

Formal derivation spine

Formal spine for Strictness/counterexample analysis for the consistency notions.

$$T \not\vdash \text{Con}(T).$$

$$T \vdash \text{Con}(T),$$

$$T^* := T + \neg\text{Con}(T)$$

The nonreverse implication from consistency to Σ_1 -soundness has a concrete witness. Let $T \supseteq I\Sigma_1$ be consistent and c.e. Gödel II gives

$$T \not\vdash \text{Con}(T).$$

If $T + \neg\text{Con}(T)$ were inconsistent, the deduction theorem would yield

$$T \vdash \text{Con}(T),$$

a contradiction. Hence

$$T^* := T + \neg\text{Con}(T)$$

is consistent. But because T is actually consistent, the standard model satisfies

$$\mathbb{N} \models \text{Con}(T),$$

so

$$\mathbb{N} \models \neg(\neg\text{Con}(T)).$$

The sentence $\neg\text{Con}(T) \equiv \text{Prov}_T(\ulcorner 0 = 1 \urcorner)$ is Σ_1 , and T^* proves it. Therefore T^* is not Σ_1 -sound. This explicitly separates ordinary consistency from 1-consistency/ Σ_1 -soundness. *Point specific to this result.* The implication diagram should not be read as a chain of equivalences. Consistent but unsound extensions supply the basic counterexamples, while separating 1- from ω -consistency requires attention to which existential matrices are decidable in the base theory. $\square$

Corollary 20.6 (PA does not prove $\text{Con}(\text{PA})$, assuming consistency). *If PA is consistent, then*

$$\text{PA} \not\vdash \text{Con}(\text{PA}).$$

Proof.

Formal derivation spine

Formal spine: specialize the preceding theorem to PA.

$$PA \supseteq I\Sigma_1, \quad PA \text{ is c.e.}, \quad \text{Prf}_{PA} \text{ satisfies D1–D3.}$$

Peano arithmetic is a computably enumerable extension of $I\Sigma_1$, and the canonical proof predicate fixed in Volume II satisfies the three derivability conditions proved above. Therefore all hypotheses of Gödel's second incompleteness theorem apply with $T = PA$. If PA is consistent, the conclusion is

$$PA \not\vdash \text{Con}(PA), \quad \text{Con}(PA) := \neg \text{Prov}_{PA}(\ulcorner 0 = 1 \urcorner).$$

This corollary does not assert that $\text{Con}(PA)$ is false. Under the external assumption that PA is consistent, $\text{Con}(PA)$ is true in the standard model; the theorem says only that the consistent theory cannot prove its own canonical consistency sentence. $\square$

How to read the symbols: set-theoretic instances are deferred

The same Gödel-II mechanism applies to recursively presented set theories once their axioms and internal natural-number arithmetic have been constructed. ZF and ZFC have not yet been defined in the series, so their formal corollary is deliberately postponed to Volume V, after those theories are available.

Worked example

Worked Example 20.7 (Why proving consistency would be too much). Suppose a consistent T satisfying the derivability conditions proved $\text{Con}(T)$. The formalized incompleteness argument proves inside T that $\text{Con}(T) \rightarrow G_T$; hence T would prove G_T . First incompleteness says a consistent T cannot do so. The contradiction is external, but the implication from consistency to the Gödel sentence is internal.

A useful warning

Gödel II is not the assertion that no one can prove $\text{Con}(T)$. A stronger metatheory may prove it; the theorem says that, under the stated hypotheses, T cannot prove its own canonical consistency statement.

Looking ahead

The next chapter, *Truth, reflection, and iterated consistency*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 21

Truth, reflection, and iterated consistency

Definitions and setup

Formal definitions used in this chapter

Arithmetical truth predicate. A formula $\text{Tr}(x)$ would define full arithmetical truth when, for every arithmetic sentence σ ,

$$\mathbb{N} \models \text{Tr}(\ulcorner \sigma \urcorner) \leftrightarrow \sigma.$$

The right-hand occurrence of σ is evaluated externally in the standard model.

Metatheory. A metatheory S is a theory in which statements about the coded syntax and proofs of another theory T are represented and reasoned about.

Local reflection over a class Γ . For a class Γ of sentences, the Γ -reflection schema for T over S consists of the individual formulas

$$\text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi \quad (\varphi \in \Gamma).$$

A schema is a family of axioms, one for each indicated sentence, not a single formula quantifying over all sentences.

Theorem set and proper deductive extension. $\text{Thm}(T) = \{\ulcorner \sigma \urcorner : T \vdash \sigma\}$. An extension $U \supseteq T$ is proper as a deductive extension when $\text{Thm}(T) \subsetneq \text{Thm}(U)$.

Finite iteration of consistency. Define

$$T_0 := T, \quad T_{n+1} := T_n + \text{Con}(T_n) \quad (n < \omega).$$

Only this finite iteration is defined here; a transfinite progression would additionally require an ordinal-notation system and an effective limit-stage rule.

Tarski's theorem concerns a single arithmetical predicate purporting to define truth for all arithmetic sentences. A local reflection principle, by contrast, is a schema of implications $\text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi$ for a restricted family. Iterated consistency theories add particular metamathematical sentences as new axioms. These notions are related by diagonalization but are not interchangeable.

How this chapter fits together

Tarski's theorem blocks a single arithmetic predicate that correctly captures truth for all arithmetic sentences. Local reflection avoids this by treating restricted instances or schemes rather than a global internal truth definition. Adding $\text{Con}(T)$ then illustrates a systematic way of strengthening a theory, while the text carefully stops before pretending that arbitrary transfinite progressions require no additional machinery.

Theorem 21.1 (Tarski undefinability of truth). *There is no arithmetic formula $\text{Tr}(x)$ such that for every arithmetic sentence σ ,*

$$\mathbb{N} \models \text{Tr}(\ulcorner \sigma \urcorner) \leftrightarrow \sigma.$$

Thus arithmetical truth is not arithmetically definable in the standard model.

Proof.

Formal derivation spine

Formal spine: diagonalize against the proposed truth predicate.

$$L \leftrightarrow \neg \text{Tr}(\ulcorner L \urcorner), \quad \text{Tr}(\ulcorner L \urcorner) \leftrightarrow L \implies L \leftrightarrow \neg L.$$

Suppose an arithmetic formula $\text{Tr}(x)$ satisfies the full Tarski biconditional for every arithmetic sentence σ :

$$\mathbb{N} \models \text{Tr}(\ulcorner \sigma \urcorner) \leftrightarrow \sigma. \tag{1}$$

Apply the diagonal lemma to the formula $\neg \text{Tr}(x)$. There is a sentence L such that the weak arithmetic used for diagonalization proves

$$L \leftrightarrow \neg \text{Tr}(\ulcorner L \urcorner). \tag{2}$$

Since that weak arithmetic is true in $\mathbb{N}$, (2) holds in $\mathbb{N}$. The instance of (1) for the sentence L gives

$$\mathbb{N} \models \text{Tr}(\ulcorner L \urcorner) \leftrightarrow L. \tag{3}$$

Combining (2) and (3) in the standard two-valued semantics yields

$$\mathbb{N} \models L \leftrightarrow \neg L,$$

which is impossible. Therefore no arithmetic formula defines the set of Gödel codes of all true arithmetic sentences.

The theorem does not prohibit partial truth predicates for bounded formula complexity, nor the satisfaction relation for a fixed set-sized structure developed in Volume V. Those predicates satisfy biconditionals only for a restricted syntactic class; diagonalization produces a sentence outside the simultaneously covered class. $\square$

Remark 21.2 (Source note). The undefinability argument is the diagonal core of Tarski's theorem on truth for formalized languages; compare [7].

Proposition 21.3 (Local reflection implies the corresponding instances of soundness). *Let S be a metatheory in which Prf_T is represented and which verifies the D1 proof-certification property*

$$T \vdash \psi \implies S \vdash \text{Prov}_T(\ulcorner \psi \urcorner).$$

Let Γ be a class of sentences. If for every $\varphi \in \Gamma$,

$$S \vdash \text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi,$$

then for every $\varphi \in \Gamma$,

$$T \vdash \varphi \implies S \vdash \varphi.$$

Proof.

Formal derivation spine

Fix $\varphi \in \Gamma$. The two hypotheses give the implication chain

$$T \vdash \varphi \implies S \vdash \text{Prov}_T(\ulcorner \varphi \urcorner) \implies S \vdash \varphi.$$

The first arrow is D1 as verified in S ; the second is modus ponens with the local reflection instance.

Let $\varphi \in \Gamma$ and assume externally that $T \vdash \varphi$. By the stated proof-certification property for S ,

$$S \vdash \text{Prov}_T(\ulcorner \varphi \urcorner).$$

By hypothesis, S also proves the corresponding reflection instance

$$S \vdash \text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi.$$

One application of modus ponens in S therefore yields

$$S \vdash \varphi.$$

Since φ was arbitrary in Γ , this proves the stated external Γ -soundness conclusion relative to the metatheory S .

The distinction between the three levels is essential. The assumption $T \vdash \varphi$ is made in the surrounding metatheory. The formula $\text{Prov}_T(\ulcorner \varphi \urcorner)$ is an arithmetical sentence inside S . The implication $\text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi$ is one member of a reflection schema, not a single global truth predicate. Löb's theorem explains why a consistent theory T satisfying the Hilbert–Bernays–Löb conditions cannot prove all of its own unrestricted reflection instances: if

$$T \vdash \text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi,$$

then Löb gives $T \vdash \varphi$. Thus local reflection is used only with the explicit metatheoretic hypotheses displayed in the statement above. $\square$

Theorem 21.4 (Strict strengthening by adjoining a consistency statement). *If T is consistent and satisfies the hypotheses of Gödel's second incompleteness theorem, then*

$$T \not\vdash \text{Con}(T).$$

Consequently $T + \text{Con}(T)$ is a proper deductive extension of T : it proves $\text{Con}(T)$, which T does not. This properness assertion does not by itself imply that the extension is consistent. If, in addition, the standard model satisfies every axiom of T , then $\mathbb{N} \models T + \text{Con}(T)$, and the extension is consistent.

Proof.

Formal derivation spine

Formal spine: nonderivability already gives a proper extension.

$$T \not\vdash \text{Con}(T), \quad T + \text{Con}(T) \vdash \text{Con}(T),$$

so

$$\text{Thm}(T) \subsetneq \text{Thm}(T + \text{Con}(T)).$$

Gödel's second incompleteness theorem gives

$$T \not\vdash \text{Con}(T). \tag{1}$$

Every theorem of T remains a theorem after the new axiom is added, so

$$\text{Thm}(T) \subseteq \text{Thm}(T + \text{Con}(T)).$$

The extension proves $\text{Con}(T)$ by the added axiom, while (1) says that T does not. Hence the inclusion is strict. This conclusion is valid whether or not the extension is consistent; an inconsistent extension is still deductively stronger.

Consistency is a separate question. If $\mathbb{N} \models T$, then no standard number codes a T -proof of contradiction, so

$$\mathbb{N} \models \text{Con}(T).$$

Thus $\mathbb{N} \models T + \text{Con}(T)$, and soundness implies that the extension is consistent. Without such a model or soundness hypothesis, consistency of T alone need not imply consistency of $T + \text{Con}(T)$: a consistent theory can prove the negation of its own canonical consistency sentence. $\square$

Proposition 21.5 (Finite and transfinite-looking iterated consistency schemes: what is proved and what is only previewed). *Define $T_0 = T$ and $T_{n+1} = T_n + \text{Con}(T_n)$. Conditional on consistency of each T_n , the inclusions $T_n \subsetneq T_{n+1}$ are strict. No claim about transfinite progressions is made without an additional ordinal-notation and recursion apparatus.*

Proof.

Formal derivation spine

Formal spine for Finite and transfinite-looking iterated consistency schemes: what is proved and what is only previewed.

$$T_0 := T, \quad T_{n+1} := T_n + \text{Con}(T_n).$$

$$T_n \not\models \text{Con}(T_n),$$

$$T_{n+1} \vdash \text{Con}(T_n).$$

Define

$$T_0 := T, \quad T_{n+1} := T_n + \text{Con}(T_n).$$

Assume inductively that each T_n is consistent and satisfies the Gödel-II hypotheses. Then

$$T_n \not\models \text{Con}(T_n),$$

while

$$T_{n+1} \vdash \text{Con}(T_n).$$

Hence

$$T_n \subsetneq T_{n+1} \quad (n < \omega).$$

This proves only the finite progression indexed by ordinary natural numbers. A transfinite progression would require, in addition, a system of ordinal notations O , an effective rule for limit stages, and a definition such as

$$T_\lambda := \bigcup_{\beta <_O \lambda} T_\beta.$$

None of that machinery has been introduced here, so no transfinite theorem is claimed. *Point specific to this result.* The finite progression $T_{n+1} = T_n + \text{Con}(T_n)$ is justified stage by stage. Extending this construction through arbitrary ordinals would require a separate effective ordinal-notation system, so no such transfinite result is being smuggled into the present statement. $\square$

Worked example

Worked Example 21.6 (Truth versus a partial satisfaction predicate). For each fixed formula-complexity class, arithmetic can define a satisfaction predicate for formulas in that class. Tarski's argument rules out one arithmetic predicate satisfying all truth biconditionals at once. The difference explains why bounded truth definitions are legitimate tools while a global internal truth predicate is impossible.

A useful warning

A scheme consisting of many formulas is not automatically equivalent to one formula asserting all of them. This distinction is crucial for reflection and truth.

End-of-volume perspective

The incompleteness results delimit formal proof without undermining ordinary mathematical reasoning: stronger theories can prove consistency statements unavailable to weaker ones, but every consistent computably enumerable theory satisfying the hypotheses of Gödel's second incompleteness theorem fails to prove its own canonical consistency sentence.

Part IV

Volume IV: ZF Set Theory and the Transfinite

Preface

Set theory is developed without silently assuming Choice. Ordinal recursion, rank, Hartogs' theorem, and collapse are the infrastructure for both constructibility and forcing.

Position in the series

Volume IV builds the ZF and transfinite infrastructure shared by constructibility and forcing: rank, ordinals, cardinal machinery without hidden Choice, and Mostowski collapse.

Sources and further reading

The set-theoretic development follows standard ZF practice; for broader context and comparison of conventions see [13, 14].

Notation for this volume

Ordinals are Greek letters; V_α will denote the cumulative hierarchy once it is defined in Chapter 3. Aleph notation is introduced by the recursion theorem in Chapter 6; beth notation is postponed to Volume V, after the Axiom of Choice has been defined.

How to read symbolic arguments in this volume

When a formula appears, read it in layers rather than as one visual block. First identify the *ambient language*: are we speaking inside a formal theory, about a structure, or in the surrounding metatheory? Next mark the objects being manipulated (formulas, codes, sets, names, conditions, ordinals) and the operation being applied to them. Finally check the side conditions—free variables, rank bounds, compatibility, genericity, transitivity, or consistency hypotheses. Whenever a displayed derivation changes level, the text says so explicitly. A displayed line is therefore meant to be read like a line of well-commented code: the symbols perform the operation, and the surrounding prose records its type, preconditions, and purpose.

Chapter 22

ZF axioms and elementary constructions

Definitions and setup

Formal definitions used in this chapter

Logical and set-theoretic abbreviations. Bounded quantifiers are abbreviations

$$\forall x \in a \varphi := \forall x (x \in a \rightarrow \varphi), \quad \exists x \in a \varphi := \exists x (x \in a \wedge \varphi).$$

Unique existence is

$$\exists! y \varphi(y) := \exists y (\varphi(y) \wedge \forall z (\varphi(z) \rightarrow z = y)),$$

with bound variables renamed first when necessary to avoid capture. Write $x \subseteq y$ for $\forall z (z \in x \rightarrow z \in y)$. The symbols $\emptyset, \{x, y\}, \bigcup x, \mathcal{P}(x)$ denote, when they exist, respectively the set with no elements, the set whose elements are exactly x, y , the set of all members of members of x , and the set of all subsets of x . Put

$$\{x\} := \{x, x\}, \quad x \cup y := \bigcup \{x, y\}, \quad x \setminus y := \{z \in x : z \notin y\}, \quad S(x) := x \cup \{x\}.$$

The difference notation is used only after Separation has supplied the indicated subset.

The axioms of ZF. ZF is the first-order theory in the language $\{\in\}$ consisting of the following axioms and schemes (all displayed free parameters are universally quantified):

Extensionality: $\forall x \forall y [\forall z (z \in x \leftrightarrow z \in y) \rightarrow x = y],$

Empty Set: $\exists x \forall z (z \notin x),$

Pairing: $\forall x \forall y \exists p \forall z [z \in p \leftrightarrow (z = x \vee z = y)],$

Union: $\forall x \exists u \forall z [z \in u \leftrightarrow \exists y \in x (z \in y)],$

Power Set: $\forall x \exists p \forall z [z \in p \leftrightarrow z \subseteq x],$

Infinity:
$$\exists I \left[\exists e (\forall z (z \notin e \wedge e \in I)) \right. \\ \left. \wedge \forall x \in I \exists s (s \in I \wedge \forall z [z \in s \leftrightarrow (z \in x \vee z = x)]) \right],$$

Foundation: $\forall x \left[(\exists z (z \in x) \rightarrow \exists y \in x \forall z (z \in x \rightarrow z \notin y)) \right].$

The displayed Infinity and Foundation lines are literal formulas of the language $\{\in\}$: in particular they do not treat $\emptyset$ or $S(x)$ as primitive terms. After Empty Set, Pairing, and Union are available, the

shorter notations $\emptyset \in I$, $S(x) \in I$, and $x \neq \emptyset$ are provably equivalent reader-level abbreviations for these pure-membership formulas. For every formula $\varphi(x, \bar{p})$, its *Separation instance* is

$$\forall a \exists b \forall x [x \in b \leftrightarrow (x \in a \wedge \varphi(x, \bar{p}))].$$

For every formula $\varphi(x, y, \bar{p})$, its *Replacement instance* is

$$\forall a \left[\forall x \in a \exists! y \varphi(x, y, \bar{p}) \rightarrow \exists b \forall y [y \in b \leftrightarrow \exists x \in a \varphi(x, y, \bar{p})] \right].$$

A *scheme* is the family of all such instances as the indicated formula varies.

Ordered pair and Cartesian product. The Kuratowski ordered pair is

$$\langle x, y \rangle := \{\{x\}, \{x, y\}\}.$$

Pairing constructs this set. The *Cartesian product* is the set

$$A \times B := \{u \in \mathcal{P}(\mathcal{P}(A \cup B)) : \exists a \in A \exists b \in B (u = \langle a, b \rangle)\}.$$

Indeed every Kuratowski pair $\langle a, b \rangle$ with $a \in A, b \in B$ lies in $\mathcal{P}(\mathcal{P}(A \cup B))$, so Power Set plus Separation supplies exactly the displayed product set. **Relation and function.** A relation from A to B is a set $R \subseteq A \times B$. A function $f : A \rightarrow B$ is such a relation satisfying

$$\forall x \in A \exists! y \in B (\langle x, y \rangle \in f).$$

Its domain and range are the Separation subsets

$$\text{dom}(f) = \{x \in \bigcup \bigcup f : \exists y \in \bigcup \bigcup f \langle x, y \rangle \in f\},$$

$$\text{ran}(f) = \{y \in \bigcup \bigcup f : \exists x \in \bigcup \bigcup f \langle x, y \rangle \in f\}.$$

For Kuratowski pairs every coordinate belongs to $\bigcup \bigcup f$, so these bounded definitions recover exactly the intended coordinates and make set existence explicit. The notation $f(x) = y$ abbreviates $\langle x, y \rangle \in f$. If $X \subseteq A$, the *image of X under $f : A \rightarrow B$* is the Separation subset

$$f[X] := \{y \in B : \exists x \in X (f(x) = y)\}.$$

Thus expressions such as $g[f[X]]$ in Cantor–Bernstein denote iterated set images and require no additional choice or recursion.

Injection, surjection, bijection, and equipotence. A function $f : A \rightarrow B$ is injective when $f(x) = f(x') \Rightarrow x = x'$, surjective when $\text{ran}(f) = B$, and bijective when both. Sets A, B are equipotent, written $|A| = |B|$, when some bijection $A \rightarrow B$ exists. We write $|A| \leq |B|$ when some injection $A \rightarrow B$ exists, and $|A| < |B|$ when $|A| \leq |B|$ but $|A| \neq |B|$.

Transitive set. A set M is *transitive* when $x \in y \in M \Rightarrow x \in M$. The transitive-closure operation is deliberately postponed until Chapter 2, after ω and recursion on ω have been constructed.

ZF consists of Extensionality, Empty Set (derivable in common presentations), Pairing, Union, Power Set, Infinity, Separation, Replacement, Foundation, and the usual logical framework. Ordered pairs are Kuratowski pairs. Relations are sets of ordered pairs, and functions are functional relations. Each construction explicitly identifies the axiom that supplies the required set. Choice is not part of ZF and is never imported silently.

How this chapter fits together

The axioms of ZF are now used as construction principles. Ordered pairs, products, relations, and functions must be built as sets; Cantor–Bernstein compares injections without Choice; Cantor’s diagonal theorem proves

that no power set is exhausted by a list; and the notion of transitivity prepares for the transitive-closure construction after ω -recursion is available. These are the set-theoretic primitives used throughout the rest of the series.

Theorem 22.1 (Construction of ordered pairs, Cartesian products, relations, and functions in ZF). *ZF proves the existence of the Kuratowski ordered pair $\langle x, y \rangle = \{\{x\}, \{x, y\}\}$, Cartesian products $A \times B$, relations $R \subseteq A \times B$, and functions represented as functional relations; each construction is a set whenever its parameters are sets.*

Proof. Pairing gives $\{x\}$ and $\{x, y\}$, hence the Kuratowski pair

$$\langle x, y \rangle := \{\{x\}, \{x, y\}\}.$$

Extensionality proves $\langle x, y \rangle = \langle x', y' \rangle$ iff $x = x'$ and $y = y'$. For sets A, B , Replacement applied to each $a \in A$ gives the set $\{\langle a, b \rangle : b \in B\}$; another Replacement followed by Union yields a set containing all such pairs. Separation extracts

$$A \times B = \{z : \exists a \in A \exists b \in B z = \langle a, b \rangle\}.$$

A relation from A to B is simply a subset of $A \times B$, hence a set by Power Set and Separation. A function is a relation $f \subseteq A \times B$ satisfying

$$\forall a \in A \exists! b \in B \langle a, b \rangle \in f.$$

Thus the graph of every function considered here is an ordinary set. Whenever a later argument needs a particular graph operation, its graph is constructed explicitly from the relevant sets by Separation, Replacement, and Union rather than being assumed as an additional primitive operation. □

Theorem 22.2 (Cantor–Bernstein theorem). *If there are injections $f : A \rightarrow B$ and $g : B \rightarrow A$, then there is a bijection $h : A \rightarrow B$.*

Proof. Let $f : A \rightarrow B$ and $g : B \rightarrow A$ be injections. We avoid any appeal to ω or recursion on ω , which is not constructed until the next chapter.

Put

$$A_0 := A \setminus \text{ran}(g).$$

Consider the subsets of A that contain A_0 and are closed under the set operation $X \mapsto g[f[X]]$:

$$\mathcal{C} := \{X \in \mathcal{P}(A) : A_0 \subseteq X \wedge g[f[X]] \subseteq X\}.$$

Power Set and Separation make $\mathcal{C}$ a set; it is nonempty because $A \in \mathcal{C}$. Define its intersection by the bounded formula

$$C := \{a \in A : \forall X \in \mathcal{C} (a \in X)\}. \tag{CB1}$$

Then $A_0 \subseteq C$, and $g[f[C]] \subseteq C$: every $X \in \mathcal{C}$ contains C , hence contains $g[f[C]]$.

In fact

$$C = A_0 \cup g[f[C]]. \tag{CB2}$$

The right-hand side is contained in C by the preceding paragraph. Conversely, let

$$D := A_0 \cup g[f[C]].$$

Then $D \subseteq C$, contains A_0 , and

$$g[f[D]] \subseteq g[f[C]] \subseteq D,$$

so $D \in \mathcal{C}$. Since C is the intersection of all members of $\mathcal{C}$, $C \subseteq D$. This proves (CB2).

Now define a relation $h \subseteq A \times B$ by

$$\langle a, b \rangle \in h \iff (a \in C \wedge b = f(a)) \vee (a \in A \setminus C \wedge g(b) = a). \tag{CB3}$$

Here $f(a) = b$ and $g(b) = a$ are the function-graph abbreviations defined at the start of the chapter. If $a \notin C$, then $a \notin A_0$, so $a \in \text{ran}(g)$; injectivity of g gives a unique b with $g(b) = a$. Thus (CB3) defines a total function $h : A \rightarrow B$.

To prove injectivity, first note that f is injective on C , while the second branch is injective because g is. The two branch images are disjoint. Indeed, if $a \in C$, $a' \notin C$, and

$$f(a) = b, \quad g(b) = a',$$

then $a' = g(f(a)) \in g[f[C]] \subseteq C$, a contradiction.

For surjectivity, take $b \in B$. If $g(b) \notin C$, then the second branch gives $h(g(b)) = b$. If $g(b) \in C$, then $g(b) \notin A_0$, and (CB2) yields some $a \in C$ with

$$g(b) = g(f(a)).$$

Injectivity of g gives $b = f(a) = h(a)$. Hence every $b \in B$ is in the range of h , so h is a bijection.

Dependency point. The familiar iterative proof uses $A_{n+1} = g[f[A_n]]$ and a union over $n < \omega$. The intersection construction (CB1) is equivalent to taking the least subset closed under $X \mapsto g[f[X]]$ containing A_0 , but it requires only Power Set and Separation. Cantor–Bernstein therefore remains available before the book constructs ω and recursion on ω . $\square$

Theorem 22.3 (Cantor’s theorem). *For every set X , no function $f : X \rightarrow \mathcal{P}(X)$ is surjective. Hence*

$$|X| < |\mathcal{P}(X)|$$

in the sense that X injects into $\mathcal{P}(X)$ but the two sets are not equipotent.

Proof. The map $x \mapsto \{x\}$ injects X into $\mathcal{P}(X)$. Suppose, toward contradiction, that $f : X \rightarrow \mathcal{P}(X)$ is surjective. Separation forms the diagonal subset

$$D = \{x \in X : x \notin f(x)\}.$$

Since $D \in \mathcal{P}(X)$, surjectivity gives $d \in X$ with $f(d) = D$. Then

$$d \in D \iff d \notin f(d) \iff d \notin D,$$

a contradiction. Hence no map from X onto $\mathcal{P}(X)$ exists. In particular there is no bijection between X and its power set. No form of Choice has been used: the argument requires only Separation and elementary logic once $\mathcal{P}(X)$ exists. $\square$

Worked example

Worked Example 22.4 (Cantor diagonalization). Given any proposed map $f : X \rightarrow \mathcal{P}(X)$, form $D = \{x \in X : x \notin f(x)\}$. If $D = f(d)$ for some d , then $d \in D$ iff $d \notin D$. Thus D is a concrete subset omitted by the range of f . Notice that no appeal to Choice occurs.

A useful warning

When working in ZF, never replace “there is an injection” by a cardinal inequality that presupposes every set is well-orderable.

Looking ahead

The next chapter, *Natural numbers and recursion*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 23

Natural numbers and recursion

Definitions and setup

Formal definitions used in this chapter

Successor and inductive set. The set-theoretic successor is $S(x) = x \cup \{x\}$. A set I is *inductive* when $0 := \emptyset \in I$ and

$$x \in I \implies S(x) \in I.$$

Least inductive set. An inductive set W is *least inductive* when $W \subseteq I$ for every inductive set I . The next theorem proves that such a set exists and is unique; only then is the symbol ω assigned to it.

Equivalence relation and quotient. An equivalence relation on A is a reflexive, symmetric, transitive relation $E \subseteq A \times A$. For $a \in A$, its equivalence class is the Separation set

$$[a]_E := \{b \in A : bEa\}.$$

The quotient is the Power-Set/Separation construction

$$A/E := \{C \in \mathcal{P}(A) : \exists a \in A (C = [a]_E)\}.$$

Thus the quotient is a set before it is used later. Later quotient constructions use exactly this definition; no later number-system symbol is presupposed here.

A set is inductive when it contains $0 = \emptyset$ and is closed under successor $x \mapsto x \cup \{x\}$. The next theorem constructs and proves uniqueness of a least inductive set; only after that proof do we call it ω . The subsequent recursion theorem then constructs arithmetic operations and finite sequences on that set. Thus the notation and the recursion principle are introduced in dependency order rather than asserted in advance.

How this chapter fits together

Infinity gives an inductive set. The next theorem will intersect all inductive subsets of one Infinity witness, prove that the resulting set is globally least and unique, and only then assign it the symbol ω . That minimality supplies induction; finite approximation then supplies recursion. Arithmetic operations are subsequently defined internally on von Neumann naturals.

Theorem 23.1 (Existence and uniqueness of a least inductive set). *ZF proves that there is a unique least inductive set W . If I is any inductive set supplied by Infinity, this W is the intersection of all inductive subsets of I .*

Proof. Infinity supplies an inductive set I , meaning $0 = \emptyset \in I$ and $n \in I \Rightarrow n \cup \{n\} \in I$. By Separation form

$$W = \{x \in I : \forall J \subseteq I [J \text{ inductive} \Rightarrow x \in J]\}.$$

Equivalently, W is the intersection of all inductive subsets of I . It is itself inductive: every inductive $J \subseteq I$ contains 0, and if all such J contain n , then they contain $n \cup \{n\}$. By definition $W \subseteq J$ for every inductive $J \subseteq I$.

Let K now be any inductive set. Then $I \cap K$ is an inductive subset of I , so $W \subseteq I \cap K \subseteq K$. Hence W is least among all inductive sets, not merely among those lying inside I . If W' is another least inductive set, then $W \subseteq W'$ and $W' \subseteq W$, so Extensionality gives $W = W'$. This proves existence and uniqueness.

Point specific to this result. The intersection is formed inside one Infinity witness I ; intersecting an arbitrary inductive set with I upgrades the local intersection property to global leastness. $\square$

Formal definitions used in this chapter

Natural-number set ω . Let ω denote the unique least inductive set whose existence and uniqueness were proved in Theorem 23.1. Its leastness yields induction: any subset of ω containing 0 and closed under successor is all of ω .

Von Neumann natural numbers and successor notation. The elements of the now-defined set ω are the *von Neumann natural numbers*. For $n \in \omega$, write

$$n + 1 := S(n) = n \cup \{n\}.$$

The later ordinal analysis proves that each such n is the set of its predecessors.

Recursion data on ω . Given a set A , an initial value $a \in A$, and a function $F : A \rightarrow A$, a *recursion solution* is a function $f : \omega \rightarrow A$ satisfying

$$f(0) = a, \quad f(n + 1) = F(f(n)) \quad (n \in \omega).$$

Finite sequence. A *finite sequence over A of length n* is a function $s : n \rightarrow A$ for $n \in \omega$. Write

$$A^n := \{s : s \text{ is a function } n \rightarrow A\}, \quad A^{<\omega} := \bigcup_{n < \omega} A^n.$$

Theorem 23.2 (Recursion theorem on omega). *Let A be a set, $a \in A$, and $F : A \rightarrow A$. There is a unique function $f : \omega \rightarrow A$ satisfying*

$$f(0) = a, \quad f(n + 1) = F(f(n)).$$

The theorem extends uniformly to recursions whose next value depends on n and the preceding value.

Proof. Let A be a set, $a \in A$, and $F : A \rightarrow A$. A finite approximation of length $n + 1$ is a function $s : n + 1 \rightarrow A$ satisfying

$$s(0) = a, \quad s(k + 1) = F(s(k)) \quad (k < n).$$

Induction on n proves existence and uniqueness of such an approximation: extend the unique approximation of length $n + 1$ by assigning $F(s(n))$ at $n + 1$. Replacement collects the approximations, and their union is a function $f : \omega \rightarrow A$ because uniqueness makes them compatible. It satisfies

$$f(0) = a, \quad f(n + 1) = F(f(n)).$$

If g satisfies the same equations, induction on n gives $f(n) = g(n)$ for every n , so $f = g$. The same argument handles recursions whose successor value depends on both n and the previous value.

Point specific to this result. Finite partial recursion graphs are unique on overlapping domains; their compatible union is therefore a function on all of ω . Replacement is used only to collect the finite approximations. $\square$

Formal definitions used in this chapter

Finite union iterates and transitive closure. With recursion on ω now established, for each set x define the sequence $\langle T_n^x : n < \omega \rangle$ by

$$T_0^x := x, \quad T_{n+1}^x := \bigcup T_n^x.$$

Define

$$\text{tc}(x) := x \cup \bigcup_{n < \omega} T_n^x.$$

Thus $\text{tc}(x)$ is a set by Replacement and Union. The next proposition proves that this explicitly constructed set is the least transitive set containing x as a subset.

Proposition 23.3 (Transitive closure exists). *For every set x , the explicitly defined set $\text{tc}(x)$ is transitive, satisfies $x \subseteq \text{tc}(x)$, and is contained in every transitive set M with $x \subseteq M$. Hence it is the unique least transitive set containing x as a subset.*

Proof. By definition,

$$\text{tc}(x) = x \cup \bigcup_{n < \omega} T_n^x, \quad T_0^x = x, \quad T_{n+1}^x = \bigcup T_n^x.$$

If $y \in z \in \text{tc}(x)$, then $z \in T_n^x$ for some $n < \omega$: the case $z \in x$ is exactly $n = 0$. Hence $y \in \bigcup T_n^x = T_{n+1}^x \subseteq \text{tc}(x)$. Thus $\text{tc}(x)$ is transitive. The displayed definition also gives $x = T_0^x \subseteq \text{tc}(x)$.

Let M be any transitive set with $x \subseteq M$. Induction on $n \in \omega$ gives $T_n^x \subseteq M$: the case $n = 0$ is the hypothesis, and transitivity of M sends $T_n^x \subseteq M$ to $\bigcup T_n^x \subseteq M$. Therefore $\text{tc}(x) \subseteq M$. Hence $\text{tc}(x)$ is the least transitive set containing x as a subset. $\square$

Proposition 23.4 (Peano arithmetic operations inside set theory). *There are unique operations $+, \cdot, (-) : \omega \times \omega \rightarrow \omega$ satisfying*

$$\begin{aligned} m + 0 &= m, & m + (n + 1) &= S(m + n), \\ m \cdot 0 &= 0, & m \cdot (n + 1) &= m \cdot n + m, \\ m^0 &= 1, & m^{n+1} &= m^n \cdot m. \end{aligned}$$

ZF proves, by induction on ω , associativity and commutativity of $+$ and $\cdot$, distributivity of multiplication over addition, and the laws $m^0 = 1$ and $m^{n+k} = m^n m^k$.

Proof. Using recursion on ω , define

$$m + 0 = m, \quad m + (n + 1) = S(m + n),$$

and then

$$m \cdot 0 = 0, \quad m \cdot (n + 1) = m \cdot n + m.$$

Exponentiation is defined similarly by $m^0 = 1$ and $m^{n+1} = m^n \cdot m$. The recursion theorem guarantees that these are functions $\omega^2 \rightarrow \omega$. The familiar arithmetic laws follow by induction. For example, associativity of addition is proved by induction on k :

$$(m + n) + 0 = m + n = m + (n + 0),$$

and if $(m + n) + k = m + (n + k)$, then

$$(m + n) + (k + 1) = S((m + n) + k) = S(m + (n + k)) = m + (n + (k + 1)).$$

Thus the set-theoretic naturals carry the standard Peano arithmetic operations constructed entirely inside ZF.

Point specific to this result. Addition and multiplication are instances of recursion on ω : $m + 0 = m$, $m + S(n) = S(m + n)$, and similarly for multiplication. Their familiar algebraic laws are then proved by induction rather than assumed from external arithmetic. $\square$

Proposition 23.5 (Finite sequences and quotient sets in ZF). *For every set A , ZF constructs the set*

$$A^{<\omega} = \bigcup_{n < \omega} A^n$$

of all finite sequences over A . More generally, if E is an equivalence relation on a set X , then the quotient $X/E = \{[x]_E : x \in X\}$ exists as a set.

Proof. A finite sequence from a set A is a function $s : n \rightarrow A$ for some $n \in \omega$. Since functions are sets and $n \times A$ is a set, all such sequences of fixed length form a subset of $\mathcal{P}(n \times A)$; Replacement over $n \in \omega$ and Union collect $A^{<\omega}$.

Now let E be an equivalence relation on a set X . For each $x \in X$, Separation forms

$$[x]_E = \{y \in X : yEx\}.$$

Replacement applied to $x \mapsto [x]_E$ gives the set $\{[x]_E : x \in X\}$; Extensionality automatically identifies repetitions. Hence the quotient X/E is a set. This is the exact quotient mechanism needed later, without introducing further number systems before their own definitions. $\square$

Worked example

Worked Example 23.6 (The first von Neumann naturals). Starting with $0 = \emptyset$ and successor $S(n) = n \cup \{n\}$, we obtain $1 = \{0\}$, $2 = \{0, 1\}$, and $3 = \{0, 1, 2\}$. Thus membership already displays the predecessor pattern $0 \in 1$, $0, 1 \in 2$, and $0, 1, 2 \in 3$. The later ordinal chapter defines the strict order on ordinals by membership and proves that this agrees with the familiar order on these finite stages. Recursive addition recovers the familiar finite calculations while remaining a set-theoretic function.

A useful warning

The existence of ω uses Infinity; the recursion theorem then uses Replacement-style collection of finite approximations. These are distinct logical steps.

Looking ahead

The next chapter, *Foundation and rank*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 24

Foundation and rank

Definitions and setup

Formal definitions used in this chapter

Well-founded relation. A relation R on a set A is well founded when every nonempty $B \subseteq A$ has an R -minimal element, that is, some $b \in B$ for which no $c \in B$ satisfies cRb .

Restriction. If f is a function and $B \subseteq \text{dom}(f)$, then $f \upharpoonright B = \{\langle x, f(x) \rangle : x \in B\}$.

Strict linear order and well-order. A strict linear order $<$ on X is irreflexive, transitive, and compares every two distinct elements. It is a well-order when every nonempty subset of X has a $<$ -least element.

Ordinal, ordinal order, successor, limit, and supremum (preliminary definitions). An ordinal is a transitive set well ordered by $\in$. For ordinals α, β , write

$$\alpha < \beta \iff \alpha \in \beta, \quad \alpha \leq \beta \iff (\alpha < \beta \text{ or } \alpha = \beta).$$

Its successor is $\alpha + 1 := \alpha \cup \{\alpha\}$. A nonzero ordinal is a *limit ordinal* when it is not a successor ordinal. For a set A of ordinals, put $\sup A := \bigcup A$; the ordinal theorems in the next chapter justify its least-upper-bound characterization. Ord denotes the class of all ordinals.

Class terminology and class function. A class is a collection specified by a first-order formula in the ambient set theory; it is a *proper class* when no set has exactly those members. The letter V denotes the ambient class of all sets. A *class function* $F : C \rightarrow D$ is a definable class relation such that for every set $x \in C$ there is exactly one set $y \in D$ with $F(x, y)$; we then write $F(x) = y$. Unlike a set-function, a class function may have a proper-class domain. **Finite predecessor closure for a relation.** Let R be a relation on a set A and let $a \in A$. Using the finite-sequence construction of Chapter 2, define

$$C_a^R := \{c \in A : c = a, \text{ or there are } n \in \omega \text{ and } s : n + 1 \rightarrow A \\ \text{with } s(0) = c, s(n) = a, \text{ and } s(i)Rs(i + 1) \text{ for every } i < n\}.$$

Thus C_a^R is the set of points that reach a by a finite R -chain, including a itself. The proof below derives the two closure identities it needs directly from this displayed definition; they are not included as part of the terminology.

Foundation says every nonempty set has an $\in$ -minimal member. The first results of this chapter derive $\in$ -induction and the set-sized well-founded recursion theorem from that axiom. Only after that recursion theorem has been proved will we define the rank class function and the cumulative hierarchy $\alpha \mapsto V_\alpha$.

How this chapter fits together

The dependency order is important. Foundation first turns membership into a well-founded relation and thereby yields induction and set-sized recursion. The chapter then uses that proved recursion theorem to define rank and the hierarchy V_α ; subsequent propositions establish the familiar rank equations. Later inner-model and forcing arguments use these proved rank bounds to keep constructions set-sized.

Theorem 24.1 (Membership induction from Foundation). *If a formula $\varphi(x)$ satisfies*

$$\forall x [(\forall y \in x \varphi(y)) \rightarrow \varphi(x)],$$

then Foundation proves $\forall x \varphi(x)$.

Proof. Assume a formula $P(x)$ satisfies

$$\forall x [(\forall y \in x P(y)) \Rightarrow P(x)].$$

Suppose P fails somewhere and form the nonempty set of counterexamples inside the transitive closure of one counterexample. By Foundation, it has an $\in$ -minimal member a . Every $y \in a$ is then not a counterexample, so $P(y)$ holds for all $y \in a$. The displayed hypothesis yields $P(a)$, contradiction. Therefore $P(x)$ holds for every set x . This is membership induction. The role of Foundation is exactly the least-counterexample step: it prevents an infinite descending membership chain from avoiding a minimal failure.

Point specific to this result. If a counterexample existed, Foundation would select an $\in$ -minimal counterexample. All of its members would satisfy the property, so the induction hypothesis would force the counterexample itself to satisfy it, contradiction. $\square$

Theorem 24.2 (Well-founded recursion). *Let R be a well-founded relation on a set A , let $\bar{p}$ be parameters, and let $\Phi(a, h, y, \bar{p})$ be a formula such that for every $a \in A$ and every function h with domain $\{b \in A : bRa\}$, there is a unique set y satisfying $\Phi(a, h, y, \bar{p})$. Then there is a unique function G with domain A such that*

$$\Phi(a, G \upharpoonright \{b \in A : bRa\}, G(a), \bar{p})$$

holds for every $a \in A$.

Proof.

Formal derivation spine

Formal spine: local solutions exist and are unique, then glue. For $a \in A$, let $C_a = C_a^R$. A *local solution at a* is a function $g : C_a \rightarrow V$ satisfying

$$\Phi(c, g \upharpoonright \{d \in A : dRc\}, g(c), \bar{p}) \quad (c \in C_a).$$

Any two local solutions agree on their common domain by well-founded induction. Existence of a local solution at every a follows by a least-counterexample argument and Replacement over the predecessor set. The compatible local solutions then glue to the desired global function.

Write $P_c := \{d \in A : dRc\}$. We first record the two finite-chain consequences used below. If bRa and $c \in C_b^R$, append the final edge bRa to a witnessing chain from c to b ; hence $c \in C_a^R$, so $C_b^R \subseteq C_a^R$. Conversely, if $c \in C_a^R \setminus \{a\}$, choose a witnessing nonempty chain from c to a and let b be its penultimate point. Then bRa and $c \in C_b^R$. Therefore

$$C_a^R \setminus \{a\} = \bigcup_{bRa} C_b^R.$$

These are proved consequences of the definition, not extra closure assumptions. First prove uniqueness on overlaps. Suppose $g : C_a \rightarrow V$ and $h : C_b \rightarrow V$ are local solutions. If they disagree somewhere on $C_a \cap C_b$,

the set of disagreement points has an R -minimal member c . Because both predecessor closures are downward closed, every dRc lies in both domains; minimality gives

$$g \upharpoonright P_c = h \upharpoonright P_c.$$

Both $g(c)$ and $h(c)$ satisfy the same instance of $\Phi(c, \cdot, y, \bar{p})$, so the uniqueness hypothesis on Φ gives $g(c) = h(c)$, a contradiction. Thus local solutions agree wherever both are defined. In particular, if a local solution at a point exists, it is unique.

Now suppose that some point has no local solution and let

$$B := \{a \in A : \text{there is no local solution on } C_a\}.$$

By Separation, B is a set; choose an R -minimal $a \in B$. Every strict predecessor bRa lies outside B , so it has a unique local solution $g_b : C_b \rightarrow V$. Because the relation “ g is the unique local solution on C_b ” is functional in b , Replacement collects

$$\mathcal{G}_a := \{g_b : bRa\}$$

as a set. The overlap argument shows that the functions in $\mathcal{G}_a$ are pairwise compatible; hence

$$g := \bigcup \mathcal{G}_a$$

is a function. Its domain is $C_a \setminus \{a\}$: every point below a by a nonempty finite R -chain belongs to C_b for the last predecessor bRa , and conversely every C_b lies below a .

In particular $g \upharpoonright P_a$ is a function with domain P_a . By the hypothesis on Φ , there is a unique set y such that

$$\Phi(a, g \upharpoonright P_a, y, \bar{p}).$$

Set $g_a := g \cup \{a, y\}$. The old recursion equations remain true on $C_a \setminus \{a\}$, and the displayed equation supplies the equation at a ; hence g_a is a local solution at a , contradicting $a \in B$. Therefore every $a \in A$ has a unique local solution g_a .

Replacement now collects the set $\{g_a : a \in A\}$. Their pairwise compatibility makes

$$G := \bigcup_{a \in A} g_a$$

a function. Since $a \in C_a$, its domain is all of A , and the local recursion equation gives

$$\Phi(a, G \upharpoonright P_a, G(a), \bar{p})$$

for every $a \in A$. If H is another global solution, the same minimal-disagreement argument gives $H = G$. Thus the recursive function exists and is unique. $\square$

Formal definitions used in this chapter

Local rank recursion and the rank class function. For a set x , the set $A_x := \text{tc}(\{x\})$ is transitive and contains x . Apply Theorem 24.2 to membership on A_x with the recursion clause

$$r_x(a) = \bigcup \{S(r_x(y)) : y \in a\}, \quad S(z) = z \cup \{z\}.$$

The resulting set-function r_x is unique. If $x \in A_z$, uniqueness makes r_z and r_x agree on the common predecessor closure of x . Hence the value $r_x(x)$ is independent of the chosen transitive set on which the local recursion is performed. Define the class function

$$\text{rank}(x) := r_x(x).$$

The next proposition proves that these values are ordinals and identifies the displayed union with the ordinal supremum.

Local construction of the cumulative hierarchy. For each ordinal α , apply Theorem 24.2 to the well-founded relation $\in$ on the set $\alpha + 1$. At $\beta \leq \alpha$ use the recursion clause

$$\begin{aligned} W(0) &= \emptyset, \\ W(\gamma + 1) &= \mathcal{P}(W(\gamma)), \\ W(\lambda) &= \bigcup_{\gamma < \lambda} W(\gamma) \quad (\lambda \neq 0 \text{ a limit ordinal}). \end{aligned}$$

Uniqueness makes the constructions for different α 's agree on overlaps. Define V_β to be their common value at β . Thus $\alpha \mapsto V_\alpha$ is a class function $\text{Ord} \rightarrow V$, obtained entirely from the already proved set-sized well-founded recursion theorem.

Proposition 24.3 (Every set has an ordinal rank). *The class function $\text{rank} : V \rightarrow \text{Ord}$ defined above satisfies*

$$\text{rank}(x) = \sup\{\text{rank}(y) + 1 : y \in x\}$$

for every set x . Moreover,

$$y \in x \implies \text{rank}(y) < \text{rank}(x).$$

It is the unique class function $R : V \rightarrow \text{Ord}$ satisfying the displayed recursion equation.

Proof. Fix a set x and consider the local recursion r_x on the transitive set $A_x = \text{tc}(\{x\})$. Membership induction on A_x proves that every value $r_x(a)$ is an ordinal. Indeed, if the predecessor values $r_x(y)$ are ordinals, then

$$r_x(a) = \bigcup_{y \in a} (r_x(y) + 1)$$

is the union of a set of ordinals and hence is an ordinal. Therefore at $a = x$,

$$\text{rank}(x) = \bigcup_{y \in x} (\text{rank}(y) + 1) = \sup\{\text{rank}(y) + 1 : y \in x\}.$$

For $y \in x$, the ordinal $\text{rank}(y) + 1$ occurs in this union, so

$$\text{rank}(y) < \text{rank}(x).$$

Finally suppose $R : V \rightarrow \text{Ord}$ is another class function satisfying the same equation. If $R(x) \neq \text{rank}(x)$ for some set x , Membership Induction (Theorem 24.1) gives an $\in$ -minimal counterexample: all members $y \in x$ then have equal R - and rank -values, so the recursion equations force equality at x as well, a contradiction. Hence the rank class function is unique. $\square$

Theorem 24.4 (Cumulative hierarchy and rank characterization). *The class family $\langle V_\alpha : \alpha \in \text{Ord} \rangle$ defined above satisfies*

$$V_0 = \emptyset, \quad V_{\alpha+1} = \mathcal{P}(V_\alpha), \quad V_\lambda = \bigcup_{\beta < \lambda} V_\beta$$

for every nonzero limit ordinal λ . The levels are transitive and monotone:

$$\alpha < \beta \implies V_\alpha \subseteq V_\beta.$$

For every set x ,

$$x \subseteq V_{\text{rank}(x)}, \quad x \in V_{\text{rank}(x)+1}.$$

Consequently every set belongs to some level and, in class notation,

$$V = \bigcup_{\alpha \in \text{Ord}} V_\alpha.$$

Proof. The three recursion equations hold by the local definition of the hierarchy. Their compatibility across different initial ordinal segments follows from uniqueness in Theorem 24.2; no appeal to the later general transfinite-recursion theorem is being made here.

We prove simultaneously, by Membership Induction restricted to ordinals β , that V_β is transitive and that

$$\alpha < \beta \implies V_\alpha \subseteq V_\beta. \quad (1)$$

For $\beta = 0$ both assertions are immediate. Suppose $\beta = \gamma + 1$ and the assertions hold at every earlier ordinal. If $x \in y \in V_\beta = \mathcal{P}(V_\gamma)$, then $y \subseteq V_\gamma$, so $x \in V_\gamma$; transitivity of V_γ gives $x \subseteq V_\gamma$, hence $x \in V_\beta$. Thus V_β is transitive. Moreover transitivity of V_γ gives

$$V_\gamma \subseteq \mathcal{P}(V_\gamma) = V_\beta.$$

If $\alpha < \gamma$, the induction hypothesis gives $V_\alpha \subseteq V_\gamma \subseteq V_\beta$; hence (1) holds at the successor stage.

If β is a nonzero limit ordinal, then

$$V_\beta = \bigcup_{\gamma < \beta} V_\gamma.$$

If $x \in y \in V_\beta$, choose $\gamma < \beta$ with $y \in V_\gamma$. Transitivity of V_γ gives $x \in V_\gamma \subseteq V_\beta$, so V_β is transitive. And for each $\alpha < \beta$, every element of V_α belongs to the displayed union, so $V_\alpha \subseteq V_\beta$. This completes the simultaneous induction and proves monotonicity without using the later transfinite-induction theorem.

Now use Membership Induction on sets to prove

$$x \subseteq V_{\text{rank}(x)}. \quad (1)$$

Assume (1) already holds for every $y \in x$. Then

$$y \in V_{\text{rank}(y)+1}$$

for every $y \in x$. Proposition 24.3 gives

$$\text{rank}(y) + 1 \leq \text{rank}(x),$$

so monotonicity gives $y \in V_{\text{rank}(x)}$. Hence every member of x lies in that level, proving (1). Therefore

$$x \in \mathcal{P}(V_{\text{rank}(x)}) = V_{\text{rank}(x)+1}.$$

Thus every set lies in some V_α . Conversely every member of every V_α is a set by construction, so the displayed class union is exactly V . $\square$

Worked example

Worked Example 24.5 (Computing small ranks). $\text{rank}(\emptyset) = 0$, $\text{rank}(\{\emptyset\}) = 1$, and $\text{rank}(\{\emptyset, \{\emptyset\}\}) = 2$. More generally $x \in V_{\alpha+1}$ exactly when every member of x lies in V_α . The recursion formula $\text{rank}(x) = \sup_{y \in x} (\text{rank}(y) + 1)$ captures this pattern.

A useful warning

A rank bound is not a cardinality bound. A low-rank set can be very large, and later arguments must not interchange these two notions.

Looking ahead

The next chapter, *Ordinals*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 25

Ordinals

Definitions and setup

Formal definitions used in this chapter

Linear order and well-order. A strict linear order $<$ on X is irreflexive, transitive, and compares every two distinct elements. It is a well-order when every nonempty subset of X has a $<$ -least element.

Order isomorphism. An order isomorphism $f : (X, <) \rightarrow (Y, <)$ is a bijection satisfying $x < x' \iff f(x) < f(x')$. The theorem below proves that every well-order has a unique ordinal representative; only after that theorem do we call that ordinal the *order type* of the well-order.

Ordinal and ordinal comparison. An ordinal is a transitive set well ordered by $\in$. For ordinals α, β , write $\alpha < \beta$ for $\alpha \in \beta$. Ord denotes the class of all ordinals.

Successor and limit ordinal. The successor of α is $\alpha + 1 := \alpha \cup \{\alpha\}$. A nonzero ordinal λ is a limit ordinal when it is not a successor.

Class function and transfinite recursion. Recalling Chapter 3, a class function is a definable class relation G such that every input in its class domain has exactly one output. A transfinite-recursion specification determines $G(\alpha)$ from the restriction $G \upharpoonright \alpha$; the theorem below proves existence and uniqueness under the displayed functional hypothesis.

An ordinal is a transitive set well ordered by membership. Every element of an ordinal is an ordinal, and two ordinals are comparable by membership. Theorem 25.2 will prove that every well-order has a unique ordinal representative; only after that proof do we define its order type. Transfinite induction and recursion are the ordinal analogues of ordinary induction/recursion, with an explicit limit stage. We use von Neumann ordinals, so $\alpha < \beta$ is literally $\alpha \in \beta$.

How this chapter fits together

Ordinals are transitive sets well-ordered by membership. Their trichotomy gives a canonical comparison; the theorem below will identify every well-order with a unique ordinal representative, after which its order type is defined; and transfinite induction/recursion extend ordinary induction beyond ω . The proofs are foundational for L , whose stages are indexed by all ordinals.

Theorem 25.1 (Trichotomy of ordinals). *For any ordinals α, β , exactly one of*

$$\alpha \in \beta, \quad \alpha = \beta, \quad \beta \in \alpha$$

holds.

Proof. An ordinal is a transitive set well ordered by $\in$. Let α, β be ordinals. Their intersection $\gamma = \alpha \cap \beta$ is an initial segment of each. If $\gamma = \alpha$, then $\alpha \subseteq \beta$, whence either $\alpha = \beta$ or $\alpha \in \beta$ because initial segments of an ordinal are precisely its elements. Symmetrically, if $\gamma = \beta$, then $\beta = \alpha$ or $\beta \in \alpha$. It is impossible for γ to be a proper initial segment of both: then $\gamma \in \alpha \cap \beta = \gamma$, contradicting Foundation/irreflexivity of ordinal membership. Therefore exactly one of

$$\alpha \in \beta, \quad \alpha = \beta, \quad \beta \in \alpha$$

holds. This is trichotomy for ordinals.

Point specific to this result. Because ordinals are transitive sets well ordered by membership, their intersection is a common initial segment. The first point of disagreement, if any, places one ordinal as an element of the other, giving trichotomy. $\square$

Theorem 25.2 (Every well-order is uniquely isomorphic to an ordinal). *Every well-ordered set $(X, <)$ is isomorphic to a unique ordinal α .*

Proof. Let $(X, <)$ be a well-ordered set. For each $x \in X$, define recursively

$$\pi(x) = \{\pi(y) : y < x\}.$$

Well-founded recursion on $<$ gives π . Membership induction on the initial segments shows every $\pi(x)$ is an ordinal, and if $y < x$ then $\pi(y) \in \pi(x)$. Conversely, every member of $\pi(x)$ has the form $\pi(y)$ for some $y < x$, so π is an order isomorphism of X onto the transitive set

$$\alpha = \{\pi(x) : x \in X\}.$$

That transitive set is an ordinal. If $\rho : X \rightarrow \beta$ is another isomorphism onto an ordinal, induction on x gives

$$\rho(x) = \{\rho(y) : y < x\} = \{\pi(y) : y < x\} = \pi(x).$$

Hence $\rho = \pi$ and $\beta = \alpha$. The ordinal representative is therefore unique.

Point specific to this result. Map each element of a well-order to the ordinal representative already assigned to its strict initial segment. Well-founded recursion gives the map, and the image is a transitive set of transitive sets, hence an ordinal; uniqueness follows from preservation of initial segments. $\square$

Formal definitions used in this chapter

Order type. For a well-ordered set $(X, <)$, define $\text{otp}(X, <)$ to be the unique ordinal α for which $(X, <) \cong (\alpha, \in)$, whose existence and uniqueness were proved in Theorem 25.2. Thus this notation is introduced only after its existence theorem.

Formal definitions used in this chapter

Class property of ordinals. In the statements below, a *class property* $P(\alpha)$ means a first-order formula $P(x, \bar{p})$ of set theory with a fixed finite tuple of set parameters $\bar{p}$. The assertion that P holds for every ordinal is therefore a scheme, one instance for each such formula and parameter tuple.

Definable functional. A formula $\Psi(s, y, \bar{p})$ is *functional in y* when ZF proves $\forall s \exists! y \Psi(s, y, \bar{p})$ on the stated domain. Writing $F(s)$ for that unique y is only an abbreviation for $\Psi(s, y, \bar{p})$; no proper-class object F is being assumed to be a set.

Theorem 25.3 (Transfinite induction). *If a class property P satisfies*

$$\forall \alpha [(\forall \beta < \alpha P(\beta)) \rightarrow P(\alpha)],$$

then $P(\alpha)$ holds for every ordinal α .

Proof. Let $P(\alpha)$ be a property of ordinals and assume

$$\forall \alpha [(\forall \beta < \alpha P(\beta)) \Rightarrow P(\alpha)].$$

If some ordinal failed P , the class of counterexample ordinals would have a least member α , because any nonempty set of candidate counterexamples has a least ordinal and one may start from a chosen counterexample and separate its predecessors. By minimality, $P(\beta)$ holds for every $\beta < \alpha$. The hypothesis then gives $P(\alpha)$, contradiction. Thus every ordinal satisfies P . This is transfinite induction. It is membership induction specialized to ordinals, where $\beta < \alpha$ means $\beta \in \alpha$.

Point specific to this result. The proof is the least-counterexample argument made possible by the well-ordering of every ordinal. If a counterexample existed, its least member would have all predecessors satisfying the property and therefore satisfy it as well. $\square$

Theorem 25.4 (Transfinite recursion). *Let $\Phi(s, y, \bar{p})$ be a formula such that ZF proves: whenever s is a function whose domain is an ordinal, there is a unique set y with $\Phi(s, y, \bar{p})$. Then there is a unique class function G with domain Ord such that, for every ordinal α ,*

$$\Phi(G \upharpoonright \alpha, G(\alpha), \bar{p}).$$

Equivalently, for a definable functional F , $G(\alpha) = F(G \upharpoonright \alpha)$.

Proof.

Formal derivation spine

Formal spine: compatible set-sized initial solutions. For every ordinal θ , there is a unique function G_θ with domain θ such that

$$\Phi(G_\theta \upharpoonright \alpha, G_\theta(\alpha), \bar{p}) \quad (\alpha < \theta).$$

Uniqueness gives

$$\theta < \eta \implies G_\eta \upharpoonright \theta = G_\theta.$$

Apply well-founded recursion to the membership relation on each ordinal θ . The hypothesis on Φ produces a unique set at every stage, so there is a unique set function G_θ on θ satisfying the displayed recursion.

If $\theta < \eta$, both G_θ and $G_\eta \upharpoonright \theta$ solve the same recursion on θ . The uniqueness part of well-founded recursion gives

$$G_\eta \upharpoonright \theta = G_\theta. \tag{1}$$

Define the class relation G by

$$G(\alpha) := G_{\alpha+1}(\alpha). \tag{2}$$

Compatibility (1) makes (2) independent of the chosen initial segment larger than α . Moreover,

$$G \upharpoonright \alpha = G_\alpha,$$

so the defining property of $G_{\alpha+1}$ gives

$$\Phi(G \upharpoonright \alpha, G(\alpha), \bar{p}).$$

Finally, if H is another class solution, transfinite induction on α shows $H \upharpoonright \alpha = G \upharpoonright \alpha$; uniqueness in the hypothesis on Φ then gives $H(\alpha) = G(\alpha)$. This proves uniqueness as a class function. The theorem is a schema indexed by the defining formula Φ , not an appeal to an undefined operation on proper classes. $\square$

Worked example

Worked Example 25.5 (Order type of a finite well-order). For a three-element chain $a < b < c$, map $a \mapsto 0$, $b \mapsto 1$, and $c \mapsto 2$. The image is the ordinal $3 = \{0, 1, 2\}$. The general proof assigns to each element the order type of its initial segment and uses well-founded recursion.

A useful warning

Ordinal comparison is about order type, not cardinality. Distinct ordinals such as ω and $\omega + 1$ can have the same cardinality.

Looking ahead

The next chapter, *Ordinal arithmetic*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 26

Ordinal arithmetic

Definitions and setup

Formal definitions used in this chapter

Successor, limit, and supremum. For an ordinal γ , $\gamma + 1 = \gamma \cup \{\gamma\}$. A nonzero nonsuccessor ordinal is a limit ordinal. For a set A of ordinals, $\sup A = \bigcup A$.

Ordinal addition. For ordinals α, β , define by recursion on β

$$\alpha + 0 = \alpha, \quad \alpha + (\beta + 1) = (\alpha + \beta) + 1, \quad \alpha + \lambda = \sup_{\beta < \lambda} (\alpha + \beta)$$

for limit λ .

Ordinal multiplication.

$$\alpha \cdot 0 = 0, \quad \alpha \cdot (\beta + 1) = \alpha \cdot \beta + \alpha, \quad \alpha \cdot \lambda = \sup_{\beta < \lambda} \alpha \cdot \beta.$$

Ordinal exponentiation. Recall that $0 = \emptyset$ and $1 = \{0\}$. For ordinals α, β , define exponentiation by recursion on the exponent:

$$\alpha^0 = 1, \quad \alpha^{\beta+1} = \alpha^\beta \cdot \alpha, \quad \alpha^\lambda = \sup_{0 < \gamma < \lambda} \alpha^\gamma$$

when λ is a nonzero limit ordinal. The restriction $0 < \gamma$ in the limit supremum is essential for the base $\alpha = 0$: these clauses give $0^0 = 1$ and $0^\beta = 0$ for every $\beta > 0$. They also give $1^\beta = 1$ for every ordinal β .

Normal ordinal function. A class function $F : \text{Ord} \rightarrow \text{Ord}$ is *normal* when it is strictly increasing and continuous at every nonzero limit ordinal, meaning

$$\beta < \gamma \implies F(\beta) < F(\gamma), \quad F(\lambda) = \sup_{\beta < \lambda} F(\beta)$$

for every nonzero limit λ .

Fixed point and epsilon number. A fixed point of an ordinal function F is an ordinal ξ with $F(\xi) = \xi$. An *epsilon number* is an ordinal ε satisfying $\omega^\varepsilon = \varepsilon$. No symbol for a least epsilon number is introduced here: the worked result below first proves existence of epsilon numbers and of a least one, and only there introduces the notation ε_0 .

Ordinal addition, multiplication, and exponentiation are defined by transfinite recursion on the right argument. They encode concatenation and lexicographic constructions of well-orders, not cardinal size. Consequently they need not commute. Suprema at limit stages are part of the definitions, so every algebraic law must be

proved by transfinite induction rather than inferred from integer arithmetic.

How this chapter fits together

Ordinal operations are defined by transfinite recursion on the right argument. They record concatenation and lexicographic order types, so they are generally not commutative. Working concrete examples before proving algebraic identities helps prevent cardinal-arithmetic intuitions from being imported incorrectly.

Theorem 26.1 (Recursive construction and laws of ordinal addition). *Ordinal addition is the unique operation satisfying*

$$\alpha + 0 = \alpha, \quad \alpha + (\beta + 1) = (\alpha + \beta) + 1, \quad \alpha + \lambda = \sup_{\beta < \lambda} (\alpha + \beta)$$

for limit λ ; it is associative and strictly increasing in the right argument, but need not be commutative.

Commented symbolic trace: ordinal recursion at successor and limit stages

The three clauses play different roles. $\alpha + 0 = \alpha$ initializes the recursion. $\alpha + (\beta + 1) = (\alpha + \beta) + 1$ appends one new point. At a limit λ , there is no immediate predecessor, so $\alpha + \lambda$ is defined as the supremum of all earlier values $\alpha + \beta$ for $\beta < \lambda$. Whenever a later proof uses transfinite induction on the right argument, check these three cases separately rather than treating “limit” as a larger successor case.

Proof. For each fixed ordinal α , transfinite recursion on β defines ordinal addition by

$$\alpha + 0 = \alpha, \quad \alpha + (\beta + 1) = S(\alpha + \beta), \quad \alpha + \lambda = \sup_{\beta < \lambda} (\alpha + \beta)$$

for limit λ . The supremum is the union of a set of ordinals and therefore an ordinal. Uniqueness follows from transfinite recursion. Associativity is proved by induction on the third argument γ . At successors,

$$(\alpha + \beta) + (\gamma + 1) = S((\alpha + \beta) + \gamma) = S(\alpha + (\beta + \gamma)) = \alpha + (\beta + (\gamma + 1)),$$

and at limits both sides are the supremum of the corresponding earlier values. The definition is order-sensitive; in general $\alpha + \beta \neq \beta + \alpha$.

Point specific to this result. Addition is recursive in the right argument and models concatenation of well-orders. The limit clause takes the supremum of earlier sums, which explains both associativity and the failure of commutativity.

Further explanation. Associativity is proved by transfinite induction on the final argument γ . The successor case uses the recursive clause twice, while the limit case uses continuity: $\alpha + (\beta + \lambda) = \sup_{\gamma < \lambda} \alpha + (\beta + \gamma)$. The same recursion shows monotonicity in the right argument. What fails is commutativity, as the worked examples below make explicit; the theorem therefore records the algebraic laws that genuinely follow from order-type concatenation rather than importing laws from integer arithmetic. $\square$

Theorem 26.2 (Recursive construction and laws of ordinal multiplication). *Ordinal multiplication is the unique operation satisfying*

$$\alpha \cdot 0 = 0, \quad \alpha(\beta + 1) = \alpha\beta + \alpha, \quad \alpha\lambda = \sup_{\beta < \lambda} \alpha\beta$$

for limit λ . It is associative and left-distributive over ordinal addition.

Proof. Fix α . Define multiplication by transfinite recursion on the right factor:

$$\alpha \cdot 0 = 0, \quad \alpha \cdot (\beta + 1) = \alpha \cdot \beta + \alpha, \quad \alpha \cdot \lambda = \sup_{\beta < \lambda} \alpha \cdot \beta.$$

Every stage is an ordinal because addition and suprema of sets of ordinals are already available. Transfinite induction proves associativity and left distributivity over ordinal addition. For example, at a successor stage,

$$\alpha(\beta + (\gamma + 1)) = \alpha((\beta + \gamma) + 1) = \alpha(\beta + \gamma) + \alpha = (\alpha\beta + \alpha\gamma) + \alpha = \alpha\beta + \alpha(\gamma + 1).$$

Limit stages follow by taking suprema. Right distributivity and commutativity generally fail, reflecting the concatenation interpretation of ordinal multiplication.

Point specific to this result. Multiplication iterates ordinal addition on the right. Its limit clause is a supremum of earlier products, so left distributivity follows by induction, while reversing the order can change the resulting order type.

Further explanation. The distributive law valid for ordinal multiplication is $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$, proved by transfinite induction on γ . For associativity, transfinite induction on the rightmost factor gives the successor identity by the recursive multiplication clause and the limit identity by continuity of both sides under suprema. At a limit λ , the definition $\alpha\lambda = \sup_{\beta < \lambda} \alpha\beta$ is the crucial continuity clause. Right distributivity generally fails, so the proof deliberately keeps track of which argument is being recursively varied. $\square$

Theorem 26.3 (Recursive construction and laws of ordinal exponentiation). *Ordinal exponentiation is the unique operation satisfying*

$$\alpha^0 = 1, \quad \alpha^{\beta+1} = \alpha^\beta \alpha, \quad \alpha^\lambda = \sup_{0 < \gamma < \lambda} \alpha^\gamma$$

for every nonzero limit ordinal λ . In particular,

$$0^0 = 1, \quad 0^\beta = 0 \ (\beta > 0), \quad 1^\beta = 1.$$

Proof. For ordinals α, β , define exponentiation by recursion on β :

$$\alpha^0 = 1, \quad \alpha^{\beta+1} = \alpha^\beta \cdot \alpha, \quad \alpha^\lambda = \sup_{0 < \gamma < \lambda} \alpha^\gamma$$

for nonzero limit λ . Apply the transfinite-recursion theorem to the displayed successor and limit clauses; it yields a unique function of the exponent, because multiplication and suprema of sets of ordinals are already defined. For $\alpha = 0$, transfinite induction gives $0^\beta = 0$ for every $\beta > 0$; at a nonzero limit all terms in the displayed supremum are therefore 0. For $\alpha = 1$, induction gives $1^\beta = 1$ at every stage. Induction proves the exponent law

$$\alpha^{\beta+\gamma} = \alpha^\beta \cdot \alpha^\gamma$$

with the convention that the recursion is on the exponent and multiplication is ordinal multiplication. At a successor, use the recursive equation and associativity of multiplication; at a limit, both sides are obtained as the supremum of the corresponding earlier powers. Again these are ordinal, not cardinal, powers and encode order type.

Point specific to this result. Exponentiation iterates multiplication with $\alpha^0 = 1$ and takes suprema at limits. Transfinite induction establishes its recursion laws; for base $\alpha > 1$ it also gives strict increase in the exponent, while the exceptional bases 0 and 1 behave exactly as stated above.

Further explanation. The recursion on the exponent gives $\alpha^0 = 1$, $\alpha^{\beta+1} = \alpha^\beta \alpha$, and $\alpha^\lambda = \sup_{0 < \gamma < \lambda} \alpha^\gamma$ at nonzero limits. Transfinite induction then proves the exponent laws in their correct ordinal form, for example $\alpha^{\beta+\gamma} = \alpha^\beta \alpha^\gamma$. Limit continuity is again essential, and it is the mechanism behind fixed points such as epsilon numbers. $\square$

Worked Theorem 26.4 (Canonical calculations with omega, omega+1, 1+omega, omega^2, and epsilon-style fixed-point preview). *The recursive definitions give*

$$1 + \omega = \omega < \omega + 1, \quad 2 \cdot \omega = \omega < \omega \cdot 2,$$

and $\omega^2 = \sup_{n < \omega} \omega \cdot n$. The ordinal function $F(\alpha) = \omega^\alpha$ is normal. It has fixed points above every prescribed ordinal; equivalently, epsilon numbers are unbounded in the ordinals. In particular a least epsilon number exists, denoted ε_0 .

Proof. The recursive definitions give concrete calculations. First,

$$1 + \omega = \sup_{n < \omega} (1 + n) = \omega,$$

whereas $\omega + 1 = S(\omega) > \omega$. Thus addition is not commutative. Using the multiplication limit clause,

$$2 \cdot \omega = \sup_{n < \omega} 2n = \omega, \quad \omega \cdot 2 = \omega + \omega > \omega.$$

For exponentiation,

$$\omega^2 = \sup_{n < \omega} \omega \cdot n,$$

which is the order type of ω many consecutive blocks of type ω . More generally $\omega^\omega = \sup_{n < \omega} \omega^n$.

We next verify the fixed-point claim rather than treating it as a preview. By the exponentiation recursion and the strict-increase observation following Theorem 26.3, the map

$$F(\alpha) = \omega^\alpha$$

is strictly increasing. If λ is a nonzero limit ordinal, then

$$F(\lambda) = \omega^\lambda = \sup_{0 < \gamma < \lambda} \omega^\gamma = \sup_{\gamma < \lambda} F(\gamma),$$

because $F(0) = 1$ is already below every $F(\gamma)$ with $0 < \gamma < \lambda$. Thus F is normal.

Fix any ordinal β . Define an increasing ω -sequence by

$$\beta_0 = \beta + 1, \quad \beta_{n+1} = F(\beta_n) = \omega^{\beta_n}, \quad \varepsilon = \sup_{n < \omega} \beta_n.$$

Since $\beta_0 > \beta$, we have $\varepsilon > \beta$. The ordinal ε is a nonzero limit, and continuity of F gives

$$F(\varepsilon) = \sup_{\gamma < \varepsilon} F(\gamma).$$

The sequence $(\beta_n)_{n < \omega}$ is cofinal in ε . Hence monotonicity of F yields

$$\sup_{\gamma < \varepsilon} F(\gamma) = \sup_{n < \omega} F(\beta_n) = \sup_{n < \omega} \beta_{n+1} = \varepsilon.$$

Therefore $\omega^\varepsilon = \varepsilon$. We have produced an epsilon number above the arbitrary ordinal β , so epsilon numbers are unbounded. Taking $\beta = 0$ gives at least one epsilon number; since every nonempty class of ordinals has a least member once a witness is fixed and Separation is applied inside that witness plus one, there is a least epsilon number, denoted ε_0 .

These computations illustrate the governing principle: ordinal operations remember the order in which blocks are concatenated, so cardinal equalities such as $|1 + \omega| = |\omega + 1|$ do not imply equality of ordinals. $\square$

Worked example

Worked Example 26.5 (Noncommutativity in one line). $1 + \omega = \sup_{n < \omega} (1 + n) = \omega$, while $\omega + 1$ has a greatest element and is strictly larger than ω . Likewise $2 \cdot \omega = \omega$, whereas $\omega \cdot 2$ consists of two consecutive copies of ω .

A useful warning

Only the stated distributive and associative laws hold. In particular, left and right distributivity are not interchangeable for ordinal multiplication.

Looking ahead

The next chapter, *Cardinals without assuming choice*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 27

Cardinals without assuming choice

Definitions and setup

Formal definitions used in this chapter

Well-orderable, equipotent, and initial ordinal. A set is well-orderable if it admits a well-order. Sets are equipotent when a bijection exists between them. An ordinal κ is initial when it is not equipotent with any smaller ordinal.

Cardinal of a well-orderable set. A cardinal is an initial ordinal. If X is well-orderable, $|X|$ denotes the unique initial ordinal equipotent with X . For cardinals κ, λ , $\kappa < \lambda$ is ordinal comparison.

Injection notation used in Hartogs' lemma. “ α injects into X ” means that there exists an injective function $f : \alpha \rightarrow X$. Hartogs' lemma below proves that some ordinal fails to inject into X ; the least such ordinal will then be denoted $h(X)$.

Power-set cardinal notation under well-orderability. If $\mathcal{P}(X)$ is well-orderable and $|X| = \kappa$, then 2^κ denotes $|\mathcal{P}(X)|$. No global use of this notation is made without a stated well-orderability hypothesis.

Aleph notation. The transfinite enumeration of the infinite initial ordinals is not assumed in advance. The Aleph Recursion Theorem below constructs a unique class function A ; after that theorem we define $\aleph_\alpha := A(\alpha)$. Beth notation is postponed to Volume V, where Choice is defined first.

Without Choice, not every set is known to be well orderable. A well-orderable cardinal is represented by its least equinumerous ordinal, an initial ordinal. Hartogs' theorem below will assign to every set an ordinal too large to inject into it, without using Choice. The aleph hierarchy then enumerates the infinite initial ordinals. Beth notation is deliberately deferred to Volume V, where Choice has first been introduced and the power-set cardinals used there are well orderable.

How this chapter fits together

Without Choice, arbitrary sets need not be comparable by well-orderable cardinal representatives. Hartogs' lemma nevertheless supplies the theorem needed to construct an ordinal too large to inject into any given set. This is enough to build the aleph hierarchy of well-orderable cardinals. The beth hierarchy is not defined in this bare-ZF chapter; it is introduced only in Volume V after Choice is available.

Theorem 27.1 (Hartogs lemma). *For every set X there is a least ordinal H_X for which no injection $H_X \rightarrow X$ exists.*

Proof. Let $\mathcal{W}_X$ be the set of pairs (Y, R) such that $Y \subseteq X$ and $R \subseteq Y \times Y$ well-orders Y . Power Set and Separation make $\mathcal{W}_X$ a set. By Theorem 25.2, every member of $\mathcal{W}_X$ has a unique ordinal order type, so

Replacement gives the set

$$A_X := \{\text{otp}(Y, R) : (Y, R) \in \mathcal{W}_X\}.$$

The set A_X is an ordinal. Its members are ordinals. If $\alpha \in A_X$ and $\beta \in \alpha$, choose a well-ordered subset (Y, R) of X of type α . Under an isomorphism $(Y, R) \cong (\alpha, \in)$, the point corresponding to β has a strict initial segment whose order type is β . That initial segment is again a well-ordered subset of X , so $\beta \in A_X$. Thus A_X is a transitive set of ordinals and hence an ordinal.

Every ordinal below A_X injects into X . If $\beta < A_X$, then $\beta \in A_X$, so some subset of X has order type β . Its witnessing isomorphism yields an injection $\beta \rightarrow X$.

The ordinal A_X itself does not inject into X . Suppose $i : A_X \rightarrow X$ were injective. Transport the membership order of A_X to $i[A_X] \subseteq X$:

$$i(\xi) R i(\eta) \iff \xi \in \eta.$$

Then $(i[A_X], R)$ is a well-ordered subset of X of order type A_X . By the definition of A_X , this implies $A_X \in A_X$, contradicting irreflexivity of ordinal membership.

Therefore A_X is exactly the least ordinal that does not inject into X . Taking $H_X = A_X$ proves the theorem.

Point specific to this result. The realized order types form a transitive set of ordinals. Every proper predecessor is therefore realized, while realizing the set itself would force the impossible relation $A_X \in A_X$. The same construction proves existence and leastness. $\square$

Formal definitions used in this chapter

Hartogs number. For every set X , define $h(X)$ to be the least ordinal H_X supplied by Theorem 27.1. Thus

$$\beta < h(X) \implies \beta \hookrightarrow X, \quad h(X) \nrightarrow X.$$

Proposition 27.2 (Initial ordinals and alephs). *A cardinal is an initial ordinal, i.e. an ordinal not equipotent with any smaller ordinal. Every well-orderable set is equipotent with a unique initial ordinal. The next theorem constructs the transfinite enumeration of the infinite initial ordinals rather than presupposing aleph notation.*

Proof. A cardinal is, by definition, an initial ordinal. If X is well-orderable, some ordinal is bijective with X ; because the ordinals are well ordered, there is a least such ordinal, and it is initial. Uniqueness is immediate from minimality. Hartogs' lemma shows that above every initial ordinal κ there is an ordinal not injecting into κ , so there is a least initial ordinal strictly above κ . This supplies the successor step needed for the transfinite enumeration constructed in the next theorem. $\square$

Theorem 27.3 (Aleph recursion). *There is a unique class function $A : \text{Ord} \rightarrow \text{Ord}$ such that $A(0) = \omega$, $A(\alpha + 1)$ is the least cardinal strictly above $A(\alpha)$, and*

$$A(\lambda) = \sup_{\beta < \lambda} A(\beta)$$

for every nonzero limit ordinal λ . From this theorem onward define

$$\aleph_\alpha := A(\alpha).$$

Proof. Apply the transfinite recursion theorem to the functional

$$F(s) = \begin{cases} \omega, & \text{dom}(s) = 0, \\ (s(\beta))^+, & \text{dom}(s) = \beta + 1, \\ \sup_{\gamma < \lambda} s(\gamma), & \text{dom}(s) = \lambda \text{ limit.} \end{cases}$$

This yields a unique class function A . Define $\aleph_\alpha := A(\alpha)$. Transfinite induction proves strict increase: at successors this is the definition of cardinal successor; at a limit λ , every $\aleph_\beta$ with $\beta < \lambda$ is below the supremum. It also proves continuity at limits,

$$\aleph_\lambda = \bigcup_{\beta < \lambda} \aleph_\beta.$$

Thus the aleph hierarchy is not merely notation: it is a transfinite recursively defined enumeration of all infinite initial ordinals.

Point specific to this result. Set $\aleph_0 = \omega$, take the next initial ordinal at successors, and take the least initial ordinal above all previous stages at limits. Transfinite recursion makes this definition uniform over all ordinals. $\square$

How to read the symbols: why beth notation is deferred

The set-valued power-set recursion is available in ZF, but the familiar beth sequence is a sequence of *cardinals*. To avoid using the still-undefined Axiom of Choice in a formal statement, the beth construction and its limit-cardinal calculation are postponed to Volume V, Chapter 2, immediately after Choice and infinite-cardinal arithmetic have been established.

Worked example

Worked Example 27.4 (Hartogs for a finite set). If X has three elements, every well-order on a subset has order type at most 3, while the ordinal 4 cannot inject into X . Hartogs' theorem generalizes this elementary observation without first well-ordering X itself.

A useful warning

In bare ZF, do not write $|X| = \aleph_\alpha$ for an arbitrary set X unless its well-orderability has been established.

Looking ahead

The next chapter, *Well-founded relations and collapse*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 28

Well-founded relations and collapse

Definitions and setup

Formal definitions used in this chapter

Field and predecessor set of a relation. For a relation R , its field is the set of all objects appearing in either coordinate. For a in the field, $\text{Pred}_R(a) = \{b : bRa\}$.

Well-founded and extensional relation. R on A is well founded when every nonempty $B \subseteq A$ has an R -minimal member. It is extensional when distinct $a, a' \in A$ have distinct predecessor sets.

Mostowski collapse and transitive collapse. A collapse of (A, R) is a bijection $\pi : A \rightarrow M$ onto a transitive set M such that

$$bRa \iff \pi(b) \in \pi(a) \quad (a, b \in A).$$

Equivalently, it is an isomorphism $\pi : (A, R) \rightarrow (M, \in)$. The Mostowski recursion is

$$\pi(a) = \{\pi(b) : bRa\}.$$

When it exists, $M = \pi[A]$ is called the transitive collapse target.

Finite theory fragment, transitive model, and external absoluteness. A finite fragment of a theory is a finite set of its axioms, including only finitely many selected instances from any schemes. A set M is a transitive model of such a fragment when it is transitive and $(M, \in)$ satisfies each axiom in the fragment. A construction is absolute between M and the ambient universe when both computations produce the same set/function for parameters from M . “Externally well founded” means well founded in the ambient universe, not merely according to M .

A binary relation R on A is well founded when every nonempty subset of A has an R -minimal member, and it is extensional when distinct points have distinct predecessor sets. The theorem below will construct the Mostowski collapse recursively; its proof establishes that well-foundedness licenses the recursion, extensionality makes the resulting map injective, and the image is transitive. None of those consequences is being assumed in this setup paragraph.

How this chapter fits together

Mostowski collapse replaces an abstract well-founded extensional relation by ordinary membership on a transitive set. The collapsing map is defined recursively from predecessor images. This theorem is the main device by which elementary hulls later become transitive models and, in L , actual constructible levels.

Theorem 28.1 (Mostowski collapse theorem). *If R is a well-founded extensional relation on a set A , then there is a unique transitive set M and a unique isomorphism $\pi : (A, R) \cong (M, \in)$, given recursively by*

$$\pi(a) = \{\pi(b) : bRa\}.$$

Proof. Let (A, R) be well founded and extensional. Define by well-founded recursion

$$\pi(a) = \{\pi(b) : bRa\}.$$

Let $M = \pi[A]$. If $x = \pi(b) \in \pi(a)$, then $x \in M$, so M is transitive. Moreover

$$bRa \iff \pi(b) \in \pi(a).$$

It remains to prove injectivity. Suppose $\pi(a) = \pi(a')$. For each bRa , $\pi(b) \in \pi(a)$, so $\pi(b) = \pi(b')$ for some $b'Ra'$. By induction on the well-founded predecessor relation, equality of collapse values implies $b = b'$. Thus a and a' have the same R -predecessors; extensionality yields $a = a'$. Hence π is an isomorphism

$$(A, R) \cong (M, \in),$$

called the Mostowski collapse.

Point specific to this result. The collapse map is forced recursively by $\pi(a) = \{\pi(b) : bRa\}$. Extensionality is exactly the hypothesis that makes this map injective; well-foundedness is exactly what makes the recursion legitimate. $\square$

Corollary 28.2 (Uniqueness of the transitive collapse). *If $\pi : (A, R) \cong (M, \in)$ and $\pi' : (A, R) \cong (M', \in)$ are collapses onto transitive sets, then $M = M'$ and $\pi = \pi'$.*

Proof. Suppose $\pi : (A, R) \rightarrow (M, \in)$ and $\rho : (A, R) \rightarrow (N, \in)$ are two transitive collapses. We prove $\pi(a) = \rho(a)$ for every $a \in A$ by well-founded induction. Assume equality holds for all bRa . Then

$$\pi(a) = \{\pi(b) : bRa\} = \{\rho(b) : bRa\} = \rho(a).$$

Therefore $\pi = \rho$, and consequently $M = \pi[A] = \rho[A] = N$. Thus both the collapse map and the resulting transitive set are unique. The proof uses only the recursive equation characterizing a collapse; once that equation is established, no additional appeal to extensionality is needed at the uniqueness stage.

Point specific to this result. Any isomorphism into a transitive membership structure must satisfy the same recursion $a \mapsto \{\pi(b) : bRa\}$. Uniqueness of well-founded recursion therefore gives uniqueness of both the collapse map and its transitive range. $\square$

Proposition 28.3 (Absoluteness of collapse data in the basic transitive setting). *Let M be a transitive set model of ZF, and suppose $(A, R) \in M$. Assume that R is externally well founded and extensional on A . Then the collapse map π^M constructed inside M is the same set function as the ambient Mostowski collapse π ; in particular, M and the ambient universe obtain the same transitive collapse target.*

Proof.

Formal derivation spine

Formal spine: uniqueness compares the two recursions.

$$\pi(a) = \{\pi(b) : bRa\}, \quad \pi^M(a) = \{\pi^M(b) : bRa\}.$$

External well-founded induction yields $\pi^M(a) = \pi(a)$ for all $a \in A$.

Because M is transitive and $A, R \in M$, the externally interpreted sets A and R are the same objects inside and outside M . Because $M \models \text{ZF}$, it can carry out the well-founded-recursion construction used in the preceding theorem and obtain a function π^M satisfying

$$\pi^M(a) = \{\pi^M(b) : bRa\} \quad (a \in A). \quad (1)$$

The ambient Mostowski theorem gives a function π satisfying

$$\pi(a) = \{\pi(b) : bRa\}. \quad (2)$$

We compare them by external well-founded induction on R . Suppose $\pi^M(b) = \pi(b)$ for every bRa . Transitivity of M makes the predecessor set $\{b \in A : bRa\}$ the same in both computations, so (1) and (2) give

$$\pi^M(a) = \{\pi^M(b) : bRa\} = \{\pi(b) : bRa\} = \pi(a).$$

Thus $\pi^M = \pi$ on A . Their ranges are consequently identical. The external well-foundedness hypothesis is explicit: a transitive model can contain a relation it incorrectly regards as well founded only when the ambient relation is actually ill founded, and the present comparison does not cover that situation. $\square$

Worked Theorem 28.4 (Explicit collapse of finite and countable coded well-founded relations). *Let (A, R) be a finite or countable coded well-founded extensional relation. Define its rank by*

$$\text{rk}_R(a) = \sup\{\text{rk}_R(b) + 1 : bRa\}.$$

Then the collapse map is computed in rank order by

$$\pi(a) = \{\pi(b) : bRa\};$$

every value is determined at stage $\text{rk}_R(a) + 1$, even when the total relation has transfinite countable rank.

Proof. Consider the finite extensional relation on $A = \{a, b, c\}$ given by

$$aRb, \quad aRc, \quad bRc.$$

Its ranks are

$$\text{rk}_R(a) = 0, \quad \text{rk}_R(b) = 1, \quad \text{rk}_R(c) = 2.$$

The recursive collapse is therefore computed in that order:

$$\pi(a) = \emptyset,$$

$$\pi(b) = \{\pi(a)\} = \{\emptyset\},$$

$$\pi(c) = \{\pi(a), \pi(b)\} = \{\emptyset, \{\emptyset\}\}.$$

Thus (A, R) collapses to the transitive set $3 = \{0, 1, 2\}$, with a, b, c corresponding to $0, 1, 2$.

For a countable coded relation the same calculation may require transfinite countable stages. Let

$$A_\alpha := \{a \in A : \text{rk}_R(a) < \alpha\}.$$

At stage $\alpha + 1$, define $\pi(a)$ for every a of rank α from values already defined on A_α . At a limit stage take the union of the previous partial functions. Well-foundedness ensures every a has an ordinal rank, so each individual value is assigned at a definite stage; no claim is made that the entire countable computation finishes after finitely many stages. $\square$

Worked example

Worked Example 28.5 (Collapsing a finite relation). Let aRb and suppose a has no predecessors. Then $\pi(a) = \emptyset$ and $\pi(b) = \{\emptyset\}$. If a third point c has predecessors a, b , then $\pi(c) = \{\emptyset, \{\emptyset\}\}$. The abstract diagram has become a transitive set with membership reproducing R .

A useful warning

Extensionality is essential for injectivity of the collapse. Two distinct points with exactly the same predecessor set would otherwise collapse to the same set.

End-of-volume perspective

The transfinite machinery now supplies the ambient language for inner models: ranks control size of constructions, ordinals index recursion, and collapse turns abstract well-founded structures into transitive sets.

Part V

Volume V: Choice, Models, Satisfaction, and Absoluteness

Preface

This volume develops the model-theoretic set theory needed later. Internal versus external viewpoints are kept explicit, especially for cardinality and satisfaction.

Position in the series

Volume V develops satisfaction, absoluteness, reflection, and elementary hulls. These tools feed directly into the construction and analysis of L , and they also clarify the model-theoretic viewpoint used in forcing.

Sources and further reading

The chapters on satisfaction, absoluteness, reflection, and elementary submodels prepare the exact model-theoretic infrastructure needed by constructibility and forcing; see [13, 14].

Notation for this volume

$M \models \varphi$ always refers to the satisfaction relation of the set-sized structure M . $M \preceq N$ denotes elementary substructure.

How to read symbolic arguments in this volume

When a formula appears, read it in layers rather than as one visual block. First identify the *ambient language*: are we speaking inside a formal theory, about a structure, or in the surrounding metatheory? Next mark the objects being manipulated (formulas, codes, sets, names, conditions, ordinals) and the operation being applied to them. Finally check the side conditions—free variables, rank bounds, compatibility, genericity, transitivity, or consistency hypotheses. Whenever a displayed derivation changes level, the text says so explicitly. A displayed line is therefore meant to be read like a line of well-commented code: the symbols perform the operation, and the surrounding prose records its type, preconditions, and purpose.

Chapter 29

Axiom of Choice and equivalents

Definitions and setup

Formal definitions used in this chapter

Choice function and Axiom of Choice (AC). If $\mathcal{A}$ is a set of nonempty sets, a choice function on $\mathcal{A}$ is a function f with

$$\text{dom}(f) = \mathcal{A}, \quad f(A) \in A \quad (A \in \mathcal{A}).$$

AC states that every such $\mathcal{A}$ has a choice function.

Partially ordered set, chain, upper bound, and maximal element. A poset $(P, \leq)$ has a reflexive, antisymmetric, transitive relation $\leq$. A chain is a subset $C \subseteq P$ whose members are pairwise comparable. An upper bound for C is $u \in P$ with $c \leq u$ for every $c \in C$. An element $m \in P$ is maximal when $m \leq p \Rightarrow p = m$. A maximal chain is a chain not properly contained in another chain.

Named Choice principles. The name *Zorn's Lemma* denotes the principle: if every chain in a nonempty poset P has an upper bound in P , then P has a maximal element. A *well-order* of X is a linear order in which every nonempty subset has a least element; the *well-ordering theorem* denotes the principle that every set admits a well-order. The *Hausdorff maximal principle* denotes the principle that every chain in a poset is contained in a maximal chain. These are definitions of named principles, not additional axioms being assumed over ZF; the following theorems prove their equivalence with AC.

The Axiom of Choice says every family of nonempty sets has a choice function. The well-ordering theorem says every set admits a well-order. Zorn's lemma says a partially ordered set in which every chain has an upper bound has a maximal element. Hausdorff's maximal principle says every chain is contained in a maximal chain. The theorems in this chapter prove over ZF that these named principles are equivalent; the definitions above do not assume those equivalences, and no proof may use one direction before it has been established.

How this chapter fits together

Choice is introduced through several equivalent maximality principles. The equivalences are useful because later arguments naturally ask for different forms: a choice function, a well-order, a maximal partial object, or a maximal chain. The proofs also provide a template for identifying exactly where Choice enters cardinal arithmetic.

Theorem 29.1 (AC implies the well-ordering theorem). *Assuming the Axiom of Choice, every set X admits a well-ordering.*

Proof. Let X be a set and assume AC. By Hartogs' theorem there is an ordinal $h(X)$ for which no injection $h(X) \rightarrow X$ exists. Apply AC to the family of nonempty subsets of X and fix a choice function

$$c : \mathcal{P}(X) \setminus \{\emptyset\} \longrightarrow X, \quad c(A) \in A.$$

We recursively choose new points of X . As long as earlier choices have not exhausted X , put

$$x_\alpha = c(X \setminus \{x_\beta : \beta < \alpha\}).$$

Transfinite recursion makes this definition legitimate at every stage at which the displayed complement is nonempty. The construction cannot continue through all $\alpha < h(X)$: if it did, the map $\alpha \mapsto x_\alpha$ would be an injection $h(X) \rightarrow X$, contradicting the defining property of the Hartogs ordinal. Hence there is a least ordinal $\gamma < h(X)$ such that

$$X = \{x_\beta : \beta < \gamma\}.$$

The map $e : \gamma \rightarrow X$, $e(\beta) = x_\beta$, is therefore bijective. Transport the ordinal order on γ to X :

$$x_\alpha <_X x_\beta \iff \alpha < \beta.$$

Every nonempty $Y \subseteq X$ corresponds under e^{-1} to a nonempty subset of the ordinal γ , hence has a least index. Its image is the $<_X$ -least element of Y . Thus $<_X$ is a well-order of X . Notice where Choice enters: it supplies one coherent selector c that can be used at every transfinite stage; Hartogs prevents the recursion from choosing distinct points forever. $\square$

Theorem 29.2 (Well-ordering theorem implies AC). *If every set can be well ordered, then every family $\mathcal{A}$ of nonempty sets has a choice function f with $f(A) \in A$ for every $A \in \mathcal{A}$.*

Proof. Assume every set can be well ordered. Let $\mathcal{A}$ be a set of nonempty sets. We need a single function f with domain $\mathcal{A}$ and $f(A) \in A$ for all $A \in \mathcal{A}$.

Form the union

$$U := \bigcup \mathcal{A}.$$

By hypothesis, choose a well-order $<$ of U . Every $A \in \mathcal{A}$ is a nonempty subset of U , so it has a unique $<$ -least element. Define

$$f(A) := \min_{<} A.$$

The graph of f is the set

$$\{\langle A, x \rangle \in \mathcal{A} \times U : x \in A \wedge \forall y \in A (x \leq y)\},$$

which exists by Separation. Uniqueness of least elements makes this graph functional, and the nonemptiness of each A makes its domain all of $\mathcal{A}$. Hence f is a choice function.

No additional use of Choice is hidden in the construction: the only global selection is the assumed well-order of U . Once that order is fixed, the value $f(A)$ is defined uniquely by a first-order property rather than selected afresh for each A . $\square$

Theorem 29.3 (AC is equivalent to Zorn's lemma). *Over ZF, the Axiom of Choice and Zorn's Lemma are equivalent. Zorn's Lemma states: if every chain in a partially ordered set P has an upper bound in P , then P has a maximal element.*

Proof. We prove both implications over ZF.

$AC \Rightarrow \text{Zorn}$. By the preceding theorem, AC well-orders every set. Let $(P, \leq)$ be a poset in which every chain has an upper bound. Well-order P as $\langle p_\alpha : \alpha < \theta \rangle$. By transfinite recursion build a chain C : at stage α , add p_α exactly when it is comparable with every element already chosen. The resulting C is a maximal chain: any $p \in P$ comparable with all of C would have been added when its index was considered. By hypothesis, C has an upper bound $u \in P$. If u were not maximal in P , choose $v > u$. Since u bounds C , the element v is comparable with every member of C , so $C \cup \{v\}$ would be a strictly larger chain, contradiction. Thus u is maximal.

$Zorn \Rightarrow AC$. Let $\mathcal{A}$ be a family of nonempty sets. Let P be the set of partial choice functions:

$$P = \{f : \text{dom}(f) \subseteq \mathcal{A}, \forall A \in \text{dom}(f) f(A) \in A\},$$

ordered by inclusion. If $C \subseteq P$ is a chain, then $\bigcup C$ is a function because members of a chain agree on overlaps, and it is again a partial choice function. Hence every chain has an upper bound. Zorn gives a maximal $f \in P$. If some $A \in \mathcal{A} \setminus \text{dom}(f)$ remained, choose one $a \in A$ (one existential choice, not AC) and extend f by $f(A) = a$, contradicting maximality. Therefore $\text{dom}(f) = \mathcal{A}$, so f is a choice function. $\square$

Theorem 29.4 (AC is equivalent to the Hausdorff maximal principle). *Over ZF, the Axiom of Choice is equivalent to the Hausdorff maximal principle: every chain in a partially ordered set is contained in a maximal chain.*

Proof. Again we prove the two directions over ZF.

Assume AC. By the equivalence just proved, Zorn's lemma is available. Given a poset $(P, \leq)$ and a chain $C_0 \subseteq P$, let

$$\mathcal{C} = \{C \subseteq P : C \text{ is a chain and } C_0 \subseteq C\},$$

ordered by inclusion. A chain $\mathcal{D} \subseteq \mathcal{C}$ has upper bound $\bigcup \mathcal{D}$: any two of its elements lie together in some larger member of the inclusion-chain, so they are comparable in P . Zorn therefore gives a maximal element of $\mathcal{C}$, namely a maximal chain containing C_0 . Taking $C_0 = \emptyset$ yields Hausdorff's maximal principle.

Conversely assume Hausdorff's maximal principle. To recover Zorn, let $(P, \leq)$ be a poset in which every chain has an upper bound. Choose a maximal chain $C \subseteq P$. Let u be an upper bound of C . If u were not maximal in P , there would exist $v \in P$ with $u < v$. Every $c \in C$ satisfies $c \leq u < v$, hence $C \cup \{v\}$ is a strictly larger chain, contradiction. Thus u is maximal and Zorn's lemma holds. The previous theorem then gives AC. Hence AC and Hausdorff's maximal principle are equivalent. $\square$

Worked example

Worked Example 29.5 (Choosing by a well-order). If a family $\mathcal{A}$ of nonempty sets has union U and U is well-ordered, define $f(A)$ to be the least element of A for each $A \in \mathcal{A}$. This one-line construction proves the easy direction from the well-ordering theorem to AC.

A useful warning

Zorn's lemma requires every chain in the chosen poset to have an upper bound in that poset. Verifying this closure condition is part of the proof, not an automatic formality.

Looking ahead

The next chapter, *Cardinal arithmetic with choice*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 30

Cardinal arithmetic with choice

Definitions and setup

Formal definitions used in this chapter

Cardinal comparison notation under Choice. For sets A, B , $|A| \leq |B|$ means that an injection $A \rightarrow B$ exists. The preceding chapter proved from AC that every set can be well ordered, so in the present AC context each cardinal is represented by its initial ordinal. Proposition 30.1 proves comparability of cardinals; comparability is not included in this definition.

Function sets and cardinal sum/product/exponentiation. For sets X, Y ,

$${}^Y X := \{f : f \text{ is a function } Y \rightarrow X\}.$$

For an indexed family of cardinals $\langle \kappa_i : i \in I \rangle$, use the initial ordinals themselves and define

$$\sum_{i \in I} \kappa_i := |\{\langle i, \alpha \rangle : i \in I, \alpha < \kappa_i\}|,$$

$$\prod_{i \in I} \kappa_i := |\{f : \text{dom}(f) = I \wedge \forall i \in I (f(i) < \kappa_i)\}|.$$

For cardinals κ, λ , define $\kappa^\lambda := |{}^\lambda \kappa|$. These definitions avoid any unproved claim about independence from arbitrarily chosen representatives.

Unbounded/cofinal subset and cofinality. For any ordinal α , a set $C \subseteq \alpha$ is *unbounded* (or *cofinal*) in α when

$$\forall \beta < \alpha \exists \gamma \in C (\beta \leq \gamma).$$

The *cofinality* $\text{cf}(\alpha)$ is the least order type of a cofinal subset of α . This least order type exists: for $\alpha > 0$, the set α itself is cofinal in α ; every subset of the ordinal α inherits a well-order and hence has an order type by Volume IV; Power Set and Replacement collect the relevant order types into a set of ordinals; and a nonempty set of ordinals has a least member. Set $\text{cf}(0) := 0$. If $\alpha = \beta + 1$, then $\{\beta\}$ is cofinal and $\text{cf}(\alpha) = 1$. The substantive regularity theorem below concerns limit ordinals.

Regular cardinal. An infinite cardinal κ is regular when $\text{cf}(\kappa) = \kappa$; otherwise it is singular.

Disjoint union. $X \sqcup Y$ denotes a tagged union, for example $(\{0\} \times X) \cup (\{1\} \times Y)$, so the two copies remain disjoint.

Under AC every set can be well ordered, so every cardinal is represented by an initial ordinal and arbitrary cardinals are comparable. Cofinality $\text{cf}(\alpha)$, defined above for every ordinal α , is the least order type of an unbounded subset of α . König's theorem is a diagonal inequality between sums and products of cardinals and is the basic obstruction controlling exponentiation. Choice assumptions are carried explicitly into every

cardinal-arithmetic statement.

How this chapter fits together

Once every set can be well-ordered, cardinals may be represented by initial ordinals and compared linearly. Infinite-cardinal addition and multiplication simplify dramatically, but exponentiation and cofinality retain genuine structure. König's theorem supplies a diagonal obstruction that will later control cardinal exponentiation.

Proposition 30.1 (Cardinal comparability under AC). *Assuming AC, for any sets A, B , either $|A| \leq |B|$ or $|B| \leq |A|$. Equivalently, the cardinal numbers are linearly ordered by injection.*

Proof. Assume AC and let A, B be arbitrary sets. By the well-ordering theorem choose well-orders of A and B . By the ordinal representation theorem from Volume IV, there are unique ordinals α, β and bijections

$$e_A : \alpha \xrightarrow{\cong} A, \quad e_B : \beta \xrightarrow{\cong} B.$$

Replace α, β , if necessary, by the least ordinals equipotent with A, B ; these are their initial-ordinal representatives, so $|A| = \alpha$ and $|B| = \beta$ as cardinals.

Ordinal trichotomy gives exactly one of

$$\alpha \in \beta, \quad \alpha = \beta, \quad \beta \in \alpha.$$

If $\alpha \leq \beta$, the inclusion $i : \alpha \hookrightarrow \beta$ is injective. Transport it through the two bijections:

$$A \xrightarrow{e_A^{-1}} \alpha \xrightarrow{i} \beta \xrightarrow{e_B} B.$$

This is an injection $A \hookrightarrow B$, so $|A| \leq |B|$. The case $\beta \leq \alpha$ is symmetric. Thus any two cardinals are comparable.

The proof shows exactly why Choice is needed: ordinal trichotomy compares only well-orderable representatives. AC is what ensures every set has such a representative. Without AC, arbitrary cardinals need not be linearly ordered by injection. $\square$

Theorem 30.2 (Arithmetic of infinite cardinals under AC). *Assume AC and let κ be an infinite cardinal. Then*

$$\kappa + \kappa = \kappa \cdot \kappa = \kappa.$$

Equivalently, if X is any infinite set, then $X \sqcup X$ and $X \times X$ are equipotent with X .

Proof. By AC it suffices to prove the assertion for an infinite initial ordinal κ . The map into either summand proves $\kappa \leq \kappa + \kappa$, and $\alpha \mapsto (\alpha, 0)$ proves $\kappa \leq |\kappa \times \kappa|$. The substantive point is an injection in the opposite direction.

We now prove by transfinite induction that for every infinite initial ordinal κ ,

$$|\kappa \times \kappa| = \kappa.$$

Here is the induction mechanism. Well-order $\kappa \times \kappa$ by first comparing $\max\{\alpha, \beta\}$, and within each level use a fixed lexicographic tie-break. For $\gamma < \kappa$, the initial segment below the level γ is contained in

$$(\gamma + 1) \times (\gamma + 1).$$

Its cardinal is strictly below κ : if $\lambda = |\gamma + 1| < \kappa$, the induction hypothesis at the smaller infinite cardinal λ gives $\lambda^2 = \lambda < \kappa$ (the finite case is trivial). Hence every proper initial segment of this well-order has cardinality $< \kappa$. Recursively assign to each pair the least unused ordinal below κ . At each stage fewer than

κ ordinals have been used, so one remains. This yields an injection $\kappa \times \kappa \hookrightarrow \kappa$. Cantor–Bernstein gives $|\kappa \times \kappa| = \kappa$.

Since $\kappa \sqcup \kappa$ injects into $\kappa \times \{0, 1\} \subseteq \kappa \times \kappa$ and κ injects into the disjoint union, Cantor–Bernstein also gives $\kappa + \kappa = \kappa$. Transporting the bijections along a well-ordering of an arbitrary infinite set X gives the set formulation. $\square$

Proposition 30.3 (Beth hierarchy after Choice). *Assume AC. Define a class function $\alpha \mapsto \beth_\alpha$ by*

$$\beth_0 = \aleph_0, \quad \beth_{\alpha+1} = 2^{\beth_\alpha}, \quad \beth_\lambda = \sup_{\beta < \lambda} \beth_\beta$$

for every nonzero limit ordinal λ . This recursion exists uniquely. Equivalently, if

$$B_0 = \omega, \quad B_{\alpha+1} = \mathcal{P}(B_\alpha), \quad B_\lambda = \bigcup_{\beta < \lambda} (\{\beta\} \times B_\beta),$$

then $|B_\alpha| = \beth_\alpha$ for all α . The displayed union is tagged by β , so its summands are pairwise disjoint.

Proof. Because AC makes every set well-orderable, the power-set cardinal $2^\kappa = |\mathcal{P}(\kappa)|$ is defined for every cardinal κ . Transfinite recursion therefore yields a unique cardinal-valued class function satisfying the displayed successor and limit clauses; call its value at α $\beth_\alpha$.

For the set-valued realization, transfinite recursion constructs B_α . At successors,

$$|B_{\alpha+1}| = |\mathcal{P}(B_\alpha)| = 2^{|B_\alpha|}.$$

At a nonzero limit λ , put $\kappa = \sup_{\beta < \lambda} |B_\beta|$. The tagged union B_λ contains a copy of every B_β , hence $\kappa \leq |B_\lambda|$. Conversely choose injections $B_\beta \hookrightarrow \kappa$; then

$$B_\lambda \hookrightarrow \lambda \times \kappa.$$

The strictly increasing sequence $\beta \mapsto |B_\beta|$ injects λ into κ , so $|\lambda| \leq \kappa$. By the infinite-cardinal multiplication theorem already proved in this chapter,

$$|\lambda \times \kappa| \leq \kappa \cdot \kappa = \kappa.$$

Thus $|B_\lambda| = \kappa$, which is exactly the limit equation. Induction gives $|B_\alpha| = \beth_\alpha$ for every α . $\square$

Proposition 30.4 (Cofinality basics). *For a limit ordinal α , $\text{cf}(\alpha)$ is the least order type of an unbounded subset of α . It is a cardinal, $\text{cf}(\alpha) \leq |\alpha|$, and $\text{cf}(\alpha)$ is regular.*

Proof. Let α be a limit ordinal. A subset $C \subseteq \alpha$ is *cofinal* if

$$\forall \beta < \alpha \exists \gamma \in C (\beta \leq \gamma).$$

Define $\text{cf}(\alpha)$ to be the least order type of a cofinal subset of α . Equivalently, it is the least ordinal κ for which there exists a cofinal map $f : \kappa \rightarrow \alpha$.

First, the least such ordinal is a cardinal. If $\kappa = \text{cf}(\alpha)$ were equipotent with a smaller ordinal $\lambda < \kappa$, compose a bijection $\lambda \rightarrow \kappa$ with a cofinal map $\kappa \rightarrow \alpha$; this would give a cofinal map from λ , contradicting minimality. Thus κ is an initial ordinal.

Second, $\text{cf}(\alpha) \leq |\alpha|$: the identity map on α is cofinal, so the least size of a cofinal domain cannot exceed the cardinality of α .

Finally, $\kappa = \text{cf}(\alpha)$ is regular. Suppose $\text{cf}(\kappa) = \lambda < \kappa$. Choose cofinal maps

$$g : \lambda \rightarrow \kappa, \quad f : \kappa \rightarrow \alpha.$$

Replacing f by the increasing map $\xi \mapsto \sup f[\xi + 1]$ if needed, we may assume it is increasing and cofinal. Then $f \circ g : \lambda \rightarrow \alpha$ is cofinal: given $\beta < \alpha$, choose $\xi < \kappa$ with $\beta \leq f(\xi)$, then choose $\eta < \lambda$ with $\xi \leq g(\eta)$, so $\beta \leq f(g(\eta))$. This contradicts minimality of κ . Hence $\text{cf}(\kappa) = \kappa$. $\square$

Theorem 30.5 (König's theorem and its cardinal-exponent corollaries). *Assume AC. If $\kappa_i < \lambda_i$ for every $i \in I$, then*

$$\sum_{i \in I} \kappa_i < \prod_{i \in I} \lambda_i.$$

Consequently, for every infinite cardinal κ ,

$$\kappa^{\text{cf}(\kappa)} > \kappa.$$

Proof. Assume $\kappa_i < \lambda_i$ for each $i \in I$. Realize the cardinal sum as a disjoint union

$$S = \bigsqcup_{i \in I} K_i, \quad |K_i| = \kappa_i,$$

and the product as $P = \prod_{i \in I} L_i$, with $|L_i| = \lambda_i$. Suppose toward contradiction that there is a surjection

$$F : S \twoheadrightarrow P.$$

For each i , look only at the i -th block $K_i \times \{i\} \subseteq S$. The set of i -th coordinates occurring among its images is

$$R_i := \{F(x, i)(i) : x \in K_i\} \subseteq L_i.$$

Because $|R_i| \leq \kappa_i < \lambda_i = |L_i|$, choose $y_i \in L_i \setminus R_i$. AC is available in this volume, so the choices y_i form a point $y = (y_i)_{i \in I} \in P$. If $F(x, i) = y$, then its i -th coordinate would be $y_i \in R_i$, contradiction. Hence F is not surjective. In cardinal notation,

$$\sum_{i \in I} \kappa_i < \prod_{i \in I} \lambda_i.$$

For the exponentiation corollary, let $\lambda = \text{cf}(\kappa)$ and choose an increasing cofinal sequence $\langle \kappa_i : i < \lambda \rangle$ of cardinals below κ . Put $\lambda_i = \kappa_i^+ \leq \kappa$. Cofinality gives

$$\sum_{i < \lambda} \kappa_i = \kappa.$$

König's inequality yields

$$\kappa < \prod_{i < \lambda} \kappa_i^+ \leq \prod_{i < \lambda} \kappa = \kappa^\lambda = \kappa^{\text{cf}(\kappa)}.$$

This is the claimed strict inequality. □

Worked example

Worked Example 30.6 (A cofinality calculation). The natural numbers are cofinal in ω , so $\text{cf}(\omega) = \omega$. By contrast, any finite subset of ω is bounded. For a successor ordinal $\alpha + 1$, the singleton $\{\alpha\}$ is cofinal, so its cofinality is 1 as an ordered set even though cardinal cofinality conventions focus on infinite initial ordinals.

A useful warning

The identity $\kappa + \kappa = \kappa$ for infinite cardinals is a theorem using well-orderability; it is not a theorem about arbitrary infinite sets in ZF.

Looking ahead

The next chapter, *Satisfaction for set-sized structures*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 31

Satisfaction for set-sized structures

Definitions and setup

Formal definitions used in this chapter

Set-sized structure and finite adequate assignment. A set-sized $\mathcal{L}$ -structure M has a set $|M|$ as its domain and set-coded interpretations of every constant, function, and relation symbol of $\mathcal{L}$. Volume I used total assignments $a : \text{Var} \rightarrow |M|$. For the set-coding construction in this volume, a *finite assignment* is a finite partial function $u : \text{Var} \rightarrow |M|$. It is *adequate for a formula* φ when $\text{FV}(\varphi) \subseteq \text{dom}(u)$. For $m \in |M|$, define

$$u[x \mapsto m] := (u \upharpoonright (\text{dom}(u) \setminus \{x\})) \cup \{\langle x, m \rangle\}.$$

By the Free-Variable Lemma from Volume I, the truth of φ under any total extension of an adequate finite assignment u is independent of the chosen extension. Thus finite adequate assignments give an equivalent, set-sized coding of satisfaction.

Term evaluation. If the finite assignment u contains every variable occurring in the term t , then $t^M[u]$ is defined recursively: a variable x is evaluated by $u(x)$, a constant by its interpretation in M , and

$$(f(t_1, \dots, t_n))^M[u] = f^M(t_1^M[u], \dots, t_n^M[u]).$$

Tarski satisfaction clauses. For every formula φ and every finite assignment u adequate for φ , define $M, u \models \varphi$ by recursion on the exact primitive syntax fixed in Volume I:

$$\begin{aligned} M, u \models t_1 = t_2 &\iff t_1^M[u] = t_2^M[u], \\ M, u \models R(t_1, \dots, t_n) &\iff (t_1^M[u], \dots, t_n^M[u]) \in R^M, \\ M, u \models \neg\psi &\iff M, u \not\models \psi, \\ M, u \models (\psi \wedge \theta) &\iff (M, u \models \psi \text{ and } M, u \models \theta), \\ M, u \models (\psi \vee \theta) &\iff (M, u \models \psi \text{ or } M, u \models \theta), \\ M, u \models (\psi \rightarrow \theta) &\iff (M, u \not\models \psi \text{ or } M, u \models \theta), \\ M, u \models \exists x \psi &\iff \exists m \in |M| \ M, u[x \mapsto m] \models \psi, \\ M, u \models \forall x \psi &\iff \forall m \in |M| \ M, u[x \mapsto m] \models \psi. \end{aligned}$$

Thus no primitive connective or quantifier is silently treated as an abbreviation. The notation $\leftrightarrow$, when it occurs inside an object-language formula, retains the Volume-I (BI) expansion into $\rightarrow$ and $\wedge$. The theorem below proves that there is a unique set relation satisfying these clauses on pairs $(\ulcorner \varphi \urcorner, u)$ with u adequate for φ . $\text{Sat}_M(\ulcorner \varphi \urcorner, u)$ is its code-level form.

Coded structure. A coded structure is a set containing a code for its domain together with set codes for the interpretations of its function, relation, and constant symbols, so the preceding recursion can be expressed uniformly by a set-theoretic formula.

Relativization. For a formula φ of set theory and a class or set M , φ^M is obtained recursively by replacing each formerly unbounded $\exists x$ by $\exists x \in M$ and each formerly unbounded $\forall x$ by $\forall x \in M$, leaving bounded quantifiers unchanged. $V \models \varphi$ denotes truth in the ambient universe.

Parameter tuple. $\bar{a} \in M^{<\omega}$ denotes a finite sequence of elements of M , coded as a set as in Volume IV.

For a set-sized structure M , the satisfaction relation is itself a set relation on formula codes and finite assignments. It is defined recursively on formula complexity. Tarski's later undefinability theorem forbids an arithmetical truth predicate for the entire standard arithmetic universe; there is no conflict because here the structure is a fixed set and satisfaction is constructed externally in set theory.

How this chapter fits together

Set theory can define satisfaction for any set-sized structure because formulas and assignments form sets and the recursion on formula complexity is set-sized. This is the technical basis for definability, relativization, reflection, and L . It does not contradict Tarski: no full truth predicate for the entire universe is being defined inside that universe.

Theorem 31.1 (Recursive construction of satisfaction for set-sized first-order structures). *For every set-sized $\mathcal{L}$ -structure M , there is a unique set relation $\text{Sat}_M(\ulcorner \varphi \urcorner, u)$ between formula codes and finite assignments adequate for those formulas, satisfying the Tarski recursion for atomic formulas, Boolean connectives, and quantifiers.*

How to read the symbols: satisfaction is recursive evaluation

Think of $\text{Sat}_M(\ulcorner \varphi \urcorner, s)$ as an interpreter whose inputs are a formula code and an assignment. Atomic formulas call the structure's interpretations of function and relation symbols; Boolean connectives recursively call the interpreter on subformulas; $\exists x$ loops over the set-sized domain. Tarski's later undefinability theorem does not conflict with this construction because the interpreter is built externally for one fixed set-sized structure.

Proof. Fix a set-sized $\mathcal{L}$ -structure M . Let $\text{Form}_{\mathcal{L}}$ be the set of formula codes and let $\text{Asn}_M^{<\omega}$ be the set of finite partial functions from variables to $|M|$. We define

$$\text{Sat}_M \subseteq \text{Form}_{\mathcal{L}} \times \text{Asn}_M^{<\omega}$$

by recursion on formula complexity.

For an atomic equality $t = u$, set

$$\text{Sat}_M(\ulcorner t = u \urcorner, s) \iff t^M[s] = u^M[s].$$

For an atomic relation $R(t_1, \dots, t_n)$, use membership of the tuple of term values in R^M . For the six primitive logical constructors fixed in Volume I impose the corresponding Tarski clauses:

$$\text{Sat}_M(\ulcorner \neg \varphi \urcorner, s) \iff \neg \text{Sat}_M(\ulcorner \varphi \urcorner, s),$$

$$\text{Sat}_M(\ulcorner \varphi \wedge \psi \urcorner, s) \iff \text{Sat}_M(\ulcorner \varphi \urcorner, s) \wedge \text{Sat}_M(\ulcorner \psi \urcorner, s),$$

$$\text{Sat}_M(\ulcorner \varphi \vee \psi \urcorner, s) \iff \text{Sat}_M(\ulcorner \varphi \urcorner, s) \vee \text{Sat}_M(\ulcorner \psi \urcorner, s),$$

$$\text{Sat}_M(\ulcorner \varphi \rightarrow \psi \urcorner, s) \iff (\neg \text{Sat}_M(\ulcorner \varphi \urcorner, s) \vee \text{Sat}_M(\ulcorner \psi \urcorner, s)),$$

$$\text{Sat}_M(\ulcorner \exists x \varphi \urcorner, s) \iff \exists a \in |M| \text{Sat}_M(\ulcorner \varphi \urcorner, s[x \mapsto a]),$$

and

$$\text{Sat}_M(\ulcorner \forall x \varphi \urcorner, s) \iff \forall a \in |M| \text{Sat}_M(\ulcorner \varphi \urcorner, s[x \mapsto a]).$$

Thus the recursive proof treats exactly the primitive syntax used by the proof calculus; the symbol $\leftrightarrow$, when displayed, is first expanded by the Volume-I convention (BI). Every recursive call is made on a proper subformula, so the recursion is well founded. Since the formula codes, assignments, and each earlier stage are sets, Replacement and Separation collect the recursive stages into a set relation.

Uniqueness is proved by induction on formula complexity. If S and S' satisfy the same Tarski clauses, they agree on atomic formulas. Assuming agreement for the immediate subformulas, each Boolean or quantifier clause determines the next truth value uniquely. Hence $S = S'$. This gives the unique satisfaction relation Sat_M . $\square$

Theorem 31.2 (Definability of the satisfaction predicate over the ambient set theory). *There is a first-order formula of set theory $\text{Sat}(M, n, u)$ that, for every set-sized coded structure M , formula code $n = \ulcorner \varphi \urcorner$, and finite assignment u adequate for φ , expresses $M, u \models \varphi$.*

Proof. The preceding construction was uniform: none of its clauses depended on a special feature of one particular structure beyond the coded interpretations of its symbols. We now internalize that recursion in the language of set theory.

Code a structure M , a formula n , and a finite adequate assignment u as sets. A *satisfaction computation up to complexity k* is a set S whose domain consists of formula/assignment pairs of complexity at most k and which satisfies the atomic, Boolean, and quantifier clauses from the previous theorem. Each clause is first-order expressible in set theory because it quantifies only over the coded structure, the finite syntax tree, assignments, and elements of the set-sized domain.

Define

$$\text{Sat}(M, n, u)$$

to mean: u is a finite assignment adequate for the formula coded by n , and there exists a satisfaction computation S up to the complexity of n such that $(n, u) \in S$. By the uniqueness part of the recursive-construction theorem, two such computations agree on their common domain, so this definition does not depend on which witness S is chosen. Conversely the set recursion constructs such an S , so the formula has exactly the intended extension.

Therefore, for every coded set-sized structure M , every formula φ , and every finite assignment u adequate for φ ,

$$V \models \text{Sat}(M, \ulcorner \varphi \urcorner, u) \iff M, u \models \varphi.$$

The point is uniform definability in the ambient set theory, not a truth predicate for the whole universe. The domain being evaluated is the fixed set M , which is why there is no conflict with Tarski's undefinability theorem. $\square$

Lemma 31.3 (Relativization lemma). *Let M be a class/set and let φ be a formula of set theory. Its relativization φ^M is obtained by replacing $\exists x$ by $\exists x \in M$ and $\forall x$ by $\forall x \in M$. For transitive set models with the inherited membership relation,*

$$(M, \in) \models \varphi(\bar{a}) \iff V \models \varphi^M(\bar{a}).$$

Proof. We prove the equivalence by induction on the construction of φ . Atomic formulas are unchanged by relativization, and because M is transitive the inherited membership and equality relations agree with the ambient ones on parameters from M . Thus the assertion is true for atomic formulas.

The Boolean steps follow immediately from the induction hypothesis. For example,

$$(M, \in) \models \neg \psi(\bar{a}) \iff \neg((M, \in) \models \psi(\bar{a})) \iff V \models \neg \psi^M(\bar{a}).$$

The quantifier step is the substantive one. Suppose $\varphi(\bar{y}) = \exists x \psi(x, \bar{y})$. By the semantics of the structure $(M, \in)$,

$$(M, \in) \models \exists x \psi(x, \bar{a}) \iff \exists b \in M (M, \in) \models \psi(b, \bar{a}).$$

By the induction hypothesis this is equivalent to

$$\exists b \in M \ V \models \psi^M(b, \bar{a}),$$

which is exactly

$$V \models (\exists x \in M) \psi^M(x, \bar{a}) = (\exists x \psi)^M(\bar{a}).$$

Universal quantifiers are analogous or follow by negation. Hence

$$(M, \in) \models \varphi(\bar{a}) \iff V \models \varphi^M(\bar{a})$$

for all parameters $\bar{a} \in M$. □

Proposition 31.4 (Parameter bookkeeping for satisfaction and definability). *The satisfaction relation is uniform in the finite tuple of parameters: if $\varphi(x, \bar{y})$ is fixed, then the set it defines over M from parameters $\bar{a} \in M$ is determined by the code $\ulcorner \varphi \urcorner$ and a finite coded tuple $\bar{a}$.*

Proof. Fix a formula $\varphi(x, y_0, \dots, y_{n-1})$. If $\bar{a} = (a_0, \dots, a_{n-1}) \in M^n$, the subset of M defined by φ with parameters $\bar{a}$ is

$$X_{\varphi, \bar{a}} := \{x \in M : M \models \varphi(x, \bar{a})\}.$$

Let the displayed free variables be $x, y_0, \dots, y_{n-1}$. For a tested element $m \in M$, form the finite adequate assignment

$$u_{m, \bar{a}} := \{\langle x, m \rangle, \langle y_0, a_0 \rangle, \dots, \langle y_{n-1}, a_{n-1} \rangle\}.$$

Then the uniform satisfaction formula gives

$$X_{\varphi, \bar{a}} = \{m \in M : \text{Sat}(M, \ulcorner \varphi \urcorner, u_{m, \bar{a}})\}.$$

Thus the defining data consist of the natural-number code $\ulcorner \varphi \urcorner$ and the finite tuple $\bar{a}$. The finite assignment $u_{m, \bar{a}}$ is uniformly set-coded from the displayed free-variable list and tuple data, so this is a single first-order relation of the ambient set theory.

This bookkeeping matters later in the construction of $\text{Def}(M)$. There are only set-many formula codes and set-many finite tuples from a set M :

$$\omega \times M^{<\omega}$$

is a set. Replacement can therefore map each pair $(e, \bar{a})$ to the corresponding definable subset. Nothing like a proper-class collection of parameters is being hidden in the phrase “definable with parameters.” □

Worked example

Worked Example 31.5 (Relativizing a quantifier). If M is a set and φ is $\forall x \exists y (x \in y)$, its relativization is $\forall x \in M \exists y \in M (x \in y)$. The satisfaction recursion interprets the latter by ranging only over elements of M . This simple change is the core of every later “ $M \models \varphi$ ” calculation.

A useful warning

A satisfaction predicate for a set-sized model is an object of the surrounding universe. Do not turn it into a global truth predicate for V .

Looking ahead

The next chapter, *Absoluteness*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 32

Absoluteness

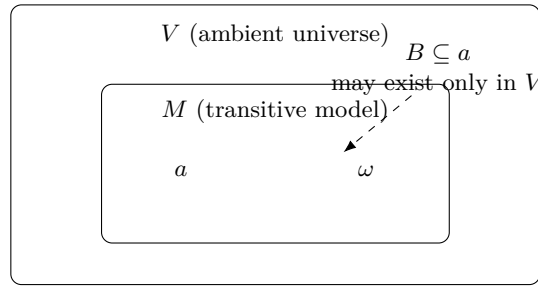


Figure 32.1: Bounded formulas are absolute because a transitive model sees the same members of an existing set. Power sets and cardinalities can differ because the ambient universe may contain additional subsets or functions.

Definitions and setup

Formal definitions used in this chapter

Transitive set/class and internal truth. M is transitive when $x \in y \in M \Rightarrow x \in M$. $M \models \varphi(\bar{a})$ means that φ is true in the structure $(M, \in)$ for parameters $\bar{a} \in M$; $V \models \varphi(\bar{a})$ denotes ambient truth.

Absoluteness. For $M \subseteq N$ and parameters $\bar{a} \in M$, a formula φ is absolute between M and N when

$$M \models \varphi(\bar{a}) \iff N \models \varphi(\bar{a}).$$

It is upward absolute when truth in M implies truth in N , and downward absolute when truth in N implies truth in M .

Bounded (Δ_0) formula. Every quantifier is of the form $\forall x \in y$ or $\exists x \in y$.

Internal computation notation. If an object is defined inside M , a superscript such as ω^M denotes the object that M computes from the same defining formula; equality $\omega^M = \omega$ is then an external comparison.

A statement is absolute between models when it has the same truth value in both. For transitive models, bounded quantifiers over an existing set range over the same elements, which is why Δ_0 formulas are absolute. Power Set and cardinality are typical nonabsolute notions because a smaller model may omit subsets or functions that exist externally. Every later absoluteness claim names its model hypotheses.

How this chapter fits together

Absoluteness asks whether an assertion has the same truth value in a transitive model and the ambient universe. Bounded formulas are absolute because their quantifiers range over elements of already present sets. Cardinality and power set are not: the smaller model may simply lack functions or subsets that exist externally.

Theorem 32.1 (Delta-0 absoluteness for transitive models). *If $M \subseteq N$ are transitive sets/classes and $\bar{a} \in M$, then every Δ_0 formula φ is absolute:*

$$M \models \varphi(\bar{a}) \iff N \models \varphi(\bar{a}).$$

Proof. Let $M \subseteq N$ be transitive and let $\bar{a} \in M$. We prove by induction on Δ_0 formulas that

$$M \models \varphi(\bar{a}) \iff N \models \varphi(\bar{a}).$$

Atomic equality and membership are absolute because both structures use the ambient equality and membership relations. Boolean connectives preserve equivalence by the induction hypothesis.

For the bounded existential step, suppose

$$\varphi(\bar{a}) \equiv \exists x \in b \psi(x, \bar{a}), \quad b \in M.$$

Transitivity of M gives $b \subseteq M$; since $M \subseteq N$, both structures quantify over exactly the same actual elements of b . Therefore

$$M \models \exists x \in b \psi(x, \bar{a})$$

iff there is $x \in b$ with $M \models \psi(x, \bar{a})$, iff by induction there is $x \in b$ with $N \models \psi(x, \bar{a})$, iff

$$N \models \exists x \in b \psi(x, \bar{a}).$$

Bounded universal quantifiers are identical or follow by negation. Since every quantifier in a Δ_0 formula is bounded, this completes the induction.

The proof also explains the boundary of the theorem. An unbounded quantifier may range over objects in $N \setminus M$, so the same argument does not apply to arbitrary formulas. Later non-absoluteness examples exploit exactly that difference. $\square$

Proposition 32.2 (Absoluteness of ordinals for transitive models). *If M is transitive and $\alpha \in M$, then $M \models \text{“}\alpha \text{ is an ordinal”}$ if and only if α is an ordinal in the ambient universe; moreover the order relation on such ordinals is the same.*

Proof. Use the standard set-theoretic definition

$$\text{Ord}(\alpha) \iff \alpha \text{ is transitive and } \in \text{ linearly orders } \alpha.$$

Every quantifier in this definition is bounded by α , so $\text{Ord}(\alpha)$ is Δ_0 . If M is transitive and $\alpha \in M$, the Δ_0 -absoluteness theorem gives

$$M \models \text{Ord}(\alpha) \iff V \models \text{Ord}(\alpha).$$

Thus M and the ambient universe agree about whether α is an ordinal.

If $\alpha, \beta \in M$ are ordinals, their order is given by membership:

$$\alpha < \beta \iff \alpha \in \beta.$$

Membership is literally the same relation inside a transitive submodel and outside it. Hence

$$M \models \alpha < \beta \iff \alpha \in \beta \iff V \models \alpha < \beta.$$

In particular, ordinal comparison and all bounded statements about initial segments of ordinals are absolute between the two settings. What may fail to be absolute is whether a given ordinal is a *cardinal*, because cardinality asks for the existence or nonexistence of functions, and an inner model can omit functions that exist outside it. $\square$

Proposition 32.3 (Absoluteness of omega and elementary finite constructions in the required setting). *Let M be a transitive model of ZF and assume $\omega \in M$. Then*

$$\omega^M = \omega.$$

Moreover, for all $m, n \in \omega$, successor, addition, and multiplication computed in M agree with the ambient operations, and every finite sequence of natural numbers belonging to M has the same domain, length, and entries in M and in the ambient universe.

Proof. Because $\omega \in M$ and M is transitive, every $n \in \omega$ belongs to M . The ambient ω is an inductive set in M , so by minimality of ω^M ,

$$\omega^M \subseteq \omega.$$

Conversely ω^M is inductive in M . The assertions $0 \in \omega^M$ and $n \in \omega^M \Rightarrow n + 1 \in \omega^M$ are absolute for the finite ordinals, so external induction yields $n \in \omega^M$ for every $n < \omega$. Hence

$$\omega \subseteq \omega^M,$$

and therefore $\omega^M = \omega$.

Addition and multiplication on ω are characterized by their recursion equations. For example,

$$m + 0 = m, \quad m + (n + 1) = (m + n) + 1.$$

The same equations hold in M , and induction on the ambient natural number n shows that the value computed in M equals the ambient value. The multiplication proof is identical after using the already established absoluteness of addition.

A finite sequence $s = \langle s(0), \dots, s(k-1) \rangle$ is coded by finitely many applications of pairing, union, and the natural-number operations. These constructions are absolute in a transitive model satisfying the indicated elementary ZF axioms. Hence M and V agree about the length k , the domain k , and every entry $s(i)$. This is the finite-syntax absoluteness used later. $\square$

Worked Theorem 32.4 (Failure of absoluteness for cardinality and power set). *Power set is not absolute even between transitive sets. With von Neumann naturals $0 = \emptyset$, $1 = \{0\}$, $2 = \{0, 1\}$, put*

$$M = \{0, 1, 2\}, \quad N = M \cup \{\{1\}\}, \quad x = 2.$$

Then $M \subsetneq N$ are transitive and

$$\mathcal{P}(x) \cap M \subsetneq \mathcal{P}(x) \cap N.$$

More generally, for transitive models $M \subseteq N$, cardinality statements can also fail to be absolute when a witnessing injection, surjection, or bijection exists in N but not in M .

Proof. Let

$$0 = \emptyset, \quad 1 = \{0\}, \quad 2 = \{0, 1\},$$

and define

$$M = \{0, 1, 2\}, \quad N = M \cup \{\{1\}\}.$$

Both M and N are transitive: every element of an element of either set is again in that set. Put $x = 2 = \{0, 1\}$. The actual power set is

$$\mathcal{P}(x) = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\} = \{0, 1, \{1\}, 2\}.$$

Hence

$$\mathcal{P}(x) \cap M = \{0, 1, 2\}, \quad \mathcal{P}(x) \cap N = \{0, 1, \{1\}, 2\},$$

so

$$\mathcal{P}(x) \cap M \subsetneq \mathcal{P}(x) \cap N.$$

This is a completely explicit failure of Power-Set absoluteness.

Cardinality fails for the same structural reason but requires an infinite setting. If $M \subseteq N$ are transitive models, $x, \kappa \in M$, and a bijection $f : x \rightarrow \kappa$ belongs to $N \setminus M$, then N may satisfy $|x| = \kappa$ while M does not, simply because M 's quantifier “there exists a bijection” ranges only over functions in M . Thus transitivity makes bounded membership facts absolute, not existence claims about arbitrary subsets or functions. $\square$

Worked example

Worked Example 32.5 (A missing subset). If M is a transitive model and $a \in M$, then every element of a is in M ; bounded quantifiers over a therefore see the same domain. But $\mathcal{P}^M(a) = \mathcal{P}(a) \cap M$ can be strictly smaller than the ambient $\mathcal{P}(a)$. “All subsets” is consequently not absolute.

A useful warning

Transitivity alone does not make arbitrary formulas absolute. Unbounded quantifiers may range over many more sets in the ambient universe.

Looking ahead

The next chapter, *Reflection*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 33

Reflection

Definitions and setup

Formal definitions used in this chapter

Rank-initial segment. V_α is the α -th level of the cumulative hierarchy from Volume IV.

Finite formula set and subformula closure. Φ is a finite set of formulas of set theory; when required, enlarge it by the finitely many subformulas needed for the induction on formula complexity. An existential subformula $\exists y \psi(y, \bar{x})$ creates a witness requirement: whenever it has a witness in V for parameters from a chosen stage, select a rank high enough to contain a witness.

Reflection for Φ . An ordinal α reflects Φ when for every $\varphi \in \Phi$ and every parameter tuple $\bar{a} \in V_\alpha$,

$$V \models \varphi(\bar{a}) \iff V_\alpha \models \varphi(\bar{a}).$$

Finite fragment of ZF. A finite fragment $T_0 \subseteq \text{ZF}$ is a finite set of particular ZF axioms, including only finitely many instances from the Separation and Replacement schemes. “Arbitrarily large α ” means that for every ordinal β there is such an $\alpha > \beta$.

Reflection is a finite-fragment theorem: given finitely many formulas and parameters, some rank V_α agrees with the universe on those formulas. The proof closes the formula set under subformulas and then ensures existential witnesses appear below a common rank. It does not produce a definable truth predicate and it does not assert that a single V_α satisfies every axiom of ZF.

How this chapter fits together

Reflection says that for every fixed finite collection Φ of formulas and every finite parameter tuple $\bar{a}$, there is an ordinal α with $\bar{a} \in V_\alpha$ such that every $\varphi \in \Phi$ is absolute between V_α and V on parameters from V_α . The construction closes under witnesses for existential subformulas and then takes a limit stage. It is a local approximation theorem, not a claim that one set-sized V_α is elementary in the entire universe for every formula.

Lemma 33.1 (Finite-fragment closure construction for formulas). *Given finitely many formulas Φ and a rank α_0 , one can construct an increasing ω -sequence of ordinals whose limit α is closed under the witness requirements for every existential subformula occurring in Φ .*

Proof. Let Φ be finite. First replace it by the finite set Φ^* of all subformulas of formulas in Φ ; reflection for Φ^* implies reflection for Φ . Fix α_0 .

Suppose α_n has been chosen. For every existential formula

$$\exists x \psi(x, \bar{y}) \in \Phi^*$$

and every parameter tuple $\bar{a} \in V_{\alpha_n}$, if $V \models \exists x \psi(x, \bar{a})$, choose one witness $b_{\psi, \bar{a}}$ and its rank. There are set-many relevant tuples and only finitely many formulas, so Replacement collects the witness ranks into a set. Choose $\alpha_{n+1} > \alpha_n$ such that every selected witness lies in $V_{\alpha_{n+1}}$.

Set

$$\alpha = \sup_{n < \omega} \alpha_n.$$

Then $V_\alpha = \bigcup_{n < \omega} V_{\alpha_n}$. The crucial closure property is:

$$\bar{a} \in V_\alpha \wedge V \models \exists x \psi(x, \bar{a}) \implies \exists b \in V_\alpha V \models \psi(b, \bar{a})$$

for every existential subformula in Φ^* . Indeed $\bar{a}$ already belongs to some V_{α_n} , and the witness selected at stage $n+1$ lies below α . This witness-closure is exactly the ingredient needed for the existential step of the Reflection Theorem. $\square$

Theorem 33.2 (Reflection theorem in the form used later). *For every finite set Φ of formulas of set theory and every set of parameters A , there is an ordinal α with $A \subseteq V_\alpha$ such that for every $\varphi \in \Phi$ and $\bar{a} \in V_\alpha$,*

$$V \models \varphi(\bar{a}) \iff V_\alpha \models \varphi(\bar{a}).$$

Commented symbolic trace: reflection is finite dependency closure

Fix finitely many formulas. Close that finite list under subformulas so every recursive truth check that may be needed is already present. Then build a sequence of ranks large enough to contain witnesses whenever one of those existential formulas becomes true. At the limit of the sequence, induction on formula complexity works because the required witnesses were deliberately placed below the chosen rank. Reflection is therefore a controlled finite closure construction, not a global truth predicate for the universe.

Proof. Let Φ^* be the finite set of subformulas of the given finite family Φ . Choose α_0 with $A \subseteq V_{\alpha_0}$, and apply the preceding closure construction to obtain $\alpha = \sup_n \alpha_n$ satisfying the witness-closure property for every existential formula in Φ^* .

We prove by induction on formula complexity that for every $\psi \in \Phi^*$ and every parameter tuple $\bar{a} \in V_\alpha$,

$$V \models \psi(\bar{a}) \iff V_\alpha \models \psi(\bar{a}).$$

Atomic formulas are absolute because V_α is transitive. Boolean connectives follow from the induction hypothesis.

For $\psi(\bar{y}) = \exists x \theta(x, \bar{y})$, suppose first that $V_\alpha \models \psi(\bar{a})$. Then some $b \in V_\alpha$ satisfies $V_\alpha \models \theta(b, \bar{a})$; by induction $V \models \theta(b, \bar{a})$, hence $V \models \psi(\bar{a})$. Conversely, if $V \models \psi(\bar{a})$, witness-closure supplies $b \in V_\alpha$ with $V \models \theta(b, \bar{a})$. The induction hypothesis gives $V_\alpha \models \theta(b, \bar{a})$, so $V_\alpha \models \psi(\bar{a})$.

Thus every formula in Φ is absolute between V and V_α on parameters from V_α . The prescribed set A is present because $A \subseteq V_{\alpha_0} \subseteq V_\alpha$. $\square$

Proposition 33.3 (Reflection with parameters). *In the Reflection Theorem, the reflecting ordinal α may be chosen above the ranks of any prescribed finite set of parameters, so the displayed equivalences hold uniformly for those parameters.*

Proof. The parameter version is obtained by changing only the starting rank in the closure construction. Let P be the prescribed finite set of parameters. Choose an ordinal α_0 satisfying

$$\forall p \in P \quad \text{rank}(p) < \alpha_0.$$

Equivalently, $P \subseteq V_{\alpha_0}$. Now run the finite-fragment witness-closure recursion from α_0 , obtaining $\alpha \geq \alpha_0$.

The proof of reflection itself is unchanged: induction on the finite subformula closure gives

$$V \models \varphi(\bar{a}) \iff V_\alpha \models \varphi(\bar{a})$$

for every formula under consideration and every $\bar{a} \in V_\alpha$. In particular all tuples formed from the prescribed parameters are allowed because those parameters lie in V_α .

The point is not a second reflection theorem but control of the reflecting level. Given any preliminary rank bound β , we may require

$$\alpha_0 > \max(\beta, \sup_{p \in P} (\text{rank}(p) + 1)).$$

The resulting reflecting α then satisfies $\alpha > \beta$ and contains all parameters. This strengthened form is what later constructions use when they need reflection above already chosen objects or ranks. $\square$

Corollary 33.4 (Set-sized approximations to finite fragments of ZF). *For every finite fragment $T_0 \subseteq \text{ZF}$, there are arbitrarily large ordinals α such that $V_\alpha \models T_0$.*

Proof. Let $T_0 \subseteq \text{ZF}$ be finite, and let β be any ordinal. Let Φ consist of the finitely many sentences in T_0 , together with all subformulas needed by the Reflection Theorem. Apply reflection with the prescribed parameter β (or simply demand the starting rank be above β). We obtain $\alpha > \beta$ such that for every $\sigma \in T_0$,

$$V \models \sigma \iff V_\alpha \models \sigma.$$

Because each σ is an axiom of ZF and we are reasoning in ZF, $V \models \sigma$. Therefore

$$V_\alpha \models \sigma \quad \text{for every } \sigma \in T_0,$$

so $V_\alpha \models T_0$.

Since the initial bound β was arbitrary, such reflecting ordinals occur above every ordinal. Thus there are arbitrarily large set-sized structures $(V_\alpha, \in)$ satisfying any fixed finite fragment of ZF. This does *not* assert that some V_α satisfies all of ZF simultaneously; that stronger conclusion would not follow from finite reflection and would conflict with the metamathematical distinctions developed later. $\square$

Worked example

Worked Example 33.5 (Reflecting one existential formula). Suppose the finite formula list contains $\exists y \psi(x, y)$. Whenever x occurs in the current rank and the universe has a witness y , choose some later rank containing such a witness. Repeating this closure for the finitely many subformulas and taking the supremum of an ω -sequence ensures the existential step of the truth induction.

A useful warning

Reflection is always for a specified finite set of formulas (or a scheme proved one formula at a time); treating it as a global truth predicate would be illegitimate.

Looking ahead

The next chapter, *Elementary hulls*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 34

Elementary hulls

Definitions and setup

Formal definitions used in this chapter

Elementary substructure. For $\mathcal{L}$ -structures $N \subseteq M$, write $N \preceq M$ (or $N \prec M$) when for every $\mathcal{L}$ -formula $\varphi(\bar{x})$ and every $\bar{a} \in N$,

$$N \models \varphi(\bar{a}) \iff M \models \varphi(\bar{a}).$$

The inclusion map is then an elementary embedding.

Expansion of a structure. An expansion M' of M has the same domain and the same interpretations for the old language symbols, with interpretations assigned to additional symbols.

Skolem function. For a formula $\exists y \varphi(y, \bar{x})$ in a structure M , a Skolem function $F_\varphi : M^{|\bar{x}|} \rightarrow M$ satisfies

$$M \models \exists y \varphi(y, \bar{a}) \implies M \models \varphi(F_\varphi(\bar{a}), \bar{a}).$$

Skolem hull. Let $\mathcal{F}$ be the set consisting of all basic function interpretations of the language together with the chosen Skolem functions. Given $A \subseteq M$, define

$$H_0 = A, \quad H_{n+1} = H_n \cup \{F(\bar{a}) : F \in \mathcal{F}, \bar{a} \in H_n^{\text{arity}(F)}\}, \quad \text{Hull}^M(A) = \bigcup_{n < \omega} H_n.$$

Here $\text{arity}(F) \in \omega$ is the finite number of inputs of F . Thus the construction does not presuppose that the language is countable. **Countable language/structure.** A language or structure is countable when its set of symbols or its domain, respectively, admits an injection into ω .

Mostowski collapse. If $(X, \in)$ is well founded and extensional, its Mostowski collapse is the unique isomorphism $\pi : X \cong \bar{X}$ onto a transitive set $\bar{X}$, as established in Volume IV.

A Skolem function chooses a witness for an existential formula whenever one exists. Closing a parameter set under all Skolem functions yields a Skolem hull. Tarski–Vaught turns the witness property into elementarity. When the ambient relation is well founded and extensional, Mostowski collapse produces a transitive copy. External countability of the hull is a metatheoretic fact and need not be recognized internally.

How this chapter fits together

Skolem functions turn existential truth into explicit closure operations. Closing a small parameter set under those functions produces a small elementary substructure by Tarski–Vaught. If the relation is well founded

and extensional, Mostowski collapse then gives a transitive copy. This two-step hull-and-collapse pattern reappears in Condensation.

Proposition 34.1 (Skolem-function expansion). *Assume the ambient metatheory has Choice. Let M be an $\mathcal{L}$ -structure. There is an expansion M^{Sk} obtained by adding, for each formula $\exists x \varphi(x, \bar{y})$, a function symbol f_φ such that for every tuple $\bar{a} \in M$,*

$$M \models \exists x \varphi(x, \bar{a}) \implies M^{\text{Sk}} \models \varphi(f_\varphi(\bar{a}), \bar{a}).$$

If the existential statement is false, the value of $f_\varphi(\bar{a})$ may be chosen arbitrarily.

Proof. Assume Choice in the ambient metatheory. For each existential $\mathcal{L}$ -formula $\exists x \varphi(x, \bar{y})$, add a new function symbol f_φ of arity $|\bar{y}|$. We must interpret it on every tuple $\bar{a} \in M^{|\bar{y}|}$.

For a fixed φ and tuple $\bar{a}$, let

$$W_{\varphi, \bar{a}} := \{b \in M : M \models \varphi(b, \bar{a})\}.$$

If this set is nonempty, the desired Skolem property requires choosing one element of it. The family of all nonempty $W_{\varphi, \bar{a}}$ is a set: there are only set-many formulas and set-many finite tuples from M . Ambient AC therefore supplies a simultaneous choice function. Define $f_\varphi^M(\bar{a})$ to be the chosen witness when $W_{\varphi, \bar{a}} \neq \emptyset$, and choose one fixed default element of the nonempty domain $|M|$ otherwise. Nonemptiness is part of the definition of an $\mathcal{L}$ -structure from Volume I.

Then by construction,

$$M \models \exists x \varphi(x, \bar{a}) \implies M^{\text{Sk}} \models \varphi(f_\varphi(\bar{a}), \bar{a}).$$

The reduct of M^{Sk} to the original language is exactly M . Thus the expansion changes no old-language truth; it merely names witnesses uniformly. If a definable well-order of M is already available, one can choose least witnesses and avoid using ambient AC; the theorem states the general metatheoretic version explicitly. $\square$

Theorem 34.2 (Skolem hull construction). *Let M be a structure with specified Skolem functions and let $A \subseteq M$. Then $\text{Hull}^M(A)$ is an elementary substructure of M . If A and the language $\mathcal{L}$ are well orderable (in particular, under ambient AC), then*

$$|\text{Hull}^M(A)| \leq \max(|A|, |\mathcal{L}|, \aleph_0).$$

Proof. Let M be expanded by Skolem functions and let $A \subseteq M$. Define an increasing sequence

$$H_0 = A, \quad H_{n+1} = H_n \cup \{f^M(\bar{a}) : f \text{ is a basic or Skolem function, } \bar{a} \in H_n^{<\omega}\},$$

and set

$$H = \bigcup_{n < \omega} H_n.$$

Then H is closed under every function symbol of the expanded language, so it is a substructure.

To prove elementarity, use the Tarski–Vaught criterion. Suppose

$$M \models \exists x \varphi(x, \bar{a}) \quad (\bar{a} \in H).$$

All entries of $\bar{a}$ lie in some common H_n . The corresponding Skolem function is one of the closure operations, hence

$$b = f_\varphi^M(\bar{a}) \in H_{n+1} \subseteq H,$$

and by the Skolem property $M \models \varphi(b, \bar{a})$. Thus every existential formula true in M with parameters from H has a witness in H , so $H \preceq M$.

For the cardinal bound, let

$$\mu = \max(|A|, |\mathcal{L}|, \aleph_0).$$

The Skolemized language has at most $\max(|\mathcal{L}|, \aleph_0)$ function symbols. If $|H_n| \leq \mu$, then there are at most $\mu^{<\omega} = \mu$ finite tuples and at most μ function applications, so $|H_{n+1}| \leq \mu$. By induction and a countable union, $|H| \leq \mu$. This H is the Skolem hull $\text{Hull}^M(A)$. $\square$

Theorem 34.3 (Countable elementary submodel lemma). *Assume ambient AC. If M is a structure in a countable language and $A \subseteq M$ is countable, then there is a countable elementary substructure $N \preccurlyeq M$ containing A .*

Proof. Let M be a structure in a countable language $\mathcal{L}$, and let $A \subseteq M$ be countable. By the Skolem-function expansion theorem, expand M to a Skolemized structure M^{Sk} . Because there are only countably many $\mathcal{L}$ -formulas, the expanded language is still countable.

Form the Skolem hull

$$H = \text{Hull}^{M^{\text{Sk}}}(A).$$

The cardinal estimate from the preceding theorem gives

$$|H| \leq \max(|A|, |\mathcal{L}|, \aleph_0) = \aleph_0.$$

Thus H is countable. Tarski–Vaught gives

$$H \preccurlyeq M^{\text{Sk}}.$$

Now take reducts to the original language. Elementarity is preserved under reduct: for every original-language formula φ and tuple $\bar{a} \in H$,

$$H \models \varphi(\bar{a}) \iff M^{\text{Sk}} \models \varphi(\bar{a}) \iff M \models \varphi(\bar{a}).$$

Therefore the original-language reduct $N = H \upharpoonright \mathcal{L}$ is a countable elementary substructure of M containing A .

The theorem is external: “countable” refers to cardinality in the surrounding universe. This distinction becomes essential in the later discussion of Skolem’s paradox. $\square$

Theorem 34.4 (Collapse of an elementary hull to a transitive model). *If $X \prec (M, \in)$ and $\in$ is well founded and extensional on X , then the Mostowski collapse $\pi : X \cong \bar{X}$ maps X isomorphically onto a transitive set $\bar{X}$, and $\bar{X}$ satisfies exactly the first-order sentences satisfied by X .*

Proof. Assume $X \prec (M, \in)$ and that $\in$ restricted to X is well founded and extensional. By the Mostowski Collapse Theorem there is a unique transitive set $\bar{X}$ and a unique isomorphism

$$\pi : (X, \in) \xrightarrow{\cong} (\bar{X}, \in)$$

satisfying the recursion

$$\pi(x) = \{\pi(y) : y \in X \wedge y \in x\}.$$

The collapse theorem supplies transitivity of $\bar{X}$ and preservation of the membership relation:

$$y \in x \text{ in } X \iff \pi(y) \in \pi(x).$$

Every isomorphism preserves first-order satisfaction. Thus, by induction on formulas, for every formula φ and tuple $\bar{a} \in X$,

$$X \models \varphi(\bar{a}) \iff \bar{X} \models \varphi(\pi(\bar{a})).$$

In particular X and $\bar{X}$ satisfy exactly the same sentences. Since $X \prec M$, we also have for sentences σ

$$M \models \sigma \iff X \models \sigma \iff \bar{X} \models \sigma.$$

The last equivalence is restricted to the theory seen by the elementary substructure; the collapse does not assert that $\bar{X}$ contains all objects of M . Its role is to replace a possibly nontransitive elementary hull by a transitive isomorphic copy in which membership has its usual set-theoretic meaning. $\square$

Worked example

Worked Example 34.5 (Building a hull in stages). Start with a countable set A . Let H_1 contain all values of the countably many Skolem functions on finite tuples from A ; then repeat with H_1 , and set $H = \bigcup_n H_n$. Each stage is countable, so H is countable. Every existential formula true with parameters in H has its Skolem witness in H , hence $H \preceq M$.

A useful warning

An elementary submodel need not be transitive. The collapse is a separate theorem, and it can move parameters unless they are known to be fixed.

Looking ahead

The next chapter, *Inner models and the Skolem paradox*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 35

Inner models and the Skolem paradox

Definitions and setup

Formal definitions used in this chapter

Inner model. A class M is an inner model of a theory $T \supseteq \text{ZF}$ when M is transitive, contains every ordinal, and $(M, \in) \models T$.

Internal versus external cardinality. For $X, Y \in M$,

$$M \models |X| \leq |Y|$$

means that some function $f \in M$ is, according to M , an injection $X \rightarrow Y$; internal equality of cardinalities uses a bijection belonging to M . Ambient cardinality permits witnessing functions from the whole universe V . Thus “ M thinks X is countable” means that M contains the relevant injection/bijection with ω^M .

Syntactic consistency, model, and transitive model. A first-order theory T is syntactically consistent when no contradiction is derivable. A model of T is a structure satisfying every axiom of T . A transitive model of a set theory is a model whose domain is a transitive set (or, when explicitly allowed, a transitive proper class).

ZF, ZFC, and their internal natural numbers. ZF is the set theory defined in Volume IV. In this volume

$$\text{ZFC} := \text{ZF} + \text{AC},$$

where AC is the Axiom of Choice defined in Chapter 1. For $T \in \{\text{ZF}, \text{ZFC}\}$, ω^T denotes the von Neumann natural-number structure whose existence and arithmetic T proves.

Recursive first-order proof calculus. A proof calculus is recursive when its finite formula, axiom, and inference-rule checks are computable; finite proof certificates and the resulting canonical predicate Prf_T are those of Volumes II–III.

Relative consistency. A statement $\text{Con}(T) \Rightarrow \text{Con}(U)$ is a metatheoretic implication saying that consistency of T entails consistency of U ; it does not by itself assert the existence of a transitive model of either theory.

Skolem-paradox terminology. “Externally countable” refers to countability in V ; “internally uncountable” means that the model contains no internal bijection with its ω . There is no contradiction because the external bijection need not belong to the model.

Corollary 35.1 (Gödel II for ZF and ZFC with the canonical proof predicate). *Let $T \in \{\text{ZF}, \text{ZFC}\}$ be presented in a fixed recursive first-order proof calculus, and let Prov_T be the canonical arithmetical provability predicate obtained by coding finite T -derivations. Interpreting that arithmetic on ω^T , the primitive-recursive proof transformations establishing D1–D3 are formalizable in T . Therefore, if T is consistent,*

$$T \not\vdash \text{Con}(T), \quad \text{Con}(T) := \neg \text{Prov}_T(\ulcorner 0 = 1 \urcorner).$$

Proof. By Volume IV, ZF constructs the von Neumann natural numbers and proves the elementary recursion needed to formalize finite computations; ZFC extends ZF. For the fixed recursive calculus, proof checking is therefore arithmetized on ω^T , giving the canonical $\text{Prf}_T(p, q)$ and $\text{Prov}_T(q) := \exists p \text{Prf}_T(p, q)$. The primitive-recursive proof transformations used in Volume III to establish D1–D3 are verified in the same internal arithmetic, as is the diagonal construction. Hence the proof of Gödel’s second incompleteness theorem from Volume III applies to this interpreted copy of arithmetic. If $T \vdash \text{Con}(T)$, the internal Gödel-II calculation yields a contradiction; therefore external consistency of T implies $T \not\vdash \text{Con}(T)$. $\square$

An inner model is a transitive proper class containing all ordinals and satisfying the relevant set-theoretic axioms internally. Internal power sets and cardinalities are computed using only subsets/functions belonging to that class. The Skolem paradox is the paradigm for this internal/external distinction. Relative consistency arguments later use inner models, but the existence of an inner model inside each model must be proved rather than inferred from consistency alone.

How this chapter fits together

An inner model is transitive and contains all ordinals, but those two facts alone do not prove every axiom of ZF inside it. The hard closure axioms must be checked. The Skolem paradox then illustrates why internal and external cardinality can disagree: functions witnessing countability externally may simply be absent internally.

Theorem 35.2 (Inner-model criterion). *Let M be a transitive class containing all ordinals. Then M is an inner model of a theory $T \supseteq \text{ZF}$ exactly when $(M, \in) \models T$. Transitivity makes membership between elements of M agree with ambient membership, but it does not by itself prove the internal Separation, Replacement, Power Set, or Choice axioms; those closure axioms must be verified separately.*

Proof. Recall the definition: an inner model is a transitive class M containing every ordinal and satisfying the specified theory in the structure $(M, \in)$. Therefore, once transitivity and inclusion of all ordinals have been established, the remaining task is exactly to prove

$$(M, \in) \models \sigma \quad \text{for every axiom } \sigma \in T.$$

If this is done, M is an inner model of T by definition.

What requires emphasis is what transitivity does *not* prove. Extensionality and many bounded statements are inherited because membership between elements of M is the ambient membership relation. But Separation asks that for $a, \bar{p} \in M$, the subset

$$\{x \in a : M \models \varphi(x, \bar{p})\}$$

belong to M . Replacement asks for the image of a set under an internally definable functional relation to lie in M . Power Set asks for the set

$$\mathcal{P}^M(a) = \{x \in M : x \subseteq a\}$$

to be an element of M , not merely to be a class externally. Choice asks for choice functions that themselves belong to M . None of these closure properties follows from transitivity alone.

Thus the criterion is deliberately axiom-by-axiom. Later, when $M = L$ or a forcing/symmetric extension is proposed, the proof must supply separate owners for precisely these nonautomatic axioms. $\square$

Proposition 35.3 (Internal versus external finiteness/countability/cardinality). *Let M be transitive and suppose $\omega^M = \omega$. For $x, \kappa \in M$, the internal assertions*

$$M \models “x \text{ is at most countable}” \quad \text{and} \quad M \models “|x| = \kappa”$$

mean respectively that M contains an injection $x \rightarrow \omega$ and that M contains a bijection $x \rightarrow \kappa$. Therefore these assertions may differ from the ambient ones precisely because the required witnessing functions may exist outside M but not inside M .

Proof. Assume M is transitive and $\omega^M = \omega$. Satisfaction of cardinality assertions is governed by which functions belong to M . For instance,

$$M \models "x \text{ is at most countable}"$$

means

$$\exists f \in M (M \models "f : x \hookrightarrow \omega \text{ is injective}").$$

Because M is transitive, if such an $f \in M$ exists then it is genuinely an injection in the ambient universe. Hence internal countability implies external countability.

The converse can fail. An external injection $g : x \rightarrow \omega$ need not be an element of M , so it need not witness the existential quantifier evaluated inside M . Likewise

$$M \models "|x| = \kappa"$$

means that some bijection $f : x \rightarrow \kappa$ belongs to M . An outer model can contain a new bijection and thereby change the cardinality it assigns to the same old set.

Finiteness is better behaved for a transitive model $M \models \text{ZF}$ with $\omega^M = \omega$. If M contains a bijection between x and some $n < \omega$, then that same finite graph is an external bijection, so x is externally finite. Conversely, if externally $x = \{a_0, \dots, a_{n-1}\}$ with $x \in M$, transitivity gives $a_i \in M$ for every $i < n$; repeated Pairing and Union construct the finite tuple $\langle a_0, \dots, a_{n-1} \rangle \in M$, and Separation constructs the graph $\{\langle i, a_i \rangle : i < n\} \in M$. Hence M also sees x as finite. The conceptual lesson is that infinite cardinality is an existential statement about functions, so its absoluteness depends on whether the relevant functions are available internally. $\square$

Worked Theorem 35.4 (Full resolution of Skolem's paradox). *Let M be a set-sized model of ZF such that $M \models "\mathbb{R} \text{ is uncountable}"$. If $N \preceq M$ is externally countable, then $N \models "\mathbb{R} \text{ is uncountable}"$ although the set $\mathbb{R}^N$ is externally countable. There is no contradiction: an external enumeration of $\mathbb{R}^N$ need not belong to N .*

Proof. Let M be a set-sized model of ZF in a countable language and suppose

$$M \models "\mathbb{R} \text{ is uncountable}."$$

By Downward Löwenheim–Skolem there is, externally, a countable elementary submodel

$$N \preceq M.$$

Elementarity transfers the sentence about the reals:

$$N \models "\mathbb{R} \text{ is uncountable}."$$

Nevertheless the domain of N is a countable set in the ambient universe. Therefore the externally viewed collection

$$\mathbb{R}^N := \{x \in N : N \models "x \text{ is a real} "\}$$

is externally countable, so the ambient universe has some surjection

$$e : \omega_{\text{ext}} \twoheadrightarrow \mathbb{R}^N,$$

where ω_{ext} denotes the ordinary external natural numbers.

This does not contradict

$$N \models "\mathbb{R} \text{ is uncountable}."$$

That internal sentence says that there is no *element of the structure* N which N , using its own membership relation and its internally defined object ω^N , regards as a surjection from ω^N onto its reals. The external map e is a metatheoretic relation on the domain of N ; it need not be represented by any element of N at all. Since M and N were not assumed transitive or well founded, one must also not identify the external set coding of a function with the function that the model's membership relation would assign to an internal element.

Thus “countable” is model-relative because the quantified candidate enumerations are internal objects of the model, whereas external countability supplies a map only in the surrounding universe. $\square$

Proposition 35.5 (Model-theoretic formulation of relative consistency and its limitations). *If a first-order theory T is syntactically consistent, completeness yields a (possibly ill-founded and nontransitive) model of T . This model-existence statement must not be confused with the stronger assertion that T has a transitive model.*

Proof. Let T be a first-order theory. Recall the language-independent definition from Volume I: T is syntactically consistent when there is no sentence θ such that both $T \vdash \theta$ and $T \vdash \neg\theta$. The Completeness Theorem gives exactly

$$T \text{ is syntactically consistent} \iff \exists M (M \models T).$$

Thus an external proof of syntactic consistency can be converted into an external existence statement for some first-order model of T . No fixed contradiction symbol is used here, because an arbitrary first-order language need not contain the nonlogical symbols occurring in somebody else’s chosen contradiction sentence.

Nothing in completeness says that the model is transitive, well founded, or a submodel of the ambient set-theoretic universe. If T is a set theory, the membership relation E^M of a model supplied by completeness can be externally ill founded, and its “natural numbers” or “ordinals” can be nonstandard. The assertion

$$\exists M (M \text{ is transitive and } M \models T)$$

is therefore strictly stronger than the bare model-existence conclusion of first-order completeness.

This distinction is essential for forcing. The pedagogical construction of a generic extension over a countable transitive model is extremely useful for understanding names and the Truth Lemma, but from $\text{Con}(T)$ alone one may not infer the existence of such a transitive model. The later Boolean-valued/finite-fragment consistency-transfer argument is included precisely to bridge that metatheoretic gap without making the invalid inference. $\square$

Worked example

Worked Example 35.6 (Skolem’s paradox in one diagram). Let M be externally countable and suppose $M \models \text{“}\mathbb{R} \text{ is uncountable.”}$ Externally enumerate the elements of $\mathbb{R}^M$. That enumeration is not required to belong to M . Internally, uncountability asserts that no bijection $\omega^M \rightarrow \mathbb{R}^M$ exists among the functions that M contains, so there is no contradiction.

A useful warning

Whenever a model says “there exists a function,” the quantified function must itself be an element of the model. This is the key to all internal/external cardinality distinctions.

End-of-volume perspective

Satisfaction, absoluteness, reflection, hulls, and collapse together form the model-theoretic toolkit used twice ahead: first to analyze L , and then to compare a forcing ground model with its extensions.

Part VI

Volume VI: The Constructible Universe

Preface

The constructible universe is built as an inner model. The difficult points are Replacement/Collection, Condensation, and the cardinal localization needed for GCH.

Position in the series

Volume VI gives the positive consistency branch: the constructible universe L satisfies ZF, Choice, and GCH. Condensation is the main structural theorem behind the cardinal calculation.

Sources and further reading

The constructible universe and the consistency of Choice and GCH originate in Gödel's monograph [8]. The treatment here also follows the modern condensation-and-cardinality organization of [13, 14].

Notation for this volume

$L_0 = \emptyset$, $L_{\alpha+1} = \text{Def}(L_\alpha)$, and $L_\lambda = \bigcup_{\beta < \lambda} L_\beta$. The symbol $<_L$ is reserved for the definable well-order constructed explicitly in the well-ordering chapter below; it is not used as an already-defined relation before that construction.

How to read symbolic arguments in this volume

When a formula appears, read it in layers rather than as one visual block. First identify the *ambient language*: are we speaking inside a formal theory, about a structure, or in the surrounding metatheory? Next mark the objects being manipulated (formulas, codes, sets, names, conditions, ordinals) and the operation being applied to them. Finally check the side conditions—free variables, rank bounds, compatibility, genericity, transitivity, or consistency hypotheses. Whenever a displayed derivation changes level, the text says so explicitly. A displayed line is therefore meant to be read like a line of well-commented code: the symbols perform the operation, and the surrounding prose records its type, preconditions, and purpose.

Chapter 36

Definability and Def(M)

Definitions and setup

Formal definitions used in this chapter

Set-sized membership structure. For a set M , $(M, \in)$ denotes the first-order structure whose domain is M and whose relation is ambient membership restricted to $M \times M$.

Normalized defining formulas and tuple satisfaction. Fix the variable list $v_0, v_1, \dots$. A *normalized one-object-variable defining formula of parameter length n* is a formula $\varphi(v_0, v_1, \dots, v_n)$ whose free variables are contained in the displayed list, with v_0 designated as the object variable and $v_1, \dots, v_n$ as parameter variables. Starting from any formula $\psi(x, \bar{y})$, finitely many applications of the capture-avoiding substitution operation from Volume I, with alpha-renaming of bound variables when required, replace its displayed free variables by distinct variables $v_0, \dots, v_n$ without changing the defined relation. Thus every finite-parameter definition has a normalized representative.

If $e = \ulcorner \varphi(v_0, \dots, v_n) \urcorner$, $\bar{a} = (a_1, \dots, a_n) \in M^n$, and $x \in M$, define the finite adequate assignment

$$u_{x, \bar{a}} := \{\langle v_0, x \rangle, \langle v_1, a_1 \rangle, \dots, \langle v_n, a_n \rangle\}.$$

The abbreviation

$$\text{Sat}_{\text{par}}(M, e, x, \bar{a}) : \Longleftrightarrow \text{Sat}(M, e, u_{x, \bar{a}})$$

therefore invokes exactly the finite-assignment satisfaction predicate constructed in Volume V; a parameter tuple is never itself treated as an assignment.

Definable subset with parameters. A set $X \subseteq M$ is *definable over $(M, \in)$ from finitely many parameters* when there are a normalized defining-formula code e and a matching finite tuple $\bar{a} \in M^{<\omega}$ such that

$$X = \{x \in M : \text{Sat}_{\text{par}}(M, e, x, \bar{a})\}.$$

Here $M^{<\omega} = \bigcup_{n < \omega} M^n$ is the set of finite sequences from M .

The operator $\text{Def}(M)$.

$$\text{Def}(M) := \{X \subseteq M : X \text{ is definable over } (M, \in) \text{ from finitely many parameters in } M\}.$$

Valid definition-code set. Define

$$C_M := \{(e, \bar{a}) \in \omega \times M^{<\omega} : e \text{ is a normalized defining-formula code of parameter length } |\bar{a}|\}.$$

The predicate inside the braces is a finite-syntax property, so C_M is a set by Separation.

Uniform defining relation. For $(e, \bar{a}) \in C_M$, write $\text{Defines}(M, e, \bar{a}, X)$ for

$$X \subseteq M \wedge \forall x \in M (x \in X \leftrightarrow \text{Sat}_{\text{par}}(M, e, x, \bar{a})).$$

Outside C_M the predicate $\text{Defines}(M, e, \bar{a}, X)$ is stipulated to be false. Hence it is a total first-order relation on set codes, not a partially defined abbreviation.

Internal computation superscript. If N is a model containing M , an expression such as $\text{Def}(M)^N$ or Sat^N denotes the object/relation computed inside N from the same defining formula. Equality of two such objects below is an external comparison in the ambient universe.

For a set-sized structure M , $\text{Def}(M)$ is the set of subsets of M definable over M using finitely many parameters from M . A definition is coded by a formula code and a finite parameter tuple. The satisfaction relation from Volume V turns this description into a genuine set-theoretic construction. Uniform coding is needed so the entire L -hierarchy can later be defined by one formula.

How this chapter fits together

Constructibility begins with a set-sized operation: $\text{Def}(M)$, the subsets of M definable over M using parameters from M . Satisfaction for set-sized structures ensures this collection is itself a set. Uniform coding of formulas and parameters is important because the same definition will be iterated through all ordinals.

Theorem 36.1 (Coding definable subsets of a set-sized structure). *For every set M , each subset of M definable over the membership structure $(M, \in)$ from finitely many parameters can, after capture-free renaming of its free variables, be written*

$$X_{e, \bar{a}} = \{x \in M : \text{Sat}_{\text{par}}(M, e, x, \bar{a})\},$$

where e is a normalized defining-formula code and $\bar{a} \in M^{<\omega}$ has the matching parameter length. Thus $(e, \bar{a}) \in C_M$ is a set-sized code for the subset.

Proof.

Formal derivation spine

Formal spine for Coding definable subsets of a set-sized structure.

$$X = X_{e, \bar{a}} := \{x \in M : \text{Sat}_{\text{par}}(M, e, x, \bar{a})\}.$$

$$(e, \bar{a}) \mapsto X_{e, \bar{a}}$$

$$(e, \bar{a}) \in C_M.$$

Fix a set M . By the normalization convention above, a subset $X \subseteq M$ is definable over $(M, \in)$ with finitely many parameters precisely when there are a normalized defining-formula code e and a matching finite tuple $\bar{a} \in M^{<\omega}$ such that

$$X = X_{e, \bar{a}} := \{x \in M : \text{Sat}_{\text{par}}(M, e, x, \bar{a})\}.$$

The map

$$(e, \bar{a}) \mapsto X_{e, \bar{a}}$$

is definable because the satisfaction relation for the fixed set-sized structure M is definable. This provides an explicit coding of every definable subset by finite syntactic data plus finitely many parameters. The present proposition is only the coding statement; the next theorem uses the fact that the code space is itself a set to prove that $\text{Def}(M)$ is a set. *Point specific to this result.* A definable subset is determined by a formula code together with finitely many parameters. Since both the syntax codes and the finite parameter tuples form sets, Replacement turns this coding into a set-sized family of subsets. $\square$

Proposition 36.2 ($\text{Def}(M)$ is a set). *For every set M ,*

$$\text{Def}(M) = \{X : \exists(e, \bar{a}) \in C_M \text{ Defines}(M, e, \bar{a}, X)\}$$

is a set.

Proof.

Formal derivation spine

Formal spine for Def(M) is a set.

$$C_M := \{(e, \bar{a}) \in \omega \times M^{<\omega} : e \text{ is a normalized defining-formula code of parameter length } |\bar{a}|\},$$

$$(e, \bar{a}) \mapsto X_{e, \bar{a}}$$

$$\text{Def}(M) = \{X \in Y : X \subseteq M \text{ and } X = X_{e, \bar{a}} \text{ for some } (e, \bar{a}) \in C_M\}.$$

The set of possible codes is

$$C_M := \{(e, \bar{a}) \in \omega \times M^{<\omega} : e \text{ is a normalized defining-formula code of parameter length } |\bar{a}|\},$$

which is a set. By the preceding coding proposition, every member of $\text{Def}(M)$ is equal to $X_{e, \bar{a}}$ for some $(e, \bar{a}) \in C_M$. Replacement applied to the definable map

$$(e, \bar{a}) \mapsto X_{e, \bar{a}}$$

produces a set Y containing all these values. Separation then defines

$$\text{Def}(M) = \{X \in Y : X \subseteq M \text{ and } X = X_{e, \bar{a}} \text{ for some } (e, \bar{a}) \in C_M\}.$$

Hence $\text{Def}(M)$ is a set. The argument never invokes a power set of a proper class or an unrestricted set of formulas. *Point specific to this result.* The previous coding gives a surjection from a set of formula/parameter codes onto $\text{Def}(M)$. Replacement followed by Separation therefore establishes sethood without appealing to an unrestricted collection of definitions. $\square$

Lemma 36.3 (Uniform coding of definable subsets). *There is a uniform coding relation $\text{Defines}(M, e, \bar{a}, X)$ for valid normalized code/parameter pairs $(e, \bar{a}) \in C_M$, and*

$$\text{Defines}(M, e, \bar{a}, X) \iff X = \{x \in M : \text{Sat}_{\text{par}}(M, e, x, \bar{a})\}.$$

Thus the definition of $\text{Def}(M)$ is uniform in M .

Proof.

Formal derivation spine

Formal spine for Uniform coding of definable subsets.

$$\text{Defines}(M, e, \bar{a}, X)$$

$$X \subseteq M \wedge \forall x \in M (x \in X \leftrightarrow \text{Sat}_{\text{par}}(M, e, x, \bar{a})).$$

$$X \in \text{Def}(M) \iff \exists (e, \bar{a}) \in C_M \text{ Defines}(M, e, \bar{a}, X).$$

Let $(e, \bar{a}) \in C_M$, so e is a normalized defining-formula code whose parameter length matches $\bar{a}$. Define

$$\text{Defines}(M, e, \bar{a}, X)$$

to mean

$$X \subseteq M \wedge \forall x \in M (x \in X \leftrightarrow \text{Sat}_{\text{par}}(M, e, x, \bar{a})).$$

All quantifiers in this definition range over sets, and Sat_{par} is the tuple abbreviation explicitly reduced above to the uniform finite-assignment satisfaction predicate from Volume V. Consequently the same first-order formula Defines works for every set M ; only the parameters change. In particular,

$$X \in \text{Def}(M) \iff \exists (e, \bar{a}) \in C_M \text{ Defines}(M, e, \bar{a}, X).$$

This uniformity, rather than mere sethood, is what later makes the hierarchy $\alpha \mapsto L_\alpha$ first-order definable. *Point specific to this result.* Uniformity means one satisfaction formula handles every formula code and every finite parameter tuple. This is what later allows the operation $M \mapsto \text{Def}(M)$ to be iterated definably through the ordinals. $\square$

Proposition 36.4 (Absoluteness properties of Def in the transitive setting). *Let M be a transitive set. Let N_0, N_1 be transitive models of ZF. If $M \in N_0 \cap N_1$, then for every formula code $e = \ulcorner \varphi \urcorner$ and every finite assignment u adequate for φ ,*

$$\text{Sat}^{N_0}(M, e, u) \iff \text{Sat}^{N_1}(M, e, u).$$

Consequently

$$\text{Def}(M)^{N_0} = \text{Def}(M)^{N_1}.$$

Proof.

Formal derivation spine

Formal spine: satisfaction for one fixed set structure is absolute. For every formula φ and every finite assignment u adequate for φ ,

$$N_0 \models \text{Sat}(M, \ulcorner \varphi \urcorner, u) \iff N_1 \models \text{Sat}(M, \ulcorner \varphi \urcorner, u).$$

Induct on the complexity of φ . Atomic equality and membership are absolute because M, N_0, N_1 are transitive and all three use the ambient membership relation on the same set M . The Boolean clauses follow directly from the induction hypothesis. For the existential step,

$$(M, \in) \models \exists x \psi(x, \bar{a})$$

means that there is a witness $b \in M$ satisfying $\psi(b, \bar{a})$. Both ambient models quantify over exactly the same set M , and the induction hypothesis gives the same truth value for each candidate b . Hence they agree on the existential formula.

Thus N_0 and N_1 compute the same satisfaction relation for $(M, \in)$. For each normalized defining-formula code e and parameter tuple $\bar{a}$, the corresponding finite assignments $u_{x, \bar{a}}$ are the same in both models, so they define the same actual subset

$$X_{e, \bar{a}} := \{x \in M : \text{Sat}_{\text{par}}(M, e, x, \bar{a})\}.$$

The valid code set C_M , being a definable subset of $\omega \times M^{<\omega}$ determined by finite syntax, is also the same in the two transitive models. It follows that both collect exactly the same family $\{X_{e, \bar{a}}\}$, namely the same $\text{Def}(M)$. The theorem concerns a fixed set-sized structure; it makes no claim that a truth predicate for an ambient universe is absolute or definable. $\square$

Worked example

Worked Example 36.5 (A definable subset of a finite structure). If $M = \{0, 1, 2\}$ with the inherited order, the subset $\{0\}$ is definable as the set of elements with no predecessor, while $\{0, 1\}$ is definable as the elements below the parameter 2. $\text{Def}(M)$ collects subsets obtainable in this way; in the constructible hierarchy the same idea is applied at every stage.

A useful warning

Definability is always relative to a structure and allowed parameters. “Definable in the universe” and “definable over L_α ” are different notions.

Looking ahead

The next chapter, *The L-hierarchy*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 37

The L-hierarchy

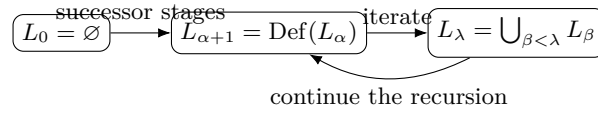


Figure 37.1: The constructible hierarchy grows by definability at successor stages and by union at limit stages.

Definitions and setup

Formal definitions used in this chapter

Successor and limit ordinal. $\alpha + 1 = \alpha \cup \{\alpha\}$. A nonzero ordinal λ is a limit ordinal when it is not a successor.

Constructible hierarchy. Using the operator Def defined in the preceding chapter, set

$$L_0 = \emptyset, \quad L_{\alpha+1} = \text{Def}(L_\alpha), \quad L_\lambda = \bigcup_{\beta < \lambda} L_\beta \quad (\lambda \text{ limit}).$$

A hierarchy $\langle L_\alpha : \alpha \in \text{Ord} \rangle$ is a class function indexed by all ordinals satisfying these clauses.

Constructible universe.

$$L := \bigcup_{\alpha \in \text{Ord}} L_\alpha = \{x : \exists \alpha \in \text{Ord} (x \in L_\alpha)\}.$$

This is a class union: it defines a class and is not asserted to be a set.

The constructible hierarchy replaces the power-set successor step of V by definability: $L_{\alpha+1} = \text{Def}(L_\alpha)$, with unions at limits. L is the union of all levels. Every constructible set therefore has a first stage of appearance and a finite definition from earlier parameters. This finiteness is the source of the canonical well-order and of the cardinal bounds used in GCH.

How this chapter fits together

The hierarchy is defined by $L_0 = \emptyset$, $L_{\alpha+1} = \text{Def}(L_\alpha)$, and unions at limits. Transfinite induction simultaneously proves transitivity and the presence of ordinals. The construction is deliberately parsimonious: only

subsets definable from earlier-stage parameters are admitted at successor stages.

Theorem 37.1 (Recursive construction of L_α). *There is a unique hierarchy $\langle L_\alpha : \alpha \in \text{Ord} \rangle$ satisfying*

$$L_0 = \emptyset, \quad L_{\alpha+1} = \text{Def}(L_\alpha), \quad L_\lambda = \bigcup_{\beta < \lambda} L_\beta \quad (\lambda \text{ limit}).$$

How to read the symbols: the constructible hierarchy is a recursive data pipeline

At stage α we have the database L_α . The successor operation $\text{Def}(L_\alpha)$ keeps only subsets definable over that database using finitely many parameters already present. At a limit stage we take the union of all previous databases. Thus L grows by definability rather than by taking every subset, which is why it can be much smaller than the ambient universe.

Proof.

Formal derivation spine

Formal spine for Recursive construction of L_α .

$$L_0 = \emptyset, \quad L_{\alpha+1} = F(L_\alpha), \quad L_\lambda = \bigcup_{\beta < \lambda} L_\beta \quad (\lambda \text{ limit}).$$

$$L_{\alpha+1} = \text{Def}(L_\alpha),$$

$$L_\lambda = \bigcup_{\beta < \lambda} L_\beta.$$

Define the successor-stage operation on sets by $F(M) = \text{Def}(M)$. We seek a class function $\alpha \mapsto L_\alpha$ satisfying

$$L_0 = \emptyset, \quad L_{\alpha+1} = F(L_\alpha), \quad L_\lambda = \bigcup_{\beta < \lambda} L_\beta \quad (\lambda \text{ limit}).$$

The operation F is definable because satisfaction for the set-sized structure $(M, \in)$ was constructed earlier, and $\text{Def}(M)$ is a set. The transfinite recursion theorem therefore gives a unique function on every ordinal γ whose values satisfy these clauses below γ . Uniqueness on overlapping initial segments makes these set-sized recursions coherent, so they determine the class hierarchy $\langle L_\alpha : \alpha \in \text{Ord} \rangle$. Nothing about transitivity is needed for existence of the recursion; that is the content of the next proposition.

Stage discipline. At a successor stage we add only subsets definable over the immediately preceding level with parameters from that level. At a limit stage we add no new definable subsets at all; we simply take the union of the earlier levels. Keeping these two clauses separate is essential in later induction arguments. $\square$

Proposition 37.2 (Every L_α is transitive). *For every ordinal α , the set L_α is transitive:*

$$x \in y \in L_\alpha \implies x \in L_\alpha.$$

Proof.

Formal derivation spine

Formal spine: transitivity of each constructible level. Induct on α . At a successor stage,

$$y \in L_{\alpha+1} = \text{Def}(L_\alpha) \implies y \subseteq L_\alpha,$$

so $x \in y \implies x \in L_\alpha \subseteq L_{\alpha+1}$. At a limit,

$$L_\lambda = \bigcup_{\beta < \lambda} L_\beta$$

is a union of an increasing chain of transitive sets.

We prove by transfinite induction on α that L_α is transitive. The claim is trivial for $L_0 = \emptyset$. Suppose L_α is transitive and let $x \in L_{\alpha+1} = \text{Def}(L_\alpha)$. By definition, x is a subset of L_α . Hence every $y \in x$ belongs to L_α , and therefore to $L_{\alpha+1}$ because the hierarchy is increasing. Thus $L_{\alpha+1}$ is transitive. If λ is a limit ordinal and $x \in L_\lambda = \bigcup_{\beta < \lambda} L_\beta$, choose $\beta < \lambda$ with $x \in L_\beta$. By the induction hypothesis $x \subseteq L_\beta \subseteq L_\lambda$, so L_λ is transitive.

It remains only to justify the monotonicity used at the successor step. By induction, every $a \in L_\alpha$ is a subset of L_α ; moreover the set a is definable over L_α from the parameter a by the formula $z \in a$. Hence $a \in \text{Def}(L_\alpha) = L_{\alpha+1}$. Therefore $L_\alpha \subseteq L_{\alpha+1}$, completing the induction.

What is special here. The successor argument uses both ingredients in the definition of $\text{Def}(L_\alpha)$: its members are subsets of L_α , and parameters from L_α ensure that the hierarchy itself is increasing. $\square$

Proposition 37.3 (alpha is a subset of L_α). *For every ordinal α ,*

$$\alpha \subseteq L_\alpha.$$

In particular, each $\beta < \alpha$ belongs to L_α .

Proof.

Formal derivation spine

Formal spine: ordinals occur by their own stage. The induction target is

$$\forall \alpha \ \alpha \subseteq L_\alpha.$$

For $\beta < \alpha$, the induction hypothesis makes β a definable subset of the earlier level containing all smaller ordinals, hence

$$\beta \in L_{\beta+1} \subseteq L_\alpha.$$

Thus every member of α lies in L_α .

We prove simultaneously by transfinite induction that $L_\alpha \cap \text{Ord} = \alpha$, which implies $\alpha \subseteq L_\alpha$. The claim is trivial at 0. At a limit λ , $L_\lambda = \bigcup_{\beta < \lambda} L_\beta$, so the ordinal members are exactly $\bigcup_{\beta < \lambda} \beta = \lambda$. At a successor $\beta + 1$, every ordinal below β already lies in L_β . The set β itself is the subset of L_β consisting of its ordinal elements, a subset definable over L_β by the first-order formula “ x is an ordinal”. Hence $\beta \in \text{Def}(L_\beta) = L_{\beta+1}$. No ordinal $\geq \beta + 1$ can belong to $L_{\beta+1}$, since every new element is a subset of L_β and transitivity would otherwise force too many ordinals below it into L_β . Thus the induction closes. *Reading the induction.* The

simultaneous statement $L_\alpha \cap \text{Ord} = \alpha$ contains both inclusions needed at the successor stage. The inclusion $\beta \subseteq L_\beta$ lets us talk about the ordinal elements already present, while the reverse inclusion says there is no unexpected ordinal in the level. Defining β as “the set of ordinal elements of L_β ” therefore creates exactly the next ordinal and no larger one. This is why the proof is stronger than merely showing $\alpha \subseteq L_\alpha$: the equality with the ordinal trace controls the induction. $\square$

Theorem 37.4 (L contains every ordinal and is transitive). *The class*

$$L := \bigcup_{\alpha \in \text{Ord}} L_\alpha$$

is transitive and contains every ordinal.

Proof.

Formal derivation spine

Formal spine: pass from the levels to the class L . Since every ordinal α satisfies $\alpha \subseteq L_\alpha$,

$$\alpha \in L_{\alpha+1} \subseteq L.$$

If $y \in L$, choose α with $y \in L_\alpha$; transitivity of L_α gives

$$x \in y \implies x \in L_\alpha \subseteq L.$$

Hence L contains all ordinals and is transitive.

Define $L_0 = \emptyset$, $L_{\alpha+1} = \text{Def}(L_\alpha)$, and $L_\lambda = \bigcup_{\beta < \lambda} L_\beta$. Transfinite recursion gives the hierarchy. By induction each L_α is transitive: a definable subset of L_α is a subset of L_α , while elements of L_α already lie at earlier stages. A simultaneous induction shows $\alpha \subseteq L_\alpha$, so every ordinal belongs to a later constructible level. Thus $L = \bigcup_\alpha L_\alpha$ is a transitive proper class containing all ordinals.

Point specific to this result. A simultaneous induction shows $\alpha \subseteq L_\alpha$ and hence $\alpha \in L_{\alpha+1}$. Taking the union over all stages therefore puts every ordinal into L , while stage transitivity makes L transitive.

Further explanation. The two assertions are proved simultaneously but play different roles. Transitivity guarantees that membership among constructible sets is inherited from the ambient universe. Containing every ordinal ensures that L has the same ordinal scale on which ranks, cardinals, and later condensation arguments are measured. At successor stages the new elements are subsets of the preceding transitive level; at limits the union of an increasing chain of transitive levels is transitive. The inclusion $\alpha \subseteq L_\alpha$ then puts every ordinal into L . $\square$

Worked example

Worked Example 37.5 (The first levels). $L_0 = \emptyset$. Since the only subset of the empty set is empty, $L_1 = \{\emptyset\}$. Subsequent finite stages build the hereditarily finite sets. By stage ω , all hereditarily finite sets have appeared; later stages begin adding infinite definable subsets.

A useful warning

Do not identify $L_{\alpha+1}$ with the full power set $\mathcal{P}(L_\alpha)$. It contains only the subsets definable over L_α from parameters in that level.

Looking ahead

The next chapter, *Uniform definability and absoluteness of L*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 38

Uniform definability and absoluteness of L

Definitions and setup

Formal definitions used in this chapter

Bounded, Σ_1 , Π_1 , and Δ_1 formulas. A quantifier is *bounded* when it has the form $\forall x \in a$ or $\exists x \in a$, abbreviating $\forall x(x \in a \rightarrow \dots)$ or $\exists x(x \in a \wedge \dots)$. A formula is Δ_0 when all of its quantifiers are bounded. A formula is Σ_1 when it is equivalent in ZF to one of the form $\exists u \psi$ with $\psi \Delta_0$; Π_1 is defined dually; and a relation is Δ_1 when ZF proves equivalent Σ_1 and Π_1 definitions of it. To separate syntax from this provable-equivalence convention, call a formula in *literal Σ_1 normal form* when it is actually written as $\exists \bar{u} \delta(\bar{u}, \bar{x})$ with $\delta \Delta_0$, and define *literal Π_1 normal form* dually.

Absoluteness terminology. If $M \subseteq N$ are transitive sets (or transitive models in the stated fragment) containing the parameters of a formula ψ , then ψ is *absolute between M and N* when

$$M \models \psi \iff N \models \psi.$$

The closure facts for $\Delta_0, \Sigma_1, \Pi_1, \Delta_1$ formulas are proved in the lemma immediately after this definition block; they are not being assumed as part of the terminology.

Uniform level-definition properties. A formula $\Theta(y, \alpha)$ is a *uniform exact-level definition for the constructible hierarchy* when ZF proves, for every ordinal α ,

$$\Theta(y, \alpha) \iff y = L_\alpha.$$

A formula $\Lambda(x, \alpha)$ is a *uniform membership definition* when ZF proves $\Lambda(x, \alpha) \leftrightarrow x \in L_\alpha$ for every ordinal α . These are properties that candidate formulas may have; this definition does not assert that any such formula exists. The theorem below proves existence and only afterward fixes the permanent symbols Level_L and Lev .

Transitive model, internal constructible level, and stage absoluteness. A transitive model $M \models \text{ZF}$ is a transitive set or class satisfying ZF. $(L_\alpha)^M$ denotes the α -th level obtained by carrying out the defining recursion for L inside M , for ordinals α that M contains. The stage construction is *absolute between specified transitive models* when those models compute the same stage as an external set. This is terminology only; Theorem 38.3 proves the relevant absoluteness statement.

Parameter-free definability. A class C is parameter-free definable when one formula $\chi(x)$ with no parameters satisfies $x \in C \iff V \models \chi(x)$. $\text{Constr}(x)$ denotes the parameter-free formula defining $x \in L$.

Internal constructible universe. L^M denotes the class that M obtains by taking the union of its internally computed constructible levels; in particular $L^L = L$ means that forming the constructible universe inside L yields the same class.

The point of the Levy-hierarchy terminology here is not cosmetic. Condensation later collapses an elementary substructure to a transitive set that is not yet known to be an L -level. The Δ_1 exact-level relation is the device that lets us compare what that transitive collapse calls a constructible level with the ambient L_α without assuming the conclusion of Condensation.

Lemma 38.1 (Levy absoluteness for transitive sets). *Let $M \subseteq N$ be transitive sets containing all parameters under discussion. Then every Δ_0 formula is absolute between M and N . Moreover, without assuming that either transitive set is a model of ZF:*

- (a) *every formula in literal Σ_1 normal form is upward absolute from M to N ;*
- (b) *every formula in literal Π_1 normal form is downward absolute from N to M .*

If, in addition, M and N are transitive models of ZF (or of the fragment proving the displayed Levy-hierarchy equivalences), then the chapter's broader provable-equivalence convention yields:

1. *If φ is Σ_1 , then*

$$M \models \varphi(\bar{a}) \implies N \models \varphi(\bar{a}).$$

2. *If φ is Π_1 , then*

$$N \models \varphi(\bar{a}) \implies M \models \varphi(\bar{a}).$$

3. *If a relation has ZF-provably equivalent Σ_1 and Π_1 definitions, then the relation is absolute between them.*

Proof. Induct on the formation of a Δ_0 formula. Atomic formulas are absolute because transitivity makes equality and membership on elements of M the same in M and N . Boolean connectives preserve the induction hypothesis. For a bounded existential,

$$M \models \exists x \in b \psi(x, \bar{a})$$

means that some $x \in b$ witnesses ψ . Since $b \in M$ and M is transitive, every such x lies in M ; the induction hypothesis therefore gives the same truth value in N . The converse is identical, because any witness satisfying $x \in b$ again belongs to M . Bounded universal quantifiers follow by negation. This proves the Δ_0 -absoluteness assertion.

If $\varphi(\bar{a})$ is literally $\exists \bar{u} \delta(\bar{u}, \bar{a})$ with δ bounded and $M \models \varphi(\bar{a})$, choose the witnesses $\bar{u} \in M$. The Δ_0 -absoluteness just proved preserves δ in N , proving literal- Σ_1 upward absoluteness. The literal- Π_1 statement is the dual argument: a counterexample in M would remain a counterexample in N .

Now assume $M, N \models \text{ZF}$ (or the indicated fragment). If φ is Σ_1 in the convention of this chapter, ZF proves it equivalent to some $\exists u \delta(u, \bar{a})$ with $\delta \Delta_0$. Both models satisfy that equivalence. A witness $u \in M$ remains a witness in N by Δ_0 -absoluteness, proving item (1). Item (2) is the dual argument for a Π_1 formula. Finally, if σ and π are provably equivalent Σ_1 and Π_1 definitions of one relation, upward absoluteness of σ and downward absoluteness of π give both directions, proving item (3). $\square$

How this chapter fits together

To use L as an inner model, the hierarchy must be definable uniformly rather than by a separate meta-definition at each ordinal. Uniform satisfaction for set-sized levels permits a single formula to express membership in L_α . Transitive models then agree on the construction in the required range, yielding $L^L = L$.

Theorem 38.2 (Uniform Δ_1 definition of the L -hierarchy). *There exist formulas $\Theta_L^\Sigma(y, \alpha)$ and $\Theta_L^\Pi(y, \alpha)$, respectively in literal Σ_1 and literal Π_1 normal form, such that ZF proves them equivalent and, for every ordinal α , proves*

$$\Theta_L^\Sigma(y, \alpha) \iff y = L_\alpha \iff \Theta_L^\Pi(y, \alpha).$$

Thus the exact constructible-level relation admits a uniform Δ_1 definition over ZF.

Proof.

Formal derivation spine

Formal spine: code one set-sized recursion history. A witness that $y = L_\alpha$ is a function h with domain $\alpha + 1$ such that

$$\begin{aligned} h(0) &= \emptyset, \\ h(\xi + 1) &= \text{Def}(h(\xi)), \end{aligned}$$

and, for every nonzero limit $\lambda \leq \alpha$,

$$h(\lambda) = \bigcup_{\xi < \lambda} h(\xi),$$

with $h(\alpha) = y$. The successor clause is expressed using the set-sized satisfaction relation from Volume V. Existence of such a coded history gives a Σ_1 definition; uniqueness of the recursion gives the complementary Π_1 definition.

Step 1: the definable-successor operation is uniformly coded. For sets u, v , let $\text{DefStage}(u, v)$ mean that $v = \text{Def}(u)$, where $\text{Def}(u)$ is the family of all subsets of u definable over the structure $(u, \in)$ from a finite tuple of parameters in u . Volume V constructed the satisfaction relation for every set-sized structure by a recursion on formula codes. Thus one can express $\text{DefStage}(u, v)$ by saying that there is a set coding the required satisfaction table and that v consists exactly of the subsets selected by formula-code/parameter-tuple pairs. All quantification after those coding sets have been supplied is bounded by the displayed sets, formula codes, or finite tuples. This gives the standard Δ_1 coding of the operation $u \mapsto \text{Def}(u)$.

Step 2: code the hierarchy only up to the requested ordinal. For an ordinal α , call a set function h an *L-history through α* when $\text{dom}(h) = \alpha + 1$ and the three clauses in the formal spine hold. Transfinite recursion, already proved in Volume IV, gives exactly one such history. Its values are

$$h(\xi) = L_\xi \quad (\xi \leq \alpha). \quad (1)$$

Hence $y = L_\alpha$ is equivalent to the existence of an *L-history through α* with $h(\alpha) = y$. After the history and the satisfaction tables needed at its successor stages have been existentially supplied, verification of the recursion is bounded. Therefore this description is Σ_1 .

Step 3: write the complementary Π_1 definition explicitly. Package into one set w all satisfaction tables and other finite coding data used to certify the successor clauses of a proposed history h . There is then a bounded predicate

$$B(\alpha, h, w)$$

saying that (h, w) is a complete certificate for an *L-history through α* . Step 2 says

$$y = L_\alpha \iff \exists h \exists w (B(\alpha, h, w) \wedge h(\alpha) = y), \quad (2)$$

which is Σ_1 . The recursion theorem proves in ZF that at least one certified history exists and that any two certified histories have the same value at every stage. Therefore ZF also proves

$$y = L_\alpha \iff \forall h \forall w (B(\alpha, h, w) \rightarrow h(\alpha) = y). \quad (3)$$

The matrix in (3) is bounded, so (3) is Π_1 . Equations (2) and (3) are provably equivalent in ZF and define the same exact-level relation. After pairing the finitely many existential witnesses in (2) into one tuple, (2) is a literal Σ_1 formula; (3) is a literal Π_1 formula. Call the two formulas obtained from (2) and (3) Θ_L^Σ and Θ_L^Π , respectively. They are in literal Σ_1 and literal Π_1 normal form, ZF proves them equivalent, and each defines exactly $y = L_\alpha$. This establishes the existence assertion in the theorem. No permanent Level_L notation or transitive-model absoluteness claim was assumed in the setup.

Why the exact-level version matters.

Merely knowing that the class L is definable is not enough for Condensation. Later arguments need one fixed first-order formula that recognizes a requested stage; the following theorem then proves the model-agreement property needed for suitable transitive models. $\square$

Formal definitions used in this chapter

Permanent exact-level and membership notation. Fix the literal- Σ_1 witness supplied by Theorem 38.2 and define

$$\text{Level}_L(y, \alpha) := \Theta_L^\Sigma(y, \alpha).$$

The theorem already proves that ZF also has the equivalent literal- Π_1 presentation Θ_L^Π , so this exact-level relation is Δ_1 . Define

$$\text{Lev}(x, \alpha) :\iff \exists y [\text{Level}_L(y, \alpha) \wedge x \in y].$$

Then, for every ordinal α , ZF proves $\text{Lev}(x, \alpha) \leftrightarrow x \in L_\alpha$.

Theorem 38.3 (Absoluteness of the L-construction between suitable transitive models). *Let M and N be transitive models of ZF with $M \subseteq N$, and let $\alpha \in M \cap \text{Ord}$. Then both models compute the same constructible level at α :*

$$(L_\alpha)^M = (L_\alpha)^N = L_\alpha.$$

In particular, $L_\alpha \subseteq M$, and the internally computed $(L_\alpha)^M$ is exactly the ambient constructible level L_α .

Proof.

Formal derivation spine

Formal spine for Absoluteness of the L-construction between suitable transitive models.

$$\text{Lev}(x, \alpha) \iff x \in L_\alpha.$$

Let $M \subseteq N$ be transitive models of ZF, and suppose that the ordinal α under discussion belongs to both. We prove by induction on $\beta \leq \alpha$ that the two models compute the same constructible level whenever all parameters required by the comparison are present.

At stage 0, both constructions give the empty set by definition, so the two hierarchies agree without using an induction hypothesis. At a successor stage, assume the two models agree on L_β . The structure $(L_\beta, \in)$ is then literally the same set-sized structure in both models. Satisfaction for a fixed set-sized structure is absolute between transitive models for the recursive definition established in Volume V; hence the same formula/parameter pairs define the same subsets of L_β . Thus both models obtain the same $\text{Def}(L_\beta) = L_{\beta+1}$. At a limit stage both take the union of the same earlier levels. This proves the desired absoluteness by transfinite induction. The restriction to suitable transitive models is essential: without transitivity, the purported earlier levels and their satisfaction relations need not be computed correctly. $\square$

Proposition 38.4 (Definability of x in L). *There is a parameter-free formula $\text{Constr}(x)$ such that*

$$V \models \text{Constr}(x) \iff x \in L.$$

One may take $\text{Constr}(x) \equiv \exists \alpha (\alpha \text{ is ordinal} \wedge x \in L_\alpha)$ using the uniform level formula.

Proof.

Formal derivation spine

Formal spine for Definability of x in L .

$$x \in L \iff \exists \alpha (\alpha \text{ is an ordinal} \wedge \exists y (\text{Level}_L(y, \alpha) \wedge x \in y)).$$

$$\text{Lev}(x, \alpha) \iff x \in L_\alpha.$$

By the uniform-definability theorem, $\text{Level}_L(y, \alpha)$ expresses exactly $y = L_\alpha$. Therefore

$$x \in L \iff \exists \alpha (\alpha \text{ is an ordinal} \wedge \exists y (\text{Level}_L(y, \alpha) \wedge x \in y)).$$

Every symbol on the right is first-order definable in the language of set theory, so membership in the constructible universe is a first-order definable class relation.

Conversely, if $x \in L$, by definition $x \in L_\alpha$ for some ordinal α , so the displayed formula holds. If the formula holds, the witness y is the uniquely determined L_α , hence x is constructible. The theorem asserts definability of the class L ; it does not assert the existence of a set collecting all constructible sets. *Reading the displayed definition.* The outer existential quantifier chooses a stage index α . The next existential quantifier chooses the unique set y that the uniform stage formula identifies as L_α . The final atomic clause $x \in y$ says that x has appeared by that stage. Thus a proper-class definition is reduced to quantification over ordinary sets and ordinals. $\square$

Theorem 38.5 (L computed inside L is L). *If the constructible hierarchy is formed internally to L , then for every ordinal α ,*

$$(L_\alpha)^L = L_\alpha,$$

and hence $L^L = L$.

Proof.

Formal derivation spine

Formal spine for L computed inside L is L .

$$\begin{aligned} L_\alpha^L &= L_\alpha \\ L^L &= \bigcup_{\alpha \in \text{Ord}^L} L_\alpha^L = \bigcup_{\alpha \in \text{Ord}} L_\alpha = L. \\ \text{Lev}(x, \alpha) &\iff x \in L_\alpha. \end{aligned}$$

The class L is transitive and contains every ordinal. Compute the constructible hierarchy internally in L , writing L_α^L for the level that L itself constructs. We prove

$$L_\alpha^L = L_\alpha$$

for every ordinal α . At the base stage both sides are empty. At a limit stage, both sides are unions of the same previously identified levels; transitivity and agreement on the relevant ordinals make those unions equal. For the successor step, the induction hypothesis identifies the underlying set-sized structure L_α in the internal and external computations. The absoluteness theorem for the L -construction then gives the same definable subsets and hence the same next level.

Taking the union over all ordinals,

$$L^L = \bigcup_{\alpha \in \text{Ord}^L} L_\alpha^L = \bigcup_{\alpha \in \text{Ord}} L_\alpha = L.$$

Thus constructibility is idempotent: once one passes to L , repeating the construction adds nothing new. $\square$

Worked example

Worked Example 38.6 (Why uniformity matters). Suppose a transitive model M contains an ordinal α and has computed every earlier L_β correctly. Because the formula defining $\text{Def}(L_\beta)$ is uniform, both M and the ambient universe apply the same set-sized satisfaction recursion at the successor step. The induction can therefore continue through α .

A useful warning

Absoluteness of the construction requires transitivity and the relevant ordinals/parameters. It is not an unrestricted statement about arbitrary nonstandard models.

Looking ahead

The next chapter, *ZF axioms in L , part I*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 39

ZF axioms in L, part I

Definitions and setup

Formal definitions used in this chapter

Relativization to L and truth in L . For a set-theoretic formula φ , φ^L is obtained by restricting every formerly unbounded quantifier to the class L . For parameters $\bar{a} \in L$,

$$L \models \varphi(\bar{a}) \iff V \models \varphi^L(\bar{a}).$$

Internal power set. For $a \in L$,

$$\mathcal{P}^L(a) := \{x \in L : x \subseteq a\} = \mathcal{P}(a) \cap L.$$

This is the collection that must be shown to belong to L in order to verify the Power Set axiom internally.

Constructible parameter and witness localization. A constructible parameter is an element of L . A witness-localization bound for a formula $\varphi(x, y, \bar{p})$ and domain $a \in L$ is an ordinal β such that whenever $L \models \exists y \varphi(x, y, \bar{p})$ for $x \in a$, some such witness lies in L_β .

Separation in L . For $a, \bar{p} \in L$, Separation asks that $\{x \in a : L \models \varphi(x, \bar{p})\} \in L$.

ZF consists of Extensionality, Empty Set (derivable in common presentations), Pairing, Union, Power Set, Infinity, Separation, Replacement, Foundation, and the usual logical framework. Ordered pairs are Kuratowski pairs. Relations are sets of ordered pairs, and functions are functional relations. Each construction explicitly identifies the axiom that supplies the required set. Choice is not part of ZF and is never imported silently.

How this chapter fits together

The easy axioms follow mostly from transitivity and closure of the hierarchy, but Separation and Power Set require genuine arguments. Separation uses definability over a sufficiently high level. Power Set is internal: L needs the set of constructible subsets of a , not every ambient subset of a . Stage bounds are therefore essential.

Theorem 39.1 (L satisfies Extensionality, Foundation, Pairing, Union, and Infinity). *The structure $(L, \in)$ satisfies Extensionality, Foundation, Pairing, Union, and Infinity.*

Proof.

Formal derivation spine

Formal spine: exhibit each required witness at a specific level. If $x, y \in L_\alpha$, then

$$\{x, y\} = \{z \in L_\alpha : z = x \vee z = y\} \in \text{Def}(L_\alpha) = L_{\alpha+1}. \quad (\text{E1})$$

If $x \in L_\alpha$, transitivity of L_α gives

$$\bigcup x = \{z \in L_\alpha : \exists u \in x (z \in u)\} \in L_{\alpha+1}. \quad (\text{E2})$$

Finally,

$$\omega = \{z \in L_\omega : z \text{ is an ordinal}\} \in \text{Def}(L_\omega) = L_{\omega+1}. \quad (\text{E3})$$

We verify exactly the five axioms named in the statement.

Extensionality. Let $x, y \in L$ and suppose that L satisfies

$$\forall z (z \in x \leftrightarrow z \in y).$$

Because L is transitive, every member of either x or y lies in L . The displayed internal statement therefore says externally that x and y have the same members. Ambient Extensionality yields $x = y$.

Foundation. Let $a \in L$ be nonempty. Ambient Foundation gives $u \in a$ with

$$u \cap a = \emptyset.$$

Since L is transitive and $u \in a \in L$, we have $u \in L$. Thus the same u witnesses Foundation in the structure $(L, \in)$.

Pairing. Choose α with $x, y \in L_\alpha$. Since L_α is transitive, both x and y are elements of the domain over which definability is taken. The subset

$$\{z \in L_\alpha : z = x \vee z = y\}$$

is definable over $(L_\alpha, \in)$ from the parameters x, y , and it is exactly $\{x, y\}$. Hence (E1) places the required pair in $L_{\alpha+1} \subseteq L$.

Union. Choose α with $x \in L_\alpha$. Transitivity gives $u \in L_\alpha$ whenever $u \in x$, and then $z \in L_\alpha$ whenever $z \in u$. Consequently the subset displayed in (E2) is exactly $\bigcup x$. It is definable over L_α from the parameter x , so $\bigcup x \in L_{\alpha+1} \subseteq L$.

Infinity. The earlier analysis of the finite constructible levels showed that every finite ordinal lies in L_ω , while no infinite ordinal lies in L_ω . Therefore the set of elements of L_ω that satisfy the already defined ordinal formula is exactly ω . It is parameter-free definable over L_ω , giving (E3). Since ω is inductive, L contains an inductive set and hence satisfies Infinity.

No use of Separation, Replacement, Power Set, or Choice is hidden in these five verifications. Pairing, Union, and Infinity are witnessed by explicit definable subsets of one already constructed level; Extensionality and Foundation use only transitivity together with the corresponding ambient ZF axioms. $\square$

Lemma 39.2 (Finite reflection along the constructible hierarchy). *Let Φ be a finite set of set-theoretic formulas closed under subformulas, and let γ be an ordinal. There is a limit ordinal $\beta > \gamma$ such that for every $\psi(\bar{x}) \in \Phi$ and every parameter tuple $\bar{a} \in L_\beta^{<\omega}$ of the appropriate length,*

$$L \models \psi(\bar{a}) \iff L_\beta \models \psi(\bar{a}).$$

Here truth in L means truth under the relativization ψ^L defined in the setup.

Proof.

Formal derivation spine

Formal spine: close stages under witnesses for finitely many formulas. Starting above γ , construct

$$\beta_0 < \beta_1 < \beta_2 < \dots$$

so that whenever $\exists y \chi(y, \bar{a}) \in \Phi$, $\bar{a} \in L_{\beta_n}$, and $L \models \exists y \chi(y, \bar{a})$, some witness belongs to $L_{\beta_{n+1}}$. Then put

$$\beta = \sup_{n < \omega} \beta_n.$$

Choose $\beta_0 > \gamma$. Suppose β_n has been chosen. There are only set many pairs consisting of an existential subformula $\exists y \chi(y, \bar{x}) \in \Phi$ and a parameter tuple $\bar{a} \in L_{\beta_n}^{<\omega}$. For each pair for which $L \models \exists y \chi(y, \bar{a})$, let $\rho(\chi, \bar{a})$ be the least constructibility stage containing some witness. Because Φ is finite, χ^L is a fixed first-order formula and membership in the constructible hierarchy is uniformly definable; ambient Replacement therefore collects these least stages. Choose $\beta_{n+1} > \beta_n$ above all of them.

Let $\beta = \sup_{n < \omega} \beta_n$. Since $\beta_{n+1} > \beta_n$ for every n , the ordinal β is a limit ordinal and $\beta > \gamma$. We prove the displayed equivalence by induction on formula complexity for formulas in Φ . Atomic formulas are absolute because L_β is transitive. Boolean connectives follow from the induction hypothesis. For the existential step, suppose $\bar{a} \in L_\beta$. Then $\bar{a} \in L_{\beta_n}$ for some n . If $L \models \exists y \chi(y, \bar{a})$, the construction places a witness in $L_{\beta_{n+1}} \subseteq L_\beta$. Conversely, if $L_\beta \models \exists y \chi(y, \bar{a})$, choose such a $y \in L_\beta$; the induction hypothesis for the proper subformula χ gives $L \models \chi(y, \bar{a})$. This proves finite reflection along the L -hierarchy. *Dependency note.* This is an external

theorem proved in the ambient ZF universe. It is not obtained by applying the earlier Reflection Theorem for V_α to L , and it does not presuppose that L has already been shown to satisfy ZF. $\square$

Theorem 39.3 (L satisfies Separation). *For every formula $\varphi(x, \bar{p})$, every $a, \bar{p} \in L$, the set*

$$\{x \in a : L \models \varphi(x, \bar{p})\}$$

belongs to L . Thus $L \models \text{Separation}$.

Commented symbolic trace: Separation inside L

Suppose $a \in L$ and parameters $\bar{p} \in L$. First choose one level L_β containing a and $\bar{p}$ and high enough for the relevant satisfaction calculation. The desired subset is then defined *over that set-sized structure* by the formula describing membership in a together with φ . Therefore it belongs to $\text{Def}(L_\beta) = L_{\beta+1}$. The proof never assumes that every ambient subset of a is constructible.

Proof.

Formal derivation spine

Formal spine for L satisfies Separation.

$$b = \{x \in a : L \models \varphi(x, \bar{p})\}.$$

$$L \models \varphi(x, \bar{p}) \iff L_\beta \models \varphi(x, \bar{p}).$$

$$b = \{x \in L_\beta : x \in a \ \& \ L_\beta \models \varphi(x, \bar{p})\}$$

Let $a, \bar{p} \in L$ and suppose the subset required by Separation is

$$b = \{x \in a : L \models \varphi(x, \bar{p})\}.$$

Choose an ordinal γ with $a, \bar{p} \in L_\gamma$. Apply the finite-reflection-along- L lemma to the finite set consisting of φ and all of its subformulas, starting above γ . It yields an ordinal $\beta > \gamma$ such that, for every $x \in a$,

$$L \models \varphi(x, \bar{p}) \iff L_\beta \models \varphi(x, \bar{p}).$$

The right-hand condition is a definition over the set-sized structure L_β using parameters $a, \bar{p} \in L_\beta$. Hence

$$b = \{x \in L_\beta : x \in a \ \& \ L_\beta \models \varphi(x, \bar{p})\}$$

belongs to $\text{Def}(L_\beta) = L_{\beta+1}$. Thus $b \in L$, proving every instance of Separation internally in L . The reflection step is the point that converts truth in the proper class L into definability over one set-sized level. *What changed between the three displayed lines.* The first line defines the desired subset using truth in the whole class L , which is not yet a set-sized definition. Reflection replaces that class-level truth test by the equivalent truth test inside one level L_β . The third line then rewrites the same subset as a definable subset of the set L_β , so the successor operation $\text{Def}(L_\beta)$ can actually create it. $\square$

Theorem 39.4 (L satisfies the internal Power Set axiom). *For every $a \in L$, the internal power set*

$$\mathcal{P}^L(a) := \{x \in L : x \subseteq a\} = \mathcal{P}(a) \cap L$$

is an element of L . Therefore $L \models \text{Power Set}$.

Proof.

Formal derivation spine

Formal spine: first form the ambient set of constructible subsets, then bound their stages.

$$C := \{x \in \mathcal{P}(a) : x \in L\}$$

is a set in the ambient universe. If α_x is the least stage with $x \in L_{\alpha_x}$, then Replacement on C gives a bound β , and

$$C = \{x \in L_\beta : x \subseteq a\} \in L_{\beta+1}.$$

Fix $a \in L$. The formula “ $x \in L$ ” is first-order definable by the uniform definability of the constructible hierarchy. Hence ambient Separation applied to the ambient set $\mathcal{P}(a)$ forms

$$C := \mathcal{P}(a) \cap L = \{x \in L : x \subseteq a\} = \mathcal{P}^L(a). \quad (1)$$

This step is essential: only after (1) do we have an ambient set over which Replacement may be applied.

For each $x \in C$, let α_x be the least ordinal such that $x \in L_{\alpha_x}$. The least stage is definable, so ambient Replacement forms the set $\{\alpha_x : x \in C\}$. Choose an ordinal β larger than its supremum and large enough that $a \in L_\beta$. Then every member of C belongs to L_β , and conversely every $x \in L_\beta$ with $x \subseteq a$ belongs to C . Therefore

$$C = \{x \in L_\beta : x \subseteq a\}. \quad (2)$$

The right-hand side of (2) is definable over the set structure L_β from the parameter a , so it belongs to

$$\text{Def}(L_\beta) = L_{\beta+1}.$$

Thus $\mathcal{P}^L(a) = C \in L$. The proof constructs the internal power set, not the full ambient power set when nonconstructible subsets of a exist. $\square$

Lemma 39.5 (Rank/localization control for witnesses used in later axioms). *For every fixed formula $\varphi(x, y, \bar{p})$ and constructible parameters $a, \bar{p}$, there is an ordinal β such that every witness y selected for $x \in a$ may be replaced by a witness lying in L_β , whenever $L \models \forall x \in a \exists y \varphi(x, y, \bar{p})$.*

Proof.

Formal derivation spine

Formal spine: bound all witness stages over one set. For each $x \in a$, define

$$\beta_x := \min\{\gamma : \exists y \in L_\gamma \ L \models \varphi(x, y, \bar{p})\}.$$

Ambient Replacement collects $\{\beta_x : x \in a\}$; with

$$\beta := \sup_{x \in a} (\beta_x + 1),$$

every $x \in a$ has a witness in L_β .

Let $a, \bar{p} \in L$ and suppose that for every $x \in a$ there exists a constructible witness y satisfying a fixed formula $\varphi(x, y, \bar{p})$ in L . For each $x \in a$, define $\rho(x)$ to be the least ordinal α for which some witness $y \in L_\alpha$ exists. The class relation defining “ $\rho(x) = \alpha$ ” is first-order definable because membership in L , the hierarchy, and satisfaction over set-sized levels are uniformly definable.

Ambient Replacement applied to the set a therefore collects the ordinals $\rho(x)$. Their supremum is bounded by some ordinal β , and every required witness may be chosen in L_β . This is only an external stage-localization statement at this point; the next chapter uses the uniform bound to prove Collection inside L without assuming Replacement in L in advance. $\square$

Worked example

Worked Example 39.6 (Internal power set). If $a \in L$, then $\mathcal{P}^L(a) = \{x \in L : x \subseteq a\}$. An ambient subset of a that is not constructible is irrelevant to the Power Set axiom as evaluated inside L . The proof must instead show that all constructible subsets of a are collected together in some later constructible level.

A useful warning

Never prove $L \models \text{Power Set}$ by claiming $\mathcal{P}(a) \subseteq L$; that assertion is generally false when the ambient universe contains nonconstructible subsets.

Looking ahead

The next chapter, *ZF axioms in L, part II*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 40

ZF axioms in L, part II

Definitions and setup

Formal definitions used in this chapter

Collection in L . For a formula $\varphi(x, y, \bar{p})$, Collection is the assertion

$$L \models \forall x \in a \exists y \varphi(x, y, \bar{p}) \implies \exists b \in L \ L \models \forall x \in a \exists y \in b \varphi(x, y, \bar{p}).$$

Functional relation and Replacement in L . A formula is functional on a in L when

$$L \models \forall x \in a \exists! y \varphi(x, y, \bar{p}).$$

Replacement asks that the corresponding image

$$\{y : \exists x \in a \ L \models \varphi(x, y, \bar{p})\}$$

be an element of L .

Definable collection bound. A collection bound is an ordinal β for which all required witnesses can be chosen in L_β , uniformly for $x \in a$. The least constructibility stage of a witness is the least γ with that witness in $L_{\gamma+1}$.

ZF consists of Extensionality, Empty Set (derivable in common presentations), Pairing, Union, Power Set, Infinity, Separation, Replacement, Foundation, and the usual logical framework. Ordered pairs are Kuratowski pairs. Relations are sets of ordered pairs, and functions are functional relations. Each construction explicitly identifies the axiom that supplies the required set. Choice is not part of ZF and is never imported silently.

How this chapter fits together

Replacement is the first serious closure test for L . Given a definable functional relation on a constructible set, witnesses may initially appear at unrelated stages. The proof first obtains a uniform ordinal bound on those witness stages; Collection then gathers candidates inside one level, and Separation extracts the actual range.

Lemma 40.1 (Definable collection bound for L). *If $L \models \forall x \in a \exists y \varphi(x, y, \bar{p})$, then there is an ordinal β such that*

$$L \models \forall x \in a \exists y \in L_\beta \varphi(x, y, \bar{p}).$$

The bound β is obtained from the least constructibility stages at which witnesses occur.

Proof.

Formal derivation spine

Formal spine: the definable Collection bound. Assuming $L \models \forall x \in a \exists y \varphi(x, y, \bar{p})$, define the least witness stages β_x and let

$$\beta = \sup\{\beta_x + 1 : x \in a\}.$$

Then

$$L \models \forall x \in a \exists y \in L_\beta \varphi(x, y, \bar{p}).$$

Proof map. The key issue is uniformity. It is not enough that each $x \in a$ has some witness somewhere in L . Associate to x the least ordinal stage at which some appropriate witness appears. This least stage is unique because ordinals are well-ordered; no choice of a particular witness is required. Ambient Replacement collects those stage ordinals and their supremum gives one level containing witnesses for all $x \in a$.

Let $a, \bar{p} \in L$ and assume $L \models \forall x \in a \exists y \varphi(x, y, \bar{p})$. Because membership in L and satisfaction by a set-sized level L_α are uniformly definable in the ambient universe, for each $x \in a$ there is a least ordinal α_x for which some $y \in L_{\alpha_x}$ satisfies the relativized witness condition. The relation $x \mapsto \alpha_x$ is definable and functional in the ambient ZF universe, so ambient Replacement produces the set $\{\alpha_x : x \in a\}$. Let β exceed its supremum and the ranks of all parameters. Then every $x \in a$ has a witness in the single set L_β . This uniform stage bound is the definable-collection bound; no choice of one witness from each fiber is required. $\square$

Theorem 40.2 (L satisfies Collection). *For every formula $\varphi(x, y, \bar{p})$, if $a, \bar{p} \in L$ and*

$$L \models \forall x \in a \exists y \varphi(x, y, \bar{p}),$$

then there is $b \in L$ such that

$$L \models \forall x \in a \exists y \in b \varphi(x, y, \bar{p}).$$

Hence $L \models \text{Collection}$.

Uses: uniform definability of the constructible hierarchy, stage localization, ambient Replacement for the set of least witness stages, and the previously proved Separation theorem inside L .

How to read the symbols: Collection needs a uniform stage bound

For each $x \in a$, existence gives at least one constructible witness y . The key move is not to choose a preferred witness; it is to choose the least ordinal stage at which *some* witness appears. Replacement in the ambient universe collects these ordinal stage numbers, whose supremum gives one level L_β containing a witness for every $x \in a$. Only after this common bound is obtained do we use Separation to collect a set of witnesses.

Proof.

Formal derivation spine

Formal spine for L satisfies Collection.

$$b = \{y \in L_\beta : \exists x \in a \varphi^L(x, y, \bar{p})\}.$$

$$\beta_x := \min\{\beta : \exists y \in L_\beta L \models \varphi(x, y, \bar{p})\}.$$

Proof map. With the stage bound in hand, Collection inside L is now local: all needed witnesses lie in one set L_β . The previously proved Separation theorem for the whole inner model L , applied to that set, selects exactly the candidate witnesses. No separate “projection” operation and no unproved Separation principle for an individual level L_γ is used.

Use the bound from the definable collection bound proved immediately above. There is an ordinal β such that for every $x \in a$ a witness $y \in L_\beta$ satisfies $\varphi^L(x, y, \bar{p})$. The level L_β is itself an element of L : for example,

$L_\beta \in L_{\beta+1}$. Apply Theorem 39.3, the already established Separation schema inside L , to the set L_β and the formula

$$\exists x \in a \varphi(x, y, \bar{p}).$$

It forms the set

$$b = \{y \in L_\beta : L \models \exists x \in a \varphi(x, y, \bar{p})\} \in L.$$

Equivalently, in the external relativized notation used in the formal spine,

$$b = \{y \in L_\beta : \exists x \in a \varphi^L(x, y, \bar{p})\}.$$

By the stage bound, L verifies that every $x \in a$ has a φ -witness in b . This is Collection. The proof is noncircular: it uses ambient Replacement only in the metatheoretic verification that the inner class L satisfies Collection, and it does not use AC in L . $\square$

Theorem 40.3 (L satisfies Replacement). *If $L \models \forall x \in a \exists! y \varphi(x, y, \bar{p})$, then the image set*

$$\{y : \exists x \in a \varphi(x, y, \bar{p})\}$$

belongs to L . Hence $L \models \text{Replacement}$.

Proof.

Formal derivation spine

Formal spine for L satisfies Replacement.

$$r = \{y \in b : \exists x \in a \varphi(x, y, \bar{p})\}.$$

$$\beta_x := \min\{\beta : \exists y \in L_\beta L \models \varphi(x, y, \bar{p})\}.$$

Proof map. Replacement is derived only after Collection. For a functional relation, Collection supplies a set containing possible values; Separation selects exactly those elements that are the unique values of members of the domain. This ordering avoids circular use of Replacement in L .

Assume in L that $\varphi(x, y, \bar{p})$ is functional on a set a : for every $x \in a$ there is a unique y with $\varphi(x, y, \bar{p})$. By Collection (the Collection theorem just proved) there is $b \in L$ containing at least one witness for every $x \in a$. Separation in L then forms

$$r = \{y \in b : \exists x \in a \varphi(x, y, \bar{p})\}.$$

Uniqueness makes r exactly the range of the functional relation. Thus the Replacement schema holds in L . This order of proof avoids the earlier circular presentation: Collection is obtained from a uniform constructibility-stage bound, and Replacement is derived only afterward from Collection plus Separation. $\square$

Theorem 40.4 (L satisfies ZF). *The inner model $(L, \in)$ satisfies every axiom of ZF.*

Proof.

Formal derivation spine

Formal spine: assemble the separately verified ZF axioms. The preceding results establish in L :

$$\text{Ext} + \text{Found} + \text{Pair} + \text{Union} + \text{Inf},$$

$$\text{Sep} + \text{Power} + \text{Collection} + \text{Replacement}.$$

Hence every axiom in the chosen ZF axiomatization holds in $(L, \in)$, i.e.

$$L \models \text{ZF}.$$

The structural axioms Extensionality, Foundation, Pairing, Union, and Infinity were verified using transitivity and the closure of the hierarchy. Separation and the internal Power Set axiom were proved by definability

and stage bounding. The Collection theorem proves Collection, and the immediately preceding Replacement theorem derives Replacement from Collection plus Separation. These are precisely the ZF axioms (in the chosen equivalent formulation with Collection/Replacement). Therefore $L \models \text{ZF}$. No appeal to the constructible well-order or to Choice is needed for this conclusion; Choice is proved only in the following chapter.

Further explanation. The conclusion is a bookkeeping theorem only after the difficult axioms have been checked separately. Extensionality, Foundation, Pairing, Union, and Infinity follow from transitivity and the level construction; Separation uses the reflecting level L_β supplied by the finite reflection lemma for the constructible hierarchy; Power Set is the internal constructible power set; and Collection/Replacement require the uniform stage-bound argument and the already established Separation theorem. Since each axiom has now been verified inside L , the conjunction of those results is exactly $L \models \text{ZF}$, with no appeal yet to Choice. $\square$

Remark 40.5 (Source note). Gödel's detailed constructibility proof verifies the ZF axioms in the inner model L ; the present organization separates the easy structural axioms from Separation, Power Set, Collection, and Replacement. Compare [8, 13, 14].

Worked example

Worked Example 40.6 (Bounding witness stages). For each $x \in a$, let β_x be the least stage containing some witness to $\varphi(x, y)$. The stage is uniquely determined, so no preferred witness and no constructible well-order is needed. Ambient Replacement collects the ordinals β_x , and their supremum gives one stage L_β containing witnesses for every $x \in a$. This uniform bound is the heart of Collection in L .

A useful warning

Do not use Replacement in L while proving Replacement in L . The stage-bound argument may use Replacement in the ambient metatheory, but the internal conclusion must be obtained from already verified closure properties.

Looking ahead

The next chapter, *Canonical well-order and Choice*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 41

Canonical well-order and Choice

Definitions and setup

Formal definitions used in this chapter

Local definition code at a successor stage. Assume that a well-order $<_{L,\alpha}$ of L_α has already been constructed. A *local definition code over L_α* is a pair $(e, \bar{a}) \in \omega \times L_\alpha^{<\omega}$, where e is the code of a normalized defining formula $\varphi_e(v_0, v_1, \dots, v_n)$, $\bar{a} \in L_\alpha^n$, and the parameter length of the formula is n . It codes the subset

$$X_{e,\bar{a}} := \{x \in L_\alpha : \text{Sat}_{\text{par}}(L_\alpha, e, x, \bar{a})\} \in \text{Def}(L_\alpha) = L_{\alpha+1}.$$

Shortlex order on finite parameter tuples. Once $<_{L,\alpha}$ is fixed, finite tuples from L_α are ordered first by length and then lexicographically using $<_{L,\alpha}$. Local definition codes are ordered lexicographically using the natural order on the formula code e and this shortlex parameter order. Lemma 41.1 below proves that these orders are well-orders whenever the underlying parameter order is a well-order.

Stage-recursive order specification. At a successor stage, keep the previously constructed order on L_α , place every old element before every new element of $L_{\alpha+1} \setminus L_\alpha$, and order the new elements by their least local definition codes. At a limit stage λ , take the union of the coherent earlier orders. The next proposition proves that this specification produces well-orders $<_{L,\alpha}$ for all ordinals α ; the notation is not presupposed before that recursion is proved.

Global constructible order. After the stage recursion has been established, define

$$<_L := \bigcup_{\alpha \in \text{Ord}} <_{L,\alpha}.$$

The following theorem proves that this is a parameter-free definable class well-order of L .

Choice function from a well-order. If $\mathcal{A}$ is a set of nonempty sets and $<$ well-orders $\bigcup \mathcal{A}$, a choice function obtained from $<$ is the function f with $f(A) = \min_{<} A$ for $A \in \mathcal{A}$.

Roadmap. The proposition below will prove, by coherent transfinite recursion, that local definition codes can be well-ordered at every stage and that each newly appearing constructible set therefore has a least such code. Only after that proposition is proved will the class union be denoted $<_L$ and proved to be a definable global well-order. Choice will then be derived from that proved well-order rather than presupposed in its construction.

How this chapter fits together

The proof order is: first verify the finite-tuple/code well-orders used at successor stages; then construct coherent set well-orders $<_{L,\alpha}$; prove their class union is a parameter-free definable well-order of L ; then derive Choice by selecting least elements. The phrases “least description” and “canonical constructible well-order” refer to outputs of those proofs, not to objects assumed in advance.

Lemma 41.1 (Shortlex and local-code well-orders). *Let $(X, <)$ be a well-ordered set. Order the finite tuples $X^{<\omega}$ first by length and, among tuples of the same length, lexicographically by $<$. Then this shortlex relation well-orders $X^{<\omega}$. Consequently, if formula codes are ordered by the usual order on ω , the lexicographic order on pairs $(e, \bar{a}) \in \omega \times X^{<\omega}$ is a well-order.*

Proof. Let $A \subseteq X^{<\omega}$ be nonempty. The set of lengths of members of A is a nonempty subset of ω , so it has a least element n . It remains to find the lexicographically least member of $A \cap X^n$. This is done coordinate by coordinate: choose the least first coordinate occurring among tuples in $A \cap X^n$, restrict to tuples having that coordinate, then choose the least second coordinate occurring in the restricted nonempty set, and continue for the finitely many coordinates. The resulting tuple belongs to $A \cap X^n$ and is lexicographically least there. Hence shortlex is a well-order.

For the code pairs, let $C \subseteq \omega \times X^{<\omega}$ be nonempty. Choose the least first coordinate e occurring in C ; among pairs with that e , choose the shortlex-least parameter tuple. This gives the least member of C in the lexicographic code order. $\square$

Proposition 41.2 (Canonical well-order of each L_α). *For every ordinal α , there is a well-order $<_{L,\alpha}$ of L_α definable uniformly from α , obtained by comparing the least formula/parameter codes defining each element at the first stage where it appears.*

Proof.

Formal derivation spine

Formal spine: a coherent transfinite recursion. At a successor stage let

$$N_\alpha := L_{\alpha+1} \setminus L_\alpha.$$

For $x \in N_\alpha$, let $c_\alpha(x)$ be the least local definition code $(e, \bar{a})$ with $X_{e,\bar{a}} = x$. Extend $<_{L,\alpha}$ by

$$L_\alpha <_{L,\alpha+1} N_\alpha, \quad x <_{L,\alpha+1} y \iff c_\alpha(x) <_{\text{code}} c_\alpha(y) \quad (x, y \in N_\alpha).$$

At a limit λ , put

$$<_{L,\lambda} = \bigcup_{\alpha < \lambda} <_{L,\alpha}.$$

We construct the orders by transfinite recursion. The order on $L_0 = \emptyset$ is empty. Assume $<_{L,\alpha}$ has been constructed and well-orders L_α . Every member of

$$L_{\alpha+1} = \text{Def}(L_\alpha)$$

has at least one local definition code $(e, \bar{a})$. The code order fixed in the setup is a well-order, so every new element $x \in N_\alpha := L_{\alpha+1} \setminus L_\alpha$ has a least code $c_\alpha(x)$. Keep the old order on L_α , put all old elements before all members of N_α , and order N_α by these least codes. The resulting relation well-orders $L_{\alpha+1}$ and extends $<_{L,\alpha}$.

If λ is limit, the earlier orders are coherent because each successor step extends its predecessor. Therefore

$$<_{L,\lambda} := \bigcup_{\alpha < \lambda} <_{L,\alpha}$$

is a well-order of $L_\lambda = \bigcup_{\alpha < \lambda} L_\alpha$: every nonempty $A \subseteq L_\lambda$ contains an element of least stage of appearance, and within that stage the corresponding earlier order has a least element of A . Transfinite recursion therefore produces $<_{L,\alpha}$ for every ordinal α .

The construction is uniform in α : the operations Def, satisfaction for a set-sized level, formula coding, finite-tuple coding, and transfinite recursion were all defined earlier. Hence the relation $<_{L,\alpha}$ is uniformly definable from α . *Why there is no circularity.* A local code at stage $\alpha + 1$ uses only the already constructed order on L_α . The order on the new stage is then defined from those codes. No “canonical code” or constructible well-order is assumed before the recursion constructs it. $\square$

Theorem 41.3 (Definable global well-order of L). *There is a parameter-free definable class relation $<_L$ that well-orders L and whose restriction to each L_α is a set well-order.*

Proof.

Formal derivation spine

Formal spine for the global order.

$$<_L = \bigcup_{\alpha \in \text{Ord}} <_{L,\alpha}.$$

For every ordinal θ , coherence gives

$$<_L \upharpoonright L_\theta = <_{L,\theta}.$$

By the preceding proposition, the relations $<_{L,\alpha}$ are coherent: if $\alpha < \beta$, then $<_{L,\beta}$ extends $<_{L,\alpha}$. Hence their class union

$$<_L := \bigcup_{\alpha \in \text{Ord}} <_{L,\alpha}$$

is a well-defined class relation on L , and its restriction to every L_θ is exactly the set well-order $<_{L,\theta}$. If $A \in L$ is nonempty, choose θ with $A \subseteq L_\theta$; then $<_{L,\theta}$ gives a least member of A . Thus $<_L$ globally well-orders L .

Uniform definability of the stage recursion, together with the parameter-free definitions of the constructible hierarchy and set-sized satisfaction, gives one parameter-free formula defining the relation $x <_L y$. No choice principle is used: the least code exists because the relevant code set is already well-ordered by the stage recursion. $\square$

Theorem 41.4 (L satisfies AC). *The definable global well-order $<_L$ yields a choice function for every constructible family of nonempty sets. Hence*

$$L \models \text{AC}.$$

Proof.

Formal derivation spine

Formal spine for L satisfies AC.

$$f(x) := \min_{<_L} x \quad (x \in A).$$

$$y = f(x) \iff y \in x \wedge \forall z \in x \neg (z <_L y)$$

$$L \models \forall x \in A [f(x) \in x],$$

Let $A \in L$ be a set of nonempty sets. Using the global well-order from the preceding theorem, define

$$f(x) := \min_{<_L} x \quad (x \in A).$$

For each $x \in A$, existence and uniqueness of $f(x)$ follow from the fact that $<_L$ well-orders L . The defining relation

$$y = f(x) \iff y \in x \wedge \forall z \in x \neg(z <_L y)$$

is first-order definable in L . Replacement in L , proved earlier without Choice, collects the graph of f . Thus

$$L \models \forall x \in A [f(x) \in x],$$

so f is a choice function and $L \models \text{AC}$. *Point specific to this result.* For any constructible family of nonempty constructible sets, the global $<_L$ -well-order selects a least element from each member. Replacement, already established inside L , collects these selected elements into a choice function. $\square$

How to read the symbols: what has been proved here, and what has not yet been inferred

The mathematical inner-model construction is now complete through Choice: the preceding chapters give, uniformly in the axioms,

$$L \models \text{ZF} + \text{AC} = \text{ZFC}.$$

This does *not* by itself constitute a proof of the metatheoretic implication $\text{Con}(\text{ZF}) \rightarrow \text{Con}(\text{ZFC})$. That implication is a statement about finite proof codes. Volume VIII first proves a general primitive-recursive proof-translation consistency lemma and only then records the formal corollary for the constructible interpretation. Until that proof-theoretic bridge has been established, the present volume uses only the inner-model conclusion above.

Worked example

Worked Example 41.5 (Choosing the least constructible element). Let $X \in L$ be nonempty. Because $<_L$ well-orders L , the subset X has a unique $<_L$ -least element. For a family $\mathcal{A} \in L$ of nonempty sets, map $A \in \mathcal{A}$ to its least element. Replacement in L collects these values into a choice function.

A useful warning

The global relation $<_L$ is definable; it is not required to be a set containing all constructible objects, since L itself is a proper class.

Looking ahead

The next chapter, *Condensation*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 42

Condensation

Definitions and setup

Formal definitions used in this chapter

Expanded constructible-level structure. Write

$$\mathcal{L}_\theta := (L_\theta, \in, <_L \upharpoonright L_\theta).$$

For $X \subseteq L_\theta$, the notation $X \prec L_\theta$ abbreviates: the induced substructure on X is an elementary substructure of $\mathcal{L}_\theta$, in the sense of the elementary-substructure definition from Volume I.

Definable Skolem functions of $\mathcal{L}_\theta$. For each formula $\exists y \varphi(y, \bar{x})$ in the finite language $\{\in, <\}$, define a partial function F_φ^θ on tuples $\bar{a} \in L_\theta^{<\omega}$ by

$$F_\varphi^\theta(\bar{a}) = \min_{<_L} \{y \in L_\theta : L_\theta \models \varphi(y, \bar{a})\}$$

whenever the displayed witness set is nonempty. The restriction $<_L \upharpoonright L_\theta$ is a set well-order by the previous chapter, so the value is unique whenever it is defined.

Skolem-hull stages. For $A \subseteq L_\theta$, define

$$H_0^\theta(A) := A,$$

and, recursively for $n < \omega$,

$$H_{n+1}^\theta(A) := H_n^\theta(A) \cup \left\{ F_\varphi^\theta(\bar{a}) : \begin{array}{l} \varphi \text{ is a formula of the language } \{\in, <\}, \\ \bar{a} \in (H_n^\theta(A))^{<\omega}, \text{ and } F_\varphi^\theta(\bar{a}) \text{ is defined} \end{array} \right\}.$$

The collection of formulas is a set and $H_n^\theta(A) \subseteq L_\theta$, so Replacement and Separation make every displayed stage a set. Define

$$\text{Hull}^{L_\theta}(A) := \bigcup_{n < \omega} H_n^\theta(A).$$

This union contains A and is closed under every defined value of every F_φ^θ : a finite tuple from the union is already contained in one common stage $H_n^\theta(A)$, so its Skolem value lies in $H_{n+1}^\theta(A)$. Conversely, any subset of L_θ containing A and closed under all these functions contains every $H_n^\theta(A)$ by induction, hence contains the displayed union. Thus the formula really constructs the least closed subset rather than using its existence as part of the definition. Zero-parameter formulas are included because the empty tuple belongs to $(H_n^\theta(A))^{<\omega}$, even when $A = \emptyset$.

Mostowski/transitive collapse. Recall that a relation is well founded and extensional as defined in Volume IV. The Mostowski collapse of $(X, \in)$ is the unique isomorphism $\pi : (X, \in) \cong (M, \in)$ onto a transitive set M , when the hypotheses of the Mostowski theorem hold.

Level-correct constructible level. A limit ordinal θ is *level-correct* (for the exact-level formula fixed in Theorem 38.2) when

$$L_\theta \models \text{Level}_L(L_\alpha, \alpha) \quad \text{for every } \alpha < \theta.$$

This is a property of the fixed literal- Σ_1 level formula, not an assertion that L_θ satisfies ZF. Lemma 42.1 below proves that level-correct limit ordinals occur arbitrarily high.

Cardinals computed in L . $|a|^L$ is the cardinality that L assigns to a ; κ is an L -cardinal when L regards κ as an initial ordinal; and $(\kappa^+)^L$ is the least L -cardinal strictly larger than κ . These are internal cardinal notions and need not equal ambient cardinalities without additional hypotheses.

Condensation identifies elementary substructures of suitably reflected constructible levels. In the form proved below, first choose a level-correct limit θ , then take $X \prec L_\theta$, collapse X to a transitive set M , and use the exact-level recognition machinery to show $M = L_\beta$ for its ordinal height β . The collapse fixes ordinals that are contained pointwise in the hull. This identification is the key localization tool for constructible subsets of a cardinal.

How this chapter fits together

Condensation identifies the transitive collapse of a suitable elementary substructure of L_θ with an actual earlier level L_β . The proof combines three ideas already developed separately: Skolem hulls, Mostowski collapse, and uniform definability of the constructible hierarchy. The identification with L_β , rather than with an arbitrary transitive model, is the decisive step.

Lemma 42.1 (Arbitrarily high level-correct constructible levels). *For every ordinal γ there is a level-correct limit ordinal $\theta > \gamma$.*

Proof. Let Φ be the finite set consisting of the fixed formula $\text{Level}_L(y, \alpha)$ and all of its subformulas. Apply the finite-reflection-along- L Lemma 39.2 above γ . It gives a limit $\theta > \gamma$ such that, for every parameter tuple from L_θ , the formulas in Φ have the same truth value in L and in L_θ . If $\alpha < \theta$, then $L_\alpha \in L_{\alpha+1} \subseteq L_\theta$, and the exact-level theorem gives

$$L \models \text{Level}_L(L_\alpha, \alpha).$$

Reflection therefore yields

$$L_\theta \models \text{Level}_L(L_\alpha, \alpha).$$

Thus θ is level-correct. Notice that this argument uses only reflection for one fixed finite family of formulas; it does not assume that $L_\theta \models \text{ZF}$. $\square$

Lemma 42.2 (Skolem hulls inside L_θ). *Let $\theta \geq 1$ be an ordinal and $A \subseteq L_\theta$. Closing A under the partial Skolem functions F_φ^θ defined above produces an elementary hull $X = \text{Hull}^{L_\theta}(A) \prec L_\theta$.*

Proof.

Formal derivation spine

Formal spine: close under the explicitly defined witness functions. Let $H_0 = A$, and recursively put

$$H_{n+1} = H_n \cup \{F_\varphi^\theta(\bar{a}) : \bar{a} \in H_n^{<\omega} \text{ and } F_\varphi^\theta(\bar{a}) \text{ is defined}\}.$$

Then

$$X := \bigcup_{n < \omega} H_n$$

is closed under all the Skolem functions, and the Tarski–Vaught criterion yields $X \prec L_\theta$.

The language $\{\in, <\}$ is countable, so its formulas can be enumerated. For every existential formula, the setup has already defined the partial function that chooses the $<_L$ -least witness whenever a witness exists in L_θ .

Build H_n by the displayed recursion and put $X = \bigcup_{n < \omega} H_n$. Every parameter tuple from X occurs in some H_n . Therefore, if

$$L_\theta \models \exists y \varphi(y, \bar{a}) \quad (\bar{a} \in X^{<\omega}),$$

the value $F_\varphi^\theta(\bar{a})$ is defined and belongs to $H_{n+1} \subseteq X$ for a sufficiently large n . This is exactly the Tarski–Vaught witness condition, so $X \prec L_\theta$.

The same construction gives the size estimate

$$|X| \leq \max(|A|, \aleph_0),$$

and, when carried out internally in L , the corresponding L -cardinality estimate. No undefined ambient structure M and no unstated family of Skolem functions is being used. $\square$

Lemma 42.3 (Mostowski collapse of the L -hull). *If $X \prec L_\theta$, the well-founded extensional structure $(X, \in)$ has a transitive Mostowski collapse $\pi : X \cong M$.*

Proof.

Formal derivation spine

Formal spine: collapse the elementary hull. Because $(X, \in)$ is well founded and extensional, define by well-founded recursion

$$\pi(x) = \{\pi(y) : y \in x \cap X\}.$$

Then $M := \pi[X]$ is transitive and

$$x \in_X y \iff \pi(x) \in \pi(y),$$

so $\pi : X \cong (M, \in)$.

Let R be well founded and extensional on A . Define recursively $\pi(a) = \{\pi(b) : bRa\}$. Well-founded recursion supplies a unique function. If $\pi(a) = \pi(a')$, then extensionality and induction on predecessors imply that a, a' have the same R -predecessors, hence $a = a'$; thus π is injective. Let $M = \pi[A]$. By definition, $x \in \pi(a)$ iff $x = \pi(b)$ for some bRa , so M is transitive and π is an isomorphism $(A, R) \cong (M, \in)$. Any other collapse satisfies the same recursion and hence equals π .

Point specific to this result. The Skolem hull is well founded because it is a substructure of the transitive L_θ . Mostowski collapse therefore produces a transitive set; the next Condensation argument identifies that set with an earlier constructible level. $\square$

Lemma 42.4 (Recognition of a constructible initial segment). *Let M be a transitive set and put*

$$\beta := M \cap \text{Ord}.$$

Assume that β is a limit ordinal and that, in the structure $(M, \in)$, the following two assertions hold:

1. *for every ordinal $\alpha \in M$ there is a set $y \in M$ with $\text{Level}_L(y, \alpha)$;*
2. *every $x \in M$ belongs to some such internally recognized constructible level, that is,*

$$M \models \forall x \exists \alpha \exists y (\alpha \in \text{Ord} \wedge \text{Level}_L(y, \alpha) \wedge x \in y).$$

Then $M = L_\beta$.

Proof. Because M is transitive, $\beta = M \cap \text{Ord}$ is an ordinal and

$$\alpha < \beta \iff \alpha \in M \cap \text{Ord}. \tag{R1}$$

By hypothesis β is limit, so the defining limit clause of the constructible hierarchy gives

$$L_\beta = \bigcup_{\alpha < \beta} L_\alpha. \tag{R1a}$$

Fix $\alpha < \beta$. By the first hypothesis, M contains a set y such that

$$M \models \text{Level}_L(y, \alpha).$$

By Theorem 38.2, the unadorned formula Level_L is in literal Σ_1 normal form. Lemma 38.1(a) therefore gives upward absoluteness from the transitive set M to the universe, so

$$V \models \text{Level}_L(y, \alpha).$$

Hence $y = L_\alpha$. Thus

$$L_\alpha \in M \quad \text{and therefore, by transitivity of } M, \quad L_\alpha \subseteq M. \quad (\text{R2})$$

Taking the union over $\alpha < \beta$ gives

$$L_\beta = \bigcup_{\alpha < \beta} L_\alpha \subseteq M. \quad (\text{R3})$$

Conversely, let $x \in M$. By the second hypothesis there are $\alpha, y \in M$ such that M regards α as an ordinal,

$$M \models \text{Level}_L(y, \alpha), \quad x \in y.$$

Ordinalhood is absolute for transitive sets, so $\alpha < \beta$. Again literal- Σ_1 upward absoluteness for the chosen Level_L formula identifies y with the ambient L_α . Therefore

$$x \in L_\alpha \subseteq L_\beta.$$

Thus $M \subseteq L_\beta$, which together with (R3) proves $M = L_\beta$.

Why this is a recognition lemma rather than Condensation itself. No elementary substructure appears in the statement. The lemma only says that a transitive set which correctly recognizes the exact L -levels it contains, and regards every one of its elements as lying in one of those levels, must be an initial segment of L . Condensation will verify those two hypotheses by elementarity and collapse. $\square$

Theorem 42.5 (Condensation lemma for L). *(Condensation in the form used below.) Let θ be a level-correct limit ordinal. If $X \prec L_\theta$ and $\pi : X \cong M$ is its transitive collapse, then there is an ordinal β such that*

$$M = L_\beta.$$

Uses: Skolem hulls inside an L -level, Mostowski collapse, and absoluteness of the L -hierarchy.

Commented symbolic trace: Condensation: hull $\rightarrow$ collapse $\rightarrow$ identify

There are three constructions to keep distinct. First take an elementary hull $X \prec L_\theta$. Second apply Mostowski collapse to obtain a transitive set M isomorphic to X . Third use elementarity plus uniform definability of the L -hierarchy to prove that this transitive set is not arbitrary: it is exactly L_β for its ordinal height β . Condensation is the third identification step, not merely the existence of a collapse.

Proof.

Formal derivation spine

Formal spine: transfer two first-order recognition statements, then use literal- Σ_1 upward absoluteness. Let $\pi : X \cong M$ be the transitive collapse and put $\beta = M \cap \text{Ord}$. Because θ is limit, L_θ has no largest ordinal and

$$L_\theta \models \forall \alpha (\alpha \in \text{Ord} \rightarrow \exists \delta (\delta = \alpha \cup \{\alpha\} \wedge \delta \in \text{Ord})), \quad (\text{C0})$$

and it satisfies

$$L_\theta \models \forall \alpha (\alpha \in \text{Ord} \rightarrow \exists y \text{Level}_L(y, \alpha)) \quad (\text{C1})$$

and

$$L_\theta \models \forall x \exists \alpha \exists y (\alpha \in \text{Ord} \wedge \text{Level}_L(y, \alpha) \wedge x \in y). \quad (\text{C2})$$

Elementarity transfers (C1)–(C2) to X , the collapse transfers them to M , and Lemma 42.4 yields $M = L_\beta$.

Step 1: verify the two recognition statements in L_θ . Let $\alpha \in L_\theta$ be an ordinal. Then $\alpha < \theta$. Since θ is level-correct,

$$L_\theta \models \text{Level}_L(L_\alpha, \alpha).$$

Because θ is limit, $L_\alpha \in L_{\alpha+1} \subseteq L_\theta$, so the displayed witness actually belongs to the structure. This proves (C1).

Now let $x \in L_\theta$. By the definition of a limit constructible level, there is some $\gamma < \theta$ with $x \in L_\gamma$. Again $L_\gamma \in L_\theta$, and level-correctness gives

$$L_\theta \models \text{Level}_L(L_\gamma, \gamma) \wedge x \in L_\gamma.$$

Thus (C2) holds as well. No downward absoluteness for a transitive set that has not been shown to satisfy ZF is being used.

Step 2: transfer the recognition statements through the hull and collapse. The notation $X \prec L_\theta$ was defined using the expanded structure $(L_\theta, \in, <_L)$. Its reduct to the language $\{\in\}$ is therefore also elementary. Since (C0), (C1), and (C2) are first-order sentences in that reduct, elementarity gives them in X . The Mostowski map $\pi : (X, \in) \cong (M, \in)$ is an isomorphism, so first-order truth is preserved and

$$M \models (\text{C0}), (\text{C1}), \text{ and } (\text{C2}). \quad (2)$$

Step 3: verify the ordinal-height hypothesis and recognize the collapse. The collapse target M is transitive by Lemma 42.3. Put

$$\beta := M \cap \text{Ord}.$$

If $\delta < \beta$, then $\delta \in M$. By (C0), M contains the von Neumann successor $\delta + 1 = \delta \cup \{\delta\}$. Transitivity makes this internal successor the ambient successor, so $\delta + 1 < \beta$. Thus β has no largest element; it is a limit ordinal. The remaining two assertions in (2) are exactly the recognition hypotheses of Lemma 42.4. Hence

$$M = L_\beta.$$

This proves Condensation.

Dependency check. The old proof implicitly identified the level computed after collapse with the ambient L_δ , which was the conclusion it needed to justify. The present proof separates the steps: Theorem 38.2 fixes a literal- Σ_1 exact-level formula; Lemma 42.1 supplies arbitrarily high levels on which that formula recognizes all earlier constructible stages; elementarity and isomorphism transfer only those first-order recognition statements; and the recognition lemma uses literal- Σ_1 upward absoluteness to identify the ambient levels. There is no circular use of Condensation and no appeal to Δ_1 -absoluteness for a collapse not known to satisfy ZF. $\square$

Remark 42.6 (Scope of the Condensation statement). Standard treatments often state Condensation for arbitrary suitable limit levels. The later arguments in this book need only arbitrarily high levels, and Lemma 42.1 supplies arbitrarily high level-correct limits. We therefore state and prove exactly that reflected form here rather than silently invoking an additional local-absoluteness theorem for the internal L -recursion of every limit level. This restriction does not weaken any subsequent localization or GCH argument in the volume; compare [8, 13, 14] for the standard broader formulations.

Corollary 42.7 (Transitive-set localization by condensation). *Let $a \in L$ be transitive, and let κ be an infinite cardinal of L such that*

$$|a|^L \leq \kappa.$$

Then

$$a \in L_{(\kappa^+)^L}. \quad (1)$$

Here both the cardinality and the successor cardinal are computed inside L .

Proof.

Formal derivation spine

Formal spine: collapse a hull containing the whole transitive set. Use Lemma 42.1 to choose a level-correct limit θ with $a, \kappa \in L_\theta$, and choose

$$X \prec L_\theta, \quad a \cup \{a, \kappa\} \subseteq X, \quad |X|^L \leq \kappa.$$

If $\pi : X \cong L_\beta$ is the collapse, then transitivity of a and $a \subseteq X$ imply

$$\pi \upharpoonright a = \text{id}_a, \quad \pi(a) = a.$$

Moreover

$$|\beta|^L \leq |X|^L \leq \kappa,$$

so $\beta < (\kappa^+)^L$.

Work inside L , where Choice and the Skolem-hull construction have already been established. Use Lemma 42.1 to choose a level-correct limit ordinal θ large enough that

$$a, \kappa \in L_\theta.$$

Because $|a|^L \leq \kappa$, the set of parameters $a \cup \{a, \kappa\}$ has L -cardinality at most κ . The Skolem-hull theorem therefore gives

$$X \prec L_\theta$$

with

$$a \cup \{a, \kappa\} \subseteq X \quad \text{and} \quad |X|^L \leq \kappa. \quad (2)$$

Let

$$\pi : X \longrightarrow M$$

be the Mostowski collapse. By Condensation,

$$M = L_\beta$$

for some ordinal β .

The reason for requiring a to be transitive is now explicit. For each $x \in a$, transitivity gives $x \subseteq a \subseteq X$. An $\in$ -induction on $x \in a$ therefore yields

$$\pi(x) = \{\pi(y) : y \in x\} = x.$$

Hence

$$\pi \upharpoonright a = \text{id}_a. \quad (3)$$

Since $a \in X$, equation (3) also gives

$$\pi(a) = \{\pi(x) : x \in a\} = a. \quad (4)$$

But $\pi(a) \in M = L_\beta$, so (4) implies

$$a \in L_\beta. \quad (5)$$

Finally, the ordinal β is a subset of the transitive collapse M , so in L

$$|\beta|^L \leq |M|^L = |X|^L \leq \kappa.$$

By the definition of the successor cardinal in L ,

$$\beta < (\kappa^+)^L.$$

Combining this with (5) and monotonicity of the constructible hierarchy gives

$$a \in L_\beta \subseteq L_{(\kappa^+)^L},$$

which is (1).

This result is deliberately about an arbitrary transitive constructible set. The next cardinal-subset theorem is a separate corollary obtained by packaging a subset $A \subseteq \kappa$ into the transitive set $\kappa \cup \{A\}$. $\square$

Worked example

Worked Example 42.8 (Collapsing a hull that contains a real). Let $A \subseteq \omega$ be constructible. Use Lemma 42.1 to choose a level-correct limit θ with $A \in L_\theta$, and take a countable elementary hull $X \prec L_\theta$ containing A . Collapse X to a transitive structure M . Since ω and its elements are fixed by the collapse, A is fixed as well. Condensation identifies $M = L_\beta$ for a countable β , showing that the real already appears at a countable constructible stage.

A useful warning

Mostowski collapse alone gives only a transitive set. Condensation additionally uses elementarity and definability of L to identify that set as a specific constructible level.

Looking ahead

The next chapter, *GCH in L*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 43

GCH in L

Definitions and setup

Formal definitions used in this chapter

L -cardinal and L -successor cardinal. An ordinal κ is an L -cardinal when $L \models$ “ κ is an initial ordinal.” Its successor cardinal $(\kappa^+)^L$ is the least L -cardinal strictly above κ .

Internal power set and exponentiation.

$$\mathcal{P}^L(\kappa) := \{A \in L : A \subseteq \kappa\}, \quad (2^\kappa)^L := |\mathcal{P}^L(\kappa)|^L.$$

For the first uncountable ordinal/cardinal computed internally in L , we write

$$\omega_1^L = (\omega^+)^L = \aleph_1^L.$$

Because $L \models \text{AC}$, $\mathcal{P}^L(\kappa)$ is well-orderable in L , so the displayed L -cardinality is defined.

CH and GCH in L . The continuum hypothesis in L is

$$L \models 2^{\aleph_0} = \aleph_1,$$

and the generalized continuum hypothesis in L is the scheme

$$L \models 2^\kappa = \kappa^+$$

for every infinite L -cardinal κ . Equivalently, the latter says $(2^\kappa)^L = (\kappa^+)^L$. When a displayed cardinal calculation in this chapter is internal to L , an L -superscript is retained whenever omitting it could create ambiguity with the ambient cardinal.

GCH in L is a localization theorem plus a counting theorem. Condensation shows that a constructible subset of an infinite constructible cardinal κ already appears below L_{κ^+} . The number of elements of that level is at most κ^+ . Cantor gives the reverse strict inequality over κ , forcing $(2^\kappa)^L = \kappa^+$. CH is written separately as the concrete $\kappa = \aleph_0$ case.

How this chapter fits together

The GCH proof has a cardinal-counting side and a localization side. First, L_α has at most $\max(\aleph_0, |\alpha|)$ elements because its members are coded by formulas and finite parameter tuples. Second, Condensation

shows that every constructible subset of an infinite cardinal κ appears before κ^+ . The two facts bound the power set from above, while Cantor supplies the strict lower bound.

Theorem 43.1 (Cardinality bounds for levels L_α). *For every infinite ordinal α ,*

$$|L_\alpha| \leq \max(\aleph_0, |\alpha|). \quad (\text{LB-V})$$

Moreover the same bound holds internally in the constructible universe:

$$L \models |L_\alpha| \leq \max(\aleph_0, |\alpha|) \quad (\alpha \text{ infinite}). \quad (\text{LB-L})$$

Consequently, if κ is an infinite L -cardinal and $\alpha < (\kappa^+)^L$, then

$$L \models |L_\alpha| \leq \kappa.$$

Proof.

Formal derivation spine

Formal spine for Cardinality bounds for levels L_α .

$$|L_\alpha| \leq \max(\aleph_0, |\alpha|).$$

$$|L_{\beta+1}| \leq \max(\aleph_0, |L_\beta|).$$

$$|L_\lambda| \leq \max(\aleph_0, |\lambda|).$$

We prove by transfinite induction on the infinite ordinal α that

$$|L_\alpha| \leq \max(\aleph_0, |\alpha|).$$

At a successor stage, every element of $L_{\beta+1} = \text{Def}(L_\beta)$ is specified by a first-order formula together with a finite tuple of parameters from L_β . There are only $\aleph_0$ formulas, and for infinite L_β there are $|L_\beta|^{<\omega} = |L_\beta|$ finite parameter tuples. Hence

$$|L_{\beta+1}| \leq \max(\aleph_0, |L_\beta|).$$

The induction hypothesis bounds the right-hand side by $\max(\aleph_0, |\beta|) \leq \max(\aleph_0, |\beta+1|)$.

If λ is an infinite limit ordinal, then $L_\lambda = \bigcup_{\beta < \lambda} L_\beta$. For infinite $\beta < \lambda$, the induction hypothesis gives $|L_\beta| \leq \max(\aleph_0, |\lambda|)$; for finite β , a finite induction on the definition of L_β gives in particular $|L_\beta| \leq \aleph_0$. Thus the union contains $|\lambda|$ many sets, each of size at most $\max(\aleph_0, |\lambda|)$, so

$$|L_\lambda| \leq \max(\aleph_0, |\lambda|).$$

This proves the ambient estimate (LB-V). The argument is a uniform ZFC proof about the constructible hierarchy: it uses only the recursive definition of L_α , formula coding, finite parameter tuples, transfinite induction, and ordinary cardinal arithmetic under Choice. We have already proved $L \models \text{ZFC}$ and Theorem 38.5 gives $L^L = L$, so the same proof may be carried out inside L . There it yields (LB-L) with every cardinality interpreted internally. If κ is an infinite L -cardinal and $\alpha < (\kappa^+)^L$, then internally $|\alpha| \leq \kappa$, giving the displayed consequence.

This theorem proves only the level-size estimate; the localization of subsets needed for GCH is the next, separate theorem. $\square$

Theorem 43.2 (Constructible subsets of a cardinal appear sufficiently early). *Let κ be an infinite cardinal of L . If*

$$A \subseteq \kappa \quad \text{and} \quad A \in L,$$

then

$$A \in L_{(\kappa^+)^L}. \quad (1)$$

Consequently

$$\mathcal{P}^L(\kappa) = \{A \in L : A \subseteq \kappa\} \subseteq L_{(\kappa^+)^L}. \quad (2)$$

Proof.

Formal derivation spine

Formal spine: reduce subset localization to transitive-set localization. For $A \subseteq \kappa$, define

$$a := \kappa \cup \{A\}.$$

Then

$$a \text{ is transitive, } |a|^L = \kappa, \quad A \in a.$$

The preceding localization corollary gives

$$a \in L_{(\kappa^+)^L},$$

and transitivity of $L_{(\kappa^+)^L}$ yields

$$A \in L_{(\kappa^+)^L}.$$

Work inside L . Let $A \subseteq \kappa$ with $A \in L$, and set

$$a := \kappa \cup \{A\}. \tag{3}$$

We check carefully why the preceding corollary applies.

First, a is transitive. If $x \in \kappa$, then x is an ordinal below κ , so

$$x \subseteq \kappa \subseteq a.$$

If $x = A$, then

$$x = A \subseteq \kappa \subseteq a.$$

Thus every element of a is a subset of a .

Second, a has L -cardinality κ . Indeed, $\kappa \subseteq a$, while adjoining the single extra element A to the infinite set κ does not change its cardinality:

$$|a|^L = \kappa. \tag{4}$$

Apply the transitive-set localization corollary to a and κ . Equations (3)–(4) give

$$a \in L_{(\kappa^+)^L}. \tag{5}$$

Every constructible level is transitive. Since $A \in a$, equation (5) therefore implies

$$A \in L_{(\kappa^+)^L},$$

which proves (1).

Because A was an arbitrary constructible subset of κ , (2) follows immediately. The logical division of labor is now clear: Condensation proves localization for transitive sets; the simple transitive package $\kappa \cup \{A\}$ converts that theorem into the subset-of-a-cardinal form used in CH and GCH. $\square$

Theorem 43.3 (CH in L). *Inside L ,*

$$2^{\aleph_0} = \aleph_1.$$

Thus $L \models \text{CH}$.

Proof.

Formal derivation spine

Formal spine for CH in L .

$$|L_{\omega_1^L}|^L \leq \aleph_1^L,$$

$$L \models 2^{\aleph_0} = \aleph_1.$$

$$A \subseteq \kappa \wedge A \in L \implies A \in L_{(\kappa^+)^L}.$$

Apply the preceding localization theorem with $\kappa = \omega$. Every real of L , viewed as a subset of ω , belongs to $L_{\omega_1^L}$. The level-size estimate gives

$$|L_{\omega_1^L}|^L \leq \aleph_1^L,$$

so L has at most $\aleph_1^L$ many reals. Cantor's theorem, which holds in L , gives $2^{\aleph_0} > \aleph_0$. Since $L \models \text{AC}$, its cardinals are linearly well ordered and there is no cardinal strictly between $\aleph_0$ and $\aleph_1$. Hence

$$L \models 2^{\aleph_0} = \aleph_1.$$

This proves CH in L as the first concrete instance of the general GCH argument.

The superscript L matters throughout: ω_1^L is the first uncountable ordinal as computed inside L . The argument is internal to that model and does not require ω_1^L to equal the ambient universe's ω_1 . This is exactly why the result is a model of CH rather than an assertion that CH holds in the ambient universe. $\square$

Theorem 43.4 (GCH in L). *For every infinite cardinal κ of L ,*

$$L \models 2^\kappa = \kappa^+.$$

Hence $L \models \text{GCH}$.

Uses: Condensation/localization, the cardinality bounds for L_α , and Cantor's theorem.

How to read the symbols: the GCH calculation is two inequalities

For an infinite constructible cardinal κ , localization shows that every constructible subset of κ appears before L_{κ^+} . The level-size estimate then gives at most κ^+ such subsets, so $2^\kappa \leq \kappa^+$ inside L . Cantor's theorem gives $2^\kappa > \kappa$. Since κ^+ is the least larger cardinal, the two inequalities force $2^\kappa = \kappa^+$.

Proof.

Formal derivation spine

Formal spine for GCH in L .

$$\mathcal{P}^L(\kappa) \subseteq L_{(\kappa^+)^L}, \quad |L_{(\kappa^+)^L}|^L \leq (\kappa^+)^L.$$

Proof map. For arbitrary infinite constructible κ , apply the same localization to subsets of κ : every such constructible subset occurs before κ^+ . The level-size estimate gives at most κ^+ many subsets; Cantor gives $2^\kappa > \kappa$. Choice in L then forces $2^\kappa = \kappa^+$.

Work inside L , where AC has already been proved, and let κ be an infinite cardinal of L . The preceding localization theorem states directly that every constructible subset $B \subseteq \kappa$ belongs to $L_{(\kappa^+)^L}$. Hence

$$\mathcal{P}^L(\kappa) \subseteq L_{(\kappa^+)^L}.$$

The internal form (LB-L) of Theorem 43.1, applied in L to the ordinal $(\kappa^+)^L$, gives

$$|L_{(\kappa^+)^L}|^L \leq (\kappa^+)^L,$$

and therefore $(2^\kappa)^L \leq (\kappa^+)^L$. Cantor's theorem holds in L , so $(2^\kappa)^L > \kappa$. Since $L \models \text{AC}$, κ^+ is the least cardinal strictly larger than κ ; consequently $(2^\kappa)^L \geq \kappa^+$. The two inequalities give $(2^\kappa)^L = \kappa^+$ for every infinite cardinal κ of L .

Detailed formal verification. Fix an infinite cardinal κ of L . Localization gives, internally to L ,

$$\mathcal{P}^L(\kappa) \subseteq L_{(\kappa^+)^L}. \tag{GCH1}$$

The level-size induction yields

$$|L_{(\kappa^+)^L}|^L \leq (\kappa^+)^L. \quad (\text{GCH2})$$

Combining (GCH1) and (GCH2),

$$(2^\kappa)^L = |\mathcal{P}^L(\kappa)|^L \leq (\kappa^+)^L. \quad (\text{GCH3})$$

Cantor's theorem is valid in L , so

$$L \models \kappa < 2^\kappa. \quad (\text{GCH4})$$

By definition, $(\kappa^+)^L$ is the least constructible cardinal strictly larger than κ . The inequalities (GCH3)–(GCH4) therefore force

$$L \models 2^\kappa = \kappa^+.$$

Notice that every cardinal symbol in this calculation is interpreted inside L ; no ambient successor cardinal is silently substituted for the constructible one.

Second detailed calculation. A useful way to see the cardinal step without suppressing the model is to write all quantities with an L -superscript:

$$(2^\kappa)^L := |\mathcal{P}(\kappa) \cap L|^L.$$

Localization says that the identity map embeds this set into $L_{(\kappa^+)^L}$, hence

$$(2^\kappa)^L \leq |L_{(\kappa^+)^L}|^L.$$

The level bound is proved by induction using only constructible cardinal arithmetic, so its value at the successor stage is

$$|L_{(\kappa^+)^L}|^L \leq (\kappa^+)^L.$$

On the other hand, Cantor's diagonal map is absolute enough to run internally in L , and therefore

$$L \models \kappa < 2^\kappa.$$

There is no L -cardinal strictly between κ and $(\kappa^+)^L$. Thus the upper and lower bounds identify the same internal cardinal. This last sentence is where successor-cardinal minimality is used; it should not be replaced by an ambient statement about κ^+ . $\square$

Remark 43.5 (Source note). Gödel's constructible universe satisfies the generalized continuum hypothesis. The proof here factors the argument into level-size bounds plus Condensation/localization; compare [8, 13, 14].

Worked example

Worked Example 43.6 (CH as the prototype). For a constructible real $A \subseteq \omega$, use Lemma 42.1 to choose a level-correct limit ordinal $\theta > \omega_1^L$ with $A \in L_\theta$, and take a countable elementary hull of L_θ containing A . Condensation places A in some L_β with $\beta < \omega_1$. There are only $\aleph_1$ such stages and each has size at most $\aleph_0$, so L has at most $\aleph_1$ reals. Cantor gives more than $\aleph_0$, hence exactly $\aleph_1$.

A useful warning

The upper bound must be established inside L with correct constructible cardinals. Ambient cardinal arithmetic cannot simply be substituted for internal cardinal arithmetic.

End-of-volume perspective

The constructible universe provides the positive consistency side: inside L , both Choice and GCH hold. The remaining task is to construct alternative universes in which selected statements fail without destroying ZF or ZFC.

Part VII

Volume VII: Forcing I: Generic Extensions and the Forcing Theorem

Preface

Forcing is first developed as a theorem about generic extensions. Atomic forcing, definability, and the Truth Lemma are all proved rather than compressed into a slogan.

Position in the series

Volume VII develops forcing as a theorem about generic extensions. It owns names, the forcing relation, the Truth Lemma, and the proof that ordinary generic extensions preserve ZFC.

Sources and further reading

The forcing theorem is presented in modern name-and-truth-lemma form. Cohen's original method is represented historically by [9, 10, 11]; systematic modern treatments include [13, 14].

Notation for this volume

The only notation used before its formal definition is the forcing-order convention: $q \leq p$ means that q is stronger. Names, their evaluation by a filter, canonical ground-model names, generic-extension notation, and the forcing relation are introduced explicitly in the chapters where their recursive definitions become available.

How to read symbolic arguments in this volume

When a formula appears, read it in layers rather than as one visual block. First identify the *ambient language*: are we speaking inside a formal theory, about a structure, or in the surrounding metatheory? Next mark the objects being manipulated (formulas, codes, sets, names, conditions, ordinals) and the operation being applied to them. Finally check the side conditions—free variables, rank bounds, compatibility, genericity, transitivity, or consistency hypotheses. Whenever a displayed derivation changes level, the text says so explicitly. A displayed line is therefore meant to be read like a line of well-commented code: the symbols perform the operation, and the surrounding prose records its type, preconditions, and purpose.

Chapter 44

Posets, dense sets, and generics

Definitions and setup

Formal definitions used in this chapter

Forcing preorder and order convention. A forcing notion is a preorder $(\mathbb{P}, \leq, 1_{\mathbb{P}})$: $\leq$ is reflexive and transitive, $1_{\mathbb{P}}$ is largest, and $q \leq p$ means that q is the stronger condition. If antisymmetry also holds, $\mathbb{P}$ is a poset.

Compatibility. Conditions $p, q \in \mathbb{P}$ are compatible when some $r \in \mathbb{P}$ satisfies $r \leq p$ and $r \leq q$; otherwise they are incompatible.

Dense and dense below. $D \subseteq \mathbb{P}$ is dense when $\forall p \in \mathbb{P} \exists q \in D (q \leq p)$. It is dense below p when every $q \leq p$ has an extension $r \leq q$ lying in D .

Filter. $G \subseteq \mathbb{P}$ is a filter when $G \neq \emptyset$, it is upward closed toward weaker conditions,

$$p \in G, p \leq q \implies q \in G,$$

and it is directed toward stronger conditions,

$$p, q \in G \implies \exists r \in G (r \leq p \text{ and } r \leq q).$$

Ambient metatheory for external generic constructions. The Rasiowa–Sikorski and externally countable generic-existence arguments below are carried out in the surrounding ZFC metatheory. More precisely, their recursive selection step uses countable dependent choice. This external choice assumption is pedagogical only: the later syntactic/Boolean-valued relative-consistency arguments do not infer a generic filter or a transitive model from a bare consistency statement.

Ground model and genericity. In the generic-extension theorems of this volume the ground model M is a transitive set model of the stated fragment of ZF, and $\mathbb{P} \in M$ is an actual forcing set with its order relation in M . If $D, \mathbb{P} \in M$ and $M \models D \subseteq \mathbb{P}$, transitivity implies that D and $\mathbb{P}$ have their literal ambient elements and that externally $D \subseteq \mathbb{P}$; this does *not* identify the internal and external power sets. A filter $G \subseteq \mathbb{P}$ is M -generic when it meets every $D \in M$ for which $M \models “D \subseteq \mathbb{P}$ is dense.” For transitive M , this density assertion is absolute because all of its set quantifiers are bounded by D and $\mathbb{P}$. For $X \in M$, define the internal power set notation

$$\mathcal{P}^M(X) := \{Y \in M : Y \subseteq X\}.$$

It need not equal the ambient $\mathcal{P}(X)$. Finally, M is *externally countable* when the set M is countable in the ambient universe.

Separative modification and quotient. Define

$$p \leq^* q \iff \forall r \leq p \exists s \leq r (s \leq q), \quad p \equiv q \iff (p \leq^* q \wedge q \leq^* p).$$

A forcing preorder is *separative* when $p \not\leq q$ implies that some $r \leq p$ is incompatible with q . The *separative quotient* is

$$\mathbb{P}_{\text{sep}} := \mathbb{P}/\equiv, \quad [p] \leq_{\text{sep}} [q] \iff p \leq^* q.$$

Dense embedding and forcing equivalence. A map $i : \mathbb{P} \rightarrow \mathbb{Q}$ is a *dense embedding* when it is injective, preserves and reflects the order,

$$p \leq q \iff i(p) \leq i(q),$$

preserves incompatibility, and has dense range $i[\mathbb{P}] \subseteq \mathbb{Q}$. A subset $D \subseteq \mathbb{P}$ is a *dense suborder* when it is dense in $\mathbb{P}$ and carries the inherited order. A set $E \subseteq \mathbb{P}$ is *predense* when every $p \in \mathbb{P}$ is compatible with some $e \in E$. Its *downward closure* is

$$E^\downarrow := \{q \in \mathbb{P} : \exists e \in E (q \leq e)\}.$$

Proposition 44.1 proves that predensity is equivalent to density of this downward closure. In this volume, two forcing notions are *forcing equivalent* when there are dense suborders of their separative quotients that are order-isomorphic. The Boolean-completion characterization is deliberately postponed until Volume VIII, where Boolean completion is defined.

A forcing notion in this volume is a preorder/partial order $(\mathbb{P}, \leq, 1_{\mathbb{P}})$ already equipped with a largest (weakest) condition $1_{\mathbb{P}}$, and $q \leq p$ means that q carries at least as much information. We do not use any top-adjoining equivalence before such an equivalence is proved. Two conditions are compatible when they have a common strengthening. A set D is dense if every condition has a strengthening in D . A filter is a nonempty subset that is directed toward stronger conditions and upward closed toward weaker conditions. An M -generic filter meets every dense subset of $\mathbb{P}$ that belongs to M .

How this chapter fits together

Forcing starts with an order convention: stronger conditions carry more information. Dense sets encode requirements that can always be met by extending a condition. Over an externally countable model, Rasiowa–Sikorski recursively meets all dense sets from that model, producing a generic filter. This countable-model construction is pedagogically central but later separated from the formal relative-consistency argument.

Proposition 44.1 (Equivalent forcing-order conventions and separativity basics). *A forcing notion is a preorder $(\mathbb{P}, \leq, 1_{\mathbb{P}})$ with a largest condition, where $q \leq p$ means that q is stronger than p . Define*

$$p \leq^* q \iff \forall r \leq p \exists s (s \leq r \text{ and } s \leq q).$$

Then $p \equiv q \iff (p \leq^ q \wedge q \leq^* p)$ is an equivalence relation, and the quotient ordered by $[p] \leq_{\text{sep}} [q] \iff p \leq^* q$ is separative and forcing equivalent to $\mathbb{P}$. Moreover, for every $E \subseteq \mathbb{P}$,*

$$E \text{ is predense} \iff E^\downarrow \text{ is dense.}$$

Proof.

Formal derivation spine

Formal spine: the separative preorder and its quotient.

$$p \leq^* q \iff \forall r \leq p \exists s \leq r (s \leq q), \quad p \equiv q \iff p \leq^* q \ \& \ q \leq^* p.$$

The quotient order is $[p] \leq_{\text{sep}} [q] \iff p \leq^* q$.

We verify the order-theoretic points rather than treating the quotient as notation. Reflexivity of $\leq^*$ follows by taking $s = r$. For transitivity, assume $p \leq^* q$ and $q \leq^* t$, and fix $r \leq p$. Choose $s \leq r, q$ from the first assumption. Since $s \leq q$, the second assumption gives $u \leq s, t$. Then $u \leq r, t$, so $p \leq^* t$.

Consequently $\equiv$ is reflexive and symmetric, and transitivity follows from transitivity of $\leq^*$. If $p \equiv p'$, $q \equiv q'$, and $p \leq^* q$, then

$$p' \leq^* p \leq^* q \leq^* q',$$

so the quotient order is well defined. Antisymmetry holds by the definition of the equivalence classes.

To prove separativity, suppose $[p] \not\leq_{\text{sep}} [q]$. Then $p \not\leq^* q$, hence there is $r \leq p$ for which no $s \leq r, q$ exists. Thus $r \perp q$, and $[r] \leq_{\text{sep}} [p]$ is incompatible with $[q]$. This is exactly the separativity condition.

For the predensity claim, suppose first that E is predense and fix $p \in \mathbb{P}$. Choose $e \in E$ compatible with p , and let $r \leq p, e$. Then $r \in E^\downarrow$ and $r \leq p$, so $E^\downarrow$ is dense. Conversely, if $E^\downarrow$ is dense and $p \in \mathbb{P}$, choose $r \leq p$ in $E^\downarrow$ and then $e \in E$ with $r \leq e$. The condition r witnesses compatibility of p with e , so E is predense.

Finally, apply the definition of forcing equivalence rather than the definition of dense embedding. The separative quotient of $\mathbb{P}$ is $\mathbb{P}_{\text{sep}}$ itself, and the separative quotient of the already separative preorder $\mathbb{P}_{\text{sep}}$ is canonically order-isomorphic to $\mathbb{P}_{\text{sep}}$. Taking the whole quotient on both sides therefore gives isomorphic dense suborders, so $\mathbb{P}$ and $\mathbb{P}_{\text{sep}}$ are forcing equivalent.

The quotient map $q(p) = [p]$ is generally *not injective*; hence it is not a dense embedding under the definition fixed above. It is instead the canonical quotient map implementing the passage to forcing equivalence. This distinction matters: no proof step below may cite injectivity of q . $\square$

Theorem 44.2 (Rasiowa–Sikorski lemma). *In the ambient metatheory, assume countable dependent choice (in particular, ZFC suffices). Let $\langle D_n : n < \omega \rangle$ be dense subsets of a poset $\mathbb{P}$, and let $p_0 \in \mathbb{P}$. There is a filter $G \subseteq \mathbb{P}$ with $p_0 \in G$ and $G \cap D_n \neq \emptyset$ for every n .*

Proof.

Formal derivation spine

Formal spine: descend through the dense sets and generate a filter.

$$p_{n+1} \leq p_n, \quad p_{n+1} \in D_n, \quad G := \{q : \exists n (p_n \leq q)\}.$$

Starting with the prescribed condition p_0 , recursively choose

$$p_{n+1} \leq p_n \quad \text{with} \quad p_{n+1} \in D_n,$$

using density of D_n . Define

$$G := \{q \in \mathbb{P} : \exists n (p_n \leq q)\}.$$

Since $p_0 \leq p_0$, the initial condition lies in G . If $q \in G$ and $q \leq r$, choose n with $p_n \leq q$. Transitivity gives $p_n \leq r$, so $r \in G$; thus G is upward closed toward weaker conditions. If $q, r \in G$, choose m, n with $p_m \leq q$ and $p_n \leq r$. The descending sequence satisfies

$$p_{\max(m,n)} \leq q, r,$$

and this common strengthening belongs to G . Hence G is directed toward stronger conditions. Finally,

$$p_{n+1} \in G \cap D_n$$

for every n . The recursive selection of p_{n+1} from the nonempty set $D_n \cap \{q : q \leq p_n\}$ is exactly where countable dependent choice in the ambient metatheory is used. Model-theoretic countability enters only later, when the dense subsets coded by a countable ground model are enumerated. $\square$

Theorem 44.3 (Existence of an M-generic filter over an externally countable transitive ground model). *Let M be an externally countable transitive model of ZF, let $\mathbb{P} \in M$, and assume $M \models \text{“}\mathbb{P} \text{ is a forcing notion”}$. Then for every $p \in \mathbb{P}$ there exists, in the ambient universe, an M-generic filter $G \subseteq \mathbb{P}$ with $p \in G$.*

Proof.

Formal derivation spine

Formal spine for Existence of an M -generic filter over an externally countable transitive ground model.

$$\{D \in M : M \models "D \subseteq \mathbb{P} \text{ is dense}"\}$$

$$G \text{ is } M\text{-generic} \iff \forall D \in M [D \subseteq \mathbb{P} \text{ dense} \Rightarrow G \cap D \neq \emptyset].$$

Assume M is externally countable and transitive and $\mathbb{P} \in M$. Because M is transitive, the elements of $\mathbb{P}$ and of every $D \in M$ with $M \models D \subseteq \mathbb{P}$ are the actual ambient elements of those sets. Moreover the assertion

$$D \subseteq \mathbb{P} \wedge \forall p \in \mathbb{P} \exists q \in D (q \leq p)$$

has all of its set quantifiers bounded by the transitive sets D and $\mathbb{P}$, so internal density agrees with ambient density for these parameters.

Externally, the family

$$\mathcal{D}_M := \{D \in M : M \models "D \subseteq \mathbb{P} \text{ is dense}"\}$$

is countable, because it is a subset of the countable set M . Enumerate it as $D_0, D_1, \dots$. Applying the Rasiowa–Sikorski construction to these actual dense subsets of $\mathbb{P}$ produces a filter $G \subseteq \mathbb{P}$ meeting every D_n , hence every dense subset of $\mathbb{P}$ coded in M . Therefore G is M -generic.

The word “externally” is essential. Nothing in the argument asserts that M itself contains an enumeration of all its dense subsets, and the existence of a countable transitive ground model is not inferred merely from $\text{Con}(\text{ZFC})$. *Why the displayed family is countable externally.* Every dense set used in the construction is itself an element of M . Since the surrounding universe sees M as countable, it can enumerate those elements of M that happen to satisfy the internal formula “dense subset of $\mathbb{P}$ ”. The enumeration is a metatheoretic tool and need not be an element of M . $\square$

Proposition 44.4 (Genericity as meeting all ground-model dense sets). *Let M be transitive and let $\mathbb{P} \in M$. A filter $G \subseteq \mathbb{P}$ is M -generic exactly when*

$$\forall D \in M [M \models "D \subseteq \mathbb{P} \text{ is dense}" \Rightarrow G \cap D \neq \emptyset].$$

For these transitive parameters, “ $D \subseteq \mathbb{P}$ is dense” is absolute, so the displayed condition is equivalently that G meets every actual dense $D \subseteq \mathbb{P}$ belonging to M .

Proof.

Formal derivation spine

Formal spine for Genericity as meeting all ground-model dense sets.

$$G \cap D \neq \emptyset$$

$$G \text{ is } M\text{-generic} \iff \forall D \in M [D \subseteq \mathbb{P} \text{ dense} \Rightarrow G \cap D \neq \emptyset].$$

By the transitivity hypothesis, the internal dense subsets of $\mathbb{P}$ coded in M are literal ambient subsets of $\mathbb{P}$, and density is absolute because its quantifiers are bounded by D and $\mathbb{P}$. Thus a filter $G \subseteq \mathbb{P}$ is M -generic exactly when

$$G \cap D \neq \emptyset$$

for every dense $D \subseteq \mathbb{P}$ with $D \in M$. If “dense” is replaced by “predense”, the condition is equivalent: from a predense set take its downward closure, which is dense, and conversely every dense set is predense.

The formulation explains why genericity depends on the chosen ground model. A filter may meet all dense sets coded in one model while failing to meet a dense set appearing in a larger model. Later forcing arguments always specify the ground model before speaking of a generic filter. $\square$

Worked Theorem 44.5 (Construction of a generic filter for a toy forcing). *Let $2 := \{0, 1\}$, let*

$$2^\omega := \{x : x \text{ is a function } \omega \rightarrow 2\},$$

and let $\mathbb{P}$ be the set of all finite partial functions $p : \omega \rightarrow 2$, ordered by reverse inclusion. Meeting the dense sets $D_n = \{p : n \in \text{dom } p\}$ produces a filter G whose union $g = \bigcup G$ is a total element of 2^ω . The proof also exhibits, for each specified old binary sequence x , a dense set ensuring that the union differs from x .

Proof.

Formal derivation spine

Formal spine for Construction of a generic filter for a toy forcing.

$$D_n = \{p : n \in \text{dom } p\}, \quad G \cap D_n \neq \emptyset \quad (n < \omega),$$

$$g := \bigcup G \in 2^\omega.$$

Take the explicitly defined poset $\mathbb{P}$ from the statement: its conditions are finite partial functions $p : \omega \rightarrow 2$, ordered by reverse inclusion. Recall that 2^ω denotes the set of total functions $\omega \rightarrow 2$. For each n , let

$$D_n = \{p : n \in \text{dom } p\}.$$

Each D_n is dense, because any finite condition can be extended by assigning a value to a previously undecided coordinate. Meeting all D_n 's makes the union of the filter a total binary sequence.

To see how genuinely generic requirements enter, fix a ground-model binary sequence $x \in 2^\omega$ and let

$$E_x = \{p : \exists n \in \text{dom } p \ (p(n) \neq x(n))\}.$$

This set is dense: extend any condition at a fresh coordinate with the opposite bit. A filter meeting every E_x for ground-model x has a union different from every old binary sequence. Thus the toy forcing already exhibits the central mechanism of Cohen forcing. $\square$

Worked example

Worked Example 44.6 (Meeting bit-deciding dense sets). With $2 = \{0, 1\}$, for finite partial functions $p : \omega \rightarrow 2$, let $D_n = \{p : n \in \text{dom}(p)\}$. Each D_n is dense because any condition can be extended by assigning a value to n . A filter meeting every D_n has a union that is a total binary sequence.

A useful warning

The statement “there exists an M -generic filter” is made externally when M is countable. It is not a theorem that an arbitrary model contains a filter generic over itself.

Looking ahead

The next chapter, *Names and valuation*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 45

Names and valuation

Definitions and setup

Formal definitions used in this chapter

Hierarchy of forcing names. Apply the transfinite-recursion theorem from Volume IV to the displayed stage operator:

$$V_0^{\mathbb{P}} = \emptyset, \quad V_\alpha^{\mathbb{P}} = \left\{ \tau : \tau \subseteq \bigcup_{\beta < \alpha} (V_\beta^{\mathbb{P}} \times \mathbb{P}) \right\}.$$

The recursion theorem supplies a unique hierarchy $\langle V_\alpha^{\mathbb{P}} : \alpha \in \text{Ord} \rangle$. Define

$$V^{\mathbb{P}} := \bigcup_{\alpha \in \text{Ord}} V_\alpha^{\mathbb{P}},$$

and call τ a $\mathbb{P}$ -name exactly when $\tau \in V^{\mathbb{P}}$. If τ is a name, choose any stage at which it occurs; Separation below that witnessing ordinal and the well-ordering of the ordinals give a least α with $\tau \in V_{\alpha+1}^{\mathbb{P}}$. Define

$$\text{rank}_{\mathbb{P}}(\tau) := \min\{\alpha : \tau \in V_{\alpha+1}^{\mathbb{P}}\}.$$

When $\mathbb{P}$ is fixed, $\text{rank}(\tau)$ abbreviates $\text{rank}_{\mathbb{P}}(\tau)$. The next theorem proves the useful constituent formula

$$\text{rank}_{\mathbb{P}}(\tau) = \sup\{\text{rank}_{\mathbb{P}}(\sigma) + 1 : \exists p \in \mathbb{P} \langle \sigma, p \rangle \in \tau\}, \quad (\text{NR})$$

and in particular proves that every constituent has strictly smaller name rank. Formula (NR) is therefore a theorem about the least-stage definition, not a second definition hidden in this setup block.

Canonical ground-model name and generic-filter name. Set-rank recursion (already available from Volume IV) defines

$$\check{x} = \{\langle \check{y}, 1_{\mathbb{P}} \rangle : y \in x\}, \quad \dot{G} = \{\langle \check{p}, p \rangle : p \in \mathbb{P}\}.$$

A set/class is transitive when $x \in y \in M \Rightarrow x \in M$. Valuation and the generic-extension notation are introduced only after Theorem 45.1 has proved that every constituent name has strictly smaller name rank.

A forcing name is a recursively stratified ground-model recipe built from lower-stage names and forcing conditions. The next theorem proves the strict rank decrease along constituents. Only after that fact is established do we define the evaluation of a name by a filter and the resulting generic-extension notation.

How this chapter fits together

The dependency order is: construct the name hierarchy and its least-stage rank; prove that constituent names have smaller rank; use that well-founded measure to define evaluation by a filter; then verify the canonical ground-model and generic-filter names and define the generic extension. No evaluation symbol is used as a definition before its recursion is justified.

Theorem 45.1 (Rank formula for forcing names). *For every $\mathbb{P}$ -name τ ,*

$$\text{rank}_{\mathbb{P}}(\tau) = \sup\{\text{rank}_{\mathbb{P}}(\sigma) + 1 : \exists p \in \mathbb{P} (\langle \sigma, p \rangle \in \tau)\}.$$

In particular, every constituent name σ of τ has $\text{rank}_{\mathbb{P}}(\sigma) < \text{rank}_{\mathbb{P}}(\tau)$.

Proof.

Formal derivation spine

Formal spine for the forcing-name rank formula.

$$V_{\alpha}^{\mathbb{P}} = \{\tau : \tau \text{ is a set of pairs } \langle \sigma, p \rangle \text{ with } \sigma \in V_{\beta}^{\mathbb{P}} \text{ for some } \beta < \alpha, p \in \mathbb{P}\}.$$

The hierarchy and least-stage rank were defined in the setup using the transfinite-recursion theorem from Volume IV. We now compute that least stage. If $\langle \sigma, p \rangle \in \tau$, then τ cannot appear before a stage strictly above the least stage containing σ . Conversely, once a stage lies strictly above every constituent rank, the stage definition places every pair of τ inside the required union, so it contains τ . Therefore

$$\text{rank}_{\mathbb{P}}(\tau) = \sup\{\text{rank}_{\mathbb{P}}(\sigma) + 1 : \exists p \in \mathbb{P} (\langle \sigma, p \rangle \in \tau)\}.$$

The supremum is 0 for the empty name. This formula is the least ordinal strictly above every constituent rank and is exactly the least-stage convention fixed in the setup; in particular it does not incorrectly add an extra successor at a limit of constituent ranks. Foundation (equivalently, descent through the stage hierarchy) makes the constituent recursion well founded.

The point of the rank is not cosmetic. Every later recursive definition—valuation, atomic forcing, and induction on names—descends to constituent names of smaller rank. Thus the name hierarchy supplies the well-founded measure that prevents circularity in the forcing definitions.

A convenient convention is to permit a name to occur at every higher stage once it has appeared. Then the hierarchy is increasing, and rank may be taken as the least stage containing the name. None of the later arguments depends on this presentational choice; what matters is only that constituents have strictly smaller rank. $\square$

Formal definitions used in this chapter

Valuation. Let $G \subseteq \mathbb{P}$ be a filter. By Theorem 45.1, every constituent σ of a name τ has smaller forcing-name rank. Hence the well-founded-recursion theorem applies to the clause

$$\tau^G := \{\sigma^G : \exists p \in G (\langle \sigma, p \rangle \in \tau)\}. \quad (\text{V})$$

The unique map supplied by that recursion is the *valuation by G* .

Theorem 45.2 (Well-founded valuation $\tau \mapsto \tau^G$). *For a filter $G \subseteq \mathbb{P}$, valuation is the unique rank-recursive map*

$$\tau^G = \{\sigma^G : \exists p \in G (\langle \sigma, p \rangle \in \tau)\}.$$

How to read the symbols: a forcing name is code; valuation executes the code

A name τ is a ground-model set of pairs $\langle \sigma, p \rangle$. It is not yet an extension object. Once a generic filter G is given, valuation keeps exactly those pairs whose condition lies in G and recursively evaluates their

lower-rank names. Thus $\tau \mapsto \tau^G$ is analogous to executing a recursively structured program with G as an external input.

Proof.

Formal derivation spine

Formal spine for Well-founded valuation $\tau \mapsto \tau^G$.

$$\tau^G = \{\sigma^G : \exists p \in G (\langle \sigma, p \rangle \in \tau)\}.$$

$$\tau^G = \{\sigma^G : \exists p \in G \langle \sigma, p \rangle \in \tau\}.$$

For a filter $G \subseteq \mathbb{P}$, define valuation by recursion on name rank:

$$\tau^G = \{\sigma^G : \exists p \in G (\langle \sigma, p \rangle \in \tau)\}.$$

Every σ occurring on the right has smaller rank than τ , so the recursive value has already been defined. The well-founded recursion theorem therefore gives existence and uniqueness of the valuation map.

Induction on rank also shows that valuation ignores pairs whose conditions are not in G , and that two names can have the same value even when they are syntactically different. This is why equality between forcing names cannot be defined as literal equality of names and must instead receive its own forcing relation. *Reading the recursive clause.* The condition $\langle \sigma, p \rangle \in \tau$ is ground-model syntax. The additional test $p \in G$ decides whether that branch of the name is active. Only after the branch is active do we use the already-defined lower-rank value σ^G . The rank decrease is the well-founded measure that prevents the definition from calling itself on the same input. $\square$

Formal definitions used in this chapter

Generic extension. If M is the ground model and G is an M -generic filter, define

$$M[G] := \{\tau^G : \tau \in M \text{ is a } \mathbb{P}\text{-name}\}.$$

This is now well-defined because valuation has been constructed by Theorem 45.2.

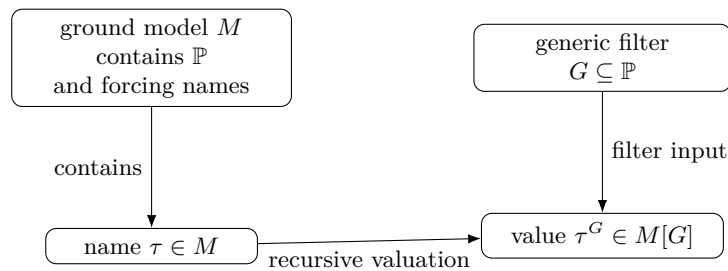


Figure 45.1: After the rank-recursive valuation has been constructed, a generic filter supplies the additional input that turns the ground-model name τ into the extension object $\tau^G \in M[G]$.

Proposition 45.3 (Canonical ground-model names $\check{x}$). *For each ground-model set x , the recursively defined canonical name*

$$\check{x} = \{\langle \check{y}, 1_{\mathbb{P}} \rangle : y \in x\}$$

satisfies $\check{x}^G = x$ for every generic filter G .

Proof.

Formal derivation spine

Formal spine for Canonical ground-model names $\text{check}(x)$.

$$\begin{aligned}\check{x} &= \{\langle \check{y}, 1_{\mathbb{P}} \rangle : y \in x\}, \\ \check{x}^G &= x. \\ \check{x}^G &= \{\check{y}^G : y \in x\} = \{y : y \in x\} = x,\end{aligned}$$

Define the canonical name of a ground-model set recursively by

$$\check{x} = \{\langle \check{y}, 1_{\mathbb{P}} \rangle : y \in x\},$$

where $1_{\mathbb{P}}$ is the weakest condition. We prove by induction on the rank of x that

$$\check{x}^G = x.$$

To verify the induction claim, expand the valuation definition for the specific name $\check{x}$. Every pair in $\check{x}$ carries the top condition, so every constituent survives valuation:

$$\check{x}^G = \{\check{y}^G : y \in x\} = \{y : y \in x\} = x,$$

using the induction hypothesis in the middle equality.

Thus every old set has a canonical name whose value is independent of the generic filter. This gives the literal embedding of the ground model into every generic extension.

The construction also preserves membership: $y \in x$ iff $\check{y}$ occurs in $\check{x}$ with the top condition. Thus check names reproduce the entire ground-model membership structure, not merely its underlying collection of objects. $\square$

Proposition 45.4 (Canonical name $\dot{G}$). *For the already-defined name*

$$\dot{G} = \{\langle \check{p}, p \rangle : p \in \mathbb{P}\},$$

one has $\dot{G}^G = G$ for every generic filter G .

Proof.

Formal derivation spine

Formal spine for Canonical name $\dot{G}$.

$$\begin{aligned}\dot{G} &= \{\langle \check{p}, p \rangle : p \in \mathbb{P}\}. \\ \dot{G}^G &= \{\check{p}^G : p \in G\} = \{p : p \in G\} = G, \\ \tau^G &= \{\sigma^G : \exists p \in G \langle \sigma, p \rangle \in \tau\}.\end{aligned}$$

Let

$$\dot{G} = \{\langle \check{p}, p \rangle : p \in \mathbb{P}\}.$$

When we evaluate this name, a pair survives exactly when its second coordinate p belongs to the chosen generic filter. Therefore

$$\dot{G}^G = \{\check{p}^G : p \in G\} = \{p : p \in G\} = G,$$

where the second equality uses the check-name theorem. Hence the generic filter itself is an element of the extension whenever the standard name $\dot{G}$ belongs to the ground model.

This name also clarifies the two viewpoints used in forcing: externally, G is chosen to meet the ground-model dense sets; internally to the extension, G is simply the value of the ground-model name $\dot{G}$.

Although G itself is new, the name $\dot{G}$ is old. This is a recurring forcing pattern: the ground model can contain a syntactic name for an object that exists only after a generic filter is supplied. $\square$

Theorem 45.5 (M is contained in $M[G]$ and $M[G]$ is transitive). *If M is transitive and G is M -generic, define*

$$M[G] = \{\tau^G : \tau \in M \text{ is a } \mathbb{P}\text{-name}\}.$$

Then $M \subseteq M[G]$, $G \in M[G]$, and $M[G]$ is transitive.

Proof.

Formal derivation spine

Formal spine for M is contained in $M[G]$ and $M[G]$ is transitive.

$$\tau^G = \{\sigma^G : \exists p \in G \langle \sigma, p \rangle \in \tau\}.$$

Every $x \in M$ belongs to $M[G]$ because $x = \check{x}^G$ and $\check{x} \in M$. Thus $M \subseteq M[G]$.

For transitivity, suppose $x \in y \in M[G]$. Choose $\tau \in M$ with $y = \tau^G$. By the definition of valuation, there are $\sigma \in M$ and $p \in G$ such that $\langle \sigma, p \rangle \in \tau$ and $x = \sigma^G$. Therefore $x \in M[G]$. So $M[G]$ is transitive.

The proof uses only that M is transitive enough for the names and their constituents to lie in M . It does not yet show that $M[G]$ satisfies any particular set-theoretic axiom; those preservation arguments come later.

The inclusion $M \subseteq M[G]$ is therefore canonical, not an arbitrary isomorphic copy. In later cardinal arguments this lets us compare the same ground-model ordinals and cardinals directly with their status in the extension. $\square$

Worked example

Worked Example 45.6 (Evaluating a simple name). If $\tau = \{\langle \check{0}, p \rangle, \langle \check{1}, q \rangle\}$, then τ^G contains 0 exactly when $p \in G$, and contains 1 exactly when $q \in G$. A name is therefore a condition-dependent recipe, not a fixed set waiting to be uncovered.

A useful warning

Two different names may have the same value in a generic extension. Forced equality is therefore an essential part of the forcing relation.

Looking ahead

The next chapter, *Atomic forcing*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 46

Atomic forcing

Definitions and setup

Formal definitions used in this chapter

Dense below a condition and order shorthand. $D \subseteq \mathbb{P}$ is dense below p when every $q \leq p$ has some $r \leq q$ in D . The notation $q \leq p, r$ abbreviates the conjunction $q \leq p$ and $q \leq r$. The notation $q \not\Vdash \Phi$ means simply that $q \Vdash \Phi$ does not hold.

Constituent name and rank recursion. A constituent name of τ is a σ appearing in some pair $\langle \sigma, r \rangle \in \tau$. Atomic forcing is defined simultaneously for membership and equality by recursion on the ranks of the names, so every recursive call involves lower-rank constituents.

Forced-subset auxiliary relation.

$$p \Vdash \sigma \subseteq \tau \iff \forall \langle \rho, r \rangle \in \sigma \forall q \leq p, r \exists s \leq q (s \Vdash \rho \in \tau).$$

Forced equality is then $p \Vdash \sigma = \tau$ iff both forced-subset directions hold.

Atomic membership forcing.

$$p \Vdash \sigma \in \tau \iff \{q \leq p : \exists \langle \rho, r \rangle \in \tau (q \leq r \wedge q \Vdash \sigma = \rho)\} \text{ is dense below } p.$$

These clauses define symbols $p \Vdash \sigma \in \tau$, $p \Vdash \sigma = \tau$, and $p \Vdash \sigma \subseteq \tau$ before they are used in the statements below.

Atomic forcing must be defined before semantic truth in the extension can be used. Membership and equality are defined simultaneously because each refers recursively to the other on lower-rank names. Monotonicity means stronger conditions preserve forced atomic facts. The atomic truth lemma is proved directly from genericity and name valuation and becomes the base case for the full Truth Lemma.

How this chapter fits together

Membership and equality of names must be defined simultaneously because each refers recursively to the other. The recursion decreases name rank, not formula complexity. Monotonicity and the atomic truth lemma then verify that the recursive definition matches actual valuation in a generic extension.

Theorem 46.1 (Simultaneous recursive definition of atomic forcing for equality and membership). *There are unique atomic forcing relations, defined simultaneously by recursion on name rank, satisfying*

$$p \Vdash \sigma \in \tau \iff \{q \leq p : \exists \langle \rho, r \rangle \in \tau (q \leq r \wedge q \Vdash \sigma = \rho)\} \text{ is dense below } p,$$

$$p \Vdash \sigma \subseteq \tau \iff \forall \langle \rho, r \rangle \in \sigma \forall q \leq p, r \exists s \leq q (s \Vdash \rho \in \tau),$$

and

$$p \Vdash \sigma = \tau \iff p \Vdash \sigma \subseteq \tau \wedge p \Vdash \tau \subseteq \sigma.$$

Every recursive reference involves a constituent name of strictly lower rank.

Commented symbolic trace: why atomic forcing is defined simultaneously

Membership and equality of names refer to one another. To decide whether $p \Vdash \sigma \in \tau$, we compare σ with names occurring inside τ , so equality is needed. To decide whether $p \Vdash \sigma = \tau$, we require mutual forced membership, so membership is needed. The apparent circularity is removed by a simultaneous recursion on name ranks: every recursive call involves constituents of strictly smaller rank.

Proof.

Formal derivation spine

Formal spine: simultaneous recursion on pairs of name ranks. Order pairs of names by

$$(\sigma', \tau') \prec (\sigma, \tau) \iff \text{rk}(\sigma') \leq \text{rk}(\sigma), \quad \text{rk}(\tau') \leq \text{rk}(\tau),$$

with at least one strict inequality. This product order is well founded. Every atomic recursive call lies at a $\prec$ -smaller pair.

Fix names σ, τ , and suppose the two atomic relations have already been defined for every smaller rank pair. In the membership clause, a recursive call has the form

$$q \Vdash \sigma = \rho \quad \text{with} \quad \langle \rho, r \rangle \in \tau.$$

Because ρ is a constituent of τ , $\text{rk}(\rho) < \text{rk}(\tau)$, so $(\sigma, \rho) \prec (\sigma, \tau)$.

For forced inclusion, a recursive call has the form

$$s \Vdash \rho \in \tau \quad \text{with} \quad \langle \rho, r \rangle \in \sigma.$$

Here $\text{rk}(\rho) < \text{rk}(\sigma)$, so $(\rho, \tau) \prec (\sigma, \tau)$. Thus the displayed membership and inclusion clauses depend only on already defined lower-rank cases. We then define equality at the current pair by

$$p \Vdash \sigma = \tau \iff p \Vdash \sigma \subseteq \tau \ \& \ p \Vdash \tau \subseteq \sigma.$$

Well-founded recursion on $\prec$ therefore gives existence. For uniqueness, suppose two pairs of atomic relations satisfy the same clauses and choose a $\prec$ -least pair at which they differ. All recursive calls in either clause occur at smaller pairs, where the two definitions agree; hence the clauses assign the same truth value at the chosen pair, a contradiction. The atomic forcing relations are therefore uniquely determined. $\square$

Lemma 46.2 (Monotonicity of atomic forcing). *If $q \leq p$ and $p \Vdash \sigma = \tau$ (respectively $p \Vdash \sigma \in \tau$), then $q \Vdash \sigma = \tau$ (respectively $q \Vdash \sigma \in \tau$).*

Proof.

Formal derivation spine

Formal spine: strengthening preserves atomic forcing. For either atomic relation $\Phi(\sigma, \tau)$,

$$q \leq p \ \& \ p \Vdash \Phi(\sigma, \tau) \implies q \Vdash \Phi(\sigma, \tau).$$

The dense set witnessing the clause below p remains dense after restricting to the cone below q .

We prove monotonicity simultaneously for atomic membership and equality by induction on the rank pair of the names. Assume $p \Vdash \sigma \in \tau$ and $q \leq p$. The definition says that below p the conditions witnessing an

appropriate constituent ρ of τ and forcing $\sigma = \rho$ are dense. Any extension of q is also an extension of p , so the same set remains dense below q ; hence $q \Vdash \sigma \in \tau$.

For equality, $p \Vdash \sigma = \tau$ means forced inclusion in both directions. Restricting from the cone below p to the smaller cone below q preserves each of those dense requirements, and the induction hypothesis applies to the lower-rank constituent equalities. Thus stronger conditions preserve atomic forcing. $\square$

Lemma 46.3 (Atomic density lemma). *For an atomic formula Φ in forcing names, let*

$$D_\Phi := \{p \in \mathbb{P} : p \Vdash \Phi \text{ or } \forall q \leq p (q \nVdash \Phi)\}.$$

Then D_Φ is dense. After the negation clause $p \Vdash \neg\Phi \iff \forall q \leq p (q \nVdash \Phi)$ is introduced, this says that conditions deciding Φ are dense.

Proof.

Formal derivation spine

Formal spine: one exhaustive dichotomy below p .

$$(\exists q \leq p \ q \Vdash \Phi) \quad \vee \quad (\forall q \leq p \ q \nVdash \Phi).$$

The first case supplies a forcing extension; the second puts p itself in D_Φ .

Fix $p \in \mathbb{P}$. There are two exhaustive cases.

If some $q \leq p$ forces Φ , then $q \in D_\Phi$ and we have found a deciding extension.

Otherwise

$$\forall q \leq p (q \nVdash \Phi).$$

Then $p \in D_\Phi$ by the second clause in the definition of D_Φ . Thus every condition has an extension in D_Φ , so D_Φ is dense. When general forcing is defined, the displayed universal condition is exactly the clause for $p \Vdash \neg\Phi$. The present lemma therefore establishes decision density without presupposing the full recursive forcing relation. $\square$

Theorem 46.4 (Atomic truth lemma). *If G is generic, then for names $\sigma, \tau \in M$,*

$$\sigma^G \in \tau^G \iff \exists p \in G (p \Vdash \sigma \in \tau),$$

and

$$\sigma^G = \tau^G \iff \exists p \in G (p \Vdash \sigma = \tau).$$

Proof.

Formal derivation spine

Formal spine for Atomic truth lemma.

$$\sigma^G \in \tau^G \iff \exists p \in G (p \Vdash \sigma \in \tau),$$

$$q \leq p \wedge p \Vdash \Phi \implies q \Vdash \Phi$$

We prove simultaneously that, for $\sigma, \tau \in M$ and an M -generic G ,

$$\sigma^G \in \tau^G \iff \exists p \in G (p \Vdash \sigma \in \tau),$$

and likewise for equality.

For membership, if $\sigma^G \in \tau^G$, choose $\langle \rho, q \rangle \in \tau$ with $q \in G$ and $\rho^G = \sigma^G$. By the inductive equality statement some condition of G forces $\rho = \sigma$; directedness of G gives a common strengthening with q , yielding a condition in G that forces $\sigma \in \tau$. Conversely, if $p \in G$ forces membership, the dense witness set in the definition meets G ; its constituent ρ has $\rho^G = \sigma^G$ by the inductive equality clause and $\rho^G \in \tau^G$ by valuation. Equality

follows by applying the membership argument in both directions and ordinary extensionality of sets. *The induction has two mutually supporting components.* Membership needs the equality statement in order to replace the constituent ρ by σ ; equality in turn is defined through mutual membership. Both references move to lower-rank names, so simultaneous induction is legitimate. This is the semantic counterpart of the simultaneous recursive definition of atomic forcing. $\square$

Proposition 46.5 (Extensionality of the forcing relation on names). *Forced equality is substitutive for atomic forcing: if $p \Vdash \sigma = \tau$, then below p the names σ and τ may be interchanged in atomic membership and equality formulas.*

Proof.

Formal derivation spine

Formal spine: substitution of forced-equal names. If $p \Vdash \sigma = \tau$ and $q \leq p$, then for every name ρ ,

$$q \Vdash \rho \in \sigma \iff q \Vdash \rho \in \tau, \quad q \Vdash \sigma \in \rho \iff q \Vdash \tau \in \rho,$$

and consequently

$$q \Vdash \rho = \sigma \iff q \Vdash \rho = \tau.$$

We prove simultaneously, by induction on the relevant name-rank tuple, that forced equality is an equivalence relation and is substitutive in both arguments of forced membership. Reflexivity is obtained by matching every constituent with itself; symmetry is built into the mutual-inclusion definition. For transitivity, assume $q \Vdash \sigma = \tau$ and $q \Vdash \tau = \nu$. A constituent of σ is densely matched with a constituent of τ , which is in turn densely matched with a constituent of ν . Directedness below q combines the two strengthenings, and the lower-rank induction hypothesis composes the two forced equalities. This proves $q \Vdash \sigma \subseteq \nu$; the reverse inclusion is symmetric.

Now assume $q \Vdash \rho \in \sigma$ and $q \Vdash \sigma = \tau$. Below an arbitrary $q_0 \leq q$, the first assertion supplies a stronger condition matching ρ with some constituent α of σ . Forced inclusion $\sigma \subseteq \tau$ then supplies a further strengthening forcing $\alpha \in \tau$. The lower-rank substitution hypothesis replaces α by the forced-equal name ρ , yielding a condition forcing $\rho \in \tau$. Since this can be done below every $q_0 \leq q$, the membership clause gives $q \Vdash \rho \in \tau$. The converse uses $\tau \subseteq \sigma$.

For membership in a third name, a witness for $q \Vdash \sigma \in \rho$ is a constituent α of ρ forced equal to σ . Transitivity of forced equality combines $\tau = \sigma$ with $\sigma = \alpha$, giving the same witness for $q \Vdash \tau \in \rho$. Finally, equality is mutual forced inclusion, so the two membership substitution laws give substitution in atomic equality formulas as well. $\square$

Worked example

Worked Example 46.6 (Why equality and membership are simultaneous). To decide whether $p \Vdash \sigma \in \tau$, one searches below p for a constituent ρ of τ forced equal to σ . But forced equality $\sigma = \rho$ itself says that every forced member of one is a forced member of the other. Neither notion can be defined first in isolation; the pair is handled by one well-founded recursion on ranks.

A useful warning

The informal clause “ p forces membership iff some pair occurs in the name” is too weak. Compatibility, strengthening, and forced equality are needed to make the definition invariant under equivalent names.

Looking ahead

The next chapter, *General forcing relation*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 47

General forcing relation

Definitions and setup

Formal definitions used in this chapter

Formula complexity and logical-basis normalization. Formula complexity is the finite parse-tree complexity fixed in Volume I. That syntax treats $\neg, \wedge, \vee, \rightarrow, \forall, \exists$ as genuine constructors. For the forcing recursion, first define a primitive-recursive normalization $\mathcal{N}$ on formulas by

$$\begin{aligned} \mathcal{N}(\alpha) &= \alpha && \text{for atomic } \alpha, \\ \mathcal{N}(\neg\varphi) &= \neg\mathcal{N}(\varphi), && \mathcal{N}(\varphi \wedge \psi) = \mathcal{N}(\varphi) \wedge \mathcal{N}(\psi), \\ \mathcal{N}(\varphi \vee \psi) &= \neg(\neg\mathcal{N}(\varphi) \wedge \neg\mathcal{N}(\psi)), && \mathcal{N}(\varphi \rightarrow \psi) = \neg(\mathcal{N}(\varphi) \wedge \neg\mathcal{N}(\psi)), \\ \mathcal{N}(\exists x \varphi) &= \exists x \mathcal{N}(\varphi), && \mathcal{N}(\forall x \varphi) = \neg\exists x \neg\mathcal{N}(\varphi). \end{aligned} \tag{N}$$

A *basis formula* is a formula using only atomic formulas, $\neg, \wedge, \exists$. The clauses below recursively define forcing for basis formulas. For an arbitrary Volume-I formula φ , define

$$p \Vdash \varphi \quad :\Longleftrightarrow \quad p \Vdash \mathcal{N}(\varphi), \tag{NF}$$

where the right-hand forcing assertion is evaluated by the basis recursion. Thus $\vee, \rightarrow, \forall$ are not being reclassified as syntactic abbreviations: (N) is an explicit preprocessing map on their primitive syntax codes. The code-level effectiveness, ordinary-truth preservation, and substitution compatibility needed from this normalization are proved in the Logical-basis Normalization Lemma immediately below.

Forcing clauses for basis formulas.

$$\begin{aligned} p \Vdash \neg\varphi &\Longleftrightarrow \forall q \leq p (q \nVdash \varphi), && p \Vdash (\varphi \wedge \psi) \Longleftrightarrow (p \Vdash \varphi \wedge p \Vdash \psi), \\ p \Vdash \exists x \varphi(x, \bar{\tau}) &\Longleftrightarrow \{q \leq p : \exists \sigma (q \Vdash \varphi(\sigma, \bar{\tau}))\} \text{ is dense below } p. \end{aligned}$$

For formulas containing the other primitive constructors $\vee, \rightarrow, \forall$, apply the normalization (N) first and then these basis clauses.

Decision. A condition p decides φ when $p \Vdash \varphi$ or $p \Vdash \neg\varphi$.

Ground-model forcing notation and definability formula. $p \Vdash_{\mathbb{P}}^M \varphi(\bar{\tau})$ means that the forcing recursion is carried out inside the transitive model M using $\mathbb{P} \in M$. $F_{\varphi}(p, \mathbb{P}, \bar{\tau})$ denotes a first-order set-theoretic formula whose truth in M expresses this forcing assertion; “constructed effectively from φ ” means effectively from its Gödel code as fixed in Volume III.

Lemma 47.1 (Logical-basis normalization lemma). *The map $\mathcal{N}$ defined in (N) has the following properties.*

-
1. If φ is a well-formed Volume-I formula, then $\mathcal{N}(\varphi)$ is a well-formed basis formula, and the map on Gödel codes

$$\ulcorner \varphi \urcorner \mapsto \ulcorner \mathcal{N}(\varphi) \urcorner$$

is primitive recursive.

2. In the exact Volume-I calculus, with every displayed biconditional expanded by (BI),

$$\vdash \varphi \leftrightarrow \mathcal{N}(\varphi). \quad (\text{N0})$$

3. For every ordinary first-order structure M , assignment a , and formula φ ,

$$M, a \models \varphi \iff M, a \models \mathcal{N}(\varphi). \quad (\text{N1})$$

4. If $\varphi[t/x]$ is capture-avoiding substitution, then

$$\mathcal{N}(\varphi[t/x]) \text{ is alpha-equivalent to } \mathcal{N}(\varphi)[t/x]. \quad (\text{N2})$$

Thus normalization commutes with substitution modulo the alpha-equivalence convention fixed in Volume I.

5. For the forcing relation defined by (NF), for every condition p ,

$$p \Vdash \varphi \iff p \Vdash \mathcal{N}(\varphi), \quad (\text{N3})$$

where the right-hand side is evaluated by the basis forcing recursion.

Uses: Volume-I unique readability, structural recursion, the exact six-constructor calculus, Tarski satisfaction, and capture-avoiding substitution.

Proof. For (1), inspect the unique outer constructor of the input formula. Atomic, negation, conjunction, and existential nodes are copied recursively; disjunction, implication, and universal nodes are replaced by the fixed finite syntax templates displayed in (N). Each output constructor belongs to the basis $\{\text{atomic}, \neg, \wedge, \exists\}$, so induction on formula construction proves well-formedness and the basis property. At code level the same case split uses only the primitive-recursive constructor and immediate-subformula functions fixed in Volume II; hence the normalization map is primitive recursive.

For (2), induct on the six primitive formula constructors of Volume I. The atomic case is reflexivity. The negation and conjunction cases follow from the induction hypotheses by propositional tautology instances. The same is true for $\vee$ and $\rightarrow$, because their outputs in (N) are precisely classically equivalent propositional contexts.

We record the quantifier step rather than hiding it under “standard equivalences.” First, the fixed calculus proves the derived *quantifier-congruence* rules

$$\vdash \chi \leftrightarrow \eta \implies \vdash \forall x \chi \leftrightarrow \forall x \eta, \quad \vdash \exists x \chi \leftrightarrow \exists x \eta. \quad (\text{QC})$$

For $\forall$, each direction is obtained from the corresponding implication $\chi \rightarrow \eta$ by generalization and the quantifier-distribution axiom; the eigenvariable condition is vacuous because there are no undischarged assumptions. For $\exists$, use the existential-duality axiom together with the already derived universal congruence and propositional contraposition. The same duality axiom, applied to $\neg\chi$, together with double-negation tautologies and (QC), gives the derived equivalence

$$\vdash \forall x \chi \leftrightarrow \neg \exists x \neg \chi. \quad (\text{UD})$$

Now the induction hypothesis for ψ and (QC) prove the existential case directly; for a universal formula they give

$$\vdash \forall x \psi \leftrightarrow \forall x \mathcal{N}(\psi) \leftrightarrow \neg \exists x \neg \mathcal{N}(\psi) = \mathcal{N}(\forall x \psi).$$

Thus (N0) is derivable in the exact calculus, including its existential-duality scheme, rather than merely semantically valid.

For (3), either apply Volume-I soundness to the theorem (N0), or verify the same induction directly from the Tarski clauses. In particular, the universal case is

$$M, a \models \forall x \psi \iff \neg \exists m \in |M| \ M, a[x \mapsto m] \models \neg \psi,$$

and the induction hypothesis replaces ψ by $\mathcal{N}(\psi)$. This gives exactly the truth clause for $\neg \exists x \neg \mathcal{N}(\psi)$.

For (4), induct simultaneously on terms and formulas after the standard alpha-renaming that makes bound variables fresh for the substituted term. The basis constructors commute literally with substitution. Each nonbasis constructor is replaced by a fixed context built from its immediate normalized constituents, so the induction hypothesis gives the same context after substitution. Different choices of fresh bound variables produce alpha-equivalent outputs by the Volume-I substitution lemma. This proves (N2).

Finally, (5) is the explicit bridge by which arbitrary primitive syntax enters forcing: equation (NF) *defines* the left-hand assertion to mean the basis-recursive assertion on the right. Item (1) guarantees that the latter is well formed, and item (4) guarantees that later substitution of names does not depend on an illicit preprocessing convention. Hence (N3) holds for every condition and every Volume-I formula. $\square$

After atomic forcing is established, Boolean connectives and quantifiers are added recursively. Negation is defined by the absence of a stronger condition forcing the positive statement. A condition decides φ when it forces φ or its negation; deciding conditions are dense. The Definability Lemma asserts that $p \Vdash \varphi(\bar{\tau})$ is a relation definable in the ground model from $p, \bar{\tau}$.

How this chapter fits together

After the atomic relation is secure, forcing extends by induction on formula complexity. Negation is formulated using all stronger conditions; quantifiers range over names; decision density guarantees that a generic filter encounters a condition settling each sentence. The Definability Lemma is crucial: forcing itself must be definable in the ground model before it can be used in ZF proofs there.

Theorem 47.2 (Recursive forcing clauses for Boolean connectives). *Extend forcing recursively to Boolean connectives by*

$$p \Vdash \varphi \wedge \psi \iff (p \Vdash \varphi \wedge p \Vdash \psi), \quad p \Vdash \neg \varphi \iff \forall q \leq p \ (q \nVdash \varphi),$$

and extend this basis recursion to the primitive constructors $\vee, \rightarrow$ by the normalization (N).

Proof.

Formal derivation spine

Formal spine for Recursive forcing clauses for Boolean connectives.

$$p \Vdash (\varphi \wedge \psi) \iff p \Vdash \varphi \wedge p \Vdash \psi,$$

$$p \Vdash \neg \varphi \iff \forall q \leq p \ (q \nVdash \varphi).$$

$$\mathcal{N}(\varphi \rightarrow \psi) = \neg(\mathcal{N}(\varphi) \wedge \neg \mathcal{N}(\psi)).$$

Once atomic forcing is fixed, define the Boolean clauses by recursion on formula complexity:

$$p \Vdash (\varphi \wedge \psi) \iff p \Vdash \varphi \wedge p \Vdash \psi,$$

$$p \Vdash \neg \varphi \iff \forall q \leq p \ (q \nVdash \varphi).$$

For the primitive propositional constructors $\vee$ and $\rightarrow$, apply $\mathcal{N}$ first; for example

$$\mathcal{N}(\varphi \rightarrow \psi) = \neg(\mathcal{N}(\varphi) \wedge \neg \mathcal{N}(\psi)).$$

The normalization is a separate structural recursion on the original syntax. The forcing recursion itself is then well founded on basis-formula complexity because every recursive forcing call is made on a proper basis

subformula. No quantifier clause is used in this theorem; those are introduced separately in the next result. *Specialization to this result.* For conjunction, force both conjuncts. For negation, require that no stronger condition force the positive formula. The clauses are chosen so that monotonicity and the Truth Lemma can be proved by induction. $\square$

Theorem 47.3 (Forcing clauses for quantifiers). *For existential formulas,*

$$p \Vdash \exists x \varphi(x, \bar{\tau})$$

iff for every $q \leq p$ there are $r \leq q$ and a name σ with $r \Vdash \varphi(\sigma, \bar{\tau})$. For the primitive universal constructor, (NF) and (N) give

$$p \Vdash \forall x \varphi(x, \bar{\tau}) \iff p \Vdash \neg \exists x \neg \mathcal{N}(\varphi)(x, \bar{\tau}).$$

Proof.

Formal derivation spine

Formal spine for Forcing clauses for quantifiers.

$$p \Vdash \exists x \varphi(x, \bar{\tau})$$

$$D := \{q \leq p : \exists \sigma (q \Vdash \varphi(\sigma, \bar{\tau}))\}$$

$$\forall r \leq p \exists q \leq r \exists \sigma q \Vdash \varphi(\sigma, \bar{\tau}).$$

For an existential formula define

$$p \Vdash \exists x \varphi(x, \bar{\tau})$$

iff the set

$$D := \{q \leq p : \exists \sigma (q \Vdash \varphi(\sigma, \bar{\tau}))\}$$

is dense below p . Equivalently,

$$\forall r \leq p \exists q \leq r \exists \sigma q \Vdash \varphi(\sigma, \bar{\tau}).$$

For a primitive universal formula, normalization gives

$$p \Vdash \forall x \varphi(x) \iff p \Vdash \neg \exists x \neg \mathcal{N}(\varphi)(x).$$

Equivalently, because forcing of $\neg\varphi$ also normalizes its argument, this is the derived notation $p \Vdash \neg \exists x \neg \varphi(x)$. This density formulation is chosen so that a generic filter meeting D supplies an actual witnessing name in the Truth Lemma. *Specialization to this result.* Quantifiers range over forcing names. Existential truth is represented densely by conditions that force some named instance; after the maximum principle is established, this can be consolidated to a single witness name below the given condition. $\square$

Lemma 47.4 (General monotonicity). *For every forcing formula φ , if $q \leq p$ and $p \Vdash \varphi$, then $q \Vdash \varphi$.*

Proof.

Formal derivation spine

Formal spine for General monotonicity.

$$p \Vdash \varphi \wedge \psi$$

$$q \leq p \wedge p \Vdash \varphi \implies q \Vdash \varphi$$

$$p \Vdash \neg \varphi \iff \neg \exists q \leq p (q \Vdash \varphi)$$

First prove the assertion for basis formulas by induction on basis-formula complexity. The atomic case is the monotonicity theorem for atomic forcing. For conjunction,

$$p \Vdash \varphi \wedge \psi$$

implies both conjuncts are forced at p , so by induction they are forced at every $q \leq p$, hence $q \Vdash \varphi \wedge \psi$. For negation, if $p \Vdash \neg\varphi$ and $q \leq p$, then no $r \leq q$ can force φ , because such an r would also satisfy $r \leq p$. Thus $q \Vdash \neg\varphi$. For an existential formula, density below p remains density below a stronger $q \leq p$. Therefore the implication holds for every basis formula. For an arbitrary primitive-syntax formula φ , apply this basis result to $\mathcal{N}(\varphi)$ and use (NF). Hence

$$q \leq p \wedge p \Vdash \varphi \implies q \Vdash \varphi$$

for all formulas. *Specialization to this result.* Strengthening is proved on the normalized basis formula: atomic forcing is monotone, and the $\neg, \wedge, \exists$ clauses preserve monotonicity. Equation (NF) then transfers the result to the original primitive syntax. $\square$

Lemma 47.5 (Forced equality is substitutive in every formula). *Let $\varphi(x, \bar{y})$ be any forcing-language formula and let $\sigma, \tau, \bar{\rho}$ be names. If*

$$p \Vdash \sigma = \tau,$$

then

$$p \Vdash \varphi(\sigma, \bar{\rho}) \iff p \Vdash \varphi(\tau, \bar{\rho}). \quad (\text{FS})$$

Uses: atomic equality substitution from Theorem 46.5 and general monotonicity.

Proof. Induct on the complexity of φ . Atomic formulas are exactly Theorem 46.5. Conjunction follows componentwise.

For negation, suppose

$$p \Vdash \neg\psi(\sigma, \bar{\rho}).$$

If some $q \leq p$ forced $\psi(\tau, \bar{\rho})$, monotonicity would also give $q \Vdash \sigma = \tau$. The induction hypothesis at q would then yield $q \Vdash \psi(\sigma, \bar{\rho})$, contradicting the negation forced at p . Thus no extension of p forces $\psi(\tau, \bar{\rho})$, so

$$p \Vdash \neg\psi(\tau, \bar{\rho}).$$

The converse follows by symmetry of forced equality.

For an existential formula, assume

$$p \Vdash \exists x \psi(x, \sigma, \bar{\rho}).$$

Given $q \leq p$, the existential clause supplies $r \leq q$ and a name ν with

$$r \Vdash \psi(\nu, \sigma, \bar{\rho}).$$

Monotonicity gives $r \Vdash \sigma = \tau$, and the induction hypothesis yields

$$r \Vdash \psi(\nu, \tau, \bar{\rho}).$$

Such witnesses exist below every $q \leq p$, so

$$p \Vdash \exists x \psi(x, \tau, \bar{\rho}).$$

Again symmetry gives the reverse implication. For a formula containing $\forall, \vee$, or $\rightarrow$, normalization commutes with substitution, so the basis-formula result applied to $\mathcal{N}(\varphi)$ gives the same conclusion. This proves (FS) for every primitive-syntax formula. $\square$

Theorem 47.6 (Density and decision lemma). *For every formula $\varphi(\bar{\tau})$ and condition p , the following two assertions hold.*

(Density characterization.)

$$p \Vdash \varphi(\bar{\tau}) \iff D_{\varphi, p} := \{q \leq p : q \Vdash \varphi(\bar{\tau})\} \text{ is dense below } p. \quad (1)$$

(Decision density.) *There is $q \leq p$ such that either $q \Vdash \varphi(\bar{\tau})$ or $q \Vdash \neg\varphi(\bar{\tau})$. Equivalently,*

$$\{q \in \mathbb{P} : q \Vdash \varphi(\bar{\tau}) \text{ or } q \Vdash \neg\varphi(\bar{\tau})\}$$

is dense.

Proof.

Formal derivation spine

Formal spine: forcing is stable under dense strengthening.

$$p \Vdash \varphi \implies \{q \leq p : q \Vdash \varphi\} \text{ is dense below } p$$

by monotonicity. Conversely, if that set is dense, induction on φ reconstructs $p \Vdash \varphi$. Finally,

$$(\exists q \leq p \ q \Vdash \varphi) \vee (\forall q \leq p \ q \nVdash \varphi)$$

gives a deciding extension.

We first prove the density characterization (1) for basis formulas by induction on their basis-formula complexity. The forward implication follows from monotonicity: if $p \Vdash \varphi$, then every $q \leq p$ also forces φ , so $D_{\varphi,p}$ is the whole cone below p .

For the converse, suppose $D_{\varphi,p}$ is dense below p .

For an atomic membership formula $\sigma \in \tau$, fix $r \leq p$. Choose $q \leq r$ with $q \Vdash \sigma \in \tau$. By the atomic membership clause, below q it is dense to find s and a pair $\langle \rho, t \rangle \in \tau$ such that

$$s \leq q, t \quad \text{and} \quad s \Vdash \sigma = \rho.$$

Hence the same witnessing set is dense below p , which is exactly the definition of $p \Vdash \sigma \in \tau$.

For atomic equality, suppose the conditions forcing $\sigma = \tau$ are dense below p . To prove $p \Vdash \sigma \subseteq \tau$, fix $\langle \rho, t \rangle \in \sigma$ and $r \leq p, t$. Choose $q \leq r$ with $q \Vdash \sigma = \tau$. Since q forces the inclusion $\sigma \subseteq \tau$, there is $s \leq q$ with $s \Vdash \rho \in \tau$. This is the required witness for the inclusion clause. The reverse inclusion is symmetric, so $p \Vdash \sigma = \tau$.

For conjunction, density of conditions forcing $\varphi \wedge \psi$ implies separately that the conditions forcing φ and those forcing ψ are dense below p . The induction hypothesis gives $p \Vdash \varphi$ and $p \Vdash \psi$, hence $p \Vdash \varphi \wedge \psi$.

For negation, assume the conditions forcing $\neg\psi$ are dense below p . If some $q \leq p$ forced ψ , density would give $r \leq q$ with $r \Vdash \neg\psi$. Monotonicity would also give $r \Vdash \psi$, contradicting the negation clause, which says that no extension of r can force ψ . Therefore

$$\forall q \leq p \ (q \nVdash \psi),$$

and by definition $p \Vdash \neg\psi$.

For an existential formula, assume the conditions forcing $\exists x \psi(x, \bar{\tau})$ are dense below p . Given $r \leq p$, first choose $q \leq r$ forcing the existential. Its defining density clause then supplies $s \leq q$ and a name σ such that

$$s \Vdash \psi(\sigma, \bar{\tau}).$$

Thus conditions forcing a named witness are dense below p , which is precisely $p \Vdash \exists x \psi(x, \bar{\tau})$. This completes the basis induction. For an arbitrary primitive-syntax formula φ , (NF) identifies its forcing set with that of $\mathcal{N}(\varphi)$, so the basis result proves (1) for every formula.

Now fix p . If some $q \leq p$ forces φ , then q decides the formula positively. If no extension of p forces φ , then the negation clause gives

$$p \Vdash \neg\varphi.$$

So every condition has a deciding extension, and the decision set is dense.

The distinction between the two conclusions matters later. Decision density merely says that some stronger condition chooses a side. The density characterization says more: if the positive side occurs densely below p , then p itself already forces the positive formula. That stronger statement is the step needed in the semantic characterization of forcing. $\square$

Theorem 47.7 (Definability lemma for the forcing relation). *Let M be a transitive set model of ZF and let $\mathbb{P} \in M$. For every formula $\varphi(\bar{x})$ of set theory there is a first-order formula*

$$F_{\varphi}(p, \mathbb{P}, \bar{\tau})$$

constructed effectively from the syntactic code of φ , such that for every $p \in \mathbb{P}$ and every tuple of $\mathbb{P}$ -names $\bar{\tau} \in M$,

$$M \models F_\varphi(p, \mathbb{P}, \bar{\tau}) \iff p \Vdash_{\mathbb{P}}^M \varphi(\bar{\tau}). \quad (1)$$

Thus the forcing relation is first-order definable over the ground model, uniformly in the formula code.

Proof.

Formal derivation spine

Formal spine: mirror the forcing recursion by first-order formulas. For example,

$$F_{\neg\psi}(p, \mathbb{P}, \bar{\tau}) \equiv \neg \exists q \in \mathbb{P} (q \leq p \wedge F_\psi(q, \mathbb{P}, \bar{\tau})),$$

and

$$F_{\exists x \psi}(p, \mathbb{P}, \bar{\tau}) \equiv \forall q \in \mathbb{P} (q \leq p \rightarrow \exists r \in \mathbb{P} \exists \sigma [r \leq q \wedge \text{Name}_{\mathbb{P}}(\sigma) \wedge F_\psi(r, \mathbb{P}, \sigma, \bar{\tau})]). \quad (2)$$

The atomic forcing relations were defined by well-founded recursion on pairs of name ranks. The well-founded-recursion theorem in ZF therefore supplies first-order formulas

$$F_=(p, \mathbb{P}, \sigma, \tau) \quad \text{and} \quad F_\in(p, \mathbb{P}, \sigma, \tau)$$

whose interpretations in M are exactly the atomic forcing relations. The predicate $\text{Name}_{\mathbb{P}}(\sigma)$, saying that σ is a $\mathbb{P}$ -name, is likewise definable by the rank recursion used to construct names.

We now recurse on formula complexity.

For conjunction, define

$$F_{\psi \wedge \theta}(p, \mathbb{P}, \bar{\tau}) \equiv F_\psi(p, \mathbb{P}, \bar{\tau}) \wedge F_\theta(p, \mathbb{P}, \bar{\tau}).$$

This is exactly the forcing clause for conjunction.

For negation, use the first formula in the formal spine:

$$F_{\neg\psi}(p, \mathbb{P}, \bar{\tau}) \equiv \neg \exists q \in \mathbb{P} (q \leq p \wedge F_\psi(q, \mathbb{P}, \bar{\tau})).$$

By the induction hypothesis,

$$M \models F_\psi(q, \mathbb{P}, \bar{\tau}) \iff q \Vdash_{\mathbb{P}}^M \psi(\bar{\tau}),$$

so the displayed first-order formula says exactly that no $q \leq p$ forces ψ , which is the definition of $p \Vdash \neg\psi$.

For an existential formula, define $F_{\exists x \psi}$ by (2). All quantifiers in (2) are ordinary first-order set quantifiers. When the formula is interpreted in the set model M , the variable σ ranges over elements of M , and the name predicate restricts it to the ground-model $\mathbb{P}$ -names. Using the induction hypothesis, (2) becomes

$$\forall q \leq p \exists r \leq q \exists \sigma \in M (\sigma \text{ is a } \mathbb{P}\text{-name} \wedge r \Vdash_{\mathbb{P}}^M \psi(\sigma, \bar{\tau})),$$

which is exactly the density clause defining

$$p \Vdash_{\mathbb{P}}^M \exists x \psi(x, \bar{\tau}).$$

For an arbitrary primitive-syntax formula φ , first compute the code of $\mathcal{N}(\varphi)$ and apply the displayed basis-formula construction. Thus primitive $\forall, \exists, \rightarrow$ nodes are handled by an explicit code transformation, not by being silently identified with different formulas.

This proves (1) for every formula. Uniformity follows from Lemma 47.1(1): the normalization map on codes is primitive recursive, and so is the basis forcing-formula recursion. No generic filter is used in this proof: definability is a theorem entirely about the ground model and the recursively defined forcing relation. $\square$

Worked example

Worked Example 47.8 (Forcing a negation). A condition p forces $\neg\varphi$ precisely when no stronger condition forces φ . Thus if some $q \leq p$ forces φ , then p cannot force its negation. The dense set of conditions deciding φ guarantees that a generic filter cannot remain permanently undecided.

A useful warning

Existential forcing cannot be reduced naively to a single witness name until the maximum principle has been justified. Keep the dense-witness clause and the maximum principle logically ordered.

Looking ahead

The next chapter, *Truth and generic extension theorem*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 48

Truth and generic extension theorem

Definitions and setup

Formal definitions used in this chapter

Countable transitive model. A countable transitive model is a transitive set model whose domain is countable in the ambient universe.

Ordinal notation. Ord^M is the class/set of objects that the transitive model M regards as ordinals; transitivity makes these genuine ambient ordinals.

Antichain and maximal antichain. An antichain is a set of pairwise incompatible conditions. It is maximal when every condition is compatible with some member of it.

How to read the symbols: main results proved in this chapter

Theorem 48.1 will prove the Truth Lemma, identifying truth in $M[G]$ with forcing by some condition in G . Combined with the already proved Definability Lemma, it yields the Fundamental/Forcing Theorem rather than defining it. Later results in the chapter prove the mixing lemma and the maximum principle, which respectively paste names along maximal antichains and consolidate locally forced existential witnesses into a single name.

The chapter now proves these semantic and antichain principles in dependency order. None of the displayed conclusions is included in the definition block: the exact truth/forcing biconditional appears first as the statement of Theorem 48.1, and the maximum and mixing assertions are owned by their later theorem statements.

How this chapter fits together

The first theorem will establish the semantic bridge between forcing and truth in a generic extension. Its negation and existential steps are the bottlenecks. Once it is proved, the earlier Definability Lemma combines with it to give the Fundamental Theorem of Forcing; the subsequent antichain arguments then prove mixing and the maximum principle.

Theorem 48.1 (Truth lemma). *Let M be a transitive set model of ZF, let $\mathbb{P} \in M$, let $G \subseteq \mathbb{P}$ be M -generic, and let $\bar{\tau} \in M$ be $\mathbb{P}$ -names. Then for every formula φ ,*

$$M[G] \models \varphi(\bar{\tau}^G) \iff \exists p \in G (p \Vdash_{\mathbb{P}}^M \varphi(\bar{\tau})). \quad (1)$$

Uses: the atomic Truth Lemma, the recursive forcing clauses, the density and decision lemma, and genericity.

How to read the symbols: the Truth Lemma is the semantic bridge

The statement

$$M[G] \models \varphi(\bar{\tau}^G) \iff \exists p \in G \ p \Vdash \varphi(\bar{\tau})$$

connects three layers: names in M , conditions in $\mathbb{P}$, and actual objects in $M[G]$. The left side is ordinary truth in the extension. The right side is a ground-model relation about names and conditions. Genericity makes the two agree by meeting the dense sets that arise in the forcing clauses.

Proof.

Formal derivation spine

Formal spine: induction on formulas, with density converted to generic witnesses. For conjunction, directedness combines two forcing conditions. For negation, genericity meets the dense decision set. For an existential forced by $p \in G$, let

$$D = \{r \leq p : \exists \sigma \ r \Vdash \psi(\sigma, \bar{\tau})\}.$$

Since D is dense below p , the set

$$D^* = D \cup \{q \in \mathbb{P} : q \perp p\}$$

is dense in $\mathbb{P}$. A generic filter containing p cannot meet the incompatible part, so it meets D .

We prove (1) by induction on the complexity of φ . The atomic membership and equality cases are exactly the Atomic Truth Lemma.

For conjunction, suppose first that

$$M[G] \models \psi(\bar{\tau}^G) \wedge \theta(\bar{\tau}^G).$$

By induction choose $p, q \in G$ with

$$p \Vdash \psi(\bar{\tau}), \quad q \Vdash \theta(\bar{\tau}).$$

Directedness of G gives $r \in G$ with $r \leq p, q$. Monotonicity yields

$$r \Vdash \psi(\bar{\tau}) \wedge \theta(\bar{\tau}).$$

Conversely, if $r \in G$ forces the conjunction, then it forces both conjuncts, and the induction hypothesis makes both true in $M[G]$.

For negation, assume

$$M[G] \models \neg\psi(\bar{\tau}^G). \tag{2}$$

The set of conditions deciding $\psi(\bar{\tau})$ is dense and belongs to M by the Definability Lemma and Separation in M . Genericity therefore gives $p \in G$ deciding the formula. The alternative $p \Vdash \psi(\bar{\tau})$ would contradict (2) by the induction hypothesis, so

$$p \Vdash \neg\psi(\bar{\tau}).$$

This proves the left-to-right implication for negation.

Conversely, suppose $p \in G$ and $p \Vdash \neg\psi(\bar{\tau})$. If $M[G] \models \psi(\bar{\tau}^G)$, the induction hypothesis gives $q \in G$ with $q \Vdash \psi(\bar{\tau})$. Choose $r \in G$ with $r \leq p, q$. By monotonicity, $r \Vdash \psi(\bar{\tau})$, contradicting $p \Vdash \neg\psi(\bar{\tau})$, since the latter says that no extension of p forces ψ . Hence

$$M[G] \models \neg\psi(\bar{\tau}^G).$$

Now consider an existential formula.

Suppose first that

$$M[G] \models \exists x \psi(x, \bar{\tau}^G).$$

Choose an actual witness $a \in M[G]$. By the definition of the generic extension there is a name $\sigma \in M$ with

$$a = \sigma^G.$$

The induction hypothesis gives $p \in G$ such that

$$p \Vdash \psi(\sigma, \bar{\tau}).$$

By monotonicity, every $q \leq p$ also forces $\psi(\sigma, \bar{\tau})$. Hence the named-witness set in the existential forcing clause contains the entire cone below p , and is therefore dense below p . The existential forcing clause now gives

$$p \Vdash \exists x \psi(x, \bar{\tau}).$$

For the converse, suppose $p \in G$ and

$$p \Vdash \exists x \psi(x, \bar{\tau}). \quad (3)$$

By definition, the set

$$D = \{r \leq p : \exists \sigma \in M [r \Vdash \psi(\sigma, \bar{\tau})]\} \quad (4)$$

is dense below p . The subtle point is that genericity was defined for sets dense in all of $\mathbb{P}$, not merely below one condition. Define

$$D^* = D \cup \{q \in \mathbb{P} : q \perp p\}. \quad (5)$$

The set D^* belongs to M : D is definable in M by the Definability Lemma, and compatibility with p is first-order definable from the poset.

We claim D^* is dense in $\mathbb{P}$. Given any q , either $q \perp p$, in which case $q \in D^*$, or q is compatible with p . In the second case choose $s \leq q, p$. Density of D below p gives $r \leq s$ with $r \in D$, and then $r \leq q$ with $r \in D^*$. This proves the claim.

Genericity gives some $r \in G \cap D^*$. Because both p and r lie in the filter G , they are compatible; hence r cannot belong to the incompatible part of (5). Thus $r \in D$. By (4) choose a name σ such that

$$r \Vdash \psi(\sigma, \bar{\tau}).$$

The induction hypothesis yields

$$M[G] \models \psi(\sigma^G, \bar{\tau}^G),$$

so

$$M[G] \models \exists x \psi(x, \bar{\tau}^G).$$

The induction just given proves the Truth Lemma for basis formulas. For an arbitrary Volume-I formula φ , Lemma 47.1(3) gives ordinary-truth equivalence with $\mathcal{N}(\varphi)$, while (NF) identifies forcing of φ with forcing of that normalized basis formula. Hence the Truth Lemma holds for every primitive-syntax formula. $\square$

Theorem 48.2 (Fundamental theorem of forcing). *Let M be a countable transitive model of ZF in the ambient universe, let $\mathbb{P} \in M$, and let $p \in \mathbb{P}$. For every formula $\varphi(\bar{x})$ and names $\bar{\tau} \in M$,*

$$p \Vdash_{\mathbb{P}}^M \varphi(\bar{\tau}) \iff \forall G \subseteq \mathbb{P} (G \text{ is } M\text{-generic and } p \in G \Rightarrow M[G] \models \varphi(\bar{\tau}^G)). \quad (1)$$

Together with the Definability Lemma, this is the Fundamental Theorem of Forcing in the countable-transitive-model presentation.

Proof.

Formal derivation spine

Formal spine: Truth Lemma plus the density characterization.

$$p \Vdash^M \varphi(\bar{\tau}) \implies \forall G \ni p \ M[G] \models \varphi(\bar{\tau}^G).$$

If $p \nVdash^M \varphi(\bar{\tau})$, then

$$\{r \leq p : r \Vdash^M \varphi(\bar{\tau})\}$$

is not dense below p . Hence some $q \leq p$ has no extension forcing φ , and therefore

$$q \Vdash^M \neg\varphi(\bar{\tau}).$$

Assume first that

$$p \Vdash^M \varphi(\bar{\tau}),$$

and let G be M -generic with $p \in G$. The forward direction of the Truth Lemma gives

$$M[G] \models \varphi(\bar{\tau}^G).$$

Thus the left side of (1) implies the right side.

Conversely, suppose

$$p \nVdash^M \varphi(\bar{\tau}). \quad (2)$$

The point requiring care is that (2) does *not* by itself say that p forces the negation, nor does decision density alone imply that an arbitrary deciding extension chooses the negative side. Use instead the density characterization from the preceding theorem. By (2), the set

$$D := \{r \leq p : r \Vdash^M \varphi(\bar{\tau})\}$$

is not dense below p . Hence there is $q \leq p$ such that

$$\forall r \leq q \ (r \nVdash^M \varphi(\bar{\tau})).$$

By the forcing clause for negation,

$$q \Vdash^M \neg\varphi(\bar{\tau}). \quad (3)$$

Because M is externally countable, Rasiowa–Sikorski supplies an M -generic filter G containing q . Since $q \leq p$ and a filter is upward closed toward weaker conditions, $p \in G$ as well. The Truth Lemma applied to (3) gives

$$M[G] \models \neg\varphi(\bar{\tau}^G).$$

Thus there is an M -generic extension through p in which $\varphi(\bar{\tau}^G)$ is false. The right side of (1) therefore fails whenever the left side fails, which proves the reverse implication.

External countability is used only in this reverse semantic direction, to construct a generic filter through q . The recursive forcing relation and its Definability Lemma are ground-model constructions and do not require external countability. $\square$

Remark 48.3 (Source note). The definability and truth lemmas are the two components customarily packaged as the Forcing Theorem. The present proof uses the modern name-recursion organization; compare [11, 13, 14].

Lemma 48.4 (Valuation does not increase rank beyond name rank). *To distinguish the two rank functions in this lemma, write $\text{rk}_V(x)$ for the set-theoretic rank of x from Volume IV and $\text{rk}_{\mathbb{P}}(\tau)$ for the forcing-name rank defined in Chapter 2 of this volume. For every forcing name τ and every filter $G \subseteq \mathbb{P}$,*

$$\text{rk}_V(\tau^G) \leq \text{rk}_{\mathbb{P}}(\tau). \quad (\text{VR})$$

Proof. Induct on $\text{rk}_{\mathbb{P}}(\tau)$. The Volume-IV rank formula and the valuation definition give

$$\text{rk}_V(\tau^G) = \sup\{\text{rk}_V(\sigma^G) + 1 : \langle \sigma, p \rangle \in \tau, p \in G\}.$$

For every active constituent σ , Theorem 45.1 gives $\text{rk}_{\mathbb{P}}(\sigma) < \text{rk}_{\mathbb{P}}(\tau)$, so the induction hypothesis applies. Therefore

$$\begin{aligned} \text{rk}_V(\tau^G) &\leq \sup\{\text{rk}_{\mathbb{P}}(\sigma) + 1 : \langle \sigma, p \rangle \in \tau, p \in G\} \\ &\leq \sup\{\text{rk}_{\mathbb{P}}(\sigma) + 1 : \exists p \in \mathbb{P} (\langle \sigma, p \rangle \in \tau)\} \\ &= \text{rk}_{\mathbb{P}}(\tau), \end{aligned}$$

where the last equality is the name-rank formula of Theorem 45.1. This proves (VR). $\square$

Proposition 48.5 (M and $M[G]$ have the same ordinals). *Let M be a transitive set model of ZF, let $\mathbb{P} \in M$ be a forcing notion, and let $G \subseteq \mathbb{P}$ be M -generic. Then*

$$\text{Ord}^{M[G]} = \text{Ord}^M.$$

Proof.

Formal derivation spine

Formal spine: check names give one inclusion; the valuation-rank bound gives the other.

For $\alpha \in \text{Ord}^M$,

$$\check{\alpha}^G = \alpha.$$

If $\gamma = \tau^G$ is an ordinal and $\alpha = \text{rk}_{\mathbb{P}}(\tau)$, then

$$\gamma = \text{rk}_V(\gamma) \leq \alpha. \quad (1)$$

Because M is transitive, every ordinal that M contains is a genuine ambient ordinal. For each $\alpha \in \text{Ord}^M$, the check-name identity gives $\check{\alpha}^G = \alpha$, so

$$\text{Ord}^M \subseteq \text{Ord}^{M[G]}.$$

Conversely, let $\gamma \in M[G]$ be an ordinal. By the definition of the generic extension, choose a name $\tau \in M$ with $\tau^G = \gamma$, and put

$$\alpha := \text{rk}_{\mathbb{P}}(\tau).$$

The name-rank recursion is absolute for the transitive ground model: its defining formula in Theorem 45.1 uses only the constituent relation on the set τ , ordinal supremum, and the forcing poset $\mathbb{P}$, all of which are computed correctly in a transitive model of ZF. Hence $\alpha \in \text{Ord}^M$.

For every ordinal γ , its Volume-IV set rank is $\text{rk}_V(\gamma) = \gamma$. Lemma 48.4 therefore gives exactly (1):

$$\gamma \leq \alpha.$$

If $\gamma = \alpha$, then $\gamma \in M$ because $\alpha \in M$. If $\gamma < \alpha$, then $\gamma \in \alpha$, and transitivity of M again gives $\gamma \in M$. Thus every ordinal of $M[G]$ already belongs to M , proving

$$\text{Ord}^{M[G]} \subseteq \text{Ord}^M.$$

Together with the first inclusion this proves equality.

Further explanation. The rank estimate is the point that a minimal-counterexample slogan leaves implicit. Forcing may add a new subset of a ground-model ordinal, but a name of ground-model rank α cannot evaluate to an ordinal above α . Once a purported new ordinal is bounded by a ground-model ordinal, transitivity puts it back inside the ground model. $\square$

Theorem 48.6 (Mixing lemma). *Let A be an antichain and let $\{\tau_a : a \in A\}$ be forcing names. There is a forcing name τ such that*

$$a \Vdash \tau = \tau_a \quad (a \in A).$$

No maximality assumption on A is needed for the mixing construction itself.

Proof.

Formal derivation spine

Formal spine: localize each name to its antichain cone, then take the union. For $a \in A$, define

$$\tau_a \restriction a := \{\langle \sigma, q \rangle : \exists r (\langle \sigma, r \rangle \in \tau_a \wedge q \leq a, r)\},$$

and put

$$\tau := \bigcup_{a \in A} \tau_a \restriction a.$$

Then below a , pieces coming from $b \neq a$ cannot occur because $a \perp b$.

The displayed localization is a set forcing name: its constituent names are constituents of τ_a , and its tags are conditions in $\mathbb{P}$. Fix $a \in A$.

We first prove

$$a \Vdash \tau \subseteq \tau_a. \quad (\text{M1})$$

Take a constituent $\langle \sigma, q \rangle \in \tau$ and a condition $s \leq a, q$. By definition of τ , there are $b \in A$ and a pair $\langle \sigma, r \rangle \in \tau_b$ such that

$$q \leq b, r.$$

Since $s \leq a, q \leq b$, the conditions a and b are compatible. Because A is an antichain, $a = b$. Hence $\langle \sigma, r \rangle \in \tau_a$ and $q \leq r$. The atomic membership clause therefore gives

$$q \Vdash \sigma \in \tau_a,$$

and monotonicity gives $s \Vdash \sigma \in \tau_a$. This is exactly the auxiliary subset clause required for (M1).

Conversely,

$$a \Vdash \tau_a \subseteq \tau. \quad (\text{M2})$$

Indeed, let $\langle \sigma, r \rangle \in \tau_a$ and let $s \leq a, r$. Then

$$\langle \sigma, s \rangle \in \tau_a \restriction a \subseteq \tau.$$

The atomic membership clause gives $s \Vdash \sigma \in \tau$. Thus the auxiliary subset clause holds for every constituent of τ_a , proving (M2).

The atomic equality clause is the conjunction of the two forced inclusions, so (M1)–(M2) yield

$$a \Vdash \tau = \tau_a.$$

Since $a \in A$ was arbitrary, the same τ works simultaneously for every member of the antichain. □

Theorem 48.7 (Maximum principle). *Assume the ground model satisfies Choice. If*

$$p \Vdash \exists x \varphi(x, \bar{\tau}),$$

then there is a name σ such that

$$p \Vdash \varphi(\sigma, \bar{\tau}).$$

Proof.

Formal derivation spine

Formal spine: choose local witnesses on a maximal antichain and mix them.

$$D = \{q \leq p : \exists \tau \ q \Vdash \varphi(\tau, \bar{\tau})\}$$

is dense below p . Choose a maximal antichain $A \subseteq D$ below p , choose one witness name τ_a for each $a \in A$, and apply the already proved Mixing Lemma 48.6.

Assume $p \Vdash \exists x \varphi(x, \bar{\tau})$. By the existential forcing clause,

$$D = \{q \leq p : \exists \tau \ q \Vdash \varphi(\tau, \bar{\tau})\}$$

is dense below p . Choice in the ground model gives a maximal antichain $A \subseteq D$ below p , and then selects, for each $a \in A$, a name τ_a satisfying

$$a \Vdash \varphi(\tau_a, \bar{\tau}). \quad (\text{MP1})$$

Apply Mixing Lemma 48.6 to obtain one name τ with

$$a \Vdash \tau = \tau_a \quad (a \in A). \quad (\text{MP2})$$

The full formula-substitution Lemma 47.5, applied to (MP1)–(MP2), gives

$$a \Vdash \varphi(\tau, \bar{\tau}) \quad (a \in A). \quad (\text{MP3})$$

Because A is maximal below p , it is predense below p . Hence for every $q \leq p$ there are $a \in A$ and $r \leq q, a$. By monotonicity of (MP3),

$$r \Vdash \varphi(\tau, \bar{\tau}).$$

Thus the conditions below p forcing $\varphi(\tau, \bar{\tau})$ are dense below p . The previously proved density characterization of forcing now yields

$$p \Vdash \varphi(\tau, \bar{\tau}).$$

This is the required single witness name.

Dependency check. The earlier version invoked an informal “mixing construction” before the Mixing Lemma had been proved. The logical order is now explicit: Mixing is proved first from the atomic clauses; Maximum then uses Mixing, Choice, predensity, and the already proved density characterization. $\square$

Worked example

Worked Example 48.8 (Using decision density in the negation step). Assume $M[G] \models \neg\varphi$. If no condition in G forced $\neg\varphi$, then for each $p \in G$ there would be a stronger condition compatible with G that can force φ . The dense decision set meets G , yielding a condition in G that forces one side; the truth of $\neg\varphi$ rules out the positive side, so the generic condition must force the negation.

A useful warning

The forcing theorem has two halves: definability in the ground model and truth in the extension. Proving only the Truth Lemma is not enough for later Replacement arguments.

Looking ahead

The next chapter, *ZFC in generic extensions*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 49

ZFC in generic extensions

Definitions and setup

Formal definitions used in this chapter

Constituent relation and bounded-name family. Recall the forcing-name hierarchy and the already proved rank theorem (Theorem 45.1). For later shorthand define

$$\sigma \prec_{\mathbb{P}} \tau \iff \exists p \in \mathbb{P} \langle \sigma, p \rangle \in \tau.$$

Theorem 45.1 then gives

$$\text{rank}_{\mathbb{P}}(\tau) = \sup\{\text{rank}_{\mathbb{P}}(\sigma) + 1 : \sigma \prec_{\mathbb{P}} \tau\}.$$

This is a recall of the earlier theorem, not a new definition of forcing-name rank. For an ordinal θ , define

$$V_{\theta}^{\mathbb{P}} := \{\tau : \tau \text{ is a forcing name and } \text{rank}_{\mathbb{P}}(\tau) < \theta\}, \quad V_{\theta}^{\mathbb{P}, M} := V_{\theta}^{\mathbb{P}} \cap M.$$

Thus $V_{\theta}^{\mathbb{P}, M}$ is the ground-model rank segment of forcing names. The later Collection/Replacement proof will establish, rather than assume here, the bounded-name lemma that places the relevant witness names inside one such set-sized segment.

Total and functional formula on a set. A formula $\varphi(x, y, \bar{p})$ is total on a in $M[G]$ when

$$M[G] \models \forall x \in a \exists y \varphi(x, y, \bar{p}),$$

and functional when the witness is unique.

Collection and Replacement. Collection asks for a single set b containing some witness for every $x \in a$; Replacement is the functional special case whose range is a set.

Internal power set.

$$\mathcal{P}^{M[G]}(a) = \{x \in M[G] : x \subseteq a\}.$$

Preservation of ZF/ZFC. Each axiom is verified inside $(M[G], \in)$; transitivity alone does not imply Separation, Collection/Replacement, Power Set, or Choice.

The statement $M[G] \models \text{ZFC}$ is a list of axiom proofs. Transitivity gives Extensionality and Foundation; canonical names give Pairing, Union, and Infinity; and the Definability Lemma gives Separation. Collection and Replacement require a set-sized family of witness names. Power Set requires a bounded family of subnames representing all relevant subsets. Choice is inherited when the ground model has a well-order of the relevant name space. The hard axioms receive separate proofs.

How this chapter fits together

Preservation of ZFC is not one theorem with a one-line proof. Separation uses definability of forcing to carve a subname. Replacement/Collection requires a ground-model bound on witness names. Power Set requires a set-sized collection of names representing all subsets of a given interpreted set. Choice uses a ground-model well-order of suitable names. These constructions are treated separately.

Theorem 49.1 (Forcing preservation of the elementary ZF axioms). *If M is a transitive set model of ZF, $\mathbb{P} \in M$, and $G \subseteq \mathbb{P}$ is M -generic, then $M[G]$ satisfies Extensionality, Empty Set, Foundation, Pairing, Union, and Infinity. Separation, Collection/Replacement, Power Set, and Choice are established separately in the following results.*

Proof.

Formal derivation spine

Formal spine: use explicit ground-model names for the elementary set operations. Extensionality and Foundation come from transitivity; Empty Set, Pairing, Union, and Infinity are obtained by explicit names whose valuations are the required sets.

We verify only the elementary ZF axioms named by this result. Extensionality and Foundation hold because $M[G]$ is a transitive class with the ambient membership relation. The empty name $\emptyset$ evaluates to $\emptyset$, so Empty Set holds. If $x = \sigma^G$ and $y = \tau^G$, define the pairing name

$$\text{Pair}(\sigma, \tau) := \{\langle \sigma, 1_{\mathbb{P}} \rangle, \langle \tau, 1_{\mathbb{P}} \rangle\}.$$

Then $\text{Pair}(\sigma, \tau)^G = \{x, y\}$, proving Pairing.

For Union one must retain the forcing conditions from both levels of the name. Given a name τ , define

$$\text{Un}(\tau) := \{\langle \rho, r \rangle : \exists \sigma, p, q (\langle \sigma, p \rangle \in \tau \wedge \langle \rho, q \rangle \in \sigma \wedge r \leq p \wedge r \leq q)\}.$$

To see that this is a set of the ground model rather than a class comprehension, form in M

$$D_0 := \text{dom}(\tau), \quad D_1 := \bigcup_{\sigma \in D_0} \text{dom}(\sigma).$$

Replacement and Union give the set D_1 , and the displayed definition of $\text{Un}(\tau)$ is then a Separation subset of the set $D_1 \times \mathbb{P}$. Thus $\text{Un}(\tau) \in M$. If $\rho^G \in \text{Un}(\tau)^G$, then for some $r \in G$ and conditions $p, q \in \mathbb{P}$ satisfying $\langle \sigma, p \rangle \in \tau$, $\langle \rho, q \rangle \in \sigma$, and $r \leq p, q$, upward closure of the filter gives $p, q \in G$; hence $\sigma^G \in \tau^G$ and $\rho^G \in \sigma^G$. Conversely, if $z \in y \in \tau^G$, choose $\langle \sigma, p \rangle \in \tau$ and $\langle \rho, q \rangle \in \sigma$ with $p, q \in G$, $y = \sigma^G$, and $z = \rho^G$. Directedness of G gives $r \in G$ with $r \leq p, q$, so $\langle \rho, r \rangle \in \text{Un}(\tau)$ and $z \in \text{Un}(\tau)^G$. Thus $\text{Un}(\tau)^G = \bigcup \tau^G$, proving Union. Finally $\check{\omega}^G = \omega$, so Infinity holds.

These direct name constructions do not establish Separation, Collection/Replacement, or Power Set. Those axioms require the definability and bounding machinery developed in the forcing theorem and are proved separately below. $\square$

Theorem 49.2 (Separation in $M[G]$). *If M is a transitive set model of ZF, $\mathbb{P} \in M$, $G \subseteq \mathbb{P}$ is M -generic, $a = \tau^G \in M[G]$, and $\varphi(x, \bar{p})$ is a formula, then the subset*

$$\{x \in a : M[G] \models \varphi(x, \bar{p})\}$$

belongs to $M[G]$.

Proof.

Formal derivation spine

Formal spine for Separation in $M[G]$.

$$\eta = \{\langle \sigma, q \rangle : \exists p (\langle \sigma, p \rangle \in \tau, q \leq p, q \Vdash \varphi(\sigma, \bar{\rho}))\}.$$

$$\eta^G = \{u \in x : M[G] \models \varphi(u, \bar{a})\}.$$

Let $x = \tau^G$ and let $\varphi(u, \bar{a})$ be a formula with parameters $\bar{a} = \bar{\rho}^G$. In the ground model define the name

$$\eta = \{\langle \sigma, q \rangle : \exists p (\langle \sigma, p \rangle \in \tau, q \leq p, q \Vdash \varphi(\sigma, \bar{\rho}))\}.$$

The Definability Lemma makes the displayed property a ground-model definable relation, so Separation in M ensures that η is a set and a forcing name.

If $u \in \eta^G$, some condition in G forces both that u comes from τ and that $\varphi(u, \bar{a})$ holds; the Truth Lemma gives $u \in x$ and $\varphi(u, \bar{a})$. Conversely, if $u \in x$ and $M[G] \models \varphi(u, \bar{a})$, choose a name for u appearing in τ ; the Truth Lemma and directedness of G give a stronger condition placing that name into η . Hence

$$\eta^G = \{u \in x : M[G] \models \varphi(u, \bar{a})\}.$$

□

Theorem 49.3 (Replacement/Collection in $M[G]$). *If M is a transitive set model of ZF, $\mathbb{P} \in M$, and $G \subseteq \mathbb{P}$ is M -generic, then $M[G]$ satisfies Collection and Replacement: whenever a formula is total on a set $a \in M[G]$, there is a single ground-model name whose valuation contains witnesses for all elements of a .*

Uses: Definability of forcing, the Truth Lemma, the name-rank hierarchy, and ground-model Replacement; no use of the Maximum Principle is made, because that principle was proved only under Choice.

Commented symbolic trace: Replacement in a generic extension

A total formula in $M[G]$ may have witnesses represented by names of many different ranks. To avoid any hidden use of Choice, the proof does *not* choose one witness name along a maximal antichain. Instead, inside M it assigns to each relevant constituent name and condition the *least ordinal rank at which some forced witness exists*, using 0 when none exists. Ground-model Replacement bounds this set of ordinals by one δ . The single name containing *all* ground-model names in $V_\delta^{\mathbb{P}, M}$ then contains a witness for every actual input after valuation. Functional Collection followed by Separation gives Replacement.

Proof.

Formal derivation spine

Formal spine: bound witness-name ranks without choosing witnesses. Let $a = \tau^G$ and let $\bar{p} = \bar{\pi}^G$. For $\langle \sigma, r \rangle \in \tau$ and $q \in \mathbb{P}$, define

$$f(\sigma, r, q) = \min\{\alpha : \exists \rho \in V_\alpha^{\mathbb{P}, M} q \Vdash \varphi(\sigma, \rho, \bar{\pi})\},$$

when the displayed set is nonempty, and put $f(\sigma, r, q) = 0$ otherwise. Ground-model Replacement gives an ordinal δ strictly above all values of f . Then

$$\dot{u} := \{\langle \rho, 1_{\mathbb{P}} \rangle : \rho \in V_\delta^{\mathbb{P}, M}\}$$

contains, after valuation, a witness for every member of a .

Let $a = \tau^G \in M[G]$, let the parameters be $\bar{p} = \bar{\pi}^G$, and assume

$$M[G] \models \forall x \in a \exists y \varphi(x, y, \bar{p}). \tag{R1}$$

We construct in the ground model one bounded family of names containing enough witnesses. The index set

$$I := \{(\sigma, r, q) : \langle \sigma, r \rangle \in \tau \text{ and } q \in \mathbb{P}\}$$

is a set in M . For $(\sigma, r, q) \in I$, let

$$A_{\sigma, r, q} := \{\alpha \in \text{Ord}^M : \exists \rho \in V_{\alpha}^{\mathbb{P}, M} \ q \Vdash \varphi(\sigma, \rho, \bar{\pi})\}.$$

Definability of the forcing relation makes this a definable class over M . If it is nonempty, its least member exists because the ordinals are well ordered; define $f(\sigma, r, q)$ to be that least member, and put $f(\sigma, r, q) = 0$ when it is empty. Thus f is a definable function on the set I , so Replacement in M gives the set $f[I]$ of ordinal values. Choose an ordinal $\delta \in M$ strictly larger than every member of $f[I]$, for example

$$\delta := \sup f[I] + \omega.$$

Now form in M the name

$$\dot{u} := \{\langle \rho, 1_{\mathbb{P}} \rangle : \rho \in V_{\delta}^{\mathbb{P}, M}\}. \quad (\text{R2})$$

We verify Collection in the fixed generic extension. Let $x \in a$. By the valuation definition there is a constituent $\langle \sigma, r \rangle \in \tau$ with $r \in G$ and $x = \sigma^G$. By (R1), choose an actual witness $y \in M[G]$ with

$$M[G] \models \varphi(x, y, \bar{p}).$$

Choose a ground-model name ρ with $y = \rho^G$. Applying the Truth Lemma to the matrix formula gives some $q \in G$ such that

$$q \Vdash \varphi(\sigma, \rho, \bar{\pi}). \quad (\text{R3})$$

Hence $A_{\sigma, r, q}$ is nonempty. By the definition of f and the choice of δ , there is a name $\rho' \in V_{\delta}^{\mathbb{P}, M}$ with

$$q \Vdash \varphi(\sigma, \rho', \bar{\pi}). \quad (\text{R4})$$

Because $q \in G$, the Truth Lemma converts (R4) to

$$M[G] \models \varphi(x, (\rho')^G, \bar{p}).$$

By (R2), $(\rho')^G \in \dot{u}^G$. Since $x \in a$ was arbitrary,

$$M[G] \models \forall x \in a \ \exists y \in \dot{u}^G \ \varphi(x, y, \bar{p}).$$

Thus $M[G]$ satisfies Collection.

If φ is functional on a , apply the Separation theorem already proved in $M[G]$ to $\dot{u}^G$ and retain exactly those y for which $\exists x \in a \ \varphi(x, y, \bar{p})$. This set is the image of a under the functional relation, so Replacement follows.

Why Choice is not hidden here. The argument never selects one witness name for each condition and never chooses a maximal antichain. It selects only the least *ordinal rank* at which some witness exists, a choice supplied by the well-ordering of the ordinals in ZF, and then takes all names below one common rank bound. This is why the proof applies to a ZF ground model even though the separate Maximum Principle above was stated under Choice. $\square$

Remark 49.4 (Source note). Preservation of Replacement is treated separately from the Truth Lemma because it requires a ground-model set bound on possible witness names; compare [13, 14].

Theorem 49.5 (Power Set in $M[G]$). *If M is a transitive set model of ZF, $\mathbb{P} \in M$, $G \subseteq \mathbb{P}$ is M -generic, and $a \in M[G]$, then*

$$\mathcal{P}^{M[G]}(a) = \{x \in M[G] : x \subseteq a\}$$

is a set belonging to $M[G]$.

Uses: the Truth Lemma, normalization of subset names, ground-model Power Set, and Separation already proved in $M[G]$.

How to read the symbols: Power Set means internal subsets

To prove Power Set for $x = \tau^G$, we need a set in $M[G]$ containing exactly the subsets of x that exist in $M[G]$. Arbitrary names can be replaced, modulo forced equality, by names of bounded rank whose pairs use constituents from τ . Because the bounded family of candidate names is a set in M , it has a name; its valuation is the internal power set. As with L , “Power Set” is always relative to the model under discussion.

Proof.

Formal derivation spine

Formal spine: normalize every subset name into one ground-model set of candidates.

$$\begin{aligned} \text{Sub}_{\dot{a}}(\dot{y}) &= \{ \langle \sigma, p \rangle \in \text{dom}(\dot{a}) \times \mathbb{P} : p \Vdash \sigma \in \dot{y} \}, \\ S &= \mathcal{P}^M(\text{dom}(\dot{a}) \times \mathbb{P}), \quad \mathcal{P}^{M[G]}(a) = \{ z \in \dot{U}^G : z \subseteq a \}. \end{aligned}$$

Let $a = \dot{a}^G \in M[G]$, with $\dot{a} \in M$. We first normalize names for subsets of a .

Suppose $y = \dot{y}^G \in M[G]$ and $y \subseteq a$. Define in M

$$\text{Sub}_{\dot{a}}(\dot{y}) := \{ \langle \sigma, p \rangle \in \text{dom}(\dot{a}) \times \mathbb{P} : p \Vdash \sigma \in \dot{y} \}. \quad (1)$$

Definability of forcing and Separation in M show that (1) is a name in M . We claim

$$\text{Sub}_{\dot{a}}(\dot{y})^G = y. \quad (2)$$

If z belongs to the left side, then $z = \sigma^G$ for some $p \in G$ with $p \Vdash \sigma \in \dot{y}$; the Truth Lemma gives $z \in y$. Conversely, if $z \in y \subseteq a$, choose $\sigma \in \text{dom}(\dot{a})$ and $r \in G$ with $z = \sigma^G$ and $\langle \sigma, r \rangle \in \dot{a}$. Since $M[G] \models \sigma^G \in \dot{y}^G$, the Truth Lemma gives $p \in G$ with $p \Vdash \sigma \in \dot{y}$, so z belongs to the left side of (2).

Now

$$S := \mathcal{P}^M(\text{dom}(\dot{a}) \times \mathbb{P})$$

is a set in M . Let

$$\dot{U} := \{ \langle \tau, 1_{\mathbb{P}} \rangle : \tau \in S \}.$$

By (2), every subset of a that belongs to $M[G]$ is an element of $\dot{U}^G$. Conversely $\dot{U}^G$ may contain values that are not subsets of a , so use the already established Separation axiom in $M[G]$ to form

$$b := \{ z \in \dot{U}^G : z \subseteq a \}.$$

Then

$$b = \mathcal{P}^{M[G]}(a).$$

Thus the internal power set is a set in $M[G]$. The argument uses the ground-model Power Set axiom only to collect the set of normalized candidate names; it does not assume that all ambient subsets of a belong to the extension. $\square$

Remark 49.6 (Source note). Power Set likewise needs bounded normalization of subset names; transitivity of the extension alone is not enough. Compare [13, 14].

Theorem 49.7 (Choice in $M[G]$ when M satisfies Choice). *If M is a transitive set model of ZFC, $\mathbb{P} \in M$, and $G \subseteq \mathbb{P}$ is M -generic, then $M[G] \models \text{AC}$. Together with the preceding preservation results, $M[G] \models \text{ZFC}$.*

Proof.

Formal derivation spine

Formal spine: well-order a set of names, then descend that well-order through the valuation map. If $x = \tau^G$, let

$$D := \text{dom}(\tau).$$

Choose in M a well-order $\triangleleft$ of D . Bound the name ranks of members of D by an ordinal $\theta \in M$, and inside $M[G]$ recursively evaluate all names in $V_\theta^{\mathbb{P}, M}$. The restriction

$$e : D \twoheadrightarrow x, \quad e(\sigma) = \sigma^G,$$

is a set surjection. For $y \in x$, let $r(y)$ be the $\triangleleft$ -least member of $e^{-1}(\{y\})$, and define

$$y <^* z \iff r(y) \triangleleft r(z).$$

Let $x = \tau^G \in M[G]$ and put

$$D := \text{dom}(\tau) = \{\sigma : \exists p \in \mathbb{P} \langle \sigma, p \rangle \in \tau\}. \quad (\text{C1})$$

The set D belongs to M , and every element of x is σ^G for some $\sigma \in D$. Since $M \models \text{AC}$, there is in M a well-order $\triangleleft$ of D . Both D and $\triangleleft$ belong to $M[G]$.

We next make the valuation map used in the argument an actual set of the extension. The name-rank function is defined in the ground model. Replacement in M bounds the ranks of members of D by some ordinal $\theta \in M$; increase θ by one if necessary so that every constituent of every name in D lies in

$$C := V_\theta^{\mathbb{P}, M}.$$

The set C belongs to $M \subseteq M[G]$ and is downward closed under taking constituents of names.

Inside $M[G]$, using the set parameter G , apply well-founded recursion on name rank to the set C . Define a set function e on C by

$$e(\sigma) := \{e(\rho) : \exists p \in G \langle \rho, p \rangle \in \sigma\}. \quad (\text{C2})$$

All recursive predecessors of σ have smaller name rank and lie in C , so the recursion is legitimate. By uniqueness of the valuation recursion established earlier,

$$e(\sigma) = \sigma^G \quad (\sigma \in C). \quad (\text{C3})$$

Consequently $e \upharpoonright D : D \rightarrow x$ is a surjection belonging to $M[G]$.

For each $y \in x$, the fiber

$$D_y := \{\sigma \in D : e(\sigma) = y\}$$

is a nonempty set. Because $\triangleleft$ is a well-order, it has a unique least element; define

$$r(y) := \min_{\triangleleft} D_y. \quad (\text{C4})$$

The graph of r is a Separation subset of $x \times D$, hence is a set of $M[G]$. Finally define

$$y <^* z \iff r(y) \triangleleft r(z). \quad (\text{C5})$$

Since r is injective and selects exactly one representative from each fiber of the surjection $e \upharpoonright D$, the relation $<^*$ is a well-order of x . Thus every set of $M[G]$ can be well ordered, so $M[G] \models \text{AC}$.

Where Choice is used. The only use of Choice is the ground-model statement $M \models \text{AC}$, which supplies the well-order $\triangleleft$ of the set D . The passage from names to their values is a set-sized recursion in $M[G]$, and the least representative of each fiber is then determined uniquely by $\triangleleft$; no second application of Choice is hidden in the construction. $\square$

Worked example

Worked Example 49.8 (Bounding names for Replacement without Choice). Suppose a condition forces that every member of τ has a unique φ -image. For each constituent name and relevant condition, consider the set of ordinal ranks at which some forced witness name first appears. When this set is nonempty, take its least ordinal. Ground-model Replacement bounds all of these least ranks by one ordinal δ . The single rank segment $V_\delta^{\mathbb{P}, M}$ then contains enough witness names for every possible input, so the range is bounded by one name without selecting witnesses along maximal antichains. This is the mechanism used in Theorem 49.3 and is valid over a ZF ground model.

A useful warning

Transitivity of $M[G]$ does not by itself prove Replacement or Power Set. Both axioms require explicit set-sized name bounds.

End-of-volume perspective

The forcing theorem turns conditions into reliable semantic information about $M[G]$. The next volume replaces dependence on an externally chosen generic with Boolean-valued semantics and then extracts formal relative-consistency conclusions.

Part VIII

Volume VIII: Forcing II: Boolean Values, Preservation, and Relative Consistency

Preface

This volume supplies the metatheoretic bridge that generic-extension pedagogy alone does not give. Boolean-valued semantics and exact preservation results are separated from countable-model intuition.

Position in the series

Volume VIII separates the metatheoretic consistency argument from countable-transitive-model intuition. Boolean-valued semantics, chain conditions, and nice names provide the transfer and preservation tools used in the final two volumes.

Sources and further reading

Boolean-valued semantics is used to separate the model-theoretic generic-extension picture from the formal relative-consistency conclusion. See [13, 14] for standard developments.

Notation for this volume

$\mathbb{B}$ is a complete Boolean algebra and $\|\varphi\| \in \mathbb{B}$ is the Boolean truth value. Ground-model cardinals carry a superscript when preservation has not yet been proved.

How to read symbolic arguments in this volume

When a formula appears, read it in layers rather than as one visual block. First identify the *ambient language*: are we speaking inside a formal theory, about a structure, or in the surrounding metatheory? Next mark the objects being manipulated (formulas, codes, sets, names, conditions, ordinals) and the operation being applied to them. Finally check the side conditions—free variables, rank bounds, compatibility, genericity, transitivity, or consistency hypotheses. Whenever a displayed derivation changes level, the text says so explicitly. A displayed line is therefore meant to be read like a line of well-commented code: the symbols perform the operation, and the surrounding prose records its type, preconditions, and purpose.

Chapter 50

Separative quotients and Boolean completion

Definitions and setup

Formal definitions used in this chapter

Separative quotient. For a forcing preorder $\mathbb{P}$, define

$$p \preceq q \iff \forall r \leq p \ (r \text{ is compatible with } q), \quad p \equiv q \iff (p \preceq q \wedge q \preceq p).$$

The *separative quotient* is $\mathbb{P}/\equiv$ with

$$[p] \leq_s [q] \iff p \preceq q.$$

Write $q_s : \mathbb{P} \rightarrow \mathbb{P}/\equiv$ for the quotient map $q_s(p) = [p]$. Its recursive action on names is

$$q_{s*}(\tau) := \{ \langle q_{s*}(\sigma), [p] \rangle : \langle \sigma, p \rangle \in \tau \}.$$

Boolean algebra and Boolean order. A Boolean algebra is a tuple

$$(\mathbb{B}, \wedge, \vee, \neg, 0_{\mathbb{B}}, 1_{\mathbb{B}})$$

in which $\wedge$ and $\vee$ are associative and commutative, satisfy the absorption laws

$$a \wedge (a \vee b) = a, \quad a \vee (a \wedge b) = a,$$

distribute over one another, $0_{\mathbb{B}}$ and $1_{\mathbb{B}}$ are respectively the identities for $\vee$ and $\wedge$, and every $a \in \mathbb{B}$ has the displayed complement:

$$a \wedge \neg a = 0_{\mathbb{B}}, \quad a \vee \neg a = 1_{\mathbb{B}}.$$

Its Boolean order is

$$a \leq b \iff a \wedge b = a.$$

Write $\mathbb{B}^+ := \mathbb{B} \setminus \{0_{\mathbb{B}}\}$.

Complete Boolean algebra. A Boolean algebra $\mathbb{B}$ is *complete* when every set $A \subseteq \mathbb{B}$ has a least upper bound $\bigvee A$ and a greatest lower bound $\bigwedge A$ in the Boolean order.

The forcing topology, interior, and closure. On a separative forcing notion $\mathbb{P}$, let

$$N_p := \{q \in \mathbb{P} : q \leq p\}.$$

Call $U \subseteq \mathbb{P}$ *open* when it is a union of sets N_p . For $A \subseteq \mathbb{P}$, define

$$\text{int}(A) := \bigcup \{U : U \text{ is open and } U \subseteq A\}, \quad \overline{A} := \mathbb{P} \setminus \text{int}(\mathbb{P} \setminus A).$$

An open set U is *regular open* when $U = \text{int}(\overline{U})$. For arbitrary $A \subseteq \mathbb{P}$, its *regular-open hull* is

$$\text{ro}(A) := \text{int}(\overline{A}).$$

Regular-open family and candidate Boolean operations. $\text{RO}(\mathbb{P})$ is the set of regular-open subsets of $\mathbb{P}$. On this set define the following candidate operations; Theorem 50.2 proves that they satisfy the complete-Boolean-algebra laws. For $U, V \in \text{RO}(\mathbb{P})$ put

$$U \wedge V := U \cap V, \quad \neg U := \text{int}(\mathbb{P} \setminus U),$$

and for an arbitrary set-indexed family $(U_i)_{i \in I} \subseteq \text{RO}(\mathbb{P})$ put

$$\bigvee_{i \in I} U_i := \text{int}\left(\overline{\bigcup_{i \in I} U_i}\right), \quad \bigwedge_{i \in I} U_i := \text{int}\left(\bigcap_{i \in I} U_i\right).$$

Finally $0 = \emptyset$ and $1 = \mathbb{P}$. The interior in the arbitrary-meet formula is essential: an infinite intersection of open sets need not be open.

Dense embedding. A map $i : \mathbb{P} \rightarrow \mathbb{Q}$ is a dense embedding when it is injective, preserves and reflects order, preserves incompatibility, and its range is dense in $\mathbb{Q}$.

Name and generic-filter translation along a dense embedding. For a dense embedding $i : \mathbb{P} \rightarrow \mathbb{Q}$, define recursively

$$i_*(\tau) := \{\langle i_*(\sigma), i(p) \rangle : \langle \sigma, p \rangle \in \tau\}.$$

If $H \subseteq \mathbb{Q}$ is a filter, put

$$G_H := \{p \in \mathbb{P} : i(p) \in H\};$$

if $G \subseteq \mathbb{P}$ is a filter, put

$$H_G := \{q \in \mathbb{Q} : \exists p \in G (i(p) \leq q)\}.$$

These are the *induced filters*. Proposition 50.3 proves that generic filters correspond in this way and that

$$(i_*(\tau))^H = \tau^{G_H}$$

for every name τ whenever H is generic.

Boolean completion and the Boolean characterization of forcing equivalence. A *Boolean completion* of a separative forcing notion $\mathbb{P}$ is a complete Boolean algebra $\mathbb{B}$ together with a dense embedding $i : \mathbb{P} \rightarrow \mathbb{B}^+$. The term *forcing equivalent* retains its Volume-VII definition: the separative quotients have order-isomorphic dense suborders. Theorem 50.2 first proves that the regular-open construction above is such a Boolean completion. Proposition 50.4 then proves that passing to these proved completions preserves exactly the previously defined forcing behavior; only after those results may isomorphic Boolean completions be used as an equivalent characterization.

Two forcing conditions are separatively equivalent when no extension can distinguish their compatibility behavior. The separative quotient removes redundant conditions. Theorem 50.2 will prove that the displayed regular-open operations make $\text{RO}(\mathbb{P})$ a complete Boolean algebra and that the separative forcing embeds densely into it. Proposition 50.3 will then prove the corresponding generic-filter/name translation. These are outputs of the chapter, not properties assumed in the setup.

How this chapter fits together

Different forcing posets can encode the same extension. The proof order is: first form and verify the separative quotient; next prove the regular-open completion theorem; then prove that dense embeddings transport generic filters and name valuations; finally identify the resulting forcing-equivalence characterization. Only

after those theorems have been proved do we use Boolean-valued semantics as an equivalent reformulation of the original forcing.

Theorem 50.1 (Separative quotient theorem). *Let $\mathbb{P}$ be a forcing preorder and define*

$$p \preceq q \iff \forall r \leq p \text{ (} r \text{ is compatible with } q \text{)}.$$

Put

$$p \equiv q \iff p \preceq q \text{ \& } q \preceq p.$$

Then $\equiv$ is an equivalence relation, the quotient $\mathbb{P}/\equiv$, ordered by

$$[p] \leq_s [q] \iff p \preceq q,$$

is a separative partial order, and $\mathbb{P}$ is forcing equivalent (in the sense defined above) to this separative quotient. The quotient map itself is generally not injective and hence is not being called a dense embedding.

Proof.

Formal derivation spine

Formal spine for the separative quotient.

$$p \preceq q \iff \forall r \leq p \text{ (} r \text{ is compatible with } q \text{)},$$

$$p \equiv q \iff p \preceq q \text{ \& } q \preceq p, \quad [p] \leq_s [q] \iff p \preceq q.$$

Hence

$$[p] \not\leq_s [q] \implies \exists r \leq p \text{ (} r \perp q \text{)},$$

which is exactly the separativity clause in the quotient.

Define $p \preceq q$ to mean that every extension of p is compatible with q , and set

$$p \equiv q \iff p \preceq q \text{ \& } q \preceq p.$$

This is an equivalence relation. Order equivalence classes by $[p] \leq_s [q]$ iff $p \preceq q$. The definition is independent of representatives because replacing either condition by an equivalent one preserves the compatibility cones.

The quotient is separative: if $[p] \not\leq_s [q]$, then some extension $r \leq p$ is incompatible with q , and $[r] \leq_s [p]$ is incompatible with $[q]$. For forcing equivalence, use the definition fixed in the setup: the separative quotient of $\mathbb{P}$ is precisely $\mathbb{P}/\equiv$, while the separative quotient of the already separative order $\mathbb{P}/\equiv$ is canonically order-isomorphic to itself. Taking the whole quotient on each side therefore gives isomorphic dense suborders. The map $p \mapsto [p]$ is merely the quotient map; when distinct conditions are equivalent it is not injective, so it is not a dense embedding under the definition used in this volume. $\square$

Theorem 50.2 (Regular-open Boolean completion of a forcing notion). *For every separative forcing notion $\mathbb{P}$, the regular-open algebra $\text{RO}(\mathbb{P})$ is a complete Boolean algebra and*

$$i(p) := \text{ro}(N_p) = \text{int}(\overline{N_p}), \quad N_p = \{q : q \leq p\},$$

defines a dense embedding $i : \mathbb{P} \rightarrow \text{RO}(\mathbb{P})^+$.

Proof.

Formal derivation spine

Formal spine: regularization supplies the complete Boolean operations, and separativity makes the cone map an embedding. For an open set $U \subseteq \mathbb{P}$, put

$$U^\perp := \text{int}(\mathbb{P} \setminus U).$$

Then

$$\text{ro}(U) = \text{int}(\overline{U}) = (U^\perp)^\perp. \quad (\text{RO})$$

For regular open sets $(U_j)_{j \in J}$,

$$\bigvee_{j \in J} U_j = \text{ro}\left(\bigcup_{j \in J} U_j\right), \quad \bigwedge_{j \in J} U_j = \text{int}\left(\bigcap_{j \in J} U_j\right), \quad \neg U = U^\perp. \quad (\text{BA})$$

Finally

$$i(p) = \text{ro}(N_p), \quad N_p = \{q : q \leq p\}. \quad (\text{DE})$$

We prove the topological and order-theoretic assertions rather than importing the regular-open theorem as a black box.

Step 1: regularization of an open set is regular open. Let $O \subseteq \mathbb{P}$ be open and set

$$R := \text{ro}(O) = \text{int}(\overline{O}).$$

Then $O \subseteq R$, so $\overline{O} \subseteq \overline{R}$ and therefore $R \subseteq \text{int}(\overline{R})$. Conversely, $R \subseteq \overline{O}$, and $\overline{O}$ is closed, so $\overline{R} \subseteq \overline{O}$. Hence

$$\text{int}(\overline{R}) \subseteq \text{int}(\overline{O}) = R.$$

Thus $R = \text{int}(\overline{R})$, so R is regular open.

For an open U , the complement $\mathbb{P} \setminus U$ is closed. Therefore

$$U^\perp = \text{int}(\mathbb{P} \setminus U)$$

is regular open by the same two-inclusion argument: its closure is contained in the closed set $\mathbb{P} \setminus U$. Also

$$(U^\perp)^\perp = \text{int}(\mathbb{P} \setminus \text{int}(\mathbb{P} \setminus U)) = \text{int}(\overline{U}),$$

which proves (RO). Notice that this identity is asserted only for the open sets to which it is applied below; the previously defined topological expression $\text{ro}(A) = \text{int}(\overline{A})$ remains the definition for arbitrary $A \subseteq \mathbb{P}$.

Step 2: arbitrary meets and joins. Let $(U_j)_{j \in J} \subseteq \text{RO}(\mathbb{P})$. Put

$$W := \text{int}\left(\bigcap_{j \in J} U_j\right).$$

Since $W \subseteq U_j$ for every j ,

$$\text{int}(\overline{W}) \subseteq \text{int}(\overline{U_j}) = U_j \quad (j \in J).$$

The left side is open, hence is contained in $\text{int}(\bigcap_j U_j) = W$. The reverse inclusion $W \subseteq \text{int}(\overline{W})$ holds for every open W . Thus W is regular open. If a regular open V satisfies $V \subseteq U_j$ for every j , then V , being open, satisfies

$$V \subseteq \text{int}\left(\bigcap_j U_j\right) = W.$$

Hence W is the greatest lower bound.

For joins put

$$O := \bigcup_{j \in J} U_j, \quad S := \text{ro}(O).$$

Step 1 makes S regular open, and every $U_j \subseteq O \subseteq S$. If a regular open V contains every U_j , then $O \subseteq V$, whence

$$S = \text{int}(\overline{O}) \subseteq \text{int}(\overline{V}) = V.$$

Thus S is the least upper bound. The empty-index cases give $0 = \emptyset$ and $1 = \mathbb{P}$.

For two regular open sets U, V , their intersection is open and

$$\text{int}(\overline{U \cap V}) \subseteq \text{int}(\overline{U}) \cap \text{int}(\overline{V}) = U \cap V,$$

so $U \cap V$ is already regular open. Thus the binary meet is ordinary intersection, as stated in the setup.

Step 3: complements and distributivity. For regular open U , let $U^* = \text{int}(\mathbb{P} \setminus U)$. Step 1 gives $U^* \in \text{RO}(\mathbb{P})$, and clearly

$$U \cap U^* = \emptyset.$$

The open set $U \cup U^*$ is dense: if a nonempty open set W were disjoint from it, then $W \subseteq \mathbb{P} \setminus U$; because W is open this would imply $W \subseteq \text{int}(\mathbb{P} \setminus U) = U^*$, a contradiction. Hence

$$U \vee U^* = \text{ro}(U \cup U^*) = \mathbb{P}.$$

So U^* is the Boolean complement.

It remains to verify finite distributivity. We use the following identity for a regular open U and an open O :

$$U \cap \text{ro}(O) = \text{ro}(U \cap O). \quad (\text{D})$$

The inclusion from right to left follows because $U \cap \text{ro}(O)$ is regular open and contains $U \cap O$. For the reverse inclusion, let $x \in U \cap \text{ro}(O)$. There is an open neighborhood W of x with

$$W \subseteq U \cap \text{ro}(O).$$

The set O is dense in $\text{ro}(O) = \text{int}(\overline{O})$; consequently $W \cap O$ is dense in W . Since $W \subseteq U$, this gives

$$W \subseteq \overline{U \cap O},$$

and hence $x \in \text{int}(\overline{U \cap O})$. This proves (D). Applying (D) to $O = V \cup W$ gives

$$\begin{aligned} U \wedge (V \vee W) &= U \cap \text{ro}(V \cup W) \\ &= \text{ro}((U \cap V) \cup (U \cap W)) \\ &= (U \wedge V) \vee (U \wedge W). \end{aligned}$$

Thus the operations in (BA) make $\text{RO}(\mathbb{P})$ a complete Boolean algebra.

Step 4: the cone map is a dense embedding. For $p \in \mathbb{P}$, the cone N_p is open and nonempty, so $i(p) = \text{ro}(N_p)$ is a nonzero regular open set. If $p \leq q$, then $N_p \subseteq N_q$, hence $i(p) \subseteq i(q)$.

Conversely suppose $p \not\leq q$. Since $\mathbb{P}$ is separative, choose $r \leq p$ with $r \perp q$. No condition below r is compatible with q , so N_q is not dense below any condition in N_r . Equivalently,

$$N_r \cap \text{ro}(N_q) = \emptyset.$$

But $N_r \subseteq N_p \subseteq \text{ro}(N_p)$. Therefore $i(p) \not\subseteq i(q)$. Thus the cone map reflects order; because $\mathbb{P}$ is a partial order, it is injective.

Compatibility is also reflected. If p, q have a common extension r , then $N_r \subseteq i(p) \cap i(q)$, so the meet is nonzero. Conversely, if $s \in i(p) \cap i(q)$, then N_p is dense below s and N_q is dense below s . First choose $t \leq s, p$, and then choose $u \leq t, q$. Hence $u \leq p, q$, so p and q are compatible. The map therefore preserves and reflects incompatibility.

Finally, let $U \in \text{RO}(\mathbb{P})$ be nonzero. Choose $p \in U$. Since U is open, $N_p \subseteq U$; since U is regular open,

$$i(p) = \text{ro}(N_p) \subseteq \text{ro}(U) = U.$$

Thus the range of i is dense in $\text{RO}(\mathbb{P})^+$. We have verified every clause in the definition of dense embedding and every complete-Boolean-algebra operation, completing the proof. $\square$

Proposition 50.3 (Dense embeddings preserve generic extensions). *Let M be a transitive ground model containing forcing notions $\mathbb{P}, \mathbb{Q}$ and a dense embedding $i : \mathbb{P} \rightarrow \mathbb{Q}$. If $H \subseteq \mathbb{Q}$ is M -generic, then $G_H = \{p : i(p) \in H\}$ is M -generic for $\mathbb{P}$; if $G \subseteq \mathbb{P}$ is M -generic, then $H_G = \{q : \exists p \in G (i(p) \leq q)\}$ is M -generic for $\mathbb{Q}$. For every $\mathbb{P}$ -name τ ,*

$$(i_*(\tau))^H = \tau^{G_H}.$$

Consequently the two presentations produce the same valued name hierarchy after the explicitly defined translation i_ .*

Proof.

Formal derivation spine

Formal spine for translating generics and names. If $H \subseteq \mathbb{Q}$ is generic and $i : \mathbb{P} \rightarrow \mathbb{Q}$ is dense, set

$$G_H = \{p \in \mathbb{P} : i(p) \in H\}.$$

Translate names recursively by

$$i_*(\tau) = \{\langle i_*(\sigma), i(p) \rangle : \langle \sigma, p \rangle \in \tau\}.$$

Then the rank induction proves

$$(i_*(\tau))^H = \tau^{G_H},$$

which gives the valuation-preserving correspondence asserted in the proposition.

Let $i : \mathbb{P} \rightarrow \mathbb{Q}$ be a dense embedding. If $H \subseteq \mathbb{Q}$ is generic over M , define

$$G = G_H = \{p \in \mathbb{P} : i(p) \in H\}.$$

Density and order preservation show that G is a filter. If $D \in M$ is dense in $\mathbb{P}$, then the downward closure of $i[D]$ is dense in $\mathbb{Q}$; genericity of H therefore implies $G \cap D \neq \emptyset$. Thus G is $\mathbb{P}$ -generic.

Conversely, if $G \subseteq \mathbb{P}$ is generic, the explicitly defined upward closure $H_G = \{q : \exists p \in G (i(p) \leq q)\}$ is the unique $\mathbb{Q}$ -generic filter generated by the dense image of G . The recursive name translation i_* satisfies $(i_*(\tau))^H = \tau^{G_H}$ by induction on name rank. Thus the two forcing presentations have the same valued names under the displayed correspondence. *What the translation preserves.* Dense embedding does more than send

filters to filters. The recursive translation of names commutes with valuation, so a set constructed from a $\mathbb{P}$ -name has exactly the same extension value as the corresponding $\mathbb{Q}$ -name. Consequently forcing-equivalent posets may look different combinatorially while presenting the same generic universe. $\square$

Proposition 50.4 (Forcing equivalence through Boolean completion). *Let $\mathbb{P}$ be a forcing notion, let $\mathbb{P}_s$ be its separative quotient, and let*

$$i : \mathbb{P}_s \longrightarrow \text{RO}(\mathbb{P}_s)^+$$

be the dense embedding of Theorem 50.2. For a $\mathbb{P}$ -name τ , put $\tau^i := i_(q_{s*}(\tau))$, using the two recursive translations defined in the setup. For every first-order formula $\varphi(\bar{x})$, every tuple of $\mathbb{P}$ -names $\bar{\tau}$, and every condition $p \in \mathbb{P}$,*

$$p \Vdash_{\mathbb{P}} \varphi(\bar{\tau}) \iff i([p]) \Vdash_{\text{RO}(\mathbb{P}_s)} \varphi(\bar{\tau}^i),$$

where $\bar{\tau}^i$ denotes that recursive name translation. Corresponding generic filters therefore yield extension structures related by the valuation-preserving bijection

$$\tau^G \longmapsto (\tau^i)^H.$$

Proof.

Formal derivation spine

Formal spine for forcing equivalence through completion. Let

$$\mathbb{P} \longrightarrow \mathbb{P}_s \xrightarrow{i} \text{RO}(\mathbb{P}_s)^+$$

be the quotient followed by the canonical dense embedding. For the recursively translated name $i_*(\tau)$,

$$p \Vdash_{\mathbb{P}} \varphi(\bar{\tau}) \iff i([p]) \Vdash_{\text{RO}(\mathbb{P}_s)} \varphi(i_*(\bar{\tau})).$$

The equivalence follows because corresponding generic filters give the same valuations of corresponding names.

Apply the dense-embedding theorem to the dense embedding of the separative quotient $\mathbb{P}_s$ into $\text{RO}(\mathbb{P}_s)$. Generics correspond in both directions, and the recursive translation of names preserves valuation. Therefore for every sentence φ and tuple of translated names,

$$p \Vdash_{\mathbb{P}} \varphi$$

holds exactly when the Boolean condition represented by p forces the translated formula in the completion.

Thus passing from a poset to its Boolean completion changes the algebraic language but not the generic extensions or the truths forced in them. This equivalence licenses the Boolean-valued semantics of the next chapter as a reformulation of the ordinary forcing developed in Volume VII. $\square$

Worked example

Worked Example 50.5 (Two forcing-equivalent conditions). If conditions p and q have exactly the same compatible extensions, no generic filter can distinguish their forcing behavior. The separative quotient identifies them. The regular-open algebra then replaces each condition by the regular open set generated by its cone, turning order-theoretic forcing into Boolean algebra.

A useful warning

Dense equivalence is about forcing behavior, not literal equality of posets. Names must be translated along the dense embedding before comparing extensions.

Looking ahead

The next chapter, *Boolean-valued semantics*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 51

Boolean-valued semantics

Definitions and setup

Formal definitions used in this chapter

Boolean-valued name hierarchy and rank. For a complete Boolean algebra $\mathbb{B}$, apply the transfinite-recursion theorem from Volume IV to

$$V_\alpha^\mathbb{B} = \left\{ \tau : \tau \text{ is a function, } \text{dom}(\tau) \subseteq \bigcup_{\beta < \alpha} V_\beta^\mathbb{B}, \text{ran}(\tau) \subseteq \mathbb{B} \right\}.$$

The recursion supplies a unique hierarchy. Define

$$V^\mathbb{B} := \bigcup_{\alpha \in \text{Ord}} V_\alpha^\mathbb{B},$$

and call τ a *Boolean name* exactly when $\tau \in V^\mathbb{B}$. For a Boolean name τ , define its *Boolean-name rank*

$$\text{rk}(\tau) := \min\{\alpha : \tau \in V_{\alpha+1}^\mathbb{B}\}.$$

The least ordinal exists because a name occurs at some stage and the ordinals are well ordered. Equivalently,

$$\text{rk}(\tau) = \sup\{\text{rk}(\rho) + 1 : \rho \in \text{dom}(\tau)\}. \quad (\text{BR})$$

Thus every member of $\text{dom}(\tau)$ has strictly smaller Boolean-name rank. Theorem 51.1 below records the hierarchy construction before the atomic truth-value recursion uses this rank as its well-founded measure.

Boolean check names. Set-rank recursion defines, for every ground-model set x , a Boolean name

$$\check{x}^\mathbb{B} := \{\langle \check{y}^\mathbb{B}, 1_\mathbb{B} \rangle : y \in x\}. \quad (\text{CK})$$

When $\mathbb{B}$ is fixed, write simply $\check{x}$. This is the Boolean-coefficient analogue of the ordinary forcing check name from Volume VII. Lemma 51.4 below proves before any load-bearing use that these names reproduce ground-model equality and membership with Boolean values $0_\mathbb{B}$ and $1_\mathbb{B}$.

Boolean operations and implication. $\neg b, b \wedge c, b \vee c$ are Boolean complement, meet, and join, and

$$a \rightarrow b := \neg a \vee b.$$

For any set $A \subseteq \mathbb{B}$, $\bigvee A$ and $\bigwedge A$ are its join and meet.

Atomic Boolean truth values.

$$\|\sigma \in \tau\| = \bigvee_{\rho \in \text{dom } \tau} \tau(\rho) \wedge \|\sigma = \rho\|,$$

$$\|\sigma = \tau\| = \left(\bigwedge_{\rho \in \text{dom } \sigma} (\sigma(\rho) \rightarrow \|\rho \in \tau\|) \right) \wedge \left(\bigwedge_{\rho \in \text{dom } \tau} (\tau(\rho) \rightarrow \|\rho \in \sigma\|) \right).$$

Boolean semantics for formulas. Extend recursively by

$$\|\neg\varphi\| = \neg\|\varphi\|, \quad \|\varphi \wedge \psi\| = \|\varphi\| \wedge \|\psi\|,$$

For fixed φ and $\bar{\tau}$, first form by Separation on the set $\mathbb{B}$ the range set

$$S_{\varphi, \bar{\tau}} := \{b \in \mathbb{B} : \exists \sigma (\sigma \text{ is a Boolean name} \wedge b = \|\varphi(\sigma, \bar{\tau})\|)\}.$$

Then define

$$\|\exists x \varphi(x, \bar{\tau})\| := \bigvee S_{\varphi, \bar{\tau}}.$$

Thus the join is explicitly taken over a set subset of $\mathbb{B}$; the proper class of Boolean names is used only inside the defining predicate for that subset. Universal quantification is defined by negation.

Complete Boolean-valued structure. A first-order structure is Boolean valued when every formula/parameter tuple receives a value in a complete Boolean algebra according to these recursive clauses; value $1_{\mathbb{B}}$ plays the role of full truth.

Boolean filters, ultrafilters, and genericity. A *proper Boolean filter* on $\mathbb{B}$ is a set $U \subseteq \mathbb{B}$ such that

$$1_{\mathbb{B}} \in U, \quad 0_{\mathbb{B}} \notin U,$$

$$a, b \in U \implies a \wedge b \in U, \quad (a \in U \ \& \ a \leq b) \implies b \in U.$$

An *ultrafilter* is a proper Boolean filter satisfying the dichotomy

$$\forall b \in \mathbb{B} \quad (b \in U \text{ or } \neg b \in U).$$

If M is a ground model containing $\mathbb{B}$, an ultrafilter U is *M-generic* when

$$U \cap D \neq \emptyset$$

for every $D \in M$ which M regards as a dense subset of the forcing poset $\mathbb{B}^+ = \mathbb{B} \setminus \{0_{\mathbb{B}}\}$, ordered by the Boolean order.

Quotient construction deferred until equality is proved. The ultrafilter will later turn Boolean equality into an ordinary equivalence relation. We do not yet write $[\tau]_U$: Theorem 51.2 and Lemma 51.3 first prove the Boolean equality laws, and Proposition 51.5 then constructs a set-sized Scott representative for each quotient class and proves quotient membership well defined.

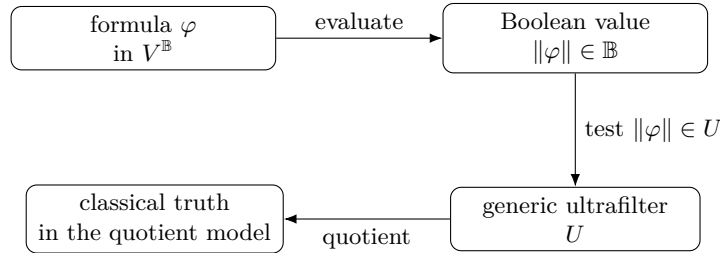


Figure 51.1: Boolean-valued semantics first assigns a Boolean element to a formula. After the Boolean equality laws are proved, the Scott-quotient construction below lets a generic ultrafilter convert that graded value into ordinary two-valued truth.

A Boolean-valued universe $V^{\mathbb{B}}$ consists of names whose membership coefficients lie in a complete Boolean algebra $\mathbb{B}$. Instead of a formula being simply true or false, it receives a Boolean truth value $\|\varphi\| \in \mathbb{B}$. Equality

and membership are defined simultaneously by rank recursion; connectives use Boolean operations and quantifiers use joins/meets. Truth value 1 plays the role of validity.

How this chapter fits together

A Boolean-valued universe assigns each atomic assertion a truth value in a complete Boolean algebra rather than merely 0 or 1. Equality and membership are again defined simultaneously by rank recursion; logical connectives become Boolean operations, and quantifiers become joins or meets. Ordinary two-valued semantics will reappear after the Scott quotient by a generic ultrafilter is constructed below.

Theorem 51.1 (Construction of the Boolean-valued universe $V^{\mathbb{B}}$). *For a complete Boolean algebra $\mathbb{B}$, define by recursion*

$$V_{\alpha}^{\mathbb{B}} = \{\tau : \text{dom}(\tau) \subseteq \bigcup_{\beta < \alpha} V_{\beta}^{\mathbb{B}}, \text{ran}(\tau) \subseteq \mathbb{B}\},$$

and $V^{\mathbb{B}} = \bigcup_{\alpha} V_{\alpha}^{\mathbb{B}}$.

Proof.

Formal derivation spine

Formal spine for Construction of the Boolean-valued universe $V^{\mathbb{B}}$.

$$V_{\alpha}^{\mathbb{B}} = \{\tau : \tau \text{ is a function, } \text{ran } \tau \subseteq \mathbb{B}, \text{dom } \tau \subseteq \bigcup_{\beta < \alpha} V_{\beta}^{\mathbb{B}}\}.$$

$$\|\sigma \in \tau\| = \bigvee_{\rho \in \text{dom } \tau} \tau(\rho) \wedge \|\sigma = \rho\|.$$

For a complete Boolean algebra $\mathbb{B}$, define by transfinite recursion

$$V_{\alpha}^{\mathbb{B}} = \{\tau : \tau \text{ is a function, } \text{ran } \tau \subseteq \mathbb{B}, \text{dom } \tau \subseteq \bigcup_{\beta < \alpha} V_{\beta}^{\mathbb{B}}\}.$$

A Boolean name therefore assigns to each possible lower-rank member a Boolean coefficient measuring the degree to which that member belongs. Set $V^{\mathbb{B}} = \bigcup_{\alpha} V_{\alpha}^{\mathbb{B}}$.

The recursion is set-sized at every stage. The rank defined in the setup satisfies

$$\text{rk}(\tau) = \sup\{\text{rk}(\rho) + 1 : \rho \in \text{dom}(\tau)\},$$

so every recursive call through the domain relation strictly decreases rank. This is the well-founded measure used in the simultaneous definitions of Boolean membership and equality.

The function presentation is equivalent to a set-of-pairs presentation of *Boolean* names: a pair $\langle \sigma, b \rangle$ records the coefficient $b = \tau(\sigma)$. Ordinary $\mathbb{P}$ -forcing names are not literally Boolean names unless the forcing conditions have first been represented in a Boolean algebra. After passing to a Boolean completion $i : \mathbb{P} \rightarrow \mathbb{B}^+$, the dense-embedding translation from the preceding chapter converts ordinary forcing names into this Boolean-coefficient presentation. Using functions is convenient because Boolean joins can combine multiple contributions to the same potential member into a single coefficient. $\square$

Theorem 51.2 (Boolean truth values for equality and membership). *For $\sigma, \tau \in V^{\mathbb{B}}$, define simultaneously*

$$\begin{aligned} \|\sigma \in \tau\| &= \bigvee_{\rho \in \text{dom } \tau} \tau(\rho) \wedge \|\sigma = \rho\|, \\ \|\sigma = \tau\| &= \left(\bigwedge_{\rho \in \text{dom } \sigma} (\sigma(\rho) \rightarrow \|\rho \in \tau\|) \right) \wedge \left(\bigwedge_{\rho \in \text{dom } \tau} (\tau(\rho) \rightarrow \|\rho \in \sigma\|) \right). \end{aligned}$$

How to read the symbols: Boolean truth values are graded truth certificates

The expression $\|\sigma \in \tau\|$ is an element of the Boolean algebra $\mathbb{B}$, not the classical number 0 or 1 in general. Its join runs over possible witnesses ρ in the domain of τ ; each term combines the Boolean weight with which τ contains ρ and the Boolean degree to which $\sigma = \rho$. Classical truth is recovered after quotienting by a generic ultrafilter.

Proof. Before carrying out the simultaneous recursion, introduce the *auxiliary Boolean inclusion value*

$$\|\sigma \subseteq \tau\|_{\text{aux}} := \bigwedge_{\rho \in \text{dom } \sigma} (\neg\sigma(\rho) \vee \|\rho \in \tau\|). \quad (\text{AUXB})$$

At this point (AUXB) is an auxiliary rank-recursive quantity used to define Boolean equality; it has not yet been identified with the Boolean semantic value of the pure set-theoretic formula $\forall x (x \in \sigma \rightarrow x \in \tau)$. Corollary 52.6, proved from the bounded-quantifier formulas before the Power Set theorem, establishes that identification. Until then the atomic equality development writes the auxiliary coefficient explicitly as $\|\sigma \subseteq \tau\|_{\text{aux}}$; only after the corollary is the subscript suppressed.

Formal derivation spine

Formal spine: simultaneous recursion in the product of the two name ranks.

$$\|\sigma \in \tau\| = \bigvee_{\rho \in \text{dom } \tau} \tau(\rho) \wedge \|\sigma = \rho\|,$$

$$\|\sigma = \tau\| = \|\sigma \subseteq \tau\|_{\text{aux}} \wedge \|\tau \subseteq \sigma\|_{\text{aux}}.$$

Every recursive call replaces at least one name by a constituent of strictly smaller rank.

Order pairs of names by the well-founded product relation

$$(\sigma', \tau') \prec (\sigma, \tau)$$

when

$$\text{rk}(\sigma') \leq \text{rk}(\sigma), \quad \text{rk}(\tau') \leq \text{rk}(\tau),$$

with at least one strict inequality. In the membership clause, $\rho \in \text{dom}(\tau)$ has $\text{rk}(\rho) < \text{rk}(\tau)$, so $(\sigma, \rho) \prec (\sigma, \tau)$. In the inclusion clause, a constituent ρ of σ gives a call at $(\rho, \tau) \prec (\sigma, \tau)$. Thus simultaneous recursion on $\prec$ defines

$$\|\sigma \in \tau\| = \bigvee_{\rho \in \text{dom } \tau} (\tau(\rho) \wedge \|\sigma = \rho\|) \quad (1)$$

and

$$\|\sigma \subseteq \tau\|_{\text{aux}} = \bigwedge_{\rho \in \text{dom } \sigma} (\neg\sigma(\rho) \vee \|\rho \in \tau\|), \quad \|\sigma = \tau\| = \|\sigma \subseteq \tau\|_{\text{aux}} \wedge \|\tau \subseteq \sigma\|_{\text{aux}}. \quad (2)$$

Completeness of $\mathbb{B}$ supplies the joins and meets. Uniqueness follows by taking a $\prec$ -least pair where two proposed definitions differ; all right-hand recursive values then agree, so (1)–(2) assign the same value there, a contradiction.

A second induction on the same product rank proves

$$\|\sigma = \sigma\| = 1$$

and the substitutivity inequalities required by first-order equality. The use of the product relation is important: ordering only by the maximum of the two ranks would not make every equality call strictly smaller when the unchanged name has the larger rank. $\square$

Lemma 51.3 (Boolean equality laws and atomic substitutivity). *For Boolean names $\sigma, \tau, \rho, \sigma', \tau'$, the recursively defined Boolean equality and membership values satisfy*

$$\begin{aligned} \|\sigma = \sigma\| &= 1_{\mathbb{B}}, & \|\sigma = \tau\| &= \|\tau = \sigma\|, \\ \|\sigma = \tau\| \wedge \|\tau = \rho\| &\leq \|\sigma = \rho\|, \end{aligned} \tag{E}$$

and

$$\|\sigma = \sigma'\| \wedge \|\tau = \tau'\| \wedge \|\sigma \in \tau\| \leq \|\sigma' \in \tau'\|. \tag{S}$$

Thus Boolean equality has the reflexivity, symmetry, transitivity, and atomic substitution properties required of first-order equality.

Proof. Reflexivity is proved by induction on name rank. Suppose reflexivity has been proved for every constituent η of σ . For each $\eta \in \text{dom}(\sigma)$, the membership join contains the summand

$$\sigma(\eta) \wedge \|\eta = \eta\| = \sigma(\eta),$$

so $\sigma(\eta) \leq \|\eta \in \sigma\|$. Every implication appearing in $\|\sigma \subseteq \sigma\|_{\text{aux}}$ therefore has value 1, and hence $\|\sigma = \sigma\| = 1$. Symmetry is immediate from the definition $\|\sigma = \tau\| = \|\sigma \subseteq \tau\|_{\text{aux}} \wedge \|\tau \subseteq \sigma\|_{\text{aux}}$.

For transitivity and substitution, perform a simultaneous induction on the finite tuple of name ranks, ordered so that replacing any displayed name by one of its constituents decreases the induction measure. The two key recursive calculations are as follows. First, meeting $\|\sigma = \tau\|$ with a coefficient $\sigma(\eta)$ gives

$$\|\sigma = \tau\| \wedge \sigma(\eta) \leq \|\eta \in \tau\|. \tag{1}$$

Meeting (1) with $\|\tau = \rho\|$ and applying the already-established lower-rank instance of membership substitution gives

$$\|\sigma = \tau\| \wedge \|\tau = \rho\| \wedge \sigma(\eta) \leq \|\eta \in \rho\|.$$

Taking the meet over all constituents η proves $\|\sigma = \tau\| \wedge \|\tau = \rho\| \leq \|\sigma \subseteq \rho\|_{\text{aux}}$; applying the same argument in the reverse direction proves the other inclusion and hence (E).

Second, expand a membership value as

$$\|\sigma \in \tau\| = \bigvee_{\eta \in \text{dom}(\tau)} \tau(\eta) \wedge \|\sigma = \eta\|. \tag{2}$$

For substitution in the first argument, meet each summand in (2) with $\|\sigma = \sigma'\|$ and use the lower-rank transitivity inequality (E) to replace $\|\sigma = \eta\|$ by $\|\sigma' = \eta\|$. For substitution in the second argument, meet each summand with $\|\tau = \tau'\|$; equation (1) sends its coefficient into $\|\eta \in \tau'\|$, and expanding that membership value together with the lower-rank transitivity inequality sends the whole summand below $\|\sigma \in \tau'\|$. Combining the two one-variable substitutions yields (S). Every appeal made in these calculations is to a tuple with at least one constituent of smaller rank, so the simultaneous induction is well founded. $\square$

Lemma 51.4 (Boolean check names reproduce ground-model membership). *For ground-model sets x, y ,*

$$\|\check{x} = \check{y}\| = \begin{cases} 1_{\mathbb{B}}, & x = y, \\ 0_{\mathbb{B}}, & x \neq y, \end{cases} \quad \|\check{x} \in \check{y}\| = \begin{cases} 1_{\mathbb{B}}, & x \in y, \\ 0_{\mathbb{B}}, & x \notin y. \end{cases} \tag{CK1}$$

In particular, the map $x \mapsto \check{x}$ is injective at Boolean truth value $1_{\mathbb{B}}$.

Proof. Use simultaneous induction on the ordinary ranks of the ground-model sets involved. The Boolean check name has constant coefficient $1_{\mathbb{B}}$ on the names of its members, so

$$\|\check{x} \in \check{y}\| = \bigvee_{z \in y} \|\check{x} = \check{z}\|. \tag{1}$$

By the lower-rank equality induction hypothesis, every term in this join is $1_{\mathbb{B}}$ exactly when $x = z$, and otherwise is $0_{\mathbb{B}}$. Hence (1) is $1_{\mathbb{B}}$ exactly when $x \in y$, proving the membership half of (CK1).

For equality, the atomic definition gives

$$\|\check{x} = \check{y}\| = \left(\bigwedge_{z \in x} \|\check{z} \in \check{y}\| \right) \wedge \left(\bigwedge_{z \in y} \|\check{z} \in \check{x}\| \right), \quad (2)$$

because every coefficient in a check name is $1_{\mathbb{B}}$. The membership part just proved shows that the first meet is $1_{\mathbb{B}}$ exactly when $x \subseteq y$, and the second exactly when $y \subseteq x$. Thus (2) is $1_{\mathbb{B}}$ exactly when $x = y$. If $x \neq y$, Extensionality in the ground universe gives an element of the symmetric difference, so one factor in (2) is $0_{\mathbb{B}}$; hence the equality value is $0_{\mathbb{B}}$. This completes the simultaneous induction. $\square$

Proposition 51.5 (Ultrafilter quotient is well defined; Scott representatives). *Let U be an ultrafilter on $\mathbb{B}$. Define a relation on Boolean names by*

$$\sigma \equiv_U \tau \iff \|\sigma = \tau\| \in U.$$

Then $\equiv_U$ is an equivalence relation. For a name τ , let $r_U(\tau)$ be the least name-rank attained by a name U -equivalent to τ , and define its Scott quotient representative

$$[\tau]_U := \{\sigma : \text{rk}(\sigma) = r_U(\tau) \text{ and } \sigma \equiv_U \tau\}. \quad (Q)$$

This is a set, and

$$[\sigma]_U = [\tau]_U \iff \sigma \equiv_U \tau.$$

Moreover the relation

$$[\sigma]_U \in_U [\tau]_U \iff \|\sigma \in \tau\| \in U \quad (M)$$

is independent of the chosen representatives. From this proposition onward $V^{\mathbb{B}}/U$ denotes the class structure whose objects are the Scott representatives $[\tau]_U$ and whose membership relation is (M).

Proof. Reflexivity and symmetry of $\equiv_U$ follow from Lemma 51.3 and $1_{\mathbb{B}} \in U$. If $\sigma \equiv_U \tau$ and $\tau \equiv_U \rho$, then the filter property gives

$$\|\sigma = \tau\| \wedge \|\tau = \rho\| \in U.$$

By (E) of Lemma 51.3, this meet is below $\|\sigma = \rho\|$; upward closure of U therefore gives $\sigma \equiv_U \rho$. Hence $\equiv_U$ is an equivalence relation.

The class of ranks of names equivalent to τ is nonempty because it contains $\text{rk}(\tau)$. To obtain its least member set-theoretically, it is enough to look at the set of ordinals $\beta \leq \text{rk}(\tau)$ for which some Boolean name of rank β is equivalent to τ ; Separation gives this nonempty set, so it has a least element $r_U(\tau)$. The names of one fixed rank form a set inside the stage hierarchy $V_{r_U(\tau)+1}^{\mathbb{B}}$; Separation therefore makes (Q) a set. If $\sigma \equiv_U \tau$, the two names have exactly the same equivalence class and hence the same least attained rank, so their Scott representatives are equal. Conversely, equal Scott representatives are nonempty; a common member and transitivity of $\equiv_U$ imply $\sigma \equiv_U \tau$.

Finally suppose $\sigma \equiv_U \sigma'$, $\tau \equiv_U \tau'$, and $\|\sigma \in \tau\| \in U$. The filter contains the meet

$$\|\sigma = \sigma'\| \wedge \|\tau = \tau'\| \wedge \|\sigma \in \tau\|.$$

By (S) of Lemma 51.3, that meet lies below $\|\sigma' \in \tau'\|$, so upward closure puts $\|\sigma' \in \tau'\|$ in U . Symmetry gives the reverse implication. Thus (M) is well defined on Scott representatives. $\square$

Theorem 51.6 (Boolean semantics for first-order formulas). *Let $\mathbb{B}$ be a complete Boolean algebra. After the atomic Boolean values are defined, extend them recursively to all first-order formulas by*

$$\|\neg\varphi\| = \neg\|\varphi\|, \quad \|\varphi \wedge \psi\| = \|\varphi\| \wedge \|\psi\|,$$

and

$$\|\exists x \varphi(x, \bar{\tau})\| = \bigvee S_{\varphi, \bar{\tau}}, \quad S_{\varphi, \bar{\tau}} = \{b \in \mathbb{B} : \exists \sigma (b = \|\varphi(\sigma, \bar{\tau})\|)\}.$$

The set $S_{\varphi, \bar{\tau}} \subseteq \mathbb{B}$ exists by Separation in the metatheory, so the join exists by completeness of $\mathbb{B}$. Universal quantification is defined by

$$\|\forall x \varphi(x, \bar{\tau})\| = \neg \|\exists x \neg \varphi(x, \bar{\tau})\|.$$

These clauses assign a unique Boolean truth value to every first-order formula and tuple of Boolean names.

Proof.

Formal derivation spine

Formal spine for Boolean semantics for first-order formulas.

$$\|\neg \varphi\| = \neg \|\varphi\|, \quad \|\varphi \wedge \psi\| = \|\varphi\| \wedge \|\psi\|.$$

$$S_{\varphi, \bar{\tau}} = \{b \in \mathbb{B} : \exists \sigma (\sigma \text{ is a Boolean name and } b = \|\varphi(\sigma, \bar{\tau})\|)\}.$$

$$\bigvee S_{\varphi, \bar{\tau}}$$

Proceed by induction on formula complexity, assuming the atomic values $\|\sigma \in \tau\|$ and $\|\sigma = \tau\|$ have already been defined. For Boolean connectives use the corresponding Boolean-algebra operations. Thus

$$\|\neg \varphi\| = \neg \|\varphi\|, \quad \|\varphi \wedge \psi\| = \|\varphi\| \wedge \|\psi\|.$$

These clauses are set-theoretic operations on elements of $\mathbb{B}$, so there is no size issue.

Now fix a formula $\exists x \varphi(x, \bar{\tau})$. By the induction hypothesis, for every Boolean name σ the value $\|\varphi(\sigma, \bar{\tau})\|$ is already an element of $\mathbb{B}$. Instead of writing an unexplained proper-class join, define

$$S_{\varphi, \bar{\tau}} = \{b \in \mathbb{B} : \exists \sigma (\sigma \text{ is a Boolean name and } b = \|\varphi(\sigma, \bar{\tau})\|)\}.$$

This is a definable subclass of the set $\mathbb{B}$; hence Separation in the ambient metatheory makes it a set. Because $\mathbb{B}$ is complete, the supremum

$$\bigvee S_{\varphi, \bar{\tau}}$$

exists in $\mathbb{B}$. We take this supremum as the Boolean value of the existential formula. Universal quantification is then defined by De Morgan duality,

$$\|\forall x \varphi(x, \bar{\tau})\| = \neg \|\exists x \neg \varphi(x, \bar{\tau})\|,$$

or equivalently as the meet of the set of attained instance values.

Uniqueness follows from the recursion on formula complexity: once the values of proper subformulas are fixed, each displayed clause determines the next value uniquely. Induction also proves invariance under Boolean-equal parameters, using the substitutivity theorem for the atomic equality relation. Finally, when $\mathbb{B} = \{0, 1\}$, joins and meets are ordinary existential and universal truth values, so the construction reduces to ordinary two-valued semantics.

The crucial size point is therefore explicit: quantification ranges over all names at the semantic level, but the values being joined live in the set $\mathbb{B}$. We take the join of the set of *attained Boolean values*, not a literal proper-class family. $\square$

Theorem 51.7 (Boolean soundness of first-order deduction). *Let Σ be a set of sentences and let φ be a sentence. If $\Sigma \models \varphi$ in first-order logic, then in every complete Boolean-valued structure*

$$\bigwedge_{\psi \in \Sigma} \|\psi\| \leq \|\varphi\|.$$

In particular, if every premise has Boolean value 1, then so does the conclusion.

Proof.

Formal derivation spine

Formal spine: finite consequence gives the Boolean inequality. Choose finite $\Gamma \subseteq \Sigma$ with $\Gamma \models \varphi$, and put

$$c := \bigwedge_{\psi \in \Gamma} \|\psi\|.$$

Boolean soundness relative to c gives $c \leq \|\varphi\|$; since

$$\bigwedge_{\psi \in \Sigma} \|\psi\| \leq c,$$

the desired inequality follows.

By first-order compactness, $\Sigma \models \varphi$ implies that some finite set

$$\Gamma = \{\psi_1, \dots, \psi_n\} \subseteq \Sigma$$

already satisfies $\Gamma \models \varphi$. Completeness of first-order deduction gives a formal derivation of φ from Γ . Let

$$c := \bigwedge_{i=1}^n \|\psi_i\|.$$

We prove by induction on that derivation that every derived formula θ satisfies $c \leq \|\theta\|$. For a premise ψ_i , this is the definition of a meet. Logical and equality axioms have value 1, so $c \leq 1 = \|\theta\|$. For modus ponens, the induction hypotheses give

$$c \leq \|\alpha\|, \quad c \leq \|\alpha \rightarrow \beta\| = \neg\|\alpha\| \vee \|\beta\|.$$

Meeting the two inequalities and using Boolean distributivity yields $c \leq \|\beta\|$. For universal generalization, suppose a derivation line $\forall x \alpha$ is obtained from α with the Volume-I eigenvariable restriction: x is not free in any undischarged premise. Since the premises here are sentences, the fixed value c is independent of the value assigned to x . The induction hypothesis therefore gives $c \leq \|\alpha(\sigma)\|$ for every Boolean name σ ; taking the meet over all attained instance values gives

$$c \leq \|\forall x \alpha\|.$$

The quantifier-axiom instances have Boolean value 1 by the existential-join/universal-meet clauses and substitutivity of Boolean equality. In particular, the primitive existential-duality axiom of the fixed Volume-I calculus is checked explicitly rather than absorbed into that sentence. For every formula ψ , complete-Boolean De Morgan duality gives

$$\begin{aligned} \|\neg \forall x \neg \psi\| &= \neg \bigwedge_{\sigma} \|\neg \psi(\sigma)\| \\ &= \neg \bigwedge_{\sigma} \neg \|\psi(\sigma)\| \\ &= \bigvee_{\sigma} \|\psi(\sigma)\| \\ &= \|\exists x \psi\|, \end{aligned} \tag{BED}$$

where the meet and join range over the set of attained Boolean values, as established in the preceding theorem. Hence the (BI)-expanded biconditional $\exists x \psi \leftrightarrow \neg \forall x \neg \psi$ has value $1_{\mathbb{B}}$. Thus every logical-axiom constructor and every inference rule of the explicitly fixed Volume-I calculus preserves the required inequality. Therefore

$$c \leq \|\varphi\|. \tag{1}$$

Finally,

$$\bigwedge_{\psi \in \Sigma} \|\psi\| \leq \bigwedge_{\psi \in \Gamma} \|\psi\| = c.$$

Together with (1) this proves

$$\bigwedge_{\psi \in \Sigma} \|\psi\| \leq \|\varphi\|.$$

When every premise has value 1, the left side is 1, so $\|\varphi\| = 1$. Thus the proof establishes the full inequality stated in the theorem, not merely the value-one special case. $\square$

Worked example

Worked Example 51.8 (A three-valued-looking picture). If $b \in \mathbb{B}$ is neither 0 nor 1, a name may satisfy $\|\check{0} \in \tau\| = b$. The assertion is not simply true or false before a generic ultrafilter is chosen. Once an ultrafilter contains b , the quotient interprets the assertion as true; if it contains $\neg b$, it becomes false.

A useful warning

Boolean truth values live in a complete algebra. Infinite quantifiers require joins and meets, so completeness is not optional bookkeeping.

Looking ahead

The next chapter, *Boolean-valued ZFC*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 52

Boolean-valued ZFC

Definitions and setup

Formal definitions used in this chapter

Validity with value one. A sentence σ is valid in the Boolean-valued universe when $\|\sigma\| = 1_{\mathbb{B}}$. An axiom scheme has value one when every instance does.

Bounded quantifier abbreviations.

$$\forall x \in a \varphi := \forall x(x \in a \rightarrow \varphi), \quad \exists x \in a \varphi := \exists x(x \in a \wedge \varphi).$$

Their Boolean values are computed using the implication, meet, join, and quantifier clauses fixed in the preceding chapter.

Boolean-valued Collection/Replacement. Collection with value one means that if $\|\forall x \in a \exists y \varphi(x, y, \bar{p})\| = 1$, some name b satisfies $\|\forall x \in a \exists y \in b \varphi(x, y, \bar{p})\| = 1$. The formula $\varphi(x, y, \bar{p})$ is *functional on a with value one* when

$$\|\forall x \in a \exists! y \varphi(x, y, \bar{p})\| = 1_{\mathbb{B}}.$$

Boolean-valued Replacement is this functional special case.

Boolean-valued Power Set. A power-set name P_a witnesses value one for $\forall y(y \in P_a \leftrightarrow y \subseteq a)$, where $y \subseteq a$ abbreviates $\forall z(z \in y \rightarrow z \in a)$.

Relation and poset notation used in the Choice proof. Ordered pairs are the Kuratowski pairs fixed in Volume IV. For set-theoretic relation names R, X , the notation

$$R(x, y) \text{ abbreviates } \langle x, y \rangle \in R.$$

The assertion that (X, R) is a *partially ordered set* means

$$R \subseteq X \times X,$$

together with reflexivity on X , transitivity on X , and antisymmetry on X . A set D is an *R -chain in X* when $D \subseteq X$ and every two members of D are R -comparable. An element $u \in X$ is an *upper bound* of D when $R(x, u)$ for every $x \in D$. A poset is *inductive* when it is nonempty and every chain has an upper bound. Thus every occurrence below of $\langle X, R \rangle$ is a nonempty inductive poset is an abbreviation for this first-order set-theoretic formula, not an additional predicate symbol.

Compatibility, maximal antichain, mixing, and maximum. Nonzero values $a, b \in \mathbb{B}^+$ are *compatible* when $a \wedge b \neq 0_{\mathbb{B}}$. A set $A \subseteq \mathbb{B}^+$ is an *antichain* when $a \wedge b = 0_{\mathbb{B}}$ for distinct $a, b \in A$, and is *maximal* when $\bigvee A = 1_{\mathbb{B}}$. More generally, if $0 < b \in \mathbb{B}$, an antichain $A \subseteq \mathbb{B}^+$ is *maximal below b* when every member of A is below b and $\bigvee A = b$. *Mixing* means combining names τ_a along an antichain into

one name τ with $a \leq \|\tau = \tau_a\|$ for every $a \in A$. The *maximum principle* is the assertion that for every formula $\varphi(x, \bar{\sigma})$ there is a name τ with

$$\|\exists x \varphi(x, \bar{\sigma})\| = \|\varphi(\tau, \bar{\sigma})\|.$$

Metatheoretic Choice convention for the maximum principle. The maximum-principle proof below is carried out in the ambient ZFC metatheory. Zorn's lemma is used to obtain an antichain maximal below a Boolean value, and ambient Collection plus Choice is used to select a set-indexed family of witness names. This is an external metatheoretic assumption; it is not the assertion that the Boolean-valued universe has already been proved to satisfy Choice.

Lemma 52.1 (Fixed meets preserve set-indexed joins). *Let $\mathbb{B}$ be a complete Boolean algebra, let $c \in \mathbb{B}$, and let $S \subseteq \mathbb{B}$ be a set. Then*

$$c \wedge \bigvee S = \bigvee_{s \in S} (c \wedge s). \quad (\text{D})$$

This is the ordinary Boolean distributive law extended to a set-indexed join by completeness; it is not the stronger property usually called complete distributivity.

Proof. Put

$$u := \bigvee_{s \in S} (c \wedge s).$$

Every summand is at most $c \wedge \bigvee S$, so

$$u \leq c \wedge \bigvee S.$$

For the reverse inequality, let $d \in \mathbb{B}$ be any upper bound of all $c \wedge s$ with $s \in S$. For every $s \in S$,

$$s = (c \wedge s) \vee (\neg c \wedge s) \leq d \vee \neg c.$$

Hence

$$\bigvee S \leq d \vee \neg c,$$

and meeting with c gives

$$c \wedge \bigvee S \leq c \wedge (d \vee \neg c) = c \wedge d \leq d.$$

Thus $c \wedge \bigvee S$ is itself the least upper bound of the family $\{c \wedge s : s \in S\}$, proving (D). $\square$

Theorem 52.2 (Mixing and maximum principles in Boolean-valued form). *Let $\mathbb{B}$ be a complete Boolean algebra. The mixing clause is purely algebraic once the antichain-indexed family of names is given. For the maximum clause, work in the ambient ZFC metatheory as specified above.*

(Mixing.) *If $A \subseteq \mathbb{B}^+$ is an antichain and $\tau_a \in V^{\mathbb{B}}$ for $a \in A$, then there is a Boolean name τ such that*

$$a \leq \|\tau = \tau_a\| \quad (a \in A).$$

(Maximum.) *For every formula $\varphi(x, \bar{y})$ and every tuple of Boolean names $\bar{\sigma}$, there is a Boolean name τ such that*

$$\|\exists x \varphi(x, \bar{\sigma})\| = \|\varphi(\tau, \bar{\sigma})\|.$$

Commented symbolic trace: mixing turns local witnesses into one name

An antichain separates mutually incompatible Boolean cases. On the piece a , the mixed name is arranged to agree with τ_a . For the maximum principle, first choose an antichain of local witness values whose join is the full existential value b ; mixing then turns those local witnesses into one global name whose truth value is exactly b .

Proof.

Formal derivation spine

Formal spine. For mixing, put

$$D := \bigcup_{a \in A} \text{dom}(\tau_a), \quad \tau(\rho) := \bigvee \{a \wedge \tau_a(\rho) : a \in A, \rho \in \text{dom}(\tau_a)\} \quad (\rho \in D).$$

Then $a \leq \|\tau = \tau_a\|$ for every $a \in A$. For maximum, with

$$b := \|\exists x \varphi(x, \bar{\sigma})\|,$$

choose an antichain A of local witness values below b with $\bigvee A = b$, mix their witnesses, and obtain

$$b \leq \|\varphi(\tau, \bar{\sigma})\| \leq b.$$

For mixing, let

$$D = \bigcup_{a \in A} \text{dom}(\tau_a).$$

For $\rho \in D$, define

$$\tau(\rho) = \bigvee \{a \wedge \tau_a(\rho) : a \in A, \rho \in \text{dom}(\tau_a)\}. \quad (1)$$

(The omitted coefficients are simply $0_{\mathbb{B}}$.) Fix $a_0 \in A$. Because distinct elements of A are disjoint, meeting (1) with a_0 kills every summand except the a_0 -summand. Thus, below a_0 , the membership coefficients of τ agree with those of τ_{a_0} . Induction on the ranks of the constituent names in the simultaneous Boolean definitions of membership and equality therefore gives

$$a_0 \leq \|\tau = \tau_{a_0}\|.$$

Since a_0 was arbitrary, this proves mixing.

For maximum, set

$$b := \|\exists x \varphi(x, \bar{\sigma})\|.$$

If $b = 0_{\mathbb{B}}$, then every witness value is $0_{\mathbb{B}}$, so any Boolean name τ works. Assume $b > 0_{\mathbb{B}}$. Let

$$D_b := \{d \in \mathbb{B}^+ : d \leq b \text{ and for some name } \rho, d \leq \|\varphi(\rho, \bar{\sigma})\|\}.$$

This set is dense below b . Let

$$S := S_{\varphi, \bar{\sigma}} = \{d \in \mathbb{B} : \exists \rho (d = \|\varphi(\rho, \bar{\sigma})\|)\},$$

the set of attained witness values from Theorem 51.6. By definition $b = \bigvee S$. If $0 < c \leq b$, Lemma 52.1 applies to the fixed meet with c and the set-indexed join $b = \bigvee S$, giving

$$c = c \wedge b = c \wedge \bigvee_{d \in S} d = \bigvee_{d \in S} (c \wedge d).$$

Since $c > 0$, some $d \in S$ satisfies $c \wedge d > 0$. Choose a Boolean name ρ witnessing $d = \|\varphi(\rho, \bar{\sigma})\|$, and put

$$e := c \wedge d > 0.$$

Then $e \in D_b$ and $e \leq c$. Thus density below b is proved using only the set $S \subseteq \mathbb{B}$; no proper-class-indexed join is being formed.

Working in the ambient metatheory ZFC, use Zorn's lemma to choose a pairwise-disjoint subset $A \subseteq D_b$ maximal by inclusion. Density below b implies

$$\bigvee A = b :$$

otherwise the nonzero value $b \wedge \neg \bigvee A$ would have a stronger member of D_b , contradicting maximality by inclusion. Thus A is an antichain maximal below b in the terminology fixed above.

For each $a \in A$ there exists at least one Boolean name ρ with $a \leq \|\varphi(\rho, \bar{\sigma})\|$. Because A is a set, ambient Collection first produces a set C of Boolean names containing at least one such witness for every $a \in A$. Ambient Choice applied to the nonempty subsets

$$C_a := \{\rho \in C : a \leq \|\varphi(\rho, \bar{\sigma})\|\} \quad (a \in A)$$

then chooses a family $\langle \tau_a : a \in A \rangle$ with

$$a \leq \|\varphi(\tau_a, \bar{\sigma})\|.$$

By mixing, there is a name τ satisfying $a \leq \|\tau = \tau_a\|$ for all $a \in A$. Substitutivity of Boolean equality gives

$$a \leq \|\varphi(\tau, \bar{\sigma})\| \quad (a \in A).$$

Taking joins,

$$b = \bigvee A \leq \|\varphi(\tau, \bar{\sigma})\|.$$

The reverse inequality is immediate from the definition of b as the join of all witness values. Therefore

$$\|\exists x \varphi(x, \bar{\sigma})\| = \|\varphi(\tau, \bar{\sigma})\|.$$

□

Formal definitions used in this chapter

Core of a Boolean name. Let $u \in V^{\mathbb{B}}$. A *core* for u is a set C of Boolean names such that

1. $\|x \in u\| = 1_{\mathbb{B}}$ for every $x \in C$;
2. whenever $\|y \in u\| = 1_{\mathbb{B}}$, there is a unique *literal member* $x \in C$ for which $\|x = y\| = 1_{\mathbb{B}}$.

The word “unique” here refers to uniqueness as an element of the ground-model set C ; Boolean equality of names remains the previously defined relation $\|x = y\|$.

Lemma 52.3 (Core and exact representative lemmas). *Work in the ambient ZFC metatheory. Every Boolean name u has a core C . Moreover, if*

$$\|u \neq \emptyset\| = 1_{\mathbb{B}}$$

and C is a core for u , then for every Boolean name y there is $x \in C$ such that

$$\|y = x\| = \|y \in u\|. \quad (\text{CR})$$

Proof. Put $I := \text{dom}(u)$. For each Boolean name x , define its *u -membership profile*

$$F_x : I \longrightarrow \mathbb{B}, \quad F_x(z) := u(z) \wedge \|z = x\|.$$

All such profiles lie in the set $\mathbb{B}^I$. By Separation,

$$R_u := \{f \in \mathbb{B}^I : \exists x (x \text{ is a Boolean name} \wedge f = F_x)\}$$

is a set. Ambient Collection produces a set W of Boolean names containing at least one representative x_f with $F_{x_f} = f$ for every $f \in R_u$.

Let

$$W_1 := \{x \in W : \|x \in u\| = 1_{\mathbb{B}}\}.$$

On W_1 define

$$x \sim y \iff \|x = y\| = 1_{\mathbb{B}}.$$

The Boolean equality laws make $\sim$ an equivalence relation. Ambient Choice selects one literal representative from every $\sim$ -class in the set W_1 ; call the resulting set C .

We verify that C is a core. Suppose $\|y \in u\| = 1$. The profile F_y belongs to R_u , so some $w \in W$ has $F_w = F_y$. Then

$$1 = \|y \in u\| = \bigvee_{z \in I} F_y(z) = \bigvee_{z \in I} F_w(z) = \|w \in u\|,$$

so $w \in W_1$. For each $z \in I$,

$$F_y(z) = F_w(z) \leq \|z = y\| \wedge \|z = w\| \leq \|y = w\|,$$

by transitivity of Boolean equality. Taking the join over z gives $\|y = w\| = 1$. The selected representative $x \in C$ of the $\sim$ -class of w therefore satisfies $\|x = y\| = 1$. If two elements of C had Boolean equality 1 with the same y , transitivity would put them in the same $\sim$ -class; the representative choice makes them literally equal. Thus C is a core.

For (CR), fix a Boolean name y and put

$$b := \|y \in u\|.$$

Since $\|u \neq \emptyset\| = 1$, the maximum principle gives a name t with $\|t \in u\| = 1$. The mixing theorem was stated for antichains contained in $\mathbb{B}^+$, so zero pieces must not be inserted silently. Put

$$A_b := \{d \in \{b, \neg b\} : d > 0_{\mathbb{B}}\}.$$

For the (possibly absent) piece b use the name y , and for the (possibly absent) piece $\neg b$ use the name t . Then A_b is an antichain with join $1_{\mathbb{B}}$; mixing gives a name w satisfying

$$b \leq \|w = y\|, \quad \neg b \leq \|w = t\|.$$

These inequalities remain literal when $b = 0$ or $b = 1$, because the corresponding zero piece is simply omitted from A_b . Substitutivity shows $\|w \in u\| = 1$. It also gives

$$\|w = y\| = \|w = y\| \wedge \|w \in u\| \leq \|y \in u\| = b,$$

so in fact $\|w = y\| = b$. Since w is a full member of u , the core property supplies the unique $x \in C$ with $\|x = w\| = 1$. Substitutivity of equality then yields $\|y = x\| = \|y = w\| = b$, proving (CR). $\square$

Corollary 52.4 (Boolean quotient truth lemma). *Work in the ambient ZFC metatheory fixed for Theorem 52.2, so that the maximum principle is available. Let U be an ultrafilter on the complete Boolean algebra $\mathbb{B}$. For every formula $\varphi(x_1, \dots, x_n)$ and Boolean names $\tau_1, \dots, \tau_n$,*

$$V^{\mathbb{B}}/U \models \varphi([\tau_1]_U, \dots, [\tau_n]_U) \iff \|\varphi(\tau_1, \dots, \tau_n)\| \in U.$$

Proof. Induct on formula complexity. The atomic cases are exactly the quotient definitions of equality and membership; substitutivity of Boolean equality makes those relations independent of the chosen representatives. The Boolean clauses for negation and conjunction agree with the ultrafilter laws. For the existential step, if a quotient witness $[\rho]_U$ satisfies φ , then the induction hypothesis gives $\|\varphi(\rho, \bar{\tau})\| \in U$, hence the larger existential value lies in U . Conversely, if $\|\exists x \varphi(x, \bar{\tau})\| \in U$, Theorem 52.2 supplies a name ρ attaining that Boolean value, so $\|\varphi(\rho, \bar{\tau})\| \in U$; the induction hypothesis then gives the quotient witness $[\rho]_U$. $\square$

Lemma 52.5 (Constituent formulas for Boolean bounded quantifiers). *Let $a \in V^{\mathbb{B}}$ and let $\psi(x, \bar{p})$ be any formula with Boolean-name parameters. Then*

$$\|\exists x \in a \psi(x, \bar{p})\| = \bigvee_{\rho \in \text{dom}(a)} (a(\rho) \wedge \|\psi(\rho, \bar{p})\|), \quad (\text{BE})$$

and

$$\|\forall x \in a \psi(x, \bar{p})\| = \bigwedge_{\rho \in \text{dom}(a)} (\neg a(\rho) \vee \|\psi(\rho, \bar{p})\|). \quad (\text{BU})$$

Consequently,

$$\|\forall x \in a \psi(x, \bar{p})\| = 1_{\mathbb{B}} \iff a(\rho) \leq \|\psi(\rho, \bar{p})\| \text{ for every } \rho \in \text{dom}(a). \quad (\text{BU1})$$

Proof. Write

$$d := \bigvee_{\rho \in \text{dom}(a)} (a(\rho) \wedge \|\psi(\rho, \bar{p})\|).$$

For each $\rho \in \text{dom}(a)$, reflexivity gives $a(\rho) \leq \|\rho \in a\|$, hence

$$a(\rho) \wedge \|\psi(\rho, \bar{p})\| \leq \|\exists x \in a \psi(x, \bar{p})\|.$$

Taking joins proves d is at most the left side of (BE).

Conversely, let σ be any Boolean name. Expanding membership and using Lemma 52.1,

$$\begin{aligned} \|\sigma \in a\| \wedge \|\psi(\sigma, \bar{p})\| &= \bigvee_{\rho \in \text{dom}(a)} (a(\rho) \wedge \|\sigma = \rho\| \wedge \|\psi(\sigma, \bar{p})\|) \\ &\leq \bigvee_{\rho \in \text{dom}(a)} (a(\rho) \wedge \|\psi(\rho, \bar{p})\|) = d, \end{aligned}$$

where the inequality is atomic/formula substitutivity of Boolean equality. Taking the existential join over σ therefore gives the reverse inequality in (BE). Notice that the existential range is the set supplied by Theorem 51.6; no proper-class join is being formed.

Since $\forall x \in a \psi$ abbreviates $\neg \exists x \in a \neg \psi$, (BE) and complete De Morgan laws yield

$$\begin{aligned} \|\forall x \in a \psi(x, \bar{p})\| &= \neg \bigvee_{\rho \in \text{dom}(a)} (a(\rho) \wedge \neg \|\psi(\rho, \bar{p})\|) \\ &= \bigwedge_{\rho \in \text{dom}(a)} (\neg a(\rho) \vee \|\psi(\rho, \bar{p})\|), \end{aligned}$$

which is (BU). The meet in (BU) is $1_{\mathbb{B}}$ exactly when every factor is $1_{\mathbb{B}}$, equivalently exactly when $a(\rho) \leq \|\psi(\rho, \bar{p})\|$ for every constituent ρ . This is (BU1). $\square$

Corollary 52.6 (Auxiliary Boolean inclusion equals Boolean formula subethood). *For all Boolean names σ, τ ,*

$$\|\sigma \subseteq \tau\|_{\text{aux}} = \|\forall x (x \in \sigma \rightarrow x \in \tau)\|. \quad (\text{SUBB})$$

Equivalently, if $x \subseteq y$ is used as the standard pure membership-language abbreviation $\forall z (z \in x \rightarrow z \in y)$, then its Boolean truth value is exactly the auxiliary inclusion value used in the recursive definition of Boolean equality. From this point onward the subscript “aux” is suppressed.

Proof. Apply the bounded-universal formula (BU) of Lemma 52.5 to $a = \sigma$ and $\psi(x)$ equal to the atomic formula $x \in \tau$. It gives

$$\|\forall x \in \sigma (x \in \tau)\| = \bigwedge_{\rho \in \text{dom}(\sigma)} (\neg \sigma(\rho) \vee \|\rho \in \tau\|).$$

The right-hand side is (AUXB), while the bounded quantifier on the left is by definition $\forall x (x \in \sigma \rightarrow x \in \tau)$. This proves (SUBB). $\square$

Lemma 52.7 (Boolean-valued set induction). *For every formula $\varphi(x, \bar{p})$ and Boolean-name parameters $\bar{p}$,*

$$\left\| \left[\forall x ((\forall y \in x \varphi(y, \bar{p})) \rightarrow \varphi(x, \bar{p})) \right] \rightarrow \forall x \varphi(x, \bar{p}) \right\| = 1_{\mathbb{B}}. \quad (\text{IND})$$

Proof. Put

$$c := \left\| \forall x ((\forall y \in x \varphi(y, \bar{p})) \rightarrow \varphi(x, \bar{p})) \right\|.$$

We prove by external induction on the Boolean-name rank $\text{rk}(\tau)$ that

$$c \leq \|\varphi(\tau, \bar{p})\| \quad (\tau \in V^{\mathbb{B}}). \quad (1)$$

Assume (1) for every constituent $\rho \in \text{dom}(\tau)$. Then for each such ρ ,

$$c \wedge \tau(\rho) \leq c \leq \|\varphi(\rho, \bar{p})\|.$$

By the bounded-universal formula (BU),

$$c \leq \|\forall y \in \tau \varphi(y, \bar{p})\|. \quad (2)$$

Since c is below every instance of the displayed universal progressive hypothesis,

$$c \leq \|(\forall y \in \tau \varphi(y, \bar{p})) \rightarrow \varphi(\tau, \bar{p})\|. \quad (3)$$

In a Boolean algebra, $d \leq A$ and $d \leq (A \rightarrow B)$ imply $d \leq B$. Applying this to (2)–(3) proves (1) for τ .

Thus $c \leq \|\varphi(\tau, \bar{p})\|$ for every Boolean name τ . By the universal-quantifier clause from Theorem 51.6,

$$c \leq \|\forall x \varphi(x, \bar{p})\|.$$

This is exactly the assertion that the implication (IND) has Boolean value $1_{\mathbb{B}}$. □

To prove $V^{\mathbb{B}} \models \text{ZFC}$ in the Boolean-valued sense, construct names witnessing every axiom with truth value 1. Separation modifies Boolean coefficients by the truth value of the defining formula. Replacement/Collection uses the mixing/maximum principle to collect a set of possible witness names. Power Set uses bounded-rank subnames. These are the same hard closure issues as in generic extensions, expressed algebraically.

How this chapter fits together

The ZF axioms are now verified with Boolean value 1. Separation modifies coefficients by the Boolean value of the defining formula; Collection/Replacement uses mixing to assemble witnesses; Power Set requires bounded-rank subnames. This mirrors the generic-extension proofs but avoids choosing an external generic filter.

Theorem 52.8 (Boolean-valued elementary ZF axioms). *In $V^{\mathbb{B}}$, the Boolean truth value of Extensionality, Empty Set, Pairing, Union, Infinity, and Foundation is $1_{\mathbb{B}}$.*

Proof. We verify the six axioms separately; none is hidden behind the phrase “direct calculation.”

Extensionality. For Boolean names a, b , the defining equality clause and (BU) give the exact identity

$$\|a = b\| = \|a \subseteq b\| \wedge \|b \subseteq a\|. \quad (\text{E1})$$

Indeed, the first half of the Boolean equality definition is

$$\bigwedge_{\rho \in \text{dom}(a)} (\neg a(\rho) \vee \|\rho \in b\|) = \|a \subseteq b\|,$$

and the second half is the reverse inclusion. Boolean propositional logic identifies $a \subseteq b \wedge b \subseteq a$ with $\forall z(z \in a \leftrightarrow z \in b)$. Hence

$$\|\forall z(z \in a \leftrightarrow z \in b) \rightarrow a = b\| = 1,$$

and universal closure gives Extensionality.

Empty Set. The Boolean check name $\check{\emptyset}$ has empty domain. Therefore

$$\|x \in \check{\emptyset}\| = 0_{\mathbb{B}} \quad \text{for every Boolean name } x,$$

so $\|\forall x(x \notin \check{\emptyset})\| = 1$.

Pairing. Given Boolean names a, b , define the coefficient-one name

$$p := \{\langle a, 1_{\mathbb{B}} \rangle, \langle b, 1_{\mathbb{B}} \rangle\},$$

with duplicate domain entries identified if $a = b$ literally. For every name x ,

$$\|x \in p\| = \|x = a\| \vee \|x = b\| = \|x = a \vee x = b\|.$$

Thus p witnesses Pairing with Boolean value 1.

Union. Given a name a , let

$$D := \bigcup_{\rho \in \text{dom}(a)} \text{dom}(\rho),$$

and for $\sigma \in D$ put

$$u(\sigma) := \bigvee_{\substack{\rho \in \text{dom}(a) \\ \sigma \in \text{dom}(\rho)}} a(\rho) \wedge \rho(\sigma). \quad (\text{U1})$$

The index set in (U1) is a set, so u is a Boolean name. For every name x , (BE), twice, gives

$$\begin{aligned} \|x \in u\| &= \bigvee_{\rho \in \text{dom}(a)} \bigvee_{\sigma \in \text{dom}(\rho)} a(\rho) \wedge \rho(\sigma) \wedge \|x = \sigma\| \\ &= \bigvee_{\rho \in \text{dom}(a)} a(\rho) \wedge \|x \in \rho\| \\ &= \|\exists y \in a (x \in y)\|. \end{aligned}$$

Hence u witnesses Union.

Infinity. Use the Boolean check name $\check{\omega}$. The check-name lemma gives

$$\|\check{\omega} \in \check{\omega}\| = 1.$$

For each $n \in \omega$, the coefficient-one Pairing and Union constructions above applied to $\check{n}$ produce a name s_n with

$$\|s_n = \check{n} \cup \{\check{n}\}\| = 1.$$

The ordinary identity $n + 1 = n \cup \{n\}$, together with the check-name lemma and substitutivity, gives

$$\|s_n = (n \check{+} 1)\| = 1, \quad \|s_n \in \check{\omega}\| = 1.$$

Since every constituent $\check{n}$ of $\check{\omega}$ has coefficient 1, (BU1) yields Boolean value 1 for closure under successor. Thus $\check{\omega}$ witnesses Infinity.

Foundation. Lemma 52.7 proves the full set-induction scheme in Boolean-valued form. Fix a Boolean name a and apply set induction to the formula

$$\varphi(x) : x \notin a.$$

In ordinary first-order logic, the assumption

$$\forall x \in a \exists y \in a (y \in x) \quad (\text{F1})$$

(the assertion that a has no $\in$ -minimal member) makes φ progressive: if every member of x is outside a and $x \in a$, (F1) produces a member of x that lies in a , a contradiction. Set induction therefore yields $\forall x x \notin a$, i.e. a is empty. This is a pure first-order derivation from the set-induction instance, so Boolean soundness, Theorem 51.7, gives

$$\|a \neq \emptyset \rightarrow \exists y \in a \forall z \in a (z \notin y)\| = 1.$$

Universal closure in a is exactly the Foundation axiom used in Volume IV. $\square$

Theorem 52.9 (Boolean-valued Separation). *For every name a and formula $\varphi(x, \bar{p})$, there is a name b such that*

$$\|\forall x (x \in b \leftrightarrow (x \in a \wedge \varphi(x, \bar{p})))\| = 1.$$

Thus Boolean-valued Separation has truth value 1.

Proof.

Formal derivation spine

Formal spine for Boolean-valued Separation.

$$\sigma(u) = \tau(u) \wedge \|\varphi(u, \bar{\rho})\|.$$

$$\|x \in \sigma\| = \|x \in \tau \wedge \varphi(x, \bar{\rho})\|$$

$$\|\mathbf{ZF}\| = 1$$

Let τ be a Boolean name and let $\varphi(x, \bar{\rho})$ be a formula. Define a new name σ with the same domain as τ by

$$\sigma(u) = \tau(u) \wedge \|\varphi(u, \bar{\rho})\|.$$

For each potential member u , its coefficient in σ is exactly the degree to which it belongs to τ and satisfies φ . Using the atomic membership calculation and distributivity of the complete Boolean algebra, one checks

$$\|x \in \sigma\| = \|x \in \tau \wedge \varphi(x, \bar{\rho})\|$$

up to Boolean equality of representatives. Therefore the Separation axiom has value 1. The construction is the Boolean analogue of restricting an ordinary forcing name by conditions that force the defining formula.

The equality caveat is important because distinct Boolean names can have equality value 1. The verification therefore uses the previously proved substitutivity of Boolean equality rather than treating names as literal set-theoretic representatives of extension objects. $\square$

Theorem 52.10 (Boolean-valued Replacement/Collection). *If*

$$\|\forall x \in a \exists y \varphi(x, y, \bar{p})\| = 1,$$

then there is a name b with

$$\|\forall x \in a \exists y \in b \varphi(x, y, \bar{p})\| = 1.$$

For functional φ , the corresponding Boolean-valued Replacement instance has value 1.

Proof.

Formal derivation spine

Formal spine: one maximum witness for each constituent, then the bounded-universal criterion. For each $\rho \in \text{dom}(a)$, put

$$b_\rho := \|\exists y \varphi(\rho, y, \bar{p})\|.$$

The hypothesis and Lemma 52.5 give

$$a(\rho) \leq b_\rho.$$

By the maximum principle choose τ_ρ with

$$b_\rho = \|\varphi(\rho, \tau_\rho, \bar{p})\|.$$

Let

$$u := \{\langle \tau_\rho, 1_{\mathbb{B}} \rangle : \rho \in \text{dom}(a)\}.$$

Then

$$a(\rho) \leq \|\exists y \in u \varphi(\rho, y, \bar{p})\|$$

for every constituent ρ , so another application of Lemma 52.5 yields Boolean Collection.

Assume

$$\|\forall x \in a \exists y \varphi(x, y, \bar{p})\| = 1_{\mathbb{B}}. \tag{1}$$

For each $\rho \in \text{dom}(a)$, define

$$b_\rho := \|\exists y \varphi(\rho, y, \bar{p})\|.$$

Applying Lemma 52.5 to the bounded universal in (1) gives

$$a(\rho) \leq b_\rho \quad (\rho \in \text{dom}(a)). \quad (2)$$

The maximum principle, Theorem 52.2, gives for each such ρ at least one name τ_ρ satisfying

$$b_\rho = \|\varphi(\rho, \tau_\rho, \bar{p})\|. \quad (3)$$

The index set $\text{dom}(a)$ is a set. Working in the ambient ZFC metatheory already fixed for the maximum principle, Collection first bounds the possible witness names and Choice selects one τ_ρ for every $\rho \in \text{dom}(a)$. Hence

$$u := \{\langle \tau_\rho, 1_{\mathbb{B}} \rangle : \rho \in \text{dom}(a)\} \quad (4)$$

is a Boolean name.

For each $\rho \in \text{dom}(a)$, reflexivity of Boolean equality gives

$$\|\tau_\rho \in u\| = 1_{\mathbb{B}},$$

because the pair $\langle \tau_\rho, 1_{\mathbb{B}} \rangle$ occurs in u . Therefore

$$\begin{aligned} \|\exists y \in u \varphi(\rho, y, \bar{p})\| &\geq \|\tau_\rho \in u\| \wedge \|\varphi(\rho, \tau_\rho, \bar{p})\| \\ &= b_\rho \\ &\geq a(\rho), \end{aligned} \quad (5)$$

where the last inequality is (2). Lemma 52.5, applied now to the formula $\exists y \in u \varphi(x, y, \bar{p})$, yields

$$\|\forall x \in a \exists y \in u \varphi(x, y, \bar{p})\| = 1_{\mathbb{B}}.$$

This proves Boolean-valued Collection.

For Replacement, assume in addition that φ is functional on a with Boolean value 1. Collection gives the witness set u just constructed. Apply Boolean Separation to u with the formula

$$\exists x \in a \varphi(x, y, \bar{p}).$$

In ordinary first-order set theory, Collection together with functionality and this Separation instance proves that the separated subset is exactly the range of the functional relation on a . The Boolean-valued soundness theorem, Theorem 51.7, transfers that first-order deduction to Boolean value 1. Thus the corresponding Replacement instance has Boolean value 1. $\square$

Theorem 52.11 (Boolean-valued Power Set and Choice). *For every Boolean name a there is a Boolean name P_a such that*

$$\|\forall y (y \in P_a \leftrightarrow y \subseteq a)\| = 1_{\mathbb{B}}. \quad (1)$$

If the ambient construction is carried out in ZFC, the Boolean-valued Choice axiom also has value $1_{\mathbb{B}}$.

Proof.

Formal derivation spine

Formal spine: canonical coefficient names give the exact Boolean power set. Let

$$I := \text{dom}(a), \quad C_a := \mathbb{B}^I.$$

For $c \in C_a$, let τ_c be the name with domain I and coefficients $\tau_c(u) = c(u)$. Define

$$P_a(\tau_c) := \|\tau_c \subseteq a\|.$$

Then for every name y ,

$$\|y \in P_a\| = \|y \subseteq a\|. \quad (2)$$

We first prove Power Set. Put

$$I := \text{dom}(a).$$

Because both I and $\mathbb{B}$ are sets, the function set

$$C_a := \{c : c : I \rightarrow \mathbb{B}\} = \mathbb{B}^I$$

is a set in the metatheory. For $c \in C_a$, let τ_c be the Boolean name with domain I defined by

$$\tau_c(u) := c(u) \quad (u \in I).$$

Replacement collects the set

$$S_a := \{\tau_c : c \in C_a\}.$$

Define the Boolean name P_a on this domain by

$$P_a(\tau_c) := \|\tau_c \subseteq a\|. \quad (3)$$

We claim that for every Boolean name y ,

$$\|y \in P_a\| = \|y \subseteq a\|. \quad (4)$$

For the first inequality, the definition of Boolean membership gives

$$\|y \in P_a\| = \bigvee_{c \in C_a} (\|\tau_c \subseteq a\| \wedge \|y = \tau_c\|). \quad (5)$$

Boolean equality is substitutive for every formula, in particular for the formula $z \subseteq a$. Hence for each c ,

$$\|\tau_c \subseteq a\| \wedge \|y = \tau_c\| \leq \|y \subseteq a\|.$$

Taking the join in (5),

$$\|y \in P_a\| \leq \|y \subseteq a\|. \quad (6)$$

For the converse inequality, set

$$b := \|y \subseteq a\|.$$

Define $c_y \in C_a$ by

$$c_y(u) := b \wedge \|u \in y\| \quad (u \in I), \quad (7)$$

and put

$$z := \tau_{c_y}.$$

We show

$$b \leq \|y = z\|. \quad (8)$$

First, z is included in y with Boolean value 1. Indeed, for each $u \in I$,

$$z(u) = c_y(u) = b \wedge \|u \in y\| \leq \|u \in y\|,$$

so the Boolean value of $z \subseteq y$ is 1.

For the reverse inclusion, fix $v \in \text{dom}(y)$. Since $b = \|y \subseteq a\|$, the definition of Boolean subethood gives

$$b \wedge y(v) \leq \|v \in a\|. \quad (9)$$

Expanding membership in a ,

$$\|v \in a\| = \bigvee_{u \in I} (a(u) \wedge \|v = u\|). \quad (10)$$

Combining (9)–(10) and distributing the meet with $b \wedge y(v)$ over the join,

$$b \wedge y(v) \leq \bigvee_{u \in I} (b \wedge y(v) \wedge a(u) \wedge \|v = u\|). \quad (11)$$

Now $y(v) \leq \|v \in y\|$, and substitutivity of equality gives

$$\|v = u\| \wedge \|v \in y\| \leq \|u \in y\|.$$

Therefore every term in the join on the right of (11) is at most

$$b \wedge \|u \in y\| \wedge \|v = u\| = c_y(u) \wedge \|v = u\|.$$

Taking the join over $u \in I$,

$$b \wedge y(v) \leq \bigvee_{u \in I} (c_y(u) \wedge \|v = u\|) = \|v \in z\|.$$

Since this holds for every $v \in \text{dom}(y)$,

$$b \leq \|y \subseteq z\|.$$

Together with $\|z \subseteq y\| = 1$, this proves (8).

Finally, substitutivity applied to the formula $w \subseteq a$ gives

$$\|y = z\| \wedge \|y \subseteq a\| \leq \|z \subseteq a\|.$$

Using (8) and the definition of b ,

$$b \leq \|z \subseteq a\|. \quad (12)$$

The particular term c_y in the join (5) therefore satisfies

$$b \leq \|z \subseteq a\| \wedge \|y = z\| \leq \|y \in P_a\|. \quad (13)$$

Equations (6) and (13) prove the claimed equality (4). Hence each instance

$$y \in P_a \leftrightarrow y \subseteq a$$

has Boolean value 1, and therefore so does the universal closure in (1). This is the Boolean-valued Power Set axiom.

For Choice, assume the ambient metatheory is ZFC. At this point every ZF axiom has already been verified with Boolean value 1. Volume V proved over ZF that Zorn's Lemma implies the Axiom of Choice. By Boolean soundness, Theorem 51.7, it is therefore enough to verify Zorn's Lemma with Boolean value 1.

Recall that an *inductive poset* is a nonempty partially ordered set in which every chain has an upper bound. We first prove the full-hypothesis form. Let X, R be Boolean names satisfying

$$\|\langle X, R \rangle \text{ is a nonempty inductive poset}\| = 1. \quad (Z1)$$

Choose a core C for X by Lemma 52.3. Since X is nonempty with Boolean value 1, the maximum principle gives a full member of X , and the core property shows that $C \neq \emptyset$.

Define an ordinary ground-model relation $\leq_C$ on the set C by

$$x \leq_C y \iff \|R(x, y)\| = 1_{\mathbb{B}}. \quad (Z2)$$

The reflexive and transitive laws for R in (Z1) immediately give reflexivity and transitivity of $\leq_C$. If $x \leq_C y$ and $y \leq_C x$, Boolean antisymmetry gives $\|x = y\| = 1$; because $x, y \in C$ are the selected literal representatives

of their Boolean-equality classes, the core uniqueness clause gives $x = y$. Thus $(C, \leq_C)$ is an ordinary partially ordered set.

We next show it is inductive. Let $D \subseteq C$ be an ordinary chain. Form the Boolean name $\dot{D}$ with

$$\text{dom}(\dot{D}) = D, \quad \dot{D}(x) = 1_{\mathbb{B}} \quad (x \in D). \quad (\text{Z3})$$

Every $x \in D$ is a full member of X , and any two members of D are $\leq_C$ -comparable. The bounded-quantifier formulas (BE)–(BU) and equality substitutivity therefore give

$$\|\dot{D} \subseteq X\| = 1, \quad \|\dot{D} \text{ is an } R\text{-chain}\| = 1.$$

By the inductiveness clause in (Z1),

$$\|\exists u \in X \forall x \in \dot{D} R(x, u)\| = 1.$$

The maximum principle chooses a name v satisfying the displayed upper-bound formula with value 1. In particular $\|v \in X\| = 1$, so the core gives $c_v \in C$ with $\|v = c_v\| = 1$. Substitutivity then yields

$$x \leq_C c_v \quad (x \in D).$$

Thus every ordinary chain in C has an ordinary upper bound in C .

Ambient ZFC now applies Zorn's Lemma to the *set* $(C, \leq_C)$. Choose a maximal element $m \in C$. We claim that m is Boolean-maximal in (X, R) :

$$\|m \in X \wedge \forall y \in X (R(m, y) \rightarrow y = m)\| = 1. \quad (\text{Z4})$$

The membership part has value 1 because $m \in C$. Suppose the bounded universal in (Z4) failed on a nonzero Boolean piece. By (BE), there would be a Boolean name y for which

$$d := \|y \in X\| \wedge \|R(m, y)\| \wedge \neg\|y = m\| > 0. \quad (\text{Z5})$$

The mixing theorem is stated for antichains of nonzero Boolean elements. Therefore set

$$A_d := \{e \in \{d, \neg d\} : e > 0_{\mathbb{B}}\}.$$

On the (present) piece d assign the name y , and on the (possibly absent) piece $\neg d$ assign m . Mixing along the positive-piece antichain A_d gives a name y^* . On d , (Z5) says that y is a member of X above m ; on $\neg d$, when that piece is nonzero, reflexivity says the same of m . Mixing and substitutivity therefore give

$$\|y^* \in X\| = 1, \quad \|R(m, y^*)\| = 1. \quad (\text{Z6})$$

The core supplies $c^* \in C$ with $\|y^* = c^*\| = 1$. Equation (Z6) implies $m \leq_C c^*$. Maximality of m in the ground poset forces $c^* = m$, hence $\|y^* = m\| = 1$. But mixing also gives $d \leq \|y^* = y\|$, so transitivity yields

$$d \leq \|y = m\|,$$

contradicting the last factor in (Z5). Therefore (Z4) holds.

It remains to pass from the full-hypothesis case to the Boolean implication at an arbitrary truth degree. For arbitrary Boolean names X, R , put

$$c := \|\langle X, R \rangle \text{ is a nonempty inductive poset}\|.$$

Let

$$X_0 := \{\check{0}\}, \quad R_0 := \{\langle \check{0}, \check{0} \rangle\},$$

the canonical names for the one-point poset. Put

$$A_c := \{e \in \{c, \neg c\} : e > 0_{\mathbb{B}}\}.$$

Mix X with X_0 , and R with R_0 , along this positive-piece antichain: on the (possibly absent) piece c use X, R , and on the (possibly absent) piece $\neg c$ use X_0, R_0 . This gives X^*, R^* satisfying

$$c \leq \|X^* = X\| \wedge \|R^* = R\|, \quad \neg c \leq \|X^* = X_0\| \wedge \|R^* = R_0\|.$$

On the first Boolean piece the antecedent holds to degree c ; on the second piece the one-point poset satisfies it outright. Hence

$$\|\langle X^*, R^* \rangle \text{ is a nonempty inductive poset}\| = 1.$$

The full-hypothesis case gives a name m which is maximal in (X^*, R^*) with value 1. On the piece c , substitutivity transfers that conclusion back to (X, R) . Consequently

$$c \leq \|\exists m \text{ } m \text{ is maximal in } (X, R)\|.$$

This is exactly the assertion that the Zorn implication has Boolean value 1; universal closure in X, R proves Boolean-valued Zorn's Lemma. Since ZF proves Zorn's Lemma $\Rightarrow$ AC and all ZF axioms already have Boolean value 1, Boolean soundness gives the Axiom of Choice with value 1. This completes the Boolean-valued ZFC verification. $\square$

Worked example

Worked Example 52.12 (Boolean separation). Given a name τ and formula $\varphi(x)$, define a new name whose coefficient at a constituent σ is $\tau(\sigma) \wedge \|\varphi(\sigma)\|$. The Boolean membership calculation then shows, with value 1, that the new name consists exactly of the members of τ satisfying φ .

A useful warning

For Replacement and Power Set, it is not enough to appeal to their truth in an eventual quotient. Their Boolean values must be shown to equal 1 by set-sized constructions in the Boolean universe.

Looking ahead

The next chapter, *Consistency transfer*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 53

Consistency transfer

Definitions and setup

Formal definitions used in this chapter

Recall of Boolean antichains and the maximum principle. Compatibility, antichains, maximal antichains, mixing, and the maximum principle were defined and proved in Theorem 52.2 before the Boolean Collection/Replacement argument. They retain exactly those meanings in this chapter. **Relative consistency.** $\text{Con}(S) \rightarrow \text{Con}(U)$ is a metatheoretic implication between the canonical syntactic consistency statements from Volume III.

Base arithmetic. The *base arithmetic* in this chapter is the fixed theory $I\Sigma_1$ from Volume II. Every proof-translation claim below is stated relative to this explicitly fixed theory; if a later argument ever requires a stronger arithmetic theory, that theory must be named in its statement.

Canonical contradiction for the set-theory presentations. The language of ZF used in this series is the pure language $\{\in\}$, so it has no primitive constant symbols $0, 1$. For the recursively axiomatized presentations of ZF, ZFC, and their extensions used from this chapter onward, fix the pure-language absurdity

$$\perp_{\in} := \exists x (x \neq x).$$

In proof-code displays, $\ulcorner 0 = 1 \urcorner$ is retained as legacy shorthand for the code $\ulcorner \perp_{\in} \urcorner$ of this fixed contradiction. Thus no undeclared constants are being added to the set-theoretic language. Any other fixed contradiction sentence could be used instead, because fixed finite derivations give primitive-recursive conversions between the corresponding contradiction-proof codes.

Primitive-recursive contradiction-proof translator. Let S, U be computably enumerable theories with fixed primitive-recursive proof predicates and fixed contradiction sentences in their respective languages. A primitive-recursive function F is a contradiction-proof translator from U to S when the base arithmetic proves

$$\forall n (\text{Prf}_U(n, \ulcorner \perp_U \urcorner) \rightarrow \text{Prf}_S(F(n), \ulcorner \perp_S \urcorner)).$$

For the set theories in this chapter, both displayed contradiction codes are written $\ulcorner 0 = 1 \urcorner$ by the preceding convention and denote $\ulcorner \perp_{\in} \urcorner$. **Interpretation/relativization translation.** If $C(x)$ is a fixed formula defining the intended interpretation domain, the relativization $\varphi \mapsto \varphi^C$ is defined recursively by leaving atomic formulas unchanged, commuting with Boolean connectives, and replacing

$$\exists x \psi \quad \text{by} \quad \exists x (C(x) \wedge \psi^C), \quad \forall x \psi \quad \text{by} \quad \forall x (C(x) \rightarrow \psi^C).$$

This translation is primitive recursive on formula codes. For the constructible interpretation below, $C(x)$ is the parameter-free constructibility formula $\text{Constr}(x)$ from Volume VI, and φ^L denotes this relativization.

Formalizable Boolean-valued interpretation. Such an interpretation consists of primitive-recursive translations of formulas/proofs together with base-arithmetic proofs that translated axioms have Boolean value one and translated inference rules preserve value one. By itself this gives a primitive-recursive translation of a target proof into a proof that its translated conclusion has Boolean value one. To obtain a *contradiction-proof translator* one must additionally prove, in the ground theory, that the Boolean value of the chosen target contradiction is $0_{\mathbb{B}}$ and that $0_{\mathbb{B}} \neq 1_{\mathbb{B}}$. Theorem 53.4 states this extra closing step explicitly.

Standard- ω transitive model. A transitive model has standard ω when its internal natural numbers are exactly the ambient ω ; this is stronger than mere existence of an arbitrary first-order model.

A countable transitive ground model is a convenient picture, not the conclusion of a consistency assumption. The formal relative-consistency theorem uses Boolean-valued soundness or forcing over models of finite fragments together with first-order completeness/compactness. If a target contradiction existed, it would use finitely many axioms; the corresponding finite forcing verification would already yield an impossible contradiction in a Boolean-valued model. This is the metatheoretic bridge used by the final independence claims.

How this chapter fits together

This chapter closes the metatheoretic gap left by the pedagogical countable-transitive-model picture. Consistency alone does not supply a transitive model. Instead, Boolean-valued semantics and compactness are applied to finite fragments: if every finite part of the target theory has the appropriate Boolean-valued model, first-order completeness converts that finite satisfiability into a genuine relative-consistency statement.

How to read the symbols: Recall: mixing and maximum were proved before Boolean Collection

Theorem 52.2, proved in the preceding chapter before its first use, states the Boolean mixing and maximum principles. The consistency-transfer arguments below may invoke it without any forward dependency.

Theorem 53.1 (Boolean-valued model theorem for a forced sentence). *If a sentence φ has Boolean value 1 in a Boolean-valued model in which every axiom of a theory T has value 1, then every first-order consequence of $T + \varphi$ has Boolean value 1.*

Proof.

Formal derivation spine

Formal spine for Boolean-valued model theorem for a forced sentence.

$$\|\varphi\| = 1.$$

$$T + \varphi \vdash \psi \implies \|\psi\| = 1_{\mathbb{B}}.$$

Assume the Boolean-valued construction has assigned value 1 to every axiom of a theory T , and suppose a forcing calculation yields

$$\|\varphi\| = 1.$$

Then $V^{\mathbb{B}}$ is a Boolean-valued model of $T + \varphi$ in the precise semantic sense that every axiom of that theory has Boolean value 1. By Boolean-valued soundness, every sentence formally derivable from any finite subset of those axioms also has value 1; in particular no finite subset can have a derivation whose conclusion has Boolean value 0.

If one works over an actual generic ultrafilter U , quotienting Boolean names by the explicitly defined relation $\sigma \equiv_U \tau \iff \|\sigma = \tau\| \in U$ gives the ordinary generic-extension semantics; Corollary 52.4 sends every value-1 sentence to a true sentence. The theorem is therefore the semantic bridge between a top Boolean value and a

model of the forced sentence. The next theorem adds the separate metatheoretic argument converting this fact into a Con-implication without assuming a transitive ground model. $\square$

Theorem 53.2 (Proof-translation consistency lemma). *Let S and U be computably enumerable theories with fixed primitive-recursive proof predicates. If there is a primitive-recursive contradiction-proof translator F from U to S , then the base arithmetic proves*

$$\text{Con}(S) \longrightarrow \text{Con}(U).$$

Uses: Primitive-recursive proof coding and the definition of syntactic consistency.

Proof. By the definition of a contradiction-proof translator, the base arithmetic proves

$$\forall n (\text{Prf}_U(n, \ulcorner 0 = 1 \urcorner) \rightarrow \text{Prf}_S(F(n), \ulcorner 0 = 1 \urcorner)). \quad (\text{PT})$$

If $\neg \text{Con}(U)$, then some n satisfies $\text{Prf}_U(n, \ulcorner 0 = 1 \urcorner)$. Formula (PT) gives $\text{Prf}_S(F(n), \ulcorner 0 = 1 \urcorner)$, hence $\neg \text{Con}(S)$. Thus the base arithmetic proves

$$\neg \text{Con}(U) \rightarrow \neg \text{Con}(S),$$

and classical contraposition gives $\text{Con}(S) \rightarrow \text{Con}(U)$. This argument concerns finite proof codes only; no model-existence or transitivity assumption is involved. $\square$

Corollary 53.3 (Relative consistency of Choice via the constructible interpretation). *For the standard computably enumerable presentations of ZF and ZFC and their fixed primitive-recursive proof predicates,*

$$\text{Con}(\text{ZF}) \longrightarrow \text{Con}(\text{ZFC}).$$

Uses: The constructible inner-model verification of Volume VI and the proof-translation consistency lemma.

Proof. Use the fixed parameter-free formula $\text{Constr}(x)$ from Volume VI and translate each set-theoretic formula φ to its constructible relativization φ^L , as defined in the setup. The map on formula codes is primitive recursive.

Volume VI proves, without using Choice in the ambient theory, the relativizations to L of Extensionality, Foundation, Pairing, Union, Infinity, Separation, Power Set, Collection/Replacement, and finally Choice. These verifications are uniform in the code of a Separation or Replacement instance: the input formula code is inserted into a fixed finite proof template, while the previously established uniform definitions of L_α , satisfaction for set-sized structures, and constructibility supply the required formulas. Hence there is a primitive-recursive procedure which, from the code of a ZFC-axiom instance, produces a ZF-proof of its L -relativization.

Now translate a finite ZFC-proof line by line. Logical axioms and inference rules are preserved by relativization, and each nonlogical axiom line is replaced by the corresponding ZF-proof just described. For the fixed pure-language contradiction

$$\perp_\epsilon = \exists x (x \neq x),$$

its constructible relativization is

$$\perp_\epsilon^L = \exists x (\text{Constr}(x) \wedge x \neq x).$$

There is a fixed first-order derivation $\perp_\epsilon^L \vdash \perp_\epsilon$: from a witness for the left-hand existential, discard the conjunct $\text{Constr}(x)$, retain $x \neq x$, and apply existential introduction. Appending this fixed derivation to the line-by-line relativized proof is primitive recursive on proof codes. Therefore the resulting construction is a primitive-recursive function F_L for which the base arithmetic verifies

$$\text{Prf}_{\text{ZFC}}(n, \ulcorner 0 = 1 \urcorner) \rightarrow \text{Prf}_{\text{ZF}}(F_L(n), \ulcorner 0 = 1 \urcorner),$$

where both contradiction codes denote the fixed $\perp_\epsilon$ by the convention above. The proof-translation consistency lemma now yields the displayed implication. *Why the corollary occurs here rather than in Volume VI.* Volume

VI establishes the mathematical inner-model interpretation. The present chapter is the first point at which the separate proof-code principle converting that interpretation into a syntactic Con-implication has been proved. $\square$

Theorem 53.4 (Formal forcing consistency transfer). *Let S and U be computably enumerable theories with fixed primitive-recursive proof predicates and chosen contradiction sentence $\perp_U$. Suppose a formalizable Boolean-valued interpretation of U in S assigns Boolean value $1_{\mathbb{B}}$ to every translated axiom, preserves Boolean value $1_{\mathbb{B}}$ under the translated inference rules, and supplies ground-theory proofs of*

$$\|\perp_U\| = 0_{\mathbb{B}} \quad \text{and} \quad 0_{\mathbb{B}} \neq 1_{\mathbb{B}}.$$

Then it supplies a primitive-recursive contradiction-proof translator from U to S , and consequently the base arithmetic proves

$$\text{Con}(S) \rightarrow \text{Con}(U).$$

In the forcing specialization used later, the target theory is $U = S + \varphi$, meaning that U is obtained from S by adjoining the sentence φ as an additional axiom, and the Boolean-valued interpretation must additionally verify $\|\varphi\| = 1_{\mathbb{B}}$.

Uses: Boolean-valued soundness, nontriviality of the Boolean algebra, and the proof-translation consistency lemma.

How to read the symbols: forcing consistency transfer has two logically separate steps

First the Boolean-valued interpretation translates each finite target-theory proof into a ground-theory proof of the corresponding Boolean-value assertion. If the target proof ended in its chosen contradiction $\perp_U$, translation gives $\|\perp_U\| = 1_{\mathbb{B}}$, while the contradiction-refutation clause gives $\|\perp_U\| = 0_{\mathbb{B}}$; nontriviality then yields a ground-theory contradiction. This constructs a primitive-recursive contradiction-proof translator. Only then does the preceding general proof-translation lemma contrapose the finite-code implication to obtain $\text{Con}(S) \rightarrow \text{Con}(U)$.

Proof. Given a coded finite U -proof, translate every line into its Boolean-value-one assertion. By the definition of a formalizable Boolean-valued interpretation in the setup, the axiom and rule verifications are uniform primitive-recursive proof transformations, not merely semantic existence claims. For the exact logical calculus fixed in Volume I, the logical-axiom recognizer has a separate branch for the primitive existential-duality scheme. That branch inserts the fixed complete-Boolean De Morgan derivation (BED) from the proof of Theorem 51.7; the universal-instantiation, quantifier-distribution, equality, propositional-tautology, modus-ponens, and generalization branches use the corresponding fixed proof templates. Thus no constructor recognized by the Volume-II proof predicate is left without a translated proof.

Translated nonlogical axioms have value $1_{\mathbb{B}}$ by hypothesis. Consequently a translated proof of the chosen contradiction $\perp_U$ ends with an S -proof of

$$\|\perp_U\| = 1_{\mathbb{B}}.$$

In the set-theory specialization, $\perp_U = \perp_{\in} = \exists x(x \neq x)$. Boolean equality reflexivity gives $\|\sigma = \sigma\| = 1_{\mathbb{B}}$ for every Boolean name σ , hence $\|\sigma \neq \sigma\| = 0_{\mathbb{B}}$. Therefore every attained witness value in the existential clause for $\perp_{\in}$ is zero and

$$\|\perp_{\in}\| = 0_{\mathbb{B}}.$$

For a different fixed target contradiction, the formalizable interpretation is required to supply the corresponding fixed ground-theory proof that its Boolean value is zero. Nontriviality gives $0_{\mathbb{B}} \neq 1_{\mathbb{B}}$. Concatenating these finitely many proof transformations produces an S -proof of contradiction. The line-by-line operations on a finite proof code are primitive recursive, so their composition is a contradiction-proof translator F from U to S .

Apply the proof-translation consistency lemma. It yields $\text{Con}(S) \rightarrow \text{Con}(U)$ in the base arithmetic. No countable transitive model is inferred from $\text{Con}(S)$; the conclusion is proof-theoretic. $\square$

Remark 53.5 (Source note). This organization separates the general finite-proof-code implication from the specifically Boolean-valued construction of the translator. The forcing/Boolean-value transfer may be compared with the standard metatheoretic treatments in [13, 14].

Proposition 53.6 (Completeness does not supply a transitive model). *For every first-order theory T , if T is syntactically consistent in the Volume-I sense—equivalently, there is no sentence θ with both $T \vdash \theta$ and $T \vdash \neg\theta$ —then there exists a first-order model:*

$$T \text{ syntactically consistent} \implies \exists M (M \models T).$$

This metatheoretic consistency condition makes sense for an arbitrary first-order language and does not smuggle in arithmetic constants $0, 1$; it also does not presuppose that T has a computable axiom set or a coded proof predicate. The Completeness Theorem does not assert that this model is well founded or transitive. Moreover, suppose T is a consistent computably enumerable theory in the language of set theory, equipped with the fixed canonical proof predicate and satisfying the hypotheses of Gödel’s second incompleteness theorem. If there exists a transitive set model $M \models T$ with $\omega^M = \omega$, then, for the induced computable presentation of $T + \text{Con}(T)$, the stronger consistency statement $\text{Con}(T + \text{Con}(T))$ holds.

Proof.

Formal derivation spine

Formal spine for completeness versus transitivity.

$$T \text{ syntactically consistent} \implies \exists M (M \models T),$$

where syntactic consistency has the language-independent Volume-I definition and no well-foundedness or transitivity conclusion occurs. Under the additional hypotheses in the second sentence,

$$\exists M [M \text{ transitive}, \omega^M = \omega, M \models T] \implies \text{Con}(T + \text{Con}(T)).$$

The first implication is exactly first-order completeness: if T is syntactically consistent, the Henkin construction produces a structure $M \models T$. Neither the construction nor the theorem says that the membership relation of such a model is the ambient membership relation, so transitivity and well-foundedness are additional external requirements.

For the second assertion, assume the additional hypotheses stated above and let M be a transitive set model of T with $\omega^M = \omega$. Because proof checking for T is primitive recursive, transitivity together with standard ω makes the statement

$$\text{Con}(T) \equiv \neg \exists p \text{Prf}_T(p, \ulcorner 0 = 1 \urcorner)$$

absolute between M and the ambient universe. Ambient consistency of T therefore gives $M \models \text{Con}(T)$, so

$$M \models T + \text{Con}(T).$$

Soundness of first-order logic now yields

$$\exists M [M \models T + \text{Con}(T)] \implies \text{Con}(T + \text{Con}(T)).$$

Gödel’s second incompleteness theorem says that, for such a consistent T , $T \not\vdash \text{Con}(T)$; equivalently, the consistency of $T + \text{Con}(T)$ is not obtained merely by applying completeness to $\text{Con}(T)$. This is why a countable-transitive-model presentation is a useful pedagogical framework but is not the formal relative-consistency theorem proved in the preceding result. $\square$

Worked example

Worked Example 53.7 (Why a finite-fragment argument is enough). Assume that for every finite subset $F \subseteq T + \varphi$, if $T_F \subseteq T$ is the finite set of axioms of T occurring in F , the Boolean construction gives a model

of $T_F \cup (F \setminus T)$. If $T + \varphi$ were inconsistent, one formal proof of contradiction would use only finitely many axioms, producing an inconsistent finite F . This contradicts its model. Thus syntactic consistency follows without ever postulating a transitive model of all of T .

A useful warning

Do not infer from syntactic consistency of T that there exists a countable transitive model of T . Completeness gives an arbitrary first-order model; transitivity and external countability are additional requirements. When T is computably enumerable and $\text{Con}(T)$ has been defined arithmetically, the same warning applies a fortiori to that coded consistency sentence.

Looking ahead

The next chapter, *Chain conditions and closure*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 54

Chain conditions and closure

Definitions and setup

Formal definitions used in this chapter

Ground model, metatheoretic Choice, and extension notation. Throughout this preservation chapter the ground model satisfies $V \models \text{ZFC}$, and $V[G]$ denotes a generic extension. In particular, Zorn's lemma, the well-ordering/cardinal-arithmetic consequences of Choice, and simultaneous selections of set-indexed families of maximal antichains are ground-model tools available in the proofs below. This standing hypothesis concerns the ordinary forcing ground model; it does not assert Choice in the symmetric inner model constructed later. Superscripts indicate where a cardinal or cofinality is computed.

Antichain and κ -chain condition. An antichain in $\mathbb{P}$ is a set of pairwise incompatible conditions. $\mathbb{P}$ is κ -cc when every antichain has cardinality $< \kappa$. The *countable chain condition* (ccc) is the ω_1 -cc.

Cofinal subset, cofinality, and regular cardinal. For an ordinal α , a set $C \subseteq \alpha$ is *cofinal* (or unbounded) when

$$\forall \beta < \alpha \exists \gamma \in C (\beta \leq \gamma).$$

The cofinality $\text{cf}(\alpha)$ is the least order type of a cofinal subset of α . Explicitly, the empty set is cofinal in 0, so $\text{cf}(0) = 0$; and for every ordinal β , the singleton $\{\beta\}$ is cofinal in $\beta + 1$, so $\text{cf}(\beta + 1) = 1$. An infinite cardinal κ is *regular* when $\text{cf}(\kappa) = \kappa$. Superscripts such as $\text{cf}^V(\alpha)$ and $\text{cf}^{V[G]}(\alpha)$ specify the universe.

$< \kappa$ -closed. $\mathbb{P}$ is $< \kappa$ -closed when every descending sequence $\langle p_i : i < \delta \rangle$ with $\delta < \kappa$ has a common lower bound, namely a condition q satisfying $q \leq p_i$ for all $i < \delta$.

Dense open and $< \kappa$ -distributive. A dense set $D \subseteq \mathbb{P}$ is *open downward* when $p \in D$ and $q \leq p$ imply $q \in D$. The forcing $\mathbb{P}$ is $< \kappa$ -distributive when, for every $\delta < \kappa$ and every family $\langle D_i : i < \delta \rangle$ of dense open sets, $\bigcap_{i < \delta} D_i$ is dense.

Preservation and “no new sequences.” A ground-model cardinal λ is preserved when the same ordinal remains a cardinal in $V[G]$. A cofinality is preserved when $\text{cf}^V(\alpha) = \text{cf}^{V[G]}(\alpha)$. Saying that forcing *adds no new sequences of ground-model ordinals of length λ* means

$${}^\lambda \text{Ord}^V \cap V[G] = {}^\lambda \text{Ord}^V \cap V.$$

A κ -chain condition bounds antichains; ccc means every antichain is countable. Closure says descending sequences of a given length have lower bounds. Distributivity says intersections of suitably few dense open sets remain dense. These properties preserve different kinds of cardinal/cofinality information. We prove only the preservation statements later used for Cohen forcing and keep closure-based comparison results separate.

How this chapter fits together

Preservation theorems turn combinatorial properties of a forcing poset into restrictions on what the extension can change. The ccc bounds antichains and prevents ω_1 from being collapsed by a countable sequence of decisions. Closure and distributivity provide different preservation mechanisms. Only the exact variants later needed for Cohen forcing are proved.

Proposition 54.1 (Antichain calculus and kappa-chain conditions). *A poset $\mathbb{P}$ satisfies the κ -chain condition if every antichain has cardinality $< \kappa$. Maximal antichains decide names locally, and the number of possible decided values of a name at one coordinate is bounded by the size of such an antichain.*

Proof.

Formal derivation spine

Formal spine: antichains bound possible decided values. If A is a maximal antichain deciding a name $\dot{x}$, define

$$S_A := \{y : \exists p \in A (p \Vdash \dot{x} = \check{y})\}.$$

Then

$$|S_A| \leq |A|,$$

so under the κ -cc every such local value set has size $< \kappa$.

An antichain is a set of pairwise incompatible conditions. A forcing notion is κ -cc if every antichain has cardinality $< \kappa$; ccc is the case $\kappa = \omega_1$. By Zorn's lemma, every antichain is contained in a maximal antichain. A maximal antichain A is predense: every condition is compatible with some member of A , for otherwise that condition could be adjoined to A .

These elementary facts turn forcing questions into cardinal bounds. If A is a maximal antichain deciding the value of a name $\dot{f}(\xi)$, then the set of values that $\dot{f}(\xi)$ can take has cardinal at most $|A|$. Under κ -cc, fewer than κ many possibilities therefore occur at each coordinate. The preservation theorems below repeatedly use exactly this “decide by an antichain, then count possible values” principle. $\square$

Theorem 54.2 (ccc preservation of ω_1). *If $\mathbb{P}$ is ccc and G is generic, then*

$$\omega_1^{V[G]} = \omega_1^V.$$

Equivalently, ccc forcing cannot add a countable sequence cofinal in the ground-model ω_1 .

Proof.

Formal derivation spine

Formal spine: ccc cannot add a countable cofinal map into ω_1^V . If $p \Vdash \dot{f} : \omega \rightarrow \omega_1^V$, choose a countable maximal antichain A_n deciding $\dot{f}(n)$ and let S_n be its possible values. Then

$$S := \bigcup_{n < \omega} S_n$$

is countable in V , hence

$$\gamma := \sup S < \omega_1^V.$$

Thus $p \Vdash \text{ran}(\dot{f}) \subseteq \gamma + 1$, contradicting cofinality.

Proof map. Assume a name is forced to be a cofinal map $\omega \rightarrow \omega_1^V$. For each input n , a maximal antichain deciding the value is countable by ccc. The union of the countably many sets of possible values is countable in the ground model, hence bounded below ω_1^V . No generic choice among those values can be cofinal.

Suppose $\mathbb{P}$ is ccc. If a name $\dot{f}$ is forced to map ω cofinally into the ground-model ω_1 , for each n take a maximal antichain deciding $\dot{f}(n)$. Each antichain is countable, so the set of possible values of all $\dot{f}(n)$ is

a countable union of countable sets and is bounded below ω_1^V , contradiction. Hence ω_1 is preserved. The general cardinal/cofinality preservation statements used later are proved analogously by bounding the set of possible values with antichains of size below the relevant cardinal. We state only the exact hypotheses needed for Cohen forcing and do not claim that arbitrary ccc forcing preserves every cofinality without qualification. $\square$

Theorem 54.3 (Cardinal preservation under the chain-condition hypotheses used later). *Let κ be regular and let $\mathbb{P}$ satisfy the κ -chain condition. If $G \subseteq \mathbb{P}$ is generic, then every ground-model cardinal $\lambda \geq \kappa$ remains a cardinal in $V[G]$, and every ground-model ordinal α with $\text{cf}^V(\alpha) \geq \kappa$ satisfies*

$$\text{cf}^{V[G]}(\alpha) = \text{cf}^V(\alpha).$$

For $\kappa = \omega_1$, this gives the cardinal/cofinality preservation needed for ccc Cohen forcing.

Proof.

Formal derivation spine

Formal spine: a κ -cc name has fewer than κ possible values at each coordinate. For a name $\dot{f}$ and ξ in its domain, choose a maximal antichain A_ξ deciding $\dot{f}(\xi)$, and let

$$B_\xi := \{\beta : \exists p \in A_\xi (p \Vdash \dot{f}(\xi) = \check{\beta})\}.$$

Then $|B_\xi| < \kappa$.

First suppose a condition forces a surjection

$$\dot{f} : \mu \longrightarrow \lambda, \quad \mu < \lambda, \quad \lambda \geq \kappa,$$

where λ is a ground-model cardinal. For every $\xi < \mu$, choose a maximal antichain A_ξ deciding $\dot{f}(\xi)$. The κ -chain condition gives $|A_\xi| < \kappa$, hence $|B_\xi| < \kappa$.

Let $B = \bigcup_{\xi < \mu} B_\xi$, and put $\nu := |\mu|^V$, the ground-model cardinality of the ordinal μ . If $\nu < \kappa$, regularity of κ and the bounds $|B_\xi| < \kappa$ give $|B| < \kappa \leq \lambda$. If $\kappa \leq \nu < \lambda$, then

$$|B| \leq \nu \cdot \sup_{\xi < \mu} |B_\xi| \leq \nu \cdot \kappa = \max(\nu, \kappa) = \nu < \lambda.$$

Here the last equalities are cardinal arithmetic for the infinite cardinals ν and κ ; no ordinal is being silently identified with its cardinality. Every possible value of $\dot{f}$ lies in B , so no condition can force $\dot{f}$ to be onto λ . Therefore no ground-model cardinal at least κ is collapsed.

For cofinality, let $\delta = \text{cf}^V(\alpha) \geq \kappa$, and suppose a condition forces a cofinal map $\dot{g} : \mu \rightarrow \alpha$ with $\mu < \delta$. Put again $\nu := |\mu|^V$; because δ is a cardinal and $\mu < \delta$, one has $\nu < \delta$. Construct sets $B_\xi \subseteq \alpha$ of possible values exactly as above. If $\nu < \kappa$, regularity of κ gives $|\bigcup_{\xi < \mu} B_\xi| < \kappa \leq \delta$. If $\kappa \leq \nu < \delta$, then

$$\left| \bigcup_{\xi < \mu} B_\xi \right| \leq \nu \cdot \kappa = \nu < \delta.$$

Since $\delta = \text{cf}^V(\alpha)$, every subset of α of cardinality less than δ is bounded in α . Thus the possible-value set is bounded, so $\dot{g}$ cannot be cofinal. Hence

$$\text{cf}^{V[G]}(\alpha) = \text{cf}^V(\alpha).$$

Taking $\kappa = \omega_1$ yields the ccc preservation theorem used in the Cohen-forcing volume. $\square$

Theorem 54.4 (Closure/distributivity preservation principles in the limited scope needed for comparison). *Let κ be regular. If $\mathbb{P}$ is $< \kappa$ -distributive, then for every $\lambda < \kappa$, every $V[G]$ -function $f : \lambda \rightarrow \text{Ord}^V$ already belongs to V . In particular, $\mathbb{P}$ adds no new sequences of ground-model ordinals of length $< \kappa$. If $\mathbb{P}$ is $< \kappa$ -closed, the same conclusion follows by recursively deciding the coordinates of a name along a descending sequence of conditions and using closure at limit stages.*

Proof.

Formal derivation spine

Formal spine: distributivity/closure blocks short new ordinal sequences. For a name $\dot{f} : \lambda \rightarrow \text{Ord}^V$ with $\lambda < \kappa$, let

$$D_\alpha := \{p : p \text{ decides } \dot{f}(\alpha)\}.$$

If $\mathbb{P}$ is $< \kappa$ -distributive, $\bigcap_{\alpha < \lambda} D_\alpha$ is dense and one condition decides all coordinates. Under $< \kappa$ -closure, recursively choose a descending deciding sequence and take lower bounds at limit stages.

Two preservation principles are needed for comparison. If $\mathbb{P}$ is $< \kappa$ -closed and $\lambda < \kappa$, then a name for a λ -sequence of ground-model ordinals can be decided coordinate by coordinate along a descending sequence of length λ ; closure supplies a common lower bound, which therefore forces the entire sequence to equal a ground-model sequence. If $\mathbb{P}$ is $< \kappa$ -distributive, intersecting fewer than κ dense open sets is dense; applying this to the dense sets deciding successive coordinates yields the same “no new shorter sequences” conclusion. Thus either hypothesis prevents new functions from any $\lambda < \kappa$ into the ground-model ordinals. This is the exact comparison principle used here; the Cohen argument itself relies instead on the separately proved chain-condition lemmas. $\square$

Worked example

Worked Example 54.5 (Why ccc preserves ω_1). If a name $\dot{f} : \omega \rightarrow \omega_1^V$ were forced cofinal, choose for each n a maximal antichain deciding $\dot{f}(n)$. Under ccc each antichain is countable, so only countably many ordinals can occur as possible values. Their supremum is still below ω_1^V , contradicting cofinality.

A useful warning

The ccc does not mean that every cardinal or cofinality is automatically preserved in every context. Use only the preservation theorem whose hypotheses have actually been proved.

Looking ahead

The next chapter, *Nice names and preservation bookkeeping*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 55

Nice names and preservation bookkeeping

Definitions and setup

Formal definitions used in this chapter

Ground-model convention (recall). Continue with the standing hypothesis $V \models \text{ZFC}$ from the preceding preservation chapter. Maximal-antichain selection and the cardinal counting below are therefore ground-model ZFC arguments.

Nice name for a subset of ω . A $\mathbb{P}$ -name is nice when

$$\dot{A} = \{\langle \check{n}, p \rangle : n < \omega, p \in A_n\},$$

where each $A_n \subseteq \mathbb{P}$ is an antichain; under ccc each A_n is countable. This paragraph defines only the shape of a nice name. Theorem 55.1 below will prove that every name forced to be a subset of ω is forced equal to a nice name.

Ground-model versus extension notation. κ^V denotes the ground-model cardinal/ordinal and $\kappa^{V[G]}$ its extension interpretation. Once preservation is proved, the same ordinal may safely be denoted simply by κ in both universes.

Cardinal counting of names. If $|\mathbb{P}| = \kappa$, counting nice names is a ground-model calculation: count possible countable antichains for each $n < \omega$, then count ω -sequences of those choices. The notation $\kappa^{\aleph_0}$ refers to this ground-model exponent unless an extension superscript is displayed.

Forcing shorthand. As in Volume VII, $\Vdash \varphi$ means $1_{\mathbb{P}} \Vdash \varphi$; thus a statement such as $\Vdash \dot{A} = \dot{B}$ is a forcing assertion at the largest condition.

A nice name for a subset of ω is determined, for each integer n , by an antichain of conditions. Under ccc each such antichain is countable, so once normalization has been proved, counting real names will reduce to counting countable ground-model antichain data. Theorem 55.1 supplies the normalization statement; it is not assumed here.

How this chapter fits together

Theorem 55.1 will show that nice names compress arbitrary names for subsets of ω into countable antichain data without changing their forced value. After that theorem, this normal form becomes the counting device behind the upper bound on the continuum in the Cohen model. The bookkeeping also keeps ground-model cardinals distinct from their extension interpretations until preservation has been established.

Theorem 55.1 (Nice-name theorem for subsets of omega under ccc forcing). *If $\mathbb{P}$ is ccc and*

$$\Vdash \dot{A} \subseteq \check{\omega},$$

then there is a nice name $\dot{B}$ with

$$\Vdash \dot{A} = \dot{B}, \quad \dot{B} = \{\langle \check{n}, p \rangle : n < \omega, p \in A_n^+\},$$

where, for each $n < \omega$, A_n is a maximal antichain deciding the statement $\check{n} \in \dot{A}$ and

$$A_n^+ := \{p \in A_n : p \Vdash \check{n} \in \dot{A}\}.$$

Under ccc every A_n , and hence every A_n^+ , is countable.

Proof.

Formal derivation spine

Formal spine: retain the positive side of one deciding antichain for each integer. For every $n < \omega$, choose a maximal antichain A_n deciding $\check{n} \in \dot{A}$, put

$$A_n^+ := \{p \in A_n : p \Vdash \check{n} \in \dot{A}\},$$

and define

$$\dot{B} := \{\langle \check{n}, p \rangle : n < \omega, p \in A_n^+\}.$$

For each $n < \omega$, decision density and Zorn's lemma give a maximal antichain A_n whose members decide the statement $\check{n} \in \dot{A}$. Retain its positive part A_n^+ and form the displayed name $\dot{B}$. Under ccc, every A_n is countable, so $\dot{B}$ is coded by a countable sequence of countable ground-model sets.

Let G be generic. If $n \in \dot{B}^G$, some $p \in G \cap A_n^+$ forces $\check{n} \in \dot{A}$, so the Truth Lemma gives $n \in \dot{A}^G$. Conversely, if $n \in \dot{A}^G$, genericity meets the maximal antichain A_n in exactly one condition p . Since p decides membership, it cannot force nonmembership; hence $p \in A_n^+$ and $n \in \dot{B}^G$. Therefore

$$\dot{A}^G = \dot{B}^G$$

for every generic G , and the Fundamental Theorem of Forcing gives

$$\Vdash \dot{A} = \dot{B}.$$

□

Proposition 55.2 (Cardinal counting of nice names). *If $|\mathbb{P}| = \kappa$ and $\mathbb{P}$ is ccc, then the number of nice names for subsets of ω is at most*

$$(\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0}.$$

Proof.

Formal derivation spine

Formal spine for Cardinal counting of nice names.

$$(\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0}$$

Assume $|\mathbb{P}| = \kappa$ and $\mathbb{P}$ is ccc. A nice name for a real is determined by a sequence $\langle A_n : n < \omega \rangle$ in which each A_n is an antichain of conditions that force n into the named set. By ccc, every A_n is countable.

There are at most $\kappa^{\aleph_0}$ countable subsets of $\mathbb{P}$, hence at most

$$(\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0}$$

countable sequences of such subsets. Therefore there are at most $\kappa^{\aleph_0}$ nice names for reals. If the ground-model arithmetic gives $\kappa^{\aleph_0} = \kappa$, as in the later GCH application, this becomes the upper bound of κ many possible extension reals.

The count is intentionally performed before evaluating any generic filter. It gives a ground-model set of canonical names that surjects onto all extension reals after valuation. Later preservation results are then used to interpret the ground-model cardinal κ as the same cardinal in the extension. $\square$

Theorem 55.3 (Cofinality preservation in the cases used by Cohen forcing). *Let $\mathbb{P} = \text{Add}(\omega, \lambda)$ for any cardinal λ . Since $\mathbb{P}$ is ccc, for every ground-model ordinal α with uncountable cofinality,*

$$\text{cf}^{V[G]}(\alpha) = \text{cf}^V(\alpha).$$

In particular,

$$\text{cf}^{V[G]}(\omega_1^V) = \omega_1^V, \quad \text{cf}^{V[G]}(\omega_2^V) = \omega_2^V,$$

and the corresponding cardinals are preserved.

Proof.

Formal derivation spine

Formal spine for Cofinality preservation in the cases used by Cohen forcing.

$$\theta := \text{cf}^V(\alpha) > \omega.$$

$$\dot{f} : \check{\mu} \longrightarrow \check{\alpha}$$

$$A_\xi = \{\beta < \alpha : \exists q \in D_\xi \ q \Vdash \dot{f}(\check{\xi}) = \check{\beta}\}.$$

Let $\mathbb{P} = \text{Add}(\omega, \lambda)$. It is ccc. Fix a ground-model ordinal α with

$$\theta := \text{cf}^V(\alpha) > \omega.$$

Suppose, toward a contradiction, that some $p \in \mathbb{P}$ forces

$$\dot{f} : \check{\mu} \longrightarrow \check{\alpha}$$

to be cofinal for an ordinal $\mu < \theta$. For each $\xi < \mu$, choose a maximal antichain D_ξ below p deciding $\dot{f}(\xi)$. Since $\mathbb{P}$ is ccc, D_ξ is countable. Let

$$A_\xi = \{\beta < \alpha : \exists q \in D_\xi \ q \Vdash \dot{f}(\check{\xi}) = \check{\beta}\}.$$

Then $|A_\xi| \leq \aleph_0$. Set $A = \bigcup_{\xi < \mu} A_\xi$. Because $\mu < \theta$ and θ is an uncountable regular cardinal,

$$|A| \leq \max(|\mu|, \aleph_0) < \theta.$$

By the definition of $\theta = \text{cf}^V(\alpha)$, a subset of α of cardinality below θ is bounded, so there is $\gamma < \alpha$ with $A \subseteq \gamma$.

Every value of $\dot{f}$ forced by a condition below p belongs to the corresponding $A_\xi \subseteq A$. Hence

$$p \Vdash \text{ran}(\dot{f}) \subseteq \check{\gamma},$$

contradicting the assumption that p forces $\dot{f}$ cofinal in α . Therefore no shorter cofinal sequence is added, and

$$\text{cf}^{V[G]}(\alpha) = \text{cf}^V(\alpha).$$

Applying this to ω_1^V and ω_2^V , together with the earlier ccc cardinal-preservation theorem, gives the two displayed special cases and preservation of the corresponding cardinals. $\square$

Proposition 55.4 (Ground-model versus extension cardinal-arithmetic bookkeeping). *When a forcing theorem states a ground-model cardinal κ remains a cardinal, the same ordinal denotes κ in the extension. Cardinal-exponent calculations must therefore separate (i) preservation of the relevant ordinals/cardinals and (ii) counting new names in the extension.*

Proof.

Formal derivation spine

Formal spine: separate preservation from counting. If forcing preserves the ground-model cardinal κ , then the same ordinal remains a cardinal in the extension:

$$\kappa^{V[G]} = \kappa^V \quad \text{as an ordinal/cardinal identity.}$$

A statement such as $2^\lambda = \kappa$ additionally requires independent bounds

$$\kappa \leq |\mathcal{P}(\lambda)|^{V[G]} \leq \#\{\text{relevant names in } V\}.$$

Forcing calculations mix two kinds of cardinal statements, and the bookkeeping must keep them separate.

First, the *number of names* is counted in the ground model. Thus an estimate such as $\kappa^{\aleph_0} = \kappa$ is used as a statement of ground-model cardinal arithmetic when bounding nice names. Second, the resulting inequality is intended to describe cardinals in the extension. To make that translation, one must already know that the relevant ground-model cardinals have not been collapsed and that the relevant cofinalities are preserved.

For Cohen forcing the ccc theorems provide this bridge. Consequently a ground-model count of at most ω_2^V real names may be read in the extension as “at most $\aleph_2$ reals” only after $\omega_2^V = \omega_2^{V[G]}$ has been established. This distinction prevents a hidden circularity in continuum computations. $\square$

Worked example

Worked Example 55.5 (Normalizing a real name). For each n , choose a maximal antichain deciding whether $n \in \dot{r}$, and keep only conditions forcing membership. The resulting name is determined by the sequence of these antichains. Under ccc each is countable, so counting reals reduces to counting countable subsets of the forcing poset.

A useful warning

Write ω_1^V or ω_2^V until the preservation theorem identifies those ordinals with the extension cardinals. Suppressing the superscript too early hides a real logical step.

End-of-volume perspective

The forcing apparatus is now strong enough for precise consistency transfer and cardinal bookkeeping. Cohen forcing can therefore be treated as an application rather than as a mysterious new construction.

Part IX

Volume IX: Cohen Forcing and Independence of CH

Preface

Cohen forcing is now an application of the general machinery. Both the lower and upper bounds for the continuum are calculated explicitly.

Position in the series

Volume IX applies the forcing machinery to Cohen forcing and computes the continuum after adding many Cohen reals, yielding the relative consistency of $\neg\text{CH}$.

Sources and further reading

The continuum calculation is Cohen's independence argument in modern forcing language; compare [9, 10, 11] and the later textbook accounts [13, 14].

Notation for this volume

The formal definition of $\text{Add}(\omega, \kappa)$ and the coordinate names is given at the start of Chapter 1 below. After a generic filter G has been fixed and totality has been proved, c_α denotes the actual real obtained from the α -th coordinate; $\dot{c}_\alpha$ denotes its forcing name.

How to read symbolic arguments in this volume

When a formula appears, read it in layers rather than as one visual block. First identify the *ambient language*: are we speaking inside a formal theory, about a structure, or in the surrounding metatheory? Next mark the objects being manipulated (formulas, codes, sets, names, conditions, ordinals) and the operation being applied to them. Finally check the side conditions—free variables, rank bounds, compatibility, genericity, transitivity, or consistency hypotheses. Whenever a displayed derivation changes level, the text says so explicitly. A displayed line is therefore meant to be read like a line of well-commented code: the symbols perform the operation, and the surrounding prose records its type, preconditions, and purpose.

Chapter 56

Adding Cohen reals

Definitions and setup

Formal definitions used in this chapter

Finite partial functions. For sets X, Y ,

$$\text{Fn}(X, Y, <\omega) := \{p : p \text{ is a function, } \text{dom}(p) \subseteq X, |\text{dom}(p)| < \omega, \text{ran}(p) \subseteq Y\}.$$

Thus a member of $\text{Fn}(X, Y, <\omega)$ is a *finite partial function* $X \rightarrow Y$.

Cohen forcing. For any cardinal κ ,

$$\text{Add}(\omega, \kappa) := \text{Fn}(\kappa \times \omega, 2, <\omega), \quad q \leq p \iff q \supseteq p.$$

The order is therefore reverse inclusion: $q \leq p$ means that q is the stronger condition. Two conditions are *compatible* when they have a common strengthening. The forcing is *separative* when $p \not\leq q$ implies that some $r \leq p$ is incompatible with q . The concrete compatibility criterion for Cohen conditions is proved in Proposition 56.1; it is not part of the definition.

Ground model, filter, dense set, and genericity (recall from Volumes VII–VIII). Throughout this volume $V \models \text{ZFC}$ is the ordinary forcing ground model. Thus the antichain-selection and cardinal-arithmetic arguments imported from Volume VIII are carried out with ground-model Choice available. V denotes this ground model throughout. A set $D \subseteq \mathbb{P}$ is *dense* when every $p \in \mathbb{P}$ has some $q \in D$ with $q \leq p$. A set $G \subseteq \mathbb{P}$ is a *filter* when $G \neq \emptyset$, it is upward closed toward weaker conditions,

$$p \in G, p \leq q \implies q \in G,$$

and it is directed toward stronger conditions,

$$p, q \in G \implies \exists r \in G (r \leq p \text{ and } r \leq q).$$

The filter G is *V-generic* when it meets every dense set $D \subseteq \mathbb{P}$ belonging to V . In this volume, “generic” without a modifier means *V-generic*. For $G \subseteq \text{Add}(\omega, \kappa)$ put

$$g := \bigcup G.$$

Binary strings, real-as-subset convention, and Cohen reals. Put

$$2^{<\omega} := \bigcup_{n < \omega} {}^n 2,$$

the set of finite binary sequences, ordered by extension: $t \leq s$ means $t \supseteq s$ as functions. A set $D \subseteq 2^{<\omega}$ is *open* in this forcing order when $s \in D$ and $t \leq s$ imply $t \in D$, and it is *dense open* when it is both dense (in the sense of Volume VII) and open.

In this forcing volume a *real* means literally a subset $r \subseteq \omega$. Its characteristic function is

$$\chi_r(n) := \begin{cases} 1, & n \in r, \\ 0, & n \notin r, \end{cases}$$

and we use the harmless bit notation $r(n) := \chi_r(n)$. Its initial-segment filter is

$$G_r := \{\chi_r \upharpoonright n : n < \omega\} \subseteq 2^{<\omega}.$$

The real r is *Cohen over* V when $G_r \cap D \neq \emptyset$ for every dense open $D \subseteq 2^{<\omega}$ belonging to V . Thus subsets of ω are the actual objects named by forcing names, while 2^ω is used through the explicitly displayed characteristic-function bijection.

Forcing names and check names (recall from Volume VII). A $\mathbb{P}$ -name is built recursively as a set of pairs $\langle \sigma, p \rangle$ with $p \in \mathbb{P}$ and σ a previously built $\mathbb{P}$ -name. For a ground-model set a , its canonical name is defined recursively by

$$\check{a} := \{\langle \check{b}, 1_{\mathbb{P}} \rangle : b \in a\}.$$

Canonical coordinate name. For $\alpha < \kappa$, define

$$\dot{c}_\alpha := \{\langle \check{n}, p \rangle : p \in \text{Add}(\omega, \kappa), (\alpha, n) \in \text{dom}(p), p(\alpha, n) = 1\}.$$

This is a forcing name. If a generic filter G is later chosen, c_α denotes the actual real $(\dot{c}_\alpha)^G$; the dot is retained whenever the forcing relation is applied.

Forcing without a displayed condition. As in Volume VII, $p \Vdash \varphi$ is the forcing relation. We write $\Vdash \varphi$ as shorthand for $1_{\mathbb{P}} \Vdash \varphi$.

The definitions above specify only the forcing poset, genericity, and the coordinate names. The assertions that the generic union is total, that its rows are Cohen reals, and that distinct rows are distinct reals are *not* being assumed here; they are the conclusions of the propositions and theorems below.

How this chapter fits together

The chapter first proves the compatibility and separativity properties of $\text{Add}(\omega, \kappa)$. It then proves, in order, totality of the generic union, Cohen genericity of each coordinate, and pairwise distinctness of the coordinate reals. Later counting arguments use only those established results.

Proposition 56.1 (Definition and separativity of $\text{Add}(\omega, \kappa)$). *For an infinite cardinal κ ,*

$$\text{Add}(\omega, \kappa) = \text{Fn}(\kappa \times \omega, 2, <\omega)$$

ordered by reverse inclusion has the following explicit compatibility criterion:

$$p, q \text{ are compatible} \iff p \upharpoonright (\text{dom } p \cap \text{dom } q) = q \upharpoonright (\text{dom } p \cap \text{dom } q).$$

When they are compatible, $p \cup q$ is a common strengthening. Moreover the forcing is separative: if $p \not\leq q$, some extension of p is incompatible with q .

Proof.

Formal derivation spine

Formal spine for Definition and separativity of $\text{Add}(\omega, \kappa)$.

$$\text{Add}(\omega, \kappa) = \text{Fn}(\kappa \times \omega, 2, <\omega),$$

For an infinite cardinal κ , define

$$\text{Add}(\omega, \kappa) = \text{Fn}(\kappa \times \omega, 2, <_\omega),$$

the set of finite partial functions from $\kappa \times \omega$ to 2, ordered by reverse inclusion: $q \leq p$ iff $q \supseteq p$. Thus a stronger condition specifies more bits.

Two finite partial functions have a common extension exactly when they agree wherever both are defined. If they agree on their common domain, their union is again a finite partial function and is a common strengthening; if they disagree at some common coordinate, no function can extend both. This proves the displayed compatibility criterion and the assertion about $p \cup q$.

The forcing is separative. If $p \not\leq q$, either p already disagrees with q on a common coordinate or q specifies a coordinate not fixed by p . In the latter case extend p at that coordinate with the opposite bit; the resulting condition is incompatible with q . Hence every failure of the order can be witnessed by a stronger condition incompatible with the alleged upper condition, exactly as separativity requires. *Why reverse inclusion is the order.* A condition is information: it specifies finitely many bits. Extending a partial function adds information and therefore makes the condition stronger, so $q \leq p$ means $q \supseteq p$. In the separativity argument, if $p \not\leq q$, the goal is to strengthen p until it explicitly conflicts with q . If they already disagree, nothing must be added; if q mentions a fresh coordinate, assigning the opposite bit at that coordinate creates the required incompatibility. $\square$

Theorem 56.2 (The generic object yields kappa Cohen reals). *If $G \subseteq \text{Add}(\omega, \kappa)$ is generic and $g = \bigcup G$, then $g : \kappa \times \omega \rightarrow 2$ is total. For each $\alpha < \kappa$, define the subset*

$$c_\alpha := \{n < \omega : g(\alpha, n) = 1\}.$$

Equivalently, in the characteristic-bit notation fixed above,

$$c_\alpha(n) = g(\alpha, n).$$

Then c_α is a Cohen real over the ground model, and the canonical name satisfies the literal set equality

$$(\dot{c}_\alpha)^G = c_\alpha.$$

Proof.

Formal derivation spine

Formal spine for The generic object yields kappa Cohen reals.

$$D_{\alpha, n} = \{p : (\alpha, n) \in \text{dom } p\}$$

Let G be generic and put $g = \bigcup G$. Directedness of G guarantees that the union is a function: two conditions in G have a common extension and therefore cannot assign conflicting values to one coordinate. For every (α, n) , the set

$$D_{\alpha, n} = \{p : (\alpha, n) \in \text{dom } p\}$$

is dense, so genericity makes g total on $\kappa \times \omega$.

For each $\alpha < \kappa$, put $c_\alpha = \{n : g(\alpha, n) = 1\}$. By the characteristic-bit convention, $c_\alpha(n) = g(\alpha, n)$. By the definition of name valuation,

$$n \in (\dot{c}_\alpha)^G \iff \exists p \in G ((\alpha, n) \in \text{dom}(p) \text{ and } p(\alpha, n) = 1) \iff c_\alpha(n) = 1.$$

Hence $(\dot{c}_\alpha)^G = c_\alpha$. If $D \subseteq 2^{<\omega}$ is a ground-model dense open set, let

$$E_{\alpha, D} := \{p \in \text{Add}(\omega, \kappa) : \exists s \in D \forall n < \text{lh}(s) [p(\alpha, n) = s(n)]\}.$$

This set is dense. Given a finite condition p , first extend its finitely many already specified α -bits to a finite initial string $s_0 \in 2^{<\omega}$; density of D gives $s \in D$ extending s_0 , and adjoining the finitely many bits of s to the

α -section produces a condition in $E_{\alpha,D}$ below p . Genericity therefore meets $E_{\alpha,D}$, so some initial segment of χ_{c_α} lies in D . Thus the initial-segment filter G_{c_α} meets every such D , and c_α is a Cohen real over the ground model. *Two separate uses of genericity.* The dense sets $D_{\alpha,n}$ make each coordinate sequence total: every bit is eventually decided. A different family of dense sets, obtained by lifting a dense open set $D \subseteq 2^{<\omega}$ to the α -coordinate of the product forcing, makes that total sequence Cohen generic. Totality and Cohen genericity are therefore distinct claims, and both are proved by meeting the appropriate dense sets. $\square$

Proposition 56.3 (Distinct coordinates yield distinct reals). *If $\alpha \neq \beta < \kappa$, then*

$$\Vdash \dot{c}_\alpha \neq \dot{c}_\beta.$$

Indeed, conditions forcing the two canonical coordinate names to disagree at some fresh n are dense.

Proof.

Formal derivation spine

Formal spine for Distinct coordinates yield distinct reals.

$$D_{\alpha,\beta} = \{q : \exists n \ q(\alpha, n) \neq q(\beta, n)\}$$

Fix distinct coordinates $\alpha \neq \beta < \kappa$. Given any condition p , choose $n \in \omega$ outside the finite set of integers occurring in the α - or β -sections of $\text{dom } p$. Extend p by assigning $p(\alpha, n) = 0$ and $p(\beta, n) = 1$. Thus the set

$$D_{\alpha,\beta} = \{q : \exists n \ q(\alpha, n) \neq q(\beta, n)\}$$

is dense. A generic filter meets it, so the union generic satisfies $c_\alpha(n) \neq c_\beta(n)$ for some n . Therefore $c_\alpha \neq c_\beta$ in every generic extension. Equivalently, the set of conditions forcing $\dot{c}_\alpha \neq \dot{c}_\beta$ is dense. To pass from density to the displayed top-condition statement, suppose toward a contradiction that $1_{\mathbb{P}} \nVdash \dot{c}_\alpha \neq \dot{c}_\beta$. By decision density there would be a condition $p \Vdash \dot{c}_\alpha = \dot{c}_\beta$; strengthening p into the dense disagreement set would then force both equality and inequality, contradicting consistency of the forcing relation. Hence

$$1_{\mathbb{P}} \Vdash \dot{c}_\alpha \neq \dot{c}_\beta.$$

After valuation, the map $\alpha \mapsto c_\alpha$ is injective in every generic extension.

Further explanation. For distinct $\alpha, \beta < \kappa$, the set of conditions that assign opposite values to (α, n) and (β, n) for some fresh n is dense. Given any finite condition, choose n outside both coordinate domains and extend the condition by setting, say, $p(\alpha, n) = 0$ and $p(\beta, n) = 1$. A generic filter meets this dense set, so $c_\alpha(n) \neq c_\beta(n)$ for some n . Hence the coordinate reals are pairwise distinct. $\square$

Worked Theorem 56.4 (One Cohen real: explicit dense sets and genericity tests). *The forcing $\text{Add}(\omega, 1)$ is naturally identified with $\text{Fn}(\omega, 2, <\omega)$, the poset of arbitrary finite partial binary functions. It is not literally $2^{<\omega}$. The initial-segment conditions*

$$D_{\text{init}} := \{p \in \text{Fn}(\omega, 2, <\omega) : \text{dom}(p) = n \text{ for some } n < \omega\}$$

form a dense suborder canonically isomorphic to $2^{<\omega}$. The sets deciding the n -th bit and the sets forcing disagreement with each ground-model real are dense. Hence a generic filter has a total union $g = \bigcup G \in 2^\omega$. The associated real in the convention of this volume is the subset

$$c := \{n < \omega : g(n) = 1\} \subseteq \omega,$$

whose characteristic function is $\chi_c = g$; this real is not equal to any ground-model real.

Proof.

Formal derivation spine**Formal spine for One Cohen real: explicit dense sets and genericity tests.**

$$E_x = \{p : \exists n \in \text{dom } p \ (p(n) \neq x(n))\}$$

Identify $1 \times \omega$ with ω , so

$$\text{Add}(\omega, 1) \cong \text{Fn}(\omega, 2, <\omega).$$

The subset D_{init} displayed in the statement is dense: from any finite partial function, choose N larger than every point of its domain and assign arbitrary bits to the missing integers below N . Ordered by reverse inclusion, D_{init} is exactly the usual initial-segment forcing $2^{<\omega}$.

The dense set $D_n = \{p : n \in \text{dom } p\}$ forces the generic union to decide the n -th bit. For a fixed ground-model real x , the set

$$E_x = \{p : \exists n \in \text{dom } p \ (p(n) \neq x(n))\}$$

is dense: extend any condition at a fresh coordinate with the opposite bit. Therefore the generic union g is total. If $c := \{n : g(n) = 1\}$, then $g = \chi_c$, and meeting E_x says exactly that $c(n) \neq x(n)$ for some n . Hence the real $c \subseteq \omega$ differs from every ground-model real.

This concrete calculation is the prototype for the many-coordinate forcing. Adding a new real, however, does not by itself determine the new value of the continuum; the later name-counting argument is still required. *Two families of dense requirements are visible here.* The sets D_n guarantee that the generic union is a total binary sequence. The sets E_x , one for each ground-model real x , guarantee novelty by forcing disagreement somewhere. A Cohen real is therefore not merely an infinite string of bits; it is a string chosen so as to meet every ground-model dense open requirement. $\square$

Worked example

Worked Example 56.5 (Forcing two coordinates to differ). Given distinct $\alpha, \beta < \kappa$ and a condition p , choose a fresh n not mentioned at either coordinate. Extend p by assigning $(\alpha, n) \mapsto 0$ and $(\beta, n) \mapsto 1$. Thus the set of conditions forcing $\dot{c}_\alpha \neq \dot{c}_\beta$ is dense.

A useful warning

The union of a generic filter is total only because the filter meets the dense set deciding each coordinate. Totality is not automatic from directedness alone.

Looking ahead

The next chapter, *The ccc*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 57

The ccc

Definitions and setup

Formal definitions used in this chapter

Δ -system and root. A family $\mathcal{A}$ of sets is a Δ -system with *root* r when

$$A \cap B = r \quad (A \neq B \in \mathcal{A}).$$

Antichain and ccc. An *antichain* in a forcing notion is a set of pairwise incompatible conditions. A forcing notion satisfies the *countable chain condition* (ccc) when every antichain is countable.

Cohen domains and restrictions. For $p \in \text{Add}(\omega, \kappa)$, $\text{dom}(p)$ is its finite set of coordinates. If $r \subseteq \text{dom}(p)$, then $p \restriction r$ is the restriction of p to r . By Proposition 56.1, two Cohen conditions p, q are compatible exactly when

$$p \restriction (\text{dom } p \cap \text{dom } q) = q \restriction (\text{dom } p \cap \text{dom } q),$$

in which case $p \cup q$ is a common strengthening.

The Δ -system lemma says that an uncountable family of finite sets has an uncountable subfamily with common pairwise intersection (the root). Applied to finite domains of Cohen conditions, and followed by thinning so root assignments agree, it produces compatible pairs. Hence Cohen forcing is ccc. This combinatorial theorem is the owner of the later cardinal-preservation input.

How this chapter fits together

The Δ -system lemma converts an uncountable family of finite domains into a large subfamily with the same pairwise intersection. After thinning once more so that conditions agree on that root, any two Cohen conditions are compatible. Hence no uncountable antichain exists.

Theorem 57.1 (Delta-system lemma). *Every uncountable family of finite sets contains an uncountable subfamily forming a Δ -system: there is a fixed finite root r such that*

$$A \cap B = r$$

for all distinct members A, B of the subfamily.

Proof.

Formal derivation spine

Formal spine: induct on the common finite size. After thinning to sets of one size n , either some point belongs to uncountably many members, reducing the remainders to size $n - 1$, or every point occurs only countably often, allowing an uncountable pairwise-disjoint subfamily. In both cases obtain an uncountable family $\mathcal{B}$ and finite r with

$$A \cap B = r \quad (A \neq B \in \mathcal{B}).$$

Thin the given uncountable family first so that every member has the same finite size n . We induct on n . For $n = 0$ the family consists only of $\emptyset$, so the assertion is trivial after deleting repetitions.

Assume the result for sets of size n , and let $\mathcal{A}$ be an uncountable family of sets of size $n + 1$. There are two cases.

If some point x belongs to uncountably many members of $\mathcal{A}$, restrict to that uncountable subfamily and remove x from every member. The resulting family consists of n -element sets, so by induction it has an uncountable Δ -subsystem with root r . Restoring x gives a Δ -system with root $r \cup \{x\}$.

Otherwise every point belongs to at most countably many members of $\mathcal{A}$. Recursively choose $A_\xi \in \mathcal{A}$ for $\xi < \omega_1$ so that A_ξ is disjoint from all earlier choices. At stage $\xi < \omega_1$, the union $U_\xi = \bigcup_{\eta < \xi} A_\eta$ is countable. Each point of U_ξ occurs in only countably many members of $\mathcal{A}$, so only countably many members of $\mathcal{A}$ meet U_ξ . Since $\mathcal{A}$ is uncountable, choose A_ξ outside them. The resulting uncountable family is pairwise disjoint, hence a Δ -system with root $\emptyset$.

The two cases complete the induction. □

Theorem 57.2 ($\text{Add}(\omega, \kappa)$ is ccc). *For every cardinal κ , $\text{Add}(\omega, \kappa)$ satisfies the countable chain condition: every antichain is countable.*

Commented symbolic trace: the Δ -system ccc argument

A Cohen condition has a finite domain. From an uncountable family of finite domains, thin to a Δ -system whose pairwise intersections are the same finite root. There are only finitely many assignments on that root, so thin again until all conditions agree there. Outside the root the domains are disjoint. Therefore any two surviving conditions have a union that is still a function, hence are compatible. This directly contradicts the possibility of an uncountable antichain.

Proof.

Formal derivation spine

Formal spine: apply the Δ -system lemma to Cohen supports. Thin an uncountable family to domains with common root r , then thin again so all conditions agree on r . For distinct survivors p, q ,

$$p \restriction r = q \restriction r, \quad (\text{dom } p \setminus r) \cap (\text{dom } q \setminus r) = \emptyset,$$

so $p \cup q$ is a condition extending both. Hence no uncountable antichain exists.

Proof map. Apply the Δ -system lemma to the finite domains of an uncountable family of Cohen conditions. There are only finitely many assignments on a fixed finite root, so thin to an uncountable subfamily agreeing there. Any two remaining conditions have disjoint non-root domains and identical root values; their union is therefore a common extension.

Let $\mathcal{A}$ be an uncountable family of conditions in $\text{Add}(\omega, \kappa)$. Their domains are finite subsets of $\kappa \times \omega$. By the Δ -system lemma, thin $\mathcal{A}$ to an uncountable subfamily whose domains have a common finite root r and are otherwise pairwise disjoint. There are only finitely many assignments $r \rightarrow 2$, so thin once more until all conditions agree on r . Any two conditions in the resulting family have union a finite partial function:

they agree on the intersection and have disjoint off-root domains. Hence they are compatible. Therefore no uncountable antichain exists and $\text{Add}(\omega, \kappa)$ is ccc.

Detailed formal verification. The ccc proof has two finite combinatorial reductions. First apply the Δ -system lemma to the finite domains of an alleged uncountable antichain. Second thin so all functions agree on the fixed root. Then for distinct conditions

$$p \cup q$$

is a common extension, contradicting antichainhood. Every compatibility assertion is therefore reduced to equality of finitely many bits on the root. $\square$

Remark 57.3 (Source note). The ccc proof for Cohen forcing is the standard finite-domain Δ -system argument; compare [9, 10, 11, 13].

Proposition 57.4 (Finite-support compatibility analysis). *If two Cohen conditions have finite domains whose intersection is a common root on which the conditions agree, then their union is a condition extending both. Thus Δ -system thinning reduces ccc to compatibility on finitely many root coordinates.*

Proof.

Formal derivation spine

Formal spine: Cohen compatibility is agreement on overlap. If

$$p \restriction (\text{dom } p \cap \text{dom } q) = q \restriction (\text{dom } p \cap \text{dom } q),$$

then

$$p \cup q \in \text{Add}(\omega, \kappa), \quad p \cup q \leq p, q.$$

For a Δ -system, the overlap is exactly the common root, so root agreement is the whole compatibility test.

A Cohen condition is a finite partial function. Two partial functions p, q have a common extension exactly when they agree on every point where both are defined. Under the stated hypothesis they agree on

$$r := \text{dom } p \cap \text{dom } q.$$

Hence $p \cup q$ is still a function. Its domain is the finite set $\text{dom } p \cup \text{dom } q$, so $p \cup q \in \text{Add}(\omega, \kappa)$. Because the forcing order is reverse inclusion,

$$p \cup q \leq p \quad \text{and} \quad p \cup q \leq q.$$

Thus p and q are compatible.

For a Δ -system of domains, every pairwise intersection is the same finite root r . Therefore, after thinning so that the conditions have the same values on r , the displayed compatibility criterion applies to every pair. No global property of the off-root coordinates is needed beyond their pairwise disjointness. $\square$

Corollary 57.5 (Preservation consequences for $\text{Add}(\omega, \kappa)$). *Because $\text{Add}(\omega, \kappa)$ is ccc, it preserves ω_1 and, under the cardinal-preservation theorem of Volume VIII, the uncountable ground-model cardinals used in the target continuum calculation.*

Proof.

Formal derivation spine

Formal spine: import the preservation theorem only after ccc is proved. From

$$\text{Add}(\omega, \kappa) \text{ is ccc}$$

and Volume VIII's ccc preservation theorem,

$$\omega_1^{V[G]} = \omega_1^V.$$

In the target case $\kappa = \omega_2^V$, the corresponding cardinal/cofinality preservation result gives

$$\omega_2^{V[G]} = \omega_2^V,$$

so later continuum inequalities may be read as inequalities involving $\aleph_1, \aleph_2$ in the extension.

The preceding theorem proves that $\text{Add}(\omega, \kappa)$ is ccc. Apply the ccc preservation theorem from Volume VIII. It gives directly

$$\omega_1^{V[G]} = \omega_1^V.$$

In the continuum calculation used later we take a ground model of GCH and $\kappa = \omega_2^V$. The cardinal-preservation result established in Volume VIII applies to this Cohen poset and yields that the ground ordinal ω_2^V is not collapsed; hence

$$\omega_2^{V[G]} = \omega_2^V.$$

These are preservation statements about the identities of ground-model ordinals as cardinals. They do not yet count reals. The lower bound $2^{\aleph_0} \geq \aleph_2$ will come from the coordinate Cohen reals, and the upper bound will come separately from nice-name counting. Keeping these steps separate prevents a cardinal-arithmetic conclusion from being smuggled into the preservation theorem itself. $\square$

Worked example

Worked Example 57.6 (A root calculation). Suppose conditions p_i have domains forming a Δ -system with root r , and all $p_i \restriction r$ are equal. For $i \neq j$, the non-root parts of the domains are disjoint, so $p_i \cup p_j$ is still a function. It extends both conditions, proving compatibility.

A useful warning

It is not enough for finite domains merely to intersect in the same root; the conditions must also agree on the values assigned on that root.

Looking ahead

The next chapter, *Lower bound for the continuum*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 58

Lower bound for the continuum

Definitions and setup

Formal definitions used in this chapter

Ground-model/extension notation. V is the ground model and $V[G]$ its generic extension. If a cardinal or ordinal is computed in a specified universe, a superscript records that universe: for example ω_2^V and $\omega_2^{V[G]}$. A ground-model cardinal κ is *preserved* when the same ordinal is still a cardinal in $V[G]$.

Continuum. In a universe W ,

$$(2^{\aleph_0})^W := |\mathcal{P}(\omega)^W|.$$

Thus $V[G] \models 2^{\aleph_0} \geq \kappa$ means that, in $V[G]$, there is an injection from the preserved cardinal κ into $\mathcal{P}(\omega)$.

CH and GCH. The Continuum Hypothesis is $2^{\aleph_0} = \aleph_1$. The Generalized Continuum Hypothesis (GCH) is the scheme

$$2^\lambda = \lambda^+$$

for every infinite cardinal λ . A cardinal κ is *regular* when $\text{cf}(\kappa) = \kappa$.

Preservation input. We use the ccc cardinal-preservation theorem of Volume VIII: ccc forcing preserves cardinals and cofinalities at the uncountable cardinals needed below. In particular, for the target forcing $\text{Add}(\omega, \omega_2^V)$, ω_1^V and ω_2^V remain ω_1 and ω_2 of the extension.

The lower bound comes from distinct generic coordinates, but cardinal language is used only after ccc preservation identifies the ground-model indexing cardinal with the corresponding extension cardinal. In the target case, ω_2^V remains $\aleph_2^{V[G]}$, and $\alpha \mapsto c_\alpha$ is an injection into the extension reals.

How this chapter fits together

The coordinate generics give an injection from κ into the reals of the extension. Turning that injection into the cardinal inequality $2^{\aleph_0} \geq \kappa$ requires knowing that the ground-model cardinal κ has not been collapsed. The preservation theorems therefore precede the continuum comparison.

Theorem 58.1 (Cardinal preservation for the specific Cohen forcing used). *In the target case $\mathbb{P} = \text{Add}(\omega, \omega_2^V)$ over a GCH ground model, ccc implies*

$$(\omega_1^V)^{V[G]} = \omega_1^{V[G]}, \quad (\omega_2^V)^{V[G]} = \omega_2^{V[G]}.$$

Proof.

Formal derivation spine**Formal spine for Cardinal preservation for the specific Cohen forcing used.**

$$\kappa \hookrightarrow 2^\omega,$$

The forcing $\text{Add}(\omega, \kappa)$ is ccc by the Δ -system argument. Volume VIII proved that ccc forcing preserves every ground-model infinite cardinal: a hypothetical new surjection from a smaller cardinal onto a larger one would, coordinate by coordinate, have only countably many possible values, and the union of those possibilities would still be too small to cover the target.

Applying that theorem to the present forcing shows that the ground-model cardinals used in the continuum calculation are not collapsed. In particular, when $\kappa = \omega_2^V$, this ordinal remains a cardinal of the extension and may later be identified with $\aleph_2^{V[G]}$. *What is being preserved.* Forcing never changes the ordinals, but an old ordinal can cease to be a cardinal if a new bijection or surjection collapses it. The ccc preservation theorem rules out the relevant new surjections. Hence the ordinal that the ground model called ω_2 is still the second uncountable cardinal in the extension. This identification is essential before the later statement “there are $\aleph_2$ Cohen reals” can be interpreted as a lower bound on the extension continuum. $\square$

Theorem 58.2 (At least kappa distinct coordinate reals in the extension). *Let $G \subseteq \text{Add}(\omega, \kappa)$ be V -generic. Then*

$$\alpha \mapsto c_\alpha = (\dot{c}_\alpha)^G$$

is an injection from the ordinal κ into $\mathcal{P}(\omega)^{V[G]}$. If κ is preserved as a cardinal, then

$$V[G] \models 2^{\aleph_0} \geq \kappa.$$

Proof.

Formal derivation spine**Formal spine for At least kappa distinct coordinate reals in the extension.**

$$V[G] \models 2^{\aleph_0} \geq \aleph_2.$$

$$\kappa \hookrightarrow 2^\omega,$$

The coordinate construction supplies an injection $\kappa \rightarrow \mathcal{P}(\omega)$, $\alpha \mapsto c_\alpha$, in the extension. The ccc preservation theorems from Volume VIII ensure that in the target case the ground-model ω_1, ω_2 remain the corresponding extension cardinals. Thus, when $\kappa = \omega_2^V$,

$$V[G] \models 2^{\aleph_0} \geq \aleph_2.$$

No cardinal comparison is made until preservation has been established.

Point specific to this result. The map $\alpha \mapsto c_\alpha$ is injective because dense sets force any two coordinate reals to differ somewhere. After preservation identifies κ with the same extension cardinal, this is the lower bound $2^{\aleph_0} \geq \kappa$.

Further explanation. The map $\alpha \mapsto c_\alpha$ is an injection from the ground-model index set κ into the set of reals of the extension because distinct coordinates give distinct reals. Once the required preservation theorem identifies the ground-model κ with the same cardinal in the extension, this injection becomes the cardinal inequality $2^{\aleph_0} \geq \kappa$. Thus the lower bound is a direct consequence of the geometry of the generic object, not of name counting. $\square$

Proposition 58.3 (Preservation of ω_1 and ω_2 in the CH/GCH ground-model case). *If the ground model satisfies CH/GCH and $G \subseteq \text{Add}(\omega, \omega_2)$ is generic, then ccc preserves the ordinals and cardinal identities of ω_1 and ω_2 .*

Proof.

Formal derivation spine

Formal spine for Preservation of ω_1 and ω_2 in the CH/GCH ground-model case.

$$\omega_1^V = \omega_1^{V[G]} \quad \text{and} \quad \omega_2^V = \omega_2^{V[G]}.$$

$$\kappa \hookrightarrow 2^\omega,$$

Because $\text{Add}(\omega, \omega_2)$ is ccc, Volume VIII's preservation theorem gives

$$\omega_1^V = \omega_1^{V[G]} \quad \text{and} \quad \omega_2^V = \omega_2^{V[G]}.$$

For ω_1 , this is also the direct cofinality argument: a name for a countable cofinal sequence would have only countably many possible values at each coordinate, whose union is still countable and therefore bounded below ω_1^V . Preservation of all infinite cardinals rules out collapsing ω_2^V onto ω_1 or any smaller cardinal.

Hence the symbols $\aleph_1$ and $\aleph_2$ in the remaining continuum calculation denote the same ordinals in the ground model and the extension.

This is the exact point where the cardinal identities needed by the next two inequalities are fixed. From here onward the notation $\aleph_2$ is unambiguous across the ground model and extension for this forcing construction. $\square$

Corollary 58.4 ($2^{\aleph_0} \geq \kappa$). *In $V[G]$ after forcing with $\text{Add}(\omega, \kappa)$,*

$$2^{\aleph_0} \geq \kappa$$

whenever the ground-model κ is preserved as a cardinal.

Proof.

Formal derivation spine

Formal spine for $2^{\aleph_0} \geq \kappa$.

$$\kappa \hookrightarrow 2^\omega,$$

For every $\alpha < \kappa$, the generic union produces a real $c_\alpha \subseteq \omega$, whose characteristic function lies in 2^ω . If $\alpha \neq \beta$, the set of conditions which choose a fresh n and assign different bits to (α, n) and (β, n) is dense, so $c_\alpha \neq c_\beta$. Hence $\alpha \mapsto c_\alpha$ is an injection $\kappa \hookrightarrow \mathcal{P}(\omega)$ in the extension. Therefore $2^{\aleph_0} \geq \kappa$.

Further explanation. This is deliberately a short corollary, but one hypothesis is worth making visible: the index cardinal must have survived the forcing. With that preservation already proved, the injection $\alpha \mapsto c_\alpha$ has domain of cardinality κ in the extension itself. Hence the extension contains at least κ reals, which is exactly the displayed lower bound.

No upper bound is contained in this argument. The opposite inequality requires the independent nice-name counting theorem, which is why the continuum calculation is deliberately split into two proofs. $\square$

Worked example

Worked Example 58.5 (The ω_2 lower bound). With $\kappa = \omega_2^V$, the family $\{c_\alpha : \alpha < \omega_2^V\}$ consists of pairwise distinct reals. Since ccc preserves ω_2 in the stated ground-model setting, the same ordinal is $\aleph_2$ in the extension. Thus $2^{\aleph_0} \geq \aleph_2$.

A useful warning

New reals alone do not determine the new continuum. A lower bound must be paired with a separate upper-bound argument.

Looking ahead

The next chapter, *Upper bound by nice names*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 59

Upper bound by nice names

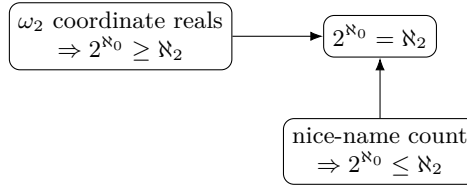


Figure 59.1: The Cohen continuum calculation is a squeeze: distinct coordinate generics give the lower bound and nice-name counting gives the upper bound.

Definitions and setup

Formal definitions used in this chapter

Real name. A $\mathbb{P}$ -name $\dot{r}$ is a *name for a real* when $1_{\mathbb{P}} \Vdash \dot{r} \subseteq \check{\omega}$. Here $\check{n}$ is the canonical ground-model name for n .

Nice name for a real. A $\mathbb{P}$ -name for a subset of ω is *nice* when it has the form

$$\dot{r} = \{ \langle \check{n}, p \rangle : n < \omega, p \in A_n \},$$

where every $A_n \subseteq \mathbb{P}$ is an antichain. A *maximal antichain below* p is an antichain contained in $\{q : q \leq p\}$ that is maximal there under inclusion. Under ccc, every A_n is countable.

Ground-model counting convention. All collections of names and all cardinal-arithmetic estimates in this chapter are formed in V . If $|\mathbb{P}| = \kappa$, a countable subset of $\mathbb{P}$ is coded by a member of $\mathbb{P}^{\omega}$, so there are at most $\kappa^{\aleph_0}$ such subsets. The hypothesis $\kappa^{\aleph_0} = \kappa$ is therefore the cardinal-arithmetic input to the upper bound.

Nice-name normal form. Saying that every real has a nice name means: for every name τ with $1_{\mathbb{P}} \Vdash \tau \subseteq \check{\omega}$, there is a nice name $\dot{r}$ with $1_{\mathbb{P}} \Vdash \tau = \dot{r}$.

A nice name for a subset of ω is determined, for each integer n , by an antichain of conditions forcing n into the set. Under ccc each such antichain is countable. Thus counting real names reduces to counting countable subsets/sequences from the forcing poset in the ground model. The construction normalizes arbitrary names without changing their forced value.

How this chapter fits together

Every real has a nice name, and a nice name is coded by a countable sequence of antichains. Under GCH and $|\text{Add}(\omega, \omega_2)| = \aleph_2$, there are only $\aleph_2$ such codes. This supplies the upper bound matching the coordinate lower bound.

Theorem 59.1 (Nice-name normal form for Cohen names for reals). *Let τ be an $\text{Add}(\omega, \kappa)$ -name with*

$$\Vdash \tau \subseteq \check{\omega}.$$

Then there is a nice name

$$\dot{r} = \{\langle \check{n}, p \rangle : n < \omega, p \in A_n^+\}$$

such that $\Vdash \tau = \dot{r}$, where A_n is a maximal antichain deciding $\check{n} \in \tau$ and

$$A_n^+ := \{p \in A_n : p \Vdash \check{n} \in \tau\}.$$

Proof.

Formal derivation spine

Formal spine: decide each integer on one maximal antichain. For every $n < \omega$, choose a maximal antichain A_n deciding $\check{n} \in \tau$, and put

$$A_n^+ := \{p \in A_n : p \Vdash \check{n} \in \tau\}, \quad \dot{r} := \{\langle \check{n}, p \rangle : n < \omega, p \in A_n^+\}.$$

Let τ be as in the statement. For each $n < \omega$, decision density gives a maximal antichain A_n every member of which forces either $\check{n} \in \tau$ or $\check{n} \notin \tau$. Retain the positive part

$$A_n^+ = \{p \in A_n : p \Vdash \check{n} \in \tau\}$$

and form

$$\dot{r} := \{\langle \check{n}, p \rangle : n < \omega, p \in A_n^+\}.$$

This is a forcing name because it is a set of pairs of check names and conditions.

We prove that the two names have the same value in every generic extension. Let G be generic. If

$$n \in \dot{r}^G,$$

then some $p \in G \cap A_n^+$ satisfies $p \Vdash \check{n} \in \tau$. The Truth Lemma yields

$$n \in \tau^G. \tag{N1}$$

Conversely, suppose $n \in \tau^G$. A generic filter meets the maximal antichain A_n in exactly one condition p . That condition decides $\check{n} \in \tau$. It cannot force nonmembership, because then the Truth Lemma would give $n \notin \tau^G$. Hence $p \in A_n^+$, and therefore

$$n \in \dot{r}^G. \tag{N2}$$

From (N1)–(N2),

$$\dot{r}^G = \tau^G$$

for every generic G . By the Fundamental Theorem of Forcing,

$$1_{\mathbb{P}} \Vdash \dot{r} = \tau.$$

Thus $\dot{r}$ is the desired nice name. The ccc is not needed for the normalization itself; it is used in the next result to make each maximal antichain A_n countable and hence to count the possible nice names. $\square$

Proposition 59.2 (Counting nice names under ground-model GCH). *Assume ground-model GCH and let $\kappa \geq \aleph_2$ be regular with $\kappa^{\aleph_0} = \kappa$. Then the set of nice $\text{Add}(\omega, \kappa)$ -names for reals has cardinality at most κ .*

How to read the symbols: counting nice names is counting structured programs

A nice name for a real is not an arbitrary set-theoretic name. For each $n < \omega$, one countable antichain records the conditions that force n into the real. Thus the name is coded by a countable sequence of countable subsets of $\mathbb{P}$. If $|\mathbb{P}| = \kappa$, there are at most $\kappa^{\aleph_0}$ choices for one countable subset and at most $(\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0}$ sequences. Under the stated GCH arithmetic this becomes κ .

Proof.

Formal derivation spine

Formal spine for Counting nice names under ground-model GCH.

$$\begin{aligned} \dot{r} &\longleftrightarrow \langle A_n^+ : n < \omega \rangle, \\ \{\text{nice real names}\} &\hookrightarrow ([\mathbb{P}]^{\leq \aleph_0})^\omega, \\ |([\mathbb{P}]^{\leq \aleph_0})^\omega| &\leq (\kappa^{\aleph_0})^{\aleph_0} = \kappa. \end{aligned}$$

Proof map. A nice name is a countable sequence of countable antichain data. When $|\mathbb{P}| = \kappa$, each antichain is a countable subset of $\mathbb{P}$, so the number of possible antichains is at most $\kappa^{\aleph_0}$. A countable sequence of them contributes another exponent by $\aleph_0$, which collapses to the same cardinal under the stated GCH calculation.

A nice real name is determined by a sequence $\langle A_n : n < \omega \rangle$ of antichains, each countable by ccc. Since κ is infinite,

$$|\mathbb{P}| = |\text{Fn}(\kappa \times \omega, 2, < \omega)| = \kappa :$$

there are κ possible finite domains and only finitely many 0/1-assignments on each fixed finite domain. Hence the number of countable subsets of $\mathbb{P}$ is at most $\kappa^{\aleph_0}$. Hence the number of countable sequences of such antichains is at most

$$(\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0} = \kappa$$

by the hypothesis stated in the proposition. Therefore there are at most κ nice names for reals. No assertion about the number of reals in the extension is made in this proposition; the next theorem obtains that upper bound by combining this ground-model name count with the nice-name representation theorem.

Detailed formal verification. The upper bound has the opposite logical direction: normalize every real name, count the normalized names in the ground model, and then use the surjection

$$\{\text{nice names}\} \rightarrow (2^\omega)^{V[G]}.$$

Under $\kappa^{\aleph_0} = \kappa$, this gives at most κ reals. Nothing in this counting step produces new reals; it bounds how many values all possible names can have.

Second detailed calculation. Write the counting argument as a chain of injections. A nice real name is determined by a sequence

$$\langle A_n^+ : n < \omega \rangle,$$

where each A_n^+ is a countable antichain of $\mathbb{P} = \text{Add}(\omega, \kappa)$. Therefore

$$\{\text{nice real names}\} \hookrightarrow ([\mathbb{P}]^{\leq \aleph_0})^\omega.$$

Since $|\mathbb{P}| = \kappa$,

$$|[\mathbb{P}]^{\leq \aleph_0}| \leq \kappa^{\aleph_0}.$$

Hence

$$\#\{\text{nice real names}\} \leq (\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0 \cdot \aleph_0} = \kappa^{\aleph_0}.$$

Under the stated hypothesis $\kappa^{\aleph_0} = \kappa$, this becomes the required κ -bound. The argument counts names in the ground model; the next theorem converts that name count into an upper bound on reals in the extension. $\square$

Theorem 59.3 (Upper bound $2^{\aleph_0} \leq \kappa$ in the target case). *Assume κ is regular, $\kappa^{\aleph_0} = \kappa$ in the ground model, and $G \subseteq \text{Add}(\omega, \kappa)$ is generic. Then every real in $V[G]$ is represented by a nice name, there are at most κ such names, and therefore*

$$V[G] \models 2^{\aleph_0} \leq \kappa.$$

Proof.

Formal derivation spine

Formal spine for Upper bound $2^{\aleph_0} \leq \kappa$ in the target case.

$$\{\text{nice names}\} \rightarrow (2^\omega)^{V[G]}.$$

By the nice-name theorem, every real in the extension is represented by a sequence $\langle A_n : n < \omega \rangle$ of antichains of $\mathbb{P}$, where the positive part of A_n decides membership of n . Under ccc each A_n is countable. Thus the number of possible nice names is at most the number of countable sequences of countable subsets of $\mathbb{P}$, bounded by $(|\mathbb{P}|^{\aleph_0})^{\aleph_0} = |\mathbb{P}|^{\aleph_0}$. Here $|\mathbb{P}| = \kappa$ for every infinite κ , and the stated hypothesis $\kappa^{\aleph_0} = \kappa$ therefore gives at most κ nice real names. Since $\text{Add}(\omega, \kappa)$ is ccc, the preservation theorem from Volume VIII keeps this κ as the same extension cardinal. Hence $V[G] \models 2^{\aleph_0} \leq \kappa$.

Further explanation. The estimate is obtained by counting names, not by trying to enumerate the actual reals after forcing. Every real has a nice name, and a nice name is coded by countably many antichains. Under ccc those antichains are countable; under the ground-model cardinal arithmetic assumed in the theorem, the total number of such codes is at most κ . Since valuation maps names onto extension reals, this bounds the continuum from above by κ . $\square$

Theorem 59.4 (Exact continuum calculation $2^{\aleph_0} = \kappa$). *Assume κ is a preserved regular ground-model cardinal with $\kappa^{\aleph_0} = \kappa$, and let $G \subseteq \text{Add}(\omega, \kappa)$ be generic. The κ distinct coordinate Cohen reals give*

$$2^{\aleph_0} \geq \kappa,$$

while nice-name counting gives

$$2^{\aleph_0} \leq \kappa.$$

Hence

$$V[G] \models 2^{\aleph_0} = \kappa.$$

Uses: the coordinate lower bound, ccc cardinal preservation, and the nice-name upper bound.

Proof.

Formal derivation spine

Formal spine for Exact continuum calculation $2^{\aleph_0} = \kappa$.

$$\alpha \mapsto c_\alpha$$

$$V[G] \models 2^{\aleph_0} \geq \kappa. \tag{C1}$$

$$\#\{\text{nice real names}\} \leq (\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0} = \kappa. \tag{C2}$$

Proof map. The equality is deliberately assembled from independent inequalities. Coordinate generics produce κ distinct reals, yielding $2^{\aleph_0} \geq \kappa$ after preservation. Nice-name counting yields $2^{\aleph_0} \leq \kappa$. Nothing about forcing alone makes the two bounds equal; the proof needs both mechanisms.

The lower-bound theorem gives $2^{\aleph_0} \geq \kappa$ from the pairwise distinct coordinate Cohen reals. The upper-bound theorem gives $2^{\aleph_0} \leq \kappa$ from nice-name counting under the stated ground-model cardinal arithmetic. The two inequalities are evaluated in the same extension, and the preservation theorem identifies the relevant ground cardinal κ with its extension cardinal. Therefore $2^{\aleph_0} = \kappa$.

Why the squeeze is legitimate. Both inequalities are statements inside the same extension $V[G]$. The lower bound counts actual pairwise distinct coordinate reals in $V[G]$. The upper bound counts all possible nice names in the ground model and uses the Truth Lemma to know that every extension real is represented by one of them. Cardinal preservation identifies the ground-model parameter κ used in the forcing definition with the extension cardinal appearing on both sides. Only after those identifications do the two inequalities combine to an equality.

Detailed formal verification. Two independent estimates determine the continuum. First, the coordinate generics are pairwise distinct, so

$$\alpha \mapsto c_\alpha$$

is an injection $\kappa \hookrightarrow \mathcal{P}(\omega)$. Equivalently, after applying the characteristic-function bijection, $\alpha \mapsto \chi_{c_\alpha}$ injects κ into 2^ω . Hence

$$V[G] \models 2^{\aleph_0} \geq \kappa. \quad (\text{C1})$$

Second, every real has a nice name. A nice name is coded by a sequence $\langle A_n : n < \omega \rangle$ of antichains, and ccc makes each A_n countable. Thus

$$\#\{\text{nice real names}\} \leq (\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0} = \kappa. \quad (\text{C2})$$

Every real in the extension is the value of one of these names, so

$$V[G] \models 2^{\aleph_0} \leq \kappa. \quad (\text{C3})$$

The preservation theorem is used before (C1)–(C3) to ensure that the ground ordinal κ is still the same cardinal in $V[G]$. Combining (C1) and (C3) gives

$$V[G] \models 2^{\aleph_0} = \kappa.$$

□

Worked example

Worked Example 59.5 (Counting names rather than values). There may be many different names for the same real, so counting all arbitrary names is wasteful. Nice-name normalization chooses a restricted form: for each n , a countable antichain records the conditions forcing n into the real. The number of these structured codes is at most $(\aleph_2^{\aleph_0})^{\aleph_0} = \aleph_2$ under the ground-model GCH calculation.

A useful warning

The equality $\kappa^{\aleph_0} = \kappa$ is a ground-model cardinal-arithmetic input here; it must not be inferred from ccc alone.

Looking ahead

The next chapter, *CH and its failure*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 60

CH and its failure

Definitions and setup

Formal definitions used in this chapter

CH and its negation. In any universe,

$$\text{CH} :\iff 2^{\aleph_0} = \aleph_1, \quad \neg\text{CH} :\iff 2^{\aleph_0} \neq \aleph_1.$$

The target Cohen extension and the preservation input already proved. In this chapter the ground model satisfies GCH and

$$\mathbb{P} = \text{Add}(\omega, \omega_2^V).$$

By Theorem 58.1, if $G \subseteq \mathbb{P}$ is V -generic, ccc preservation identifies the ground-model ordinal ω_2^V with $\omega_2^{V[G]}$. Thus the assertion

$$V[G] \models 2^{\aleph_0} = \aleph_2$$

has an unambiguous extension-internal meaning. The preservation statement here is an explicit recall of Theorem 58.1, not part of the definition of the target forcing.

Forcing a sentence. $\mathbb{P} \Vdash \varphi$ and $\Vdash_{\mathbb{P}} \varphi$ both mean $1_{\mathbb{P}} \Vdash \varphi$.

Adding one Cohen real and adding $\aleph_2$ Cohen reals behave differently. A countable forcing over CH has only $\aleph_1$ nice real names, so CH can survive. By contrast $\text{Add}(\omega, \omega_2)$ supplies $\aleph_2$ distinct reals and still admits an $\aleph_2$ nice-name upper bound under GCH. The exact continuum calculation, not merely novelty of the generic real, yields $\neg\text{CH}$.

How this chapter fits together

The contrast between one Cohen real and ω_2 Cohen reals is pedagogically important. Adding a real need not destroy CH: over a CH ground model, the count of nice names for one Cohen real remains $\aleph_1$. Adding ω_2 many coordinate reals over GCH forces the lower bound up to $\aleph_2$, while the same nice-name machinery holds the upper bound there.

Worked Theorem 60.1 (Adding one Cohen real over CH can preserve CH). *If $V \models \text{CH}$ and $G \subseteq \text{Add}(\omega, 1)$ is generic, then*

$$V[G] \models \text{CH}.$$

Thus adding a single Cohen real need not destroy CH.

Proof.

Formal derivation spine

Formal spine: one Cohen real over CH still leaves only $\aleph_1$ reals. Since $\text{Add}(\omega, 1)$ is countable, every nice name for a real is coded by a countable sequence of subsets of a countable set. Under CH,

$$\#\{\text{nice real names}\} \leq (2^{\aleph_0})^{\aleph_0} = \aleph_1.$$

ccc preserves ω_1 , while the old reals give the reverse inequality, hence

$$V[G] \models 2^{\aleph_0} = \aleph_1.$$

Assume the ground model satisfies CH and force with the countable poset $\text{Add}(\omega, 1)$. It is ccc, so ω_1 is preserved. Every extension real has a nice name, and because the forcing is countable, a nice name is coded by a countable sequence of subsets of a countable set, hence by a ground-model real. Under CH there are only $\aleph_1$ such codes. Thus the extension has at most $\aleph_1$ reals, while it contains the old $\aleph_1$ reals. Consequently it still satisfies $2^{\aleph_0} = \aleph_1$. The example proves that adding a new real does not by itself force failure of CH.

Why this does not contradict the existence of a new real. CH says that the total number of reals is $\aleph_1$; it does not say that every real already belongs to the ground model. The forcing adds at least one genuinely new real, but a set of size $\aleph_1$ can acquire new elements while still having cardinality $\aleph_1$, provided the old and new reals together are still bounded by $\aleph_1$. Nice-name counting supplies exactly that global upper bound. $\square$

Theorem 60.2 ($\text{Add}(\omega, \omega_2)$ over GCH forces $2^{\aleph_0} = \aleph_2$). *If $V \models \text{GCH}$ and $G \subseteq \text{Add}(\omega, \omega_2^V)$ is generic, then*

$$V[G] \models 2^{\aleph_0} = \aleph_2.$$

Commented symbolic trace: the continuum is pinned between the same cardinal

The lower bound and upper bound come from different mechanisms. Coordinate generics give an injection $\omega_2 \rightarrow \mathcal{P}(\omega)$, so $2^{\aleph_0} \geq \aleph_2$. Nice-name counting gives at most $\aleph_2$ reals, so $2^{\aleph_0} \leq \aleph_2$. The ccc preservation theorem is needed before writing these inequalities with the extension's $\aleph_2$: it guarantees that the ground-model ω_2 has not collapsed.

Proof.

Formal derivation spine

Formal spine for $\text{Add}(\omega, \omega_2)$ over GCH forces $2^{\aleph_0} = \aleph_2$.

$$2^{\aleph_0} = \aleph_2,$$

$$\kappa := \omega_2^V.$$

$$\kappa^{\aleph_0} = \aleph_2,$$

Let the ground model satisfy GCH and force with $\mathbb{P} = \text{Add}(\omega, \omega_2)$. By the ccc theorem above, $\mathbb{P}$ is ccc; by the preservation results, the ground ω_1 and ω_2 remain $\aleph_1$ and $\aleph_2$ in the extension. The ω_2 coordinate generics are pairwise distinct reals, giving $2^{\aleph_0} \geq \aleph_2$. Conversely every real has a nice name, and the number of nice names is at most $(\aleph_2)^{\aleph_0} = \aleph_2$ by GCH, giving $2^{\aleph_0} \leq \aleph_2$. Hence

$$2^{\aleph_0} = \aleph_2,$$

so CH fails. The formal relative-consistency conclusion is taken only after applying the metatheoretic transfer theorem of Volume VIII.

Further explanation. The theorem is the point at which all four ingredients meet: ccc preserves the relevant cardinals, coordinate generics give $2^{\aleph_0} \geq \aleph_2$, nice-name counting gives $2^{\aleph_0} \leq \aleph_2$, and the ground-model GCH

hypothesis supplies the cardinal arithmetic needed in that count. Therefore the extension computes the continuum exactly as $\aleph_2$. Since $\aleph_2 > \aleph_1$, CH fails there.

Detailed formal verification. The one-real and many-real cases must be calculated separately. For $\text{Add}(\omega, 1)$, the forcing is countable, so under CH the set of nice names has size at most $\aleph_1$; ccc preserves $\aleph_1$, hence CH remains true. For $\text{Add}(\omega, \omega_2)$, the coordinate lower bound is $\aleph_2$, while GCH gives the matching nice-name upper bound. Thus the amount of Cohen forcing, not merely the existence of a new real, determines the continuum.

Second detailed calculation. In the target application put

$$\kappa := \omega_2^V.$$

GCH gives

$$\kappa^{\aleph_0} = \kappa,$$

and ccc preserves ω_1^V and ω_2^V , so in the extension

$$\kappa = \aleph_2^{V[G]}.$$

The lower-bound theorem gives

$$V[G] \models 2^{\aleph_0} \geq \aleph_2,$$

while the nice-name theorem and the preceding cardinal calculation give

$$V[G] \models 2^{\aleph_0} \leq \aleph_2.$$

Therefore

$$V[G] \models 2^{\aleph_0} = \aleph_2.$$

Since CH is the equation $2^{\aleph_0} = \aleph_1$, and ccc also preserves $\aleph_1$, the same extension satisfies $\neg\text{CH}$. $\square$

Remark 60.3 (Source note). Cohen's independence proof combines cardinal preservation, many generic reals, and an upper bound on real names. Compare [9, 10, 11].

Corollary 60.4 (The Cohen extension satisfies not-CH). *In the extension of the preceding theorem,*

$$V[G] \models \neg\text{CH}.$$

Proof.

Formal derivation spine

Formal spine for The Cohen extension satisfies not-CH.

$$2^{\aleph_0} = \aleph_2.$$

$$2^{\aleph_0} \neq \aleph_1,$$

The preceding theorem gives, in the Cohen extension,

$$2^{\aleph_0} = \aleph_2.$$

The preservation theorem gives the usual strict inequality $\aleph_1 < \aleph_2$ in that same extension. Therefore

$$2^{\aleph_0} \neq \aleph_1,$$

which is exactly the negation of CH. No new forcing argument is needed here; this corollary records the logical consequence of the exact continuum computation.

The role of this result is therefore logical rather than combinatorial: the preceding forcing computation has already done all of the set-theoretic work. Here we merely translate the strict cardinal inequality into the standard sentence $\neg\text{CH}$ that will be used by the consistency-transfer theorem. *Logical translation.* In ZFC, CH is the equation $2^{\aleph_0} = \aleph_1$. The previous theorem establishes the incompatible equation $2^{\aleph_0} = \aleph_2$, and cardinal preservation guarantees $\aleph_1 \neq \aleph_2$ in the same model. Thus the inference to $\neg\text{CH}$ is simply substitution into the definition of CH; all forcing-specific work has already been discharged. $\square$

Worked example

Worked Example 60.5 (One real versus ω_2 reals). For $\text{Add}(\omega, 1)$, the forcing is countable, so nice real names are coded by reals from the ground model; CH bounds them by $\aleph_1$. For $\text{Add}(\omega, \omega_2)$, the coordinate family already has size $\aleph_2$. The two forcings use the same local finite conditions but produce different continuum sizes because the number of coordinates changes the lower bound.

A useful warning

The slogan “Cohen forcing makes CH false” is too vague. The specific forcing and ground-model cardinal arithmetic determine the resulting continuum.

Looking ahead

The next chapter, *Homogeneity and relative consistency*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 61

Homogeneity and relative consistency

Definitions and setup

Formal definitions used in this chapter

Automorphism and coordinate permutation. An automorphism of a forcing notion $(\mathbb{P}, \leq)$ is a bijection $\pi : \mathbb{P} \rightarrow \mathbb{P}$ satisfying

$$p \leq q \iff \pi(p) \leq \pi(q).$$

For a permutation ρ of the coordinate set $\kappa \times \omega$, define the transported condition map π_ρ by

$$\pi_\rho(p)(\rho(\xi)) := p(\xi) \quad (\xi \in \text{dom } p).$$

Theorem 61.1 verifies that these maps are forcing automorphisms and uses them to prove weak homogeneity.

Weak homogeneity. A forcing notion is *weakly homogeneous* when for every $p, q \in \mathbb{P}$ there is an automorphism π such that $\pi(p)$ is compatible with q . This is the homogeneity notion used in this chapter.

Action on forcing names and the symmetry lemma. If π is a forcing automorphism, its action on names is defined recursively by

$$\pi\tau := \{\langle \pi\sigma, \pi p \rangle : \langle \sigma, p \rangle \in \tau\}.$$

Recall the forcing *symmetry lemma* proved in Volume VII: for every formula φ and tuple of forcing names $\bar{\tau}$,

$$p \Vdash \varphi(\bar{\tau}) \iff \pi p \Vdash \varphi(\pi\bar{\tau}).$$

Canonical ground-model parameters. For $a \in V$, $\check{a}$ denotes its canonical forcing name. The rank-induction fact $\pi\check{a} = \check{a}$ is verified in the proof of Theorem 61.1 before it is used there.

Relative consistency and independence. $\text{Con}(T)$ abbreviates the arithmetized statement that the recursively axiomatized theory T has no proof of contradiction. A sentence φ is *independent of T relative to $\text{Con}(T)$* when

$$\text{Con}(T) \Rightarrow \text{Con}(T + \varphi) \quad \text{and} \quad \text{Con}(T) \Rightarrow \text{Con}(T + \neg\varphi).$$

The transfer theorems from Volumes VI and VIII turn model/forcing constructions into these syntactic consistency implications.

Cohen forcing admits many coordinate permutations. Homogeneity says conditions can be moved by automorphisms into compatible positions, so parameter-free/ground-parameter statements are not decided differently on unrelated local pieces. For relative consistency, however, homogeneity is secondary: the main load-bearing steps are ccc, name counting, and the Volume VIII consistency-transfer theorem.

How this chapter fits together

Automorphisms of Cohen forcing show that no condition can encode privileged ground-model-free information about the generic coordinates. Homogeneity is then combined with the formal consistency-transfer machinery of Volume VIII. Finally, the positive constructible model and the negative Cohen model give the two sides of CH independence.

Theorem 61.1 (Homogeneity of Cohen forcing). *Let $\mathbb{P} = \text{Add}(\omega, \kappa)$ with largest condition $1_{\mathbb{P}}$. For any $p, q \in \mathbb{P}$ there is a coordinate permutation π inducing an automorphism of $\mathbb{P}$ such that $\pi(p)$ is compatible with q . Consequently, if $\varphi(\check{a}_1, \dots, \check{a}_n)$ is a sentence whose parameters are canonical ground-model names and some condition forces it, then*

$$1_{\mathbb{P}} \Vdash \varphi(\check{a}_1, \dots, \check{a}_n).$$

Proof.

Formal derivation spine

Formal spine: permute the entire countable coordinate set used by conditions. For finite domains $d_p, d_q \subseteq \kappa \times \omega$, choose a permutation π of $\kappa \times \omega$ such that

$$\pi[d_p] \cap d_q$$

contains only protected coordinates on which the two conditions already agree. Then $\pi(p) \cup q$ is a condition.

An arbitrary permutation π of the coordinate set $\kappa \times \omega$ induces an automorphism of $\text{Add}(\omega, \kappa)$ by

$$(\pi p)(\pi(\alpha, n)) := p(\alpha, n).$$

Let p, q be conditions. Their domains are finite. Fix any finite protected set of coordinates that must remain unchanged. Because $\kappa \times \omega$ is infinite even when $\kappa = 1$, choose a permutation fixing the protected set pointwise and moving every unprotected coordinate in $\text{dom}(p)$ away from $\text{dom}(q)$. On any protected overlap the values were required to agree. Hence

$$\pi(p) \cup q$$

is a finite partial function and therefore a common extension of $\pi(p)$ and q . This proves weak homogeneity. No preservation of the second coordinate is required; imposing that unnecessary restriction would fail to prove the $\kappa = 1$ case.

Before using ground-model parameters, verify that every forcing automorphism fixes check names. By induction on the rank of a ,

$$\pi \check{a} = \{ \langle \pi \check{b}, \pi 1_{\mathbb{P}} \rangle : b \in a \} = \{ \langle \check{b}, 1_{\mathbb{P}} \rangle : b \in a \} = \check{a},$$

because any order automorphism fixes the unique largest condition $1_{\mathbb{P}}$. Now suppose

$$p \Vdash \varphi(\check{a}_1, \dots, \check{a}_n) \tag{1}$$

for canonical ground-model names. The check-name calculation and the symmetry lemma recalled in the chapter definitions give

$$\pi(p) \Vdash \varphi(\check{a}_1, \dots, \check{a}_n). \tag{2}$$

If the largest condition did not force φ , decision density would give $q \Vdash \neg \varphi$. Choose π so that $\pi(p)$ is compatible with q , and let $r \leq \pi(p), q$. Monotonicity applied to (2) gives $r \Vdash \varphi$, while $r \leq q$ gives $r \Vdash \neg \varphi$, a contradiction. Thus

$$1_{\mathbb{P}} \Vdash \varphi(\check{a}_1, \dots, \check{a}_n).$$

The restriction to canonical ground-model parameters is essential because arbitrary names need not be fixed by the automorphism group. $\square$

Theorem 61.2 ($\text{Con}(\text{ZFC})$ implies $\text{Con}(\text{ZFC} + \text{not-CH})$ using Volume VIII transfer). *Using the formal forcing consistency-transfer theorem of Volume VIII,*

$$\text{Con}(\text{ZFC}) \implies \text{Con}(\text{ZFC} + \neg\text{CH}).$$

Proof.

Formal derivation spine

Formal spine for $\text{Con}(\text{ZFC})$ implies $\text{Con}(\text{ZFC} + \text{not-CH})$ using Volume VIII transfer.

$$\text{Con}(\text{ZFC} + \text{GCH}).$$

$$\text{Con}(\text{ZFC} + \text{GCH}) \implies \text{Con}(\text{ZFC} + \neg\text{CH}).$$

$$\text{Con}(\text{ZFC}) \implies \text{Con}(\text{ZFC} + \neg\text{CH}).$$

Assume $\text{Con}(\text{ZFC})$. The constructible-universe argument of Volume VI, together with the formal metatheoretic transfer now available in Volume VIII, gives

$$\text{Con}(\text{ZFC} + \text{GCH}).$$

Over that strengthened but equiconsistent ground theory, the forcing calculation of this volume proves that $\text{Add}(\omega, \omega_2)$ forces ZFC together with $2^{\aleph_0} = \aleph_2$, hence $\neg\text{CH}$.

Apply the formal forcing consistency-transfer theorem of Volume VIII to this calculation. It yields

$$\text{Con}(\text{ZFC} + \text{GCH}) \implies \text{Con}(\text{ZFC} + \neg\text{CH}).$$

The first consistency implication moved us to an equiconsistent ground theory where the cardinal arithmetic is convenient; the second moved from that ground theory to the Cohen extension. Ordinary transitivity of implication now composes these two metatheoretic statements:

$$\text{Con}(\text{ZFC}) \implies \text{Con}(\text{ZFC} + \neg\text{CH}).$$

At no stage is a countable transitive model inferred merely from consistency.

The intermediate use of GCH is a convenience for the clean name-counting calculation, not an additional consistency assumption. Gödel's constructible-universe theorem supplies that ground theory at no extra consistency strength relative to ZFC. $\square$

Theorem 61.3 (Relative independence of CH from ZFC after combining with L). *If ZFC is consistent, then neither CH nor $\neg\text{CH}$ is provable from ZFC: constructibility gives*

$$\text{Con}(\text{ZFC}) \Rightarrow \text{Con}(\text{ZFC} + \text{CH}),$$

while Cohen forcing gives

$$\text{Con}(\text{ZFC}) \Rightarrow \text{Con}(\text{ZFC} + \neg\text{CH}).$$

Uses: the constructible model of CH, the Cohen extension with $\neg\text{CH}$, and formal consistency transfer.

Proof.

Formal derivation spine

Formal spine for Relative independence of CH from ZFC after combining with L.

$$\text{Con}(\text{ZFC} + \text{CH}),$$

$$\text{Con}(\text{ZFC} + \neg\text{CH}).$$

Assume ZFC is consistent. The constructible universe gives the positive branch

$$\text{Con}(\text{ZFC} + \text{CH}),$$

indeed the stronger $\text{Con}(\text{ZFC} + \text{GCH})$. The preceding Cohen-forcing theorem gives the negative branch

$$\text{Con}(\text{ZFC} + \neg\text{CH}).$$

If ZFC proved CH, then adding $\neg\text{CH}$ to ZFC would make an inconsistent theory, contradicting the negative branch. Hence $\text{ZFC} \not\vdash \text{CH}$. If ZFC proved $\neg\text{CH}$, then $\text{ZFC} + \text{CH}$ would be inconsistent, contradicting the constructible positive branch. Hence $\text{ZFC} \not\vdash \neg\text{CH}$. Therefore CH is independent of ZFC, conditional on the consistency of ZFC.

This final step uses consistency only to ensure that the two model constructions really witness compatible extensions of the same base theory. It does not claim that ZFC can internally prove its own consistency or the displayed relative-consistency statements. $\square$

Worked example

Worked Example 61.4 (Moving finite support by a permutation). Given two finite Cohen conditions, a permutation of the unused coordinate indices can move the non-root support of one away from the other while fixing a prescribed finite root. The image becomes compatible with the second condition. This finite-support mobility is the concrete source of Cohen homogeneity.

A useful warning

Relative consistency is the conclusion $\text{Con}(\text{ZFC}) \Rightarrow \text{Con}(\text{ZFC} + \neg\text{CH})$, not an absolute proof that ZFC is consistent.

End-of-volume perspective

Together with L , the Cohen extension gives the two relative-consistency directions for CH. The final volume applies a symmetry restriction to a generic extension to separate the Axiom of Choice from ZF.

Part X

Volume X: Symmetric Extensions and Independence of Choice

Preface

Symmetric extensions are treated independently from ordinary forcing. The hard ZF axioms in the symmetric submodel receive their own proved results before Choice is allowed to fail.

Position in the series

Volume X passes from ordinary generic extensions to symmetric submodels. The support calculus preserves ZF while allowing Choice to fail, completing the second Cohen independence branch.

Sources and further reading

The permutation/symmetric-model method is developed as the pure-set counterpart of the Fraenkel–Mostowski idea. Standard treatments and the basic Cohen model may be compared with [12, 13].

Notation for this volume

$\pi\tau$ is the recursive action of a forcing automorphism on a name; $\text{sym}(\tau)$ is its stabilizer; N is the class of values of hereditarily symmetric names.

How to read symbolic arguments in this volume

When a formula appears, read it in layers rather than as one visual block. First identify the *ambient language*: are we speaking inside a formal theory, about a structure, or in the surrounding metatheory? Next mark the objects being manipulated (formulas, codes, sets, names, conditions, ordinals) and the operation being applied to them. Finally check the side conditions—free variables, rank bounds, compatibility, genericity, transitivity, or consistency hypotheses. Whenever a displayed derivation changes level, the text says so explicitly. A displayed line is therefore meant to be read like a line of well-commented code: the symbols perform the operation, and the surrounding prose records its type, preconditions, and purpose.

Chapter 62

Fraenkel–Mostowski precursor

Definitions and setup

Formal definitions used in this chapter

Atoms, sets, and the ZFA language used here. Work in the one-sorted language $\{\in, \text{At}\}$, where $\text{At}(x)$ means that x is an *atom* (urelement), and abbreviate

$$\text{Set}(x) := \neg \text{At}(x).$$

The theory *ZFA* used in this chapter consists of the following modification of ZF. Atoms have no members,

$$\forall x (\text{At}(x) \rightarrow \forall y y \notin x),$$

and only sets may have members,

$$\forall x \forall y (y \in x \rightarrow \text{Set}(x)).$$

Extensionality is restricted to sets:

$$\forall x \forall y (\text{Set}(x) \wedge \text{Set}(y) \wedge \forall z (z \in x \leftrightarrow z \in y) \rightarrow x = y).$$

Pairing, Union, Power Set, Infinity, Separation, Replacement, and Foundation have the same membership clauses as in ZF, but every object asserted to exist by a set-forming axiom is additionally required to satisfy *Set*; atoms are nevertheless allowed to occur as elements of those sets. Separation and Replacement are schemes over formulas of the expanded language $\{\in, \text{At}\}$. This paragraph fixes the exact meaning of “ZFA” used below; Choice is not included unless explicitly added.

Permutation group. For a set A , $\text{Sym}(A)$ is the group of all bijections $A \rightarrow A$. We fix $G \leq \text{Sym}(A)$.

Hierarchy over atoms and induced action. Put

$$V_0(A) := A, \quad V_{\alpha+1}(A) := A \cup \mathcal{P}(V_\alpha(A)), \quad V_\lambda(A) := \bigcup_{\alpha < \lambda} V_\alpha(A)$$

for limit λ . A permutation $\pi \in G$ acts on atoms by the given permutation and recursively on non-atoms by

$$\pi x := \{\pi y : y \in x\}.$$

Stabilizers and support. For $S \subseteq A$ and an object x ,

$$\text{Fix}(S) := \{\pi \in G : \forall s \in S \pi(s) = s\}, \quad \text{Stab}(x) := \{\pi \in G : \pi x = x\}.$$

A finite $S \subseteq A$ *supports* x when $\text{Fix}(S) \leq \text{Stab}(x)$. Recall the transitive closure operation from Volume IV. An object is *hereditarily finitely supported* when it has a finite support and every object in its

transitive closure has a finite support. The *basic Fraenkel–Mostowski class* in this chapter is the class of hereditarily finitely supported objects for a countably infinite atom set A with the full permutation group $G = \text{Sym}(A)$.

Dedekind-finite. In this volume an infinite set X is called *Dedekind-finite* when there is no injection $\omega \rightarrow X$. This explicit definition is the only formulation used in the proofs below.

The Fraenkel–Mostowski method begins with atoms and a permutation group. A finite support controls an object’s invariance. The hereditarily finitely supported class isolates the combinatorial mechanism by which Choice can fail in a full Fraenkel–Mostowski development. In this chapter it is used only as a precursor: we do not yet claim that the displayed class satisfies every axiom of ZFA. The later symmetric-forcing construction supplies the load-bearing pure-set model verification.

How this chapter fits together

The atom calculation is a warm-up for symmetric forcing. Finite supports make the full set of atoms invariant while preventing any finitely supported enumeration from being invariant. This produces an infinite Dedekind-finite set inside the basic finite-support class and reveals the mechanism by which Choice will later fail. No model-theoretic conclusion is drawn here until the relevant axioms have been verified; the pure-ZF symmetric-forcing construction later in the volume performs that verification.

Proposition 62.1 (Permutation actions on sets with atoms). *Let A be a set of atoms and $G \leq \text{Sym}(A)$. The action of G extends recursively to the cumulative hierarchy over atoms by*

$$\pi x = \{\pi y : y \in x\}$$

for non-atoms, while πa is the given permutation on atoms.

Proof.

Formal derivation spine

Formal spine for Permutation actions on sets with atoms.

$$\pi(x) = \{\pi(y) : y \in x\}.$$

$$\pi(x \cup y) = \pi(x) \cup \pi(y), \quad x \in y \iff \pi(x) \in \pi(y),$$

$$\text{Fix}(S) \leq \text{Stab}(x).$$

Let A be a set of atoms and let G be a group of permutations of A . Extend each $\pi \in G$ recursively to the cumulative hierarchy over the atoms by

$$\pi(x) = \{\pi(y) : y \in x\}.$$

For atoms the value is given by the original permutation; for ordinary sets the displayed clause recursively moves every member. Foundation makes that recursion well founded. We now verify that this recursive action respects the set-theoretic operations and membership relation:

$$\pi(x \cup y) = \pi(x) \cup \pi(y), \quad x \in y \iff \pi(x) \in \pi(y),$$

and $(\pi \circ \sigma)(x) = \pi(\sigma(x))$. Thus G acts by automorphisms of the set-with-atoms universe. The permutation model is obtained not by changing membership, but by restricting attention to objects whose behavior under this action is controlled by small supports. $\square$

Theorem 62.2 (Finite-support closure rules in the atom construction). *Let $S, T \subseteq A$ be finite. If S supports x and T supports y , then $S \cup T$ supports the unordered pair $\{x, y\}$, the Kuratowski ordered pair $(x, y) = \{\{x\}, \{x, y\}\}$, and the union $x \cup y$ whenever the latter is defined. If S supports x and $\pi \in G$, then*

$\pi[S]$ supports πx . Consequently any object obtained from finitely many finitely supported parameters by finitely many of these operations has a finite support, namely the union of the transported parameter supports required by the construction.

Proof.

Formal derivation spine

Formal spine for Finite-support symmetry calculus in the atom model.

$$\text{Fix}(S) = \{\pi \in G : \pi(a) = a \text{ for every } a \in S\}.$$

$$\text{Fix}(S) \leq \text{Stab}(x).$$

For a finite set of atoms S , let

$$\text{Fix}(S) = \{\pi \in G : \pi(a) = a \text{ for every } a \in S\}.$$

Say that S supports x if every permutation in $\text{Fix}(S)$ fixes x . If S supports x and T supports y , then $S \cup T$ supports $\{x, y\}$, ordered pairs built from x, y , and every finite construction from them. If S supports x , then $\pi[S]$ supports $\pi(x)$.

Call an object hereditarily finitely supported when it has a finite support and every object in its transitive closure does as well, as defined above. Rank induction shows that these objects are closed under the finite set operations verified in this theorem. This is the finite-support calculus that the later forcing construction abstracts using a normal filter of subgroups. $\square$

Theorem 62.3 (Dedekind-finiteness of the atom set in the finite-support class). *Let A be countably infinite, let $G = \text{Sym}(A)$, and let the basic Fraenkel–Mostowski class be the hereditarily finitely supported objects defined above. Then A belongs to this class and is infinite, but no injection $f : \omega \rightarrow A$ belongs to the class. Thus A is Dedekind-finite relative to the functions present in the finite-support class.*

Proof.

Formal derivation spine

Formal spine for Dedekind-finiteness of the atom set in the finite-support class.

$$f(n) = \pi(f)(n) = \pi(f(n)) = b,$$

$$\text{Fix}(S) \leq \text{Stab}(x).$$

The whole atom set A has empty support, because every permutation sends A to itself. It is infinite, since for every n it contains n distinct atoms. Suppose there were an injection $f : \omega \rightarrow A$ in the finite-support universe. Let S be a finite support for f . Choose n with $f(n) \notin S$ and an atom $b \notin S \cup \{f(n)\}$. Let π swap $f(n)$ and b while fixing S pointwise.

Because S supports f , $\pi(f) = f$, and π fixes the natural number n . Hence

$$f(n) = \pi(f)(n) = \pi(f(n)) = b,$$

contradiction. Thus A is infinite and Dedekind-finite in the precise relative sense stated for the finite-support class. *Why the contradiction uses a permutation rather than cardinal arithmetic.* The support S records every atom on which the name of f is allowed to depend. Once an output value lies outside S , a permutation can move that value without changing the name of the function. Evaluating the unchanged function at the unchanged input must then give both the old and moved atoms, which is impossible. $\square$

Proposition 62.4 (Why the atom construction is not yet a pure-ZF construction). *The preceding Fraenkel–Mostowski precursor uses a distinguished set A of atoms satisfying $\text{At}(a)$. Therefore its ambient structures are structures for the expanded ZFA language $\{\in, \text{At}\}$, not structures for pure set theory in which every object is a set. In particular, the existence of the Dedekind-finite atom set in this chapter by itself yields no model, and hence no relative-consistency conclusion, for pure $\text{ZF} + \neg\text{AC}$.*

Proof. An atom $a \in A$ satisfies $\text{At}(a)$ and has no members. It is not identified with the empty set, because the predicate At distinguishes urelements from sets. Thus the ontology and language of the precursor contain objects that are not pure sets. Forgetting the predicate At would collapse the distinction between an atom and the empty set and would not preserve the intended structure.

The support argument proving that A is Dedekind-finite is therefore only the combinatorial precursor. The later symmetric-forcing chapters repeat the support mechanism on ordinary forcing names, define a class consisting entirely of pure sets, and verify the ZF axioms there one by one before deriving $\neg\text{AC}$. That later verification, not the atom warm-up, is the load-bearing pure-ZF construction. $\square$

Worked example

Worked Example 62.5 (Moving an unsupported atom). If a function $f : \omega \rightarrow A$ has finite support S , choose an output atom $a \notin S$ and swap it with another atom $b \notin S$. The permutation fixes f and the input integer but moves the output, a contradiction. The support argument is exactly the one later transferred to Cohen names.

A useful warning

The atom calculation displays the support mechanism behind failure of Choice, but it does not by itself establish $\text{Con}(\text{ZF} + \neg\text{AC})$ because ZF has no atoms.

Looking ahead

The next chapter, *Automorphisms of forcing and names*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 63

Automorphisms of forcing and names

Definitions and setup

Formal definitions used in this chapter

Ground-model convention for this volume. Whenever genericity or a symmetric model is discussed, $M \models \text{ZFC}$ denotes the transitive forcing ground model and $\mathbb{P} \in M$ is the forcing notion. A *ground-model $\mathbb{P}$ -name* means a set $\tau \in M$ that M recognizes as a $\mathbb{P}$ -name; write $M^{\mathbb{P}}$ for the class of these names. Unless an explicitly external syntactic calculation is being made, “forcing name” in the semantic symmetric-model chapters means a member of $M^{\mathbb{P}}$. The automorphisms used in the construction belong to M . Thus if $\mathcal{G}$ is used as a ground-model automorphism group, then $\mathcal{G} \in M$ and M verifies that every $\pi \in \mathcal{G}$ is an automorphism of $\mathbb{P}$. Purely algebraic assertions about automorphisms of a poset do not require M .

Recall: check names. For every ground-model set x , its canonical check name is the rank-recursively defined forcing name

$$\check{x} := \{ \langle \check{y}, 1_{\mathbb{P}} \rangle : y \in x \}.$$

Volume VII proved $\check{x}^G = x$ for every generic G . The present chapter uses only the displayed recursive definition until the valuation theorem below is invoked.

Automorphism group of forcing. For a forcing notion $(\mathbb{P}, \leq, 1_{\mathbb{P}})$,

$$\text{Aut}(\mathbb{P}) := \{ \pi : \mathbb{P} \rightarrow \mathbb{P} : \pi \text{ is a bijection and } p \leq q \iff \pi p \leq \pi q \}.$$

Fix a subgroup $\mathcal{G} \leq \text{Aut}(\mathbb{P})$.

Action on names and filters. Recursively on name rank,

$$\pi \tau := \{ \langle \pi \sigma, \pi p \rangle : \langle \sigma, p \rangle \in \tau \}.$$

For a filter $G \subseteq \mathbb{P}$ set $\pi G := \{ \pi p : p \in G \}$. The facts that this recursion is a genuine group action, that M -genericity is preserved, and that valuation commutes with simultaneous transport are proved in the first two results below.

Group action and equivariance. A group action satisfies $1x = x$ and $(\pi\rho)x = \pi(\rho x)$. A map F is *equivariant* when $F(\pi x) = \pi F(x)$ (with the relevant actions understood).

Name stabilizer. For a name τ ,

$$\text{sym}(\tau) := \{ \pi \in \mathcal{G} : \pi \tau = \tau \}.$$

A subgroup $H \leq \mathcal{G}$ *supports* τ when $H \leq \text{sym}(\tau)$.

An automorphism of a forcing poset acts recursively on names by applying the automorphism to every attached condition and constituent name. The symmetry lemma states that forcing is equivariant under this

action. The stabilizer of a name is the subgroup fixing it. These definitions let group-theoretic supports control semantic facts about the eventual interpretations.

How this chapter fits together

A forcing automorphism acts first on conditions and then recursively on names. Valuation is covariant under this action, and the symmetry lemma transports forcing truth along automorphisms. These three layers must match before support arguments can be made inside a forcing construction.

Theorem 63.1 (Recursive automorphism action on forcing names). *If π is an automorphism of a forcing notion $\mathbb{P}$, define recursively*

$$\pi\tau = \{\langle \pi\sigma, \pi p \rangle : \langle \sigma, p \rangle \in \tau\}.$$

This gives an action of the automorphism group on all forcing names.

How to read the symbols: automorphisms act on every layer of a name

If π moves forcing conditions, it must also move all conditions occurring inside a name, and recursively all lower-rank names appearing there. The definition

$$\pi\tau = \{\langle \pi\sigma, \pi p \rangle : \langle \sigma, p \rangle \in \tau\}$$

performs that recursive renaming. The covariance identity $(\pi\tau)^{\pi G} = \tau^G$ says that renaming both the program (the name) and its input (the generic filter) leaves the resulting object unchanged.

Proof.

Formal derivation spine

Formal spine for Recursive automorphism action on forcing names.

$$\pi\tau = \{\langle \pi\sigma, \pi p \rangle : \langle \sigma, p \rangle \in \tau\}.$$

$$(\pi\tau)^{\pi G} = \tau^G.$$

Let π be an automorphism of the forcing poset. Define its action on names by rank recursion:

$$\pi\tau = \{\langle \pi\sigma, \pi p \rangle : \langle \sigma, p \rangle \in \tau\}.$$

Every constituent σ has smaller name rank, so the recursion is well founded. Induction gives $\text{id}(\tau) = \tau$ and $(\pi\rho)(\tau) = \pi(\rho\tau)$, so this is genuinely a group action. The construction preserves name rank and sends ground check names to themselves because their conditions are the top condition and their constituents are recursively fixed.

Further explanation. The recursion is on name rank. Once $\pi\sigma$ has been defined for every name occurring below τ , the displayed set of pairs defines $\pi\tau$. Rank is preserved because automorphisms change conditions but not the membership depth of names. A second induction proves $\pi(\rho$ occurring in $\tau)$ occurs in $\pi\tau$, so the action respects the entire hereditary name structure needed later for symmetric names. $\square$

Proposition 63.2 (Generic transport and valuation under automorphisms). *Let M be the forcing ground model and let $\pi \in \text{Aut}(\mathbb{P})$ belong to M . If $G \subseteq \mathbb{P}$ is M -generic, then πG is M -generic. Moreover, for every forcing name τ ,*

$$(\pi\tau)^{\pi G} = \tau^G.$$

Thus valuation is equivariant with respect to simultaneous transport of names and the generic filter.

Proof.

Formal derivation spine

Formal spine for Valuation commutes with the automorphism action.

$$\begin{aligned} (\pi\tau)^{\pi G} &= \tau^G. \\ (\pi\tau)^{\pi G} &= \{(\pi\sigma)^{\pi G} : \exists p \in G \langle \pi\sigma, \pi p \rangle \in \pi\tau\}. \end{aligned}$$

First suppose $D \in M$ is dense in $\mathbb{P}$. Because $\pi \in M$, the inverse image

$$\pi^{-1}[D] = \{p \in \mathbb{P} : \pi p \in D\}$$

belongs to M and is dense: given p , choose $d \leq \pi p$ in D and then $\pi^{-1}d \leq p$. Since G meets $\pi^{-1}[D]$, the filter πG meets D . Order preservation also shows directly that πG is a filter. Hence πG is M -generic.

We now prove by induction on name rank that

$$(\pi\tau)^{\pi G} = \tau^G.$$

Indeed, $(\pi\tau)^{\pi G}$ consists of the values $(\pi\sigma)^{\pi G}$ for pairs $\langle \pi\sigma, \pi p \rangle \in \pi\tau$ with $\pi p \in \pi G$. This is equivalent to $\langle \sigma, p \rangle \in \tau$ with $p \in G$, and the induction hypothesis turns the constituent value into σ^G . Thus the two sets have exactly the same elements. This covariance is the semantic counterpart of the syntactic symmetry lemma for forcing.

Further explanation. The induction compares the same pair at every rank. We have

$$(\pi\tau)^{\pi G} = \{(\pi\sigma)^{\pi G} : \exists p \in G \langle \pi\sigma, \pi p \rangle \in \pi\tau\}.$$

By the induction hypothesis $(\pi\sigma)^{\pi G} = \sigma^G$, and the condition clause is equivalent to $\langle \sigma, p \rangle \in \tau$. The resulting set is exactly τ^G . This covariance identity is why automorphisms of names correctly model symmetry of the interpreted objects. $\square$

Theorem 63.3 (Symmetry lemma for forcing). *For every forcing formula φ , automorphism π , condition p , and names $\bar{\tau}$,*

$$p \Vdash \varphi(\bar{\tau}) \iff \pi p \Vdash \varphi(\pi\bar{\tau}).$$

Commented symbolic trace: the symmetry lemma is equivariance of forcing

The equivalence

$$p \Vdash \varphi(\bar{\tau}) \iff \pi p \Vdash \varphi(\pi\bar{\tau})$$

says that forcing commutes with a relabelling of coordinates. Atomic forcing is checked first using the recursive action on names; Boolean connectives preserve the equivalence immediately from the induction hypothesis; quantifiers use that π is a bijection on names. This equivariance is what later allows finite-support information to control which objects survive in the symmetric model.

Proof.

Formal derivation spine

Formal spine for Symmetry lemma for forcing.

$$\begin{aligned} \pi\tau &= \{\langle \pi\sigma, \pi p \rangle : \langle \sigma, p \rangle \in \tau\}. \\ p \Vdash \varphi(\bar{\tau}) &\iff \pi p \Vdash \varphi(\pi\bar{\tau}). \\ (\pi\tau)^{\pi G} &= \tau^G. \end{aligned}$$

For an automorphism π of $\mathbb{P}$, define recursively

$$\pi\tau = \{\langle \pi\sigma, \pi p \rangle : \langle \sigma, p \rangle \in \tau\}.$$

Induction on name rank gives $(\pi\tau)^{\pi G} = \tau^G$. A simultaneous induction on the definition of forcing proves the symmetry lemma

$$p \Vdash \varphi(\bar{\tau}) \iff \pi p \Vdash \varphi(\pi\bar{\tau}).$$

The stabilizer $\text{sym}(\tau) = \{\pi : \pi\tau = \tau\}$ is a subgroup. These facts make support arguments compatible with forcing truth rather than merely with raw names.

Point specific to this result. The proof is an induction on the forcing relation: automorphisms preserve order, compatibility, and the recursive action on names. Therefore applying an automorphism to both a condition and every parameter preserves exactly what that condition forces.

Further explanation. For atomic formulas, the claim follows from the recursive definitions of forcing together with preservation of compatibility under automorphisms. Boolean connectives follow immediately by induction. In the existential step, if p forces that some name is a witness, applying π to that witness and to every stronger condition shows that πp forces the transformed existential statement. Applying π^{-1} gives the reverse implication. Thus the forcing relation is equivariant under the entire automorphism group. $\square$

Remark 63.4 (Source note). The symmetry lemma is the formal bridge between automorphisms of forcing and invariance of symmetric names; compare the symmetric-model treatments in [11, 12, 13].

Proposition 63.5 (Stabilizers and support calculus for names). *For a name τ ,*

$$\text{sym}(\tau) = \{\pi : \pi\tau = \tau\}$$

is a subgroup of the automorphism group. A support condition may be expressed by requiring a designated subgroup such as $\text{Fix}(S)$ to lie inside $\text{sym}(\tau)$.

Proof.

Formal derivation spine

Formal spine for Stabilizers and support calculus for names.

$$\text{sym}(\tau) := \{\pi \in \mathcal{G} : \pi\tau = \tau\}.$$

$$(\pi\rho)\tau = \pi(\rho\tau) = \pi\tau = \tau;$$

$$H \leq \text{sym}(\tau).$$

For a name τ , define

$$\text{sym}(\tau) := \{\pi \in \mathcal{G} : \pi\tau = \tau\}.$$

It is a subgroup: the identity fixes τ ; if π, ρ fix τ , then

$$(\pi\rho)\tau = \pi(\rho\tau) = \pi\tau = \tau;$$

and if $\pi\tau = \tau$, then applying π^{-1} gives $\pi^{-1}\tau = \tau$. A subgroup $H \leq \mathcal{G}$ supports τ when

$$H \leq \text{sym}(\tau).$$

If H supports τ , then the conjugate subgroup $\pi H \pi^{-1}$ supports $\pi\tau$, because

$$(\pi h \pi^{-1})(\pi\tau) = \pi(h\tau) = \pi\tau.$$

This support transport is the group-theoretic input used by normal filters. *Point specific to this result.*

A support is a small set of coordinates whose pointwise stabilizer fixes the name. Intersections/unions of supports correspond to intersections of stabilizer subgroups, providing the closure calculations used for pairing, separation, and the Cohen family.

Further explanation. If $H = \text{sym}(\tau)$, then H records exactly the automorphisms under which the name is unchanged. A support S is a concrete device for producing a subgroup contained in H : every permutation fixing S pointwise fixes the name. Intersections of supports support finite tuples, and conjugating a support subgroup corresponds to transporting the support by the automorphism. These elementary facts are precisely what normality of the filter packages abstractly. $\square$

Worked example

Worked Example 63.6 (Preview: a coordinate-indexed family of names). Suppose a later forcing construction supplies a family of names $(\tau_n)_{n < \omega}$ and coordinate permutations satisfying

$$\pi\tau_n = \tau_{\pi(n)}.$$

Then the valuation covariance theorem gives

$$(\pi\tau_n)^{\pi G} = \tau_n^G.$$

This is only a schematic preview of the automorphism mechanism. The actual Cohen coordinate names, their definition, and the identity describing their transport are introduced and proved in the later chapter *The basic Cohen symmetric model*; no property of those as-yet-undefined names is used here.

A useful warning

Fixing a name syntactically and fixing its value in one generic extension are different statements. The support theory is formulated at the level of names and justified semantically by Theorem 63.3, the ordinary forcing symmetry lemma.

Looking ahead

The next chapter, *Symmetric and hereditarily symmetric names*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 64

Symmetric and hereditarily symmetric names

Definitions and setup

Formal definitions used in this chapter

Domain and constituent closure of a name. For a forcing name τ ,

$$\text{dom}(\tau) := \{\sigma : \exists p \in \mathbb{P} \langle \sigma, p \rangle \in \tau\}.$$

Define finite constituent stages by

$$C_0(\tau) := \{\tau\}, \quad C_{n+1}(\tau) := C_n(\tau) \cup \bigcup_{\rho \in C_n(\tau)} \text{dom}(\rho) \quad (n < \omega),$$

and define the *constituent closure*

$$\text{TC}_{\text{name}}(\tau) := \bigcup_{n < \omega} C_n(\tau).$$

Thus every member of $\text{TC}_{\text{name}}(\tau)$ is reached from τ by finitely many constituent steps. Conversely the displayed union is closed under $\rho \mapsto \text{dom}(\rho)$, so it is the least set of names containing τ with that closure property. The phrase “a name in the transitive closure of τ ” below always means a name in this constituent closure.

Normal filter of subgroups. A nonempty family $\mathcal{F}$ of subgroups of $\mathcal{G}$ is a *normal filter* when

$$H \in \mathcal{F}, H \leq K \leq \mathcal{G} \implies K \in \mathcal{F},$$

$$H, K \in \mathcal{F} \implies H \cap K \in \mathcal{F},$$

and

$$H \in \mathcal{F}, \pi \in \mathcal{G} \implies \pi H \pi^{-1} \in \mathcal{F}.$$

A subgroup belonging to $\mathcal{F}$ is an *allowed support subgroup*.

Forcing-name rank. Recall from Volume VII that the *forcing-name rank* of a $\mathbb{P}$ -name τ is

$$\text{rank}_{\mathbb{P}}(\tau) := \min\{\alpha : \tau \in V_{\alpha+1}^{\mathbb{P}}\} = \sup\{\text{rank}_{\mathbb{P}}(\sigma) + 1 : \exists p \in \mathbb{P} \langle \sigma, p \rangle \in \tau\}.$$

The empty name has rank 0, and every immediate constituent σ of τ satisfies $\text{rank}_{\mathbb{P}}(\sigma) < \text{rank}_{\mathbb{P}}(\tau)$. The recursion defining forcing names, their valuations, and the automorphism action is therefore well founded on this rank.

Symmetric system and symmetric names. Abstractly, a *symmetric system* is a triple $(\mathbb{P}, \mathcal{G}, \mathcal{F})$ in which $\mathcal{G} \leq \text{Aut}(\mathbb{P})$ and $\mathcal{F}$ is a normal filter of subgroups of $\mathcal{G}$. A *ground-model symmetric system over M* is a triple $\mathbb{P}, \mathcal{G}, \mathcal{F} \in M$ such that M verifies the preceding symmetric-system conditions. All semantic symmetric-forcing arguments below use a ground-model symmetric system over the fixed $M \models \text{ZFC}$. For a ground-model name $\tau \in M^{\mathbb{P}}$, Separation in M gives the stabilizer $\text{sym}(\tau) \in M$. Such a name τ is *symmetric* when $\text{sym}(\tau) \in \mathcal{F}$. Hereditary symmetry is defined by recursion on the forcing-name rank within $M^{\mathbb{P}}$: τ is *hereditarily symmetric* when

$$\tau \in \text{HS} \iff \text{sym}(\tau) \in \mathcal{F} \wedge \forall \langle \sigma, p \rangle \in \tau (\sigma \in \text{HS}).$$

The equivalent constituent-closure characterization is proved below rather than built into the definition. **Dense below, common extensions, bounded quantifiers, and the symmetric forcing relation.** Recall explicitly that $D \subseteq \mathbb{P}$ is *dense below p* when every $q \leq p$ has some $r \leq q$ in D . The notation

$$q \leq p, r$$

means simultaneously $q \leq p$ and $q \leq r$; thus it asserts that q is a common strengthening of p and r . As formula abbreviations we use

$$\forall x \in \tau \psi(x) := \forall x (x \in \tau \rightarrow \psi(x)), \quad \exists x \in \tau \psi(x) := \exists x (x \in \tau \wedge \psi(x)).$$

For formulas whose name parameters lie in HS, use the logical-basis normalization proved in Volume VII. Because Volume I treats $\vee, \rightarrow, \forall$ as primitive syntax constructors, recall its defining recursion

$$\begin{aligned} \mathcal{N}(\alpha) &= \alpha && \text{for atomic } \alpha, \\ \mathcal{N}(\neg \varphi) &= \neg \mathcal{N}(\varphi), && \mathcal{N}(\varphi \wedge \psi) = \mathcal{N}(\varphi) \wedge \mathcal{N}(\psi), \\ \mathcal{N}(\varphi \vee \psi) &= \neg(\neg \mathcal{N}(\varphi) \wedge \neg \mathcal{N}(\psi)), && \mathcal{N}(\varphi \rightarrow \psi) = \neg(\mathcal{N}(\varphi) \wedge \neg \mathcal{N}(\psi)), \\ \mathcal{N}(\exists x \varphi) &= \exists x \mathcal{N}(\varphi), && \mathcal{N}(\forall x \varphi) = \neg \exists x \neg \mathcal{N}(\varphi). \end{aligned} \tag{SN}$$

The symmetric forcing recursion below is defined on the basis syntax $\{\neg, \wedge, \exists\}$ after the atomic clauses. For an arbitrary Volume-I formula set

$$p \Vdash_{\text{sym}} \varphi \iff p \Vdash_{\text{sym}} \mathcal{N}(\varphi), \tag{SNF}$$

with the right-hand side evaluated by the basis recursion. This is a semantic preprocessing definition, not a claim that the primitive syntax nodes $\vee, \rightarrow, \forall$ are abbreviations. Volume VII's Logical-basis Normalization Lemma proves explicitly that the code map is primitive recursive, that φ and $\mathcal{N}(\varphi)$ have the same ordinary truth, and that normalization commutes with capture-avoiding substitution up to alpha-equivalence. Now introduce the *auxiliary forced-inclusion relation*

$$p \Vdash_{\text{sym}}^{\text{aux}} \sigma \subseteq \tau \iff \forall \langle \rho, r \rangle \in \sigma \forall q \leq p, r \exists s \leq q (s \Vdash_{\text{sym}} \rho \in \tau). \tag{AUX}$$

Forced equality is defined from this auxiliary relation by

$$p \Vdash_{\text{sym}} \sigma = \tau \iff p \Vdash_{\text{sym}}^{\text{aux}} \sigma \subseteq \tau \wedge p \Vdash_{\text{sym}}^{\text{aux}} \tau \subseteq \sigma,$$

$$p \Vdash_{\text{sym}} \sigma \in \tau \iff \{q \leq p : \exists \langle \rho, r \rangle \in \tau (q \leq r \wedge q \Vdash_{\text{sym}} \sigma = \rho)\} \text{ is dense below } p.$$

For basis formulas use

$$p \Vdash_{\text{sym}} (\varphi \wedge \psi) \iff (p \Vdash_{\text{sym}} \varphi \wedge p \Vdash_{\text{sym}} \psi),$$

$$p \Vdash_{\text{sym}} \neg \varphi \iff \neg \exists q \leq p (q \Vdash_{\text{sym}} \varphi),$$

$$p \Vdash_{\text{sym}} \exists x \varphi(x, \bar{r}) \iff \{q \leq p : \exists r \leq q \exists \sigma \in \text{HS} (r \Vdash_{\text{sym}} \varphi(\sigma, \bar{r}))\} \text{ is dense below } p.$$

Primitive occurrences of $\vee, \rightarrow, \forall$ are handled by (SN)–(SNF), not by reclassifying those syntax nodes as abbreviations. We write $\Vdash_{\text{sym}} \varphi$ for $1_{\mathbb{P}} \Vdash_{\text{sym}} \varphi$. Separately, inside the pure membership language,

$$\sigma \subseteq \tau \text{ abbreviates } \forall x (x \in \sigma \rightarrow x \in \tau). \tag{SUBF}$$

Lemma 64.13 below proves that forcing the formula (SUBF) is equivalent to the auxiliary relation (AUX). Until that lemma is reached, the phrase “auxiliary subset clause” refers specifically to (AUX); afterward the superscript “aux” is suppressed.

Canonical HS name operations. For $\sigma, \tau \in \text{HS}$ the canonical pair name is

$$\text{Pair}(\sigma, \tau) := \{\langle \sigma, 1_{\mathbb{P}} \rangle, \langle \tau, 1_{\mathbb{P}} \rangle\}.$$

For a name τ , a canonical union name is

$$\text{Union}(\tau) := \{\langle \rho, r \rangle : \exists \langle \sigma, p \rangle \in \tau \exists \langle \rho, q \rangle \in \sigma (r \leq p \text{ and } r \leq q)\}.$$

For an HS name σ , define its *canonical successor name* by

$$\text{Succ}(\sigma) := \sigma \cup \{\langle \sigma, 1_{\mathbb{P}} \rangle\}.$$

Its intended value is $\sigma^G \cup \{\sigma^G\}$. The HS closure and the valuation identities for these names are proved in Theorem 64.14; they are not assumed as part of the definitions.

Candidate symmetric class. If $G \subseteq \mathbb{P}$ is M -generic, define

$$N_G := \{\tau^G : \tau \in \text{HS}^M\}.$$

At this point N_G is only a defined class between the ground model and the full generic extension; later theorems prove transitivity and the ZF axioms. After those verifications we call it the *symmetric model*. When the generic filter is fixed, we abbreviate N_G by N .

Bounded subname. For $\tau, \bar{\rho} \in \text{HS}$,

$$\tau \restriction \varphi(\bar{\rho}) := \{\langle \eta, p \rangle \in \tau : p \Vdash_{\text{sym}} \varphi(\eta, \bar{\rho})\}.$$

Fix a normal filter $\mathcal{F}$ of subgroups of the automorphism group. A name is symmetric when its stabilizer belongs to $\mathcal{F}$; it is hereditarily symmetric when every name in its constituent closure is symmetric. For a fixed generic filter, the candidate class $N = N_G$ was defined above as the values of hereditarily symmetric ground-model names. Normality is what makes symmetry stable under conjugation and hence under the forcing automorphism action.

How this chapter fits together

A normal filter of subgroups $\mathcal{F}$ determines exactly which stabilizers are admitted: a name τ is symmetric precisely when $\text{sym}(\tau) \in \mathcal{F}$. A symmetric name has an allowed stabilizer; a hereditarily symmetric name requires the same property all the way down its constituent closure. Evaluating these names gives the candidate class N . Theorems below will prove that it is transitive, contains the ground model and all ordinals, and lies inside the full generic extension; only after the ZF verification do we use the term *symmetric model* as a proved model-theoretic description.

Proposition 64.1 (Normal filters of subgroups and symmetric names). *Let $\mathcal{G} \leq \text{Aut}(\mathbb{P})$. A normal filter of subgroups is a nonempty family $\mathcal{F}$ such that*

$$H \in \mathcal{F}, H \leq K \leq \mathcal{G} \Rightarrow K \in \mathcal{F}, \quad H, K \in \mathcal{F} \Rightarrow H \cap K \in \mathcal{F},$$

and

$$H \in \mathcal{F}, \pi \in \mathcal{G} \Rightarrow \pi H \pi^{-1} \in \mathcal{F}.$$

A name τ is symmetric when $\text{sym}(\tau) := \{\pi \in \mathcal{G} : \pi \tau = \tau\} \in \mathcal{F}$, and it is hereditarily symmetric when it is symmetric and every constituent name occurring recursively in it is hereditarily symmetric.

Proof.

Formal derivation spine

Formal spine for Normal filters of subgroups and symmetric names.

$$H \in \mathcal{F} \implies \pi H \pi^{-1} \in \mathcal{F} \quad (\pi \in \mathcal{G}).$$

$$\tau \text{ symmetric} \iff \text{sym}(\tau) \in \mathcal{F}.$$

$$\tau \in \text{HS} \iff \tau \text{ is symmetric and } \forall \langle \sigma, p \rangle \in \tau \ (\sigma \in \text{HS}).$$

A normal filter $\mathcal{F}$ on subgroups of $\mathcal{G}$ satisfies: if $H \in \mathcal{F}$ and $H \leq K$, then $K \in \mathcal{F}$; if $H, K \in \mathcal{F}$, then $H \cap K \in \mathcal{F}$; and

$$H \in \mathcal{F} \implies \pi H \pi^{-1} \in \mathcal{F} \quad (\pi \in \mathcal{G}).$$

Define

$$\tau \text{ symmetric} \iff \text{sym}(\tau) \in \mathcal{F}.$$

Define hereditary symmetry recursively by

$$\tau \in \text{HS} \iff \tau \text{ is symmetric and } \forall \langle \sigma, p \rangle \in \tau \ (\sigma \in \text{HS}).$$

Thus symmetry is a stabilizer condition, while hereditary symmetry additionally constrains every constituent name. *Point specific to this result.* Normality under conjugation is what makes symmetry invariant under the group action on names. Closure upward and under finite intersections is what permits several supported parameters to be used together in one construction. $\square$

Lemma 64.2 (Constituent-closure characterization of hereditary symmetry). *For a forcing name τ in a symmetric system,*

$$\tau \in \text{HS} \iff \text{every name } \rho \in \text{TC}_{\text{name}}(\tau) \text{ is symmetric.}$$

In particular, every constituent name in the closure of an HS name is itself HS.

Proof. Assume first that $\tau \in \text{HS}$. The recursive definition says that τ is symmetric and that every immediate constituent $\sigma \in \text{dom}(\tau)$ is HS. Repeating the same statement down the well-founded name-rank recursion shows that every name reached after finitely many constituent steps is symmetric. These are exactly the members of $\text{TC}_{\text{name}}(\tau)$.

Conversely, suppose every member of $\text{TC}_{\text{name}}(\tau)$ is symmetric. We prove by induction on name rank that every $\rho \in \text{TC}_{\text{name}}(\tau)$ is HS. Such a ρ is symmetric by hypothesis; each immediate constituent of ρ also belongs to the same constituent closure and has smaller name rank, so the induction hypothesis makes every immediate constituent HS. The recursive definition then gives $\rho \in \text{HS}$. Taking $\rho = \tau$ proves the converse. The final assertion follows by applying the same induction to any constituent of an HS name. $\square$

Lemma 64.3 (Automorphisms preserve hereditary symmetry). *Let $(\mathbb{P}, \mathcal{G}, \mathcal{F})$ be a symmetric system. If $\tau \in \text{HS}$ and $\pi \in \mathcal{G}$, then $\pi\tau \in \text{HS}$. More precisely, if $H \in \mathcal{F}$ supports τ , then $\pi H \pi^{-1} \in \mathcal{F}$ supports $\pi\tau$.*

Proof. If $h = \pi k \pi^{-1}$ with $k \in H$, then

$$h(\pi\tau) = \pi(k\tau) = \pi\tau.$$

Thus $\pi H \pi^{-1} \leq \text{sym}(\pi\tau)$, and normality of $\mathcal{F}$ gives $\pi H \pi^{-1} \in \mathcal{F}$. Hence $\pi\tau$ is symmetric. Apply the same argument by induction down the constituent closure $\text{TC}_{\text{name}}(\tau)$: every constituent σ of τ is HS, so every transported constituent $\pi\sigma$ is HS. Therefore $\pi\tau$ is hereditarily symmetric. $\square$

Lemma 64.4 (Agreement with ordinary forcing on atomic HS formulas). *Let $\sigma, \tau \in \text{HS}$ and $p \in \mathbb{P}$. Then the symmetric and ordinary forcing relations agree on the atomic formulas built from these names:*

$$p \Vdash_{\text{sym}} \sigma \in \tau \iff p \Vdash \sigma \in \tau, \quad p \Vdash_{\text{sym}} \sigma = \tau \iff p \Vdash \sigma = \tau,$$

and likewise for the auxiliary forced-subset relation.

Proof. The ordinary atomic forcing relation of Volume VII and the symmetric atomic relation displayed above satisfy the same simultaneous rank-recursive clauses for membership, forced inclusion, and equality. Restrict the ordinary relation to pairs of HS names. If a constituent name occurs in an HS name, it is again HS by hereditary symmetry, so every recursive call stays inside this restricted domain. The uniqueness part of the simultaneous rank-recursion theorem for atomic forcing therefore identifies the two restricted relations. Concretely, one may equivalently prove the displayed equivalences simultaneously by induction on the pair of name ranks; at each step the two sides invoke identical lower-rank clauses. No assertion about compound formulas is made here. $\square$

Lemma 64.5 (Monotonicity and equivariance of symmetric forcing). *Let $\bar{\tau} \in \text{HS}$ and let $\varphi(\bar{x})$ be a formula. Then:*

1. *if $p \Vdash_{\text{sym}} \varphi(\bar{\tau})$ and $q \leq p$, then $q \Vdash_{\text{sym}} \varphi(\bar{\tau})$;*
2. *for every $\pi \in \mathcal{G}$,*

$$p \Vdash_{\text{sym}} \varphi(\bar{\tau}) \iff \pi p \Vdash_{\text{sym}} \varphi(\pi \bar{\tau}).$$

Proof. First replace an arbitrary input formula by its normalized basis formula using (SNF). Since the normalization is independent of conditions and name values, it suffices to prove both assertions for basis formulas. Proceed simultaneously by the atomic name-rank recursion and then by induction on basis-formula complexity. For atomic formulas the clauses are exactly the rank-recursive clauses displayed above; the monotonicity argument is the one from Volume VII, and equivariance follows because π preserves the forcing order, compatibility, density below a condition, and the recursive action on names.

For conjunction both assertions follow componentwise. For negation, suppose $p \Vdash_{\text{sym}} \neg\psi$ and $q \leq p$. If some $r \leq q$ forced ψ , then $r \leq p$, contradicting the defining negation clause; hence $q \Vdash_{\text{sym}} \neg\psi$. Equivariance of negation follows because $r \leq p$ iff $\pi r \leq \pi p$ and the induction hypothesis transports forcing of ψ in both directions.

For $\exists x \psi(x, \bar{\tau})$, monotonicity follows because a set dense below p is dense below every $q \leq p$. For equivariance, Lemma 64.3 gives

$$\sigma \in \text{HS} \iff \pi \sigma \in \text{HS},$$

and $q \mapsto \pi q$ is an order isomorphism from the cone below p onto the cone below πp . The induction hypothesis therefore carries the dense witness set in the existential clause bijectively onto the corresponding dense witness set for $\psi(x, \pi \bar{\tau})$. This proves both assertions for basis formulas, and (SNF) transfers them to every primitive-syntax formula. $\square$

Theorem 64.6 (Definability and effective uniformity of symmetric forcing). *Let M be a transitive set model of ZFC and let $(\mathbb{P}, \mathcal{G}, \mathcal{F}) \in M$ be a ground-model symmetric system. There is a first-order formula $\text{HS}_{\mathbb{P}, \mathcal{G}, \mathcal{F}}(\tau)$ which defines in M the hereditarily symmetric $\mathbb{P}$ -names. Moreover, for every formula $\varphi(\bar{x})$ of set theory there is a first-order formula*

$$F_{\varphi}^{\text{sym}}(p, \mathbb{P}, \mathcal{G}, \mathcal{F}, \bar{\tau})$$

constructed primitive-recursively from the Gödel code of φ , such that for $p \in \mathbb{P}$ and $\bar{\tau} \in \text{HS}^M$,

$$M \models F_{\varphi}^{\text{sym}}(p, \mathbb{P}, \mathcal{G}, \mathcal{F}, \bar{\tau}) \iff p \Vdash_{\text{sym}}^M \varphi(\bar{\tau}). \quad (\text{DEF})$$

Thus, for fixed parameters, the symmetric forcing relation is a ground-model definable relation, uniformly and effectively in the formula code.

Uses: well-founded recursion on forcing-name rank and the displayed recursive clauses for hereditary symmetry and symmetric forcing.

Proof. The predicate saying that τ is a $\mathbb{P}$ -name is defined by the forcing-name rank recursion. On the same well-founded constituent relation, the displayed recursive clause

$$\tau \in \text{HS} \iff \text{sym}(\tau) \in \mathcal{F} \wedge \forall \langle \sigma, p \rangle \in \tau (\sigma \in \text{HS})$$

therefore gives one first-order definition of hereditary symmetry over M . The atomic symmetric forcing relations are likewise simultaneous well-founded recursions on pairs of name ranks; the well-founded-recursion theorem gives fixed first-order formulas $F_{\equiv}^{\text{sym}}$ and $F_{\subseteq}^{\text{sym}}$ for them (and for the auxiliary inclusion relation).

Now recurse on the finite parse tree of φ . Conjunction is translated by conjunction of the previously constructed formulas, while negation is translated by

$$F_{\neg\psi}^{\text{sym}}(p, \dots) \equiv \neg \exists q \in \mathbb{P} (q \leq p \wedge F_{\psi}^{\text{sym}}(q, \dots)).$$

For an existential formula use

$$F_{\exists x \psi}^{\text{sym}}(p, \dots) \equiv \forall q \in \mathbb{P} (q \leq p \rightarrow \exists r \in \mathbb{P} \exists \sigma [r \leq q \wedge \text{HS}_{\mathbb{P}, \mathcal{G}, \mathcal{F}}(\sigma) \wedge F_{\psi}^{\text{sym}}(r, \dots, \sigma)]).$$

These are exactly the defining clauses of the basis relation. For an arbitrary primitive-syntax formula φ , first compute $\mathcal{N}(\varphi)$ and use the basis formula just constructed. Hence (DEF) holds for all formulas without identifying primitive $\vee, \rightarrow, \forall$ nodes with abbreviation strings.

Finally, the operation on Gödel codes is primitive recursive: first apply the primitive-recursive normalization map $\varphi \mapsto \mathcal{N}(\varphi)$, then send atomic and basis-compound codes to the fixed forcing-formula templates. No forcing name is numerically encoded in this claim; names remain ordinary set variables in the output formula. $\square$

Theorem 64.7 (Decision density and its effective uniformity for symmetric forcing). *For every formula $\varphi(\bar{x})$, every tuple $\bar{\tau} \in \text{HS}$, and every condition $p \in \mathbb{P}$, there is $q \leq p$ which symmetrically decides $\varphi(\bar{\tau})$. Equivalently,*

$$D_{\varphi, \bar{\tau}} := \{q \in \mathbb{P} : q \Vdash_{\text{sym}} \varphi(\bar{\tau}) \text{ or } q \Vdash_{\text{sym}} \neg \varphi(\bar{\tau})\}$$

is dense. If $(\mathbb{P}, \mathcal{G}, \mathcal{F}) \in M$ is a ground-model symmetric system over a transitive $M \models \text{ZFC}$ and $\bar{\tau} \in \text{HS}^M$, then this dense set belongs to M .

Fix the Gödel coding of formulas and finite ZFC proofs used later in this volume. There is also a primitive-recursive map which, from the code of φ , produces the code of a ZFC proof of the universally closed internal decision-density instance for φ , conditional only on the displayed first-order assertions that $(\mathbb{P}, \mathcal{G}, \mathcal{F})$ is a symmetric system and that the parameters are HS. Thus “decision density” is available later as an effective proof template, not merely as a metatheoretic existence theorem.

Uses: Theorem 64.6, atomic decision density for the ordinary forcing recursion, atomic agreement Lemma 64.4, and monotonicity Lemma 64.5.

Proof. The set $D_{\varphi, \bar{\tau}}$ belongs to M by Theorem 64.6 and Separation. By (SNF), deciding φ is equivalent to deciding the normalized basis formula $\mathcal{N}(\varphi)$, so for density it suffices to induct on basis-formula complexity. For atomic formulas the symmetric relation agrees with ordinary atomic forcing on HS names by Lemma 64.4; the rank-recursive atomic forcing theorem gives a deciding extension.

For conjunction, first strengthen to decide the first conjunct and then strengthen again to decide the second. The final condition decides the conjunction. For negation, a condition deciding ψ also decides $\neg\psi$: if it forces $\neg\psi$ there is nothing to prove, while if it forces ψ , monotonicity prevents any extension from forcing $\neg\psi$, so it forces $\neg\neg\psi$.

Let φ be $\exists x \psi(x, \bar{\tau})$ and fix p . Put

$$W := \{q \leq p : \exists r \leq q \exists \sigma \in \text{HS} (r \Vdash_{\text{sym}} \psi(\sigma, \bar{\tau}))\}.$$

If W is dense below p , then the existential clause gives $p \Vdash_{\text{sym}} \exists x \psi$. If W is not dense below p , choose $q \leq p$ such that no extension of q lies in W . No extension of q can force the existential formula, since its defining dense witness set would then place a still stronger condition in W . Hence the negation clause gives $q \Vdash_{\text{sym}} \neg \exists x \psi$. Thus every p has a deciding extension.

For the final effectiveness assertion, inspect the induction just given as a construction of finite proof codes. The atomic case inserts the fixed ZFC proof of atomic decision density; conjunction and negation append fixed finite proof templates to the proof codes already produced for the immediate subformulas; and the existential case appends the fixed proof of the displayed dichotomy for W . After first computing the primitive-recursive code of $\mathcal{N}(\varphi)$ using the Volume-VII normalization lemma, passing to immediate basis subformulas, substituting their codes into the fixed templates, and concatenating finite derivations are primitive-recursive syntactic operations. Primitive recursion on the normalized formula code, composed with normalization, therefore yields the claimed primitive-recursive proof-code constructor. No forcing name is an input to that constructor: name parameters remain universally quantified set variables in the output ZFC formula. $\square$

Lemma 64.8 (Dense forcing lifts to the original condition). *Let $\varphi(\bar{\tau})$ be any formula with HS name parameters. If*

$$D = \{q \leq p : q \Vdash_{\text{sym}} \varphi(\bar{\tau})\}$$

is dense below p , then $p \Vdash_{\text{sym}} \varphi(\bar{\tau})$.

Uses: the recursive definition of symmetric forcing and its monotonicity.

Proof. Proceed by the same simultaneous recursion used to define $\Vdash_{\text{sym}}$. For the atomic relation $\sigma \in \tau$, each extension of p can first be strengthened into D and then, by the membership clause at that stronger condition, further strengthened to a condition witnessing equality with a constituent of τ ; hence the witness set in the membership clause is dense below p . For the auxiliary forced-inclusion relation, fix $\langle \rho, r \rangle \in \sigma$ and $q \leq p, r$. Density gives $q' \leq q$ with $q' \Vdash_{\text{sym}}^{\text{aux}} \sigma \subseteq \tau$; its auxiliary subset clause then gives $s \leq q'$ forcing $\rho \in \tau$. Equality follows from the two auxiliary inclusion directions.

It is enough to prove the assertion for basis formulas, because both the hypothesis set and conclusion for an arbitrary φ are, by (SNF), exactly those for $\mathcal{N}(\varphi)$. For conjunction, intersect the two dense forcing sets by two successive strengthenings and use monotonicity. For negation, if some $r \leq p$ forced ψ , density would give $q \leq r$ forcing $\neg\psi$, while monotonicity would keep $q \Vdash_{\text{sym}} \psi$, a contradiction. Thus no extension of p forces ψ , which is exactly $p \Vdash_{\text{sym}} \neg\psi$. For an existential formula, first strengthen an arbitrary $q \leq p$ into D ; the existential clause there supplies, densely below that strengthening, a further condition and an HS witness name. Hence the existential witness set is dense below p . This proves the basis case, and (SNF) proves the lemma for every primitive-syntax formula. $\square$

Lemma 64.9 (Disjunction is dense local choice). *For formulas φ, ψ with HS name parameters and a condition p ,*

$$p \Vdash_{\text{sym}} (\varphi \vee \psi) \iff D_{\vee}(p) := \{q \leq p : q \Vdash_{\text{sym}} \varphi \text{ or } q \Vdash_{\text{sym}} \psi\} \text{ is dense below } p. \quad (\text{OR})$$

By (SN), forcing the primitive formula $\varphi \vee \psi$ means forcing the normalized basis formula $\neg(\neg\mathcal{N}(\varphi) \wedge \neg\mathcal{N}(\psi))$. Since forcing φ itself means forcing $\mathcal{N}(\varphi)$, the proof may write φ, ψ for their normalized basis forms without changing the statement.

Proof. Assume first that $p \Vdash_{\text{sym}} \varphi \vee \psi$. If $D_{\vee}(p)$ were not dense below p , choose $q \leq p$ such that no extension of q lies in $D_{\vee}(p)$. Decision density for φ gives $r \leq q$ deciding φ . It cannot force φ , so it forces $\neg\varphi$. Decision density for ψ below r gives $s \leq r$ deciding ψ ; again it cannot force ψ , so

$$s \Vdash_{\text{sym}} \neg\varphi \wedge \neg\psi.$$

This contradicts monotonicity of $p \Vdash_{\text{sym}} \neg(\neg\varphi \wedge \neg\psi)$. Hence $D_{\vee}(p)$ is dense.

Conversely suppose $D_{\vee}(p)$ is dense. If some $q \leq p$ forced $\neg\varphi \wedge \neg\psi$, density would give $r \leq q$ forcing either φ or ψ . Monotonicity would keep the corresponding negation forced at r , a contradiction. Thus no extension of p forces $\neg\varphi \wedge \neg\psi$, which by the negation clause is exactly $p \Vdash_{\text{sym}} \varphi \vee \psi$. $\square$

Lemma 64.10 (First-order rules and bounded-name forcing). *Let $p \in \mathbb{P}$, let all displayed name parameters belong to HS, and let φ, ψ be formulas.*

1. Reflexivity and substitution for forced equality:

$$p \Vdash_{\text{sym}} \sigma = \sigma,$$

and if $p \Vdash_{\text{sym}} \sigma = \tau$ and $p \Vdash_{\text{sym}} \varphi(\sigma, \bar{\rho})$, then $p \Vdash_{\text{sym}} \varphi(\tau, \bar{\rho})$.

2. Modus ponens: if $p \Vdash_{\text{sym}} \varphi$ and $p \Vdash_{\text{sym}} (\varphi \rightarrow \psi)$, then $p \Vdash_{\text{sym}} \psi$.
3. Universal instantiation: if $p \Vdash_{\text{sym}} \forall x \varphi(x, \bar{\rho})$, then $p \Vdash_{\text{sym}} \varphi(\sigma, \bar{\rho})$ for every $\sigma \in \text{HS}$.
4. Constituent membership: if $\langle \rho, r \rangle \in \tau$, then $r \Vdash_{\text{sym}} \rho \in \tau$.
5. Bounded-name criterion: if for every $\langle \rho, r \rangle \in \tau$ and every $q \leq p, r$ one has $q \Vdash_{\text{sym}} \psi(\rho, \bar{\eta})$, then

$$p \Vdash_{\text{sym}} \forall x \in \tau \psi(x, \bar{\eta}).$$

Uses: the recursive atomic clauses, decision density, monotonicity, and Lemma 64.8.

Proof. Equality reflexivity and substitution are proved simultaneously by induction on the ranks of the names for atomic formulas and then on basis-formula complexity. At the atomic stage the two subset clauses defining equality give reflexivity, and the membership/subset recursions are unchanged when a forced-equal name is replaced by the other one. The clauses for $\neg$, $\wedge$, and $\exists$ preserve this interchangeability. For arbitrary primitive-syntax formulas, $\mathcal{N}$ commutes with substitution, so (SNF) transfers the basis result to every formula.

For modus ponens, use (SNF): forcing the primitive implication $\varphi \rightarrow \psi$ means forcing $\neg(\mathcal{N}(\varphi) \wedge \neg\mathcal{N}(\psi))$, while forcing φ and ψ means forcing their normalized basis forms. Suppress the symbols $\mathcal{N}$ for this paragraph. If some $r \leq p$ forced $\neg\psi$, then monotonicity would give $r \Vdash_{\text{sym}} \varphi$ as well, hence $r \Vdash_{\text{sym}} \varphi \wedge \neg\psi$, contradicting the forced implication. Decision density for ψ therefore shows that conditions forcing ψ are dense below p ; Lemma 64.8 yields $p \Vdash_{\text{sym}} \psi$.

For universal instantiation, (SN) defines forcing of the primitive $\forall x \varphi$ through the normalized basis formula $\neg\exists x \neg\mathcal{N}(\varphi)$. If an extension $q \leq p$ forced $\neg\varphi(\sigma, \bar{\rho})$, then the existential clause would make conditions witnessing $\exists x \neg\varphi(x, \bar{\rho})$ dense below a further extension of q , contradicting the negation clause at p . Hence no extension of p forces $\neg\varphi(\sigma, \bar{\rho})$. Decision density and Lemma 64.8 now imply $p \Vdash_{\text{sym}} \varphi(\sigma, \bar{\rho})$.

If $\langle \rho, r \rangle \in \tau$, reflexivity gives $q \Vdash_{\text{sym}} \rho = \rho$ for every $q \leq r$. Thus the witness set in the atomic membership clause for $\rho \in \tau$ is dense below r , proving constituent membership.

Finally assume the hypothesis of the bounded-name criterion. By definition,

$$\forall x \in \tau \psi(x, \bar{\eta}) \quad \text{abbreviates} \quad \neg\exists x (x \in \tau \wedge \neg\psi(x, \bar{\eta})).$$

Thus it is enough to show that no extension of p forces the displayed existential. Suppose toward contradiction that some $q \leq p$ does force it. By the existential clause there are a stronger condition $s \leq q$ and an HS name ξ with

$$s \Vdash_{\text{sym}} \xi \in \tau \wedge \neg\psi(\xi, \bar{\eta}).$$

The atomic membership clause lets us strengthen once more to $t \leq s$ and find a constituent $\langle \rho, r \rangle \in \tau$ such that

$$t \leq r, \quad t \Vdash_{\text{sym}} \xi = \rho.$$

Since $t \leq p, r$, the hypothesis gives $t \Vdash_{\text{sym}} \psi(\rho, \bar{\eta})$; equality substitution gives $t \Vdash_{\text{sym}} \psi(\xi, \bar{\eta})$. Monotonicity also preserves $t \Vdash_{\text{sym}} \neg\psi(\xi, \bar{\eta})$, a contradiction. Hence no extension of p forces the existential, and the negation clause is exactly the required bounded universal statement. $\square$

Lemma 64.11 (Conditionwise implication and quantifier closure). *Let all displayed name parameters be HS.*

1. If, for every $q \leq p$,

$$q \Vdash_{\text{sym}} \alpha \implies q \Vdash_{\text{sym}} \beta, \tag{CI}$$

then

$$p \Vdash_{\text{sym}} (\alpha \rightarrow \beta).$$

2. If $p \Vdash_{\text{sym}} \varphi(\sigma, \bar{\tau})$ for an HS name σ , then

$$p \Vdash_{\text{sym}} \exists x \varphi(x, \bar{\tau}). \quad (\text{EI})$$

3. If for every $q \leq p$ and every HS name σ ,

$$q \Vdash_{\text{sym}} \varphi(\sigma, \bar{\tau}), \quad (\text{UC0})$$

then

$$p \Vdash_{\text{sym}} \forall x \varphi(x, \bar{\tau}). \quad (\text{UC})$$

Proof. For (1), use (SNF) and (SN):

$$p \Vdash_{\text{sym}} (\alpha \rightarrow \beta) \iff p \Vdash_{\text{sym}} \neg(\mathcal{N}(\alpha) \wedge \neg\mathcal{N}(\beta)). \quad (\text{CI-N})$$

Assume that some $r \leq p$ forces the basis conjunction on the right. By (SNF), this says that $r \Vdash_{\text{sym}} \alpha$ and $r \Vdash_{\text{sym}} \neg\beta$. Hypothesis (CI), applied to r , gives $r \Vdash_{\text{sym}} \beta$, contradicting the negation clause. Hence no extension of p forces that conjunction, and the basis negation clause together with (CI-N) gives $p \Vdash_{\text{sym}} (\alpha \rightarrow \beta)$.

For (2), let $q \leq p$. Monotonicity gives $q \Vdash_{\text{sym}} \varphi(\sigma, \bar{\tau})$. In the existential witness set below p , take the stronger condition $r = q$ and the same HS witness σ . Thus the witness set is dense below p , proving (EI).

For (3), by (SNF) and (SN), forcing the primitive universal $\forall x \varphi(x, \bar{\tau})$ is forcing the basis formula

$$\neg \exists x \neg \mathcal{N}(\varphi)(x, \bar{\tau}). \quad (\text{UC-N})$$

Suppose toward contradiction that some $r \leq p$ forced $\exists x \neg \mathcal{N}(\varphi)(x, \bar{\tau})$. The existential clause yields a stronger $s \leq r$ and an HS name σ such that

$$s \Vdash_{\text{sym}} \neg \mathcal{N}(\varphi)(\sigma, \bar{\tau}).$$

But (UC0), applied to $s \leq p$ and this σ , gives $s \Vdash_{\text{sym}} \varphi(\sigma, \bar{\tau})$, hence by (SNF) $s \Vdash_{\text{sym}} \mathcal{N}(\varphi)(\sigma, \bar{\tau})$, a contradiction. Therefore no extension of p forces the existential in (UC-N). The basis negation clause and (UC-N) give exactly $p \Vdash_{\text{sym}} \forall x \varphi(x, \bar{\tau})$. $\square$

Corollary 64.12 (Effective forcing-logic proof templates). *Fix the formula/proof coding used above. From the codes of the formulas occurring in an instance of Lemma 64.8, Lemma 64.9, Lemma 64.10, or Lemma 64.11, one can primitive-recursively construct a ZFC proof code for the corresponding universally closed internal forcing-logic instance, conditional on the symmetric-system and HS guards.*

Proof. The atomic parts are fixed rank-recursion theorems. Every compound-formula step in the four proofs is obtained by syntactic substitution into one of finitely many displayed templates, together with monotonicity and the effective Decision Density Theorem 64.7. Formula parsing, capture-avoiding substitution, finite concatenation of proofs, and insertion of previously produced proof codes are primitive recursive. Structural recursion on the input formula codes therefore produces the required ZFC proof codes. This corollary is the precise effective interface used by the later condition-relative derivation translator. $\square$

Lemma 64.13 (Auxiliary forced inclusion equals forcing the subset formula). *For HS names σ, τ and every condition p ,*

$$p \Vdash_{\text{sym}}^{\text{aux}} \sigma \subseteq \tau \iff p \Vdash_{\text{sym}} \forall x (x \in \sigma \rightarrow x \in \tau). \quad (\text{SUB})$$

From this lemma onward we write simply $p \Vdash_{\text{sym}} \sigma \subseteq \tau$ for either equivalent condition.

Uses: the dense-forcing lemma, constituent membership, the bounded-name criterion, universal instantiation, and modus ponens.

Proof. Assume first

$$p \Vdash_{\text{sym}}^{\text{aux}} \sigma \subseteq \tau. \quad (1)$$

Fix a constituent $\langle \rho, r \rangle \in \sigma$ and a condition $q \leq p, r$. For every $u \leq q$, (1) supplies $s \leq u$ with $s \Vdash_{\text{sym}} \rho \in \tau$. Thus conditions forcing $\rho \in \tau$ are dense below q , and Lemma 64.8 gives

$$q \Vdash_{\text{sym}} \rho \in \tau.$$

The bounded-name criterion Lemma 64.10(5), with $\psi(x)$ equal to $x \in \tau$, now yields

$$p \Vdash_{\text{sym}} \forall x \in \sigma (x \in \tau),$$

which is exactly the right side of (SUB) by the bounded-quantifier abbreviation.

Conversely suppose

$$p \Vdash_{\text{sym}} \forall x (x \in \sigma \rightarrow x \in \tau). \quad (2)$$

Fix $\langle \rho, r \rangle \in \sigma$ and $q \leq p, r$. Constituent membership gives $q \Vdash_{\text{sym}} \rho \in \sigma$. Monotonicity of (2) and universal instantiation give

$$q \Vdash_{\text{sym}} (\rho \in \sigma \rightarrow \rho \in \tau),$$

and modus ponens gives $q \Vdash_{\text{sym}} \rho \in \tau$. In the auxiliary clause (AUX) we may therefore choose the required strengthening to be $s = q$ itself. This proves $p \Vdash_{\text{sym}}^{\text{aux}} \sigma \subseteq \tau$, completing the equivalence. $\square$

Theorem 64.14 (Hereditary symmetry and the required name operations). *Let $(\mathbb{P}, \mathcal{G}, \mathcal{F})$ be a symmetric system, let HS be the class of hereditarily symmetric names, and let $\tau, \sigma, \rho_1, \dots, \rho_n \in \text{HS}$. Then the canonical pair, union, and successor names built from these names lie in HS, and $\pi\tau \in \text{HS}$ for every $\pi \in \mathcal{G}$. For every generic filter G ,*

$$\text{Pair}(\sigma, \tau)^G = \{\sigma^G, \tau^G\}, \quad \text{Union}(\tau)^G = \bigcup \tau^G, \quad \text{Succ}(\sigma)^G = \sigma^G \cup \{\sigma^G\}.$$

Moreover, for any formula $\varphi(x, \bar{y})$, the bounded subname

$$\tau \restriction \varphi(\bar{\rho}) := \{\langle \eta, p \rangle \in \tau : p \Vdash_{\text{sym}} \varphi(\eta, \bar{\rho})\}$$

is hereditarily symmetric. Every name in $\text{TC}_{\text{name}}(\tau)$ is HS by Lemma 64.2.

Proof.

Formal derivation spine

Formal spine: combine supports by intersection and transport them by conjugation. If H supports τ and K supports σ , then $H \cap K \in \mathcal{F}$ supports the pair/union constructions. If H supports τ , then $\pi H \pi^{-1}$ supports $\pi\tau$.

Choose support subgroups in $\mathcal{F}$ for the finitely many names under consideration. Closure of $\mathcal{F}$ under finite intersections gives a common support H . Every element of H fixes each input name, hence fixes the canonical pair name and the canonical union name obtained from them. Their constituent names are already HS, so the resulting names are hereditarily symmetric.

For the valuation formulas, every filter contains $1_{\mathbb{P}}$, so the pair name evaluates immediately to $\{\sigma^G, \tau^G\}$. For the union name, if $x \in \text{Union}(\tau)^G$, choose a pair $\langle \rho, r \rangle$ in the union name with $r \in G$ and $x = \rho^G$. By definition there are $\langle \sigma, p \rangle \in \tau$ and $\langle \rho, q \rangle \in \sigma$ with $r \leq p, q$. Upward closure of the filter gives $p, q \in G$, hence $\sigma^G \in \tau^G$ and $x \in \sigma^G$, so $x \in \bigcup \tau^G$. Conversely, if $x \in \bigcup \tau^G$, choose $\langle \sigma, p \rangle \in \tau$ and $\langle \rho, q \rangle \in \sigma$ with $p, q \in G$, $\sigma^G \in \tau^G$, and $x = \rho^G$. Directedness of G supplies $r \in G$ with $r \leq p, q$, so $\langle \rho, r \rangle \in \text{Union}(\tau)$ and $x \in \text{Union}(\tau)^G$.

For the successor name, any support of σ fixes both the old pairs of σ and the added pair $\langle \sigma, 1_{\mathbb{P}} \rangle$; all constituents are HS. Thus $\text{Succ}(\sigma) \in \text{HS}$. Its valuation is immediate from the definition:

$$\text{Succ}(\sigma)^G = \sigma^G \cup \{\sigma^G\}.$$

The assertion that $\pi\tau \in \text{HS}$ for $\pi \in \mathcal{G}$ is exactly Lemma 64.3; we record it here because the name operations used later include automorphism transport.

For the bounded subname, let $H \in \mathcal{F}$ be a common support for τ and the parameter names $\bar{\rho}$. If $h \in H$, Lemma 64.5 gives

$$p \Vdash_{\text{sym}} \varphi(\eta, \bar{\rho}) \iff hp \Vdash_{\text{sym}} \varphi(h\eta, h\bar{\rho}) = \varphi(h\eta, \bar{\rho}).$$

Since $h\tau = \tau$, the action of h permutes exactly the pairs in $\tau \restriction \varphi(\bar{\rho})$. Hence H supports this subname. Every constituent η belongs to $\text{TC}_{\text{name}}(\tau)$, so Lemma 64.2 makes it HS. Therefore the bounded subname is hereditarily symmetric. $\square$

Theorem 64.15 (Root forcing of Extensionality, Empty Set, set induction, and Foundation). *For every ground-model symmetric system over $M \models \text{ZFC}$, the symmetric forcing relation satisfies the following at the root.*

1. Extensionality:

$$\Vdash_{\text{sym}} \forall x \forall y [\forall z (z \in x \leftrightarrow z \in y) \rightarrow x = y]. \quad (\text{EX})$$

2. Empty Set:

$$\Vdash_{\text{sym}} \exists e \forall z (z \notin e). \quad (\text{EM})$$

3. Set induction: for every formula $\varphi(x, \bar{u})$ and every HS parameter tuple $\bar{\rho}$,

$$\Vdash_{\text{sym}} [\forall x ((\forall y \in x \varphi(y, \bar{\rho})) \rightarrow \varphi(x, \bar{\rho})) \rightarrow \forall x \varphi(x, \bar{\rho})]. \quad (\text{SI})$$

4. Foundation:

$$\Vdash_{\text{sym}} \forall a [a \neq \emptyset \rightarrow \exists y \in a \forall z \in a (z \notin y)]. \quad (\text{FD})$$

The proofs are uniform in the code of φ in (SI).

Proof. Extensionality. Fix HS names σ, τ and a condition p with

$$p \Vdash_{\text{sym}} \forall z (z \in \sigma \leftrightarrow z \in \tau). \quad (1)$$

We prove $p \Vdash_{\text{sym}} \sigma \subseteq \tau$. Let $\langle \rho, r \rangle \in \sigma$ and $q \leq p, r$. Constituent membership gives $q \Vdash_{\text{sym}} \rho \in \sigma$. Monotonicity of (1), universal instantiation, and modus ponens give $q \Vdash_{\text{sym}} \rho \in \tau$. In the atomic subset clause we may therefore take the required strengthening to be q itself. Hence

$$p \Vdash_{\text{sym}} \sigma \subseteq \tau.$$

The symmetric argument gives $p \Vdash_{\text{sym}} \tau \subseteq \sigma$, so by the definition of forced equality,

$$p \Vdash_{\text{sym}} \sigma = \tau. \quad (2)$$

Lemma 64.11(1) converts the conditionwise implication (1) $\Rightarrow$ (2) into the forced implication. Applying Lemma 64.11(3) twice gives the universal closure (EX).

Empty Set. Let $\dot{e} := \emptyset$, the empty forcing name. It is HS because it is fixed by every automorphism and has no constituents. No condition can force $\eta \in \dot{e}$: the witness set in the atomic membership clause is empty and therefore is not dense below any condition. Thus every condition forces $\eta \notin \dot{e}$ for every HS name η . Lemma 64.11(3) gives

$$\Vdash_{\text{sym}} \forall z (z \notin \dot{e}),$$

and Lemma 64.11(2) gives the existential witness $\dot{e}$. This proves (EM).

Set induction. Fix a formula $\varphi(x, \bar{u})$, HS parameters $\bar{\rho}$, and a condition p satisfying

$$p \Vdash_{\text{sym}} \forall x ((\forall y \in x \varphi(y, \bar{\rho})) \rightarrow \varphi(x, \bar{\rho})). \quad (3)$$

We claim

$$p \Vdash_{\text{sym}} \forall x \varphi(x, \bar{\rho}). \quad (4)$$

Suppose otherwise. By Lemma 64.11(3), failure of (4) means that for some $q \leq p$ and some HS name τ , $q \not\Vdash_{\text{sym}} \varphi(\tau, \bar{\rho})$. Decision density supplies $r \leq q$ with

$$r \Vdash_{\text{sym}} \neg \varphi(\tau, \bar{\rho}). \quad (5)$$

Among all pairs (s, ξ) with $s \leq p$, $\xi \in \text{HS}$, and

$$s \Vdash_{\text{sym}} \neg \varphi(\xi, \bar{\rho}),$$

choose one, still denoted (r, τ) , for which the forcing-name rank of τ is least. This minimum exists: the class of such ranks is nonempty by (5), and after choosing any one witness rank α , Separation on the set $\alpha + 1$ produces the nonempty set of witness ranks at most α , which has a least ordinal.

Monotonicity of (3) and universal instantiation at r give

$$r \Vdash_{\text{sym}} (\forall y \in \tau \varphi(y, \bar{\rho})) \rightarrow \varphi(\tau, \bar{\rho}). \quad (6)$$

If $r \Vdash_{\text{sym}} \forall y \in \tau \varphi(y, \bar{\rho})$, then modus ponens with (6) contradicts (5). Hence

$$r \not\Vdash_{\text{sym}} \forall y \in \tau \varphi(y, \bar{\rho}). \quad (7)$$

Apply the contrapositive of the bounded-name criterion in Lemma 64.10: from (7) there must exist a constituent $\langle \xi, u \rangle \in \tau$ and a condition $s \leq r, u$ for which

$$s \not\Vdash_{\text{sym}} \varphi(\xi, \bar{\rho}).$$

Decision density gives $t \leq s$ with

$$t \Vdash_{\text{sym}} \neg \varphi(\xi, \bar{\rho}). \quad (8)$$

But every constituent has strictly smaller forcing-name rank than the name containing it, so

$$\text{rank}_{\mathbb{P}}(\xi) < \text{rank}_{\mathbb{P}}(\tau),$$

contradicting the minimal choice of τ . Thus (4) holds. Lemma 64.11(1) turns (3) $\Rightarrow$ (4) into the forced implication, proving (SI) after root closure.

Foundation. Fix an HS name τ . Suppose toward contradiction that a condition p forces both

$$\tau \neq \emptyset \quad (9)$$

and the negation of

$$\exists y \in \tau \forall z \in \tau (z \notin y). \quad (10)$$

We use set induction with the formula

$$\psi(x) := x \notin \tau.$$

We first verify its induction premise below p . Let $q \leq p$, let $\sigma \in \text{HS}$, and assume

$$q \Vdash_{\text{sym}} \forall y \in \sigma (y \notin \tau). \quad (11)$$

We claim

$$q \Vdash_{\text{sym}} \sigma \notin \tau. \quad (12)$$

If not, decision density gives $r \leq q$ with

$$r \Vdash_{\text{sym}} \sigma \in \tau. \quad (13)$$

We show that

$$D := \{s \leq r : s \Vdash_{\text{sym}} \exists z (z \in \tau \wedge z \in \sigma)\}$$

is dense below r . Suppose not. Then some $u \leq r$ has no extension in D . Fix a constituent $\langle \rho, a \rangle \in \tau$ and a condition $v \leq u, a$. If an extension $w \leq v$ forced $\rho \in \sigma$, then constituent membership gives $w \Vdash_{\text{sym}} \rho \in \tau$, and Lemma 64.11(2), applied to the witness ρ , would give

$$w \Vdash_{\text{sym}} \exists z (z \in \tau \wedge z \in \sigma),$$

contrary to the choice of u . Hence no extension of v forces $\rho \in \sigma$, so by the negation clause

$$v \Vdash_{\text{sym}} \rho \notin \sigma.$$

The bounded-name criterion, applied to every constituent of τ , yields

$$u \Vdash_{\text{sym}} \forall z \in \tau (z \notin \sigma). \quad (14)$$

Monotonicity of (13) gives $u \Vdash_{\text{sym}} \sigma \in \tau$. Thus Lemma 64.11(2), with witness σ , gives

$$u \Vdash_{\text{sym}} \exists y \in \tau \forall z \in \tau (z \notin y),$$

contradicting the negation forced by p . Therefore D is dense below r , and Lemma 64.8 gives

$$r \Vdash_{\text{sym}} \exists z (z \in \tau \wedge z \in \sigma). \quad (15)$$

The existential clause supplies $s \leq r$ and an HS name ξ with

$$s \Vdash_{\text{sym}} \xi \in \tau \wedge \xi \in \sigma. \quad (16)$$

But monotonicity of (11) and universal instantiation at s give

$$s \Vdash_{\text{sym}} \xi \notin \tau,$$

contradicting (16). Hence (12) holds.

Lemma 64.11(1) and then Lemma 64.11(3) show that, below p , the set-induction premise for ψ is forced. By (SI),

$$p \Vdash_{\text{sym}} \forall x (x \notin \tau). \quad (17)$$

On the other hand, (9) is the abbreviation $p \Vdash_{\text{sym}} \exists x (x \in \tau)$. Its existential clause yields a stronger condition and an HS witness forcing membership in τ , contradicting (17). Thus no condition can force both nonemptiness of τ and the negation of the Foundation conclusion. Lemma 64.11(1) gives the Foundation implication for each τ , and Lemma 64.11(3) gives the universal closure (FD).

All constructions in the set-induction proof are recursive in the formula code of φ : the formula-dependent steps are only formation of the corresponding symmetric-forcing formulas and insertion into the fixed minimal-rank argument. $\square$

Proposition 64.16 (Evaluation of hereditarily symmetric names gives a transitive class N). *For an M -generic filter G , define*

$$N = \{\tau^G : \tau \in \text{HS}^M\}.$$

Then N is a transitive class.

Proof.

Formal derivation spine

Formal spine for Evaluation of hereditarily symmetric names gives a transitive class N.

$$\pi H \pi^{-1} \in \mathcal{F}$$

Let $N = \{\tau^G : \tau \text{ is hereditarily symmetric}\}$. If $x \in y \in N$, choose a hereditarily symmetric name τ with $y = \tau^G$. By definition of valuation, $x = \sigma^G$ for some constituent name σ in the transitive closure of τ and some condition in G . Hereditary symmetry of τ implies that σ is hereditarily symmetric, so $x \in N$. Hence N is transitive. Check names are fixed by every automorphism and are hereditarily symmetric, so the ground model is contained in N ; in particular every ordinal, represented by its check name, lies in N .

Further explanation. Take $x \in y \in N$. Choose a hereditarily symmetric name τ with $\tau^G = y$. By the definition of valuation, $x = \sigma^G$ for some constituent name σ appearing below τ with a condition in G . Hereditary symmetry of τ implies hereditary symmetry of σ , so $x \in N$. Hence N is transitive. This argument also explains why ordinary symmetry of the top name is insufficient: symmetry must persist through all constituents. $\square$

Theorem 64.17 (N contains the ground model and all ordinals). *The symmetric model N contains the ground model and all ordinals:*

$$M \subseteq N \subseteq M[G], \quad \text{Ord}^N = \text{Ord}^M.$$

Proof.

Formal derivation spine

Formal spine for N contains the ground model and all ordinals.

$$\pi \check{x} = \check{x}.$$

$$x = \check{x}^G \in N.$$

$$\pi H \pi^{-1} \in \mathcal{F}$$

For every ground-model set x , the check name $\check{x}$ is fixed by every automorphism: induction on rank gives

$$\pi \check{x} = \check{x}.$$

Its stabilizer is therefore the whole automorphism group, which belongs to every normal filter. The same induction shows that all constituents of $\check{x}$ are hereditarily symmetric. Hence

$$x = \check{x}^G \in N.$$

Thus the entire ground model lies in the symmetric submodel.

In particular every ground-model ordinal belongs to N . Ordinary forcing adds no new ordinals, and $N \subseteq M[G]$; therefore N and the ground model have exactly the same ordinals.

Thus the symmetric model changes which sets are available without changing the ordinal scale. This fact is used repeatedly when comparing finite supports, cardinal statements, and the ambient generic extension. $\square$

Theorem 64.18 (Symmetric Truth Lemma). *Fix an M -generic filter $G \subseteq \mathbb{P}$ and put $N = \{\tau^G : \tau \in \text{HS}^M\}$. For every formula $\varphi(\bar{x})$ and $\bar{\tau} \in \text{HS}^M$,*

$$N \models \varphi(\bar{\tau}^G) \iff \exists p \in G (p \Vdash_{\text{sym}} \varphi(\bar{\tau})). \quad (\text{TL})$$

Uses: Definability Theorem 64.6, Decision Density Theorem 64.7, atomic agreement, monotonicity, and the definition of N .

Proof. First prove (TL) for basis formulas by induction on basis-formula complexity. Lemma 64.4 identifies symmetric atomic forcing with ordinary atomic forcing on HS names, so the atomic forcing theorem gives the base case. Conjunction is immediate.

For negation, if some $p \in G$ forces $\neg\psi$, then the induction hypothesis cannot give a condition in G forcing ψ , since directedness of G and monotonicity would produce a common stronger condition forcing both. Conversely, if $N \models \neg\psi$, genericity makes G meet the dense decision set from Theorem 64.7; it cannot meet the positive side by the induction hypothesis, so it meets the side forcing $\neg\psi$.

Suppose $N \models \exists x \psi(x, \bar{\tau}^G)$. Choose $b \in N$ witnessing this and choose $\sigma \in \text{HS}^M$ with $b = \sigma^G$. By induction some $r \in G$ forces $\psi(\sigma, \bar{\tau})$. Monotonicity shows that below r the same σ is a witness on every stronger condition, so the existential clause gives $r \Vdash_{\text{sym}} \exists x \psi(x, \bar{\tau})$.

Conversely, if $p \in G$ forces the existential statement, its dense witness set is an element of M by Theorem 64.6 and is dense below p . Genericity gives $r \in G$, $r \leq p$, and $\sigma \in \text{HS}^M$ with $r \Vdash_{\text{sym}} \psi(\sigma, \bar{\tau})$. The induction hypothesis yields $N \models \psi(\sigma^G, \bar{\tau}^G)$, proving the existential statement in N . This completes the basis-formula induction. For an arbitrary Volume-I formula φ , the Volume-VII normalization lemma gives ordinary truth equivalence $\varphi \leftrightarrow \mathcal{N}(\varphi)$, while (SNF) identifies its symmetric forcing assertion with that of $\mathcal{N}(\varphi)$. Hence (TL) holds for every primitive-syntax formula. $\square$

Lemma 64.5 and Theorem 64.18 are the two properties used below under the names *symmetric forcing symmetry* and *symmetric Truth Lemma*; both now occur before the first ZF-axiom verification that invokes them.

Theorem 64.19 (Condition-relative Pairing, Union, successor, and root Infinity). *Let $\sigma, \tau, \eta \in \text{HS}^M$ and $p \in \mathbb{P}$. Then*

$$p \Vdash_{\text{sym}} \eta \in \text{Pair}(\sigma, \tau) \iff p \Vdash_{\text{sym}} (\eta = \sigma \vee \eta = \tau), \quad (\text{P})$$

$$p \Vdash_{\text{sym}} \eta \in \text{Union}(\tau) \iff p \Vdash_{\text{sym}} \exists x (x \in \tau \wedge \eta \in x), \quad (\text{U})$$

and

$$p \Vdash_{\text{sym}} \eta \in \text{Succ}(\sigma) \iff p \Vdash_{\text{sym}} (\eta \in \sigma \vee \eta = \sigma). \quad (\text{S})$$

Consequently the symmetric forcing relation forces at the root the Pairing and Union axioms,

$$\Vdash_{\text{sym}} \forall x \forall y \exists z \forall w [w \in z \leftrightarrow (w = x \vee w = y)], \quad (\text{PAIR})$$

$$\Vdash_{\text{sym}} \forall a \exists u \forall z [z \in u \leftrightarrow \exists x (x \in a \wedge z \in x)], \quad (\text{UN})$$

and the following standard membership-language form of Infinity:

$$\Vdash_{\text{sym}} \exists I [\exists e (e \in I \wedge \forall z (z \notin e)) \wedge \forall x \in I \exists y \in I \forall z (z \in y \leftrightarrow (z \in x \vee z = x))]. \quad (\text{INF})$$

All three root proofs are finite templates; the only varying objects are the displayed name parameters.

Uses: Theorem 64.14; Lemmas 64.8, 64.9, 64.10, and 64.11; and Theorem 64.17 for hereditary symmetry of check names.

Proof. Pairing. The only constituents of $\text{Pair}(\sigma, \tau)$ are $\langle \sigma, 1_{\mathbb{P}} \rangle$ and $\langle \tau, 1_{\mathbb{P}} \rangle$. Hence the atomic membership clause says that $p \Vdash_{\text{sym}} \eta \in \text{Pair}(\sigma, \tau)$ exactly when conditions forcing $\eta = \sigma$ or $\eta = \tau$ are dense below p . Lemma 64.9 identifies that dense-local-choice condition with $p \Vdash_{\text{sym}} (\eta = \sigma \vee \eta = \tau)$, proving (P).

Because (P) holds below every condition, Lemma 64.11(1) gives both forced implications in the biconditional, and conjunction gives the biconditional itself. Lemma 64.11(3) universally closes the variable w , while Lemma 64.11(2) uses $\text{Pair}(\sigma, \tau)$ as the existential witness. Two further applications of Lemma 64.11(3), first to y and then to x , give (PAIR).

Union. Assume first that $p \Vdash_{\text{sym}} \eta \in \text{Union}(\tau)$. By the atomic membership clause, densely below p there are a condition t and a constituent $\langle \rho, r \rangle \in \text{Union}(\tau)$ with

$$t \leq r, \quad t \Vdash_{\text{sym}} \eta = \rho. \quad (1)$$

By the definition of the union name there are $\langle \xi, a \rangle \in \tau$ and $\langle \rho, b \rangle \in \xi$ with $r \leq a, b$. Constituent membership and monotonicity therefore give

$$t \Vdash_{\text{sym}} \xi \in \tau, \quad t \Vdash_{\text{sym}} \rho \in \xi.$$

Equality substitution in (1) gives $t \Vdash_{\text{sym}} \eta \in \xi$, and existential introduction with the HS witness ξ yields

$$t \Vdash_{\text{sym}} \exists x (x \in \tau \wedge \eta \in x).$$

These conditions are dense below p , so Lemma 64.8 proves the forward implication in (U).

Conversely suppose

$$p \Vdash_{\text{sym}} \exists x (x \in \tau \wedge \eta \in x). \quad (2)$$

Fix $q \leq p$. The existential clause supplies $r \leq q$ and an HS name ξ with

$$r \Vdash_{\text{sym}} \xi \in \tau \wedge \eta \in \xi. \quad (3)$$

Apply the atomic membership clause first to $\xi \in \tau$: after strengthening, find $\langle \xi_0, a \rangle \in \tau$ with

$$s \leq r, a, \quad s \Vdash_{\text{sym}} \xi = \xi_0.$$

Equality substitution in (3) gives $s \Vdash_{\text{sym}} \eta \in \xi_0$. Apply the atomic membership clause again: after a further strengthening, find $\langle \rho, b \rangle \in \xi_0$ with

$$t \leq s, b, \quad t \Vdash_{\text{sym}} \eta = \rho.$$

Since $t \leq a, b$, the pair $\langle \rho, t \rangle$ is literally a constituent of $\text{Union}(\tau)$. It therefore witnesses the atomic membership clause for $\eta \in \text{Union}(\tau)$ below the original q . Such witnesses are dense below p , proving the reverse implication in (U).

As for Pairing, (U) holds at every condition, so Lemma 64.11(1) gives the two forced implications, Lemma 64.11(3) universally closes z , Lemma 64.11(2) introduces the witness $\text{Union}(\tau)$, and one more universal closure gives (UN).

Successor. A constituent of $\text{Succ}(\sigma)$ is either a constituent already occurring in σ or the added pair $\langle \sigma, 1_{\mathbb{P}} \rangle$. Thus a dense atomic-membership witness for $\eta \in \text{Succ}(\sigma)$ yields densely either $\eta \in \sigma$ (by constituent membership plus equality substitution) or $\eta = \sigma$. Lemma 64.9 gives the forward implication in (S). Conversely, conditions forcing $\eta = \sigma$ use the added constituent, while the atomic witness set for $\eta \in \sigma$ is contained in the atomic witness set for $\eta \in \text{Succ}(\sigma)$. Lemma 64.8 therefore gives the reverse implication. Hence (S) holds at every condition.

Infinity. Recall explicitly the ground check-name recursion from Volume VII:

$$\check{x} := \{ \langle \check{y}, 1_{\mathbb{P}} \rangle : y \in x \}. \quad (4)$$

Theorem 64.17 shows that every $\check{x}$ is HS. For the von Neumann natural numbers, the definition of the successor name gives the literal identity

$$\text{Succ}(\check{n}) = (n \check{+} 1) \quad (n < \omega). \quad (5)$$

Using (S), the same implication-and-universal-closure argument as above gives, for every $n < \omega$,

$$\Vdash_{\text{sym}} \forall z [z \in (n \check{+} 1) \leftrightarrow (z \in \check{n} \vee z = \check{n})]. \quad (6)$$

Also $\langle \check{n}, 1_{\mathbb{P}} \rangle \in \check{\omega}$, so constituent membership gives

$$\Vdash_{\text{sym}} \check{n} \in \check{\omega} \quad (n < \omega). \quad (7)$$

Fix a constituent $\langle \check{n}, 1_{\mathbb{P}} \rangle \in \check{\omega}$ and a condition q . Equations (6)–(7), conjunction, and existential introduction with witness $(n \check{+} 1)$ give

$$q \Vdash_{\text{sym}} \exists y \in \check{\omega} \forall z [z \in y \leftrightarrow (z \in \check{n} \vee z = \check{n})].$$

The bounded-name criterion therefore yields

$$\Vdash_{\text{sym}} \forall x \in \check{\omega} \exists y \in \check{\omega} \forall z [z \in y \leftrightarrow (z \in x \vee z = x)]. \quad (8)$$

Finally $\check{0} = \emptyset$, so it has no constituents and every condition forces $z \notin \check{0}$ for every HS z . Universal closure gives $\Vdash_{\text{sym}} \forall z (z \notin \check{0})$, while (7) with $n = 0$ gives $\Vdash_{\text{sym}} \check{0} \in \check{\omega}$. Existential introduction supplies the empty member $e = \check{0}$; combine this with (8) and finally use $I = \check{\omega}$ as the existential witness. This proves (INF). $\square$

Worked example

Worked Example 64.20 (Why hereditary symmetry is needed). A name τ could itself be fixed by many automorphisms while containing a constituent name σ with no acceptable support. If $\sigma^G \in \tau^G$, transitivity of the intended symmetric model would then fail. Requiring symmetry recursively for every constituent prevents this problem.

A useful warning

Symmetry of a top-level name alone is not enough. The hereditary clause is what makes the evaluated class transitive.

Looking ahead

The next chapter, *ZF in a symmetric extension, elementary axioms*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 65

ZF in a symmetric extension, elementary axioms

Definitions and setup

Formal definitions used in this chapter

Symmetric model. Let M be the ground model and $G \subseteq \mathbb{P}$ be M -generic. If HS^M denotes the class of names that M regards as hereditarily symmetric, define

$$N := \{\tau^G : \tau \in \text{HS}^M\}.$$

The preceding chapter proves that N is transitive and that $M \subseteq N \subseteq M[G]$.

Recalled symmetric-forcing theorems. The symmetric Truth Lemma was proved in Theorem 64.18: for $\bar{\tau} \in \text{HS}^M$,

$$N \models \varphi(\bar{\tau}^G) \iff \exists p \in G \, p \Vdash_{\text{sym}} \varphi(\bar{\tau}).$$

The symmetry theorem for this relation was proved in Lemma 64.5:

$$p \Vdash_{\text{sym}} \varphi(\bar{\tau}) \iff \pi p \Vdash_{\text{sym}} \varphi(\pi \bar{\tau}) \quad (\pi \in \mathcal{G}).$$

These displayed equivalences are recalled dependencies, not definitions introduced in the present block.

Separation, Collection, and Replacement. For a transitive class N , *Separation* is the schema asserting that $\{x \in a : N \models \varphi(x, \bar{p})\}$ is a set in N . *Collection* asserts that

$$N \models \forall x \in a \, \exists y \varphi(x, y, \bar{p})$$

implies the existence of $b \in N$ with

$$N \models \forall x \in a \, \exists y \in b \, \varphi(x, y, \bar{p}).$$

Replacement is the functional case, collecting the unique y associated to each $x \in a$.

Because N is a transitive subclass of a generic extension, Extensionality and Foundation are inherited, but set-formation axioms need names that remain hereditarily symmetric. Pairing/Union/Infinity use canonical supported name operations. Separation uses symmetric-forcing equivariance (Lemma 64.5) to show the defining subname has the common support. Collection is already nontrivial and prepares Replacement.

How this chapter fits together

The symmetric model is transitive, so Extensionality and Foundation are inherited, but the constructive axioms still require symmetric names. Pairing and Union combine finitely many supports. Separation uses Lemma 64.5 to define an invariant subname. Collection is subtler because witnesses chosen independently may have unrelated supports; the proof closes a set of witnesses under the relevant group action.

Theorem 65.1 (Extensionality/Foundation/Pairing/Union/Infinity in N). *The transitive symmetric model N satisfies Extensionality, Foundation, Pairing, Union, and Infinity.*

Proof.

Formal derivation spine

Formal spine: the elementary axioms are preserved by HS name operations. If $\sigma, \tau \in \text{HS}$, the canonical pair and union names are again HS, with stabilizers containing a finite intersection of the original stabilizers. Check names satisfy

$$\check{x} \in \text{HS}, \quad \check{\omega}^G = \omega.$$

Since $N \subseteq M[G]$ is transitive, Extensionality and Foundation are inherited; the HS constructions give Pairing, Union, and Infinity.

Extensionality and Foundation hold in N because N is transitive and has the ambient membership relation. If $x = \sigma^G$ and $y = \tau^G$ are in N , choose support subgroups for σ and τ ; their intersection belongs to the normal filter and supports the canonical pair name, so $\{x, y\} \in N$. The same closure under finite intersections supports the canonical union name, yielding Union. The check name $\check{\omega}$ is fixed by the whole group, so Infinity holds.

These constructions prove exactly the elementary axioms listed. Separation, Collection/Replacement, and Power Set require Lemma 64.5 and separate support-bounding arguments and are treated in the next two chapters. $\square$

Theorem 65.2 (Symmetric Separation in condition-relative and semantic form). *Let $\tau, \bar{\rho} \in \text{HS}^M$, and let $\varphi(x, \bar{y})$ be a formula. Define*

$$\text{Sep}_\varphi(\tau, \bar{\rho}) := \{\langle \xi, q \rangle : \exists r (\langle \xi, r \rangle \in \tau \wedge q \leq r \wedge q \Vdash_{\text{sym}} \varphi(\xi, \bar{\rho}))\}. \quad (\text{SEP0})$$

Then $\text{Sep}_\varphi(\tau, \bar{\rho}) \in \text{HS}^M$, and for every $\eta \in \text{HS}^M$ and every condition p ,

$$p \Vdash_{\text{sym}} \eta \in \text{Sep}_\varphi(\tau, \bar{\rho}) \iff p \Vdash_{\text{sym}} (\eta \in \tau \wedge \varphi(\eta, \bar{\rho})). \quad (\text{SEPF})$$

Consequently

$$\Vdash_{\text{sym}} \forall x (x \in \text{Sep}_\varphi(\tau, \bar{\rho}) \leftrightarrow (x \in \tau \wedge \varphi(x, \bar{\rho}))). \quad (\text{SEP1})$$

For every M -generic G , if $a = \tau^G$ and $\bar{p} = \bar{\rho}^G$, then

$$\text{Sep}_\varphi(\tau, \bar{\rho})^G = \{x \in a : N \models \varphi(x, \bar{p})\}. \quad (\text{SEPG})$$

Thus $N \models \text{Separation}$.

Proof.

Formal derivation spine

Formal spine: the separating name has an exact forcing membership law.

$$\text{Sep}_\varphi(\tau, \bar{\rho}) = \{\langle \xi, q \rangle : \exists r (\langle \xi, r \rangle \in \tau, q \leq r, q \Vdash_{\text{sym}} \varphi(\xi, \bar{\rho}))\},$$

$$p \Vdash_{\text{sym}} \eta \in \text{Sep}_\varphi(\tau, \bar{\rho}) \iff p \Vdash_{\text{sym}} (\eta \in \tau \wedge \varphi(\eta, \bar{\rho})).$$

Definability of the symmetric forcing relation and Separation in the ground model make (SEP0) a set-name. Let $H \in \mathcal{F}$ support τ and every parameter in $\bar{\rho}$. If $\pi \in H$, then equivariance (Lemma 64.5) gives

$$q \Vdash_{\text{sym}} \varphi(\xi, \bar{\rho}) \iff \pi q \Vdash_{\text{sym}} \varphi(\pi\xi, \bar{\rho}),$$

and $\pi\tau = \tau$. Thus H fixes the name in (SEP0). Every constituent ξ is a constituent of the HS name τ , hence is HS. Therefore

$$\text{Sep}_\varphi(\tau, \bar{\rho}) \in \text{HS}^M. \quad (1)$$

We prove (SEPF). Suppose first that

$$p \Vdash_{\text{sym}} \eta \in \text{Sep}_\varphi(\tau, \bar{\rho}). \quad (2)$$

Fix $q \leq p$. By the atomic membership clause there are

$$t \leq q, \quad \langle \xi, s \rangle \in \text{Sep}_\varphi(\tau, \bar{\rho}), \quad t \leq s, \quad t \Vdash_{\text{sym}} \eta = \xi. \quad (3)$$

By the definition of the separating name, choose r with

$$\langle \xi, r \rangle \in \tau, \quad s \leq r, \quad s \Vdash_{\text{sym}} \varphi(\xi, \bar{\rho}). \quad (4)$$

Since $t \leq s \leq r$, constituent membership gives $t \Vdash_{\text{sym}} \xi \in \tau$; monotonicity of (4) gives $t \Vdash_{\text{sym}} \varphi(\xi, \bar{\rho})$. Equality substitution using (3) yields

$$t \Vdash_{\text{sym}} \eta \in \tau \wedge \varphi(\eta, \bar{\rho}).$$

Thus conditions forcing the right-hand side of (SEPF) are dense below p , and Lemma 64.8 gives the desired forcing statement at p .

Conversely suppose

$$p \Vdash_{\text{sym}} \eta \in \tau \wedge \varphi(\eta, \bar{\rho}). \quad (5)$$

Fix $q \leq p$. The atomic membership clause for $\eta \in \tau$ supplies

$$t \leq q, \quad \langle \xi, r \rangle \in \tau, \quad t \leq r, \quad t \Vdash_{\text{sym}} \eta = \xi. \quad (6)$$

By monotonicity of (5), $t \Vdash_{\text{sym}} \varphi(\eta, \bar{\rho})$; equality substitution in (6) gives $t \Vdash_{\text{sym}} \varphi(\xi, \bar{\rho})$. Hence

$$\langle \xi, t \rangle \in \text{Sep}_\varphi(\tau, \bar{\rho}). \quad (7)$$

The condition t itself, together with (6)–(7), is a witness in the atomic membership clause for $\eta \in \text{Sep}_\varphi(\tau, \bar{\rho})$. Such witnesses are dense below p , so (2) follows. This proves (SEPF).

For fixed η , (SEPF) rules out at every condition either direction of a forced failure of the biconditional. Hence every condition forces

$$\eta \in \text{Sep}_\varphi(\tau, \bar{\rho}) \leftrightarrow (\eta \in \tau \wedge \varphi(\eta, \bar{\rho})).$$

This holds for every condition and every HS name η , so the universal-closure criterion Lemma 64.11(3), with $p = 1_{\mathbb{P}}$, gives (SEP1) directly. No conversion of a forced negated universal into an existential counterexample is used.

Finally let G be M -generic. Apply the symmetric Truth Lemma to (SEP1). Since the separating name is HS by (1), its valuation lies in N , and for every $x = \eta^G \in N$,

$$x \in \text{Sep}_\varphi(\tau, \bar{\rho})^G \iff x \in \tau^G \text{ and } N \models \varphi(x, \bar{\rho}^G).$$

This is (SEPG) and proves Separation in N . □

Lemma 65.3 (Symmetric collection lemma, condition-relative form). *Let $p \in \mathbb{P}$ and let $\tau, \bar{\rho} \in \text{HS}$. If*

$$p \Vdash_{\text{sym}} \forall x \in \tau \exists y \varphi(x, y, \bar{\rho}),$$

then there is an HS name σ such that

$$p \Vdash_{\text{sym}} \forall x \in \tau \exists y \in \sigma \varphi(x, y, \bar{\rho}).$$

Proof.

Formal derivation spine

Formal spine: choose HS witnesses on predense antichains, then close them under one support subgroup. For each constituent $\langle \xi, r \rangle \in \tau$, conditions below both p and r have a dense set of HS witnesses. Choose a maximal antichain from that dense set, collect one witness for every antichain element in the ground model, close the resulting set of names under a common support subgroup, and use the bounded-name criterion of Lemma 64.10.

Work in the ground model $M \models \text{ZFC}$ fixed for the symmetric construction. Choose $H \in \mathcal{F}$ supporting τ and every parameter in $\bar{\rho}$. Fix a constituent $\langle \xi, r \rangle \in \tau$. Define

$$D_{\xi, r} := \{s \in \mathbb{P} : s \leq p, r \text{ and } \exists \eta \in \text{HS } s \Vdash_{\text{sym}} \varphi(\xi, \eta, \bar{\rho})\}. \quad (1)$$

This set is dense in the common cone below p and r . Indeed, let $q \leq p, r$. By Lemma 64.10, constituent membership gives $q \Vdash_{\text{sym}} \xi \in \tau$. Monotonicity preserves the assumed bounded universal statement at q ; universal instantiation and modus ponens from the same lemma therefore give

$$q \Vdash_{\text{sym}} \exists y \varphi(\xi, y, \bar{\rho}).$$

The defining existential clause now says exactly that every extension of q can be strengthened to a member of $D_{\xi, r}$.

The set $D_{\xi, r}$ belongs to M : it is obtained by Separation from the set $\mathbb{P}$, because hereditary symmetry and the symmetric forcing relation are definable in the ground model. Inside the set poset $\{q \in \mathbb{P} : q \leq p, r\}$, Zorn's lemma in M therefore gives an antichain $A_{\xi, r} \subseteq D_{\xi, r}$ maximal among antichains contained in $D_{\xi, r}$. Here an *antichain* means a set of pairwise incompatible conditions, and maximality together with density of $D_{\xi, r}$ means that $A_{\xi, r}$ is *predense* in the common cone: every $q \leq p, r$ is compatible with some $a \in A_{\xi, r}$. Since the constituent set of τ is a set and, for each constituent, the collection of eligible maximal antichains is a nonempty subset of $\mathcal{P}(\mathbb{P})$, Choice in M makes these selections simultaneously for all constituents of τ .

Let

$$I := \{(\xi, r, a) : \langle \xi, r \rangle \in \tau, a \in A_{\xi, r}\}.$$

This is a ground-model set. For every $i = (\xi, r, a) \in I$, membership of a in $D_{\xi, r}$ gives at least one HS name η satisfying $a \Vdash_{\text{sym}} \varphi(\xi, \eta, \bar{\rho})$. Because this is a set-indexed relation, Collection in M first produces a set containing at least one such witness name for every $i \in I$; Choice in M then selects one, denoted η_i , for each i . Thus

$$a \Vdash_{\text{sym}} \varphi(\xi, \eta_{\xi, r, a}, \bar{\rho}) \quad ((\xi, r, a) \in I). \quad (2)$$

Put

$$S := \{\eta_i : i \in I\}, \quad S^* := H \cdot S := \{\pi \eta : \pi \in H, \eta \in S\},$$

and define

$$\sigma := \{\langle \eta, 1_{\mathbb{P}} \rangle : \eta \in S^*\}. \quad (3)$$

The group H permutes S^* , hence supports σ . Automorphisms preserve hereditary symmetry by Lemma 64.3, so every constituent of σ is HS. Therefore $\sigma \in \text{HS}$.

We verify the forcing conclusion directly. Fix $\langle \xi, r \rangle \in \tau$ and $q \leq p, r$. To see that $q \Vdash_{\text{sym}} \exists y \in \sigma \varphi(\xi, y, \bar{\rho})$, it suffices by Lemma 64.8 to show that conditions forcing this formula are dense below q . Given any $u \leq q$, predensity of $A_{\xi, r}$ supplies $a \in A_{\xi, r}$ compatible with u ; choose $t \leq u, a$. By (2) and monotonicity,

$$t \Vdash_{\text{sym}} \varphi(\xi, \eta_{\xi, r, a}, \bar{\rho}).$$

By (3) and constituent membership from Lemma 64.10, $t \Vdash_{\text{sym}} \eta_{\xi, r, a} \in \sigma$. Conjunction and the existential clause therefore give

$$t \Vdash_{\text{sym}} \exists y \in \sigma \varphi(\xi, y, \bar{\rho}).$$

Hence these conditions are dense below q , and Lemma 64.8 lifts the formula to q itself.

We have proved the hypothesis of the bounded-name criterion for every constituent of τ and every common strengthening below p . Lemma 64.10 now yields

$$p \Vdash_{\text{sym}} \forall x \in \tau \exists y \in \sigma \varphi(x, y, \bar{\rho}),$$

as required. $\square$

Corollary 65.4 (Root-forcing form of the Collection schema). *For every formula $\varphi(x, y, \bar{z})$ and every HS parameter tuple $\bar{\rho}$,*

$$\Vdash_{\text{sym}} \forall a [(\forall x \in a \exists y \varphi(x, y, \bar{\rho})) \rightarrow \exists b \forall x \in a \exists y \in b \varphi(x, y, \bar{\rho})]. \quad (\text{COLR})$$

The proof is uniform in the code of φ .

Proof. Fix an HS name τ for the variable a , and abbreviate

$$A(\tau) := \forall x \in \tau \exists y \varphi(x, y, \bar{\rho}),$$

$$C(\tau, \sigma) := \forall x \in \tau \exists y \in \sigma \varphi(x, y, \bar{\rho}).$$

Suppose $p \Vdash_{\text{sym}} A(\tau)$. Lemma 65.3 supplies an HS name σ with

$$p \Vdash_{\text{sym}} C(\tau, \sigma). \quad (1)$$

For every $q \leq p$, monotonicity preserves (1), so q itself belongs to the witness set in the existential clause for $\exists b C(\tau, b)$. Hence

$$p \Vdash_{\text{sym}} \exists b C(\tau, b). \quad (2)$$

Therefore no condition can force

$$A(\tau) \wedge \neg \exists b C(\tau, b),$$

so every condition forces the implication

$$A(\tau) \rightarrow \exists b C(\tau, b). \quad (3)$$

Since (3) holds for every condition and every HS name τ , Lemma 64.11(3), applied at the root, gives the universal closure (COLR).

The construction is uniform in φ : the only formula-dependent operations are the primitive-recursive formation of the symmetric-forcing formula for φ and its insertion into the fixed witness-antichain construction of Lemma 65.3. This is the exact uniformity used later in the proof-code audit. $\square$

Theorem 65.5 (Collection in N). *The symmetric model N satisfies Collection: if*

$$N \models \forall x \in a \exists y \varphi(x, y, \bar{p}),$$

then there is $b \in N$ such that

$$N \models \forall x \in a \exists y \in b \varphi(x, y, \bar{p}).$$

Uses: the symmetric Truth Lemma, symmetric-forcing equivariance (Lemma 64.5), and the symmetric Collection construction.

Proof. Choose HS names $\tau, \bar{\rho}$ with $a = \tau^G$ and $\bar{p} = \bar{\rho}^G$. By the symmetric Truth Lemma applied to the assumed statement in N , there is a condition $p \in G$ such that

$$p \Vdash_{\text{sym}} \forall x \in \tau \exists y \varphi(x, y, \bar{\rho}).$$

Apply Lemma 65.3 at this same condition p . It gives an HS name σ satisfying

$$p \Vdash_{\text{sym}} \forall x \in \tau \exists y \in \sigma \varphi(x, y, \bar{\rho}).$$

Set $b := \sigma^G$. Since $\sigma \in \text{HS}$, we have $b \in N$; because $p \in G$, the symmetric Truth Lemma gives

$$N \models \forall x \in a \exists y \in b \varphi(x, y, \bar{p}).$$

This is exactly Collection in N . $\square$

Worked example

Worked Example 65.6 (A symmetric separating subname). Given a hereditarily symmetric name τ and formula φ with supported parameters, retain a constituent $\langle \sigma, p \rangle$ only along conditions that force $\varphi(\sigma)$. Any automorphism fixing the common supports preserves both τ and the forcing relation, so it preserves the new subname. The result is therefore hereditarily symmetric.

A useful warning

Transitivity of N does not automatically give Separation or Collection. The required subsets and witness sets must themselves have hereditarily symmetric names.

Looking ahead

The next chapter, *ZF in a symmetric extension, hard axioms*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 66

ZF in a symmetric extension, hard axioms

Definitions and setup

Formal definitions used in this chapter

Ground-model interpretation of HS. HS^M is the class, computed inside the ground model M , of hereditarily symmetric $\mathbb{P}$ -names. For a name σ ,

$$\text{dom}(\sigma) = \{\rho : \exists p \in \mathbb{P} \langle \rho, p \rangle \in \sigma\}.$$

Bounded candidate set. For $\sigma \in \text{HS}^M$ define the ground-model set

$$B_\sigma := \{\tau \subseteq \text{dom}(\sigma) \times \mathbb{P} : \tau \in \text{HS}^M\}.$$

A name in B_σ is a *bounded subname* of σ : all of its possible members are drawn from the constituent names already occurring in σ .

Normalization of an internal subset. If $\eta \in \text{HS}^M$ and $\eta^G \subseteq \sigma^G$, set

$$\eta^* := \{\langle \rho, p \rangle \in \text{dom}(\sigma) \times \mathbb{P} : p \Vdash_{\text{sym}} \rho \in \eta\}.$$

The bounding lemma proves $\eta^* \in B_\sigma$ and $(\eta^*)^G = \eta^G$.

Internal power set. For $a \in N$,

$$\mathcal{P}^N(a) := \{x \in N : x \subseteq a\} = \mathcal{P}^{M[G]}(a) \cap N.$$

The Power Set problem is to construct, inside N , a set whose members are exactly these internal subsets.

Orbit closure. If $H \leq \mathcal{G}$ and A is a set of names,

$$H \cdot A := \{\pi\sigma : \pi \in H, \sigma \in A\}.$$

This set-sized closure is used in the symmetric Collection argument to put witnesses under a common allowed support.

Replacement follows from symmetric Collection for functional relations. Power Set is subtler: for $a \in N$, one must collect exactly the subsets of a that also belong to N . A bounded-rank normalization of hereditarily symmetric subnames produces a ground-model set of candidate names, and symmetry of that candidate family gives an internal power set. Only after these two gates pass may we conclude $N \models \text{ZF}$.

How this chapter fits together

Replacement and Power Set are the main closure tests for a symmetric model. Replacement follows from symmetric Collection plus Separation for a functional relation. Power Set requires a rank bound: every subset of a given symmetric set that lies in N must have an equivalent hereditarily symmetric subname within one set-sized family. Only after this is proved can one assert $N \models \text{ZF}$.

Theorem 66.1 (Replacement in N). *The symmetric model N satisfies Replacement. If $N \models \forall x \in a \exists! y \varphi(x, y, \bar{p})$, then the image*

$$\{y : \exists x \in a \varphi(x, y, \bar{p})\}$$

belongs to N .

Proof.

Formal derivation spine

Formal spine: functional Collection followed by Separation. From

$$N \models \forall x \in a \exists! y \varphi(x, y, \bar{p}),$$

Collection gives $b \in N$ with

$$N \models \forall x \in a \exists y \in b \varphi(x, y, \bar{p}).$$

Then Separation forms

$$r := \{y \in b : \exists x \in a \varphi(x, y, \bar{p})\}.$$

Assume

$$N \models \forall x \in a \exists! y \varphi(x, y, \bar{p}). \quad (1)$$

The symmetric Collection lemma, interpreted through its Truth Lemma, supplies a set $b \in N$ such that

$$N \models \forall x \in a \exists y \in b \varphi(x, y, \bar{p}). \quad (2)$$

Apply the already proved Separation schema in N to the set b :

$$r := \{y \in b : \exists x \in a \varphi(x, y, \bar{p})\} \in N. \quad (3)$$

By (2), every value associated with a member of a lies in r . Conversely, the definition (3) says that every member of r is associated with some $x \in a$. Functionality in (1) ensures that this is exactly the image of a under the relation φ , with no extraneous alternative values for any input. Thus r is the Replacement set. Power Set and Choice play no role in this deduction; the substantive symmetry work was already carried by Collection and Separation. $\square$

Lemma 66.2 (Condition-relative bounded normalization for symmetric subsets). *Let $\sigma, \eta \in \text{HS}^M$, and define the ground-model set*

$$B_\sigma := \{\tau \subseteq \text{dom}(\sigma) \times \mathbb{P} : \tau \in \text{HS}^M\}.$$

Define the normalization of η relative to σ by

$$\eta_\sigma^* := \{\langle \rho, q \rangle \in \text{dom}(\sigma) \times \mathbb{P} : q \Vdash_{\text{sym}} \rho \in \eta\}. \quad (\text{BN0})$$

Then $\eta_\sigma^ \in B_\sigma$. Moreover, for every condition $p \in \mathbb{P}$,*

$$p \Vdash_{\text{sym}} \eta \subseteq \sigma \implies p \Vdash_{\text{sym}} \eta = \eta_\sigma^*. \quad (\text{BN})$$

Consequently, if G is M -generic and $\eta^G \subseteq \sigma^G$, then

$$(\eta_\sigma^*)^G = \eta^G. \quad (\text{BNG})$$

Finally, if $\pi\sigma = \sigma$, then $\pi[B_\sigma] = B_\sigma$.

How to read the symbols: the symmetric Power Set problem

The ambient extension $M[G]$ may have many subsets of $x \in N$ that are not in the symmetric model N . The target is therefore $\mathcal{P}^N(x) = \{y \in N : y \subseteq x\}$. The strengthened bounding lemma does more than normalize actual subsets after a generic is chosen: below every condition that forces $\eta \subseteq \sigma$, it forces η equal to one bounded HS name $\eta_\sigma^* \in B_\sigma$. That condition-relative statement is the form needed both for the Power Set forcing proof and for the later proof-code formalization.

Proof.

Formal derivation spine

Formal spine: normalize a forced subset to a subname of σ .

$$\eta_\sigma^* = \{\langle \rho, q \rangle \in \text{dom}(\sigma) \times \mathbb{P} : q \Vdash_{\text{sym}} \rho \in \eta\},$$

$$\Vdash_{\text{sym}} \eta_\sigma^* \subseteq \eta, \quad p \Vdash_{\text{sym}} \eta \subseteq \sigma \implies p \Vdash_{\text{sym}} \eta \subseteq \eta_\sigma^*.$$

Hence $p \Vdash_{\text{sym}} \eta = \eta_\sigma^*$.

Because $\text{dom}(\sigma) \times \mathbb{P}$ is a set in M and the symmetric forcing relation is definable in the ground model, Separation in M forms the set (BN0). Let $H_\sigma, H_\eta \in \mathcal{F}$ support σ and η . If $\pi \in H_\sigma \cap H_\eta$, then π preserves $\text{dom}(\sigma)$ and Lemma 64.5 gives

$$q \Vdash_{\text{sym}} \rho \in \eta \iff \pi q \Vdash_{\text{sym}} \pi \rho \in \eta.$$

Thus $H_\sigma \cap H_\eta$ supports η_σ^* . Every constituent ρ of η_σ^* is a constituent of the HS name σ , hence is HS. Therefore

$$\eta_\sigma^* \in \text{HS}^M, \quad \eta_\sigma^* \subseteq \text{dom}(\sigma) \times \mathbb{P},$$

so $\eta_\sigma^* \in B_\sigma$.

We next prove the two forced inclusions. First,

$$\Vdash_{\text{sym}} \eta_\sigma^* \subseteq \eta. \tag{1}$$

Indeed, let $\langle \rho, q \rangle \in \eta_\sigma^*$ and let $s \leq q$. By definition of η_σ^* , $q \Vdash_{\text{sym}} \rho \in \eta$; monotonicity gives $s \Vdash_{\text{sym}} \rho \in \eta$. In the atomic subset clause we may therefore take the required strengthening to be s itself. This proves (1).

Now assume

$$p \Vdash_{\text{sym}} \eta \subseteq \sigma. \tag{2}$$

We show

$$p \Vdash_{\text{sym}} \eta \subseteq \eta_\sigma^*. \tag{3}$$

Fix a constituent $\langle \xi, r \rangle \in \eta$ and a common strengthening $q \leq p, r$. By the subset clause applied to (2), there is $s \leq q$ such that

$$s \Vdash_{\text{sym}} \xi \in \sigma. \tag{4}$$

By the atomic membership clause, the set of conditions below s that witness equality with a constituent of σ is dense below s . Choose

$$t \leq s, \quad \langle \rho, u \rangle \in \sigma, \quad t \leq u, \quad t \Vdash_{\text{sym}} \xi = \rho. \tag{5}$$

Since $t \leq q \leq r$, constituent membership (Lemma 64.10) gives

$$t \Vdash_{\text{sym}} \xi \in \eta.$$

Equality substitution using (5) yields

$$t \Vdash_{\text{sym}} \rho \in \eta. \tag{6}$$

Because $\rho \in \text{dom}(\sigma)$, equation (6) implies

$$\langle \rho, t \rangle \in \eta_\sigma^*.$$

Constituent membership again gives

$$t \Vdash_{\text{sym}} \rho \in \eta_\sigma^*,$$

and (5) plus substitution gives

$$t \Vdash_{\text{sym}} \xi \in \eta_\sigma^*.$$

Thus every $q \leq p, r$ has a stronger t forcing $\xi \in \eta_\sigma^*$, which is exactly the atomic subset clause (3).

Combining (1) and (3) with the definition of forced equality proves (BN).

For (BNG), suppose $\eta^G \subseteq \sigma^G$. Both values belong to the transitive symmetric model N , and subethood is the bounded formula $\forall x \in \eta^G (x \in \sigma^G)$; hence it is absolute between N and the ambient generic extension. Thus $N \models \eta^G \subseteq \sigma^G$. The symmetric Truth Lemma, together with Lemma 64.13 identifying forced subethood with the auxiliary inclusion clause, yields some $p \in G$ satisfying

$$p \Vdash_{\text{sym}} \eta \subseteq \sigma.$$

By (BN), $p \Vdash_{\text{sym}} \eta = \eta_\sigma^*$; applying the Truth Lemma to equality gives

$$\eta^G = (\eta_\sigma^*)^G.$$

Finally, if $\pi\sigma = \sigma$, then π preserves $\text{dom}(\sigma) \times \mathbb{P}$. Automorphisms preserve hereditary symmetry by Lemma 64.3, so $\tau \mapsto \pi\tau$ maps B_σ into itself; applying π^{-1} gives surjectivity. Hence $\pi[B_\sigma] = B_\sigma$. $\square$

Theorem 66.3 (Power Set in N and its condition-relative forcing form). *Let $\sigma \in \text{HS}^M$, and define*

$$\dot{\mathcal{P}}_N(\sigma) := \{\langle \tau, r \rangle \in B_\sigma \times \mathbb{P} : r \Vdash_{\text{sym}} \tau \subseteq \sigma\}. \quad (\text{PS0})$$

Then $\dot{\mathcal{P}}_N(\sigma) \in \text{HS}^M$, and for every $\eta \in \text{HS}^M$ and every $p \in \mathbb{P}$,

$$p \Vdash_{\text{sym}} \eta \in \dot{\mathcal{P}}_N(\sigma) \iff p \Vdash_{\text{sym}} \eta \subseteq \sigma. \quad (\text{PSF})$$

Consequently

$$\Vdash_{\text{sym}} \forall y (y \in \dot{\mathcal{P}}_N(\sigma) \leftrightarrow y \subseteq \sigma). \quad (\text{PS1})$$

If G is M -generic and $a = \sigma^G$, then

$$\dot{\mathcal{P}}_N(\sigma)^G = \mathcal{P}^N(a) = \{x \in N : x \subseteq a\}. \quad (\text{PSG})$$

Thus $N \models \text{Power Set}$.

Uses: the condition-relative bounded normalization lemma, monotonicity, the dense-forcing lemma, and symmetric-forcing equivariance.

Proof.

Formal derivation spine

Formal spine: Power Set is proved before choosing a generic.

$$\dot{\mathcal{P}}_N(\sigma) = \{\langle \tau, r \rangle \in B_\sigma \times \mathbb{P} : r \Vdash_{\text{sym}} \tau \subseteq \sigma\},$$

$$p \Vdash_{\text{sym}} \eta \subseteq \sigma \xrightarrow{\text{Lemma 66.2}} p \Vdash_{\text{sym}} \eta = \eta_\sigma^* \quad \text{with } \eta_\sigma^* \in B_\sigma,$$

which yields the reverse implication in (PSF); the forward implication follows from the atomic membership clause and density.

Because $B_\sigma \times \mathbb{P}$ is a ground-model set and the symmetric forcing relation is definable, Separation in M forms (PS0). Let $H \in \mathcal{F}$ support σ . If $h \in H$, then $h\sigma = \sigma$, Lemma 66.2 gives $h[B_\sigma] = B_\sigma$, and equivariance (Lemma 64.5) gives

$$r \Vdash_{\text{sym}} \tau \subseteq \sigma \iff hr \Vdash_{\text{sym}} h\tau \subseteq \sigma.$$

Thus H fixes $\dot{\mathcal{P}}_N(\sigma)$. Every constituent name lies in $B_\sigma \subseteq \text{HS}^M$, so the new name is hereditarily symmetric. We prove (PSF). First suppose

$$p \Vdash_{\text{sym}} \eta \in \dot{\mathcal{P}}_N(\sigma). \quad (1)$$

Fix $q \leq p$. By the atomic membership clause, the set of witness conditions is dense below p , so there are

$$t \leq q, \quad \langle \tau, r \rangle \in \dot{\mathcal{P}}_N(\sigma), \quad t \leq r, \quad t \Vdash_{\text{sym}} \eta = \tau. \quad (2)$$

By (PS0), $r \Vdash_{\text{sym}} \tau \subseteq \sigma$; monotonicity gives the same statement at t . Equality substitution in (2) yields

$$t \Vdash_{\text{sym}} \eta \subseteq \sigma.$$

Thus conditions forcing $\eta \subseteq \sigma$ are dense below p . Lemma 64.8 gives

$$p \Vdash_{\text{sym}} \eta \subseteq \sigma. \quad (3)$$

Conversely assume (3). Lemma 66.2 gives

$$\eta_\sigma^* \in B_\sigma, \quad p \Vdash_{\text{sym}} \eta = \eta_\sigma^*. \quad (4)$$

Substitutivity applied to (3)–(4) gives

$$p \Vdash_{\text{sym}} \eta_\sigma^* \subseteq \sigma,$$

so by (PS0)

$$\langle \eta_\sigma^*, p \rangle \in \dot{\mathcal{P}}_N(\sigma). \quad (5)$$

For every $q \leq p$, monotonicity of (4) gives $q \Vdash_{\text{sym}} \eta = \eta_\sigma^*$, and (5) has tag p with $q \leq p$. Hence every $q \leq p$ itself belongs to the witness set in the atomic membership clause for $\eta \in \dot{\mathcal{P}}_N(\sigma)$. That witness set is therefore dense below p , proving

$$p \Vdash_{\text{sym}} \eta \in \dot{\mathcal{P}}_N(\sigma).$$

This completes (PSF).

We next derive the root-forcing statement (PS1) from (PSF), without appealing to a generic extension. Fix an HS name η . For any condition p , if an extension $q \leq p$ forced

$$\eta \in \dot{\mathcal{P}}_N(\sigma) \wedge \neg(\eta \subseteq \sigma),$$

then (PSF) at q would also give $q \Vdash_{\text{sym}} \eta \subseteq \sigma$, contradicting the negation clause. Thus every condition forces the implication from membership to subethood. The same argument using the reverse direction of (PSF) proves the converse implication. Hence every condition forces the biconditional for this η .

The biconditional has therefore been forced at every condition for every HS name η . Lemma 64.11(3), applied with $p = 1_{\mathbb{P}}$, directly gives the universal closure (PS1).

Finally let G be M -generic and put $a = \sigma^G$. By the symmetric Truth Lemma applied to (PS1), for every $b = \eta^G \in N$,

$$b \in \dot{\mathcal{P}}_N(\sigma)^G \iff b \subseteq a.$$

Since $\dot{\mathcal{P}}_N(\sigma) \in \text{HS}^M$, its value lies in N . This proves (PSG) and the Power Set axiom in N . $\square$

Theorem 66.4 (Symmetric-model theorem: N satisfies ZF). *The class N obtained from hereditarily symmetric names is a transitive model of every axiom of ZF.*

Proof.

Formal derivation spine**Formal spine: axiom-by-axiom ledger for $(N, \in)$.**

ZF axiom/schema	proved by
Extensionality, Foundation	N transitive inside $M[G]$
Pairing, Union, Infinity	canonical HS name constructions
Separation	$\Vdash_{\text{sym}} + \text{Lemma 64.5}$
Collection	symmetric collection lemma
Replacement	Collection + Separation
Power Set	B_σ and the condition-tagged Power Set name

The class N is transitive and contains the ground model and all ordinals. Because its membership relation is the ambient one, Extensionality is inherited immediately. Any nonempty $a \in N$ has an ambient $\in$ -minimal element by Foundation in $M[G]$; transitivity puts that minimal element in N , so N satisfies Foundation.

For Pairing, Union, and Infinity, the preceding canonical-name constructions produce HS names from HS inputs and evaluate to the required sets. Separation was proved by forming a subname with the symmetric forcing relation and showing that a common support fixes it. The symmetric Collection lemma supplies one HS set containing witnesses for every element of a given domain. Functional Collection followed by symmetric Separation gives Replacement.

For Power Set, fix $a = \sigma^G \in N$. The bounding lemma gives a ground-model set B_σ containing an HS normalized name for every $b \in N$ with $b \subseteq a$. The condition-tagged name

$$\dot{P}_N(\sigma) = \{ \langle \tau, p \rangle \in B_\sigma \times \mathbb{P} : p \Vdash_{\text{sym}} \tau \subseteq \sigma \}$$

is HS by Lemma 64.5 and evaluates exactly to

$$\{b \in N : b \subseteq a\}.$$

Thus N satisfies the internal Power Set axiom.

These arguments establish every axiom and schema of ZF. No construction supplies a global well-order or a choice function, so the theorem deliberately concludes $N \models \text{ZF}$, not $N \models \text{ZFC}$. $\square$

Remark 66.5 (Source note). The basic symmetric-extension construction is a pure-set analogue of permutation-model arguments with atoms. The closure proof for ZF and the failure of Choice should be compared with [11, 12, 13].

Worked example

Worked Example 66.6 (Internal power set of a symmetric set). Let $x = \tau^G \in N$. The ambient extension may contain many subsets of x whose names have no acceptable support. The Power Set axiom in N asks for the set $\{y \in N : y \subseteq x\}$. The bounding lemma supplies a set of hereditarily symmetric subnames of τ representing exactly those internal subsets, and a symmetric name for that set.

A useful warning

Do not import the ambient power set $\mathcal{P}^{M[G]}(x)$ into N . Many of those subsets are deliberately excluded by symmetry.

Looking ahead

The next chapter, *The basic Cohen symmetric model*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 67

The basic Cohen symmetric model

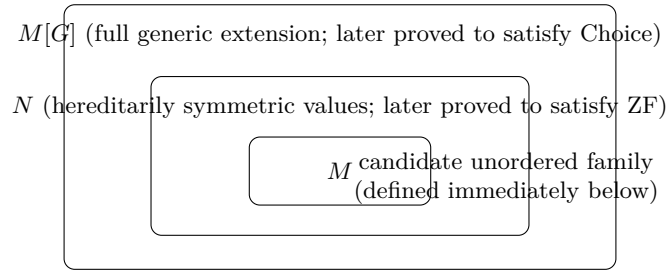


Figure 67.1: Roadmap for the construction. The candidate class N has already been defined, while the unordered Cohen-family name is defined immediately after this figure. Later theorems prove $M \subseteq N \subseteq M[G]$, prove $N \models \text{ZF}$, prove that the resulting unordered family is an infinite Dedekind-finite member of N , and hence prove failure of Choice in N .

Definitions and setup

Formal definitions used in this chapter

Basic Cohen permutation data. Let

$$\mathbb{P} := \text{Add}(\omega, \omega) = \text{Fn}(\omega \times \omega, 2, <\omega).$$

For a permutation $\pi : \omega \rightarrow \omega$, define its support by

$$\text{supp}(\pi) := \{n < \omega : \pi(n) \neq n\}.$$

Call π a *finite-support* (or *finitary*) permutation when $\text{supp}(\pi)$ is finite, and put

$$\mathcal{G} := \{\pi \in \text{Sym}(\omega) : |\text{supp}(\pi)| < \omega\},$$

where $\text{Sym}(\omega)$ is the group of all bijections $\omega \rightarrow \omega$ under composition. For $\pi \in \mathcal{G}$ and $p \in \mathbb{P}$, define the transported condition by

$$(\pi p)(\pi(n), m) := p(n, m)$$

on the transported finite domain. The lemma immediately below proves that these maps are forcing automorphisms.

Finite coordinate supports for names. Write

$$[\omega]^{<\omega} := \{S \subseteq \omega : S \text{ is finite}\}.$$

For $S \in [\omega]^{<\omega}$ put

$$\text{Fix}(S) := \{\pi \in \mathcal{G} : \forall n \in S \pi(n) = n\}.$$

For a ground-model name $\tau \in M^{\mathbb{P}}$, a finite set $S \subseteq \omega$ *supports* τ in this Cohen system when

$$\text{Fix}(S) \leq \text{sym}(\tau).$$

The phrase *empty support* means that $\emptyset$ supports the ground-model name; since $\text{Fix}(\emptyset) = \mathcal{G}$, this says that every element of $\mathcal{G}$ fixes the name. Before the filter verification below, we call such a name simply *empty-supported*. Lemma 67.1 proves that, for the basic Cohen filter, every empty-supported ground-model name is symmetric in the formal sense defined above.

Finite-support subgroup family in the ground model. Work with the fixed transitive ground model $M \models \text{ZFC}$. Write $\mathcal{P}^M(\mathcal{G})$ for the collection of subsets of $\mathcal{G}$ that belong to M (once the next lemma verifies $\mathcal{G} \in M$, this is exactly the power set of $\mathcal{G}$ computed in M). Define the intended ground-model filter by

$$\mathcal{F} := \{H \in \mathcal{P}^M(\mathcal{G}) : H \leq \mathcal{G} \text{ and } \exists S \in [\omega]^{<\omega} \text{Fix}(S) \leq H\}.$$

This is the candidate M -filter generated by the pointwise stabilizers of finite coordinate sets. The next lemma proves that M verifies the normal-filter axioms, so $(\mathbb{P}, \mathcal{G}, \mathcal{F})$ is a ground-model symmetric system over M .

Coordinate support of a condition. For $p \in \mathbb{P}$ define its set of first coordinates by

$$\text{coord}(p) := \{n < \omega : \exists m < \omega ((n, m) \in \text{dom}(p))\}.$$

Because every condition in $\text{Fn}(\omega \times \omega, 2, <\omega)$ has finite domain, $\text{coord}(p)$ is finite.

Coordinate Cohen names. For $n < \omega$ define

$$\dot{c}_n := \{\langle \check{m}, p \rangle : m < \omega, p \in \mathbb{P}, (n, m) \in \text{dom}(p), p(n, m) = 1\}.$$

For an M -generic filter G , put $c_n := \dot{c}_n^G$.

Unordered Cohen family. Define

$$\dot{A} := \{\langle \dot{c}_n, 1_{\mathbb{P}} \rangle : n < \omega\}, \quad A := \dot{A}^G.$$

The action, support, and valuation properties of these names are proved below before they are used in the main theorem.

Use countably many Cohen coordinates together with the group of finite coordinate permutations and the candidate ground-model filter generated by pointwise stabilizers of finite coordinate sets. Lemma 67.1 verifies that this candidate is an M -normal filter and hence gives the required ground-model symmetric system. After that verification, individual Cohen real names have one-coordinate supports while the unordered family has empty support.

How this chapter fits together

The basic construction uses countably many Cohen coordinates and finite coordinate permutations. The first lemma verifies the ground-model normal filter and the support calculations; only after that lemma do we use those facts in the symmetric-model argument. The resulting asymmetry between the unordered family and any attempted ordering of it is what destroys Choice.

Lemma 67.1 (The basic Cohen data form a ground-model symmetric system). *For the data in the preceding definition block, formed over the fixed transitive $M \models \text{ZFC}$:*

1. $\mathbb{P}, \mathcal{G}, \mathcal{F} \in M$; M verifies that $\mathcal{G}$ is a subgroup of $\text{Sym}(\omega)$ and that each $\pi \in \mathcal{G}$ induces an automorphism of $\mathbb{P}$ by the displayed coordinate action.
2. M verifies that $\mathcal{F}$ is a normal filter of subgroups of $\mathcal{G}$.
3. A ground-model name $\tau \in M^{\mathbb{P}}$ is symmetric for this filter exactly when it has a finite coordinate support S , i.e.

$$\text{Fix}(S) \leq \text{sym}(\tau)$$

for some finite $S \subseteq \omega$.

4. For every $n < \omega$, the names $\dot{c}_n$ and $\dot{A}$ belong to $M^{\mathbb{P}}$, and for every $\pi \in \mathcal{G}$,

$$\pi \dot{c}_n = \dot{c}_{\pi(n)}, \quad \pi \dot{A} = \dot{A}.$$

Hence $\{n\}$ supports $\dot{c}_n$ and $\emptyset$ supports $\dot{A}$.

5. For every generic G ,

$$\dot{A}^G = \{c_n : n < \omega\}.$$

Proof. If π, ρ have finite supports, then

$$\text{supp}(\pi\rho) \subseteq \text{supp}(\rho) \cup \rho^{-1}[\text{supp}(\pi)],$$

which is finite, and $\text{supp}(\pi^{-1}) = \pi[\text{supp}(\pi)]$ is finite. Thus $\mathcal{G}$ is a subgroup. Transporting the finite domain of a Cohen condition by the bijection $(n, m) \mapsto (\pi(n), m)$ preserves reverse inclusion, so each π acts by a forcing automorphism.

Because M is transitive and satisfies ZFC, it contains every finite subset of ω , every finite partial function used as a Cohen condition, and every finitary permutation of ω ; hence $\mathbb{P}, \mathcal{G} \in M$. The displayed definition of $\mathcal{F}$ is a Separation definition over the ground-model set $\mathcal{P}^M(\mathcal{G})$, so $\mathcal{F} \in M$. The following verification is carried out in M (and is absolute for the displayed finite-group calculations). The family $\mathcal{F}$ is nonempty because $\mathcal{G} = \text{Fix}(\emptyset) \in \mathcal{F}$, and it is upward closed among the subgroups of $\mathcal{G}$ belonging to M by definition. If $\text{Fix}(S) \leq H$ and $\text{Fix}(T) \leq K$, then

$$\text{Fix}(S \cup T) \leq H \cap K,$$

so $H \cap K \in \mathcal{F}$. Finally,

$$\pi \text{Fix}(S) \pi^{-1} = \text{Fix}(\pi[S]),$$

and $\pi[S]$ is finite, proving normality. This proves (1) and (2). Statement (3) is a literal unwinding of the definitions of symmetry and $\mathcal{F}$.

The canonical check names $\check{m}$ belong to M for $m < \omega$. Separation and Replacement in M , applied to the ground-model sets ω and $\mathbb{P}$, therefore give $\dot{c}_n \in M^{\mathbb{P}}$ for every $n < \omega$ and then $\dot{A} \in M^{\mathbb{P}}$. For (4), apply the recursive action on names. Transporting a pair $\langle \check{m}, p \rangle$ occurring in $\dot{c}_n$ gives $\langle \check{m}, \pi p \rangle$, and the coordinate-action definition says

$$(n, m) \in \text{dom}(p), p(n, m) = 1 \iff (\pi(n), m) \in \text{dom}(\pi p), (\pi p)(\pi(n), m) = 1.$$

Hence $\pi \dot{c}_n = \dot{c}_{\pi(n)}$. Therefore a permutation fixing n fixes $\dot{c}_n$, so $\{n\}$ is a support. Also

$$\pi \dot{A} = \{\langle \dot{c}_{\pi(n)}, 1_{\mathbb{P}} \rangle : n < \omega\} = \dot{A},$$

because π permutes ω bijectively. Thus $\dot{A}$ has empty support.

For (5), evaluation of the displayed name gives

$$\dot{A}^G = \{\dot{c}_n^G : n < \omega\} = \{c_n : n < \omega\},$$

since $1_{\mathbb{P}} \in G$. □

Theorem 67.2 (The unordered Cohen family A belongs to N). *In the basic Cohen symmetric system, if $\dot{c}_n$ is the n -th Cohen real name, then*

$$\dot{A} = \{\langle \dot{c}_n, 1 \rangle : n < \omega\}$$

is hereditarily symmetric, so

$$A = \dot{A}^G = \{c_n : n < \omega\} \in N.$$

Proof.

Formal derivation spine

Formal spine for The unordered Cohen family A belongs to N .

$$\dot{A} = \{\langle \dot{c}_n, 1 \rangle : n < \omega\}.$$

$$\pi \dot{A} = \dot{A}.$$

$$A = \dot{A}^G = \{c_n : n < \omega\}$$

In the basic Cohen symmetric system, let $\dot{c}_n$ be the name of the n -th coordinate Cohen real and define

$$\dot{A} = \{\langle \dot{c}_n, 1 \rangle : n < \omega\}.$$

This is an *unordered* name: the index n is used only to list the constituents, not as structure that must survive in the symmetric model. A permutation π of the coordinate set sends $\dot{c}_n$ to $\dot{c}_{\pi(n)}$, and therefore merely permutes the constituents of $\dot{A}$. Hence

$$\pi \dot{A} = \dot{A}.$$

The entire family is therefore fixed even though its individual coordinate names are moved around. Consequently the family has empty support; $\dot{A}$ has empty support.

Each coordinate name $\dot{c}_n$ has the finite support $\{n\}$, and its lower-rank constituents have finite supports as well. Thus $\dot{A}$ is hereditarily symmetric. Consequently

$$A = \dot{A}^G = \{c_n : n < \omega\}$$

belongs to the symmetric submodel N , even though the sequence $\langle c_n : n < \omega \rangle$ need not belong to N . $\square$

Theorem 67.3 (A is infinite in N). *The unordered Cohen family $A = \{c_n : n < \omega\}$ is infinite in N : for every $k < \omega$, N contains at least k distinct members of A .*

Proof.

Formal derivation spine

Formal spine for A is infinite in N .

$$c_0, c_1, \dots, c_{m-1}$$

$$A = \{c_n : n < \omega\}$$

For each natural number m , the coordinate reals

$$c_0, c_1, \dots, c_{m-1}$$

are elements of A . Volume IX proves their pairwise distinctness in the full generic extension; since $N \subseteq M[G]$ is transitive and equality of sets is absolute between the two structures, they are pairwise distinct in N as well. Their finite tuple and its range are built from finitely many individually supported objects, so the corresponding finite set belongs to N . Therefore N proves that A has at least m elements for every $m < \omega$.

If A were finite in N , it would be in bijection in N with some natural number m , contradicting the existence in N of $m + 1$ distinct coordinate reals. Hence $N \models "A \text{ is infinite}."$ This does not produce an enumeration of A ; that distinction is the point of the later Dedekind-finiteness theorem. $\square$

Lemma 67.4 (Finite-support theorem for symmetric objects). *In the basic Cohen symmetric system, every hereditarily symmetric name τ has a finite support $S \subseteq \omega$ such that*

$$\text{Fix}(S) \leq \text{sym}(\tau).$$

Proof.

Formal derivation spine

Formal spine for Finite-support theorem for symmetric objects.

$$\text{Fix}(S)$$

$$\text{Fix}(S) \leq \text{sym}(\tau).$$

$$A = \{c_n : n < \omega\}$$

In the basic symmetric system, the normal filter is generated by the pointwise stabilizers

$$\text{Fix}(S)$$

of finite coordinate sets $S \subseteq \omega$. If a name τ is symmetric, its stabilizer $\text{sym}(\tau)$ lies in this generated filter. By definition of a generated filter, some finite S satisfies

$$\text{Fix}(S) \leq \text{sym}(\tau).$$

Thus every permutation fixing S pointwise fixes τ ; S is a finite support for the name.

A hereditarily symmetric name has the same property at every constituent. Hence every object of the symmetric model has some representing name controlled by finitely many coordinates. This finite-support theorem is the exact input used in the permutation contradiction showing that no enumeration of the unordered Cohen family can belong to N . $\square$

Worked example

Worked Example 67.5 (The unordered family has empty support). For $\dot{A} = \{\langle \dot{c}_n, 1 \rangle : n < \omega\}$, a coordinate permutation merely rearranges the constituents, so $\pi \dot{A} = \dot{A}$ for every π . Thus $\dot{A}$ has empty support even though each individual $\dot{c}_n$ is moved by permutations that move n .

A useful warning

A set can be highly symmetric even when none of its individual elements is fixed. That distinction is the key feature of the unordered Cohen family.

Looking ahead

The next chapter, *Failure of Choice*, uses the constructions just established. Keep the hypotheses of the present chapter explicit when carrying them forward.

Chapter 68

Failure of Choice

Definitions and setup

Formal definitions used in this chapter

Finite, infinite, countably infinite subset, and Dedekind-finiteness. Recall the von Neumann natural numbers from Volume IV. A set X is *finite* when

$$\text{Fin}(X) :\Longleftrightarrow \exists n \in \omega \exists f (f : n \longrightarrow X \text{ is a bijection}),$$

and it is *infinite* when $\neg \text{Fin}(X)$. A set X has a countably infinite subset exactly when there is an injection $f : \omega \rightarrow X$ (the range of f is such a subset). An infinite set with no such injection is *Dedekind-finite* in the sense used here. Inside the forcing language, $\text{Fin}(\dot{X})$ uses the HS parameter $\dot{\omega}$ and the same first-order definition of “function”, “bijection”, and ordered pair recalled below.

Forced function-value notation. Recall from Volume IV that the ordered pair is the Kuratowski pair $\langle x, y \rangle = \{\{x\}, \{x, y\}\}$, and a set-theoretic function is a relation with a unique second coordinate for each first coordinate in its domain. The pure language of set theory has no function-application term and no ordered-pair term. Accordingly,

$$\langle x, y \rangle \in f$$

is a definitional abbreviation for the membership-language formula asserting that the Kuratowski pair of x, y exists and belongs to f . When a name $\dot{f}$ is forced to be a function and $n, k < \omega$, we further abbreviate that formula by

$$\dot{f}(\dot{n}) = \dot{c}_k \quad :\Longleftrightarrow \quad \langle \dot{n}, \dot{c}_k \rangle \in \dot{f}.$$

Under the forced functionality hypothesis this says exactly that the unique value of $\dot{f}$ at $\dot{n}$ is $\dot{c}_k$. Expressions such as “ $\dot{f}(\dot{n}) \in \dot{A}$ ” are *not* used as formulas; totality into $\dot{A}$ is written

$$\exists y (y \in \dot{A} \wedge \langle \dot{n}, y \rangle \in \dot{f}).$$

Consistency notation, proof predicates, and base arithmetic. For a recursively axiomatized theory T , $\text{Prf}_T(n, m)$ is the fixed primitive-recursive proof predicate from Volumes II–III. Recall the Volume-VIII set-theory convention

$$\perp_{\in} := \exists x (x \neq x) :$$

for the pure language $\{\in\}$, the legacy display “ $0 = 1$ ” denotes “ $\perp_{\in}$ ”. Thus $\text{Con}(T)$ abbreviates “ $\neg \exists n \text{Prf}_T(n, \ulcorner \perp_{\in} \urcorner)$ ” for the set theories in this chapter. The *base arithmetic* is the theory $I\Sigma_1$ fixed in Volume VIII for primitive-recursive proof-code reasoning. Put

$$U_{\neg \text{AC}} := \text{ZF} + \neg \text{AC}.$$

Internal ZFC convention for the proof-code section. Up to this point the symbol M was useful for the *semantic* presentation: after choosing an external transitive $M \models \text{ZFC}$ and an M -generic G , the values of HS names form the class N_G . A relative-consistency proof cannot assume inside ZFC that such a transitive set model exists. Therefore, from this paragraph through the final proof-code argument, we work *inside ZFC itself* with the same explicitly definable data

$$\mathbb{P} = \text{Fn}(\omega \times \omega, 2, <\omega), \quad \mathcal{G} = \{\pi \in \text{Sym}(\omega) : \text{supp}(\pi) \text{ is finite}\},$$

and

$$\mathcal{F}_{\text{int}} = \{H \subseteq \mathcal{G} : H \leq \mathcal{G} \wedge \exists S \in [\omega]^{<\omega} \text{Fix}(S) \leq H\}.$$

The forcing-name class, the HS predicate, and $p \Vdash_{\text{sym}} \varphi(\bar{\tau})$ are the class relations defined by the same first-order recursion formulas proved earlier. In this proof-theoretic section we reuse the shorter symbols $\mathcal{F}, \text{HS}, \Vdash_{\text{sym}}$ for these *internal ZFC* definitions. Thus a root-forcing assertion below is literally a formula/theorem scheme of ZFC. The external M, G, N_G notation is used only when a separate semantic corollary is being discussed.

Root symmetric translation. Fix this internal basic Cohen symmetric system. For a sentence θ of set theory, its *root symmetric translation* is the ground-theory assertion

$$\theta^{\text{sym}} := (1_{\mathbb{P}} \Vdash_{\text{sym}} \theta).$$

If $\theta(x_1, \dots, x_n)$ has free variables, introduce ordinary ground-theory set variables $\tau_1, \dots, \tau_n$ for forcing names and define the guarded root translation

$$\theta^{\text{sym}}(\tau_1, \dots, \tau_n) := \left(\bigwedge_{i=1}^n \tau_i \in \text{HS} \right) \longrightarrow (1_{\mathbb{P}} \Vdash_{\text{sym}} \theta(\tau_1, \dots, \tau_n)). \quad (\text{RT})$$

Thus the syntactic translation acts on the *formula code*; arbitrary names are values of set variables and are not themselves encoded by natural numbers. The effective Definability Theorem 64.6 converts the code of θ primitive-recursively into the code of the ground-theory forcing formula, and fixed syntactic wrapping then gives the code of (RT).

Formalizable symmetric interpretation. A symmetric interpretation of a recursively axiomatized theory U in a ground theory S is *formalizable* when (i) the map from a formula code to the code of its guarded root symmetric translation is primitive recursive; (ii) there are primitive-recursive S -proof transformations validating every logical axiom and inference rule of the fixed Volume-I first-order calculus under this translation; (iii) from the code of each nonlogical axiom instance of U one can primitive-recursively construct an S -proof of its root symmetric translation; and (iv) for the fixed contradiction sentence $\perp_U$, the ground theory has a fixed proof of $\neg(\perp_U^{\text{sym}})$, together with a fixed finite proof template which combines a proof of $\perp_U^{\text{sym}}$ with that refutation to produce a proof of the ground-theory contradiction $\perp_S$. Clause (iv) is essential: clauses (i)–(iii) translate a target contradiction proof only into a proof that the root forces the contradiction; they do not by themselves yield a contradiction in the ground theory. Only finite syntactic objects—formula codes, derivation codes, and fixed proof templates—are inputs to these primitive-recursive maps; forcing names remain quantified set objects inside the translated formulas.

Symmetric contradiction-proof translator. A primitive-recursive function F_{sym} is a symmetric contradiction-proof translator from $U_{\neg\text{AC}}$ to ZFC when the base arithmetic verifies

$$\text{Prf}_{U_{\neg\text{AC}}}(n, \ulcorner 0 = 1 \urcorner) \longrightarrow \text{Prf}_{\text{ZFC}}(F_{\text{sym}}(n), \ulcorner 0 = 1 \urcorner).$$

The final theorem constructs such a translator from a formalizable symmetric interpretation and then applies the proof-translation consistency lemma of Volume VIII. The separate implication $\text{Con}(\text{ZF}) \rightarrow \text{Con}(\text{ZFC})$ is the constructible-interpretation corollary proved in Volume VIII.

Lemma 68.1 (Root-forced infinitude of the unordered Cohen family). *For the basic Cohen symmetric system,*

$$\Vdash_{\text{sym}} \neg \text{Fin}(\dot{A}). \quad (\text{AINF})$$

Equivalently, no condition can force $\dot{A}$ to be bijective with a natural number.

Uses: the definition of $\dot{A}$, atomic agreement Lemma 64.4, constituent membership, decision density, and the Volume-IX forced distinctness of coordinate Cohen names.

Proof. First record two root facts. Since $\langle \dot{c}_i, 1_{\mathbb{P}} \rangle \in \dot{A}$, constituent membership gives

$$\Vdash_{\text{sym}} \dot{c}_i \in \dot{A} \quad (i < \omega). \quad (1)$$

If $i \neq j$, Volume IX proves in ordinary forcing that no condition forces $\dot{c}_i = \dot{c}_j$. Atomic agreement (Lemma 64.4) says the same for symmetric atomic forcing; hence the symmetric negation clause gives

$$\Vdash_{\text{sym}} \dot{c}_i \neq \dot{c}_j \quad (i \neq j). \quad (2)$$

Suppose toward contradiction that some condition p forces $\text{Fin}(\dot{A})$. By the existential clause, after strengthening there are an HS name ν and then an HS name $\dot{f}$ such that

$$q \Vdash_{\text{sym}} \nu \in \check{\omega}, \quad q \Vdash_{\text{sym}} "\dot{f} : \nu \longrightarrow \dot{A} \text{ is a bijection}." \quad (3)$$

The atomic membership clause for $\nu \in \check{\omega}$ lets us strengthen once more and choose an actual $m < \omega$ with

$$r \Vdash_{\text{sym}} \nu = \check{m}.$$

Equality substitution in (3) gives

$$r \Vdash_{\text{sym}} "\dot{f} : \check{m} \longrightarrow \dot{A} \text{ is a bijection}." \quad (4)$$

For each $i \leq m$, root fact (1) and the forced surjectivity in (4) give the first-order statement

$$\exists x \in \check{m} \langle x, \dot{c}_i \rangle \in \dot{f}. \quad (5)$$

Starting from r , apply the existential clause successively for $i = 0, \dots, m$. At the i -th stage choose an HS witness name ξ_i forced to belong to $\check{m}$ and to satisfy $\langle \xi_i, \dot{c}_i \rangle \in \dot{f}$. The atomic membership clause for $\xi_i \in \check{m}$ then gives a further strengthening and an actual ground integer $k_i < m$ with $\xi_i = \check{k}_i$ forced there. Equality substitution in the ordered-pair membership formula yields, after all $m + 1$ finite steps, one condition s such that

$$s \Vdash_{\text{sym}} \dot{f}(\check{k}_i) = \dot{c}_i \quad (0 \leq i \leq m). \quad (6)$$

This finite construction is formalizable by induction on $i \leq m$ in the ground model; no function-value object is treated as a forcing name.

There are $m + 1$ indices k_i but only m possible values, so the finite pigeonhole theorem gives $i < j \leq m$ with $k_i = k_j$. The forced functionality of $\dot{f}$ in (4), together with (6) and the first-order equality rules of Lemma 64.10, yields

$$s \Vdash_{\text{sym}} \dot{c}_i = \dot{c}_j,$$

contradicting (2). Thus no condition forces $\text{Fin}(\dot{A})$. By the negation clause at the root, (AINF) follows. $\square$

Proof architecture for the support contradiction. If $\dot{f} \in \text{HS}$ has finite support S of size m , every coordinate permutation fixing S fixes $\dot{f}$. To refute an alleged injection $\omega \rightarrow A$, the proof below first decides the values of $\dot{f}$ on $m + 1$ distinct natural numbers. Injectivity makes the resulting coordinate indices distinct, so at least one output index k lies outside S . One then transposes k with a fresh $k' \notin S \cup \text{coord}(p)$ and uses compatibility together with the symmetric-forcing equivariance Lemma 64.5 to derive a forced equality of two distinct coordinate Cohen reals. This paragraph is a roadmap; every forcing step is verified in Theorem 68.2.

The finite-support contradiction targets a putative injection/enumeration of the unordered Cohen family. A finite support cannot protect every output coordinate; swapping a fresh output coordinate while fixing the support must both fix the function and move that output. This proves the family is infinite Dedekind-finite. Since AC implies every infinite set has a countably infinite subset, Choice fails in the symmetric model.

How this chapter fits together

The final contradiction assumes an injection from ω into the unordered Cohen family. A name for that function has finite support, so one can move an output coordinate outside the support while fixing the function name and input. Compatibility of the original and permuted conditions then forces two distinct coordinate reals to be equal. The resulting infinite Dedekind-finite set witnesses failure of Choice.

Theorem 68.2 (Root-forced Dedekind-finiteness and its semantic consequence). *In the basic Cohen symmetric system,*

$$\Vdash_{\text{sym}} \neg \exists f \text{ “} f : \check{\omega} \longrightarrow \dot{A} \text{ is injective.”} \quad (\text{DINJ})$$

Consequently, in every resulting symmetric model,

$$N \models \text{“there is no injection } \omega \rightarrow A \text{”}.$$

Together with Lemma 68.1, this says that A is forced to be infinite and Dedekind-finite.

Uses: the finite-support theorem and a coordinate permutation fixing the support but moving one Cohen real.

Commented symbolic trace: finite support versus an alleged injection

Assume $f : \omega \rightarrow A$ belongs to the symmetric model and let a name for f have finite support S , with $|S| = m$. Decide the forced values of the function on $m + 1$ distinct inputs. Injectivity makes the corresponding coordinate indices pairwise distinct, so one of them, say k , lies outside S . Choose a fresh $k' \notin S \cup \{k\}$ outside the finite coordinate support of the deciding condition, then swap k and k' while fixing S . Support fixes the function name and the chosen input; equivariance transports the decided output from $\dot{c}_k$ to $\dot{c}_{k'}$; compatibility lets one common strengthening see both conclusions. That condition forces $c_k = c_{k'}$, contradicting the already proved distinctness of coordinate Cohen reals.

Proof.

Formal derivation spine

Formal spine for A is Dedekind-finite in N .

$$\text{Fix}(S) \leq \text{sym}(\dot{f}). \quad (\text{D1})$$

$$f(n) = c_k. \quad (\text{D2})$$

$$\pi \dot{f} = \dot{f},$$

Proof map. Use forcing conditions rather than only values in one extension. If the finite support has size m , decide the values of $\dot{f}$ on $m + 1$ distinct inputs. Injectivity forces the corresponding Cohen-coordinate indices to be distinct, so at least one lies outside the support. Choose a fresh coordinate k' outside the deciding condition's finite coordinate set. The symmetric-forcing equivariance Lemma 64.5 makes the permuted condition decide the same function value as $\dot{c}_{k'}$; a common strengthening then contradicts distinctness of the coordinate reals.

Suppose, toward contradiction, that $f : \omega \rightarrow A$ is injective in the symmetric model, and choose a hereditarily symmetric name $\dot{f}$ supported by a finite coordinate set S , say $|S| = m$. Below a condition forcing that $\dot{f}$ is such an injection, successively decide the values on $m + 1$ distinct inputs. The resulting coordinate indices are pairwise distinct by injectivity, hence one index k lies outside S . Let p be the final deciding condition and choose $k' \notin S \cup \{k\} \cup \text{coord}(p)$. If π swaps k and k' while fixing S , then $\pi \dot{f} = \dot{f}$ and $\pi \check{n} = \check{n}$ for the selected input n . Lemma 64.5 transports the decision from $\dot{c}_k$ to $\dot{c}_{k'}$. Since $k' \notin \text{coord}(p)$, the conditions p and πp are compatible; a common strengthening forces $\dot{c}_k = \dot{c}_{k'}$. By Lemma 64.4, this is also an ordinary atomic forcing assertion, contradicting the Volume-IX theorem that distinct coordinate Cohen names are forced unequal. Thus no injection $\omega \rightarrow A$ belongs to N , while A is infinite. Hence A is Dedekind-finite.

Detailed formal verification. Suppose, toward contradiction, that an HS name $\dot{f}$ and a condition p_0 satisfy

$$p_0 \Vdash_{\text{sym}} \text{“}\dot{f} : \check{\omega} \rightarrow \dot{A} \text{ is injective.”}$$

Choose a finite support $S \subseteq \omega$ with

$$\text{Fix}(S) \leq \text{sym}(\dot{f}). \quad (\text{D1})$$

Put $m := |S|$ and choose the $m + 1$ distinct inputs

$$n_i := i \quad (0 \leq i \leq m).$$

Starting with p_0 , construct successively stronger conditions

$$p_0 \geq p_1 \geq \cdots \geq p_{m+1}$$

and indices $k_0, \dots, k_m < \omega$ such that

$$p_{i+1} \Vdash_{\text{sym}} \dot{f}(\check{n}_i) = \dot{c}_{k_i} \quad (0 \leq i \leq m). \quad (\text{D2})$$

To perform the i -th step, the forced assertion that $\dot{f}$ is a total function from $\check{\omega}$ into $\dot{A}$, together with $\check{n}_i \in \check{\omega}$, gives

$$p_i \Vdash_{\text{sym}} \exists y (y \in \dot{A} \wedge \langle \check{n}_i, y \rangle \in \dot{f}). \quad (\text{D2a})$$

The existential clause first supplies a stronger condition and an HS value name η_i forcing both conjuncts. Apply the atomic membership clause to $\eta_i \in \dot{A}$, where $\dot{A} = \{\langle \dot{c}_k, 1 \rangle : k < \omega\}$: after a further strengthening there is some $k_i < \omega$ forced equal to η_i . Equality substitution in $\langle \check{n}_i, \eta_i \rangle \in \dot{f}$ yields

$$p_{i+1} \Vdash_{\text{sym}} \langle \check{n}_i, \dot{c}_{k_i} \rangle \in \dot{f},$$

which is exactly the abbreviation in (D2). No expression $\dot{f}(\check{n}_i)$ is treated as a forcing name.

Let $p := p_{m+1}$. The indices $k_0, \dots, k_m$ are pairwise distinct. Indeed, if $i \neq j$ and $k_i = k_j$, monotonicity of the decisions in (D2) would make p force equal outputs at the distinct inputs $n_i \neq n_j$, contradicting the injectivity forced by p_0 . Since S has only m elements, the pigeonhole principle gives an index $i \leq m$ with $k_i \notin S$. Fix such an i , and write

$$n := n_i, \quad k := k_i.$$

Then, by monotonicity,

$$p \Vdash_{\text{sym}} \dot{f}(\check{n}) = \dot{c}_k. \quad (\text{D3})$$

The finite set $\text{coord}(p)$ records the first coordinates mentioned by the condition p . Choose

$$k' \notin S \cup \{k\} \cup \text{coord}(p),$$

and let π transpose k and k' while fixing every member of S . By (D1),

$$\pi \dot{f} = \dot{f}, \quad \pi \check{n} = \check{n}, \quad \pi \dot{c}_k = \dot{c}_{k'}.$$

Lemma 64.5 applied to (D3) gives

$$\pi p \Vdash_{\text{sym}} \dot{f}(\check{n}) = \dot{c}_{k'}. \quad (\text{D4})$$

Because $k' \notin \text{coord}(p)$, the conditions p and πp agree wherever their domains overlap; hence they are compatible. Let $q \leq p, \pi p$. Monotonicity applied to (D3) and (D4) yields

$$q \Vdash_{\text{sym}} \dot{c}_k = \dot{c}_{k'}.$$

By Lemma 64.4, the last equality is also forced by the ordinary atomic forcing relation. Volume IX proved that $\Vdash \dot{c}_k \neq \dot{c}_{k'}$ whenever $k \neq k'$, so no condition can ordinarily force their equality. This contradiction was derived from an *arbitrary* pair consisting of an HS name $\dot{f}$ and a condition forcing that it is an injection $\check{\omega} \rightarrow \dot{A}$. Hence no such pair exists. If some condition forced the existential statement

$$\exists f \text{ “} f : \check{\omega} \rightarrow \dot{A} \text{ is injective,”}$$

then its existential clause would supply a stronger condition and an HS witness name of exactly the forbidden kind. Therefore no condition forces that existential, and the negation clause at the root gives (DINJ). The symmetric Truth Lemma then yields the displayed semantic statement in every generic symmetric model. Lemma 68.1 supplies root-forced infinitude, so the unordered family is Dedekind-finite in the intended sense. $\square$

Remark 68.3 (Source note). The finite-support permutation contradiction is the characteristic mechanism of the basic Cohen symmetric model; compare [11, 12].

Corollary 68.4 (N does not satisfy AC). *The symmetric model N does not satisfy the Axiom of Choice, because AC implies that every infinite set has a countably infinite subset, while $A \in N$ is infinite and Dedekind-finite.*

Proof.

Formal derivation spine

Formal spine for N does not satisfy AC.

$$\omega \longrightarrow A$$

The previous theorem gives an infinite set $A \in N$ for which there is no injection $\omega \rightarrow A$ in N . If N satisfied the Axiom of Choice, then A could be well ordered. Because A is infinite, its well-order type would be an infinite ordinal and hence would contain an initial segment of order type ω . Composing that inclusion with the well-ordering bijection would give an injection

$$\omega \longrightarrow A$$

belonging to N , contradicting Dedekind-finiteness.

Therefore $N \not\models \text{AC}$. The contradiction uses only the well-ordering consequence of AC proved in Volume V and the already established ZF model N . $\square$

Theorem 68.5 (Condition-relative logical proof translation for symmetric forcing). *Work in the internal ZFC convention fixed above, and restrict the Volume-I calculus to the pure membership language of set theory (so its terms are variables). There is a primitive-recursive proof transformation with the following stronger, condition-relative invariant. Given a coded finite derivation*

$$\delta_0, \dots, \delta_\ell$$

and ground-theory proofs of the symmetric-forcing translations of its nonlogical-axiom lines, the transformation produces a ground-theory proof which, for an arbitrary condition variable p and arbitrary HS substitutions for the free variables, verifies that whenever the translated undischarged assumptions are forced at p , the translated current line is forced at p . In particular, for a derivation of a sentence from nonlogical axioms it produces a proof of its root symmetric translation.

The map on derivation codes and the accompanying proof codes is primitive recursive, and the base arithmetic $I\Sigma_1$ proves its line-by-line correctness.

Uses: the effective Decision Density Theorem 64.7, the effective forcing-logic templates in Corollary 64.12, and monotonicity/equivariance.

Proof. The transformation is defined by primitive recursion on the length of the finite Volume-I derivation. Its induction invariant is explicit. If $\bar{x} = (x_1, \dots, x_k)$ are the free variables occurring up to the current line, replace them by arbitrary ground-theory set variables $\bar{\tau} = (\tau_1, \dots, \tau_k)$, add the guard $\bigwedge_i \tau_i \in \text{HS}$, and retain an arbitrary condition variable p . For a line δ_i , the generated proof establishes

$$\left[\bigwedge_{\gamma \in \Gamma_i} p \Vdash_{\text{sym}} \gamma(\bar{\tau}) \right] \longrightarrow p \Vdash_{\text{sym}} \delta_i(\bar{\tau}), \quad (\text{LI})$$

under the HS guard, where Γ_i is the finite set of undischarged nonlogical assumptions on which that line depends. For a theory proof, nonlogical-axiom lines are supplied by their axiom-translation templates and hence may be inserted into (LI) by monotonicity from the root.

Propositional-tautology axioms. From the code of a propositional substitution instance, first compute its normalized basis formula by $\mathcal{N}$. The Logical-basis Normalization Lemma from Volume VII proves that this normalized formula has the same ordinary truth table as the original primitive-syntax formula. Compute the finite list of immediate subformulas that play the role of propositional variables in the normalized formula. Iterated decision density gives a dense set below p on which all these subformulas are decided. On a condition in that dense set, their forced truth values form an ordinary Boolean assignment. The finite truth-table certificate therefore verifies the normalized tautology using only the basis clauses for $\neg$ and $\wedge$, and (SNF) identifies this with forcing the original tautology. Lemma 64.8 lifts the conclusion from the dense deciding set to p . The effective part of Theorem 64.7 and Corollary 64.12 provide primitive-recursive ZFC proof codes for the normalization, finite decision iteration, and dense lifting; the truth-table certificate itself is primitive recursive from the propositional-axiom code.

Universal instantiation. In the pure membership language every term is a variable, so the relevant axiom has the form $\forall x \varphi(x) \rightarrow \varphi(y)$ with the usual capture-avoidance condition. It is an instance of Lemma 64.10(3), followed by Lemma 64.11(1). Capture-avoiding substitution of the coded variable is the primitive-recursive syntactic operation fixed in Volume I.

Existential duality. Volume I includes the logical axiom

$$\exists x \varphi \leftrightarrow \neg \forall x \neg \varphi. \quad (\text{ED})$$

First note the forcing double-negation law

$$p \Vdash_{\text{sym}} \chi \iff p \Vdash_{\text{sym}} \neg \neg \chi. \quad (\text{DN})$$

The forward implication follows directly from monotonicity and the negation clause. For the reverse implication, decision density for χ gives below every $q \leq p$ a deciding extension; such an extension cannot force $\neg \chi$, so it forces χ , and dense lifting yields $p \Vdash_{\text{sym}} \chi$.

Using (DN) inside the existential clause gives

$$p \Vdash_{\text{sym}} \exists x \varphi \iff p \Vdash_{\text{sym}} \exists x \neg \neg \varphi. \quad (\text{E-DN})$$

By (SN), the normalized basis formula for the primitive expression $\forall x \neg \varphi$ is

$$\mathcal{N}(\forall x \neg \varphi) = \neg \exists x \neg \neg \mathcal{N}(\varphi).$$

Hence the normalized right side of (ED) is $\neg \neg \exists x \neg \neg \mathcal{N}(\varphi)$, which is forcing-equivalent by (DN), (E-DN), and (SNF) to the primitive formula $\exists x \varphi$. Lemma 64.11(1) forces both implications; the conjunction clause therefore forces their conjunction, which is exactly the Volume-I (BI) expansion of the displayed biconditional. The proof uses only the effective decision-density/dense-lifting templates, so its proof code is primitive recursive in the code of φ .

Quantifier distribution. Consider

$$\forall x(\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \forall x \psi), \quad x \notin \text{FV}(\varphi). \quad (\text{QD})$$

Suppose $q \leq p$ forces the antecedent of the outer implication and also forces φ . For any HS name σ , universal instantiation gives

$$q \Vdash_{\text{sym}} \varphi \rightarrow \psi(\sigma).$$

Because x is not free in φ , the same forced formula φ is used for every substitution σ ; modus ponens yields $q \Vdash_{\text{sym}} \psi(\sigma)$. Lemma 64.11(3) gives $q \Vdash_{\text{sym}} \forall x \psi$, and two applications of Lemma 64.11(1) produce (QD).

Equality axioms. Reflexivity and the substitutivity scheme are exactly Lemma 64.10(1), whose proof is a simultaneous rank/formula induction. The coded replacement of the displayed free argument position is primitive recursive.

Modus ponens. If the generated proofs for two earlier lines establish at p

$$p \Vdash_{\text{sym}} \alpha, \quad p \Vdash_{\text{sym}} (\alpha \rightarrow \beta),$$

Lemma 64.10(2) gives $p \Vdash_{\text{sym}} \beta$. Thus the proof-code constructor simply concatenates the two earlier certificates with the fixed certificate for that lemma.

Generalization. Suppose the derivation infers $\forall x \varphi$ from φ , and the Volume-I eigenvariable condition says that x is not free in any undischarged assumption on which this occurrence depends. Fix $q \leq p$ and an arbitrary HS name σ . Substitute σ for x in the induction invariant. The undischarged assumptions are unchanged because of the eigenvariable condition, so the induction hypothesis yields $q \Vdash_{\text{sym}} \varphi(\sigma)$. Since this holds for every such q, σ , Lemma 64.11(3) gives $p \Vdash_{\text{sym}} \forall x \varphi$.

These seven cases are precisely the logical axioms and inference rules of the fixed Volume-I calculus. Each output proof is obtained by concatenating previously produced finite proof codes with one of finitely many fixed forcing-logic templates and by primitive-recursive substitution of formula/term codes. Arbitrary forcing names never appear as numerical inputs: they remain set variables guarded by the first-order predicate “HS.”

Finally, proof checking, formula substitution, finite sequence recursion, and the function sending an input derivation to the sequence of output proof codes are primitive recursive. The assertion that the i -th generated object is a correct ground-theory proof is a primitive-recursive proof-checking assertion. Induction on $i \leq \ell$, formalizable in IS_1 , verifies the construction line by line. At $i = \ell$ this is exactly the claimed condition-relative proof transformation; for a sentence proof with root-forced nonlogical axioms, take $p = 1_{\mathbb{P}}$. $\square$

Proposition 68.6 (Root ZF and root failure of Choice). *In the basic Cohen symmetric system,*

$$\Vdash_{\text{sym}} \text{ZF} \quad \text{and} \quad \Vdash_{\text{sym}} \neg \text{AC}, \quad (\text{RZF})$$

where the first display means: every individual ZF axiom, and every instance of the Separation and Replacement schemes, has its root symmetric-forcing proof. For the two schemes these proofs are uniform in the code of the defining formula.

Uses: Theorems 64.15, 64.19, 65.2, and 66.3; Corollary 65.4; Lemma 68.1; Theorem 68.2; and the condition-relative logical translator Theorem 68.5.

Proof. The root proofs of Extensionality, Empty Set, and Foundation are in Theorem 64.15. Pairing, Union, and Infinity are in Theorem 64.19. The condition-relative Separation theorem (Theorem 65.2) gives every Separation instance at the root, uniformly in the formula code. Corollary 65.4 gives the Collection schema at the root, again uniformly in the formula code. The condition-relative Power Set theorem (Theorem 66.3) gives Power Set at the root.

For Replacement, use the fixed first-order derivation in ZF which obtains the functional range from Collection and then separates exactly those collected witnesses satisfying the defining functional formula. Given a code for a Replacement instance, primitive- recursively form the corresponding Collection and Separation instances, insert their root proofs, and apply Theorem 68.5 to that fixed derivation. This produces the root proof of the Replacement instance uniformly in its formula code. Thus every ZF axiom has a root proof.

It remains to prove the second assertion in (RZF) syntactically. Lemma 68.1 gives

$$\Vdash_{\text{sym}} \text{“}\dot{A} \text{ is infinite,”} \quad (1)$$

and Theorem 68.2 gives

$$\Vdash_{\text{sym}} \neg \exists f \text{ “} f : \check{\omega} \rightarrow \dot{A} \text{ is injective.”} \quad (2)$$

Volume V proves, over ZF, that AC implies the well-ordering theorem. Append the fixed ZF argument already used in Corollary 68.4: if an infinite set X is well-ordered, its order type is an infinite ordinal, whose first ω positions give an injection $\omega \rightarrow X$. Thus the fixed Volume-V derivation of the well-ordering theorem extends by finitely many first-order/set-theoretic steps to a ZF proof of

$$\text{AC} \longrightarrow \forall X (X \text{ infinite} \rightarrow \exists f (f : \omega \rightarrow X \text{ is injective})). \quad (3)$$

Fix this finite derivation once and for all.

Suppose some condition p forced AC. Every root-forced ZF axiom remains forced at p by monotonicity. Apply the condition-relative transformation of Theorem 68.5 to the fixed derivation (3), taking the AC line as an undischarged nonlogical assumption. It yields at p

$$p \Vdash_{\text{sym}} \forall X (X \text{ infinite} \rightarrow \exists f (f : \check{\omega} \rightarrow X \text{ is injective})).$$

Universal instantiation at $\dot{A}$, root fact (1), monotonicity, and modus ponens give

$$p \Vdash_{\text{sym}} \exists f \text{ “} f : \check{\omega} \rightarrow \dot{A} \text{ is injective.”}$$

This contradicts (2): by the symmetric negation clause, (2) says that no condition can force the displayed existential statement. Hence no condition forces AC. Applying the negation clause at the root gives $\Vdash_{\text{sym}} \neg \text{AC}$, as required. $\square$

Proposition 68.7 (Internalization bridge for the root-forcing proofs). *Under the internal ZFC convention above, the basic Cohen data form a symmetric system and the root-forcing arguments used in Proposition 68.6 are ZFC proof schemes, not consequences of the existence of a transitive set model. More precisely:*

1. ZFC proves the internal analogues of the definability, monotonicity, equivariance, decision-density, dense-forcing, and first-order forcing-logic theorems used above;
2. from the code of a Separation or Collection formula, and from the code of a Replacement instance, one can primitive-recursively construct the corresponding ZFC proof code of the root-forcing instance;
3. Pairing, Union, Infinity, Extensionality, Empty Set, Foundation, Power Set, and the basic finite-support proof of $\neg \text{AC}$ use fixed finite ZFC proof templates together with primitive-recursive syntactic substitution.

Consequently all nonlogical-axiom proof codes required for the final symmetric interpretation are genuine ZFC proof codes.

Proof. The earlier transitive-model parameter M served two semantic purposes: it supplied a set universe in which to choose a generic filter externally, and it let the text say that names/antichains/witness families “belong to the ground model.” Neither purpose is needed for a root-forcing proof inside ZFC.

Internally, $\mathbb{P}, \mathcal{G}, \mathcal{F}_{\text{int}}$ are sets by Power Set, Separation, and Replacement, and the finite-support calculations proving the symmetric-system axioms are the same displayed ZFC calculations as in Lemma 67.1. The predicates “ τ is a forcing name,” “ τ is HS,” and the forcing formulas are first-order class definitions obtained by the well-founded-recursion/definability theorem. Whenever an earlier root proof formed a set by “Separation in M ,” the internal proof uses ordinary Separation in ZFC; whenever it bounded a set-indexed family by Collection/Replacement or selected set-indexed antichains/witnesses by Choice, the internal proof uses the corresponding ZFC axiom directly. No generic filter occurs in any of the condition-relative root proofs.

For effectiveness, the only varying syntactic input in Separation and Collection is the code of the displayed formula. The effective Definability Theorem 64.6, effective Decision Density Theorem 64.7, and Corollary 64.12 provide primitive-recursive proof-code constructors for the forcing-logic pieces; inserting the input formula into the fixed name-construction and support arguments is primitive-recursive syntactic substitution. Replacement is obtained from the fixed finite derivation “functional Collection + Separation $\Rightarrow$ Replacement” by Theorem 68.5. The remaining axioms and the finite-support contradiction have no formula-schema input and therefore contribute fixed proof templates. Hence the proof codes asserted in the proposition are literal ZFC proof codes, with no hidden assumption “there exists a transitive model of ZFC.” $\square$

Proposition 68.8 (Root contradiction is uniformly refutable). *With*

$$\perp_{\in} := \exists x (x \neq x),$$

ZFC proves

$$\neg(1_{\mathbb{P}} \Vdash_{\text{sym}} \perp_{\in}). \tag{RC0}$$

Moreover there is a fixed primitive-recursive proof-code wrapper $R_\perp$ such that $I\Sigma_1$ verifies

$$\text{Prf}_{\text{ZFC}}(m, \ulcorner 1_{\mathbb{P}} \Vdash_{\text{sym}} \perp_{\in} \urcorner) \longrightarrow \text{Prf}_{\text{ZFC}}(R_\perp(m), \ulcorner \perp_{\in} \urcorner). \quad (\text{RC1})$$

Thus a proof that the root forces the chosen target contradiction can be converted uniformly into an actual contradiction proof in the ground theory.

Uses: equality reflexivity for symmetric forcing, the existential and negation clauses, the internalization bridge, and primitive-recursive proof concatenation.

Proof. Work in the internal ZFC convention. Suppose, toward contradiction, that

$$1_{\mathbb{P}} \Vdash_{\text{sym}} \exists x (x \neq x).$$

By the existential forcing clause, the set of conditions below the root which can be strengthened to a condition forcing $\sigma \neq \sigma$ for some HS name σ is dense below $1_{\mathbb{P}}$. Since the root condition exists, this dense set is nonempty; hence there are $r \in \mathbb{P}$ and $\sigma \in \text{HS}$ with

$$r \Vdash_{\text{sym}} \sigma \neq \sigma. \quad (1)$$

Lemma 64.10(1) gives the forced equality reflexivity

$$r \Vdash_{\text{sym}} \sigma = \sigma. \quad (2)$$

But $\sigma \neq \sigma$ abbreviates $\neg(\sigma = \sigma)$, and the symmetric negation clause says that (1) is equivalent to there being no extension of r which forces $\sigma = \sigma$. Condition r itself is such an extension and satisfies (2), a contradiction. This proves (RC0).

All formulas in this argument are fixed except for bound set variables, and the internal forcing clauses, equality-reflexivity template, and proof concatenation are available as literal ZFC proofs by Proposition 68.7 and Corollary 64.12. Fix once and for all a ZFC proof d of the negation in (RC0). Given a ZFC proof code m of the positive root assertion, concatenate m , the fixed code d , and the fixed propositional derivation

$$A \rightarrow (\neg A \rightarrow \perp_{\in})$$

with two applications of modus ponens, where A is the fixed sentence $1_{\mathbb{P}} \Vdash_{\text{sym}} \perp_{\in}$. The resulting proof-code operation $R_\perp$ is primitive recursive, and $I\Sigma_1$ verifies its correctness by the same finite-sequence proof checking used throughout this chapter. This is (RC1). $\square$

Proposition 68.9 (The basic Cohen symmetric construction is a formalizable interpretation). *The basic Cohen symmetric system developed in this volume gives a formalizable symmetric interpretation of*

$$U_{\neg\text{AC}} = \text{ZF} + \neg\text{AC}$$

in ZFC in the sense defined above.

Uses: the internalization bridge Proposition 68.7, the root-ZF/failure-of-Choice Proposition 68.6, the root-contradiction refutation Proposition 68.8, and the logical proof-translation theorem.

Proof. By Proposition 68.7, all proof-theoretic claims in this section are internal ZFC statements; no existence of a transitive set model is an extra premise. The forcing order, automorphism action, finite-support subgroups, and HS predicate are uniformly first-order definable, and Theorem 64.6 proves the needed effective uniformity of the symmetric forcing relation. Its primitive-recursive map on formula codes, followed by the fixed guard/root wrapper, produces the guarded root symmetric-forcing formula primitive-recursively. Name parameters remain set variables ranging over HS; no coding of arbitrary forcing names by natural numbers is assumed.

The nonlogical-axiom ownership is now entirely at the root, not merely semantic after a generic has been chosen. Theorem 64.15 supplies fixed root templates for Extensionality, Empty Set, and Foundation; Theorem 64.19 supplies Pairing, Union, and Infinity. Theorem 65.2 inserts the input formula code into the condition-relative Separation construction. Corollary 65.4 does the same for Collection, and Theorem 66.3 supplies the fixed bounded-name normalization and root Power Set proof. Proposition 68.6 explains the uniform primitive-recursive passage from Collection plus Separation to each Replacement instance and also supplies the fixed root proof of $\neg AC$. Thus each code of a nonlogical axiom of $U_{\neg AC}$ is sent, by primitive-recursive syntactic substitution into these finite root-forcing templates, to a ZFC proof of its guarded root symmetric translation.

The logical axioms and inference rules are handled by the condition-relative primitive-recursive transformer of Theorem 68.5. Its stronger invariant also verifies the eigenvariable restriction during generalization and lets fixed finite first-order derivations—such as Replacement from Collection plus Separation and the Volume-V implication used to refute AC—be reused without a semantic generic-extension argument. Finally, Proposition 68.8 supplies the fixed ZFC refutation of the root translation of the chosen contradiction and the primitive-recursive wrapper converting any proof of that root translation into a ZFC contradiction proof. Hence all four clauses in the definition of a formalizable symmetric interpretation are satisfied exactly as stated. $\square$

Theorem 68.10 (Relative consistency of failure of Choice). *For the recursively axiomatized theories fixed above,*

$$\text{Con}(\text{ZF}) \implies \text{Con}(\text{ZF} + \neg AC).$$

More precisely, the symmetric construction yields a primitive-recursive contradiction-proof translator from $U_{\neg AC} = \text{ZF} + \neg AC$ to ZFC, so

$$\text{Con}(\text{ZFC}) \implies \text{Con}(U_{\neg AC}),$$

and the displayed theorem follows by composing this with the constructible-interpretation implication $\text{Con}(\text{ZF}) \rightarrow \text{Con}(\text{ZFC})$ proved in Volume VIII.

Uses: the formalizable symmetric interpretation, the root-contradiction extraction Proposition 68.8, the proof-translation consistency lemma, and the constructible relative-consistency corollary.

Proof.

Formal derivation spine

Formal spine: keep the two consistency transfers separate.

$$\text{Con}(\text{ZF}) \implies \text{Con}(\text{ZFC}) \implies \text{Con}(\text{ZF} + \neg AC).$$

The second arrow comes from a primitive-recursive map

$$F_{\text{sym}} : \text{contradiction proofs in } U_{\neg AC} \mapsto \text{contradiction proofs in ZFC}.$$

The ground theory for the proof translation is literally ZFC, in the internal sense fixed before the root translation and verified by Proposition 68.7. The earlier external transitive-model presentation is used only for semantic explanations of generic extensions; it is not a hypothesis of the proof-code translator. The root-forcing chapters establish, uniformly in the code of the relevant formula instance, the ZF axiom schemes and the basic finite-support proof of $\neg AC$. Consequently a finite formal proof in

$$U_{\neg AC} = \text{ZF} + \neg AC$$

can be translated line by line into its symmetric-forcing verification over ZFC.

Theorem 68.5 gives the primitive-recursive transformations for every logical axiom and inference rule, and Proposition 68.9 supplies primitive-recursive ZFC proof templates for every nonlogical axiom line of $U_{\neg AC}$. Let T_{sym} be the resulting primitive-recursive line-by-line translation of proof codes. If a target proof ends in the designated contradiction $\perp_{\in}$, this first stage yields a ZFC proof of its *root translation*,

$$1_{\mathbb{P}} \Vdash_{\text{sym}} \perp_{\in};$$

it does not yet yield a ZFC proof of $\perp_{\in}$ itself. Proposition 68.8 supplies the missing fixed post-processing map $R_{\perp}$. Define

$$F_{\text{sym}} := R_{\perp} \circ T_{\text{sym}}.$$

Composition of primitive-recursive functions is primitive recursive, and the base arithmetic verifies

$$\text{Prf}_{U_{\neg\text{AC}}}(n, \ulcorner 0 = 1 \urcorner) \rightarrow \text{Prf}_{\text{ZFC}}(F_{\text{sym}}(n), \ulcorner 0 = 1 \urcorner), \quad (\text{SCT})$$

where both displayed codes denote $\perp_{\in}$. Thus the final contradiction-extraction step is explicit rather than hidden inside the phrase “translate line by line.” The proof-translation consistency lemma of Volume VIII applied to (SCT) gives

$$\text{Con}(\text{ZFC}) \implies \text{Con}(\text{ZF} + \neg\text{AC}). \quad (1)$$

Separately, the constructible interpretation proved in Volume VI and converted into a proof-code implication in Volume VIII gives

$$\text{Con}(\text{ZF}) \implies \text{Con}(\text{ZFC}). \quad (2)$$

Composing (2) with (1) yields

$$\text{Con}(\text{ZF}) \implies \text{Con}(\text{ZF} + \neg\text{AC}),$$

as required. *Why the old direct ZF-ground argument was removed.* The symmetric construction used above was developed over a ZFC ground model. The formal consistency proof must therefore pass through $\text{Con}(\text{ZFC})$; silently replacing that ground theory by ZF would change the hypotheses of the construction. $\square$

Worked example

Worked Example 68.11 (Finite support kills an enumeration). Suppose $p \Vdash \dot{f}(\check{n}) = \dot{c}_k$ with k outside the finite support S of $\dot{f}$. Swap k with a fresh k' while fixing S , choosing k' so that p and πp are compatible. Symmetry gives $\pi p \Vdash \dot{f}(\check{n}) = \dot{c}_{k'}$. A common strengthening therefore forces $c_k = c_{k'}$, contradicting their previously proved distinctness.

A useful warning

The failure of Choice occurs in the symmetric intermediate model N , not in the full Cohen extension $M[G]$, which still satisfies Choice when the ground model does.

End-of-volume perspective

The basic symmetric Cohen model completes the second independence arc: ordinary forcing supplies a rich ambient ZFC extension, while hereditary symmetry selects an inner model of ZF in which an infinite Dedekind-finite set witnesses failure of Choice.

Glossary of recurring terms and notation

Axiom of Choice (AC)

The assertion that every family of nonempty sets has a choice function; in ZF it is equivalent to the well-ordering theorem and to Zorn's lemma.

Boolean-valued model

A semantics in which formulas receive values in a complete Boolean algebra rather than only 0 or 1. It provides a formal bridge from forcing calculations to relative consistency.

ccc

The countable chain condition: every antichain is countable. For Cohen forcing it is proved by a Δ -system argument.

CH / GCH

The Continuum Hypothesis is $2^{\aleph_0} = \aleph_1$. GCH asserts $2^\kappa = \kappa^+$ for every infinite cardinal κ .

Cohen real

A binary sequence generic for the finite-partial-function forcing $\text{Add}(\omega, 1)$ over the chosen ground model.

Completeness

In first-order logic, semantic consequence implies formal derivability. This is distinct from a theory being complete, which means it decides every sentence of its language.

Condensation

The theorem identifying the transitive collapse of suitable elementary substructures of L_θ with earlier levels L_β .

Consistency

A theory is syntactically consistent if it does not derive a contradiction. Stronger notions used in Volume III include 1-consistency, ω -consistency, and Σ_1 -soundness.

Constructible hierarchy

$L_0 = \emptyset$, $L_{\alpha+1} = \text{Def}(L_\alpha)$, and $L_\lambda = \bigcup_{\beta < \lambda} L_\beta$ at limits; L is their union over all ordinals.

Dedekind-finite

A set with no injection from ω into it. In ZF such a set can be infinite; the basic symmetric Cohen model supplies the key example in Volume X.

Dense set

A subset D of a forcing poset such that every condition has a stronger extension in D . Generic filters meet every dense set belonging to the ground model.

Diagonal lemma

For a formula $\varphi(x)$, a suitable arithmetic theory proves $\theta \leftrightarrow \varphi(\ulcorner \theta \urcorner)$ for some sentence θ .

Elementary substructure

$A \preceq M$ means that every first-order formula with parameters from A has the same truth value in A and M .

Forcing name	A recursively built set of pairs $\langle \sigma, p \rangle$ whose value is interpreted relative to a generic filter.
Forcing relation	$p \Vdash \varphi(\bar{\tau})$ is the ground-model relation expressing that every relevant generic extension containing p satisfies $\varphi(\bar{\tau}^G)$.
Generic filter	A directed, upward-closed filter meeting every dense subset of the forcing poset that belongs to the ground model under discussion.
Gödel numbering	An effective coding of finite syntactic objects by natural numbers, chosen so that formation, substitution, and proof checking are arithmetically representable.
Inner model	A transitive class containing every ordinal and satisfying the specified fragment of set theory with the inherited membership relation.
Mostowski collapse	A well-founded extensional relation is uniquely isomorphic to membership on a transitive set.
Nice name	Under ccc forcing, a normalized name for a subset of a ground-model set, organized by antichains deciding membership. Nice names make counting arguments possible.
Primitive recursive	A class of number-theoretic functions generated from basic functions by composition and primitive recursion. Much of syntactic coding is kept inside this class.
Provability predicate	$\text{Prov}_T(y) \equiv \exists p \text{Prf}_T(p, y)$, where Prf_T represents the effective proof relation for T .
Reflection	A finite collection of formulas, with chosen parameters, agrees between the universe and a sufficiently large rank-initial segment V_α .
Relative consistency	A metamathematical implication such as $\text{Con}(T) \Rightarrow \text{Con}(T + \varphi)$. It is not an assertion that T proves its own consistency.
Satisfaction	The recursively defined relation $M \models \varphi[\bar{a}]$ between a set-sized structure, a formula, and an assignment.
Skolem hull	The closure of a parameter set under chosen Skolem functions; Tarski–Vaught then gives an elementary substructure.
Support	A finite or otherwise controlled set of coordinates whose pointwise stabilizer fixes a symmetric name. Support is the mechanism that makes the failure of Choice visible in Volume X.
Symmetric name	A forcing name whose stabilizer lies in a fixed normal filter of subgroups; hereditarily symmetric names are symmetric throughout their transitive name structure.
Transitive model	A set or class M is transitive if $x \in y \in M$ implies $x \in M$. Transitivity gives important absoluteness properties but does not by itself imply closure under Power Set or Replacement.
Truth Lemma	In a generic extension, $M[G] \models \varphi(\bar{\tau}^G)$ iff some $p \in G$ forces $\varphi(\bar{\tau})$.
ω-consistency	A theory does not prove an existential statement while proving the negation of every standard numeral instance of its matrix.

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Index

- Absoluteness, 147
- Adding Cohen reals, 294
- Atomic forcing, 217
- Automorphisms of forcing and names, 327
- Axiom of Choice, 380
 - failure in symmetric model, 372
- Axiom of Choice and equivalents, 136

- Boolean-valued model, 278, 380
- Boolean-valued semantics, 254
- Boolean-valued ZFC, 263

- Canonical well-order and Choice, 188
- Cantor theorem, 106
- Cardinal arithmetic with choice, 139
- Cardinals without assuming choice, 126
- ccc, 380
- CH, 380
- CH and its failure, 312
- Chain conditions and closure, 283
- Coding finite objects, 42
- Cohen forcing
 - ccc, 300
 - failure of CH, 313
- Cohen real, 380
- compactness
 - first-order, 31
- Compactness and Löwenheim–Skolem, 30
- completeness, 380
 - first-order, 28
 - propositional, 10
- Condensation, 192
- Condensation lemma, 195, 380
- consistency, 380
- Consistency and soundness notions, 82
- Consistency transfer, 277
- constructible universe, 380
 - Choice in L , 190
 - GCH in L , 202
 - ZF in L , 186
- continuum
 - Cohen forcing calculation, 310

- Dedekind-finite set, 370, 380
- Definability and $\text{Def}(M)$, 165
- dense set, 380

- diagonal lemma, 69, 380

- Elementary hulls, 154
- elementary substructure, 380

- Failure of Choice, 367
- First-order syntax, substitution, and equality, 12
- fixed-point lemma, 69
- forcing
 - name, 381
 - Power Set, 241
 - relation, 381
 - Replacement, 239
- Formal deduction, 20
- Formalized provability, 86
- Foundation and rank, 112
- Fraenkel–Mostowski precursor, 323
- Fundamental Theorem of Forcing, 232

- GCH, 202, 380
- GCH in L , 199
- General forcing relation, 222
- generic filter, 381
- Gödel completeness theorem, 28
- Gödel incompleteness theorem
 - first, 75
 - second, 92
- Gödel numbering, 381
- Gödel numbering of syntax, 57
- Gödel’s first theorem, 72

- Henkin completeness, 25
- Homogeneity and relative consistency, 316

- independence
 - Axiom of Choice, 377
 - CH, 318
- inner model, 381
- Inner models and the Skolem paradox, 158

- Lower bound for the continuum, 303
- Löb and second incompleteness, 90
- Löb theorem, 91
- Löwenheim–Skolem theorem
 - downward, 33
 - upward, 33

- Mostowski collapse theorem, 130, 381
- Names and valuation, 212
- Natural numbers and recursion, 108
- nice name, 288, 381
- Nice names and preservation bookkeeping, 287
- omega-consistency, 381
- Ordinal arithmetic, 121
- Ordinals, 117
- Posets, dense sets, and generics, 207
- Primitive recursion, 38
- primitive recursive, 381
- Proof and provability predicates, 61
- Propositional proof systems, 8
- provability predicate, 63, 381
- Recursive and computably enumerable relations, 46
- Reflection, 151
- Reflection theorem, 152, 381
- relative consistency, 381
 - forcing transfer, 280
 - not CH, 318
- relativization, 145
- Representability, 53
- Rosser and essential undecidability, 77
- Rosser incompleteness theorem, 79
- satisfaction, 381
- Satisfaction for set-sized structures, 143
- Self-reference, 67
- Separative quotients and Boolean completion, 247
- Skolem hull, 155, 381
- Skolem paradox, 160
- Structures and satisfaction, 16
- support, 381
- Symmetric and hereditarily symmetric names, 332
- symmetric model
 - Power Set, 357
 - Replacement, 356
- symmetric name, 381
- Syntax and propositional semantics, 3
- Tarski undefinability theorem, 96
- Tarski–Vaught criterion, 32
- The basic Cohen symmetric model, 362
- The ccc, 299
- The L-hierarchy, 169
- transitive model, 381
- Truth and generic extension theorem, 230
- Truth Lemma
 - forcing, 231, 381
- Truth, reflection, and iterated consistency, 95
- Uniform definability and absoluteness of L, 173
- Upper bound by nice names, 307
- Weak arithmetic, 49
- Well-founded relations and collapse, 129
- ZF axioms and elementary constructions, 103
- ZF axioms in L, part I, 179
- ZF axioms in L, part II, 184
- ZF in a symmetric extension, elementary axioms, 349
- ZF in a symmetric extension, hard axioms, 355
- ZFC in generic extensions, 237